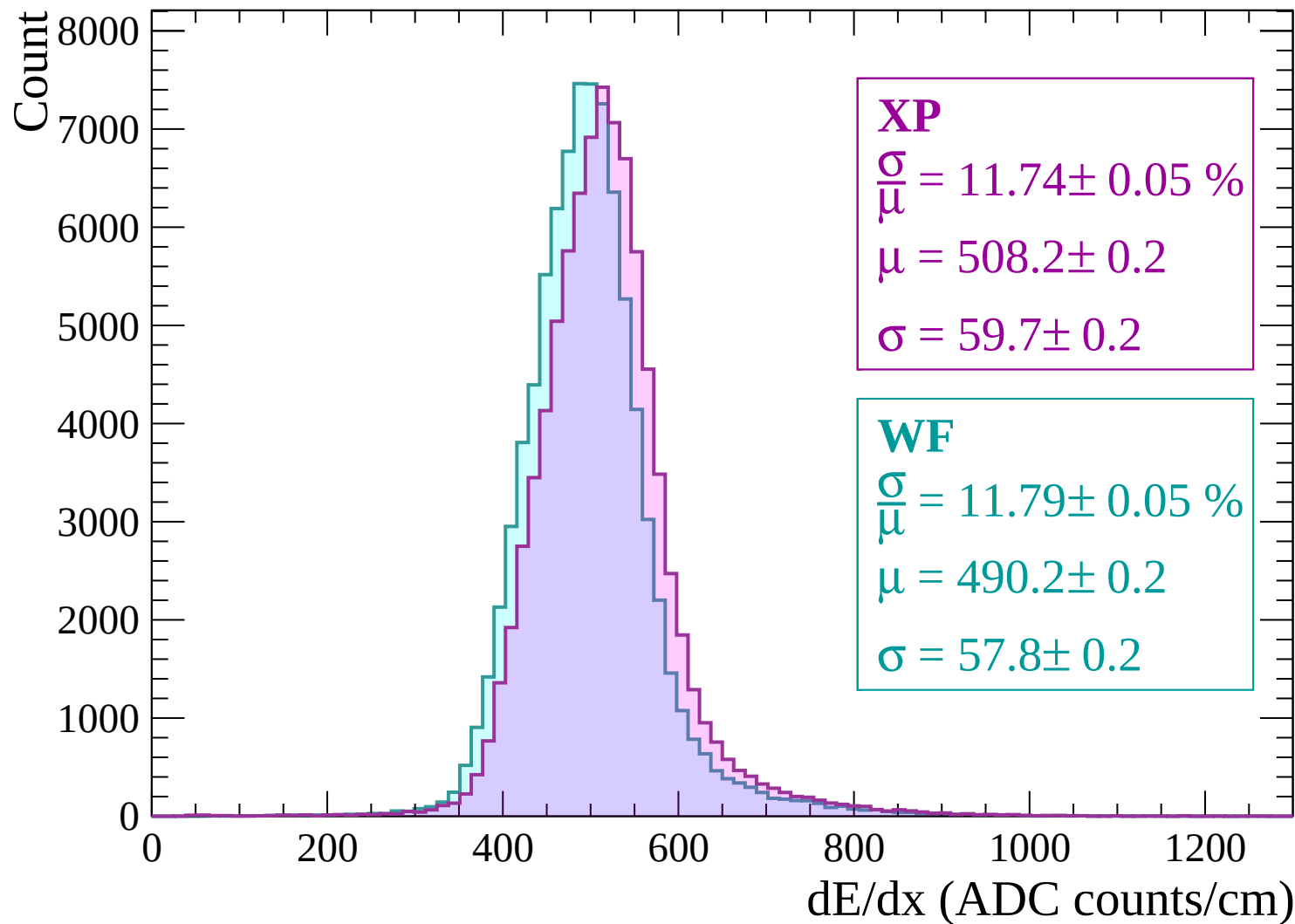
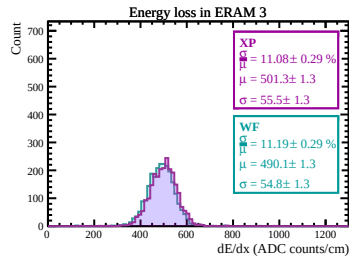
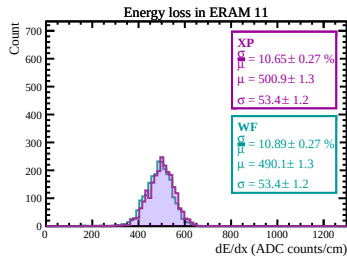
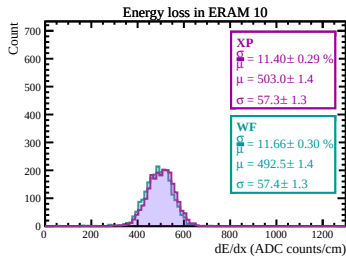
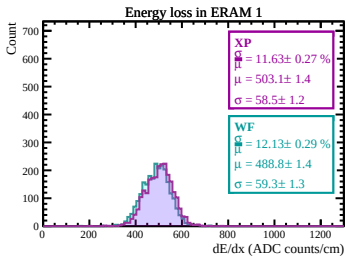
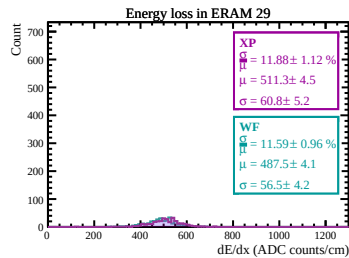
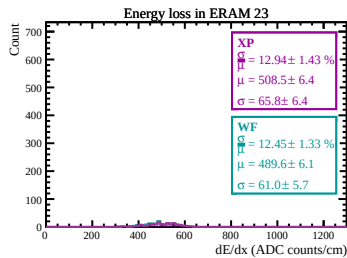
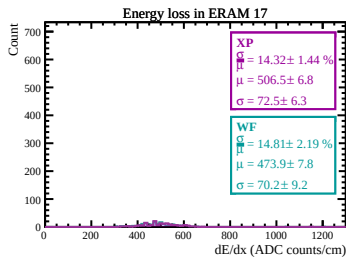
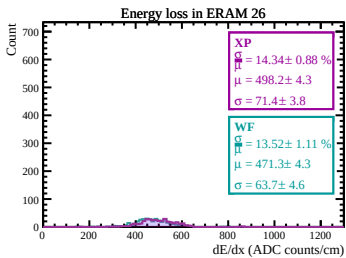
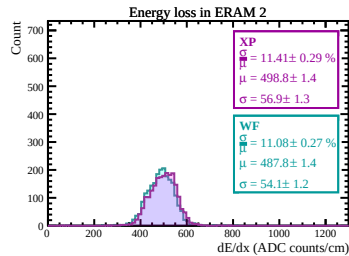
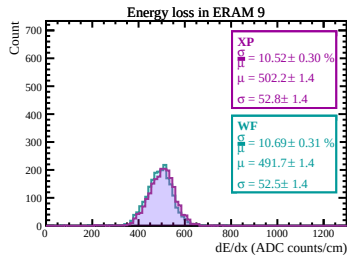
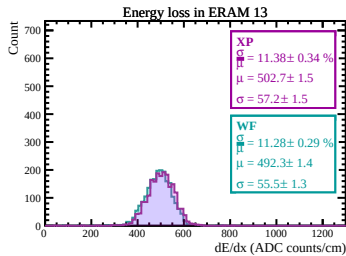
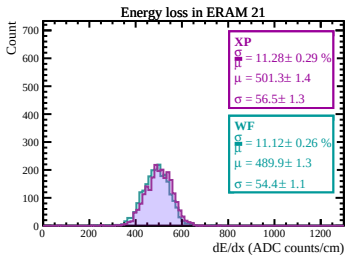
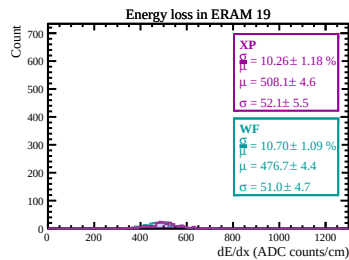
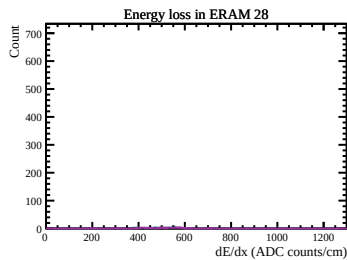
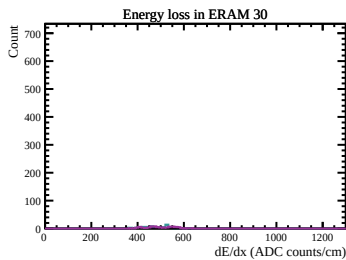
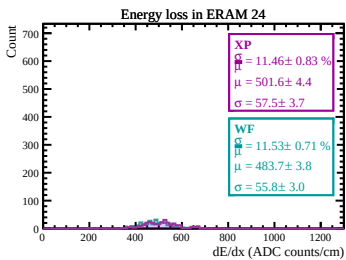
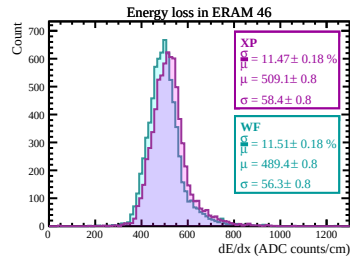
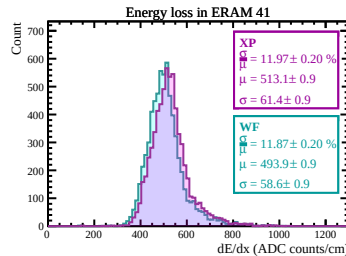
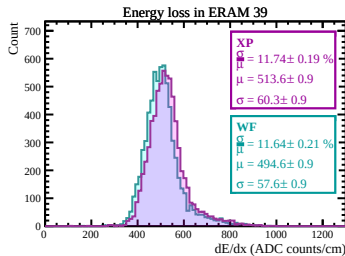
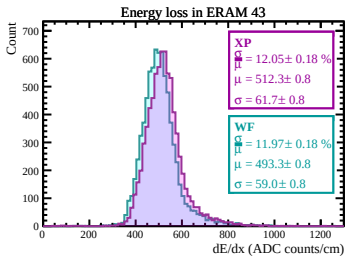
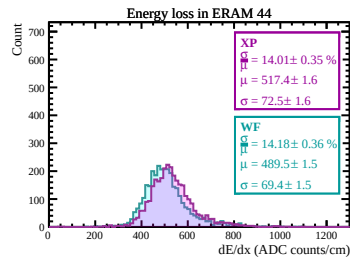
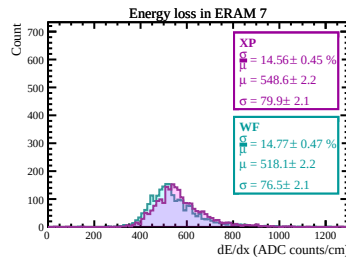
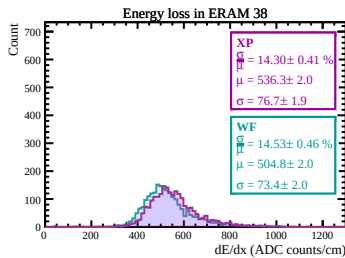
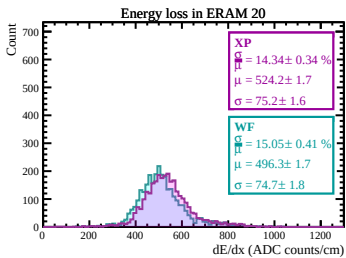
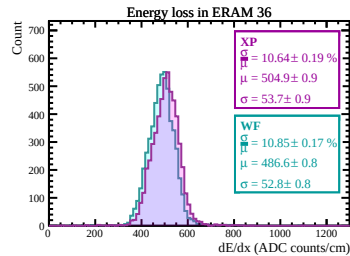
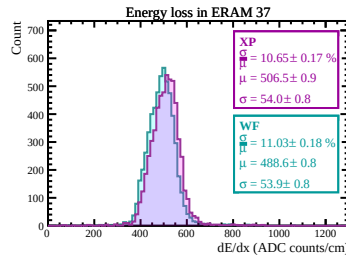
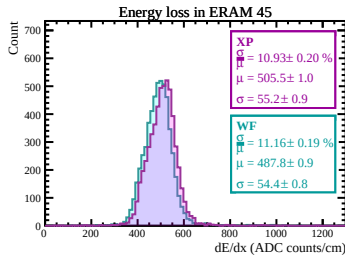
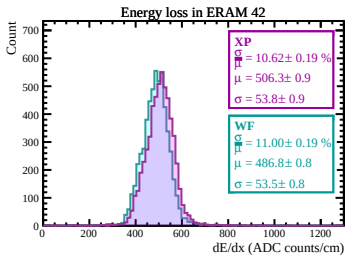
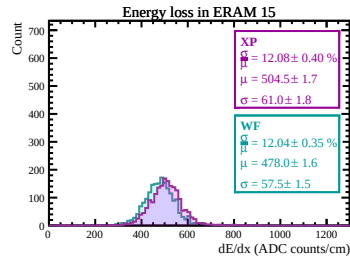
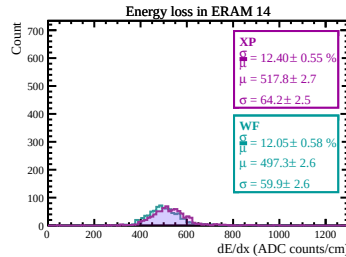
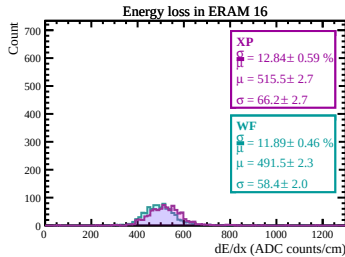
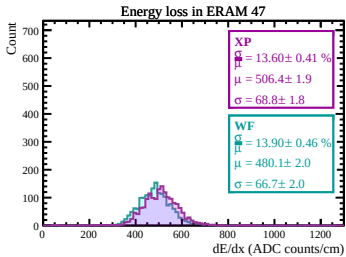
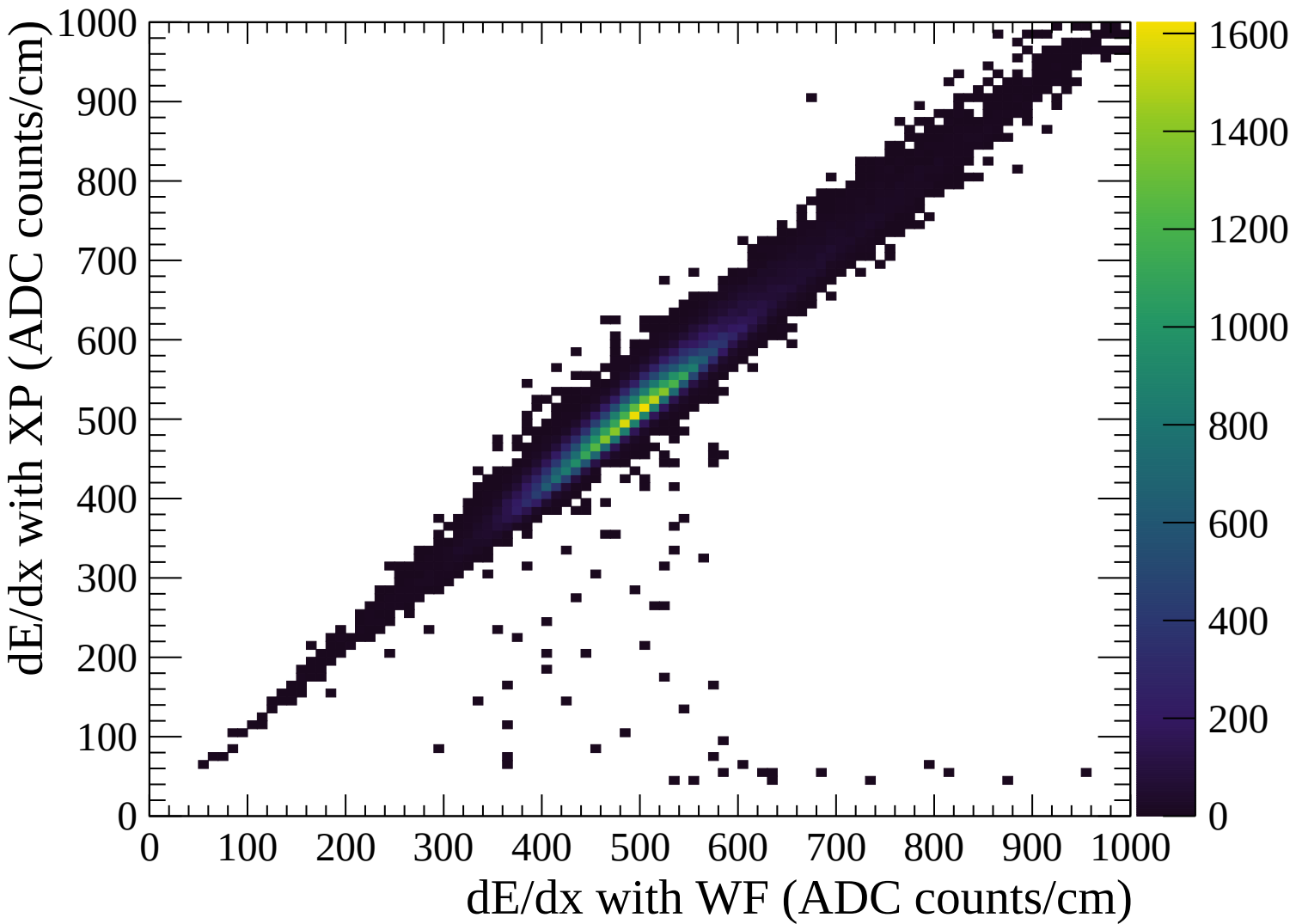
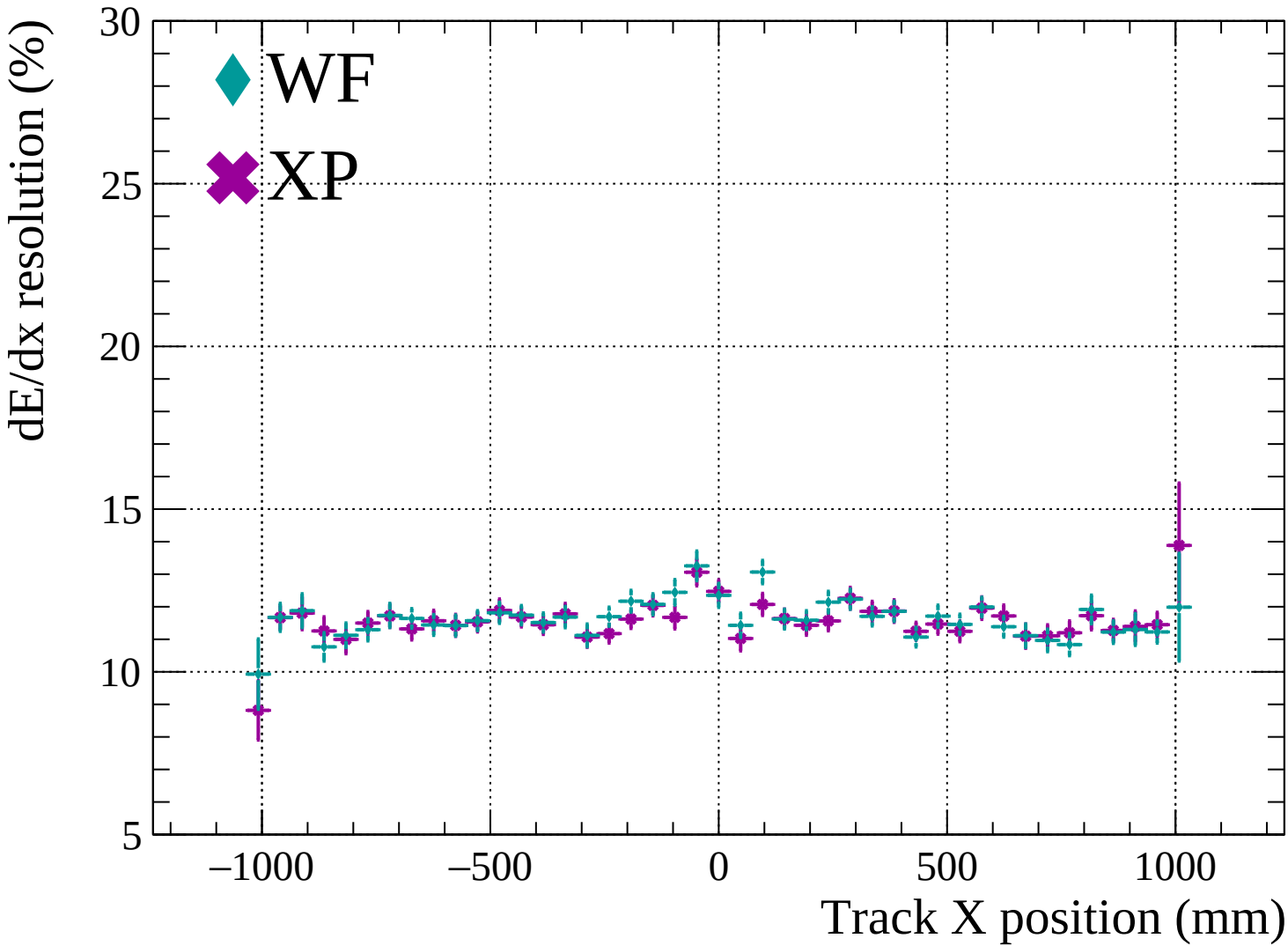


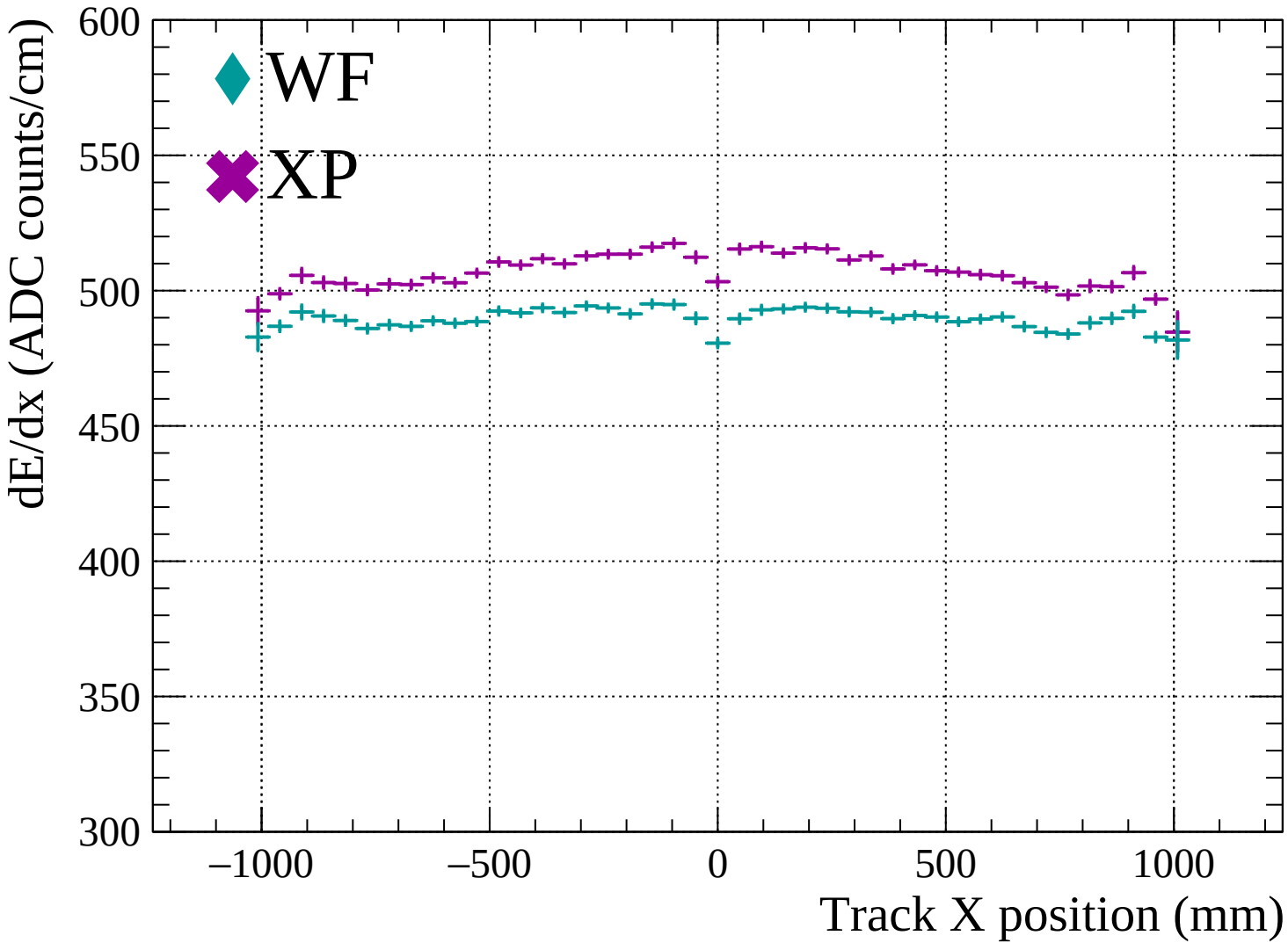


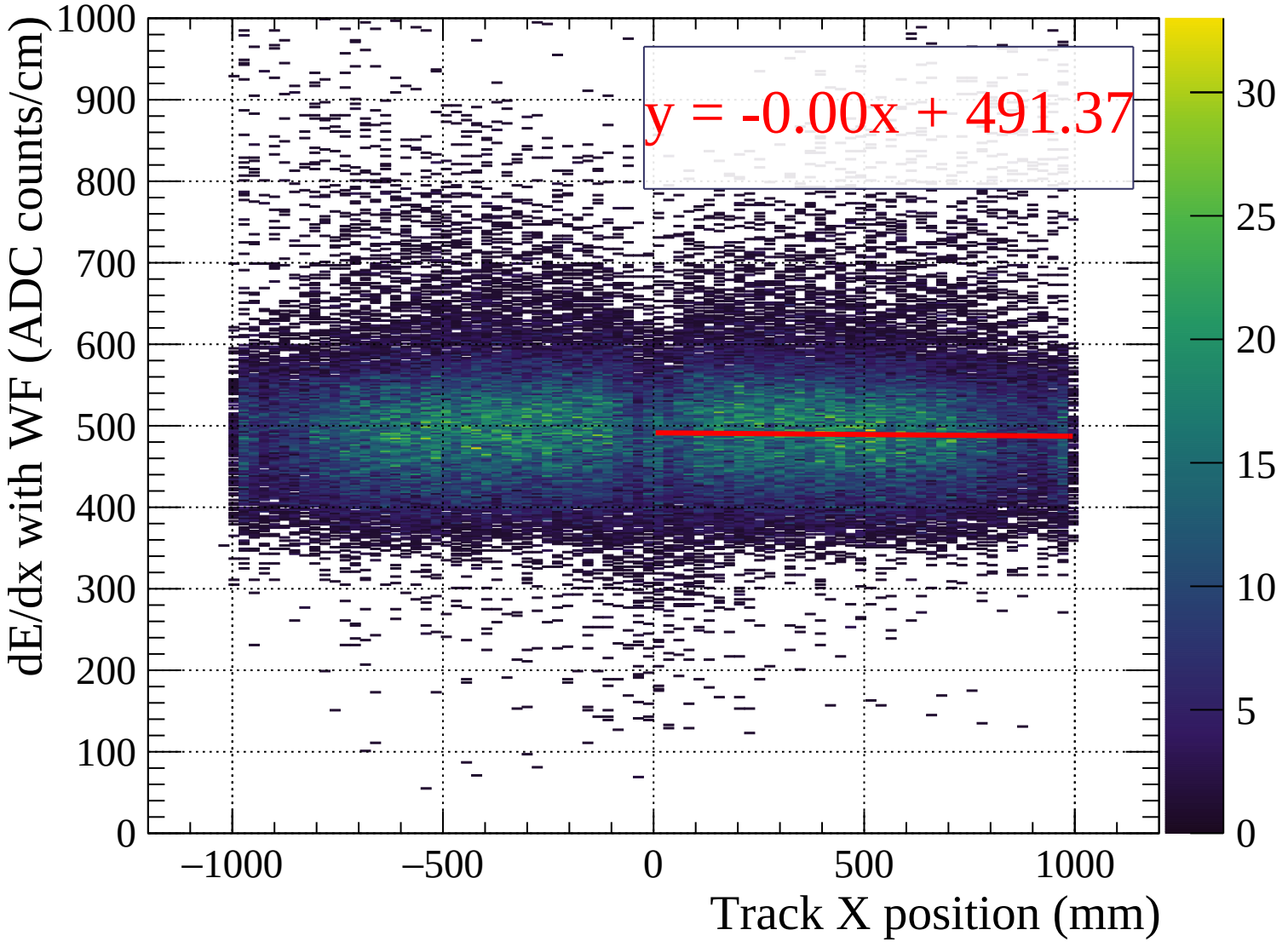


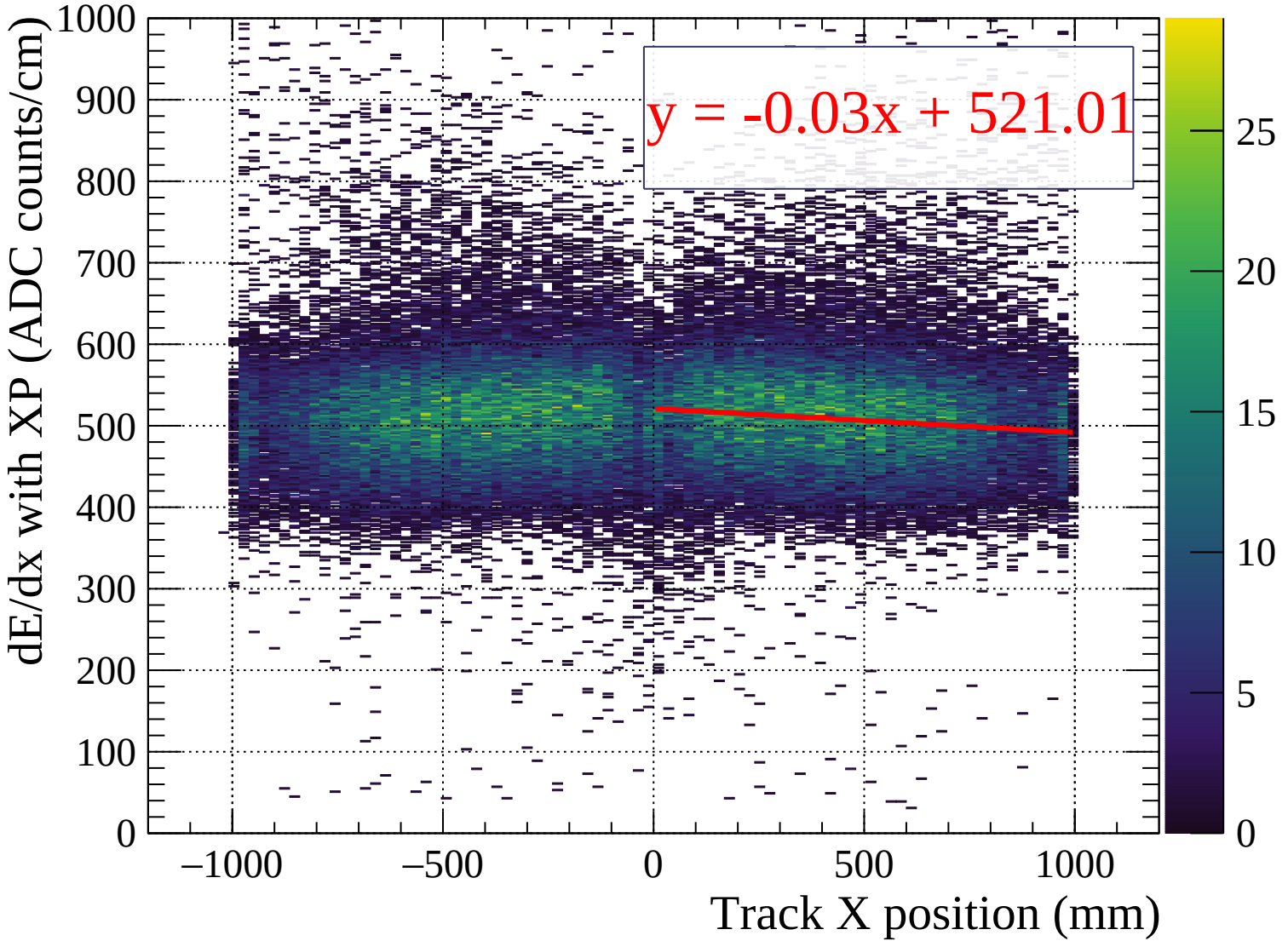


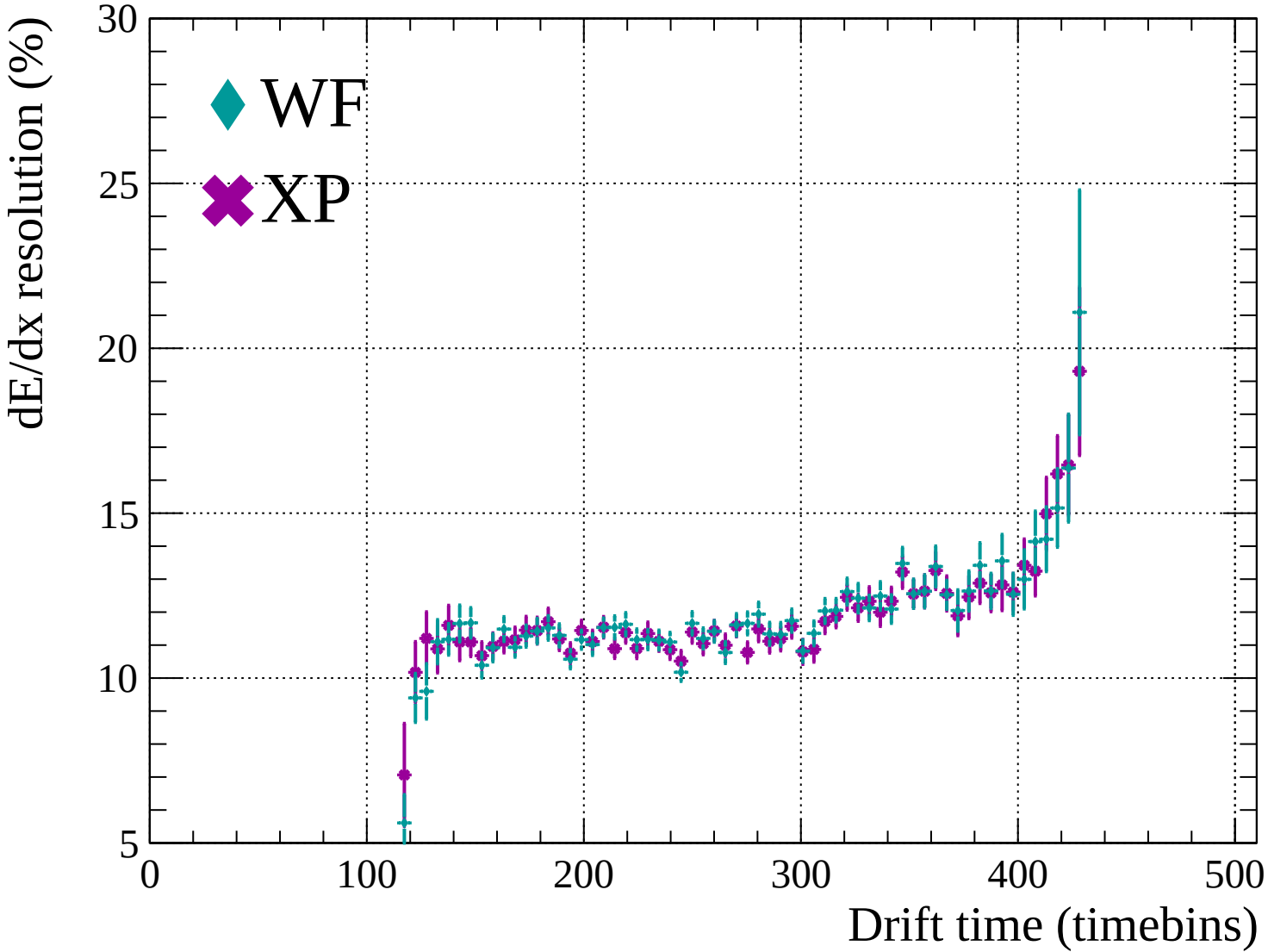


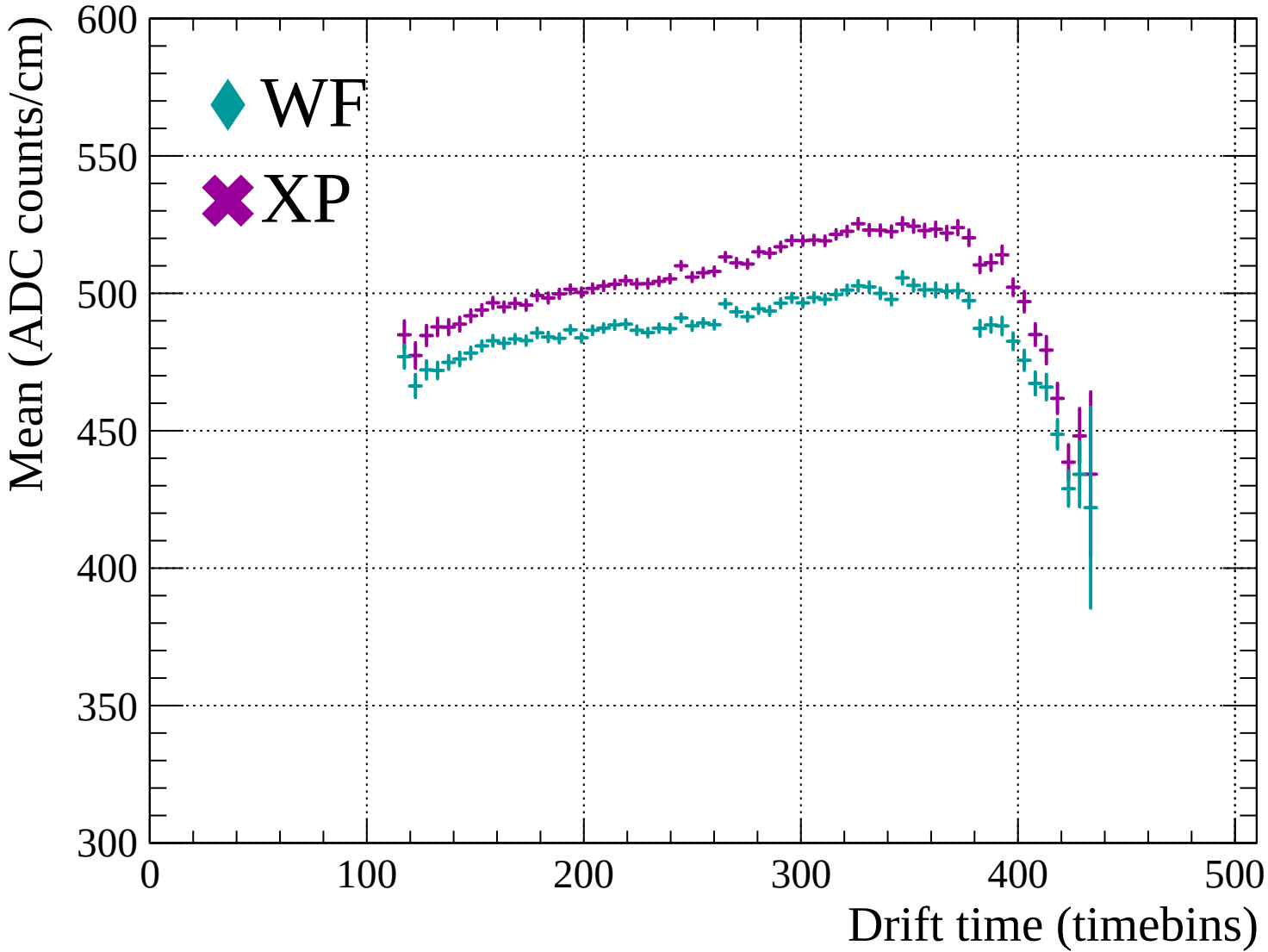


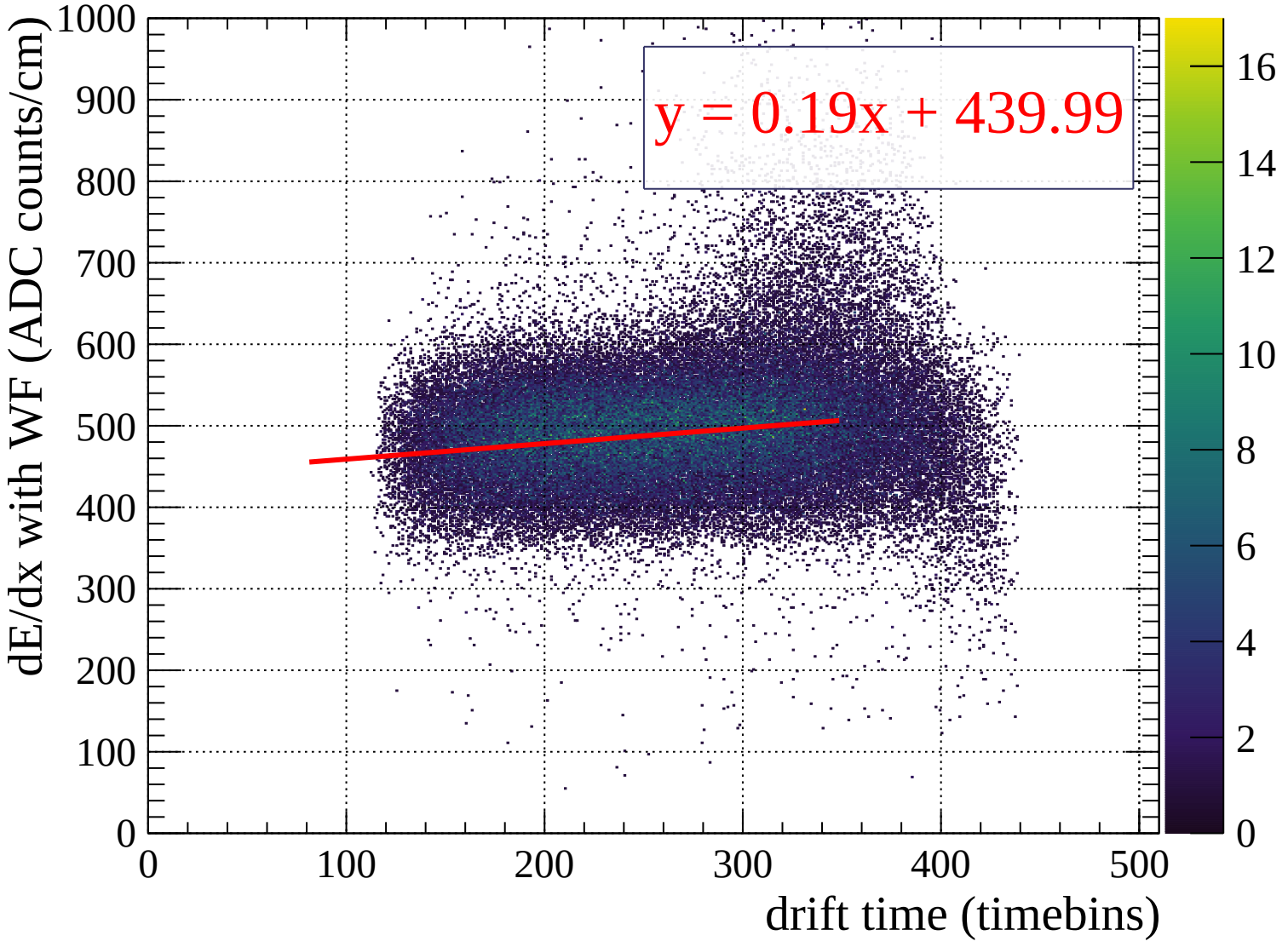


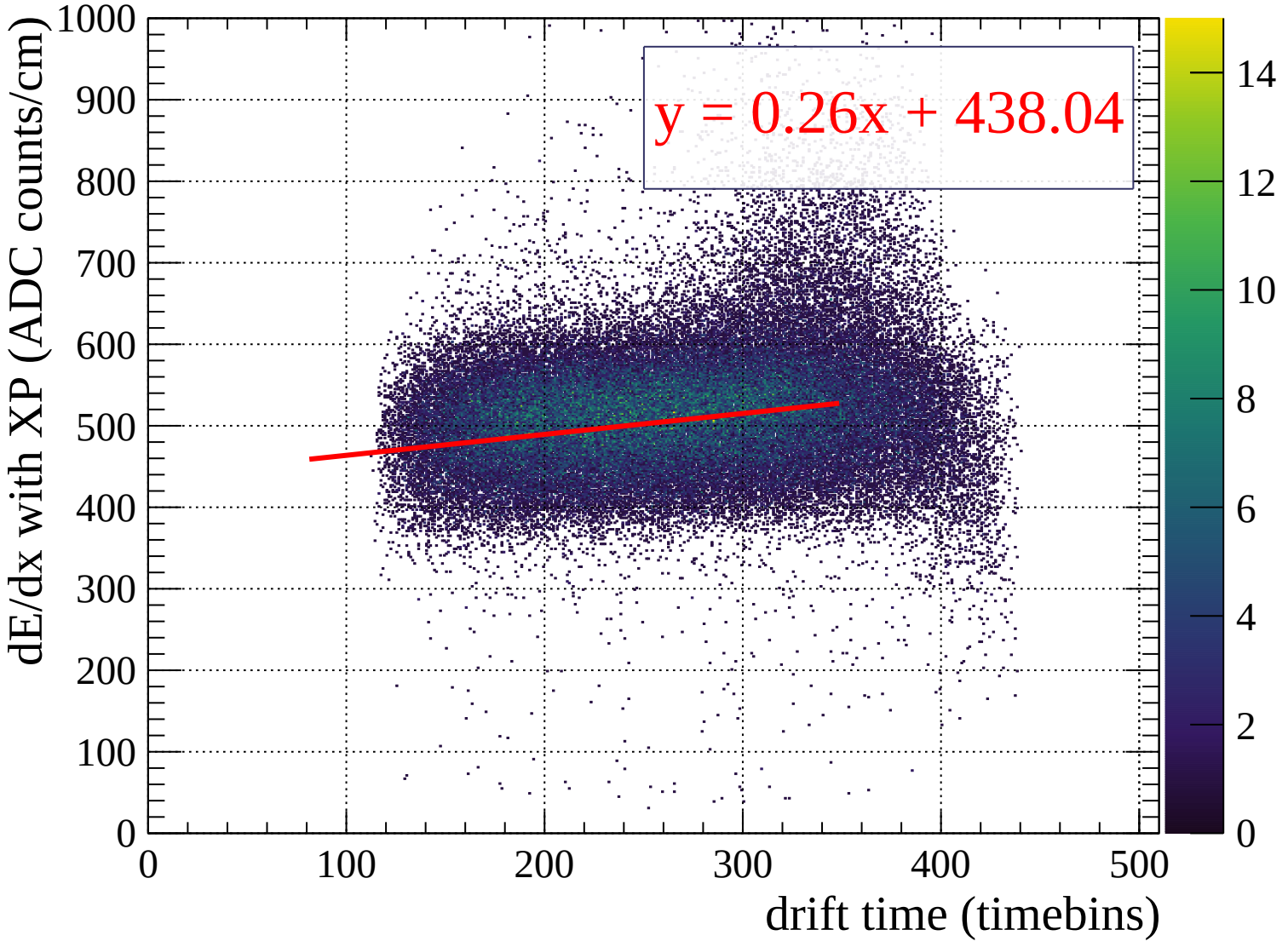


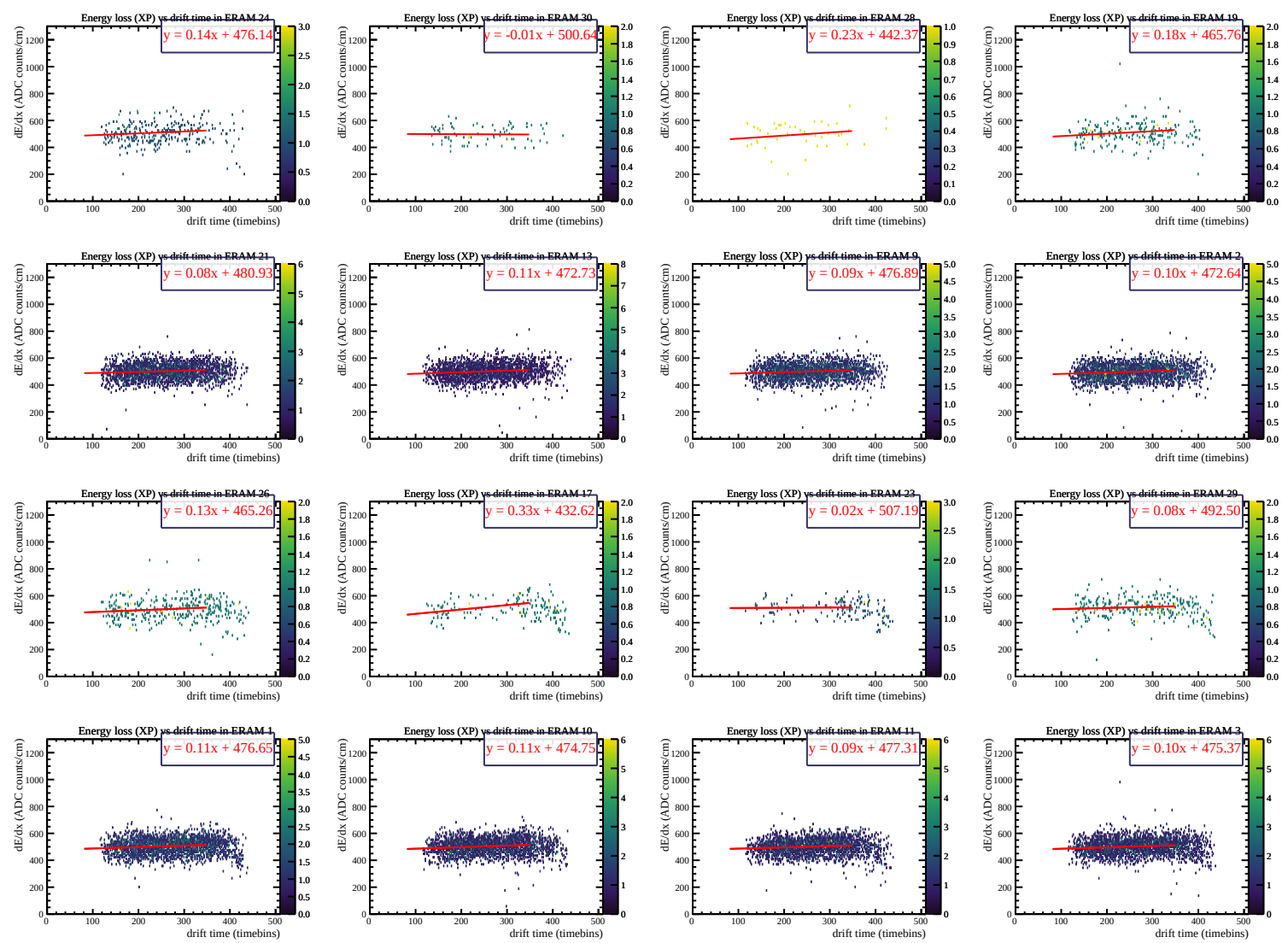


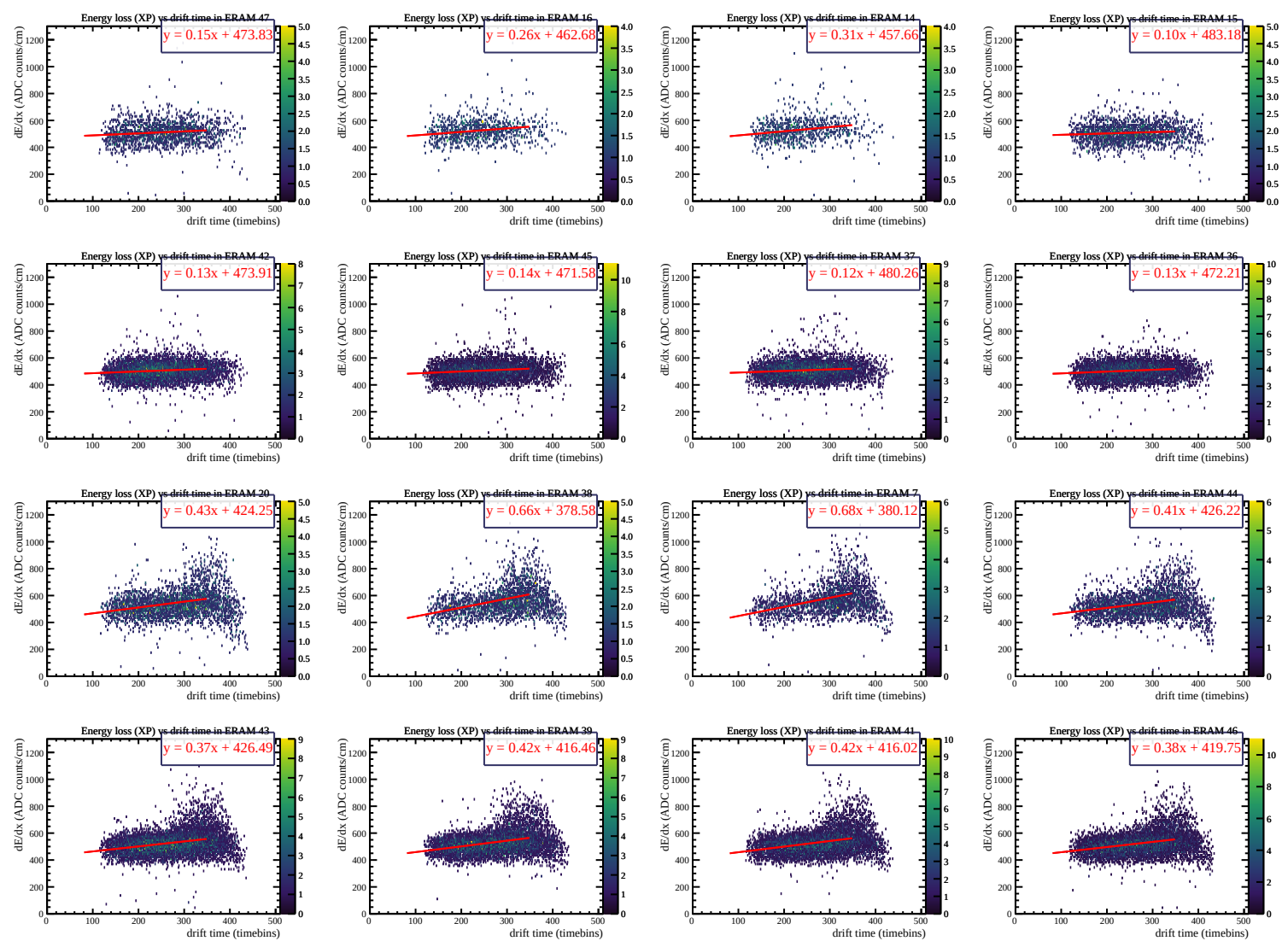


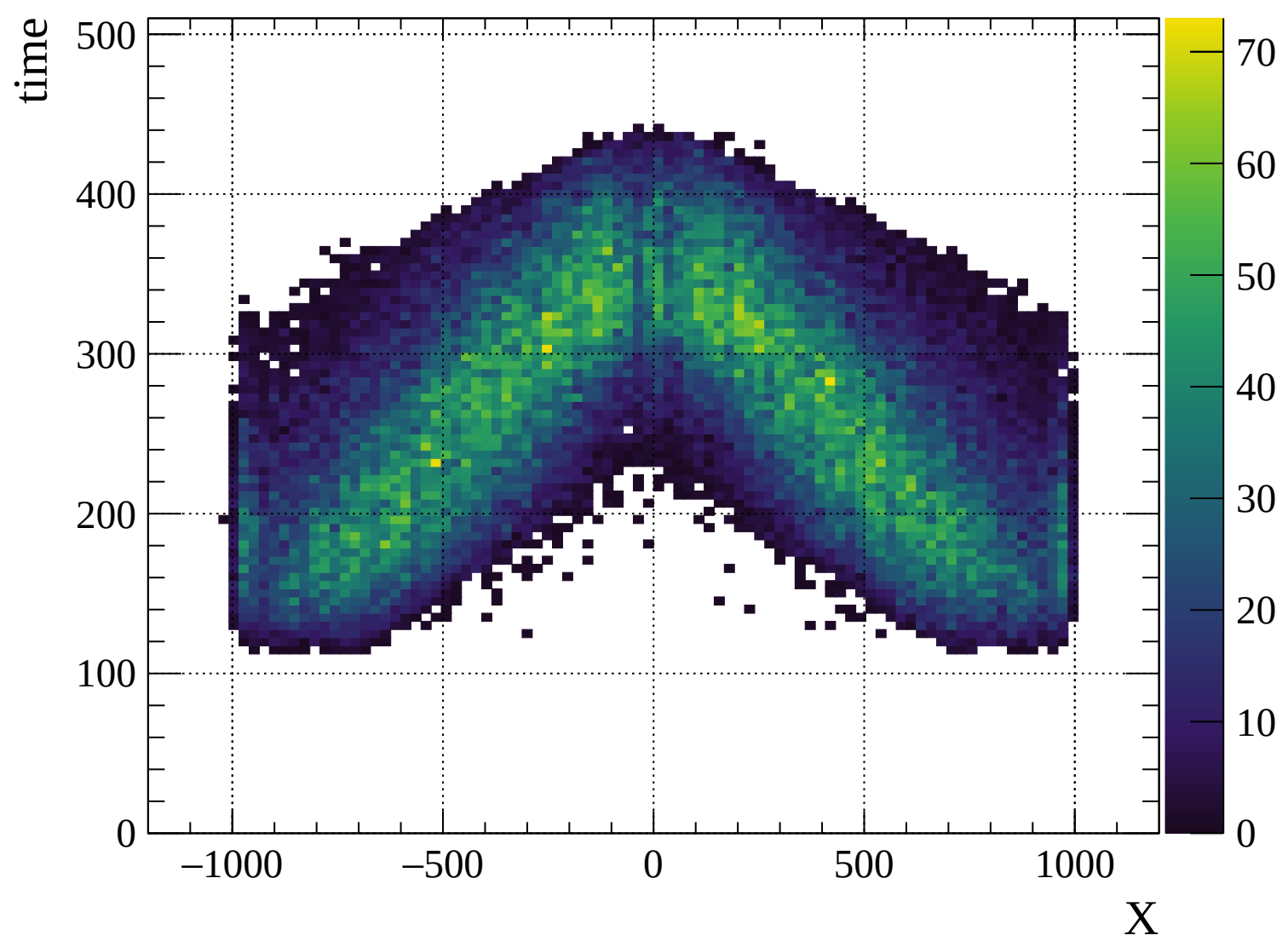


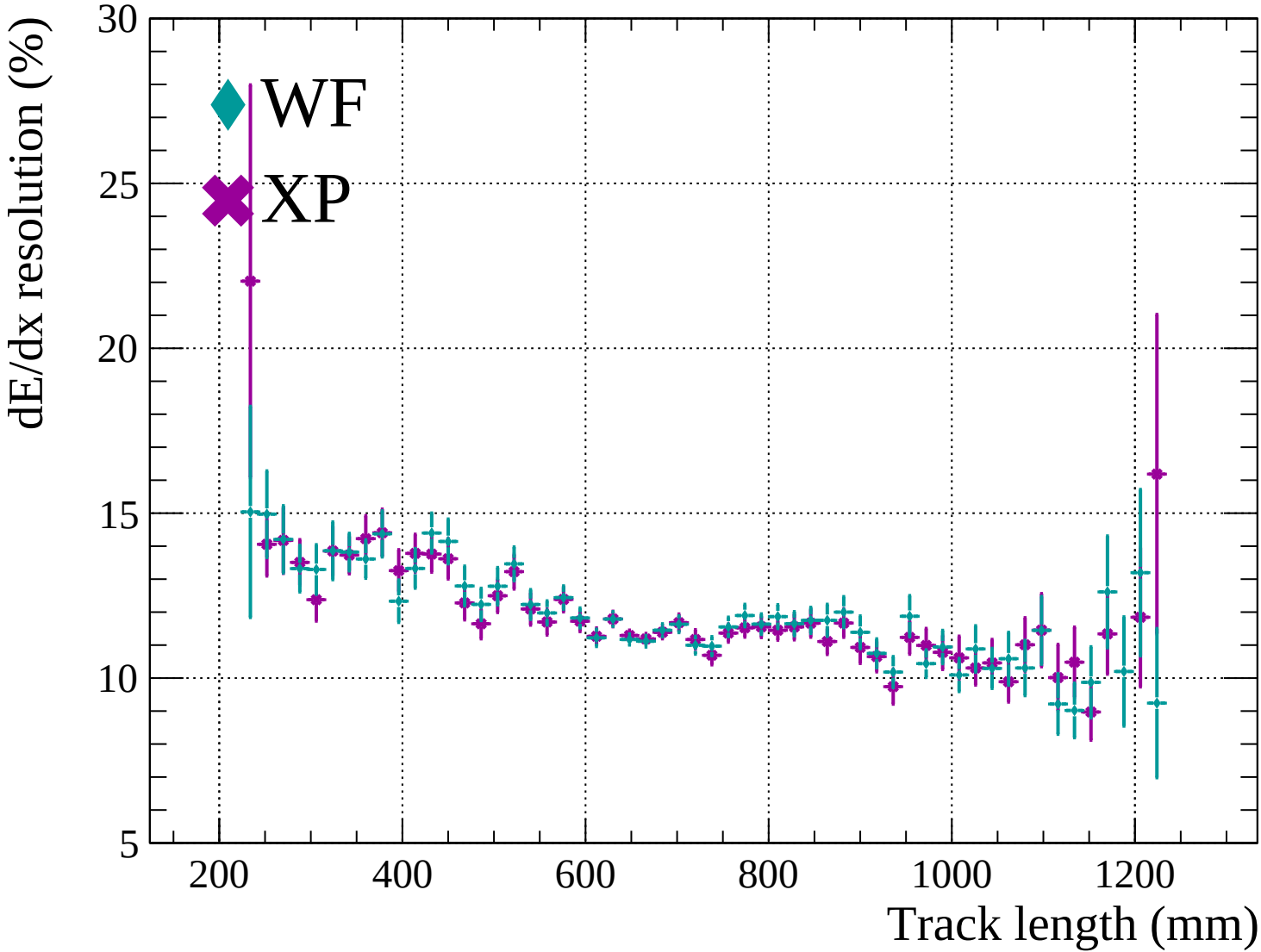


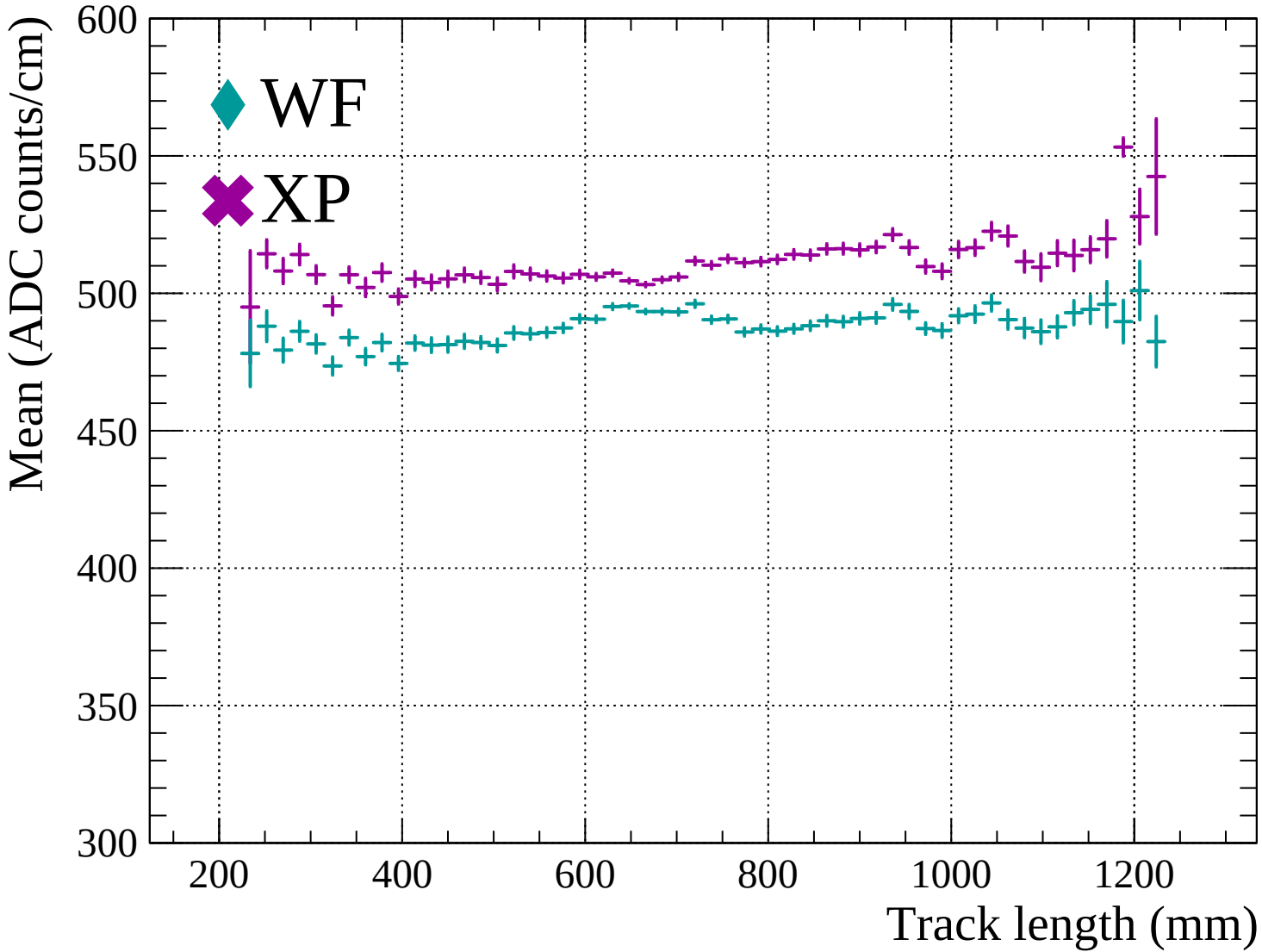


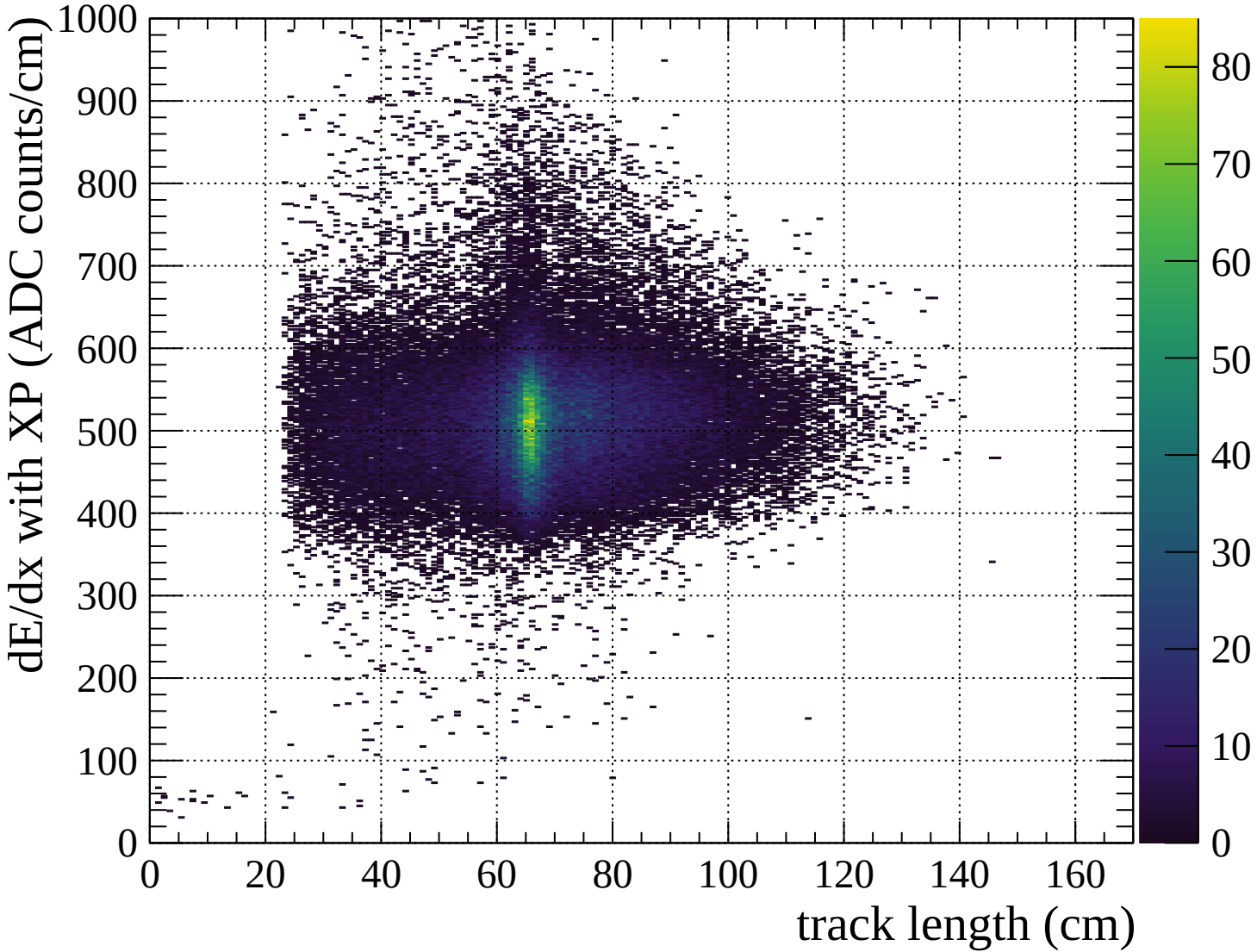


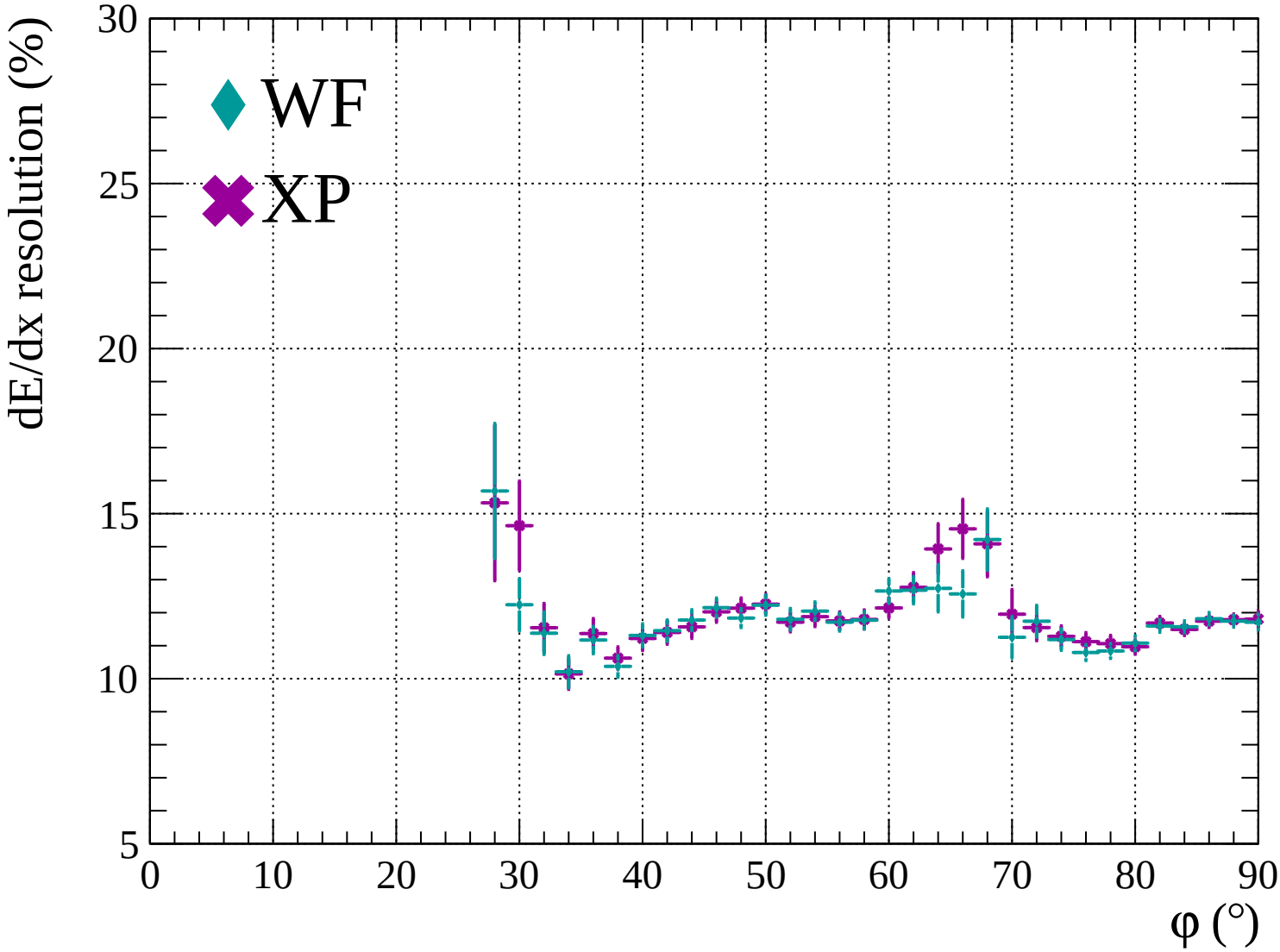


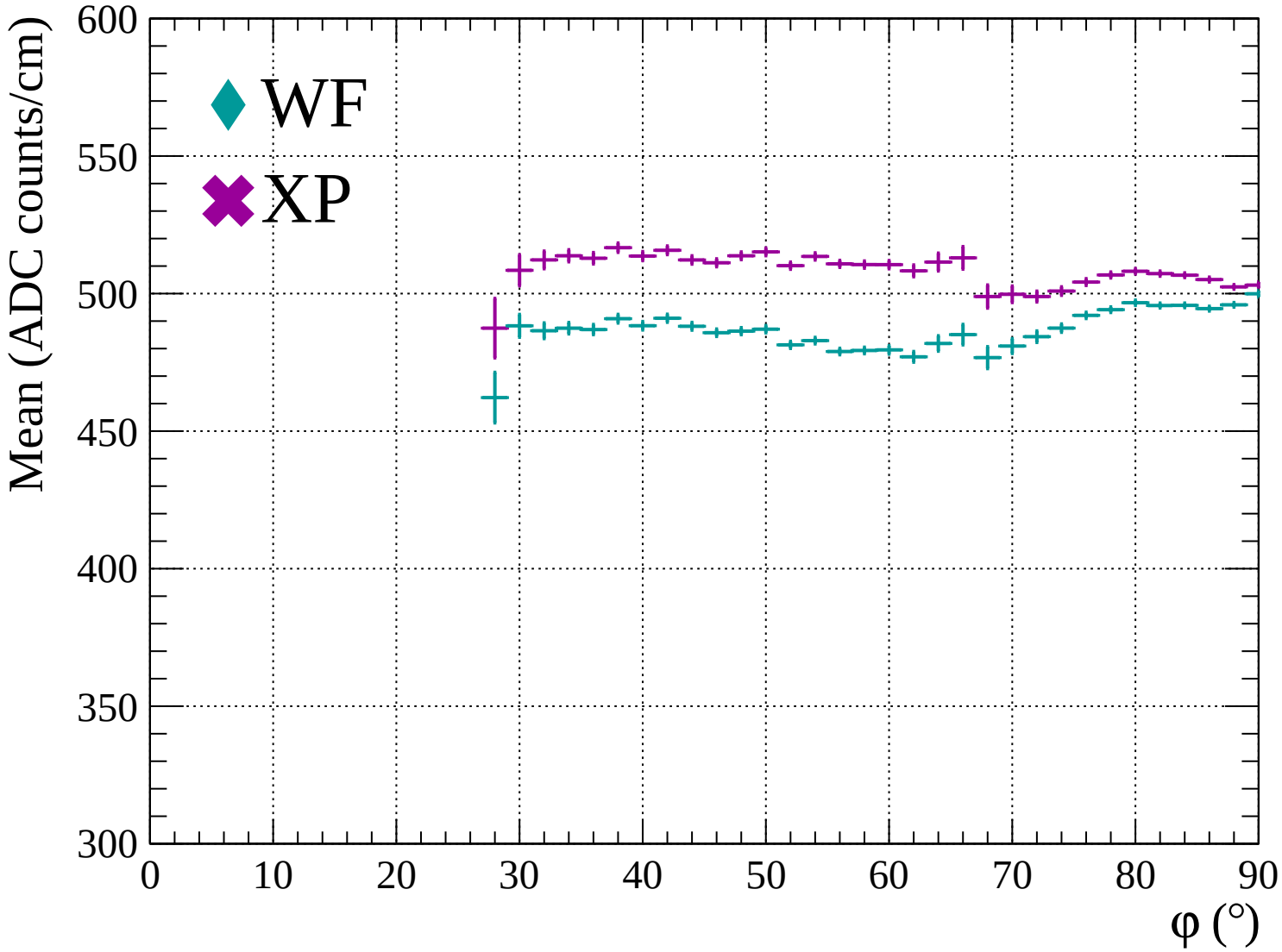


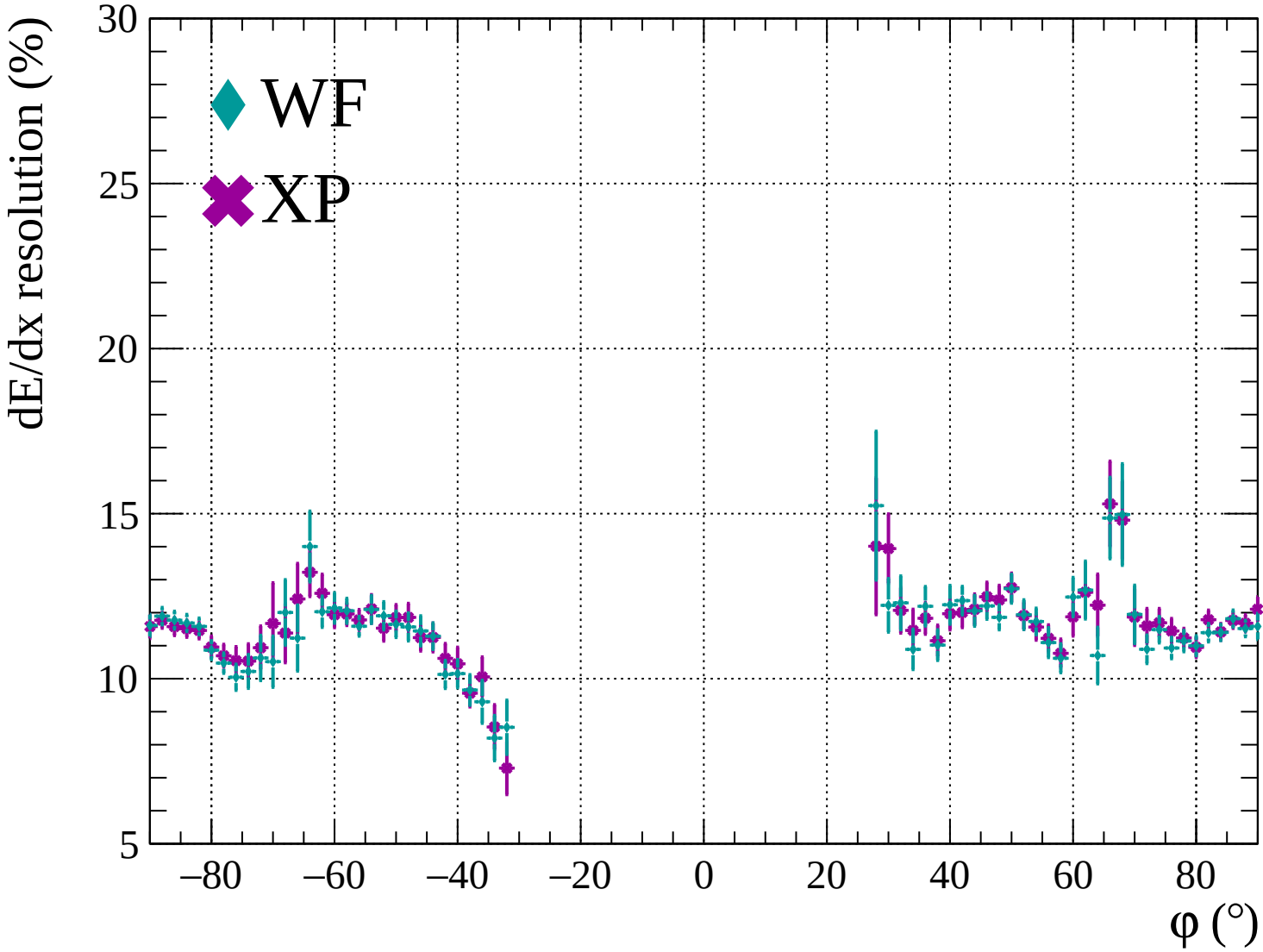


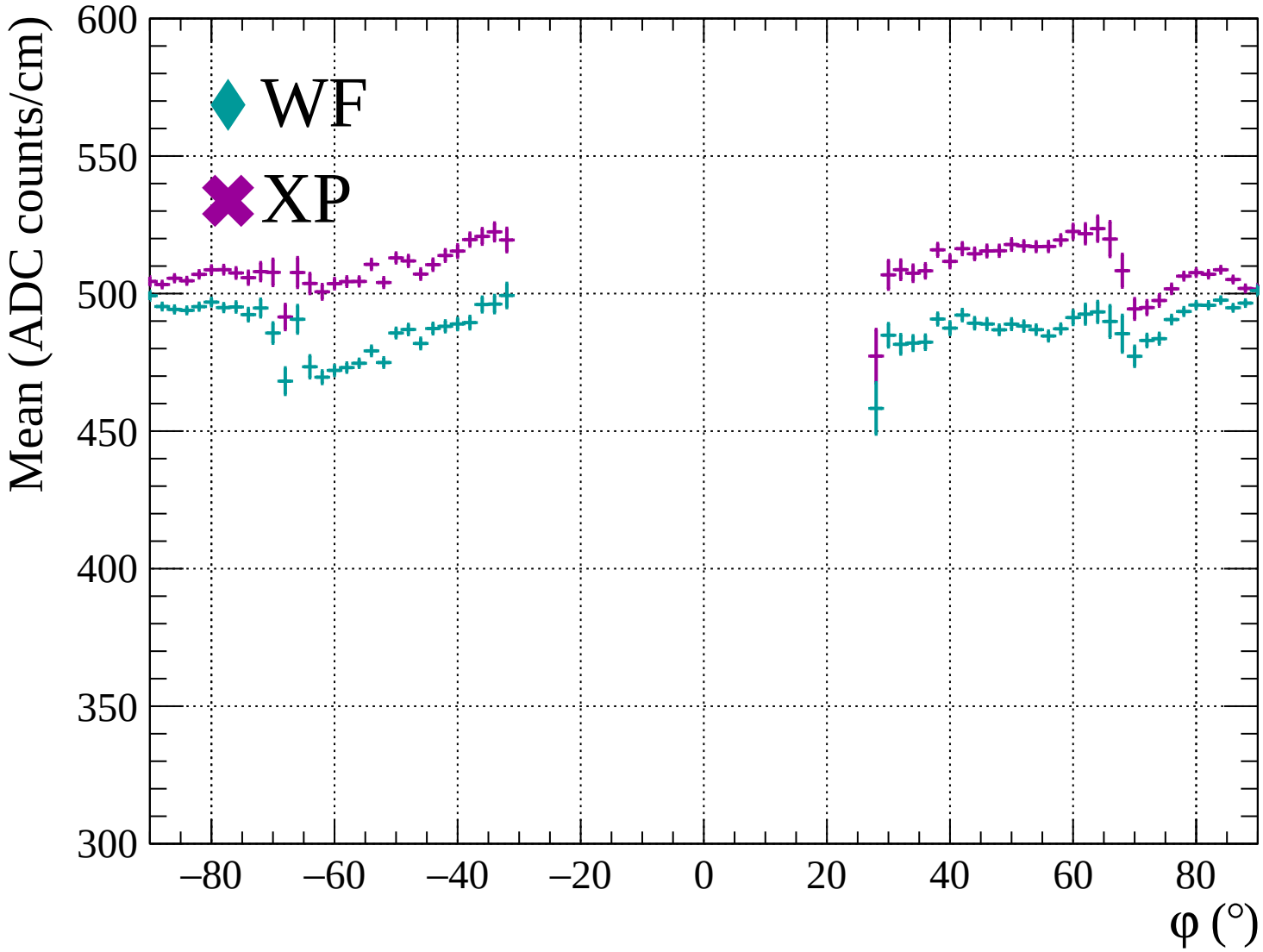


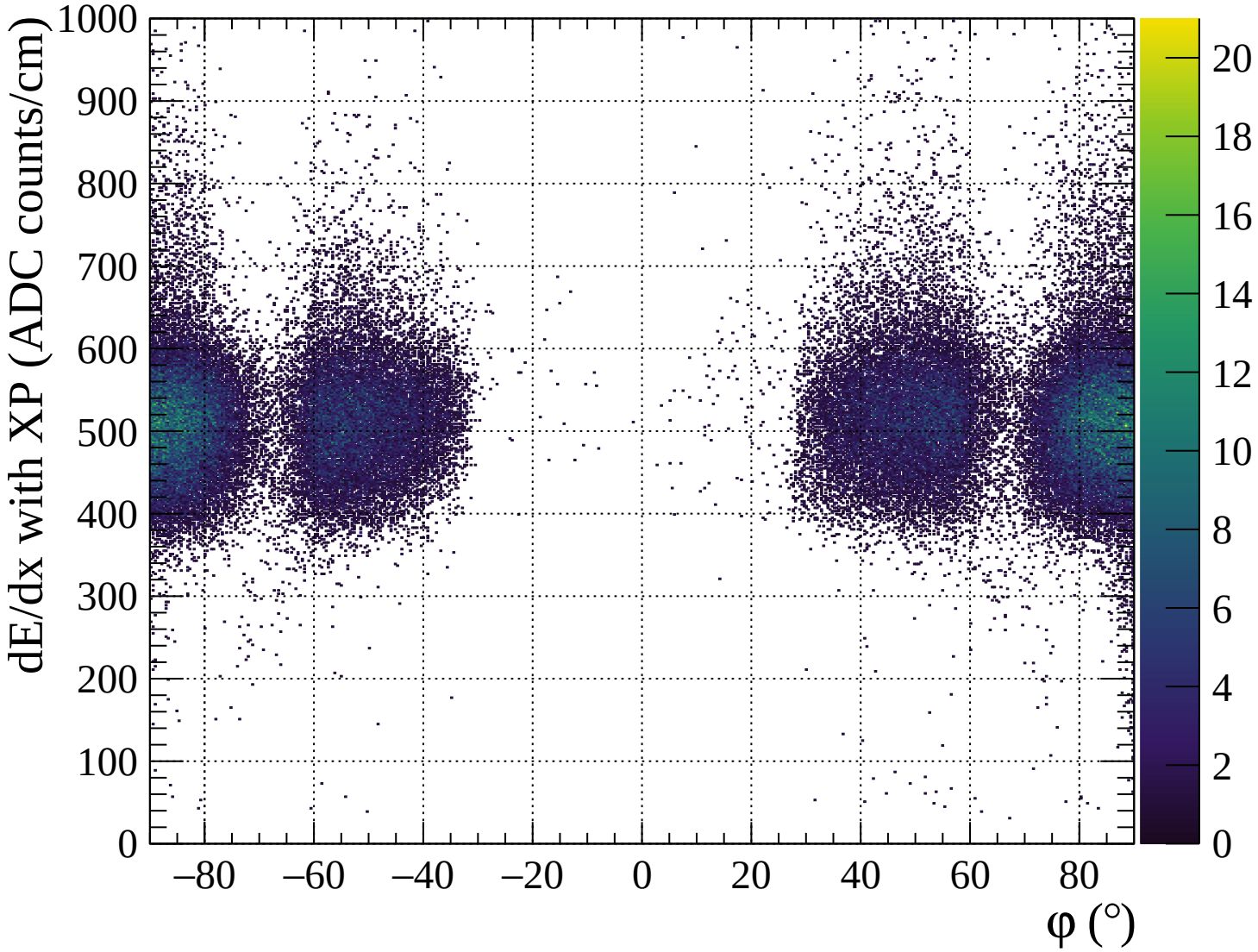


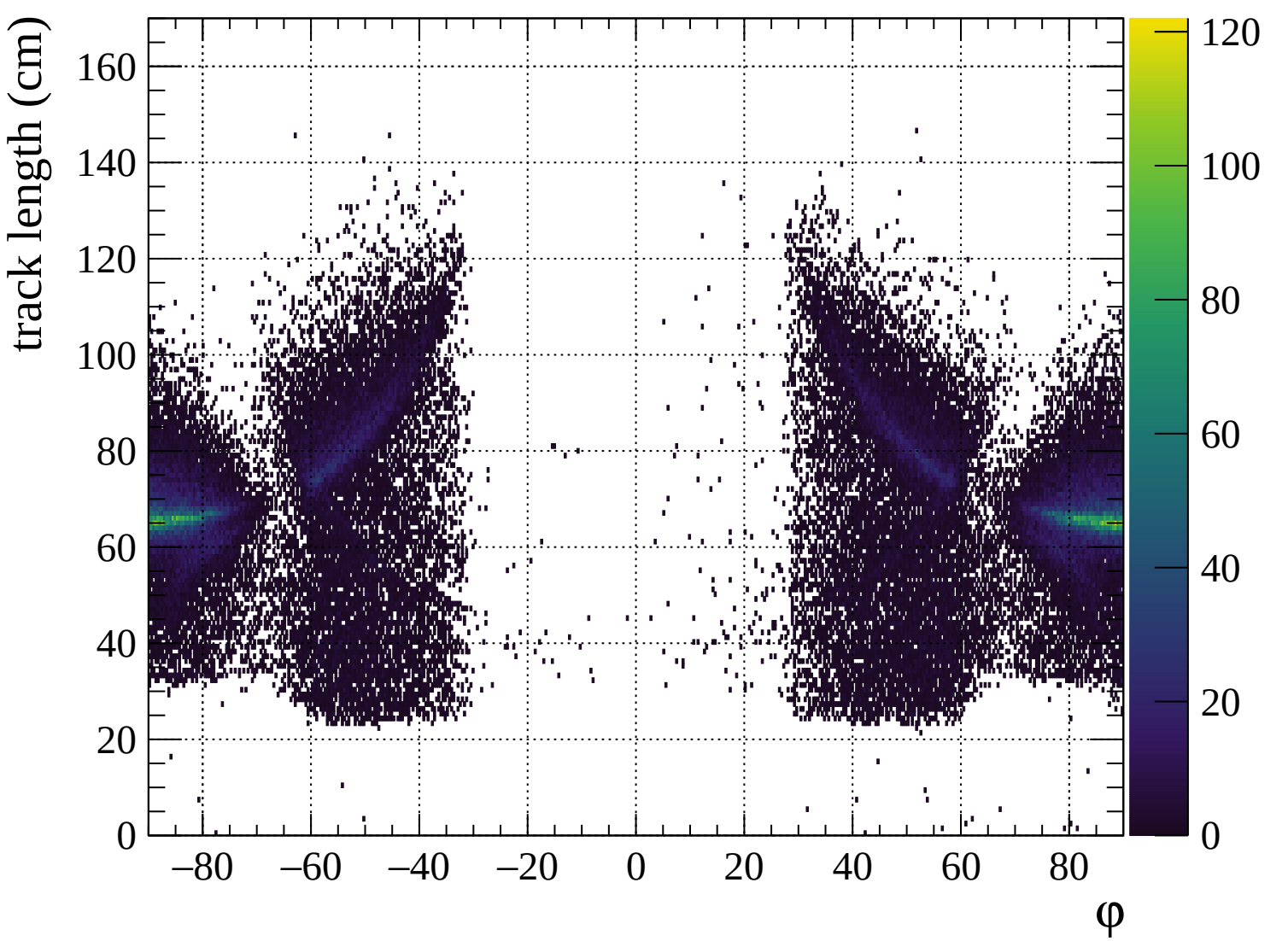


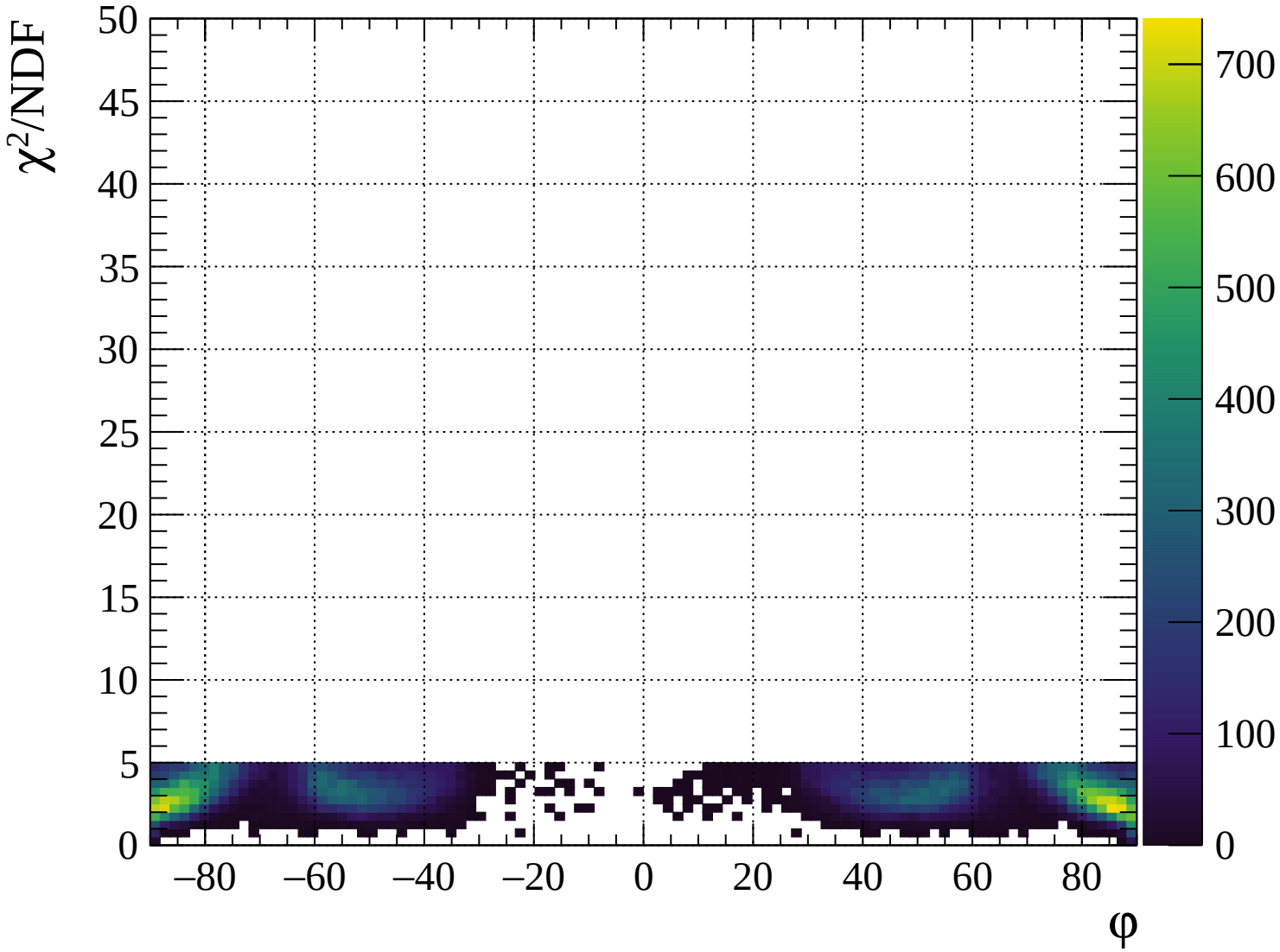


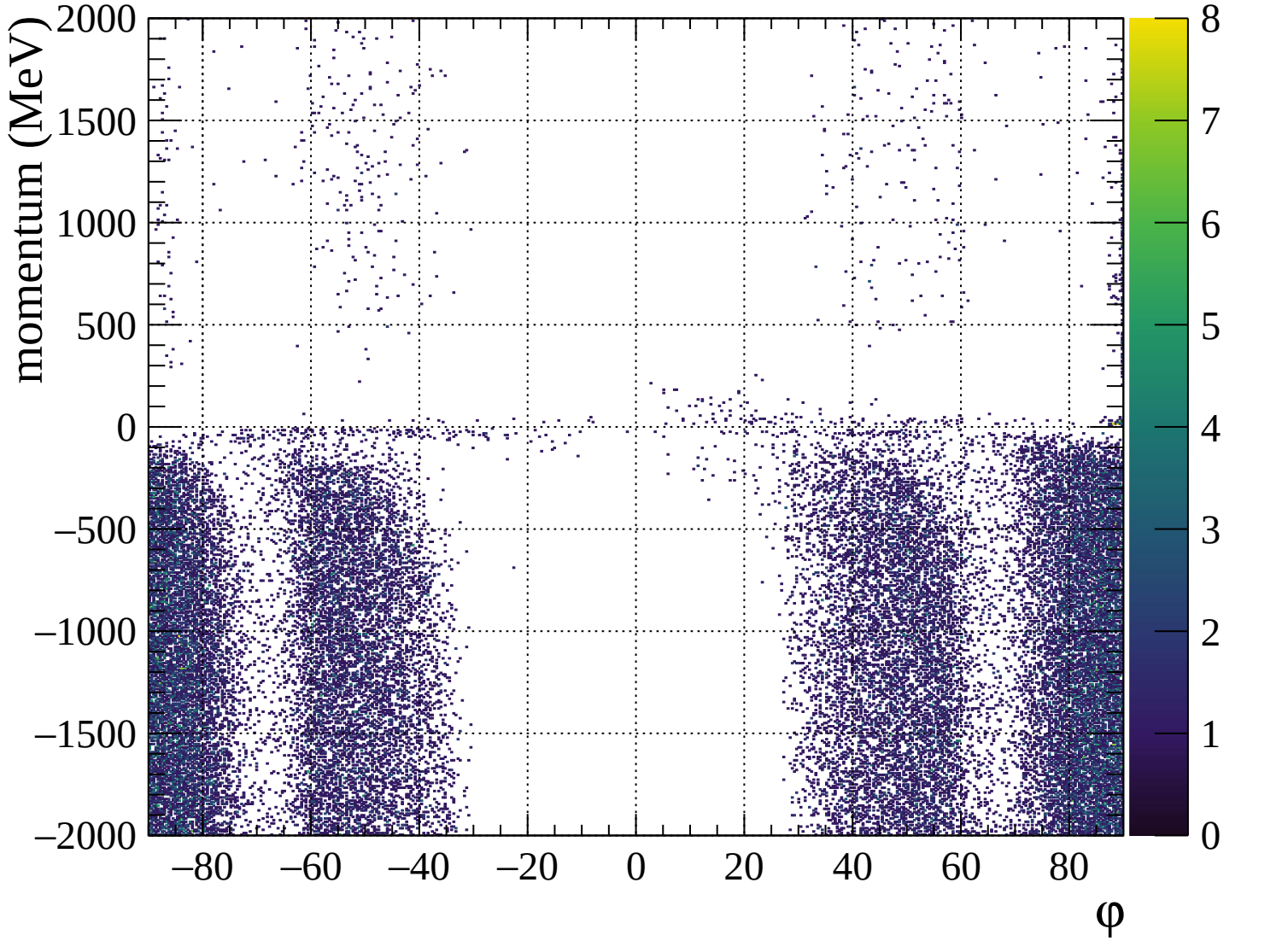


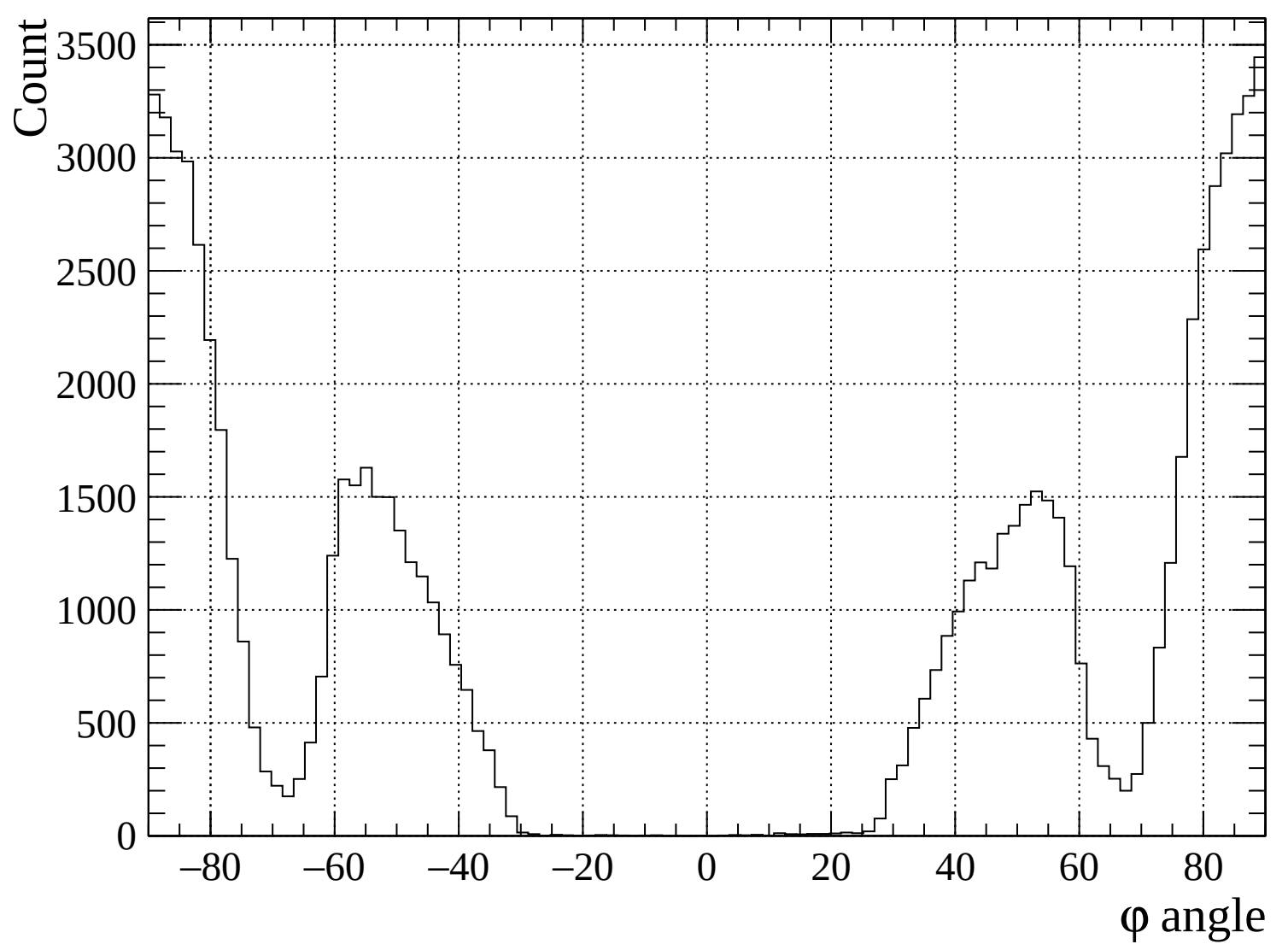


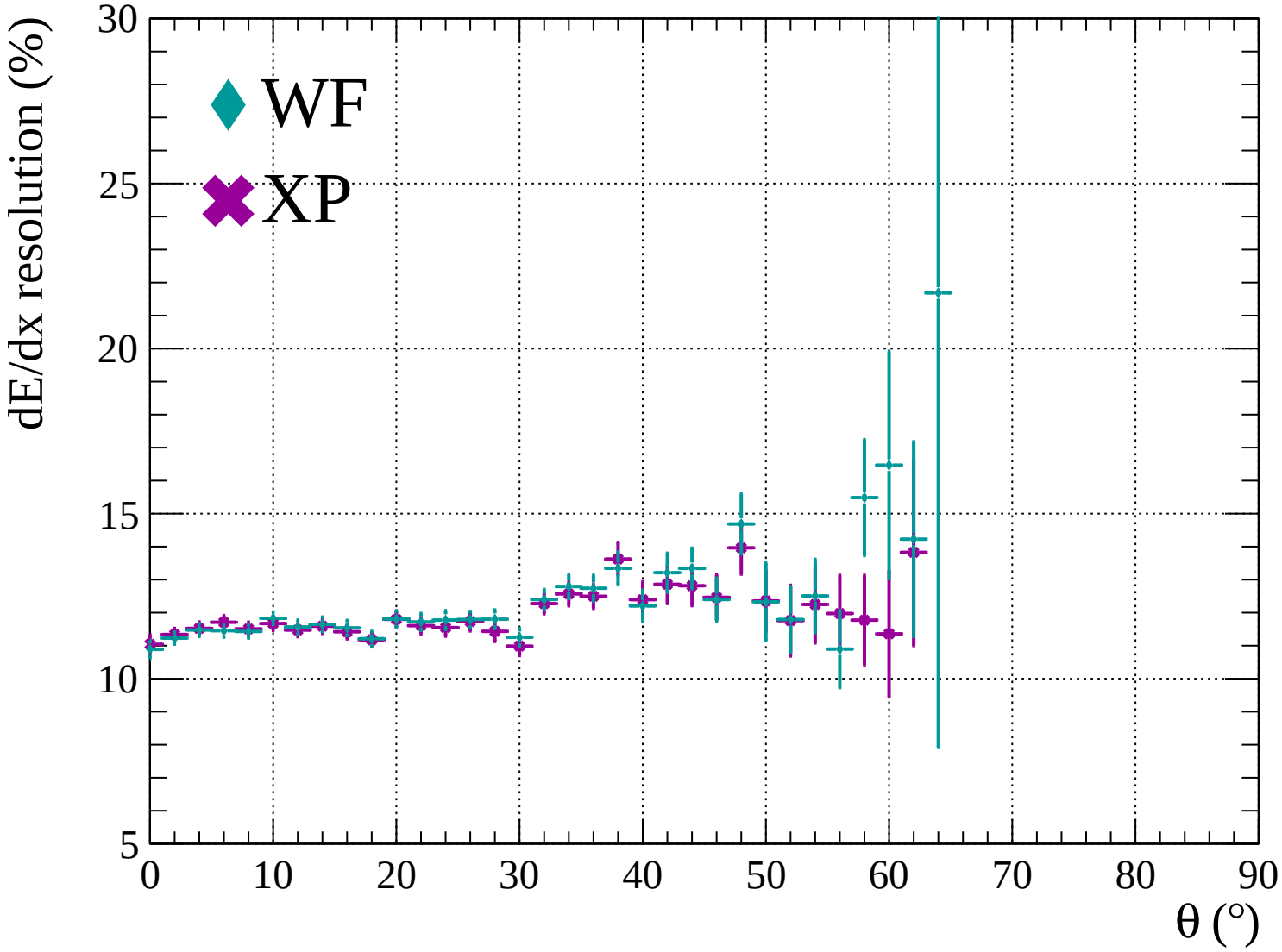


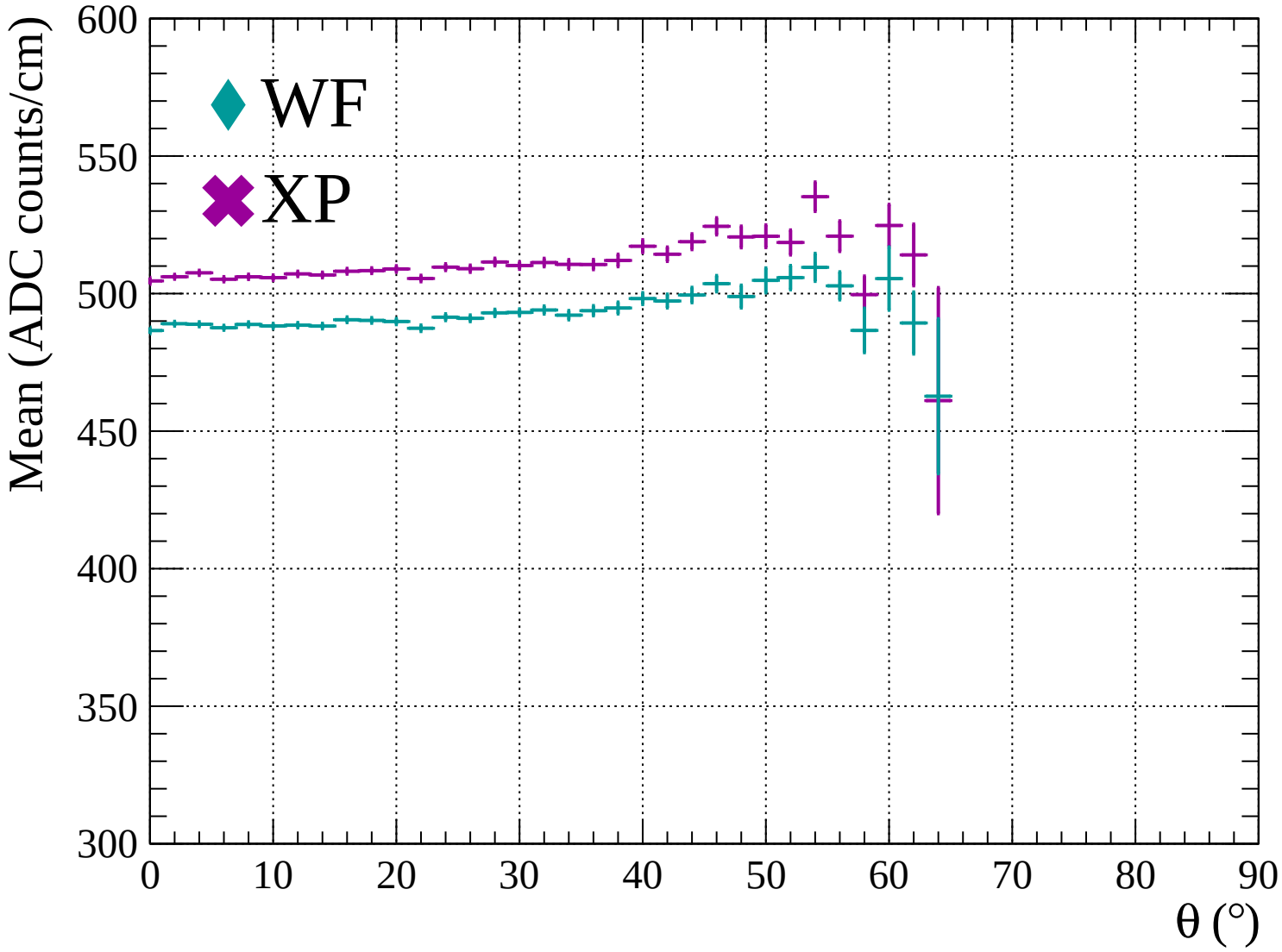


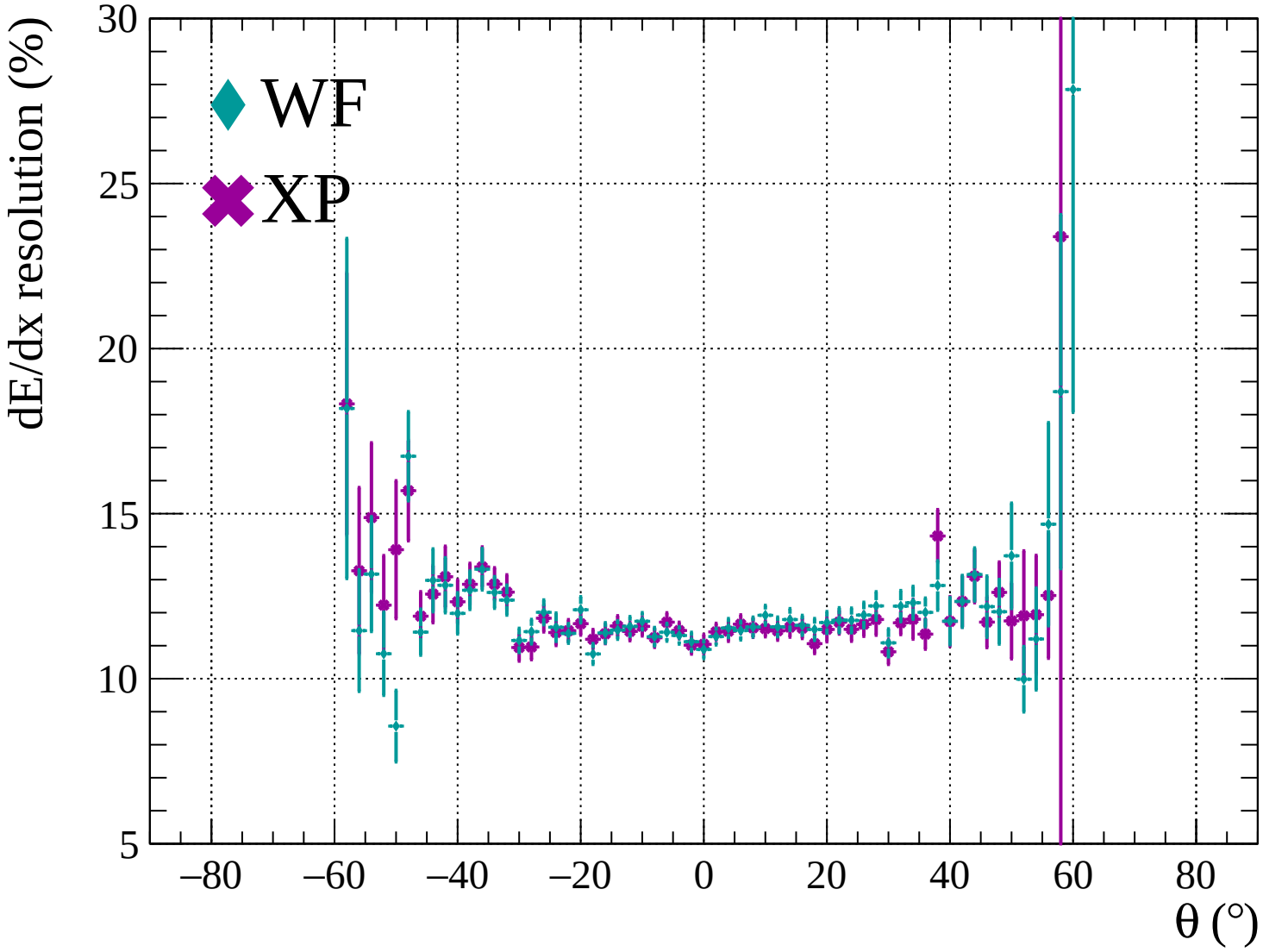


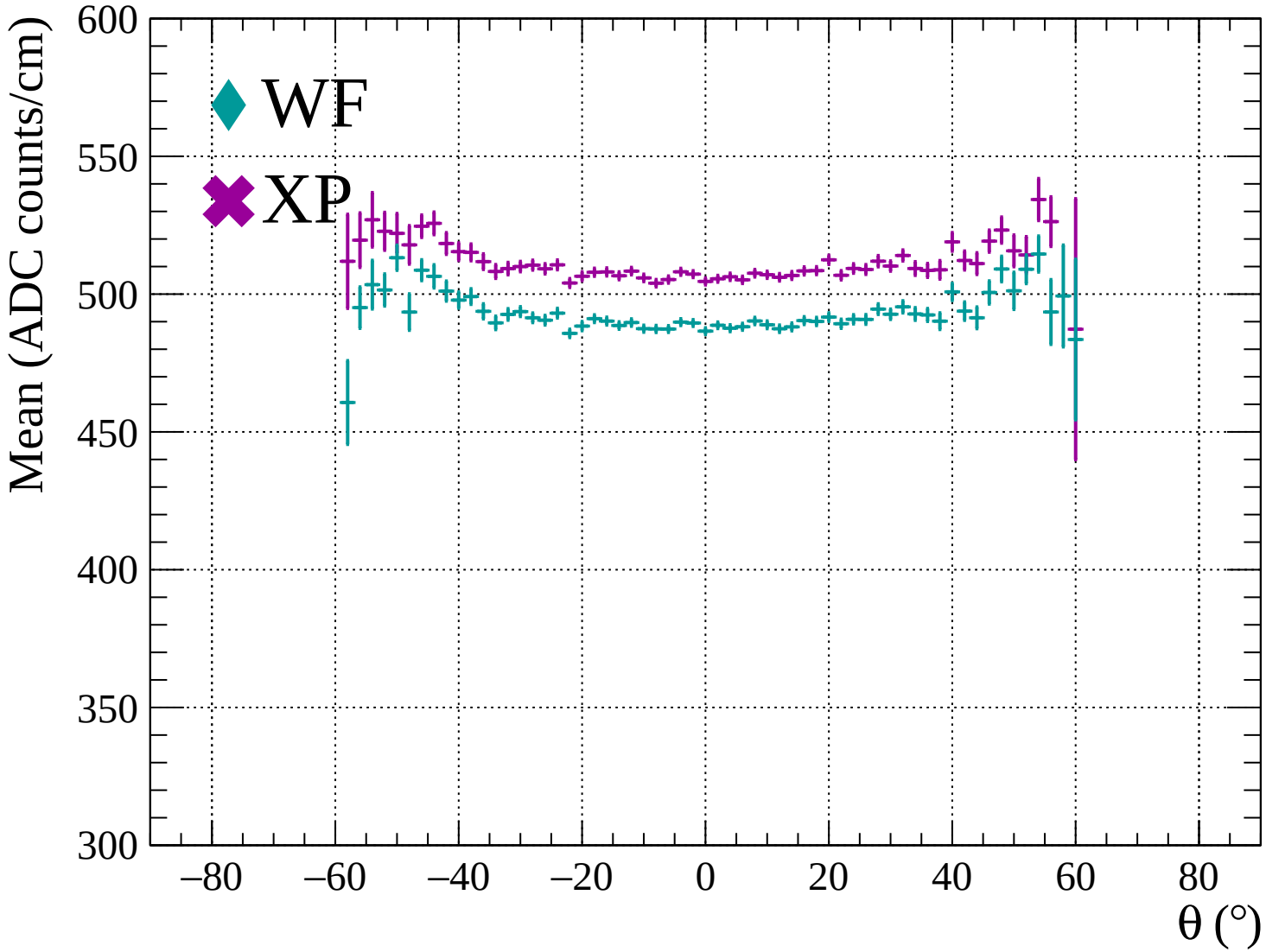


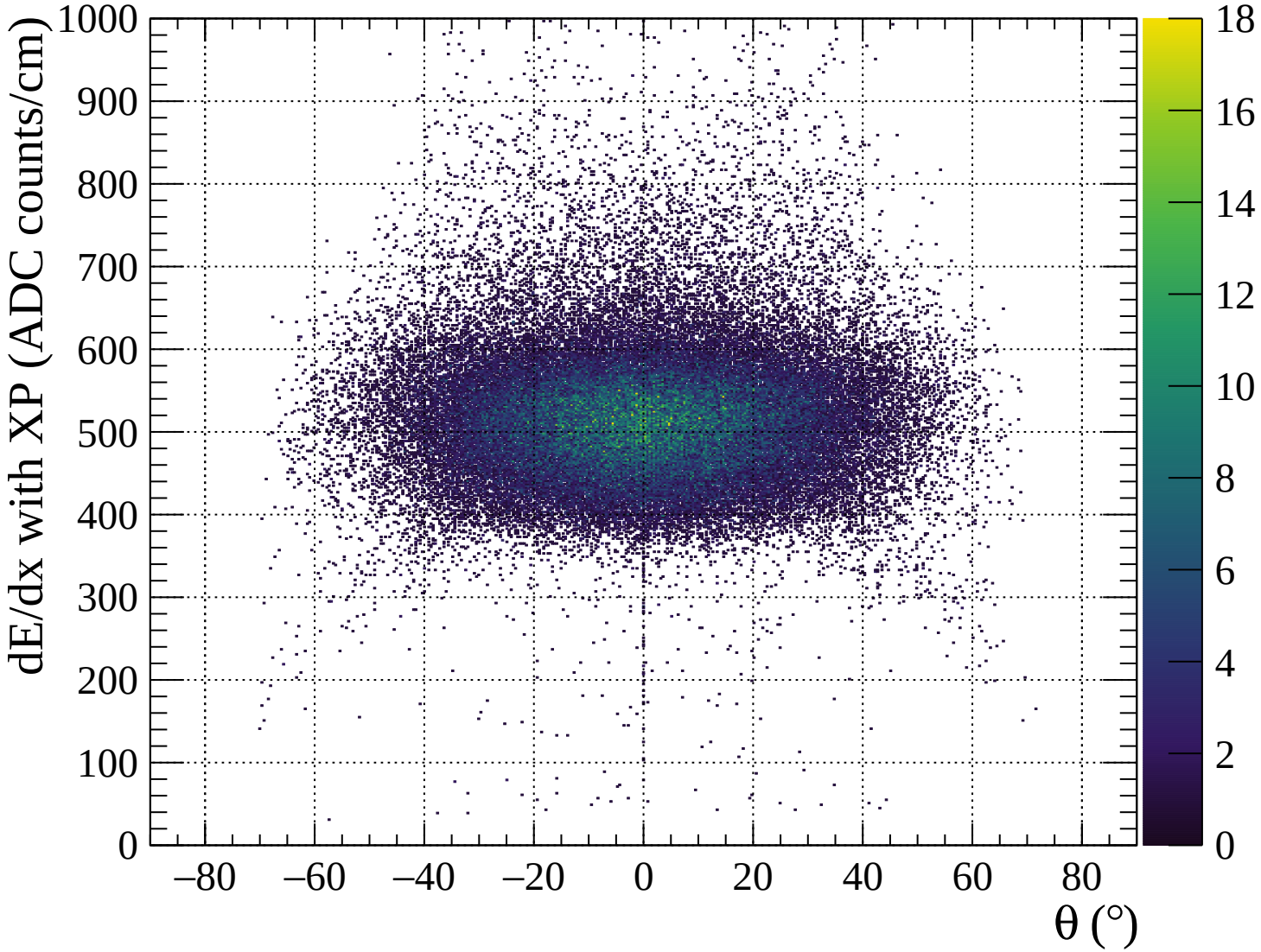


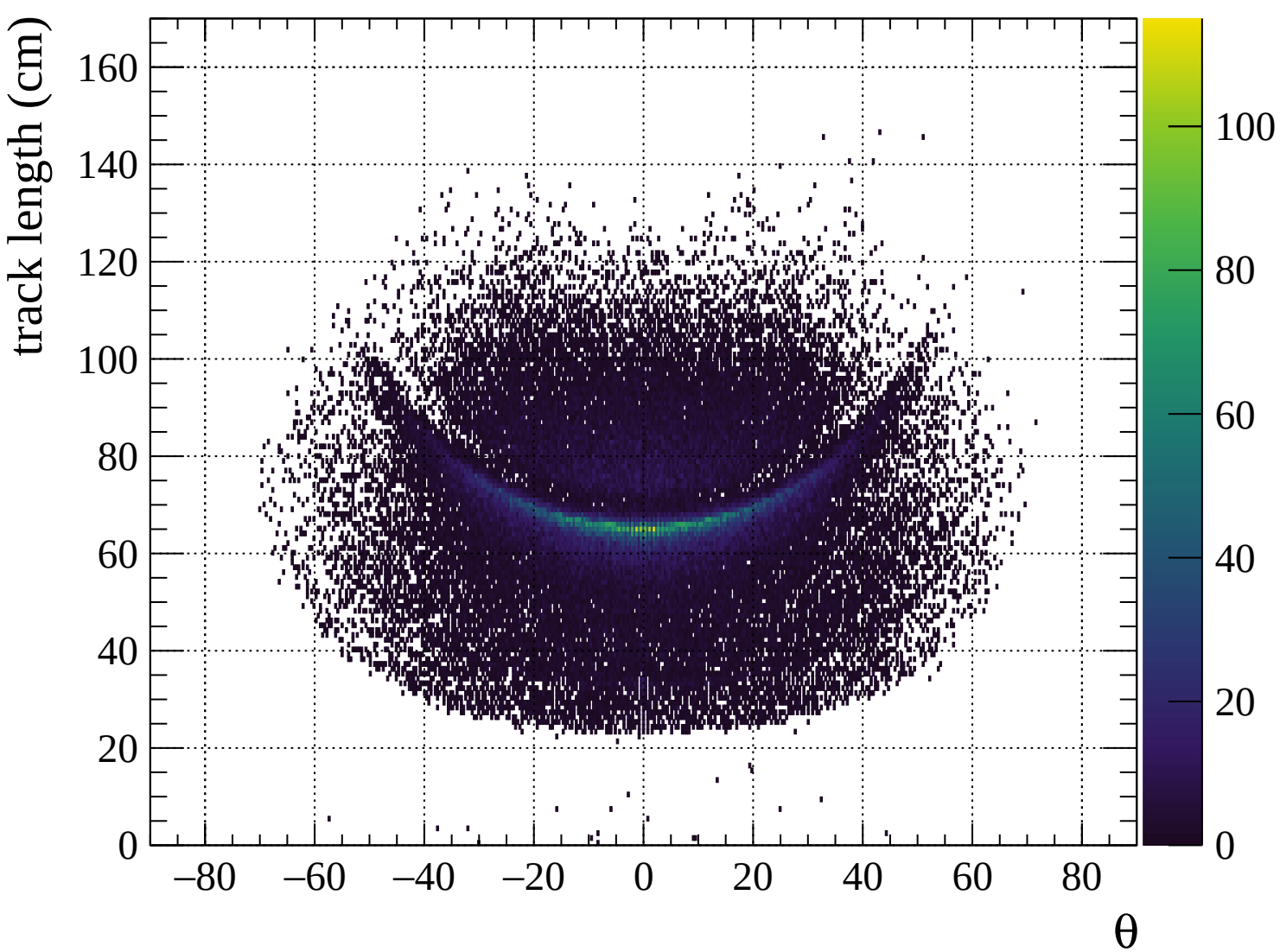


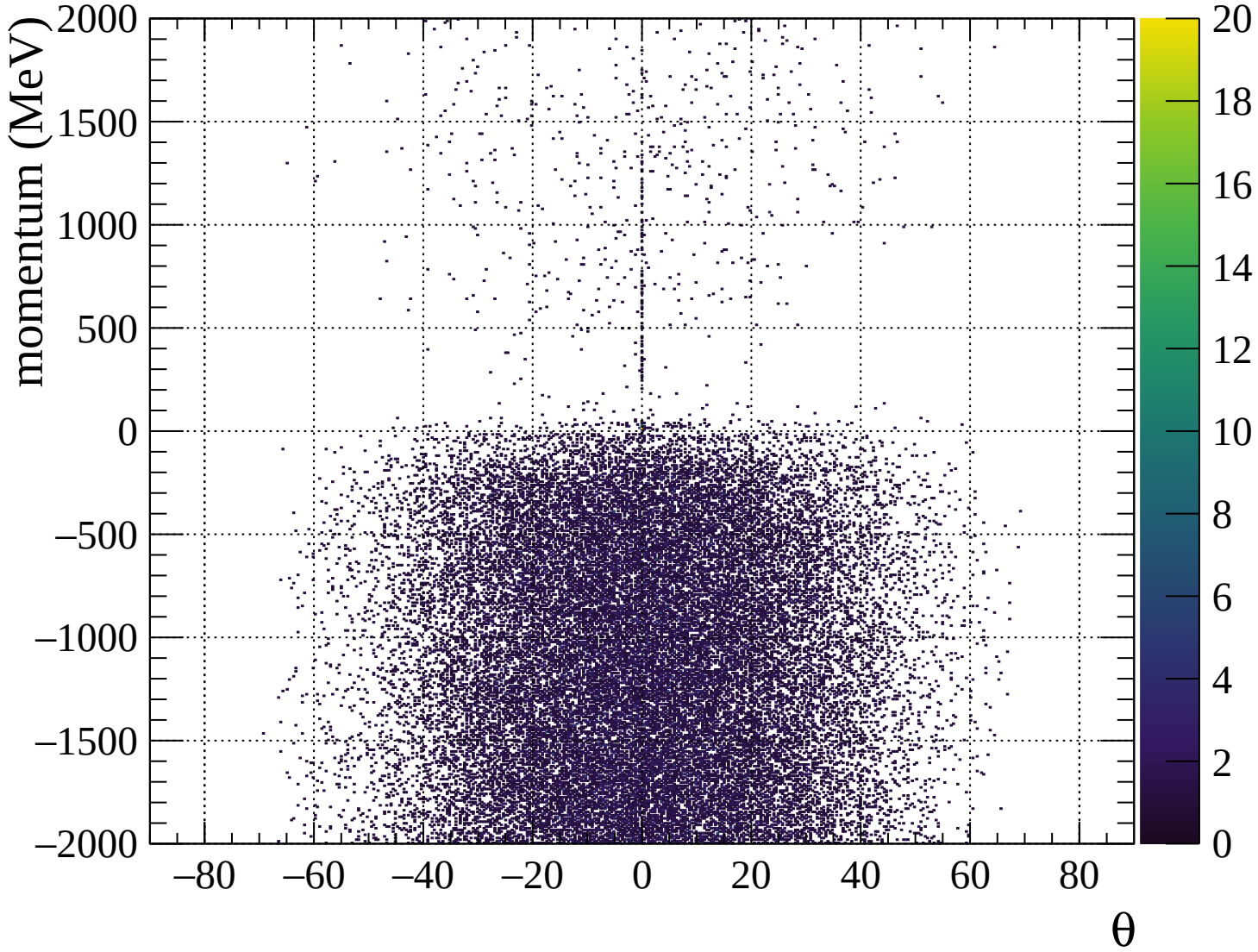


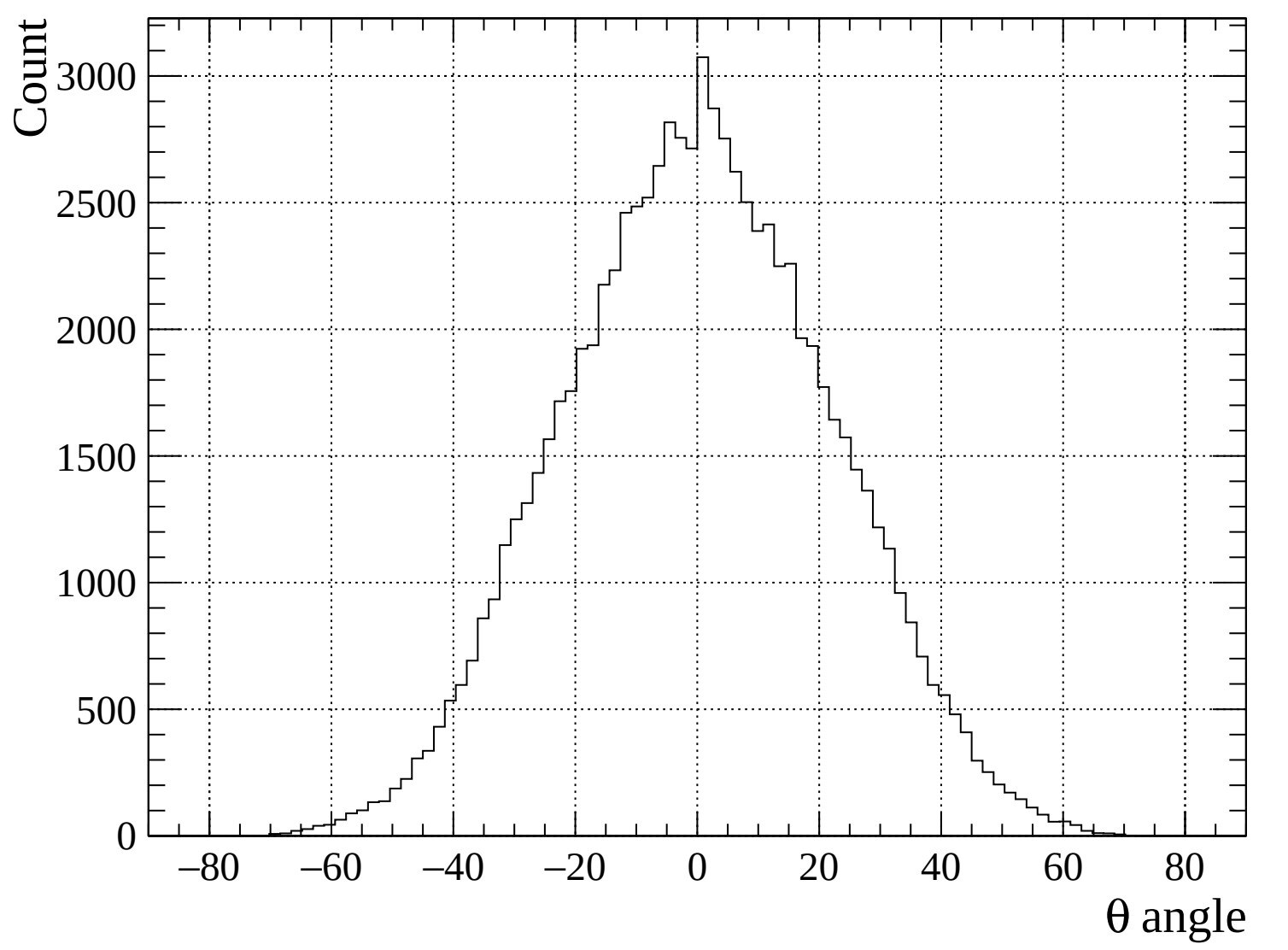


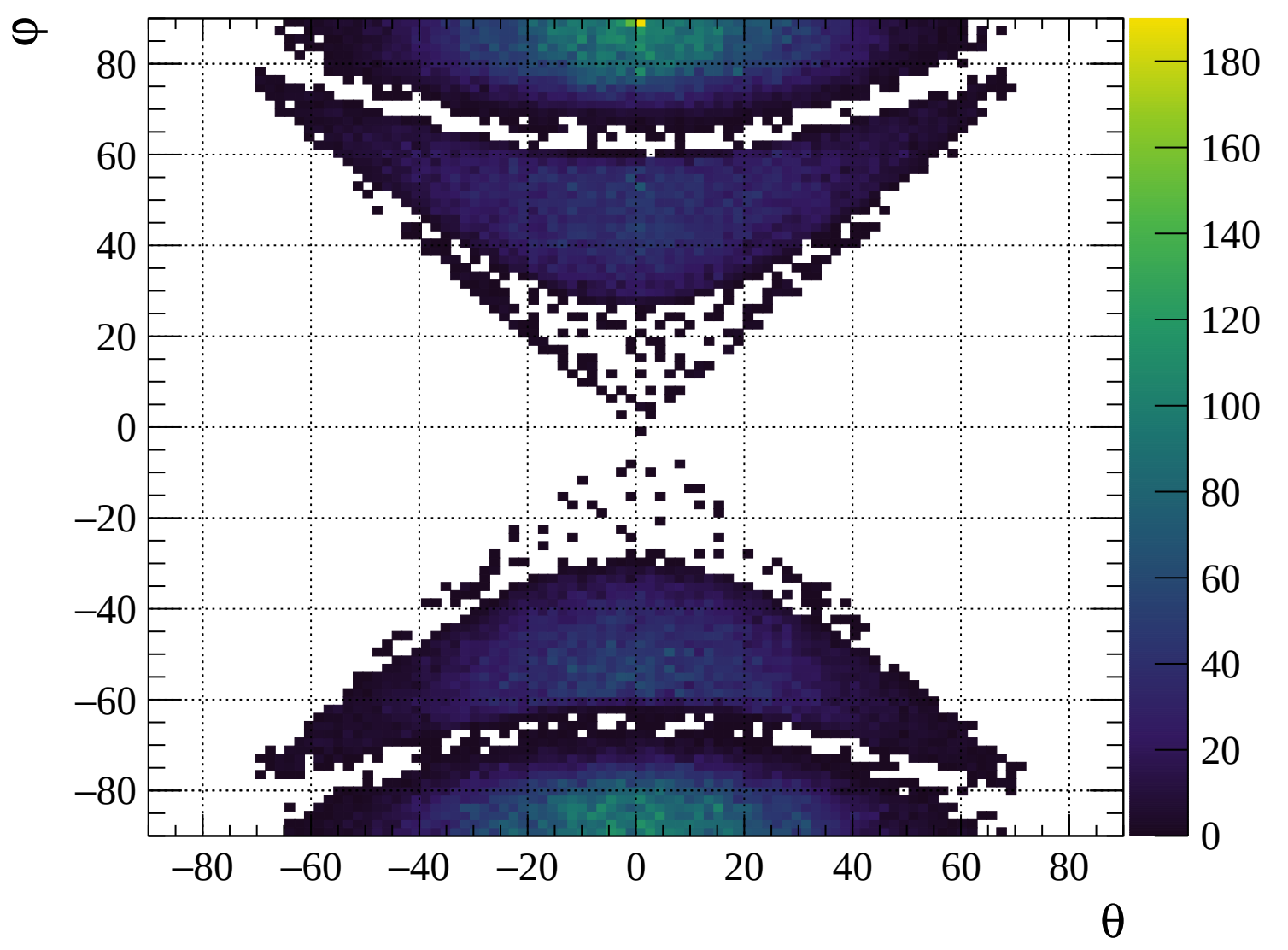


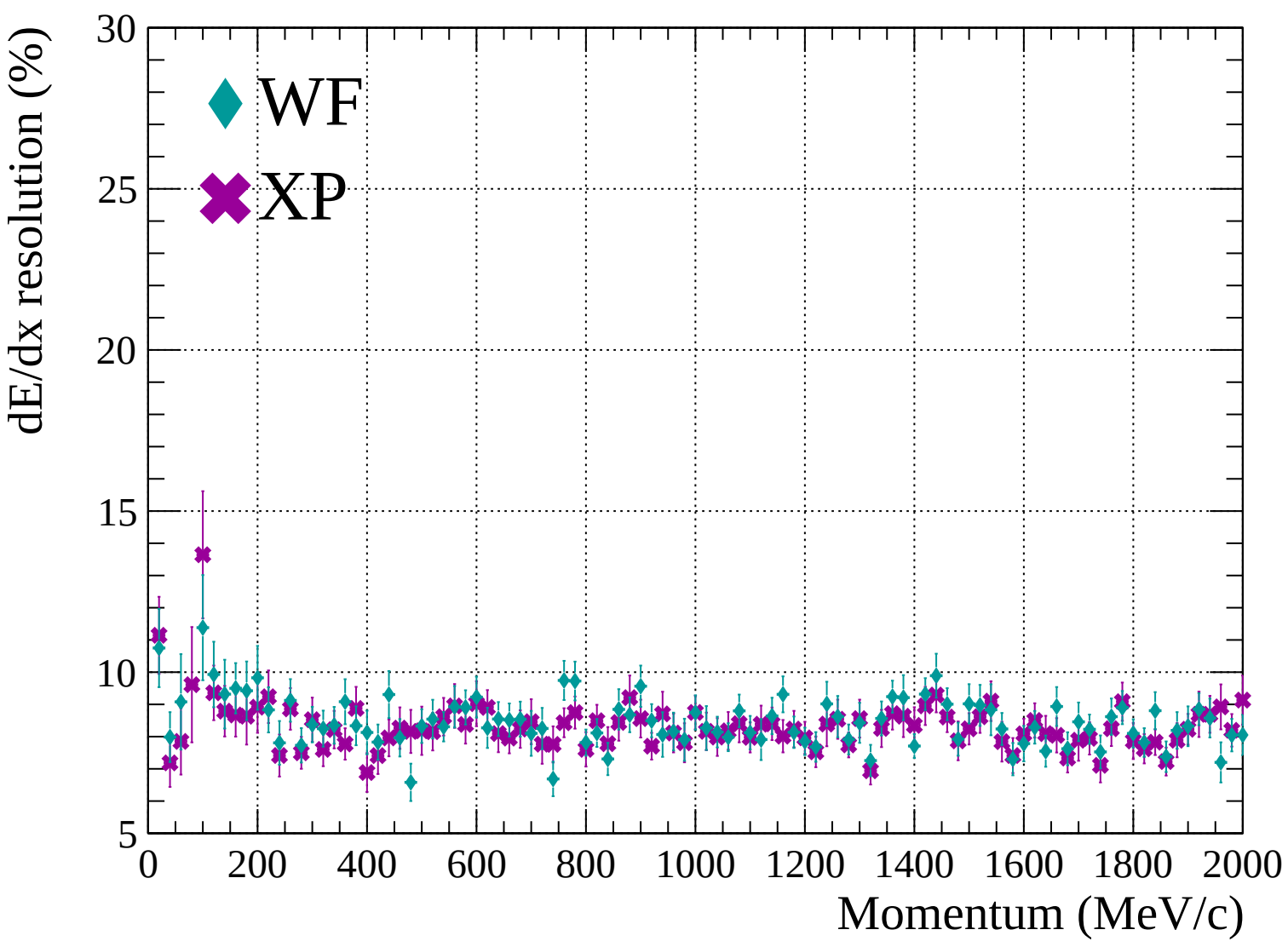


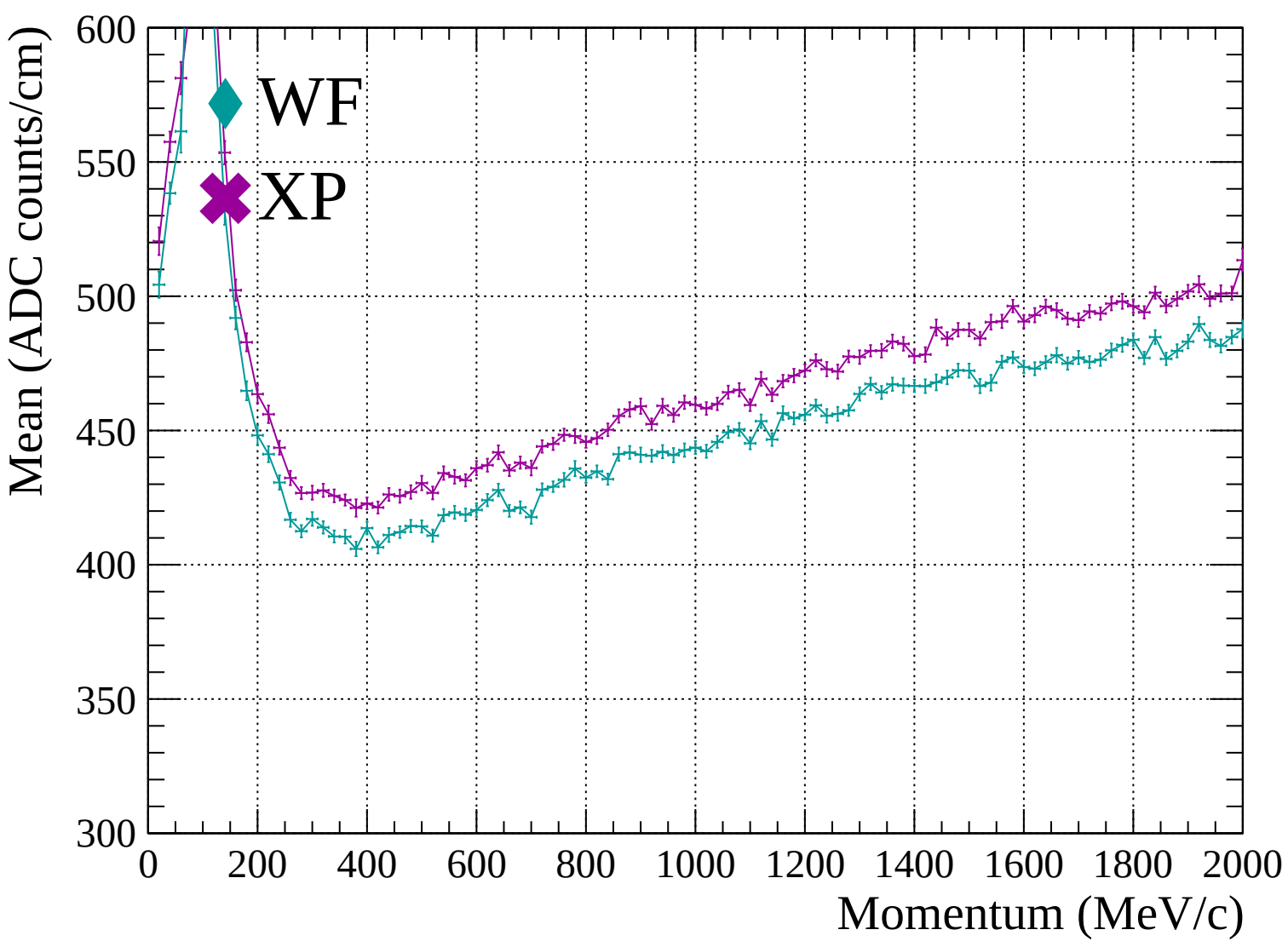


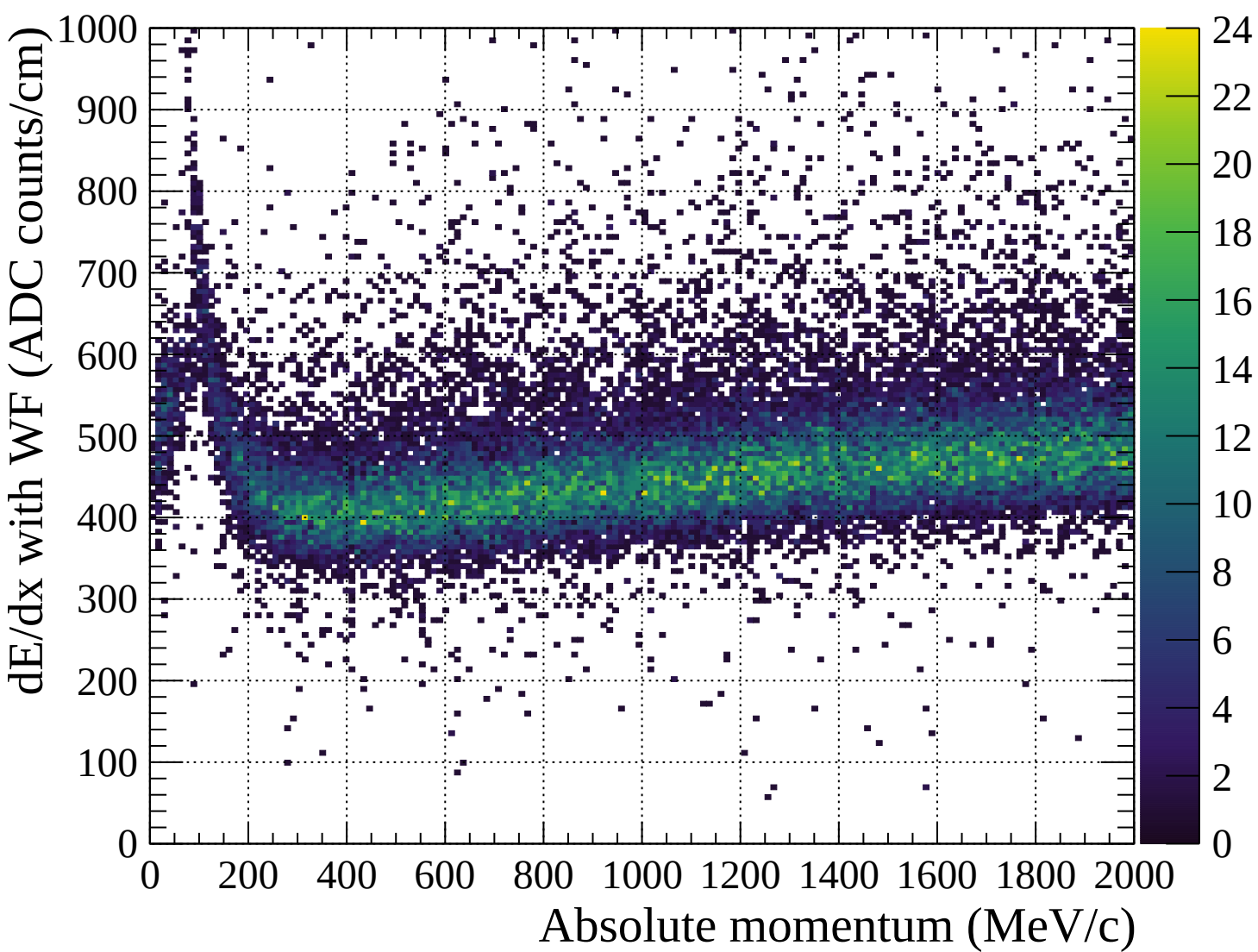


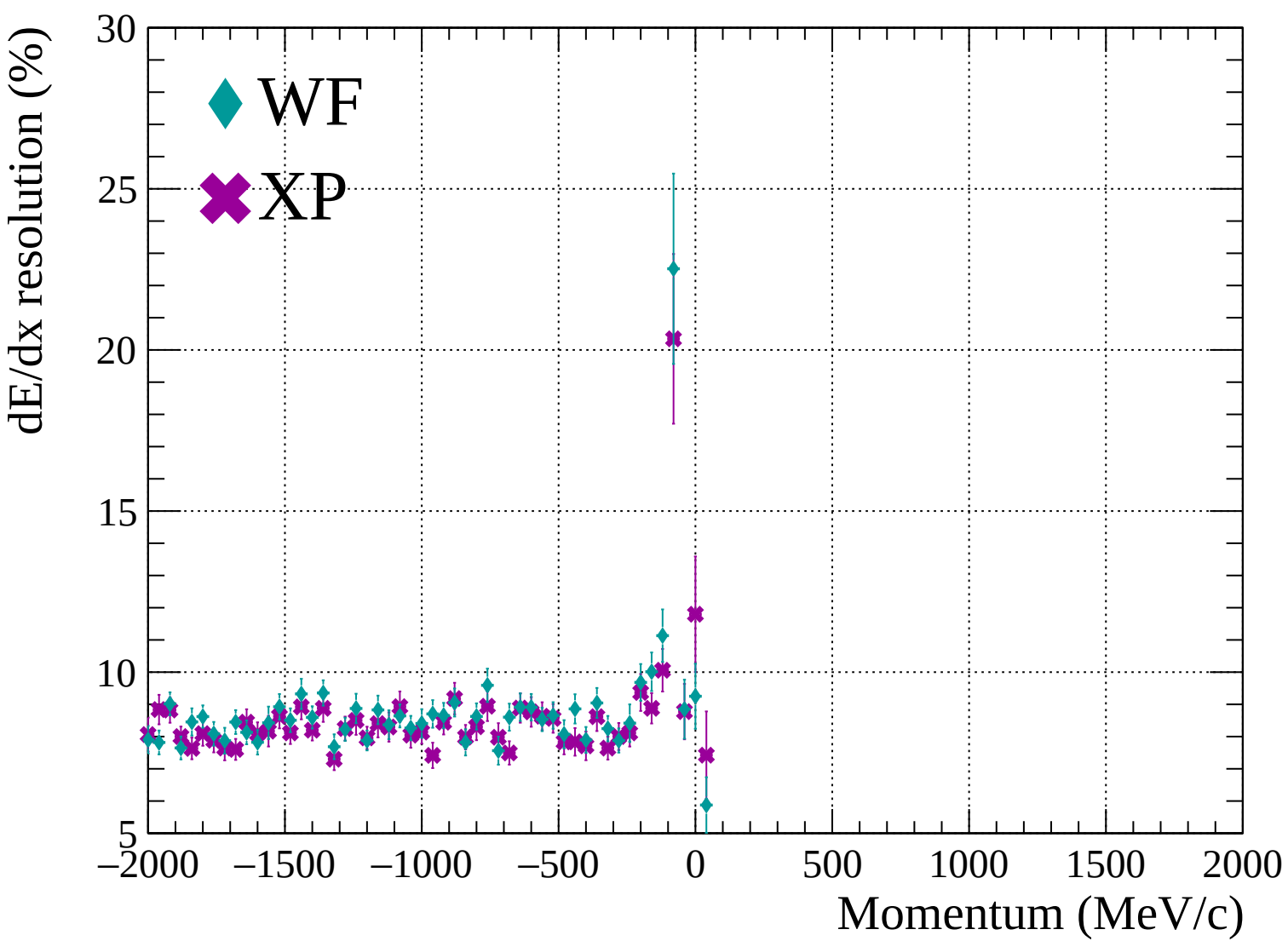


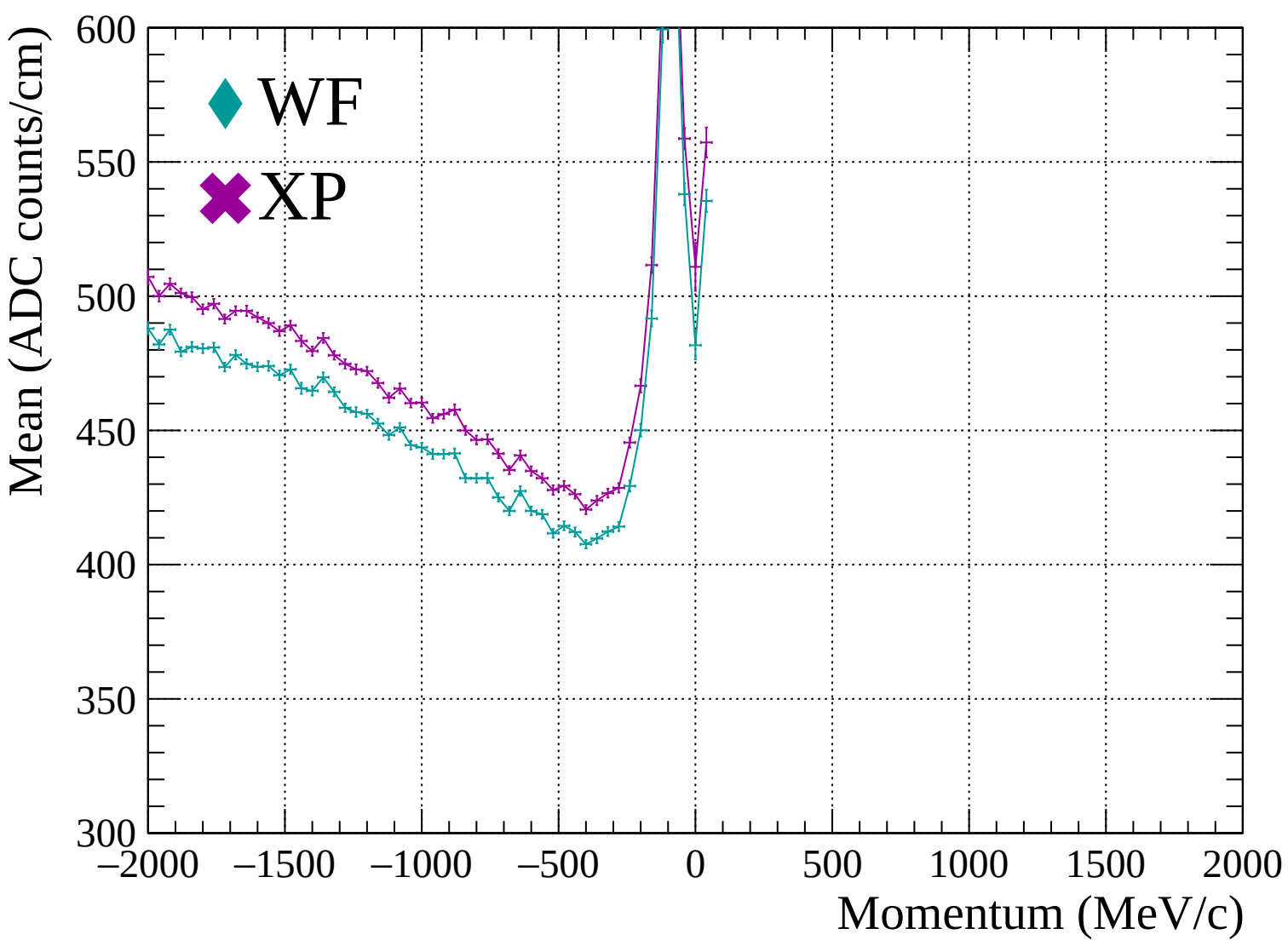


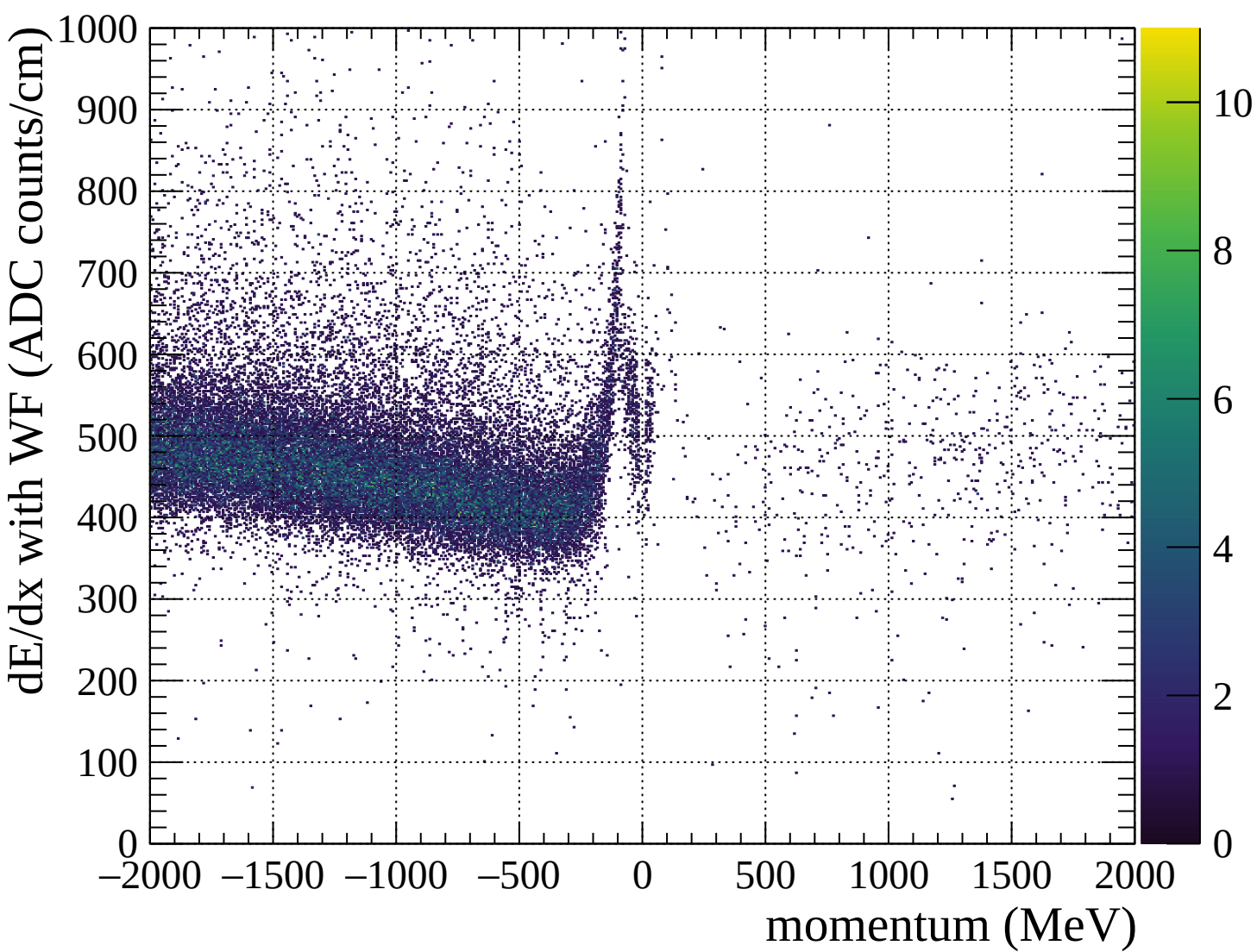


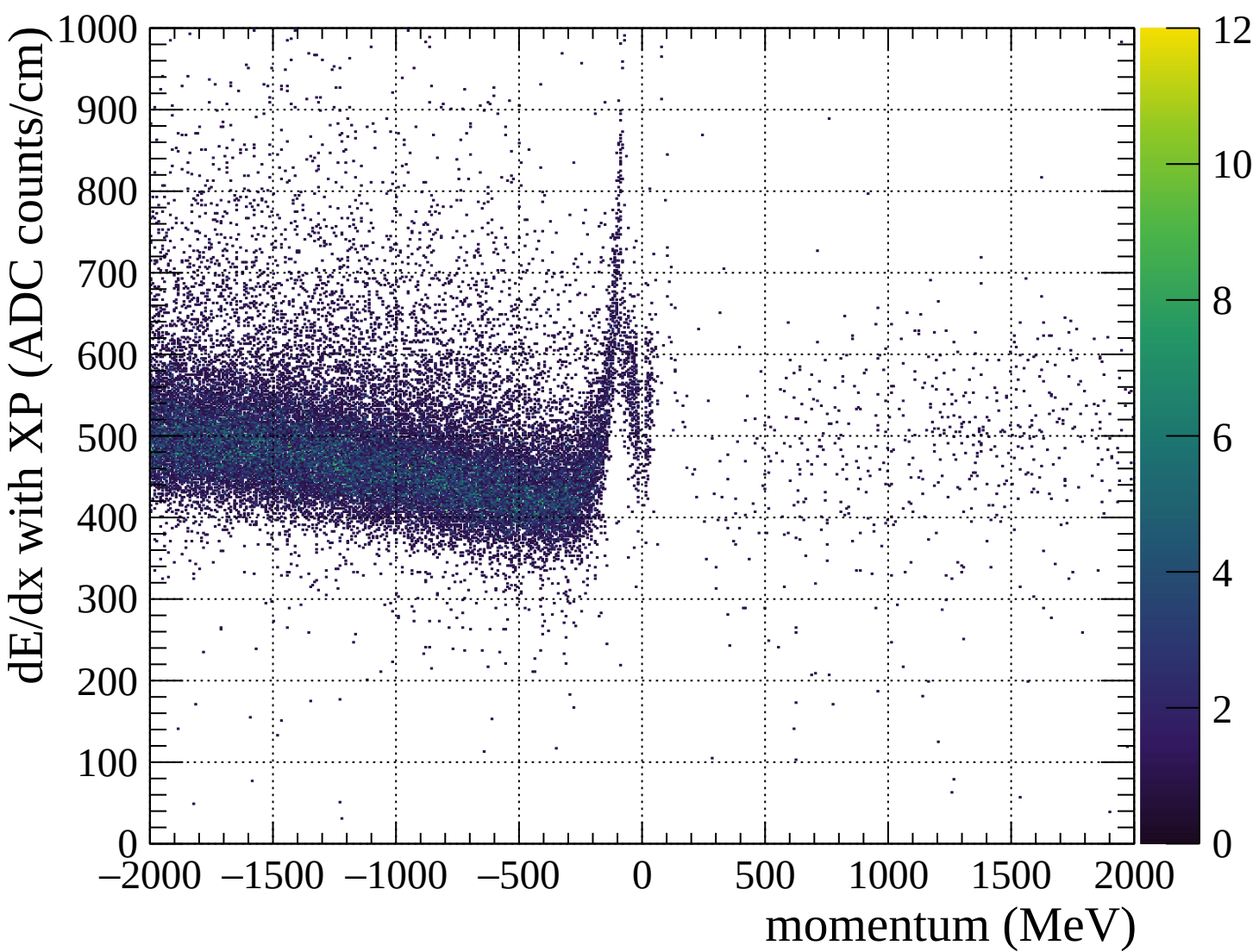


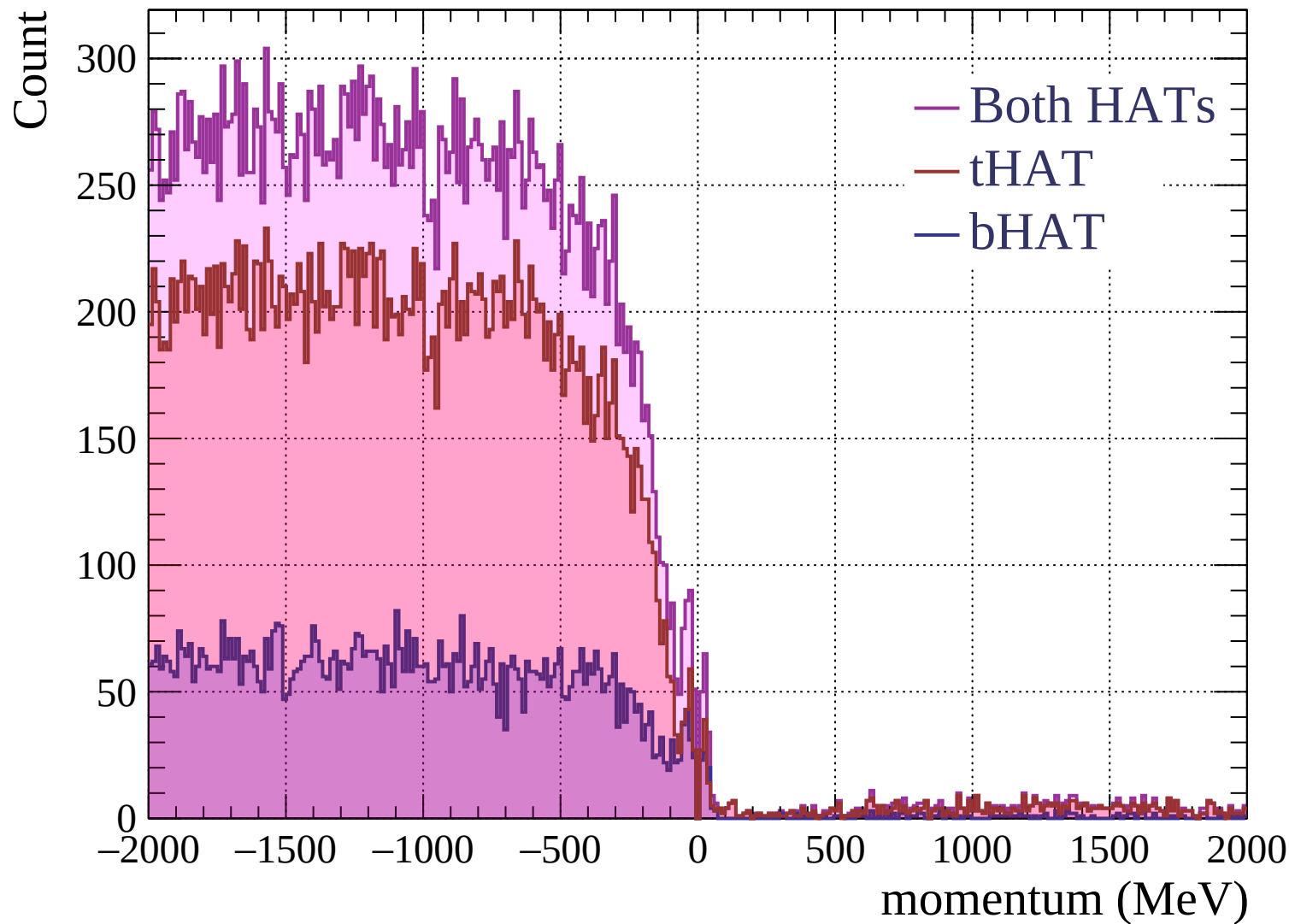


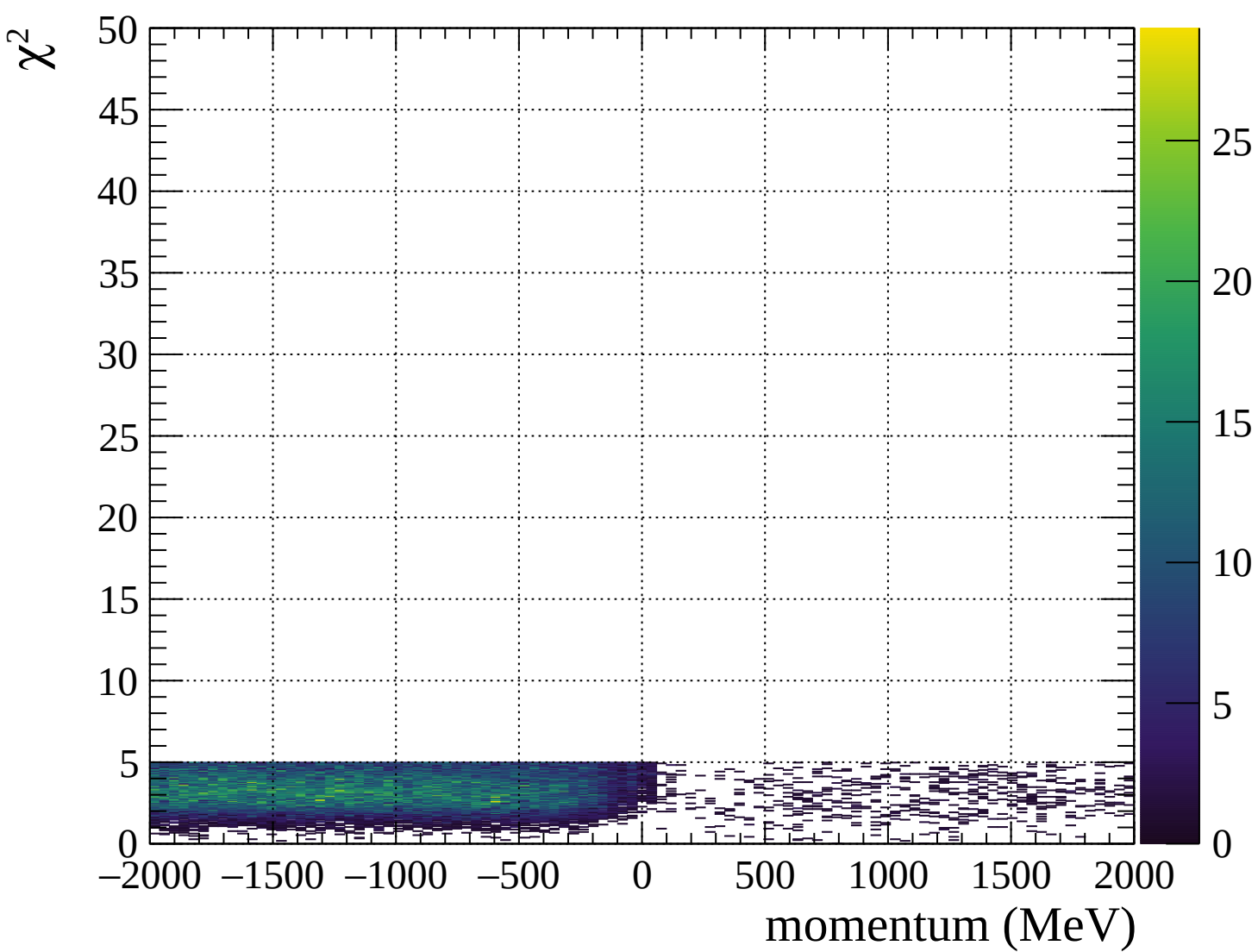




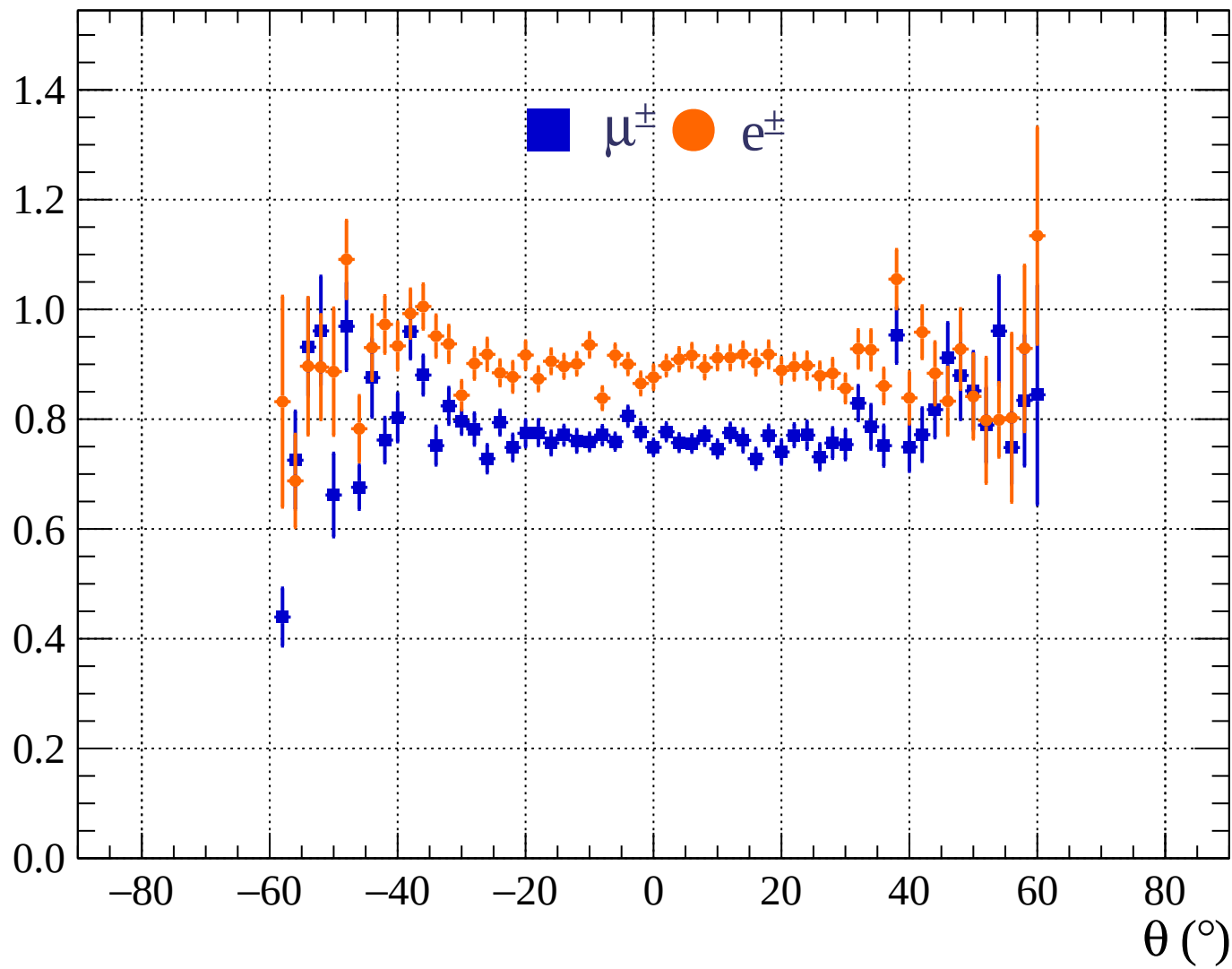




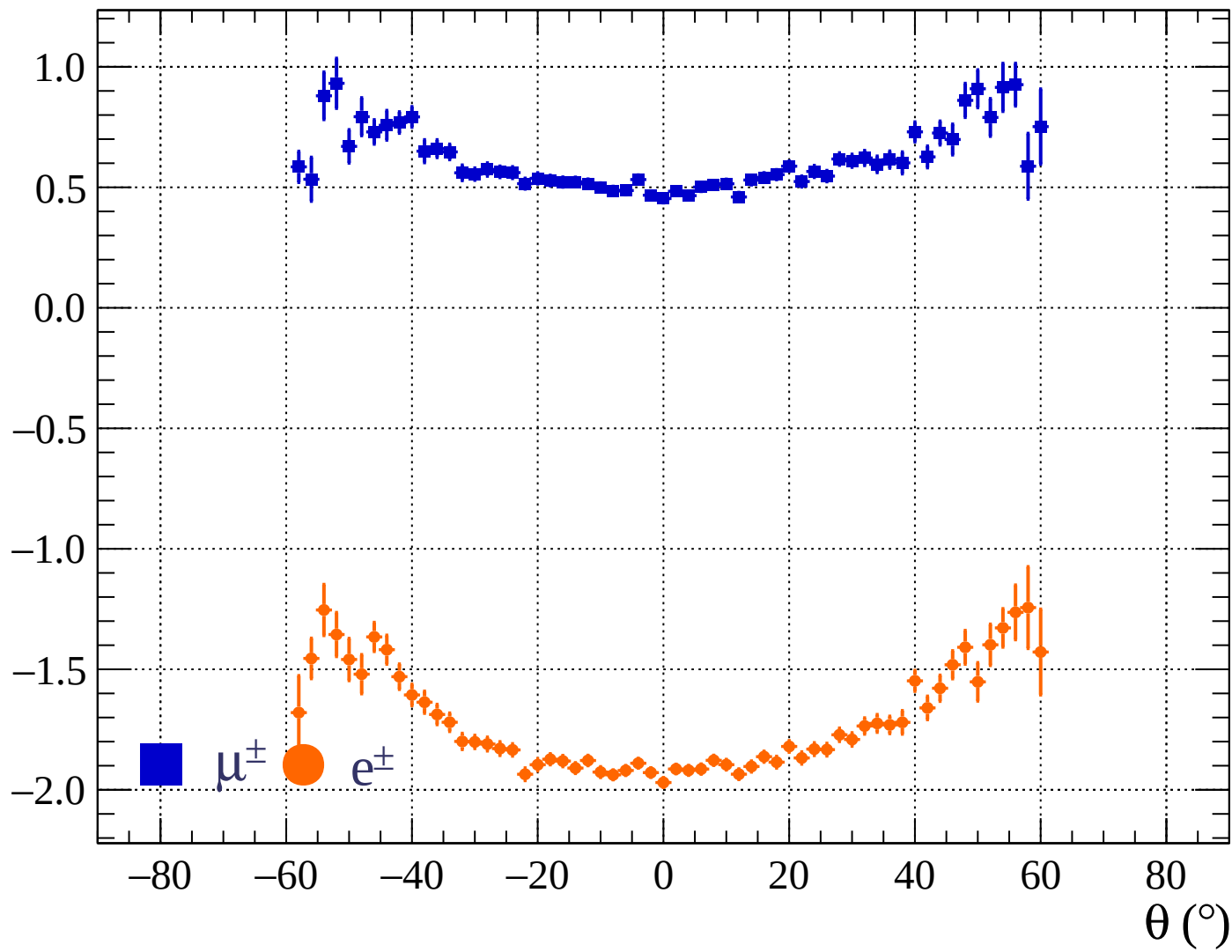




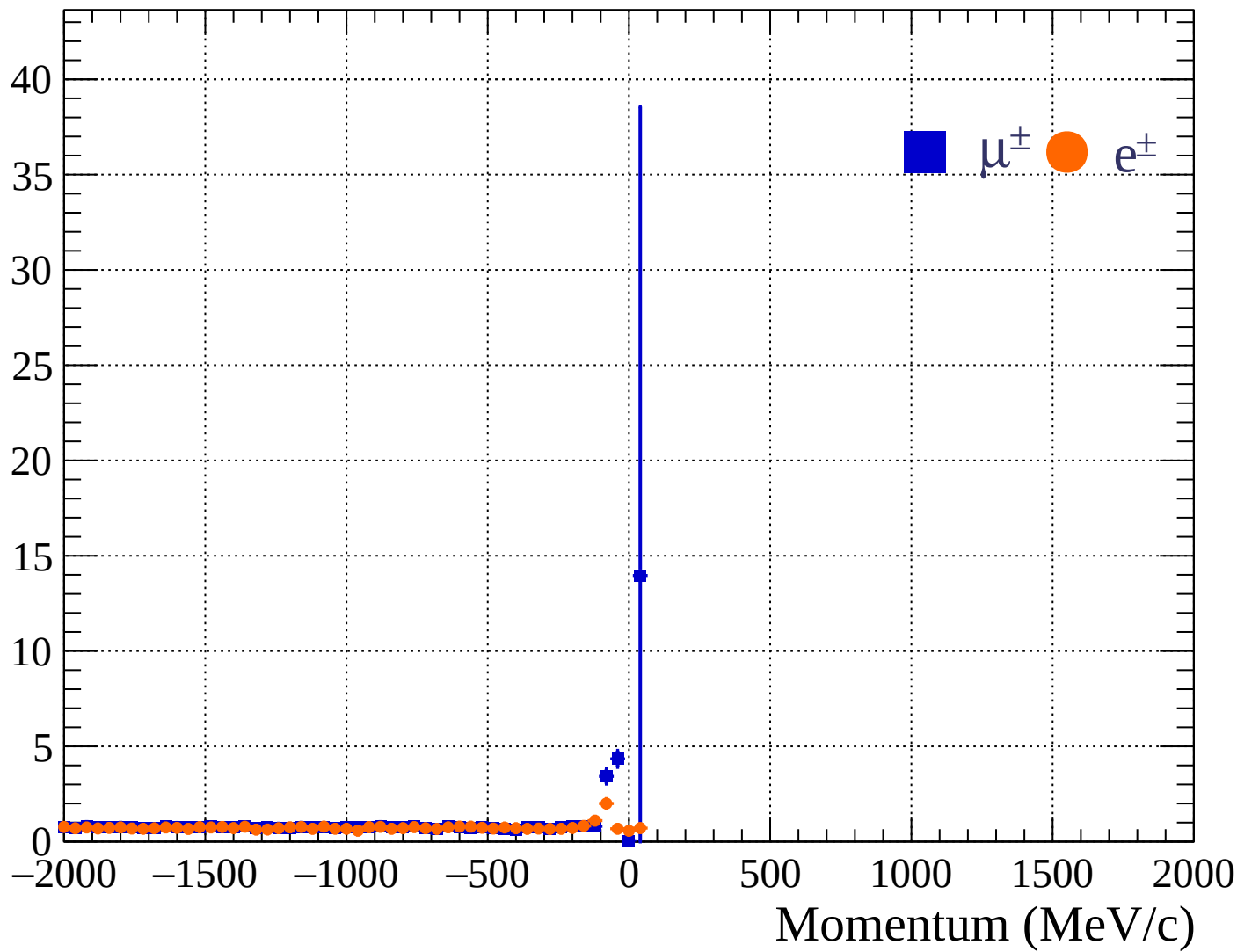
Pulls standard deviation



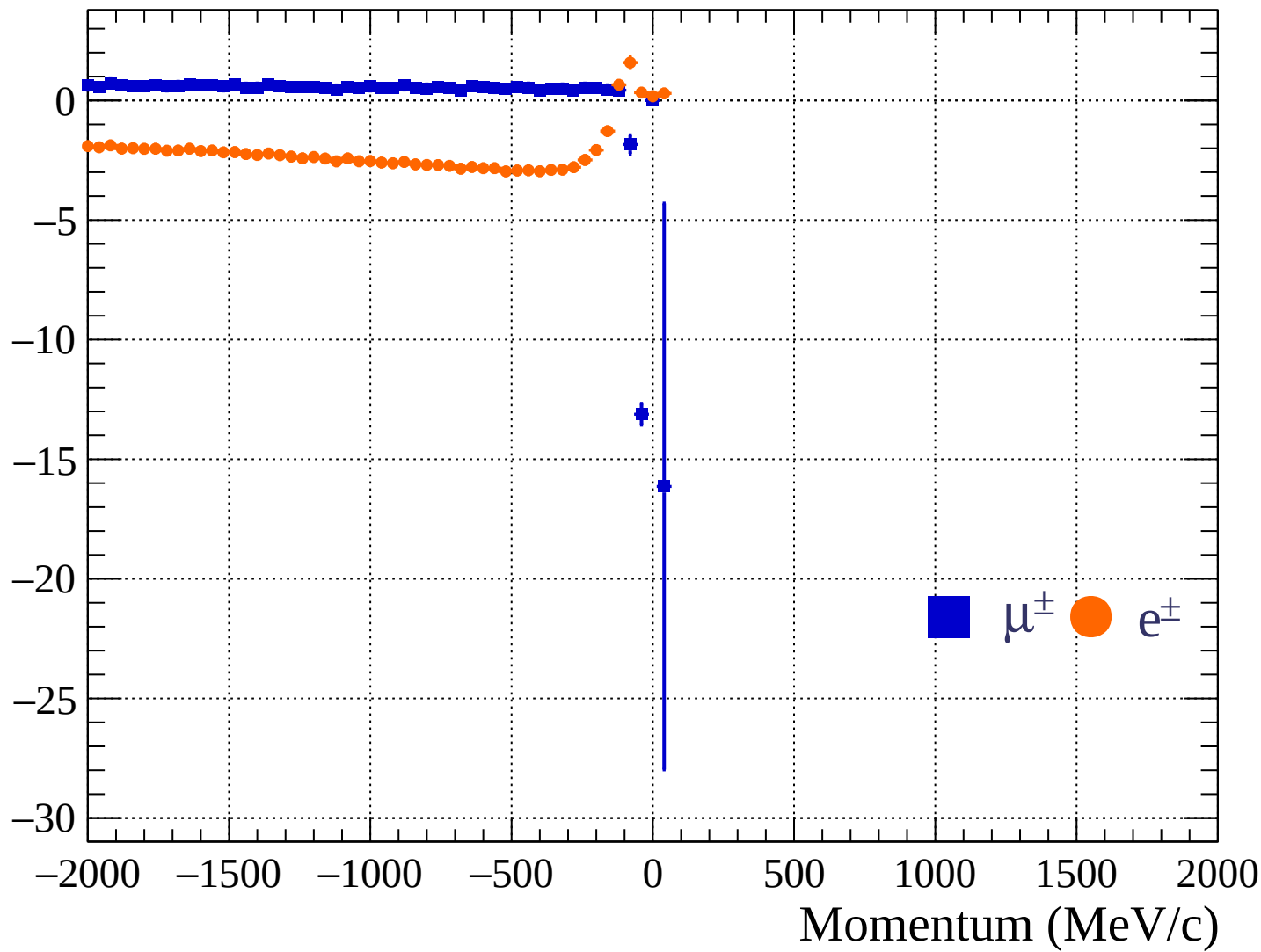
Pulls mean

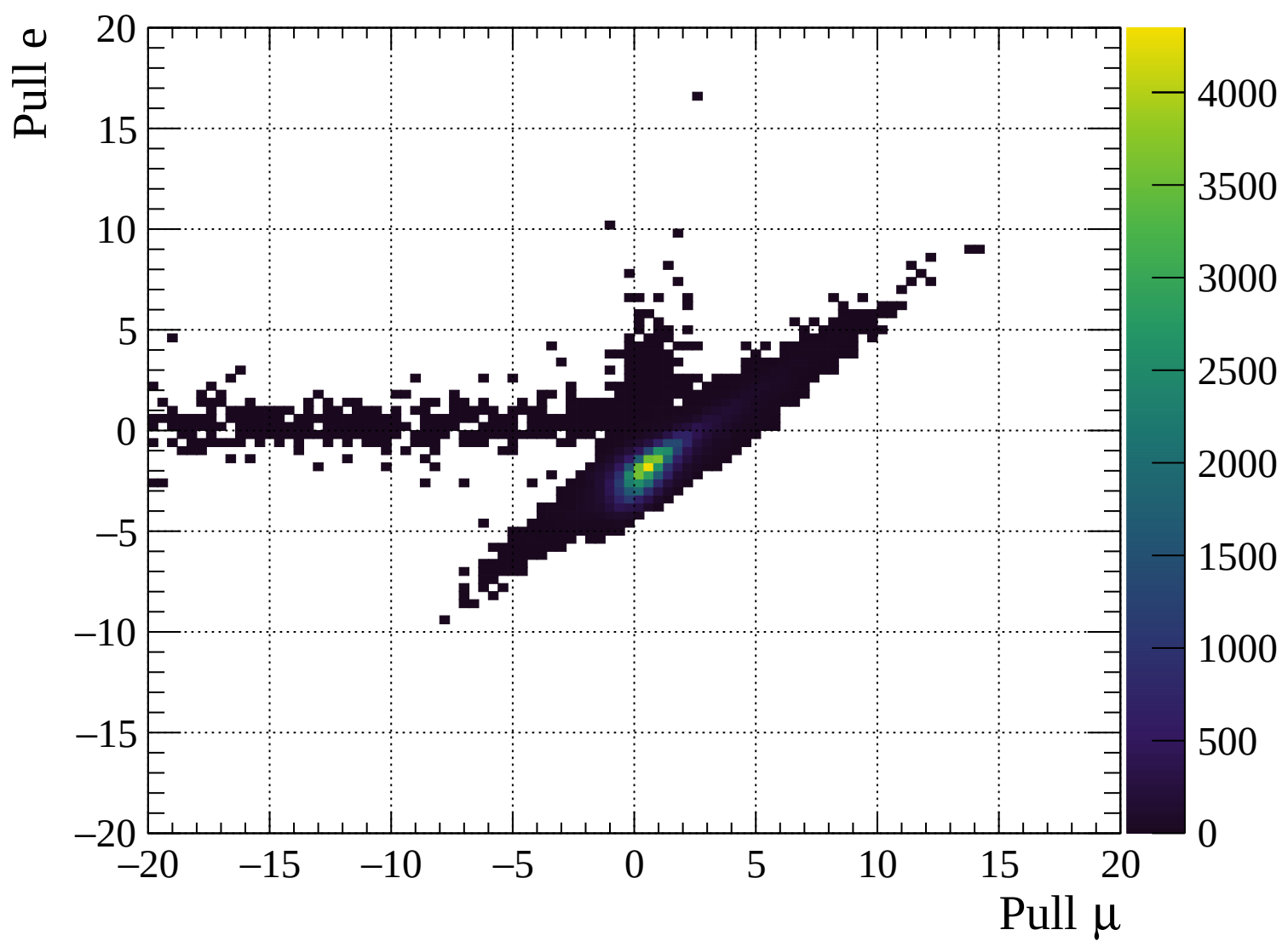


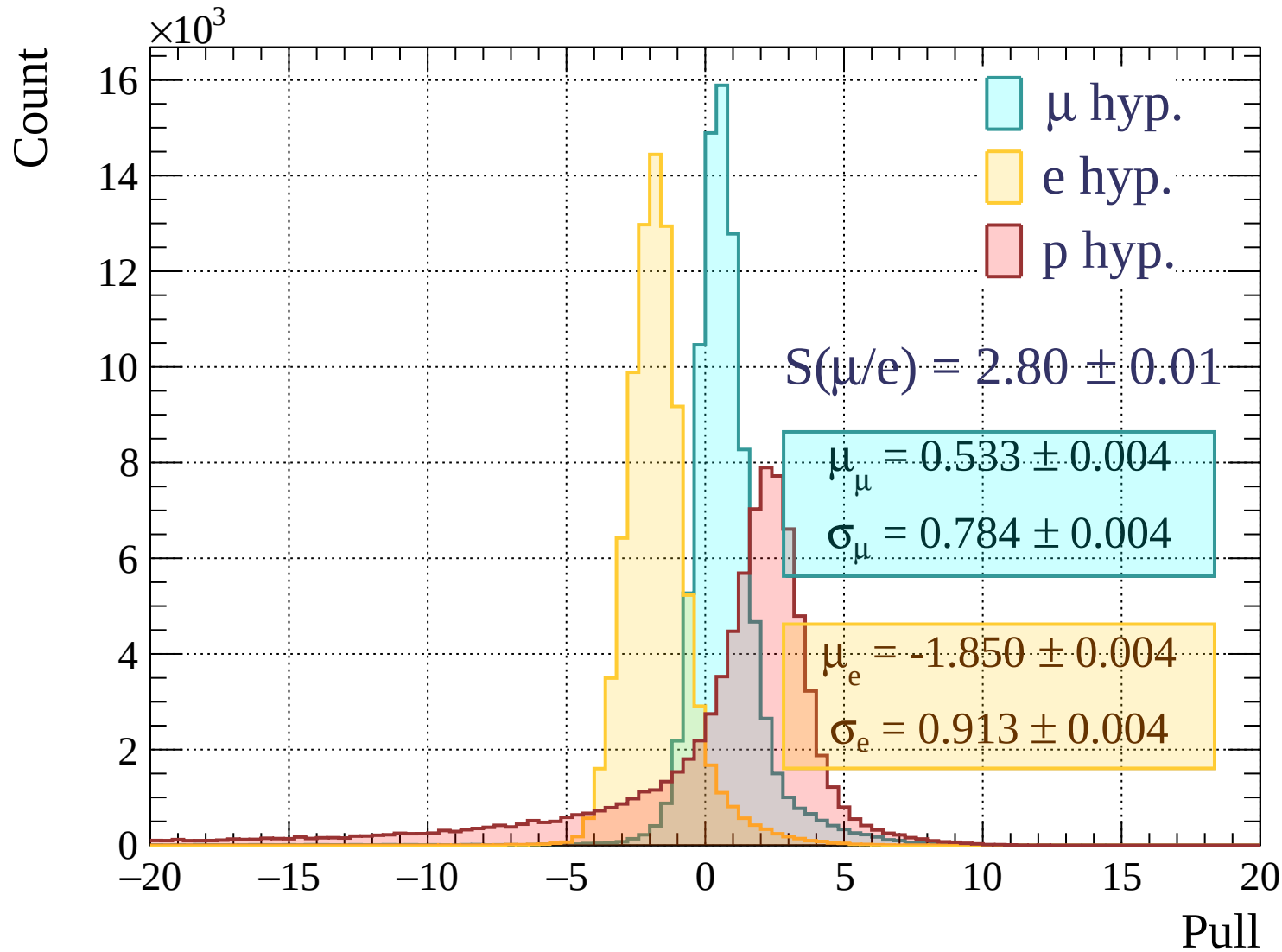
Pulls standard deviation

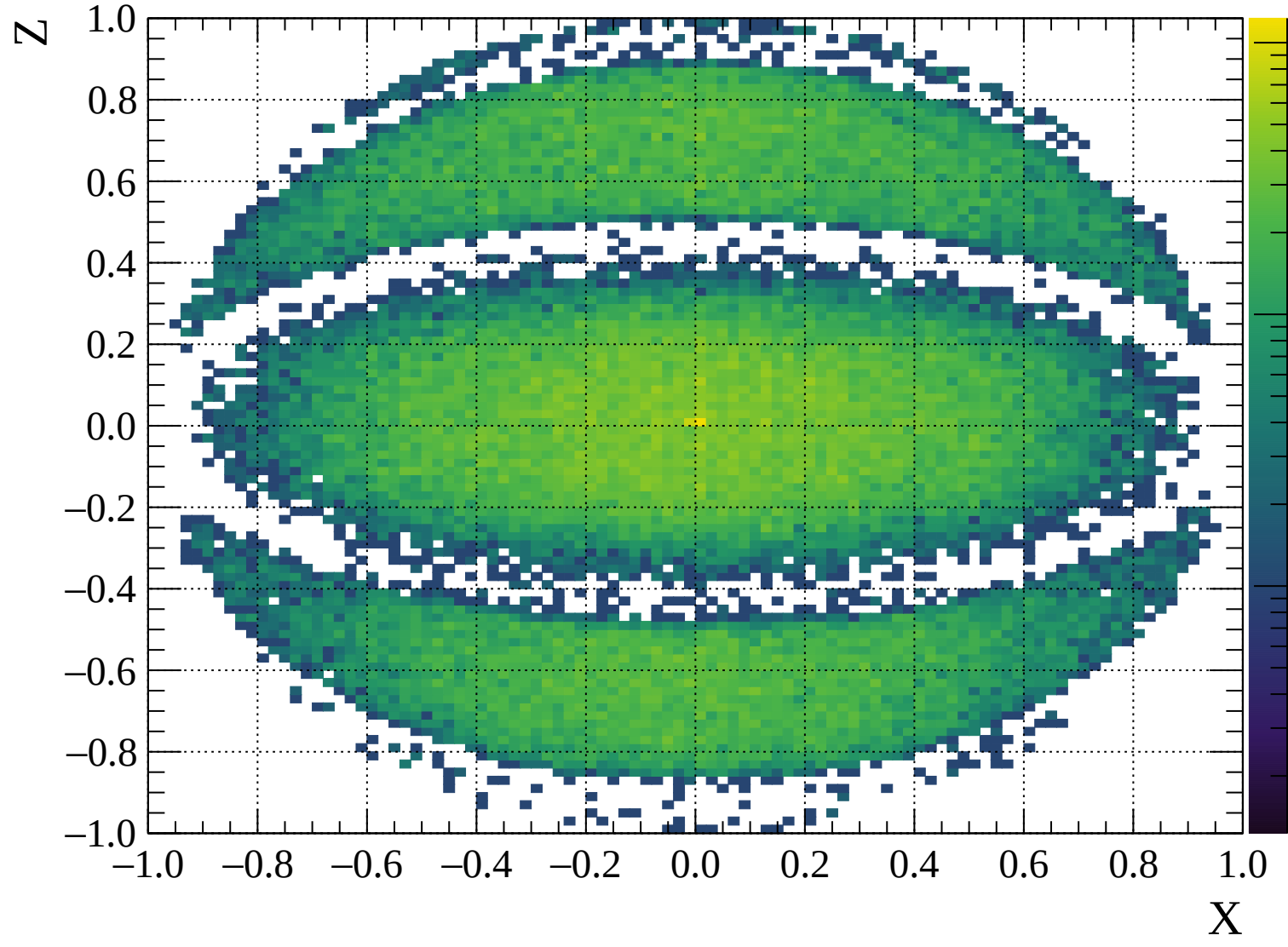


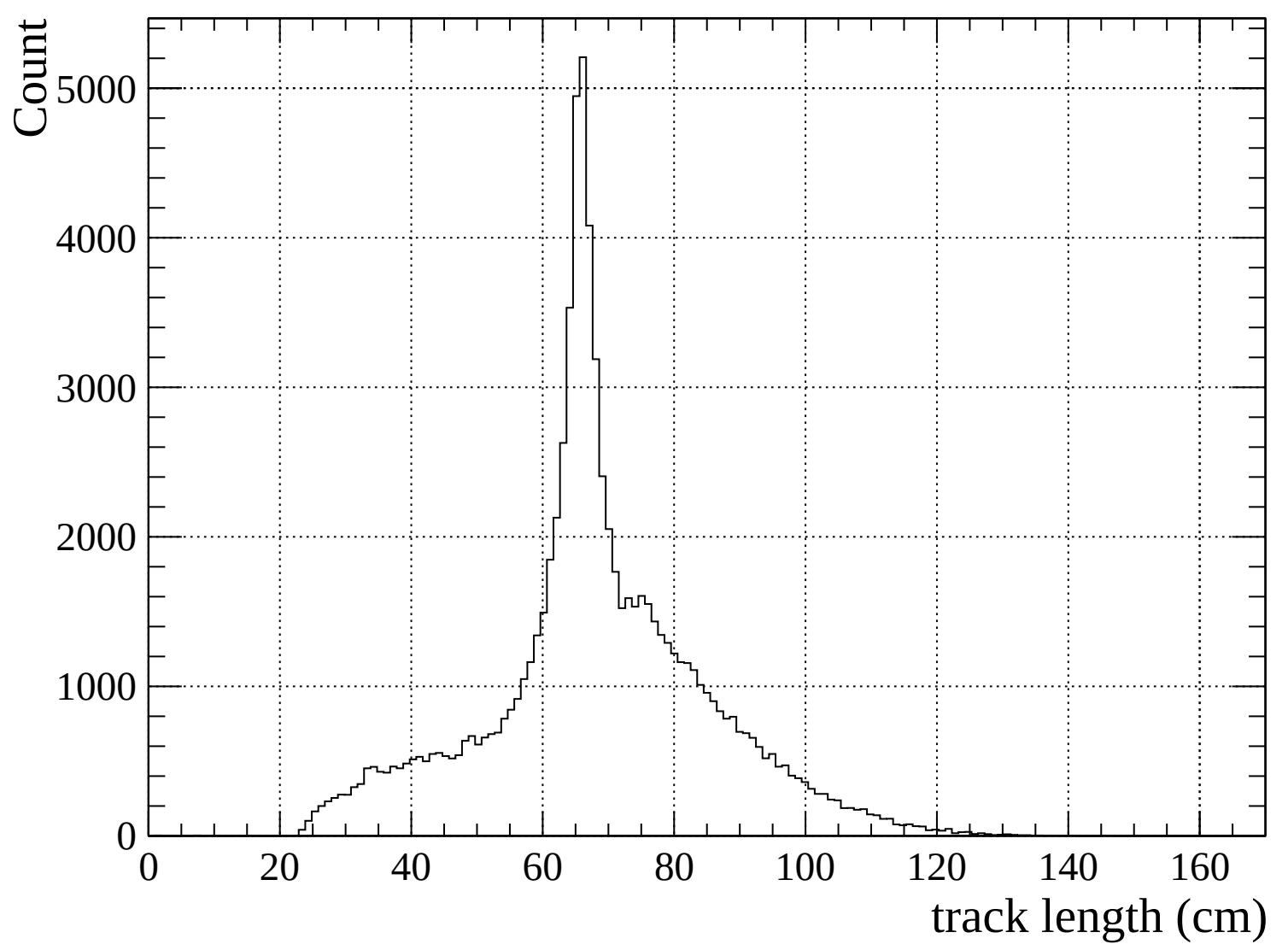
Pulls mean

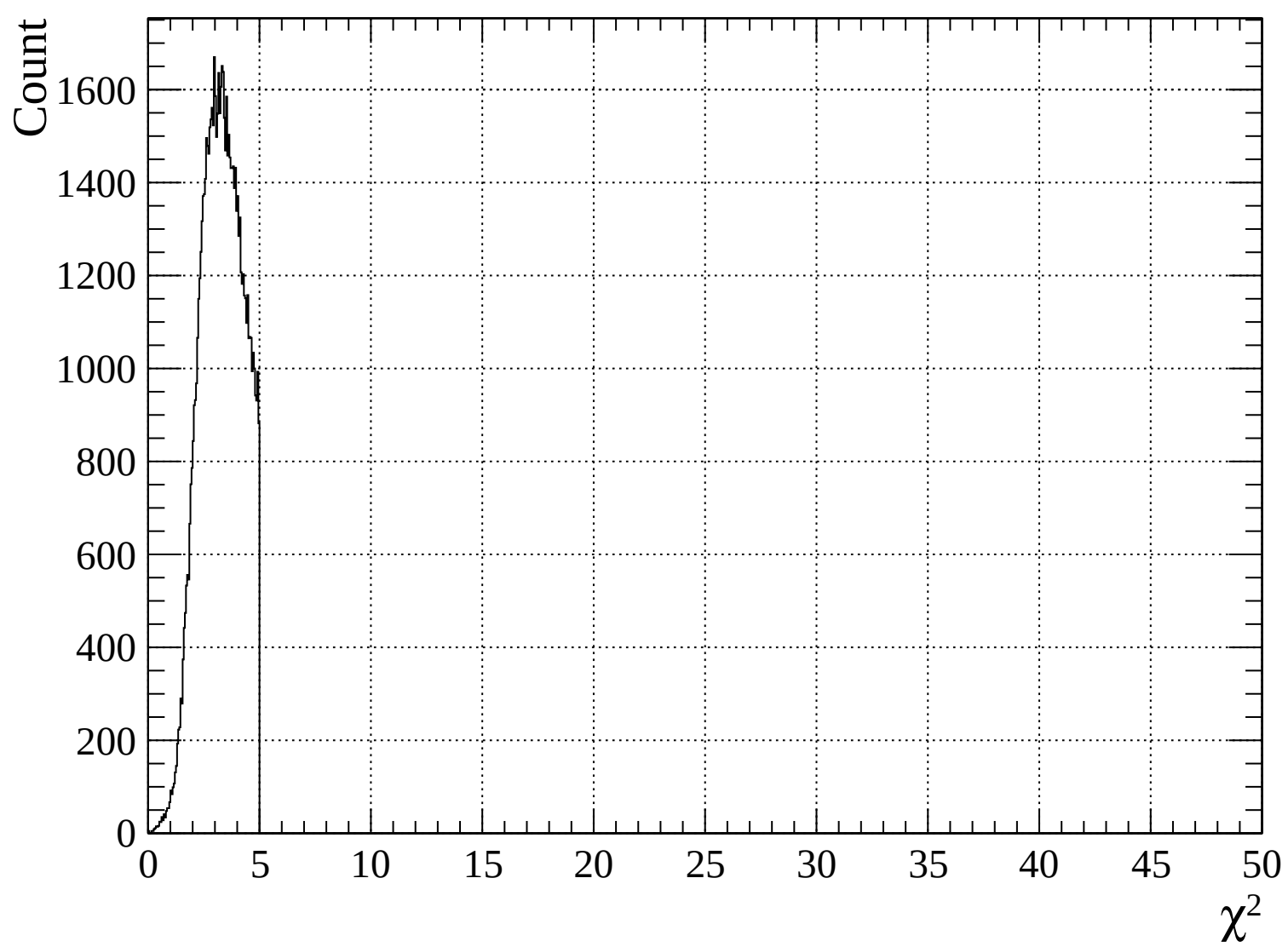


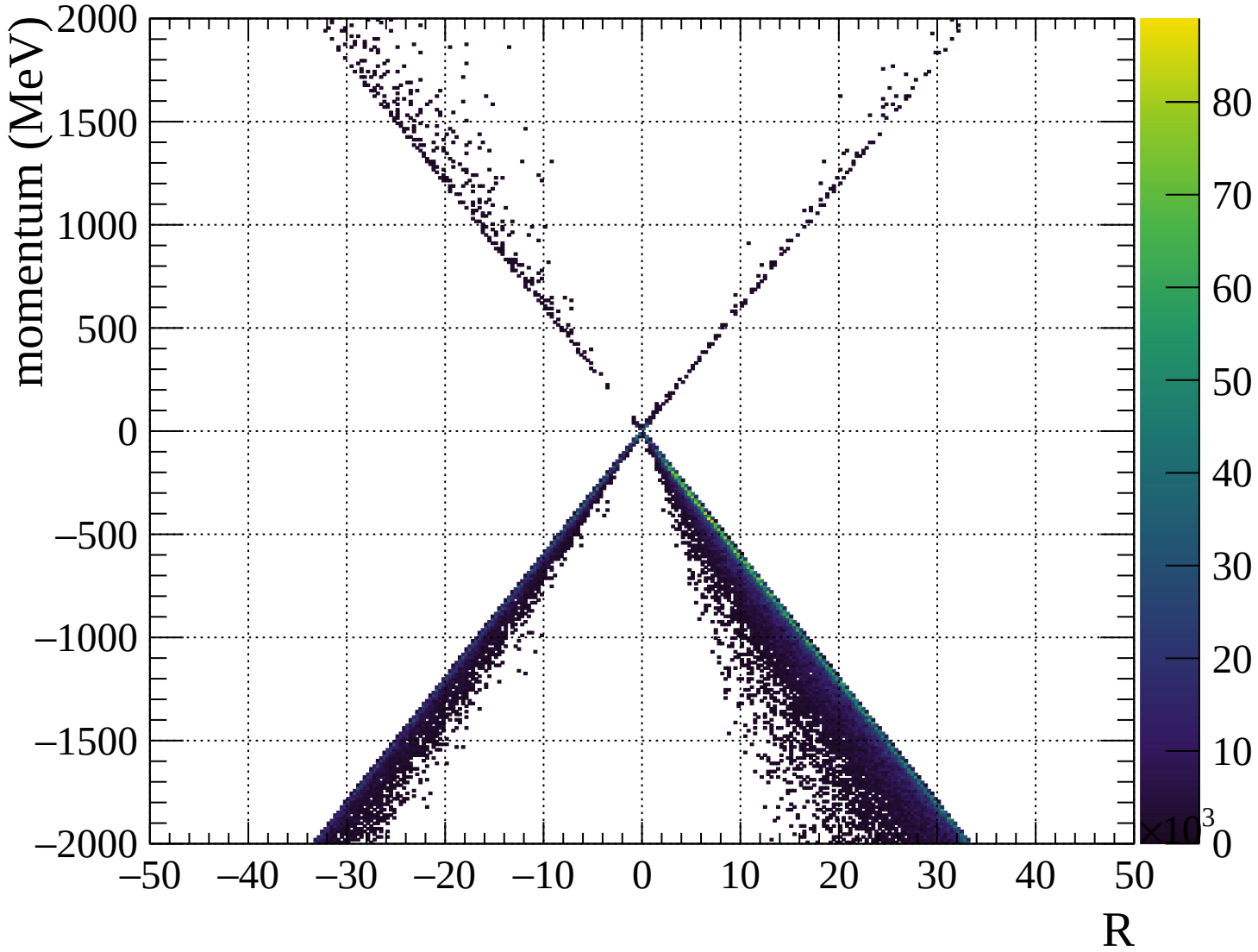


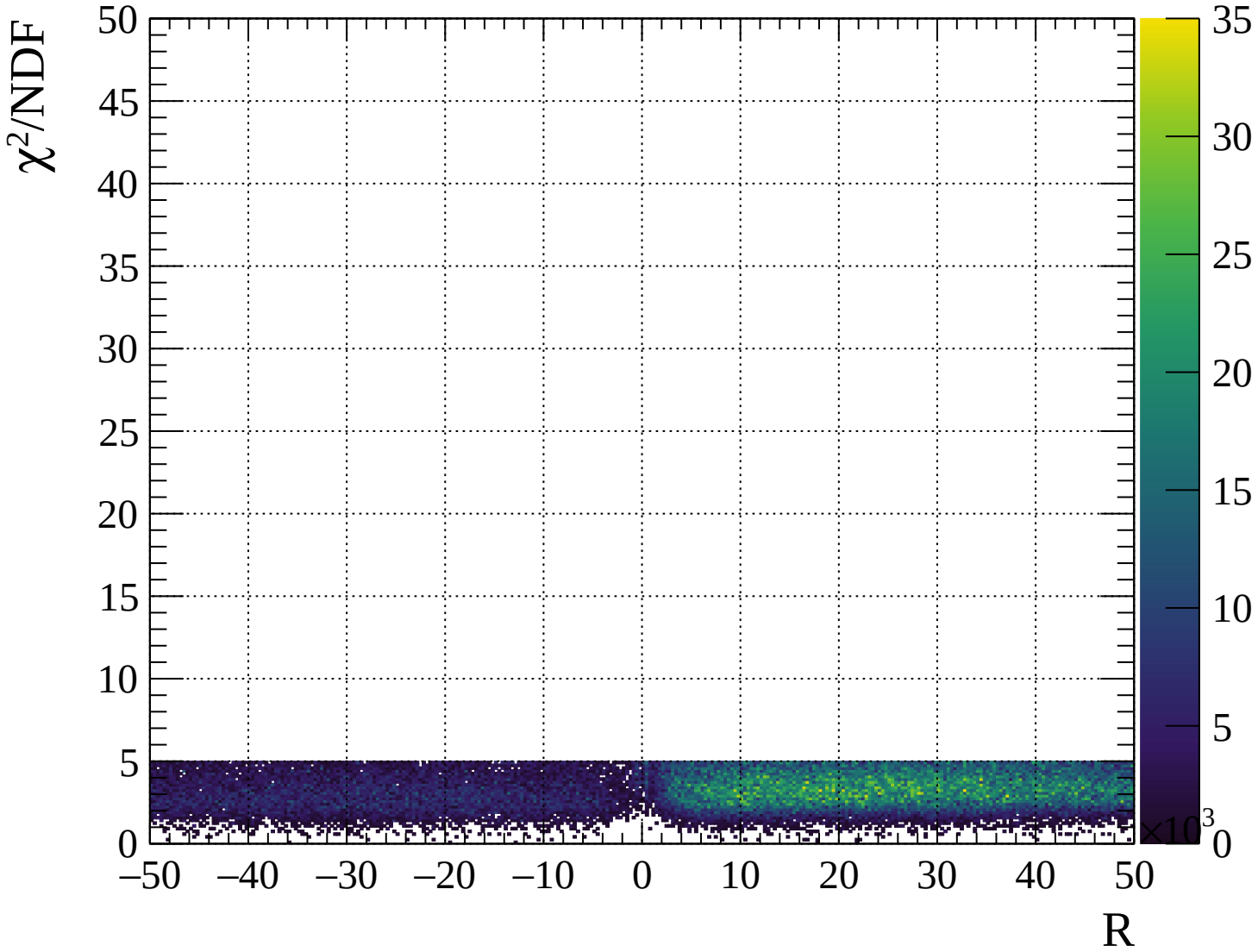






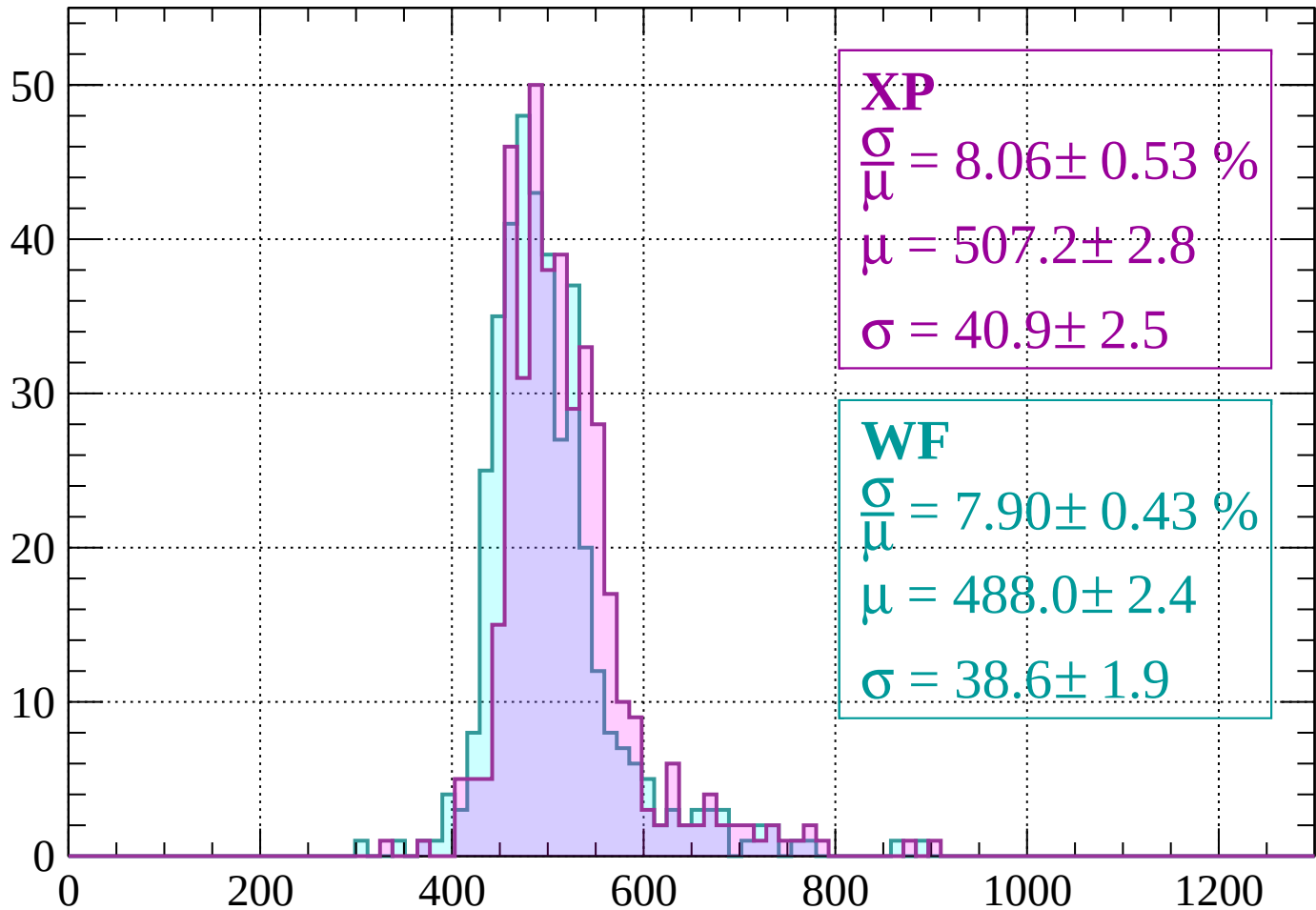






Energy loss | -2000 < p < -1960

Count



XP

$$\frac{\sigma}{\mu} = 8.06 \pm 0.53 \%$$

$$\mu = 507.2 \pm 2.8$$

$$\sigma = 40.9 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 7.90 \pm 0.43 \%$$

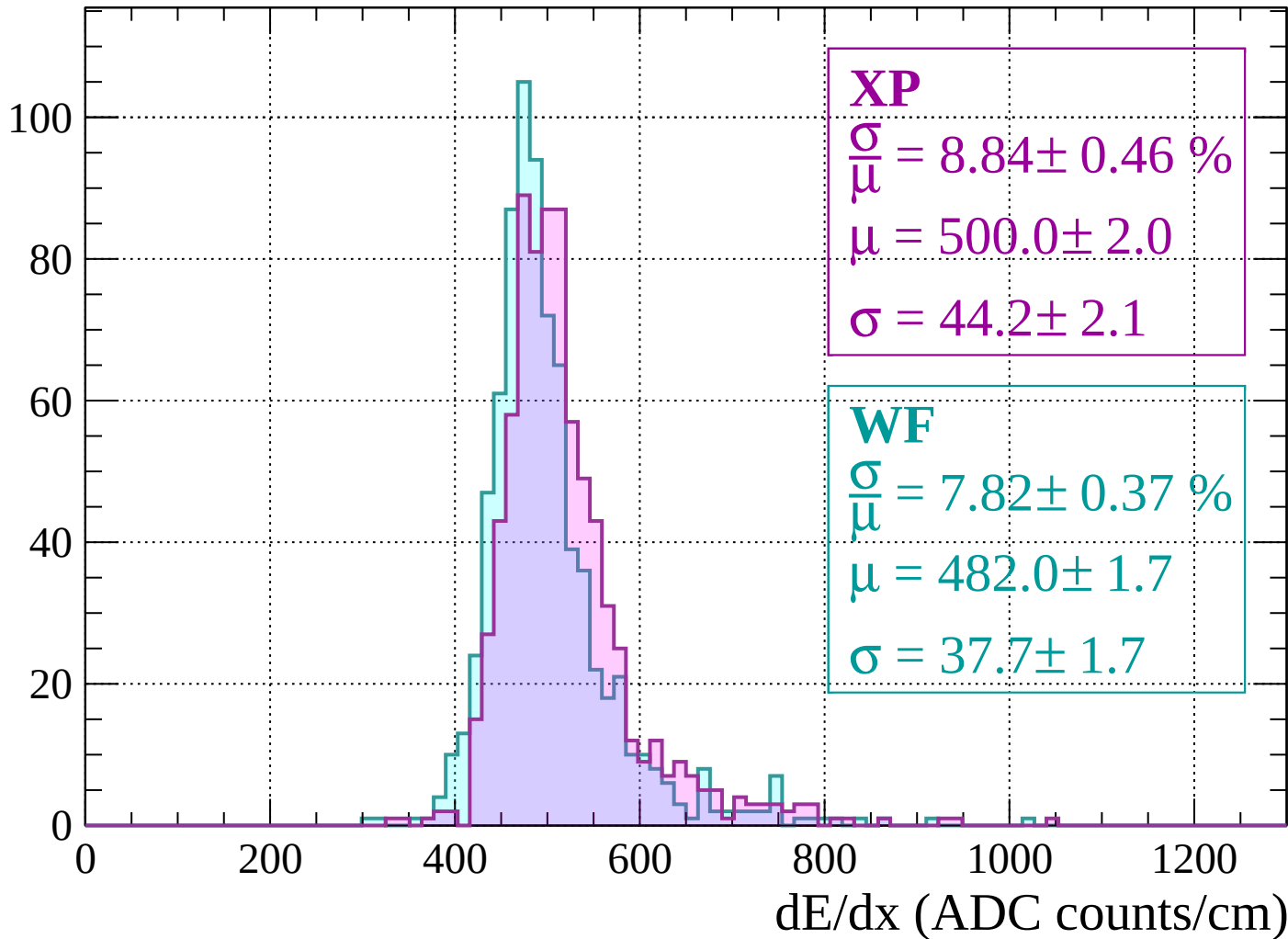
$$\mu = 488.0 \pm 2.4$$

$$\sigma = 38.6 \pm 1.9$$

dE/dx (ADC counts/cm)

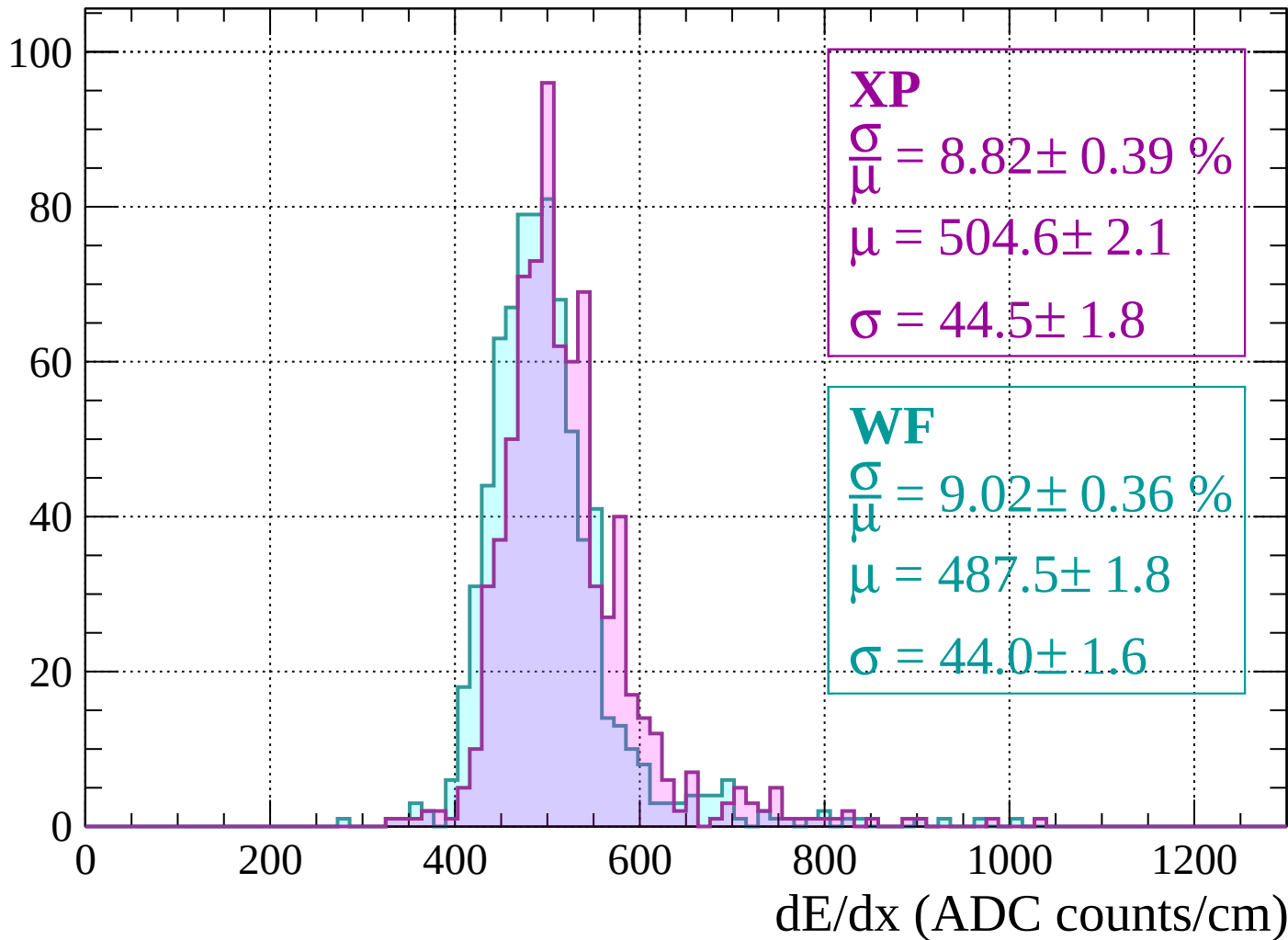
Energy loss | -1960 < p < -1920

Count



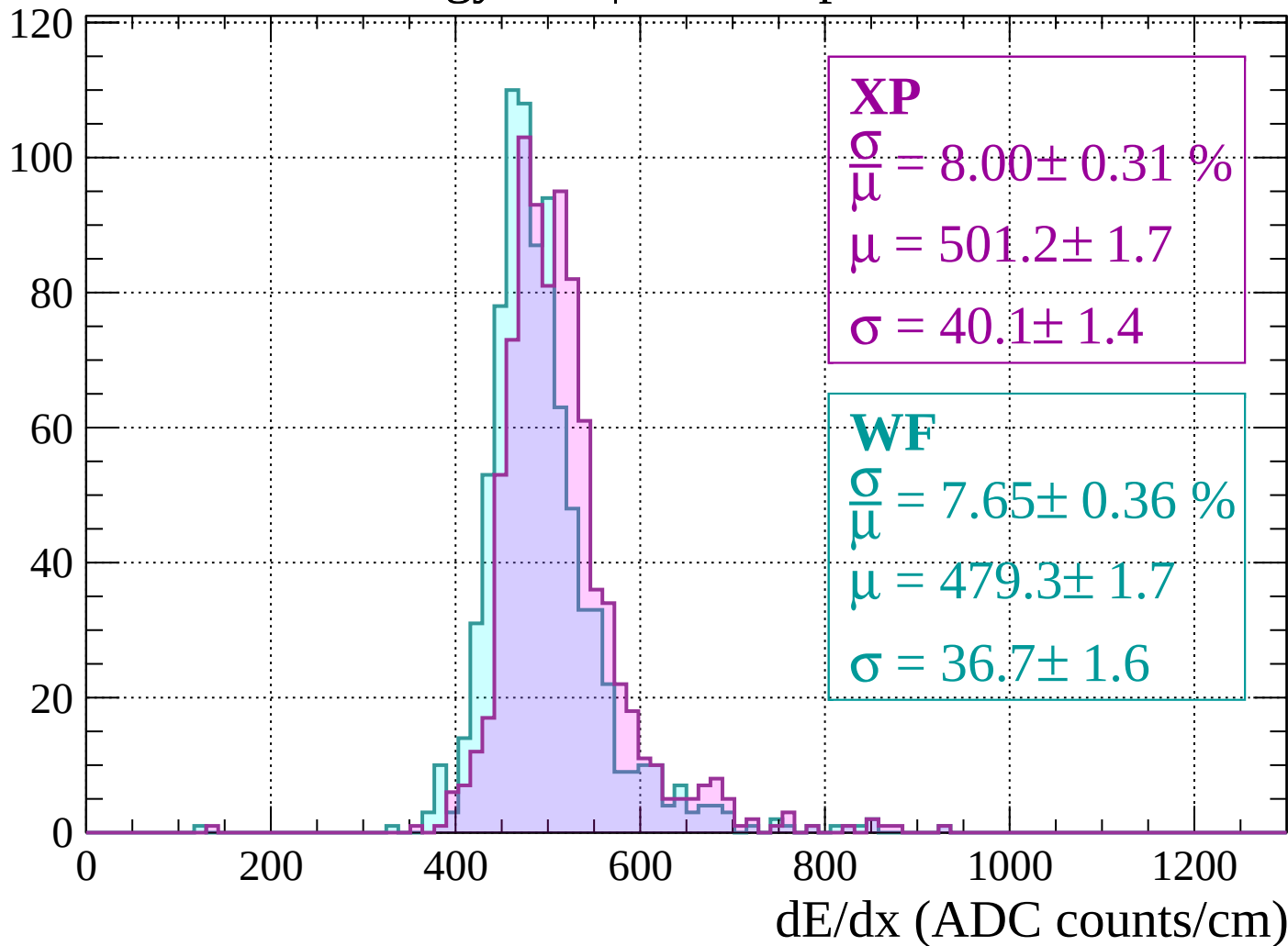
Energy loss | -1920 < p < -1880

Count



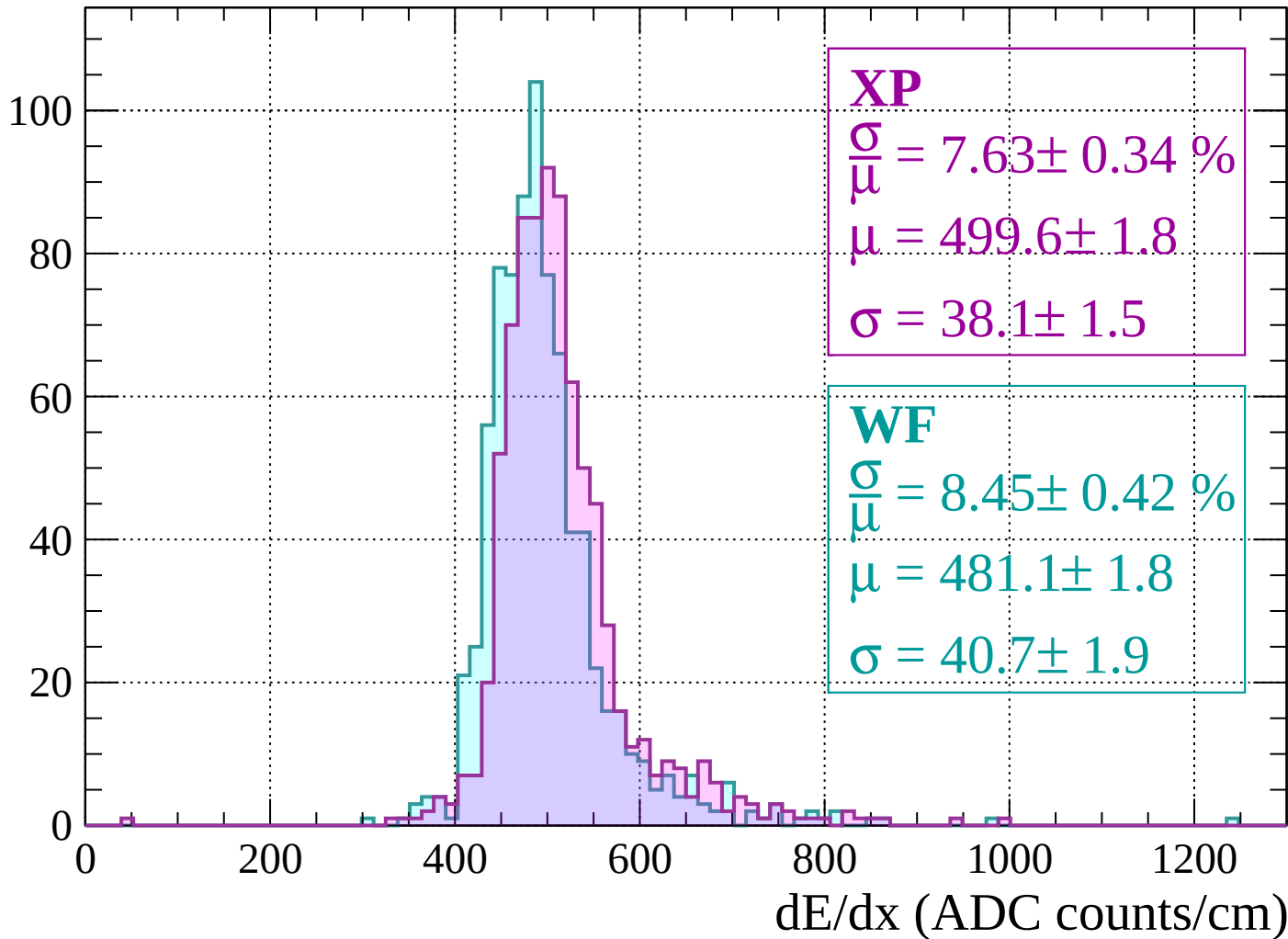
Energy loss | $-1880 < p < -1840$

Count



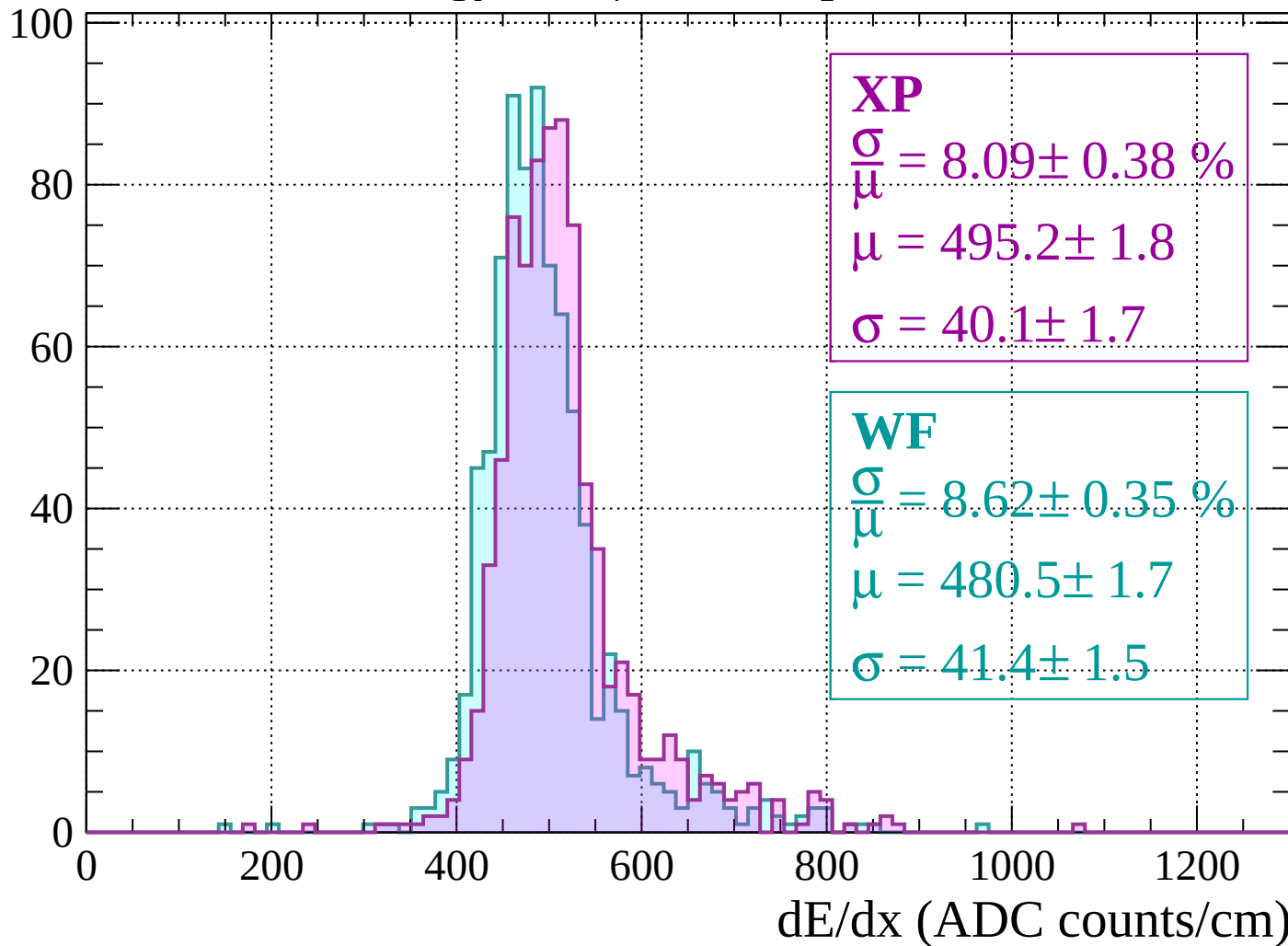
Energy loss | $-1840 < p < -1800$

Count



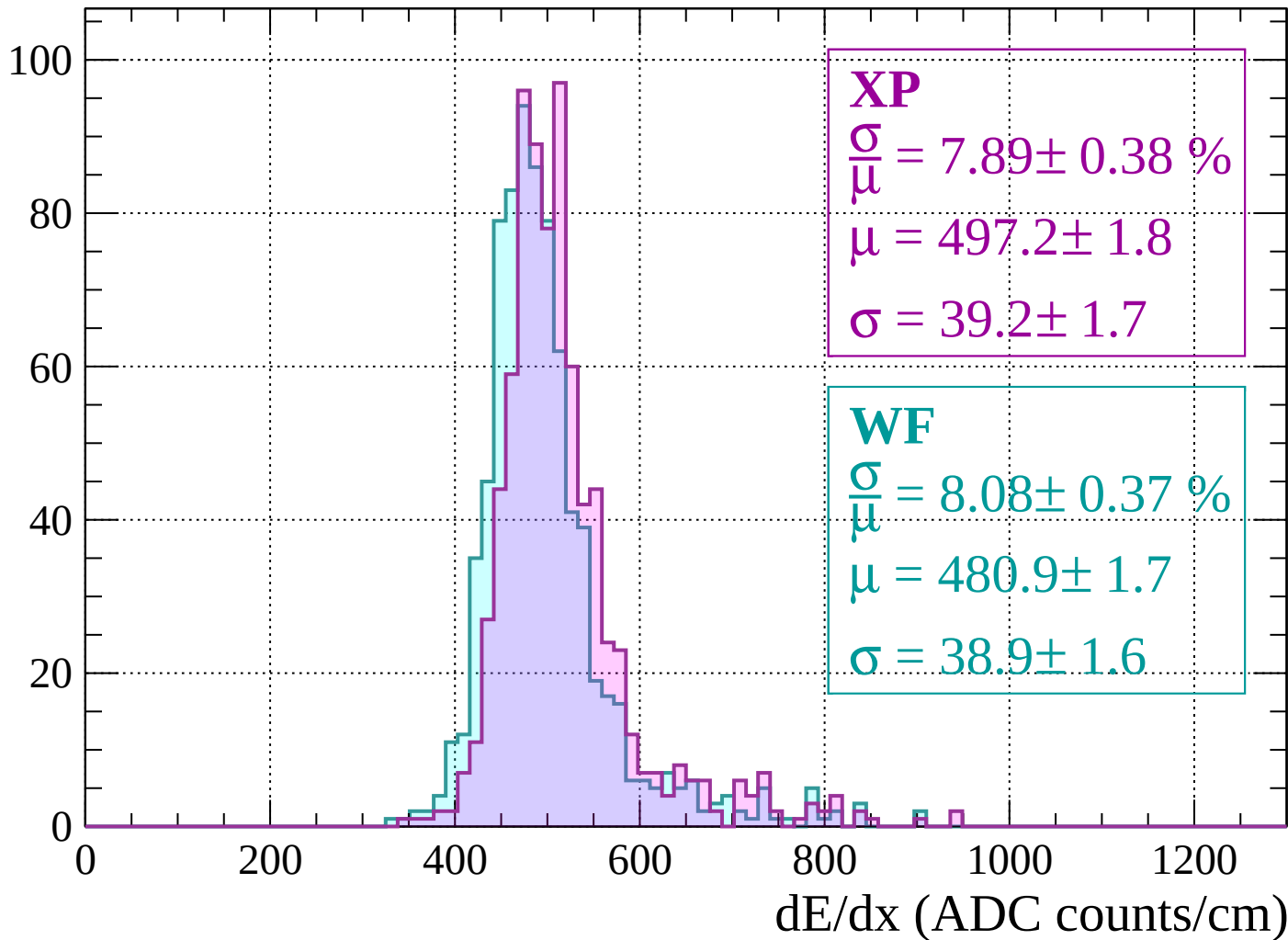
Energy loss | -1800 < p < -1760

Count



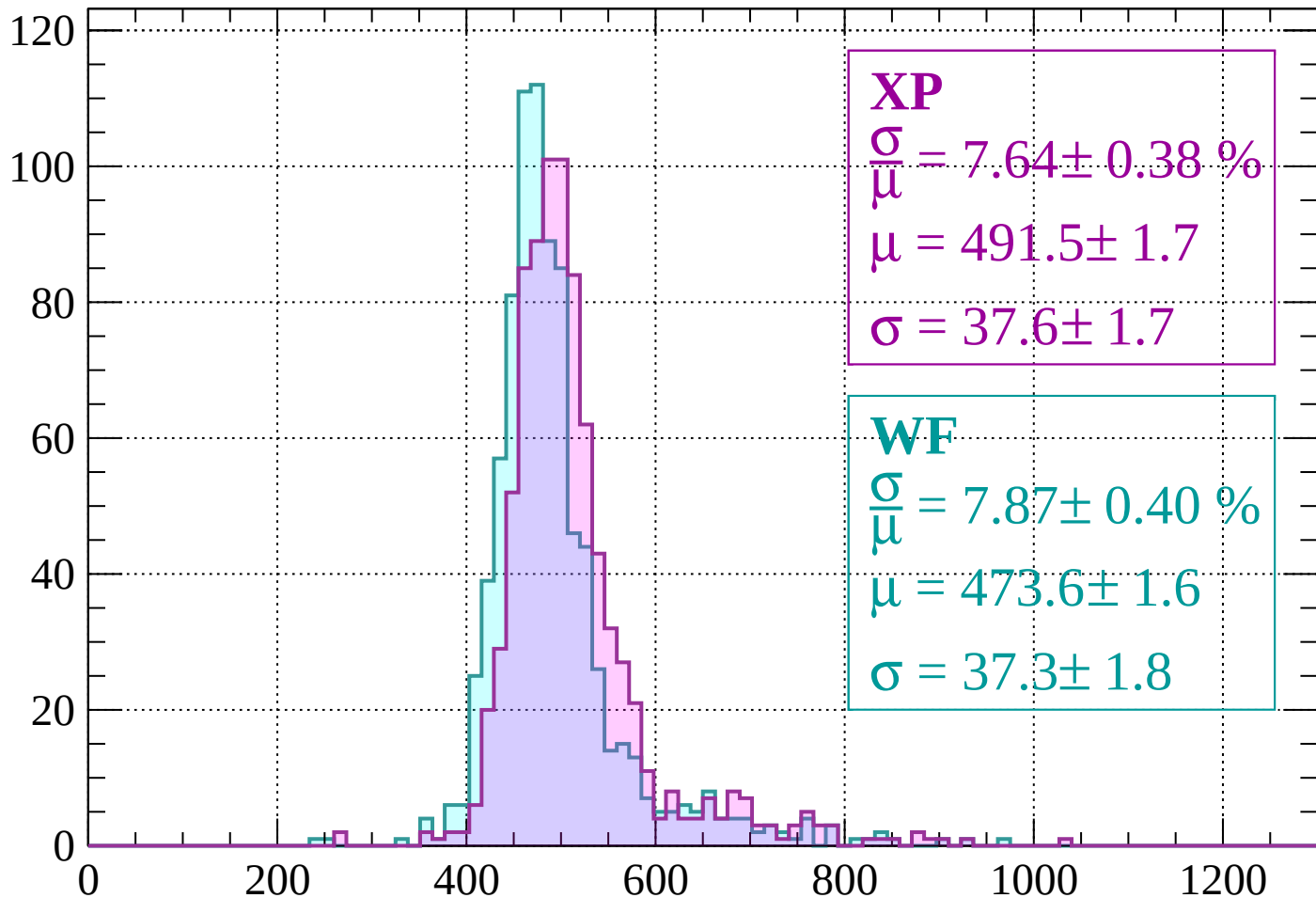
Energy loss | $-1760 < p < -1720$

Count



Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 7.64 \pm 0.38 \%$$

$$\mu = 491.5 \pm 1.7$$

$$\sigma = 37.6 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.87 \pm 0.40 \%$$

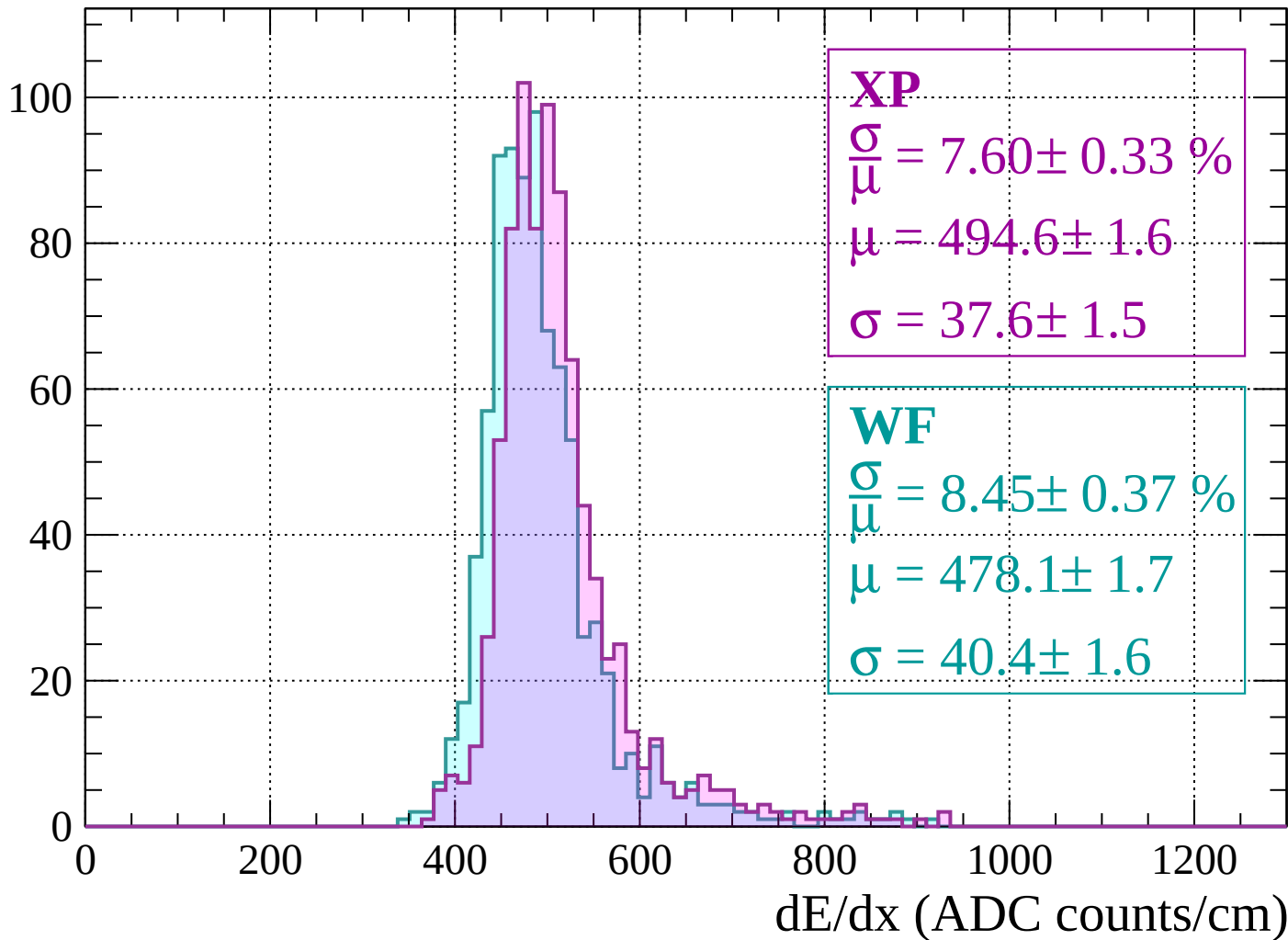
$$\mu = 473.6 \pm 1.6$$

$$\sigma = 37.3 \pm 1.8$$

dE/dx (ADC counts/cm)

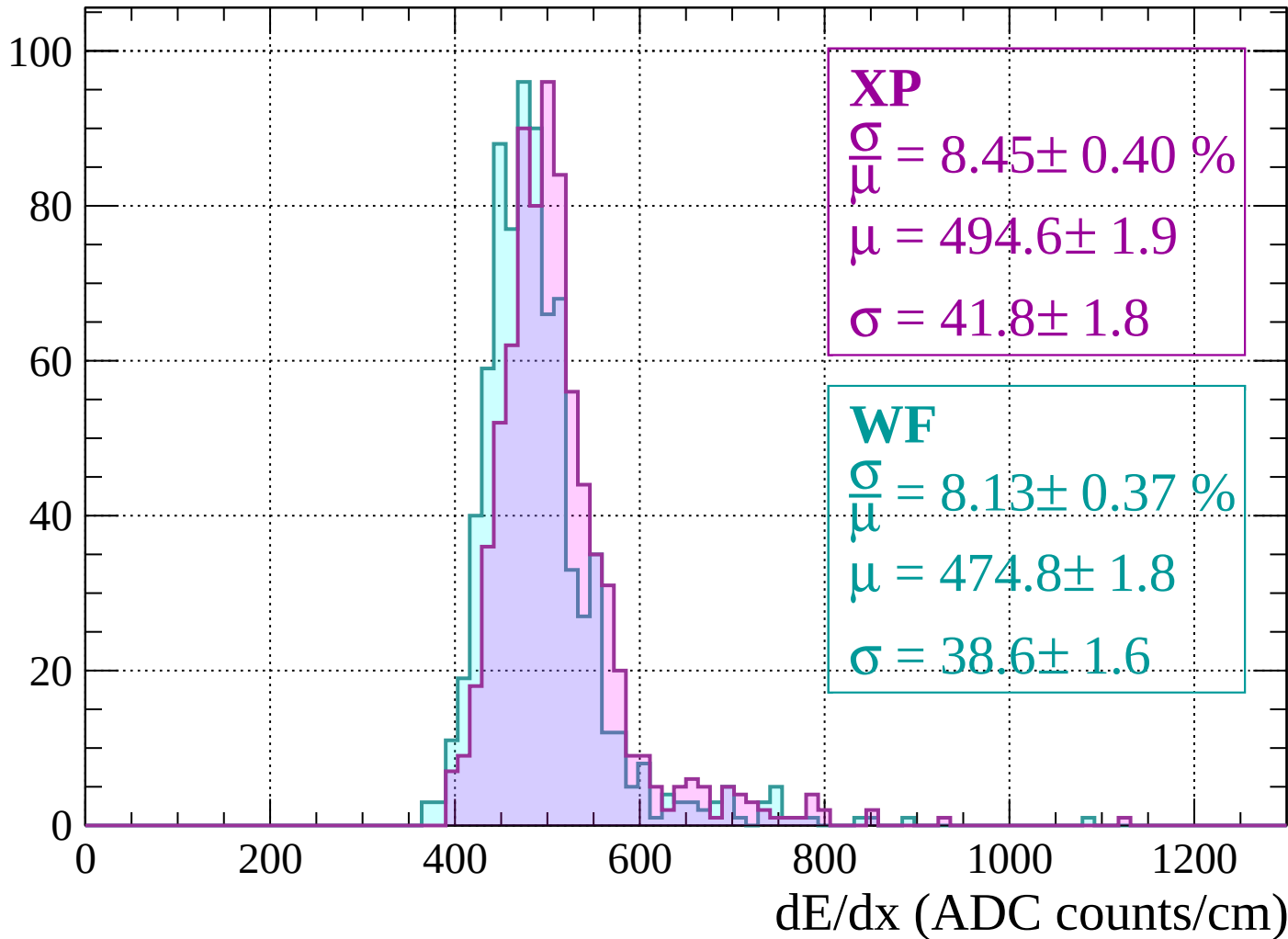
Energy loss | $-1680 < p < -1640$

Count



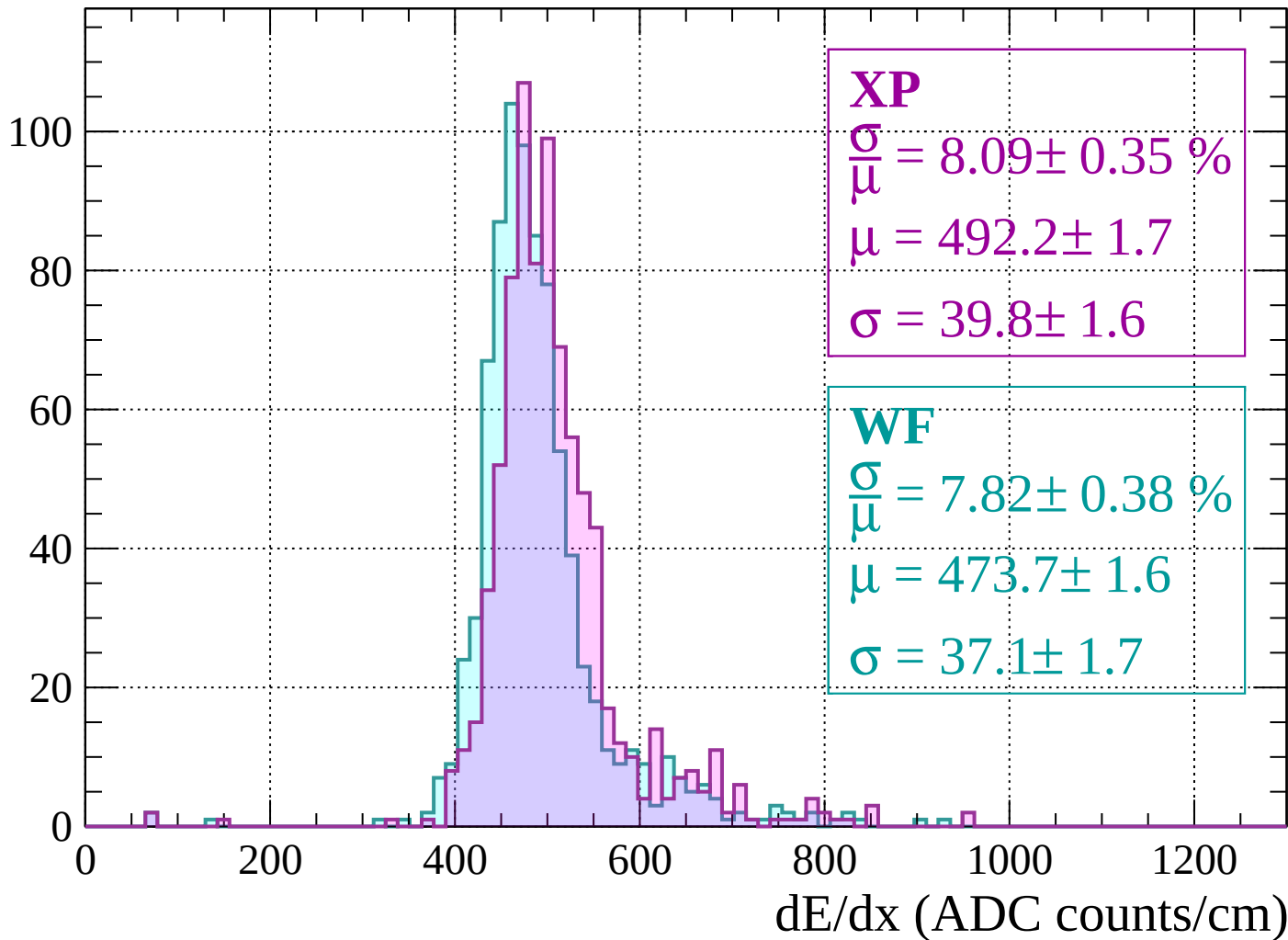
Energy loss | $-1640 < p < -1600$

Count



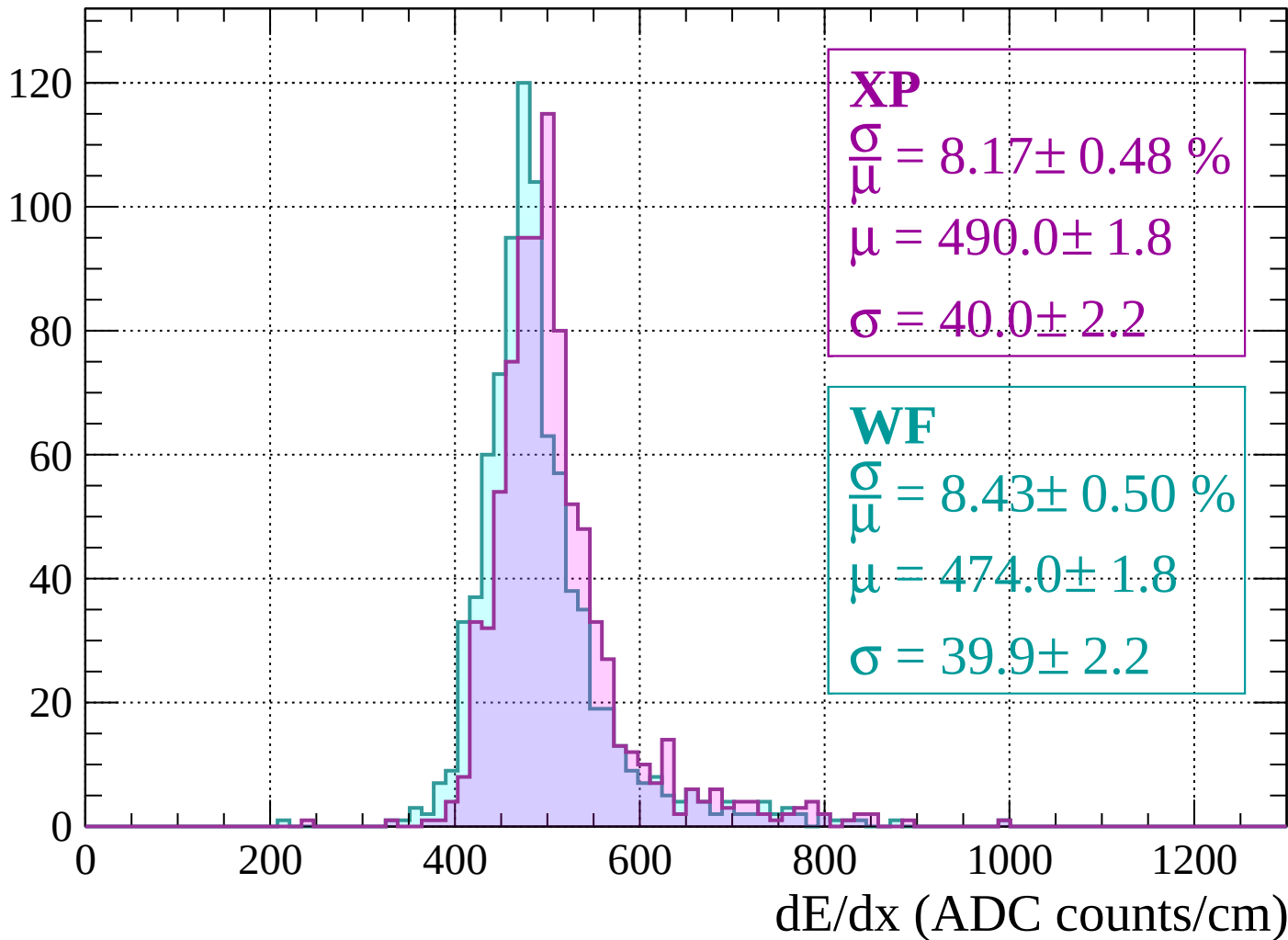
Energy loss | $-1600 < p < -1560$

Count



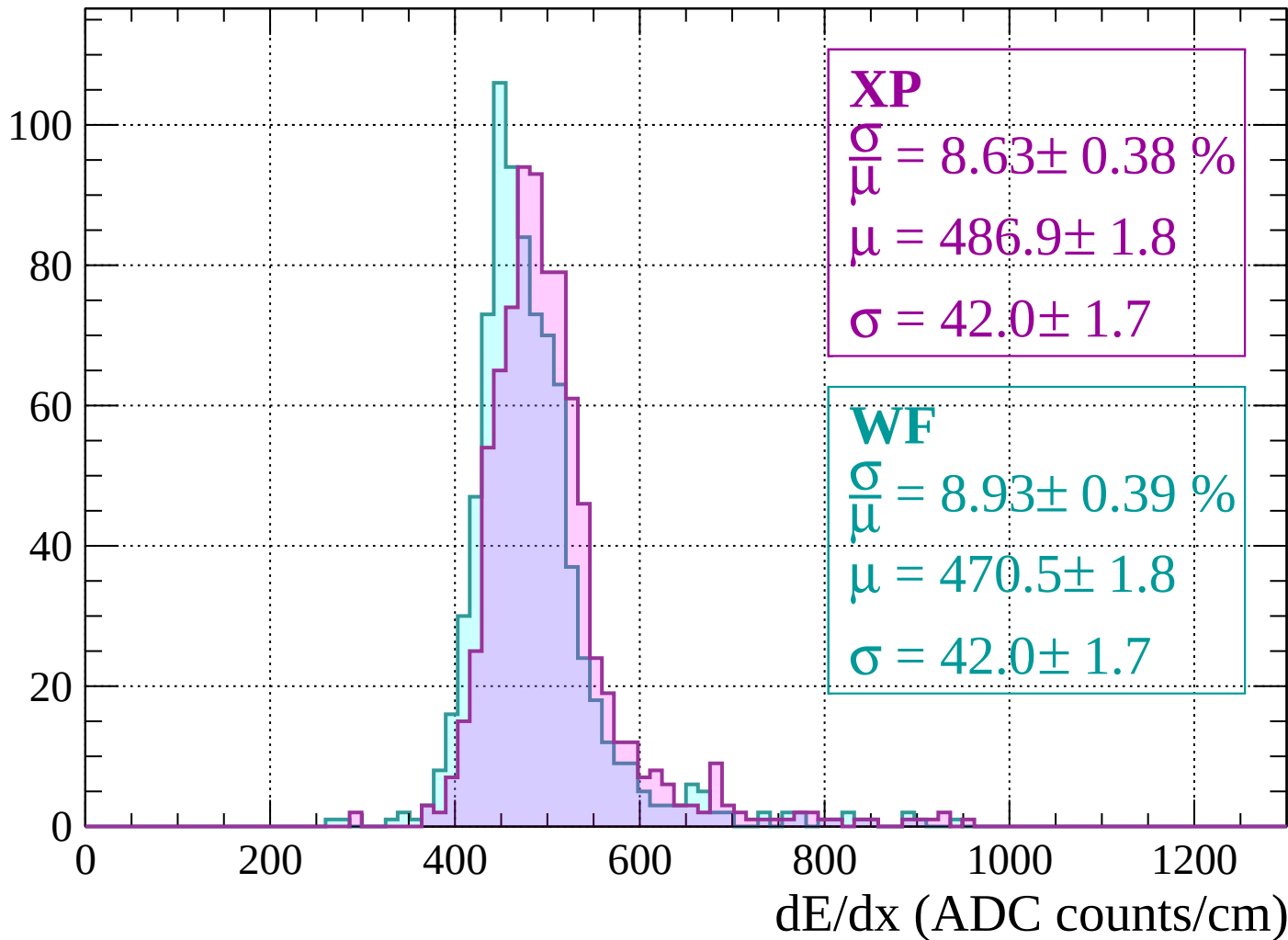
Energy loss | $-1560 < p < -1520$

Count



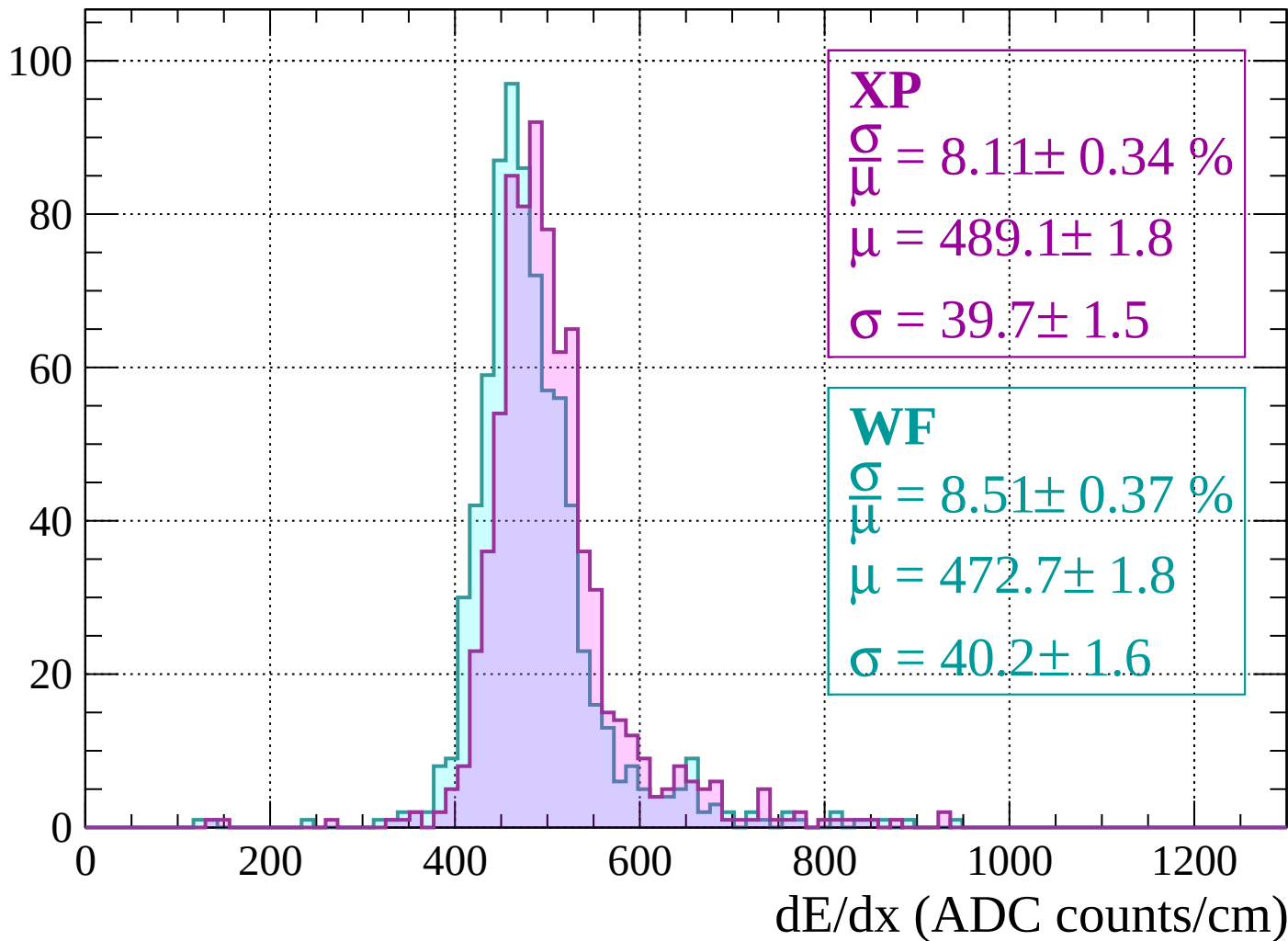
Energy loss | $-1520 < p < -1480$

Count



Energy loss | $-1480 < p < -1440$

Count



Energy loss | $-1440 < p < -1400$

Count

100

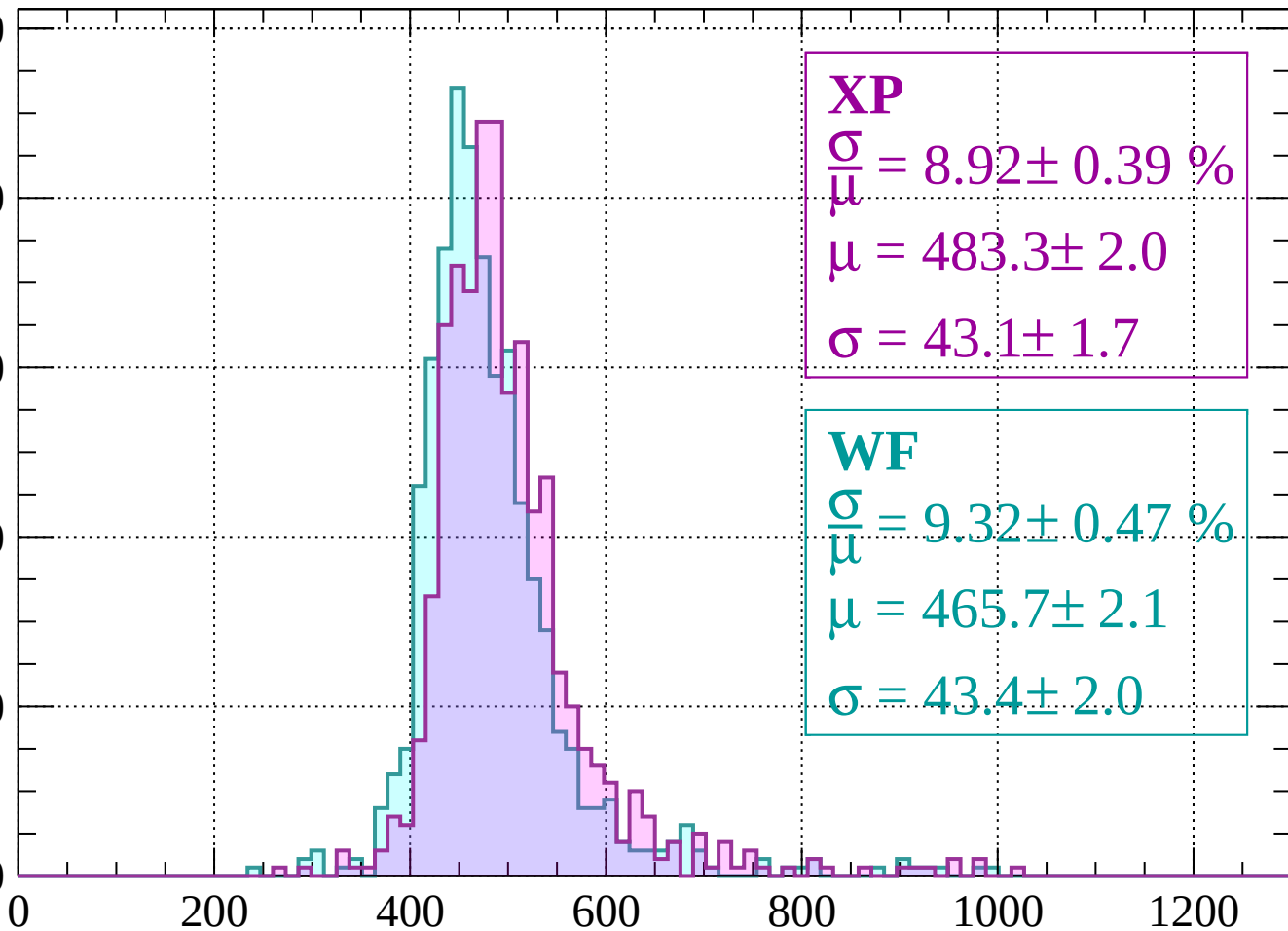
80

60

40

20

0



XP

$$\frac{\sigma}{\mu} = 8.92 \pm 0.39 \%$$

$$\mu = 483.3 \pm 2.0$$

$$\sigma = 43.1 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 9.32 \pm 0.47 \%$$

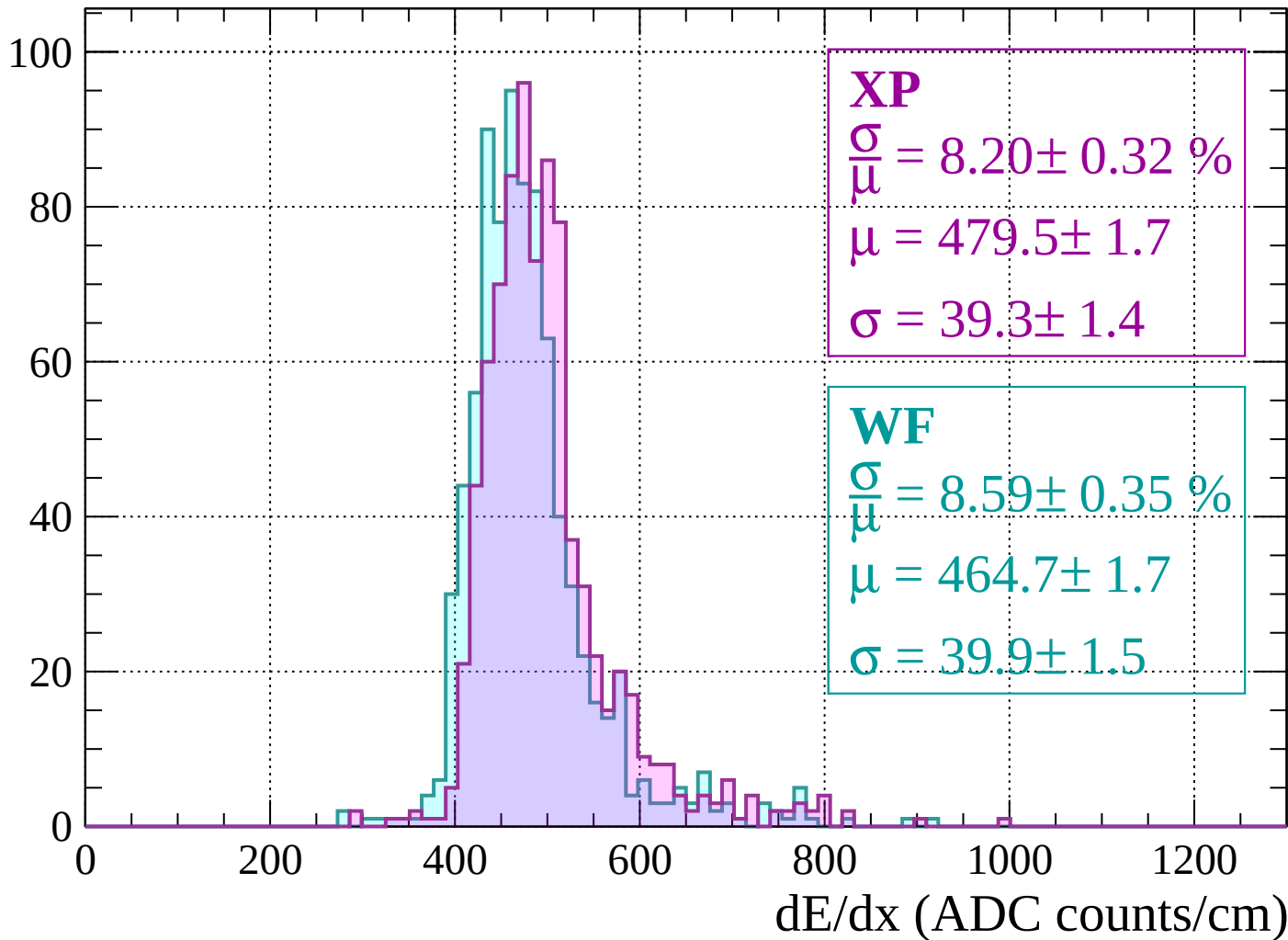
$$\mu = 465.7 \pm 2.1$$

$$\sigma = 43.4 \pm 2.0$$

dE/dx (ADC counts/cm)

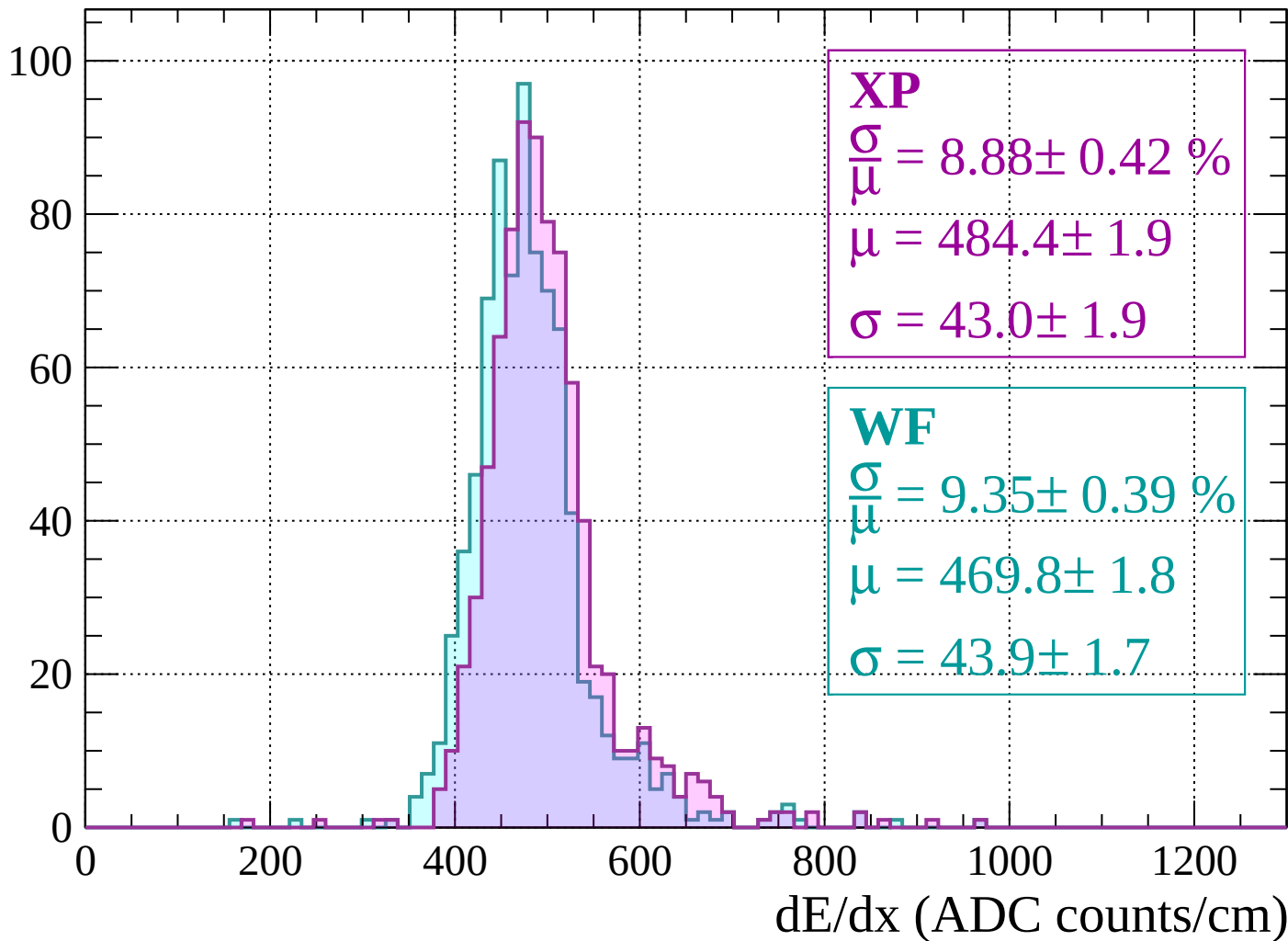
Energy loss | $-1400 < p < -1360$

Count



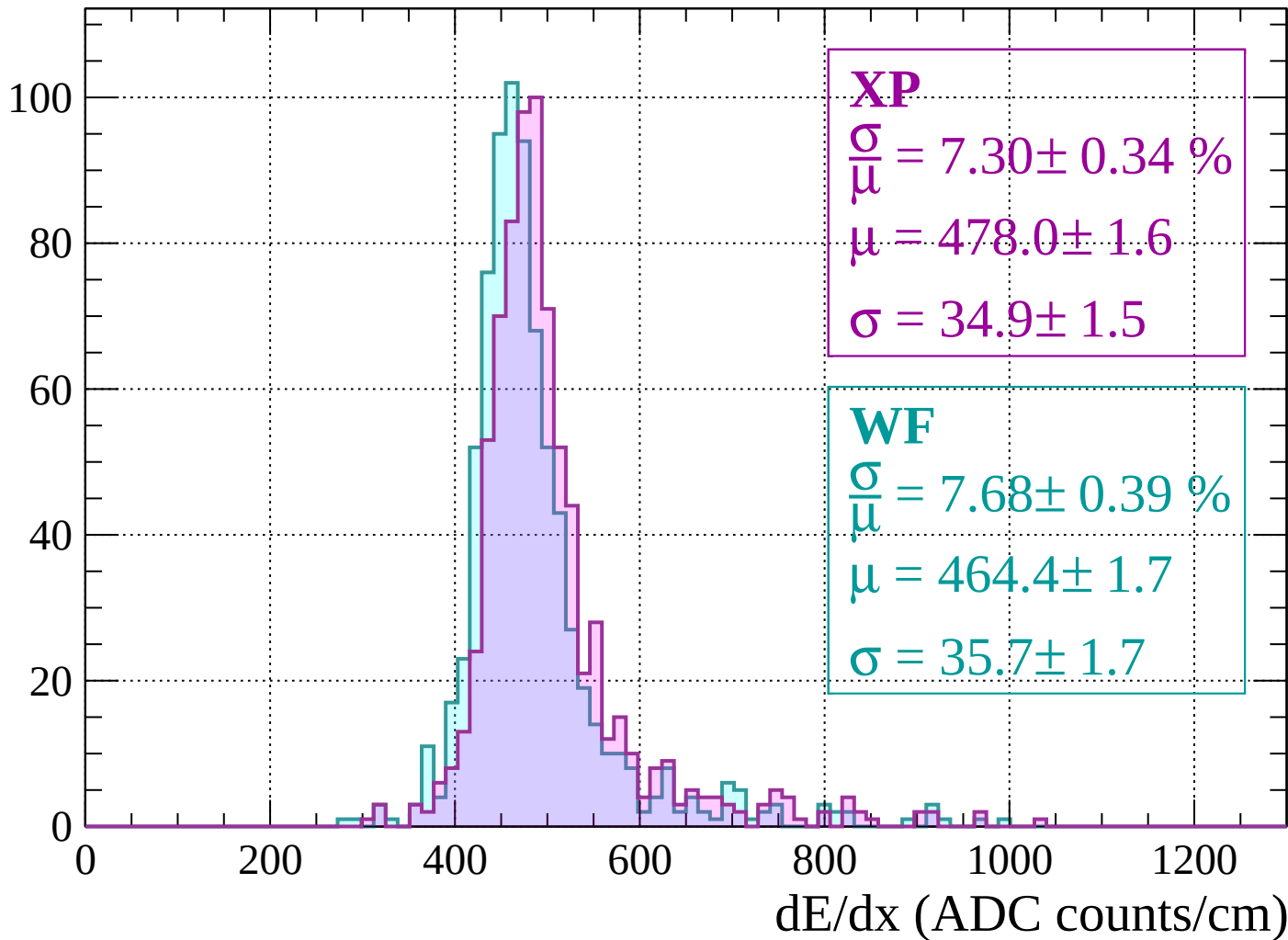
Energy loss | $-1360 < p < -1320$

Count



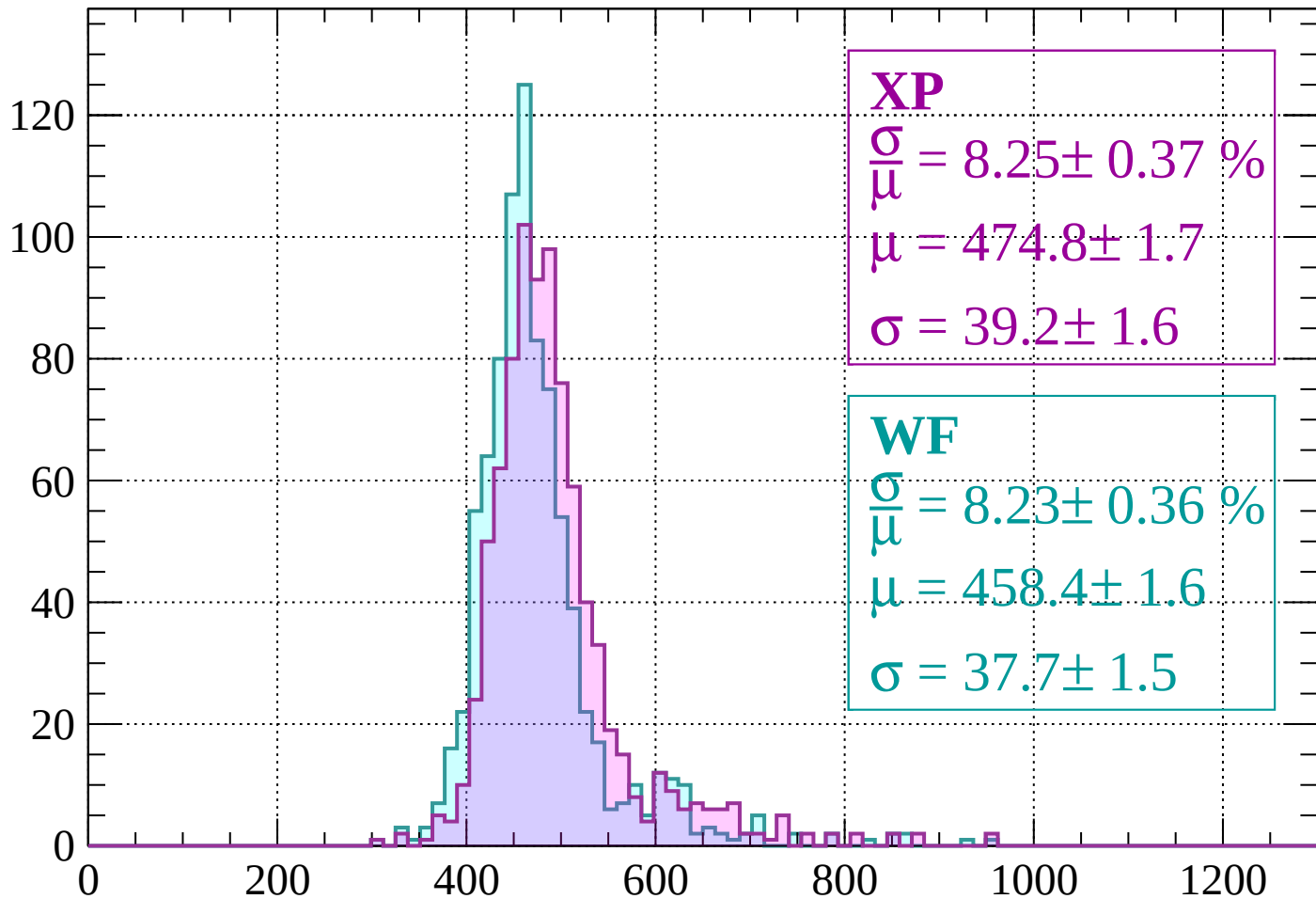
Energy loss | $-1320 < p < -1280$

Count



Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 8.25 \pm 0.37 \%$$

$$\mu = 474.8 \pm 1.7$$

$$\sigma = 39.2 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.23 \pm 0.36 \%$$

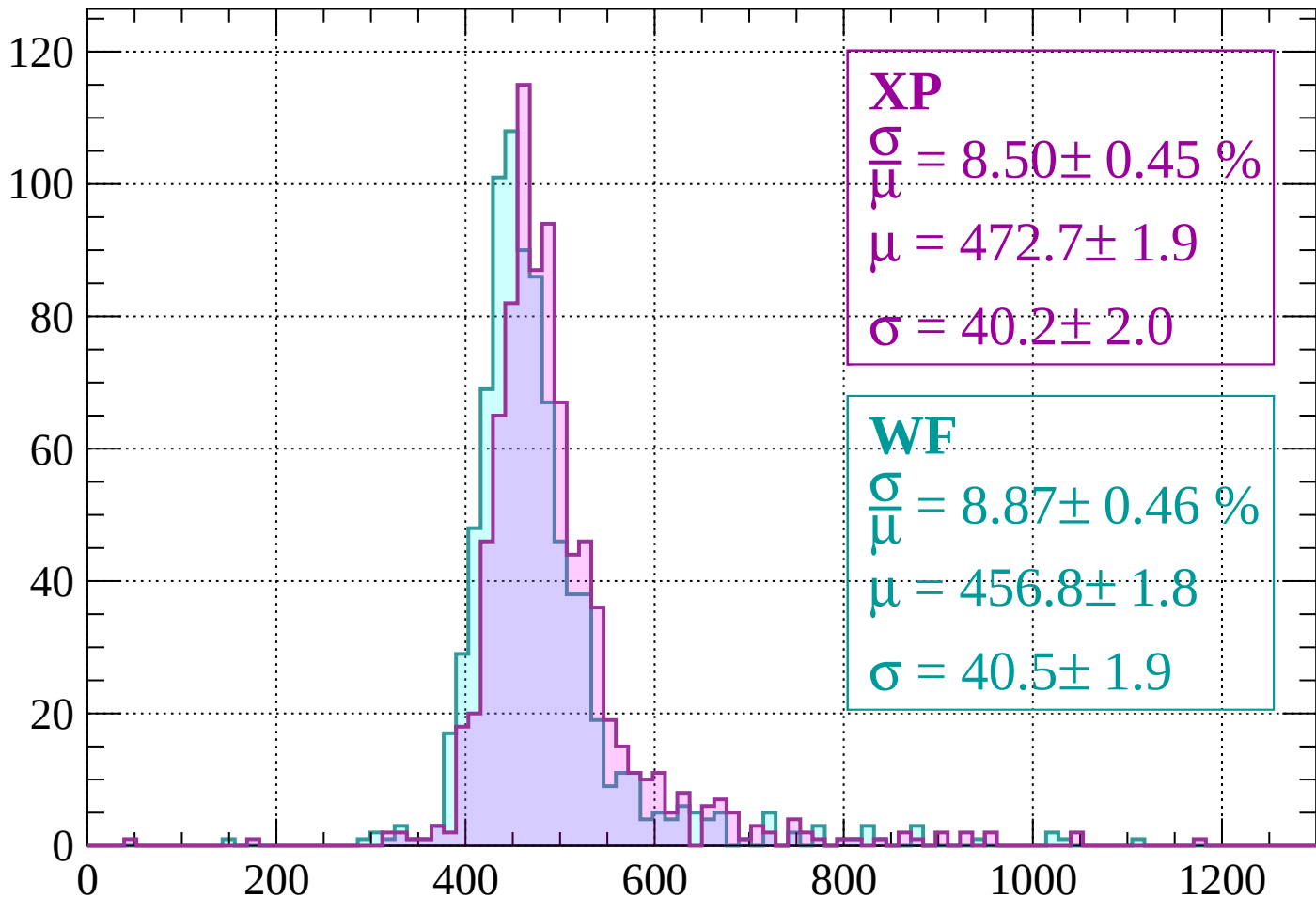
$$\mu = 458.4 \pm 1.6$$

$$\sigma = 37.7 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 8.50 \pm 0.45 \%$$

$$\mu = 472.7 \pm 1.9$$

$$\sigma = 40.2 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 8.87 \pm 0.46 \%$$

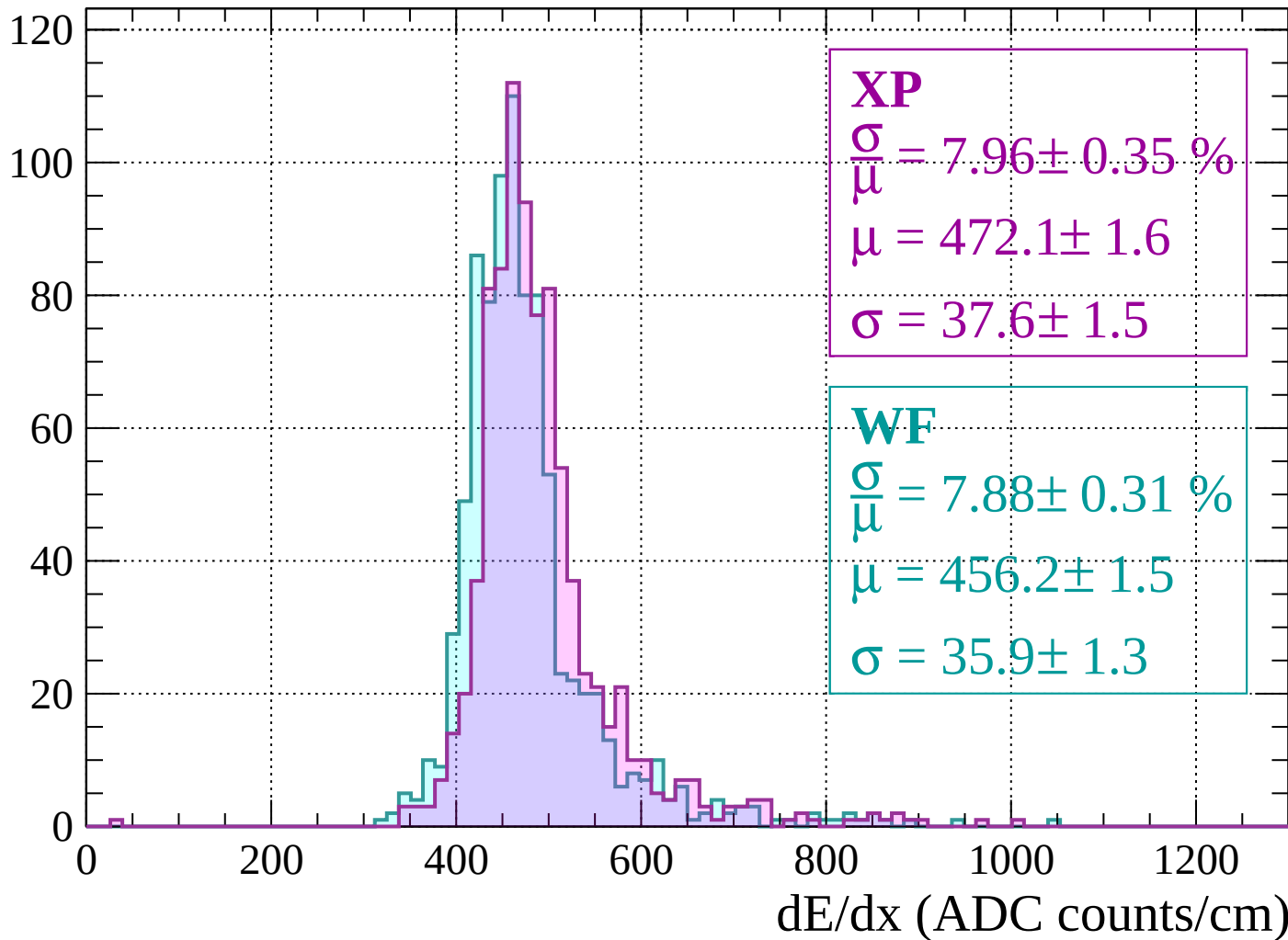
$$\mu = 456.8 \pm 1.8$$

$$\sigma = 40.5 \pm 1.9$$

dE/dx (ADC counts/cm)

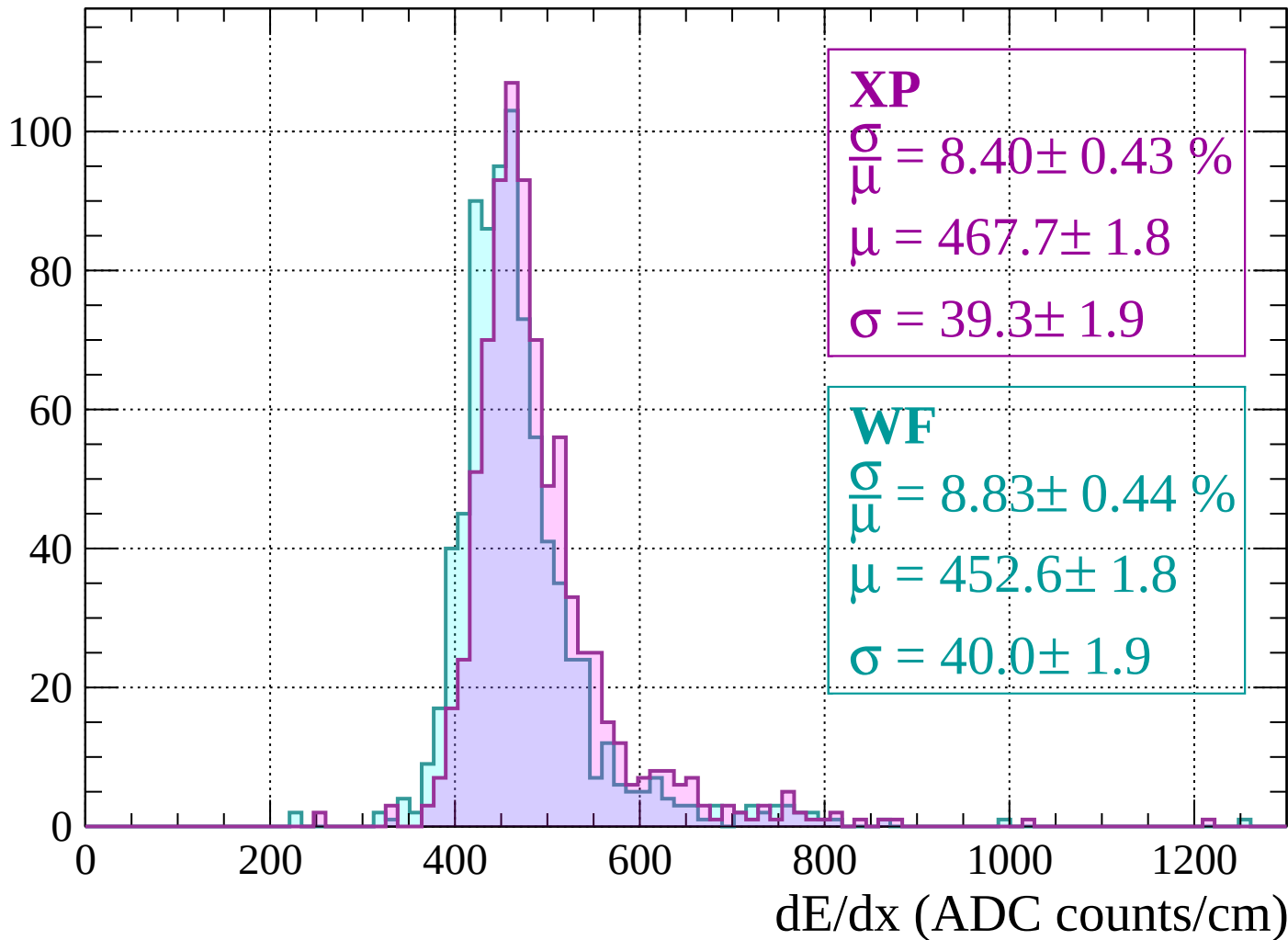
Energy loss | -1200 < p < -1160

Count



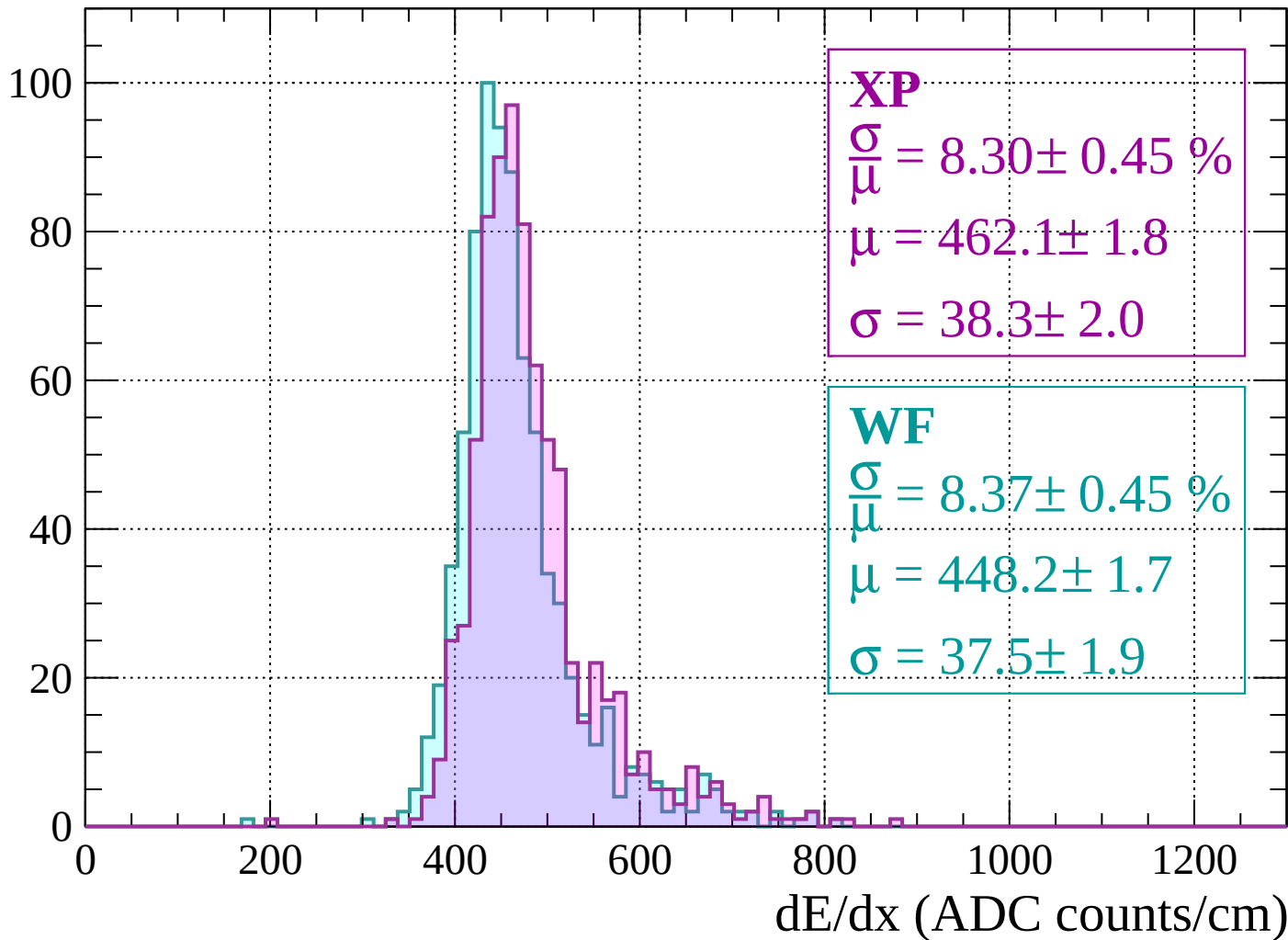
Energy loss | $-1160 < p < -1120$

Count



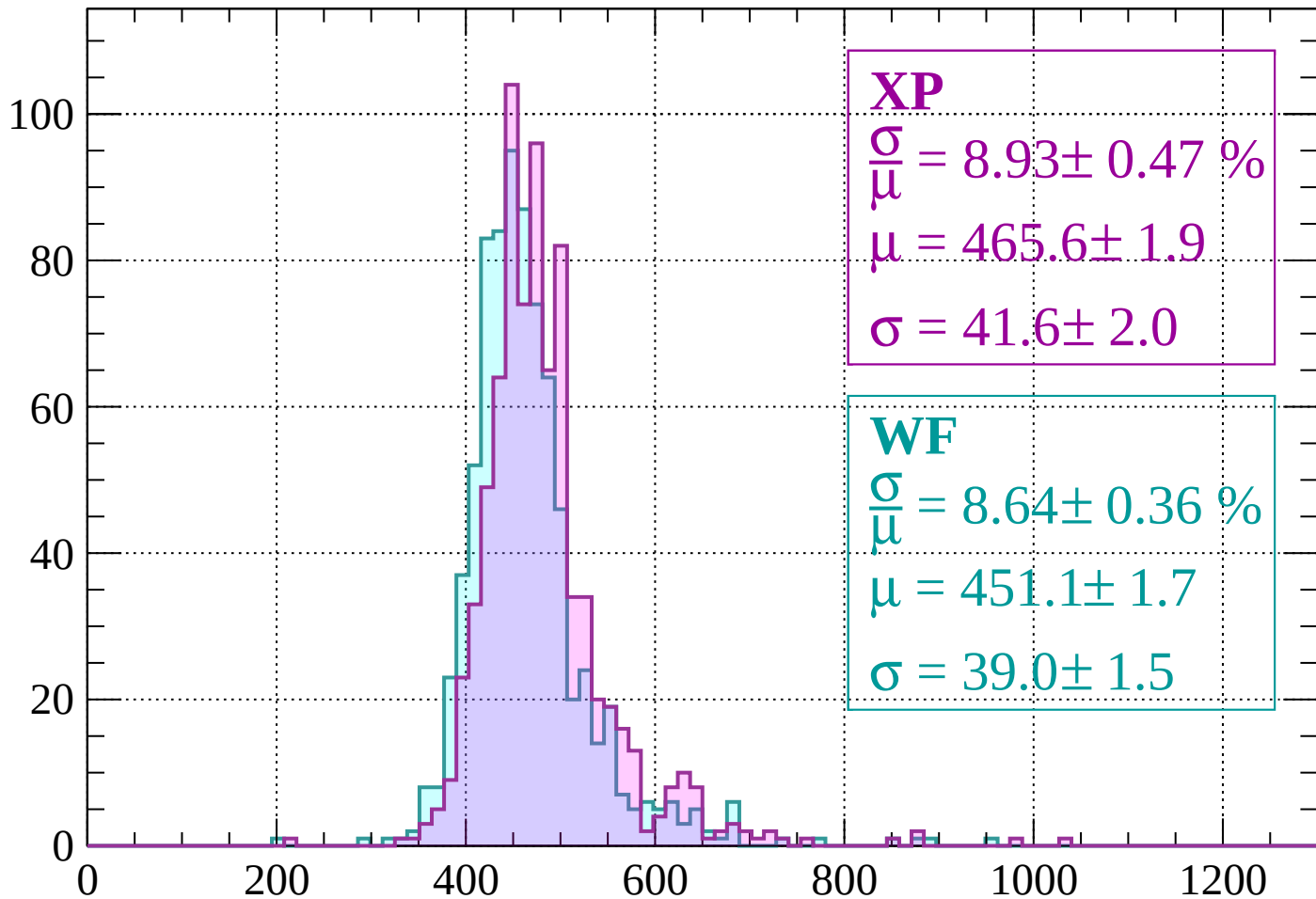
Energy loss | $-1120 < p < -1080$

Count



Energy loss | $-1080 < p < -1040$

Count



XP

$$\frac{\sigma}{\mu} = 8.93 \pm 0.47 \%$$

$$\mu = 465.6 \pm 1.9$$

$$\sigma = 41.6 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 8.64 \pm 0.36 \%$$

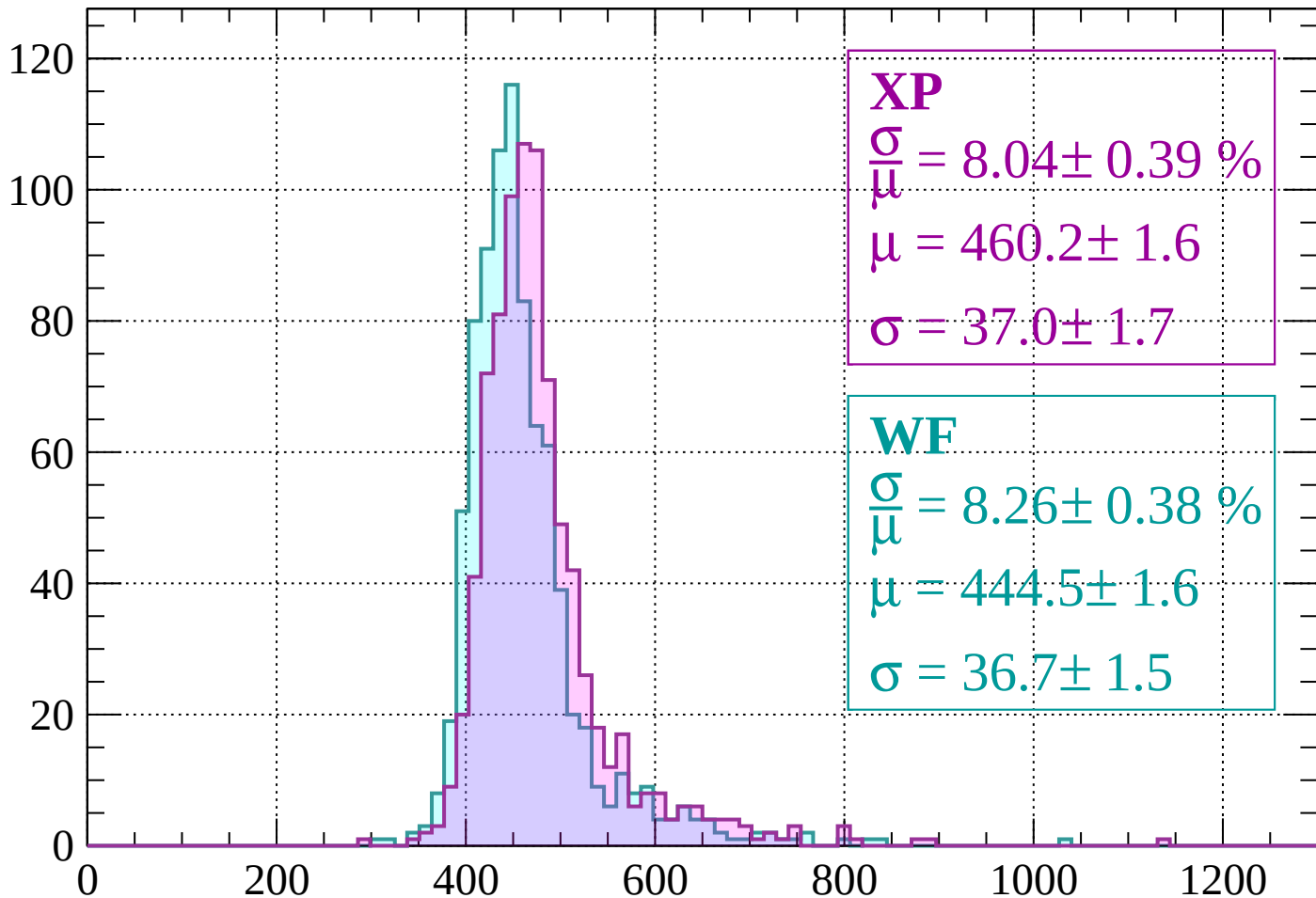
$$\mu = 451.1 \pm 1.7$$

$$\sigma = 39.0 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\mu} = 8.04 \pm 0.39 \%$$

$$\mu = 460.2 \pm 1.6$$

$$\sigma = 37.0 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.26 \pm 0.38 \%$$

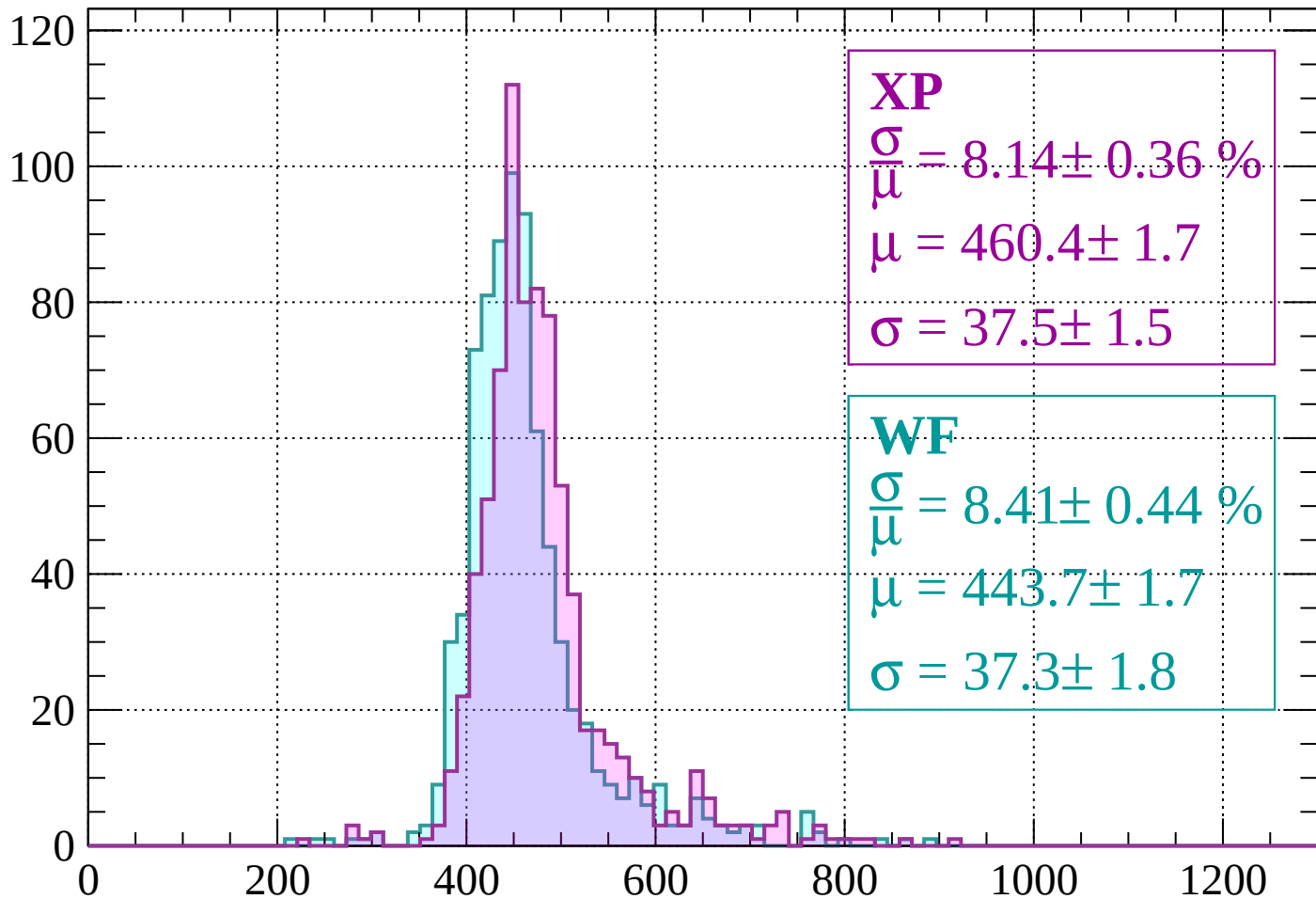
$$\mu = 444.5 \pm 1.6$$

$$\sigma = 36.7 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.36 \%$$

$$\mu = 460.4 \pm 1.7$$

$$\sigma = 37.5 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.41 \pm 0.44 \%$$

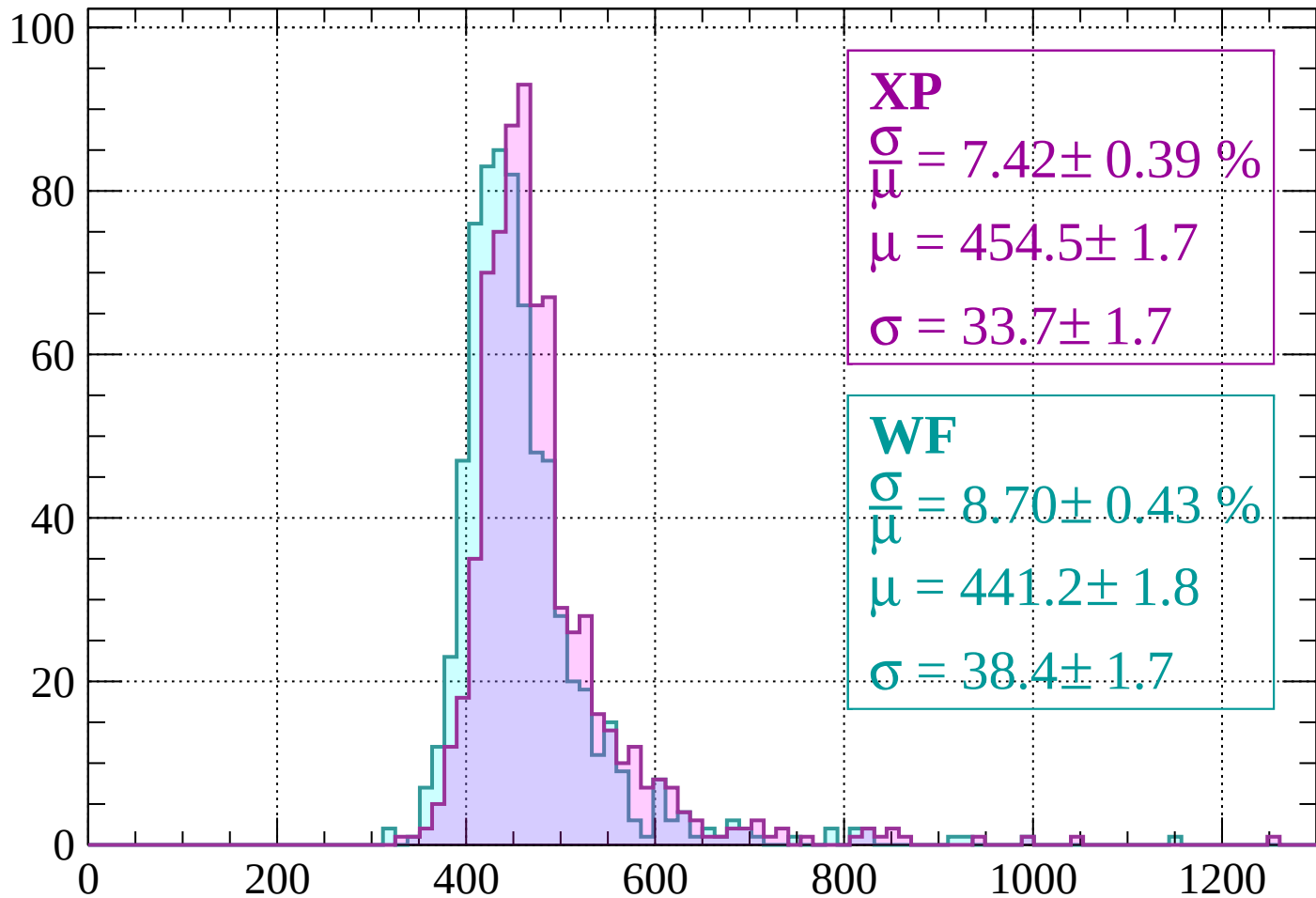
$$\mu = 443.7 \pm 1.7$$

$$\sigma = 37.3 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 7.42 \pm 0.39 \%$$

$$\mu = 454.5 \pm 1.7$$

$$\sigma = 33.7 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.70 \pm 0.43 \%$$

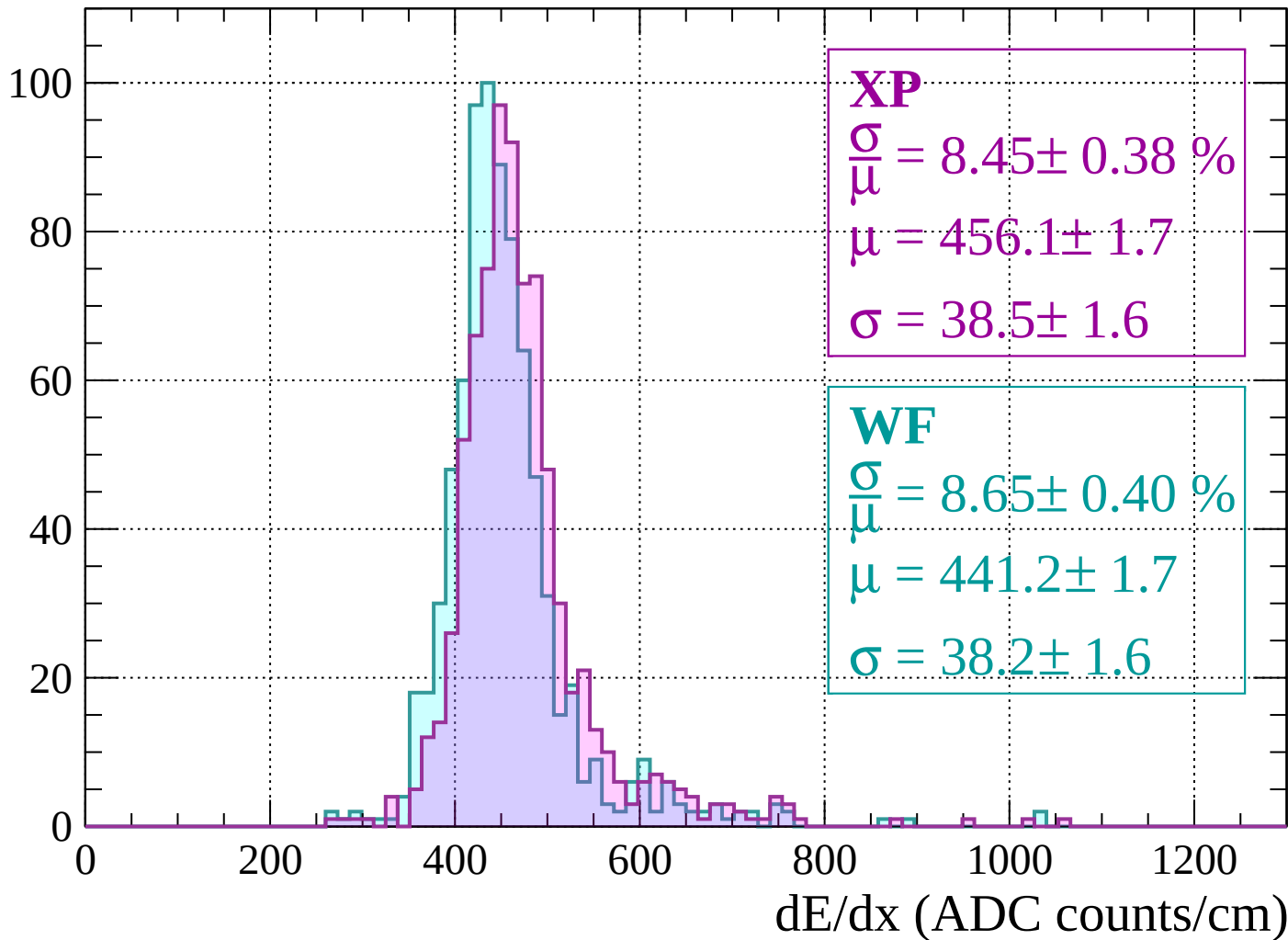
$$\mu = 441.2 \pm 1.8$$

$$\sigma = 38.4 \pm 1.7$$

dE/dx (ADC counts/cm)

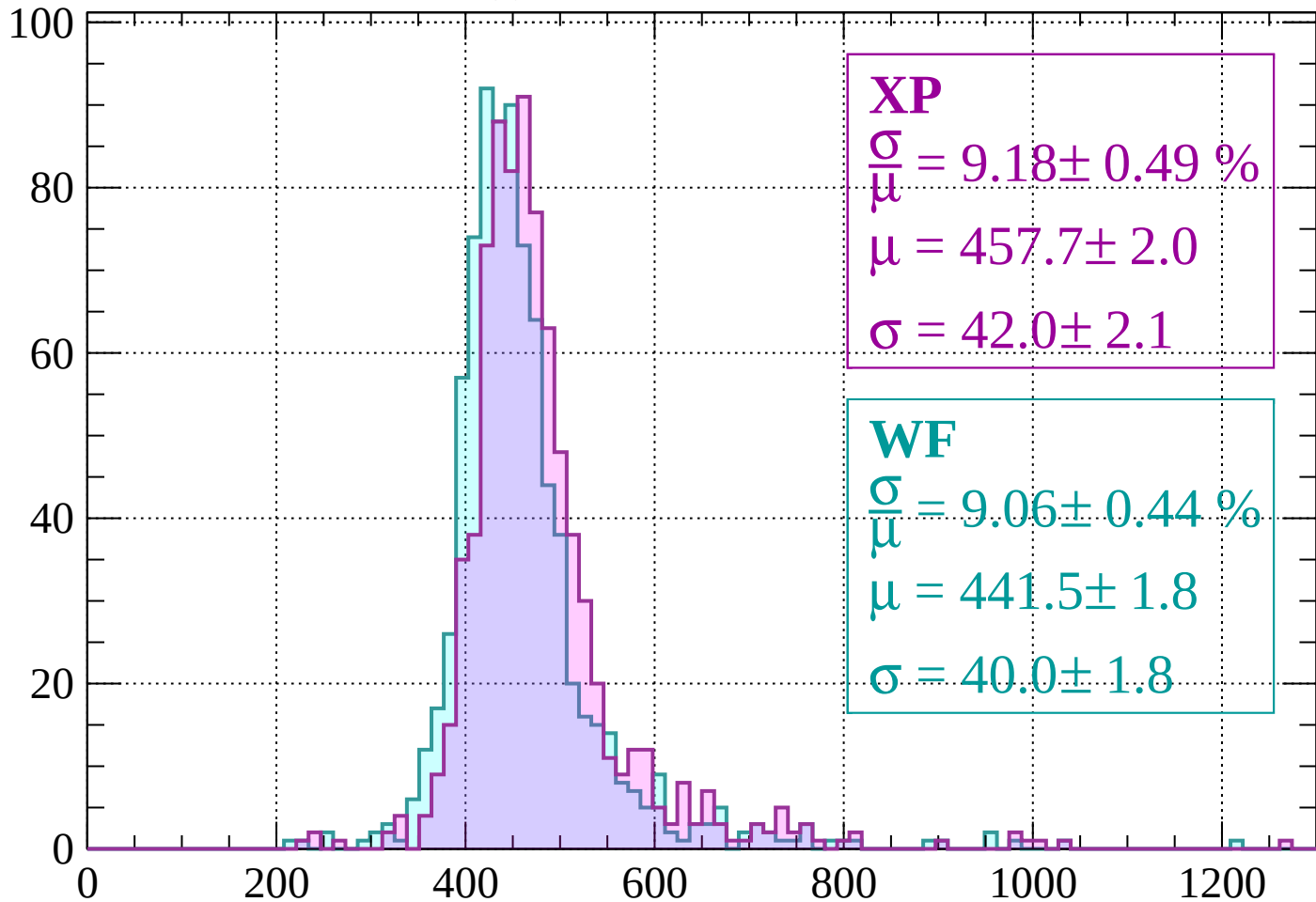
Energy loss | $-920 < p < -880$

Count



Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 9.18 \pm 0.49 \%$$

$$\mu = 457.7 \pm 2.0$$

$$\sigma = 42.0 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 9.06 \pm 0.44 \%$$

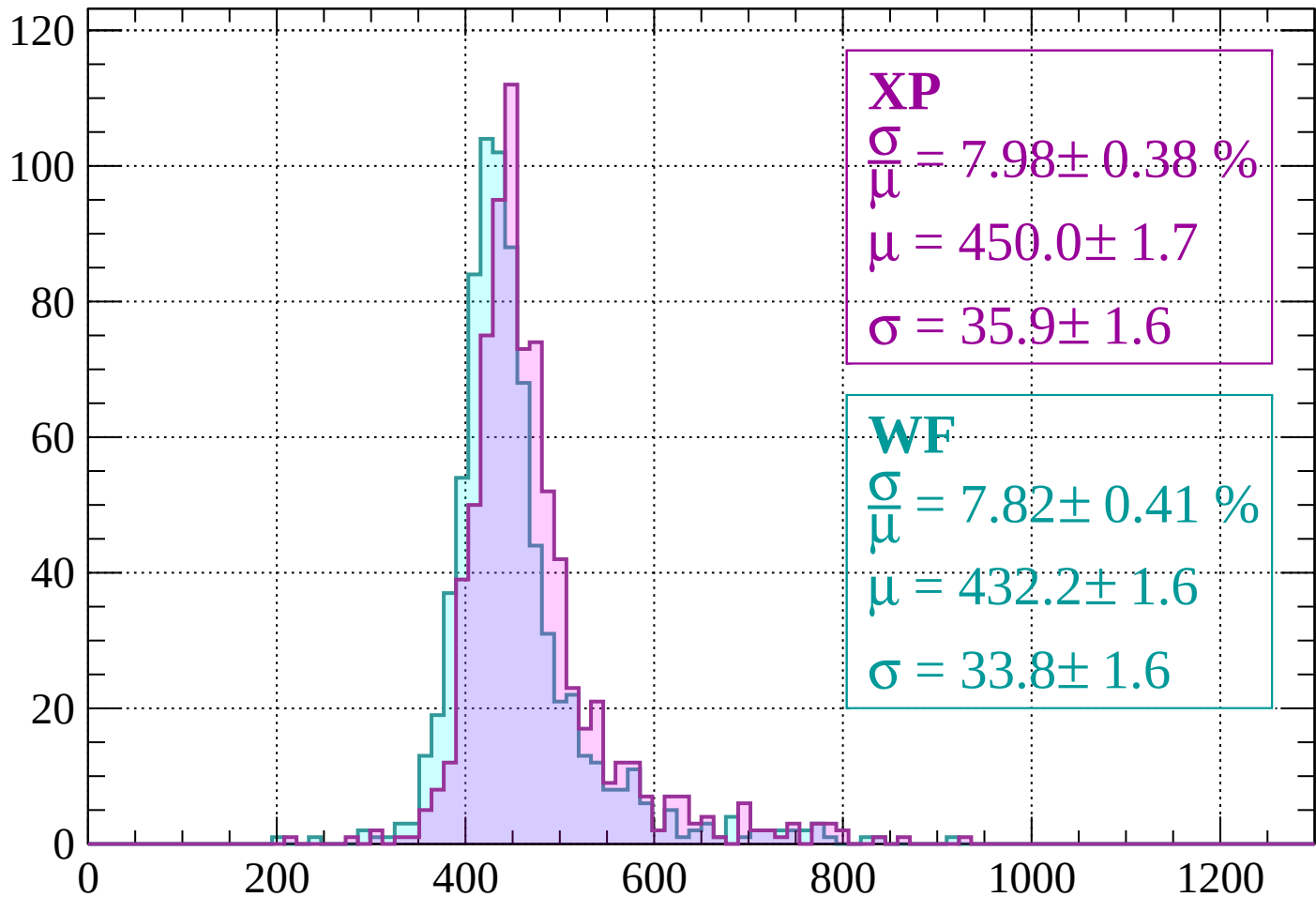
$$\mu = 441.5 \pm 1.8$$

$$\sigma = 40.0 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.98 \pm 0.38 \%$$

$$\mu = 450.0 \pm 1.7$$

$$\sigma = 35.9 \pm 1.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 7.82 \pm 0.41 \%$$

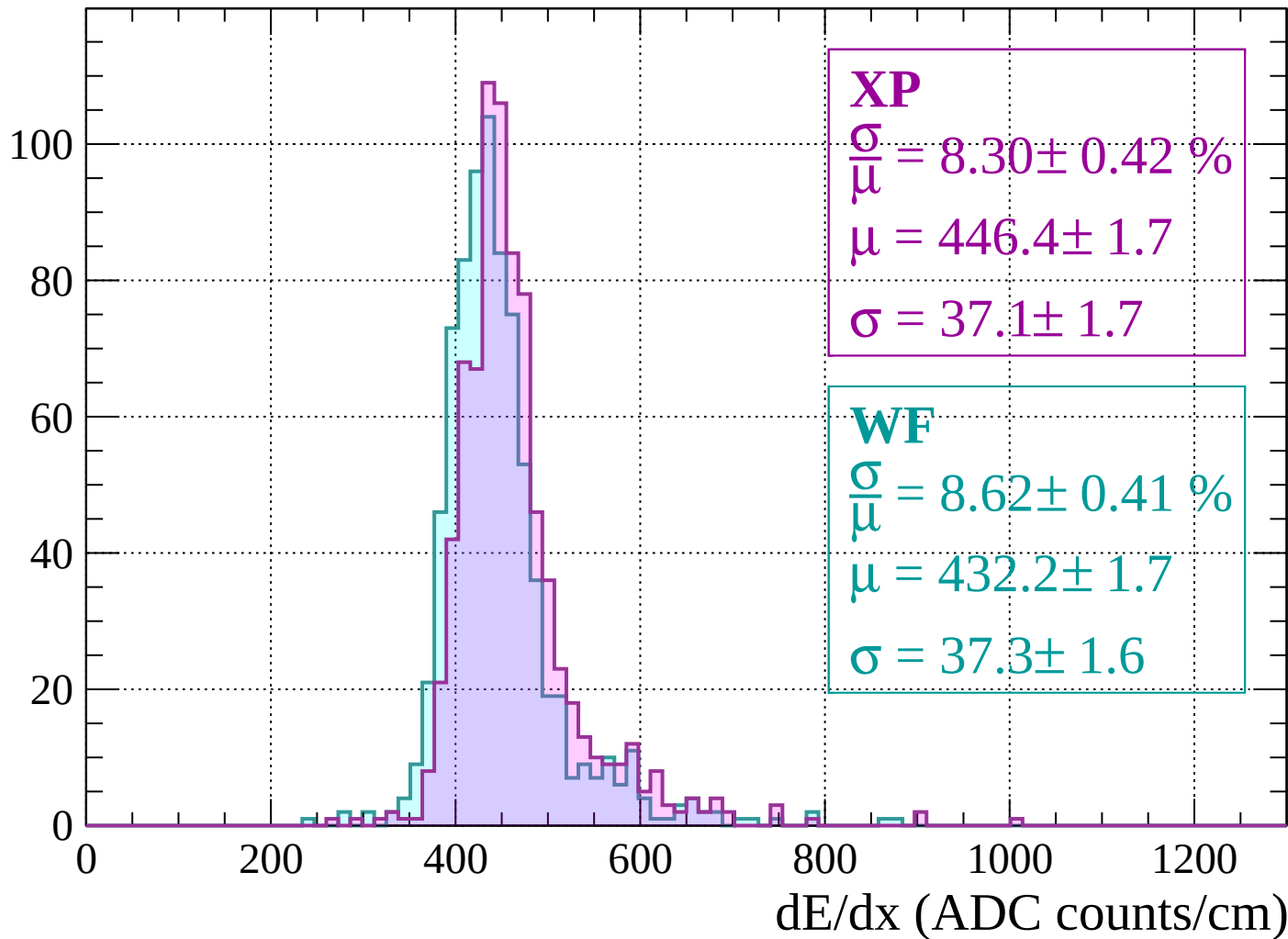
$$\mu = 432.2 \pm 1.6$$

$$\sigma = 33.8 \pm 1.6$$

dE/dx (ADC counts/cm)

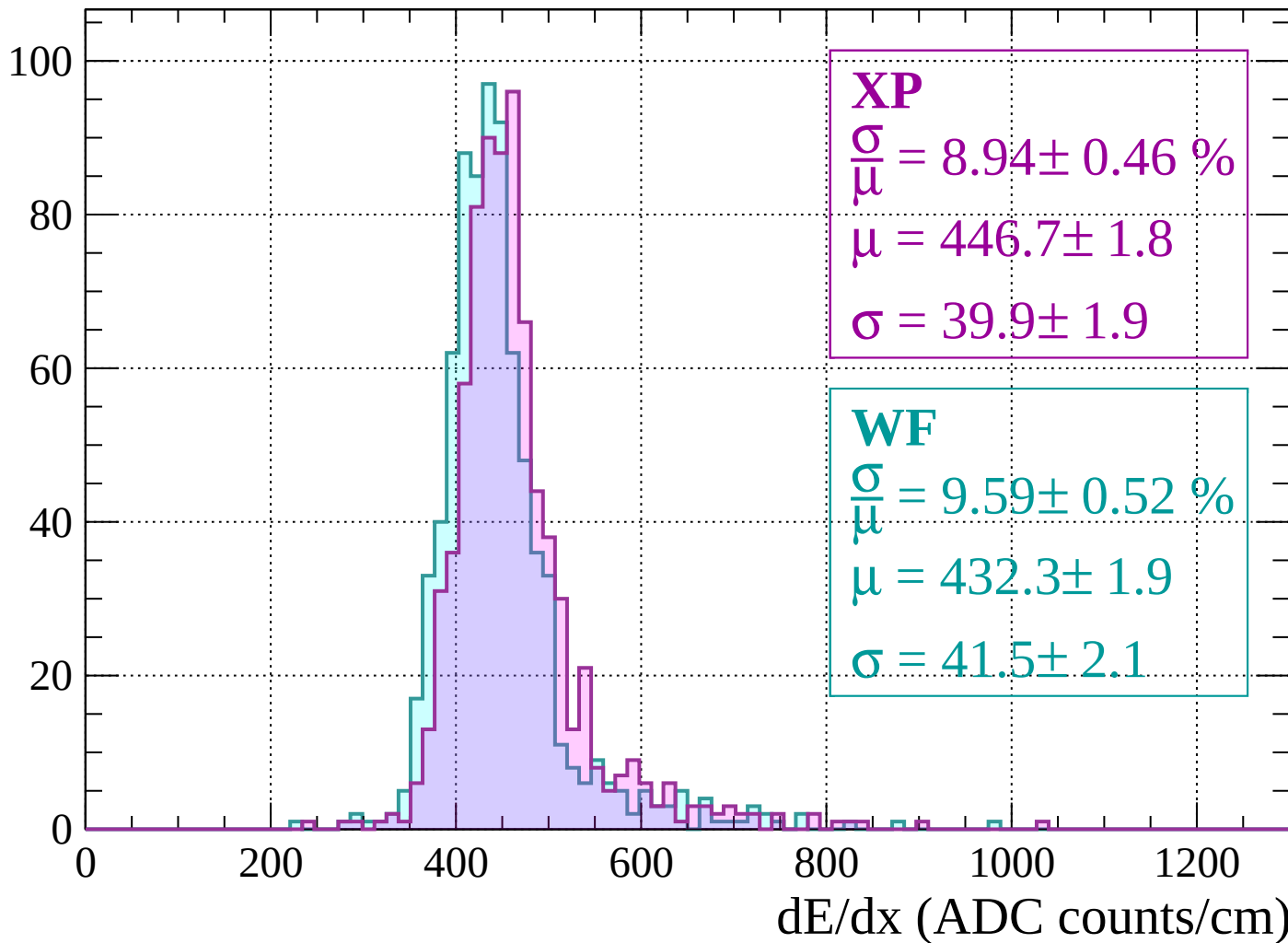
Energy loss | $-800 < p < -760$

Count



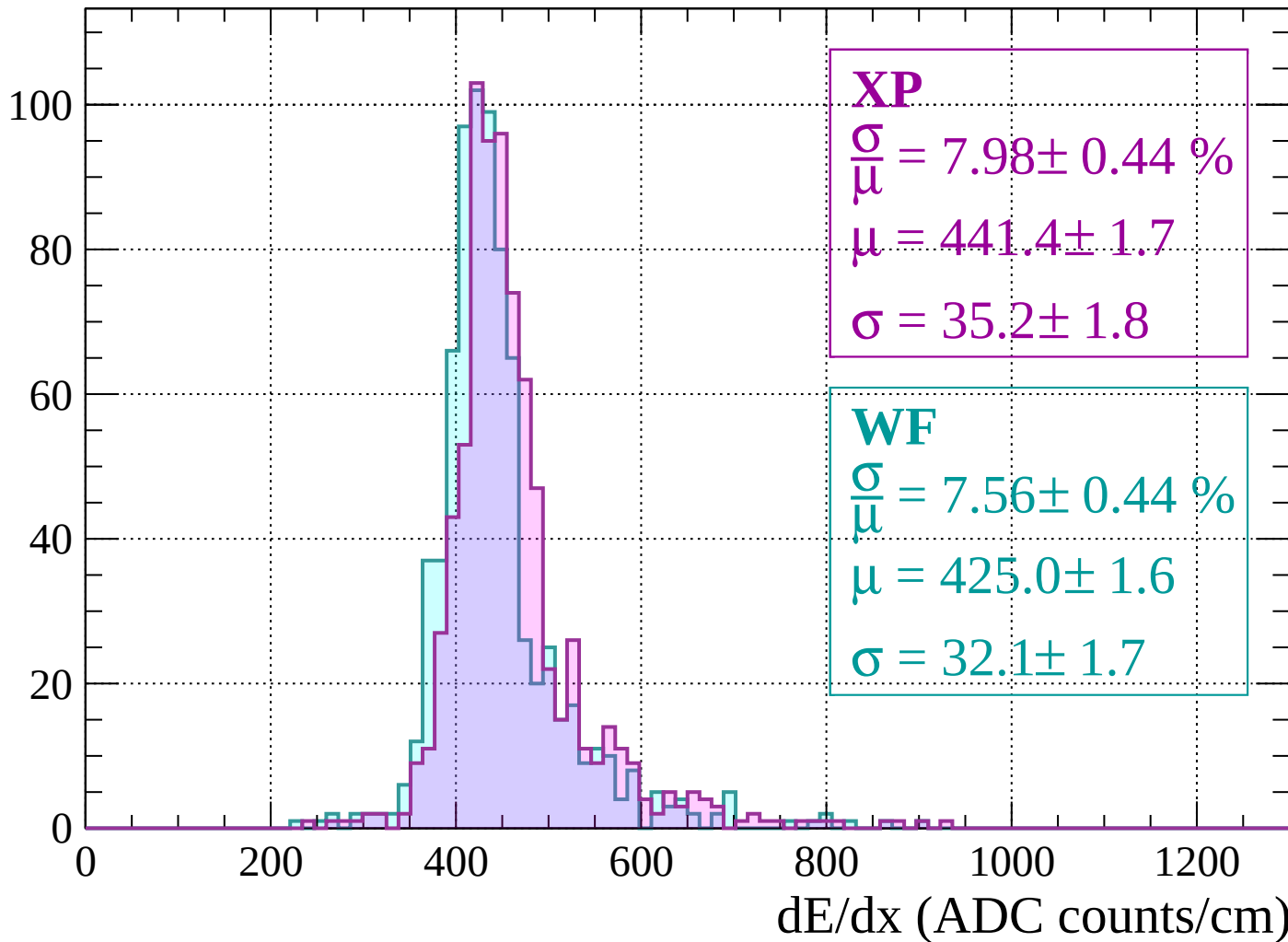
Energy loss | $-760 < p < -720$

Count



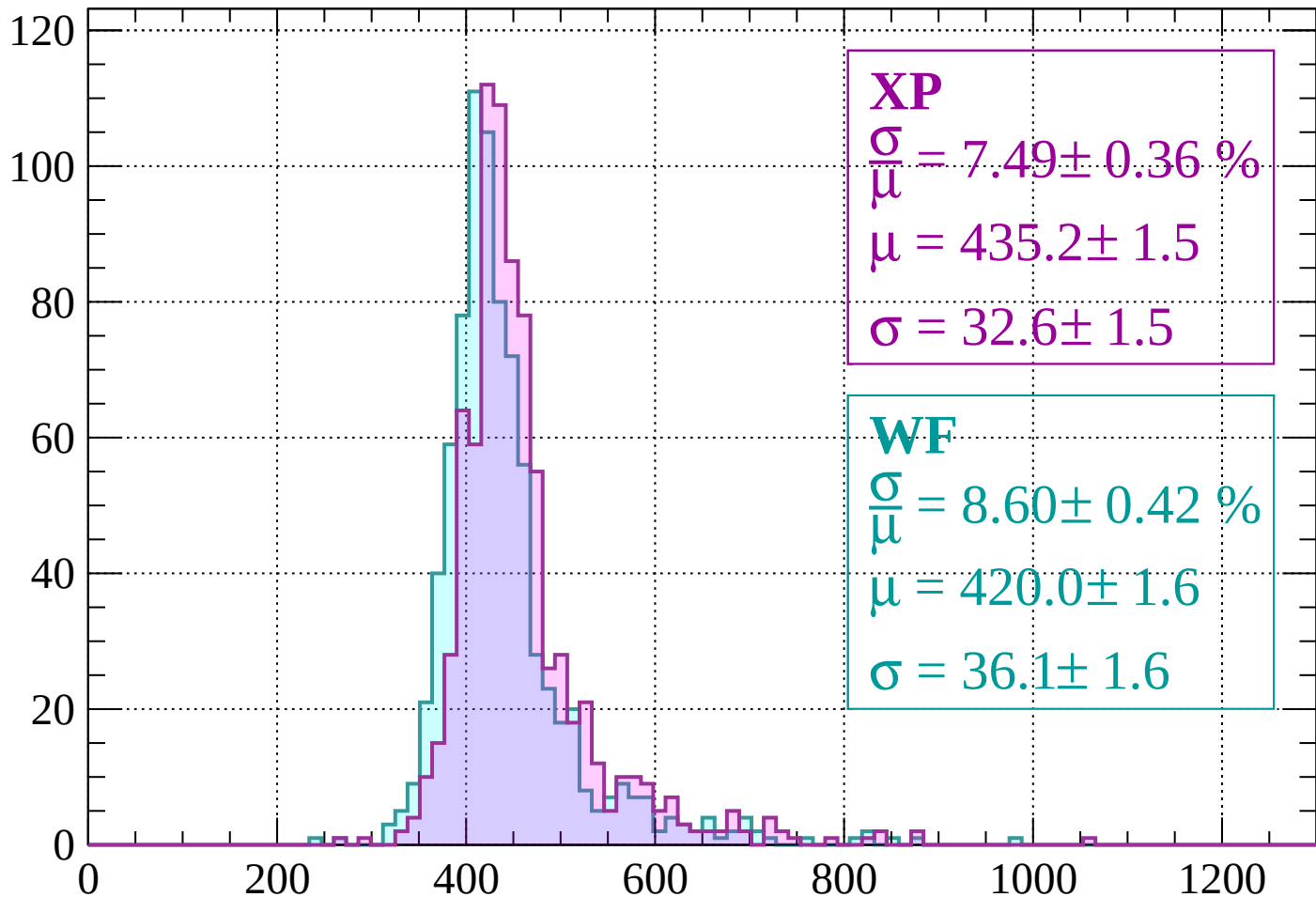
Energy loss | $-720 < p < -680$

Count



Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 7.49 \pm 0.36 \%$$

$$\mu = 435.2 \pm 1.5$$

$$\sigma = 32.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.60 \pm 0.42 \%$$

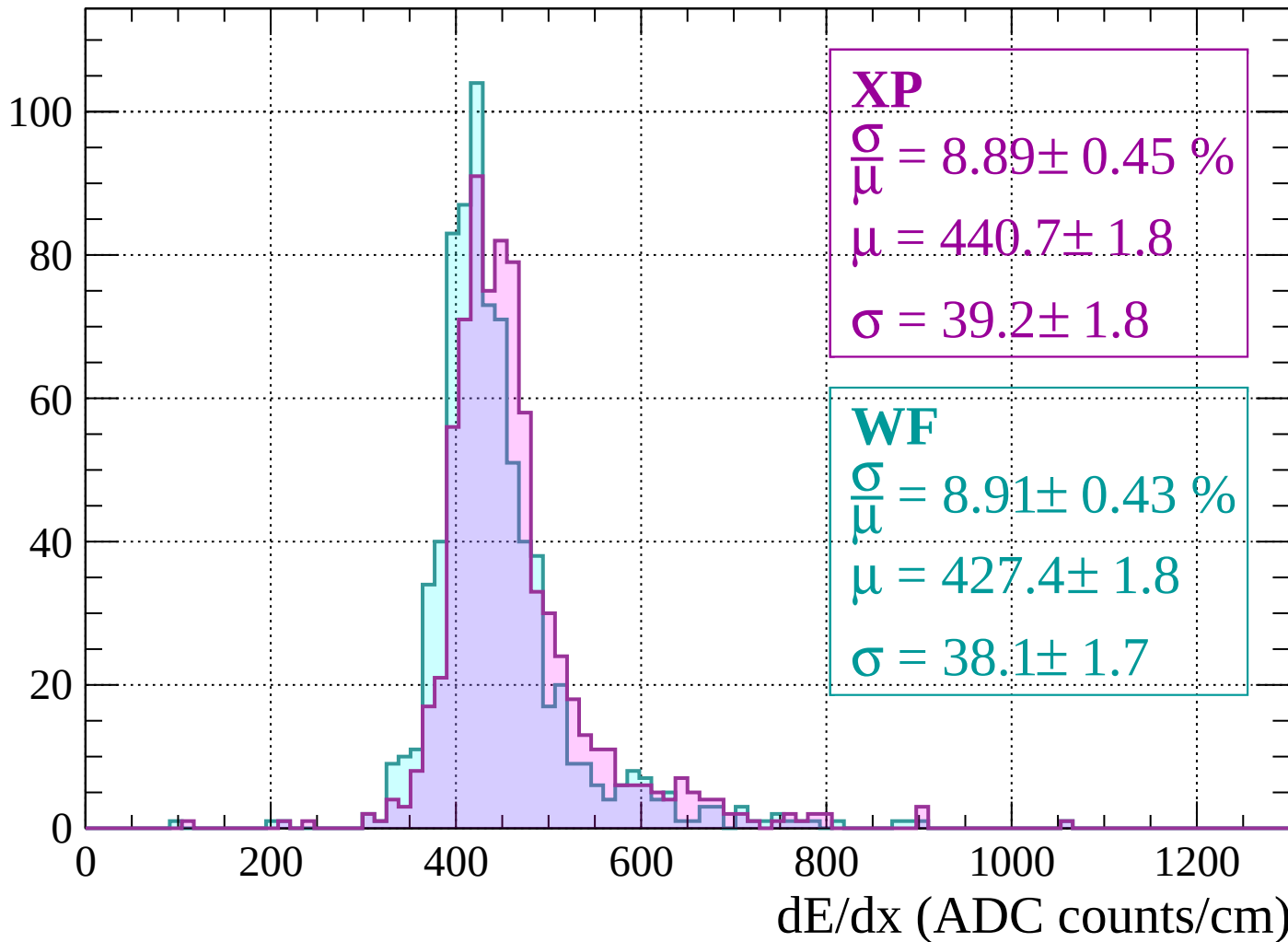
$$\mu = 420.0 \pm 1.6$$

$$\sigma = 36.1 \pm 1.6$$

dE/dx (ADC counts/cm)

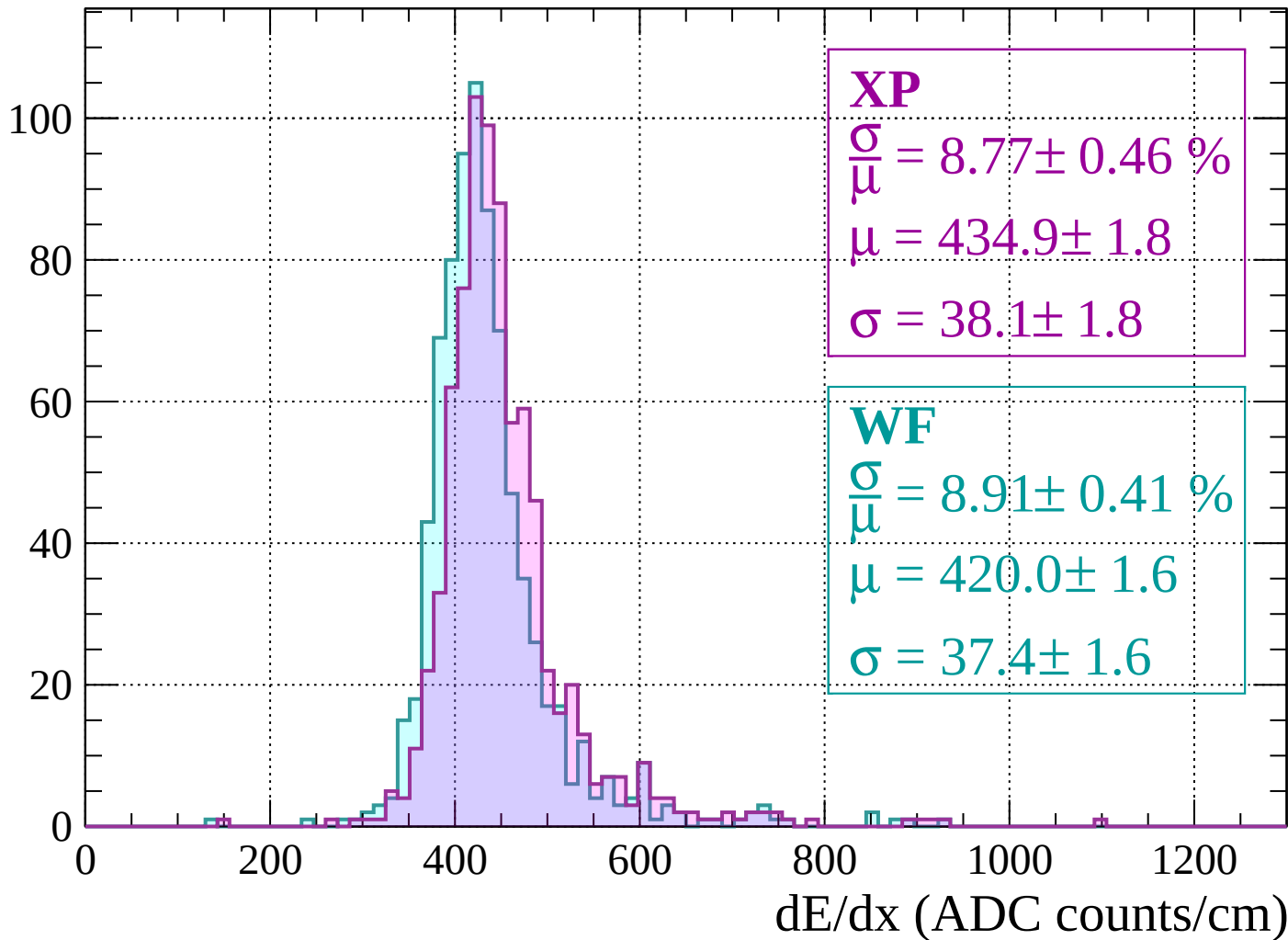
Energy loss | $-640 < p < -600$

Count



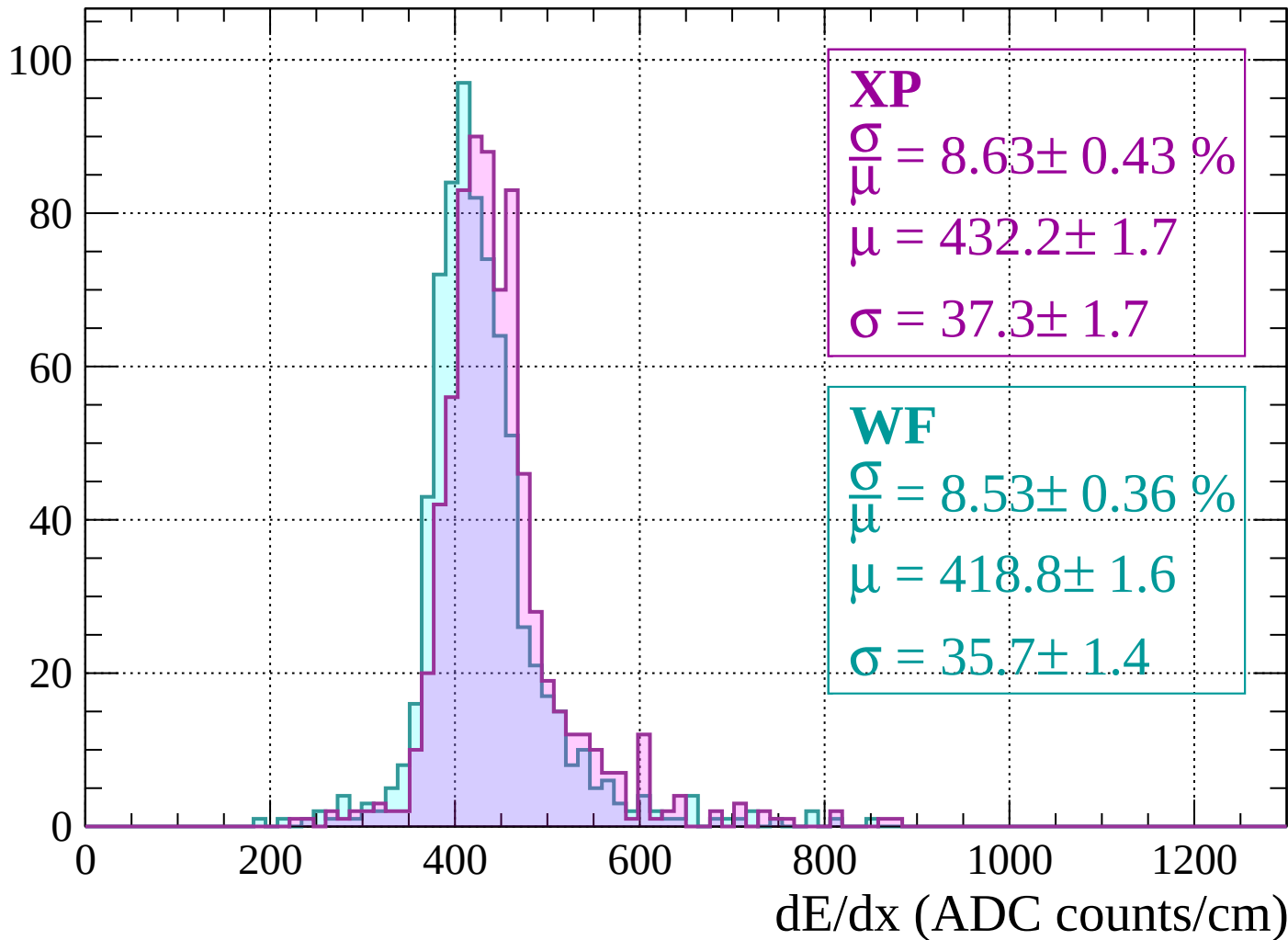
Energy loss | $-600 < p < -560$

Count



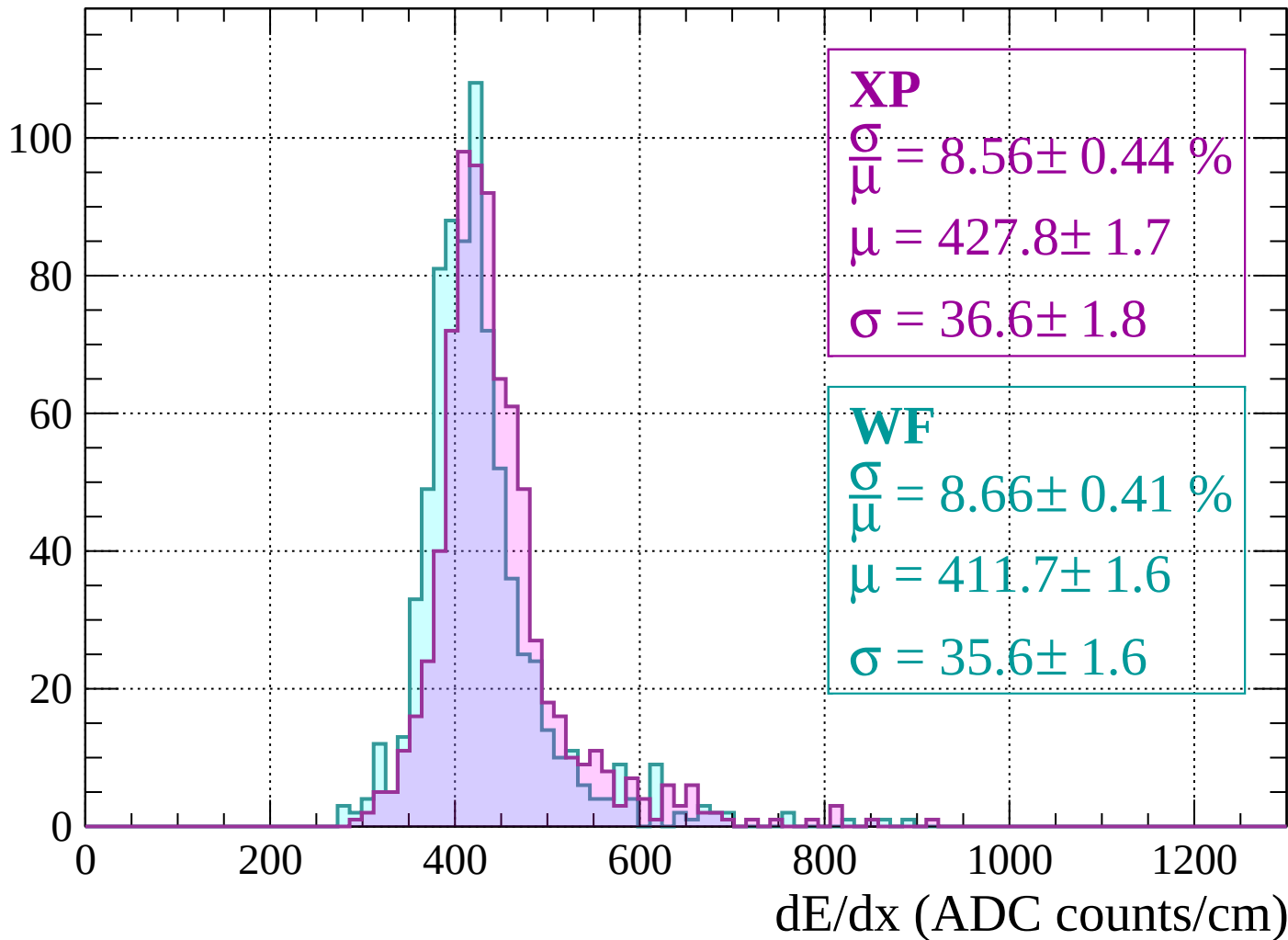
Energy loss | $-560 < p < -520$

Count



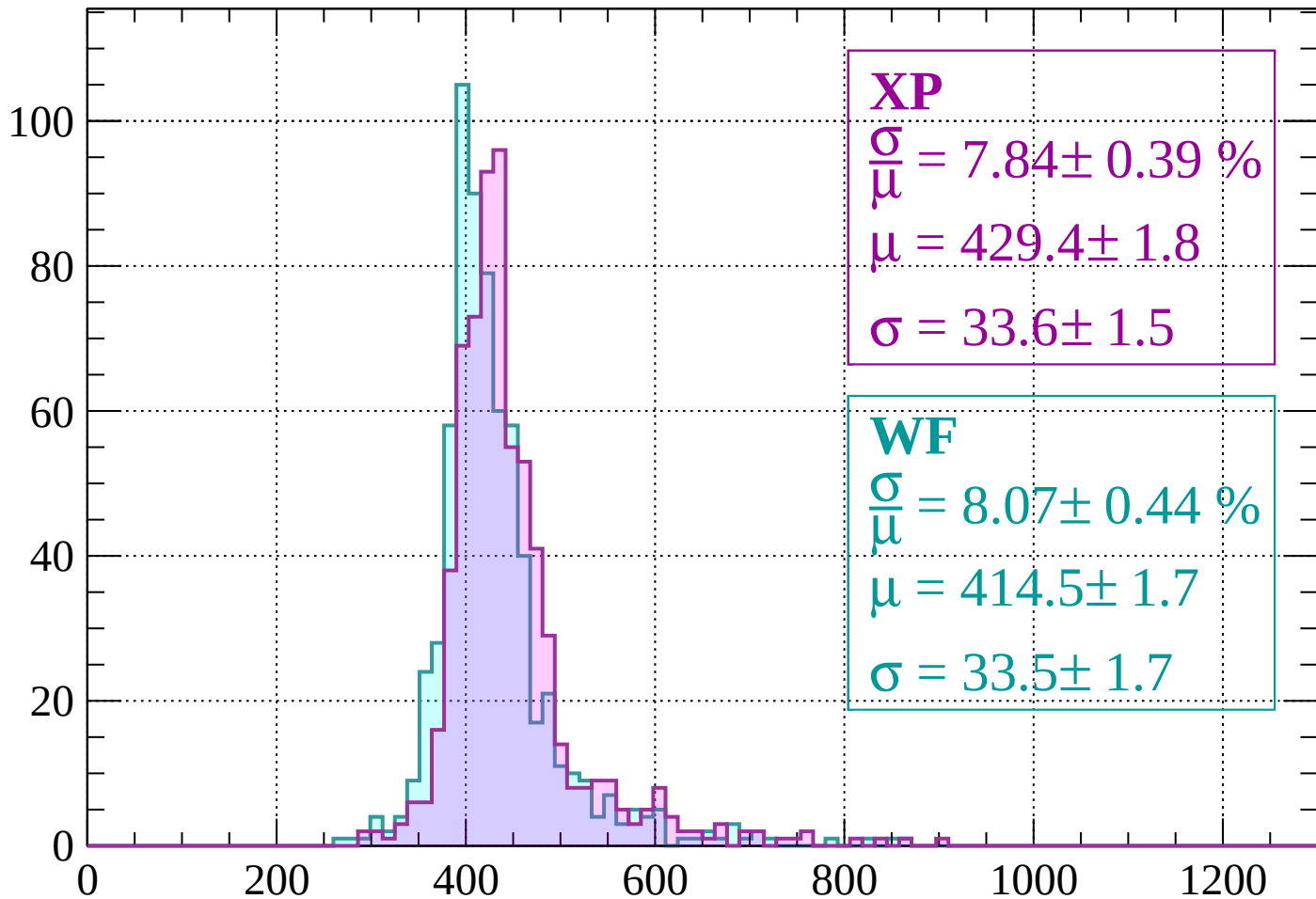
Energy loss | $-520 < p < -480$

Count



Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 7.84 \pm 0.39 \%$$

$$\mu = 429.4 \pm 1.8$$

$$\sigma = 33.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.07 \pm 0.44 \%$$

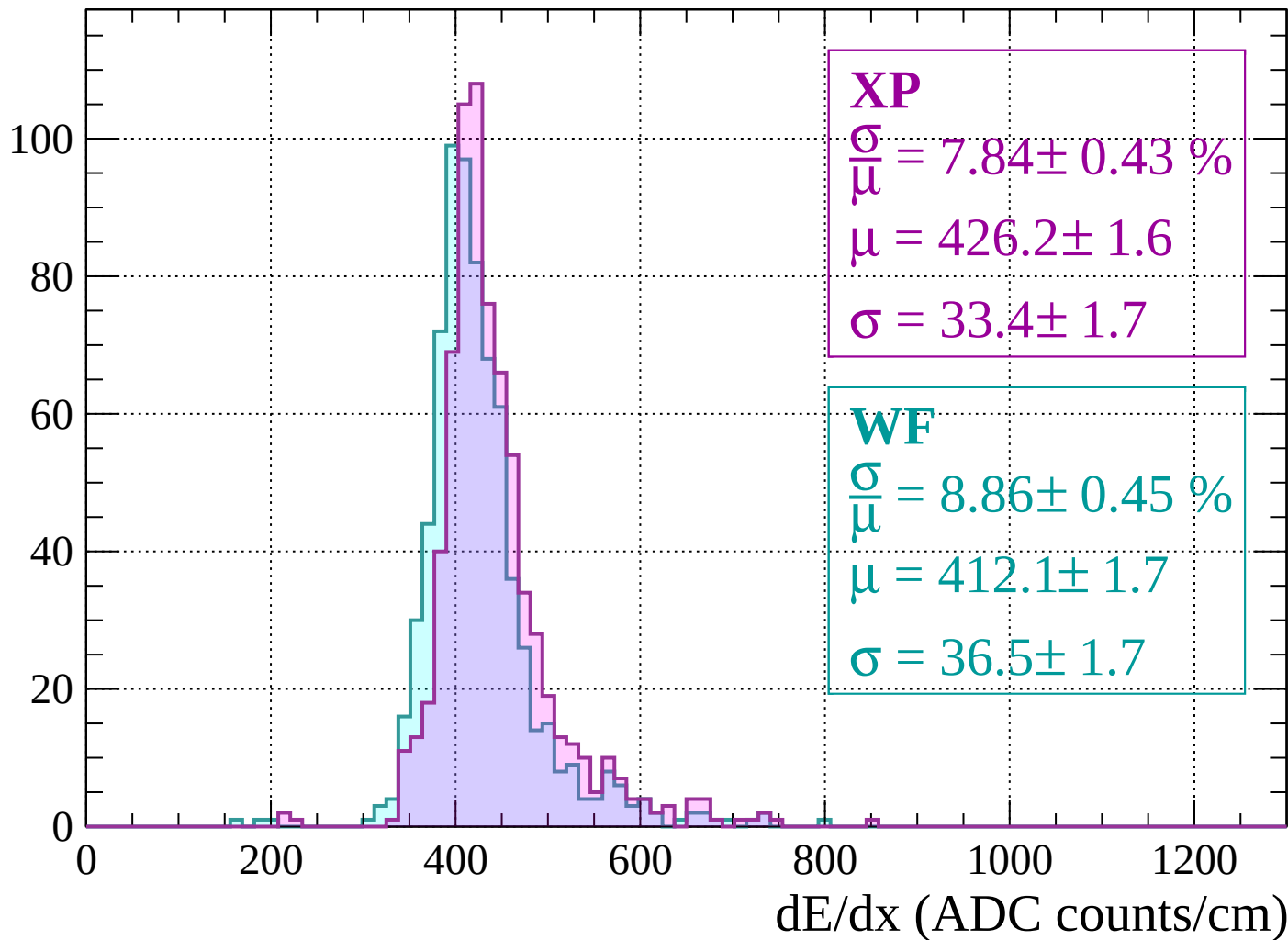
$$\mu = 414.5 \pm 1.7$$

$$\sigma = 33.5 \pm 1.7$$

dE/dx (ADC counts/cm)

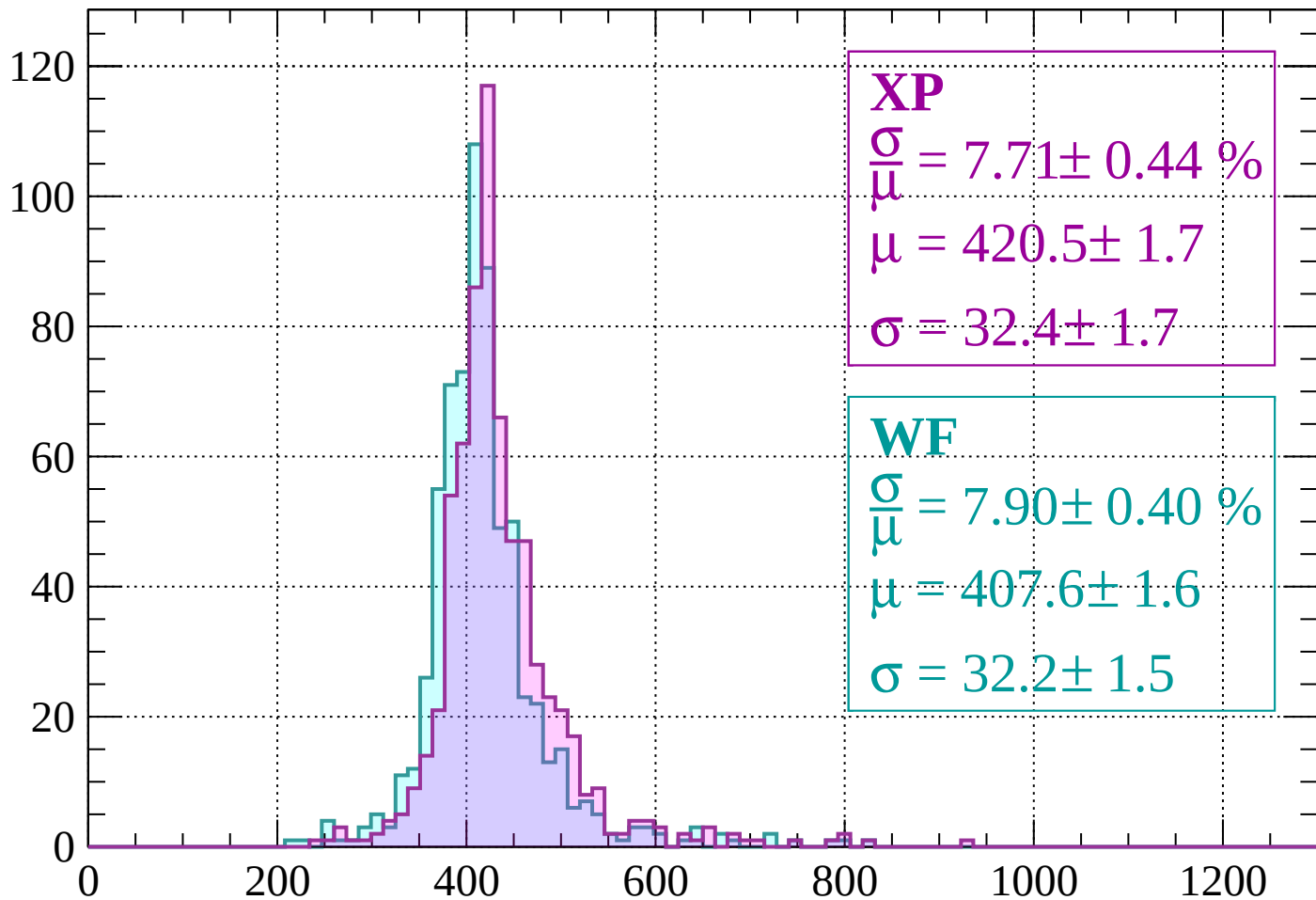
Energy loss | $-440 < p < -400$

Count



Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 7.71 \pm 0.44 \%$$

$$\mu = 420.5 \pm 1.7$$

$$\sigma = 32.4 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.90 \pm 0.40 \%$$

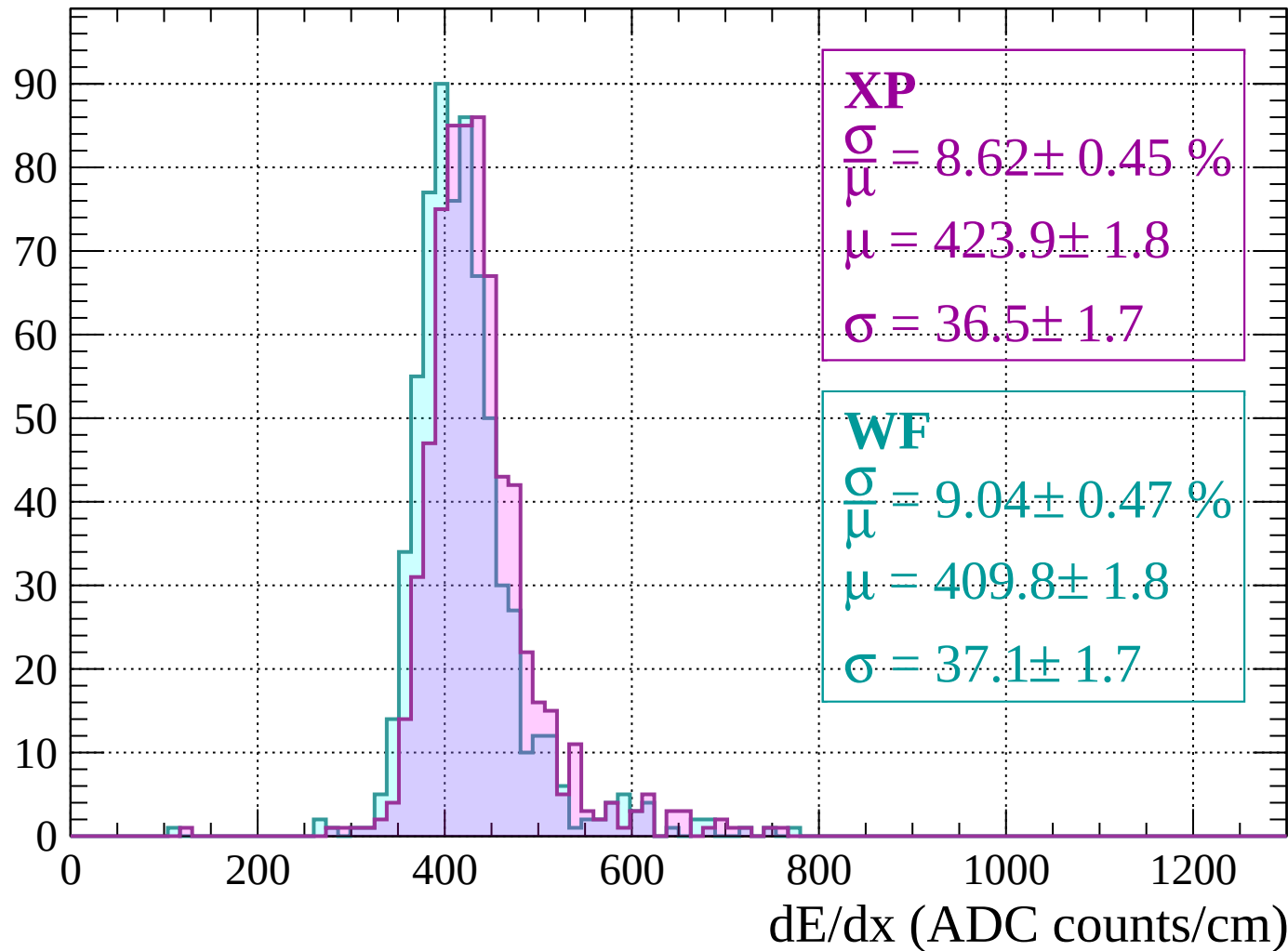
$$\mu = 407.6 \pm 1.6$$

$$\sigma = 32.2 \pm 1.5$$

dE/dx (ADC counts/cm)

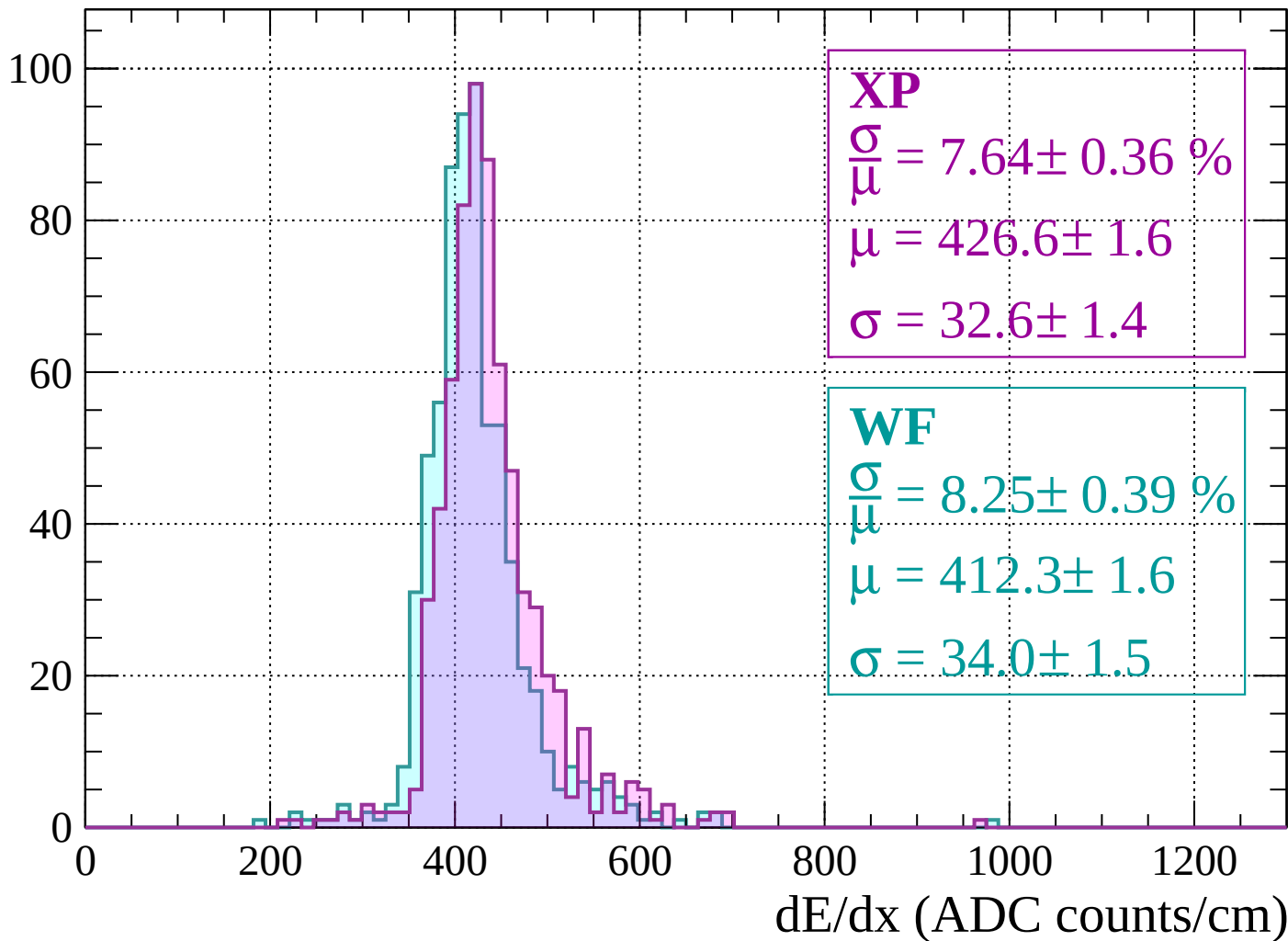
Energy loss | $-360 < p < -320$

Count



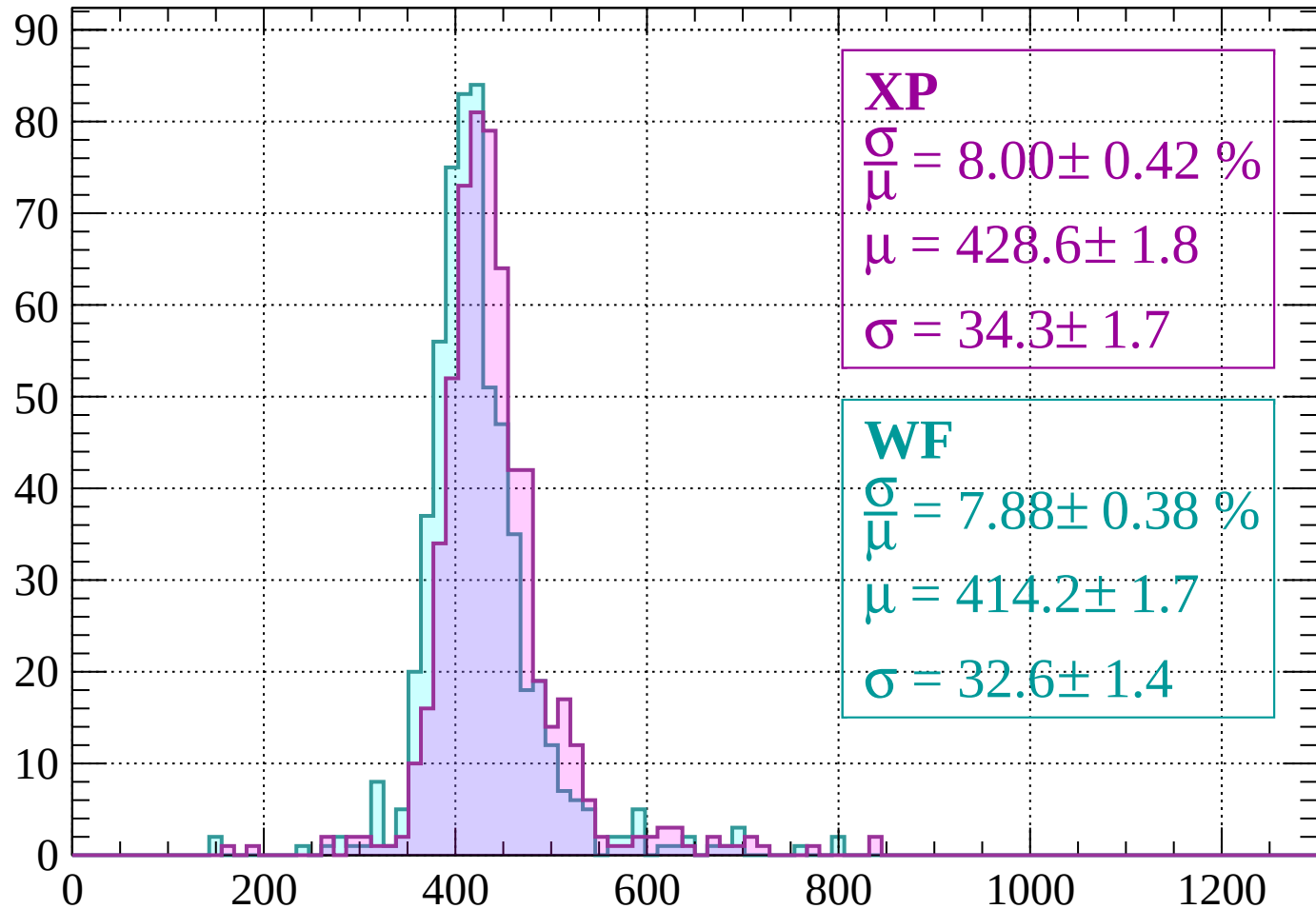
Energy loss | $-320 < p < -280$

Count



Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 8.00 \pm 0.42 \%$$

$$\mu = 428.6 \pm 1.8$$

$$\sigma = 34.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.88 \pm 0.38 \%$$

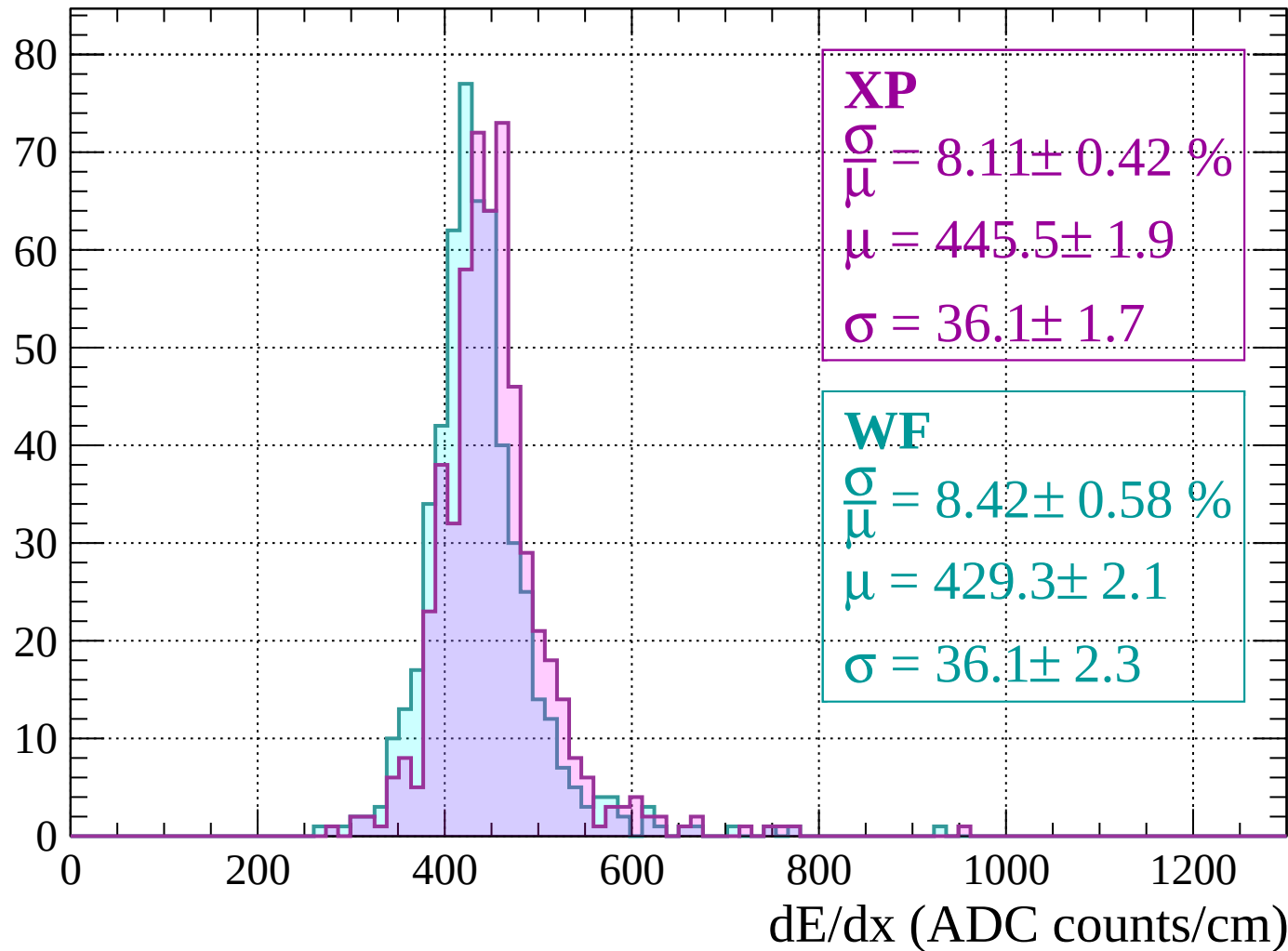
$$\mu = 414.2 \pm 1.7$$

$$\sigma = 32.6 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



Energy loss | $-200 < p < -160$

Count

XP

$$\frac{\sigma}{\mu} = 9.37 \pm 0.57 \%$$

$$\mu = 466.6 \pm 2.5$$

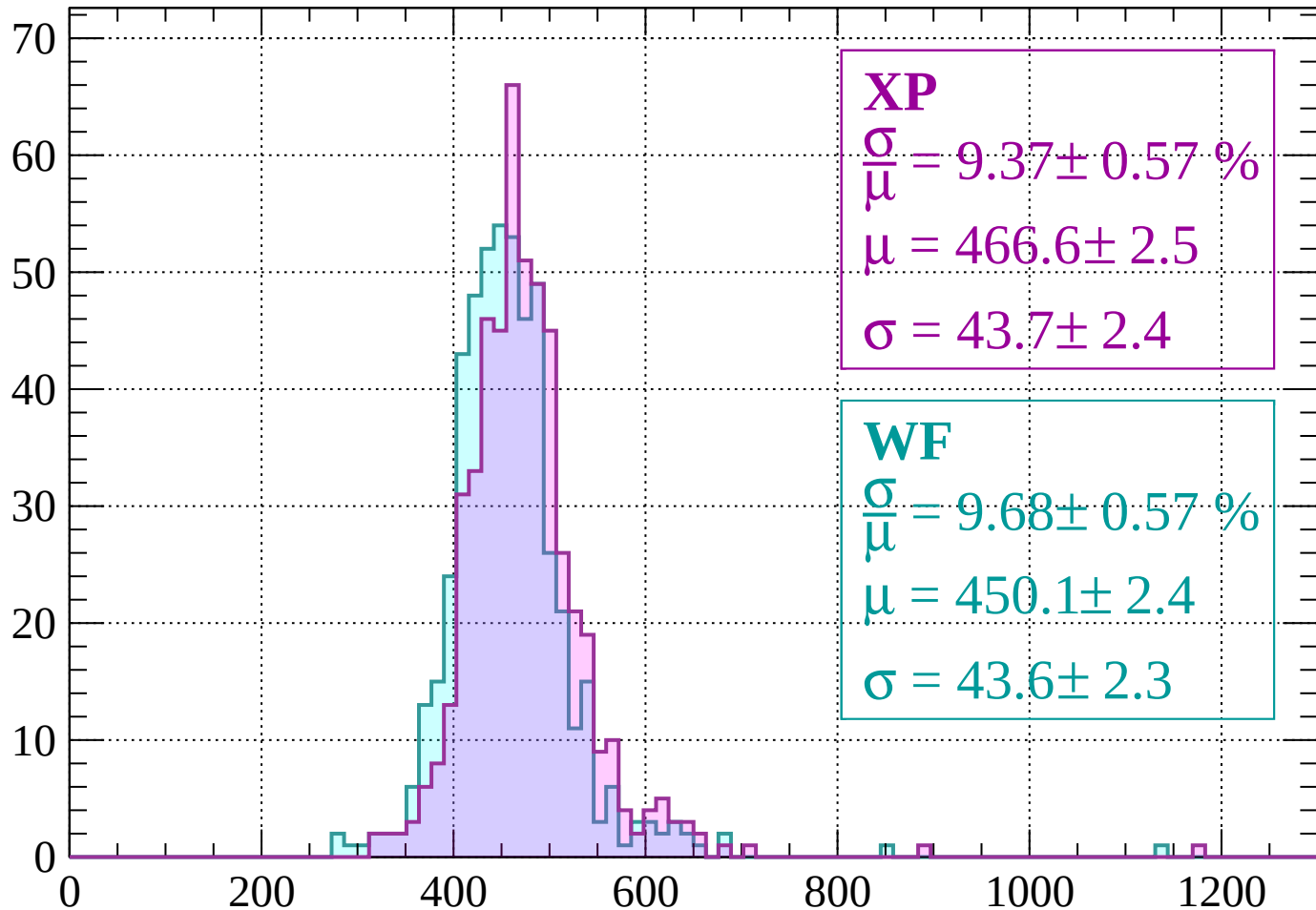
$$\sigma = 43.7 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 9.68 \pm 0.57 \%$$

$$\mu = 450.1 \pm 2.4$$

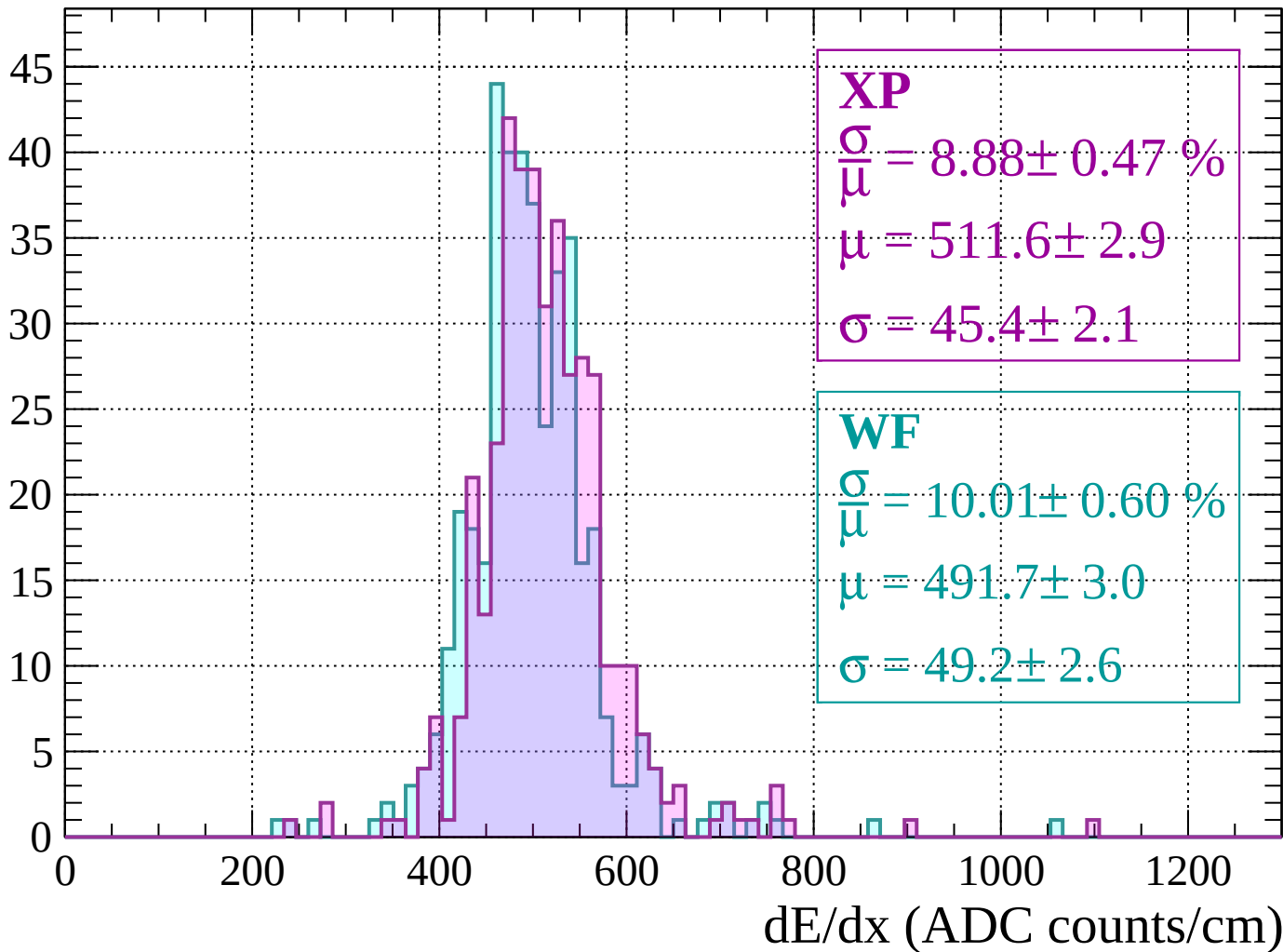
$$\sigma = 43.6 \pm 2.3$$



dE/dx (ADC counts/cm)

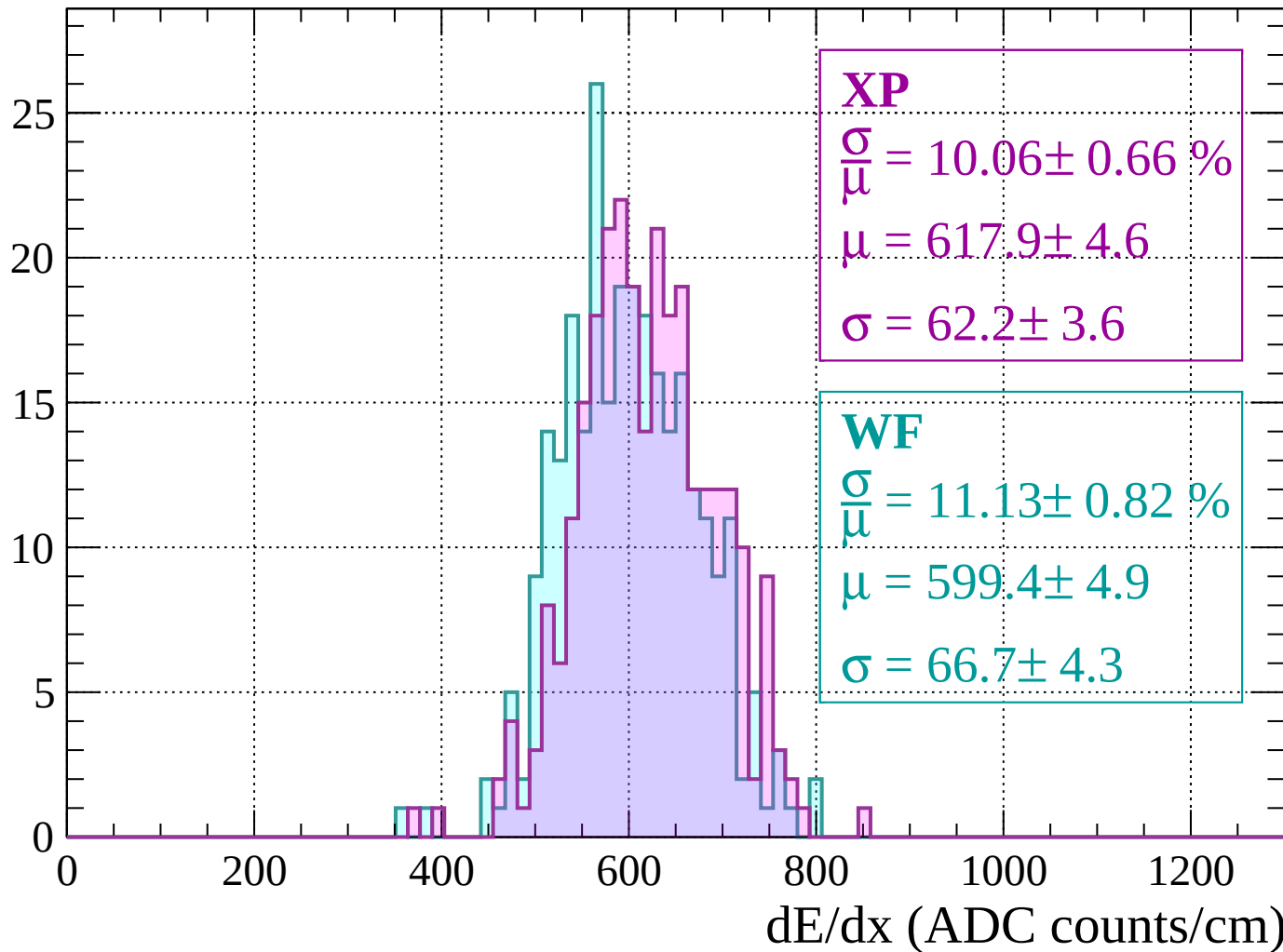
Energy loss | $-160 < p < -120$

Count



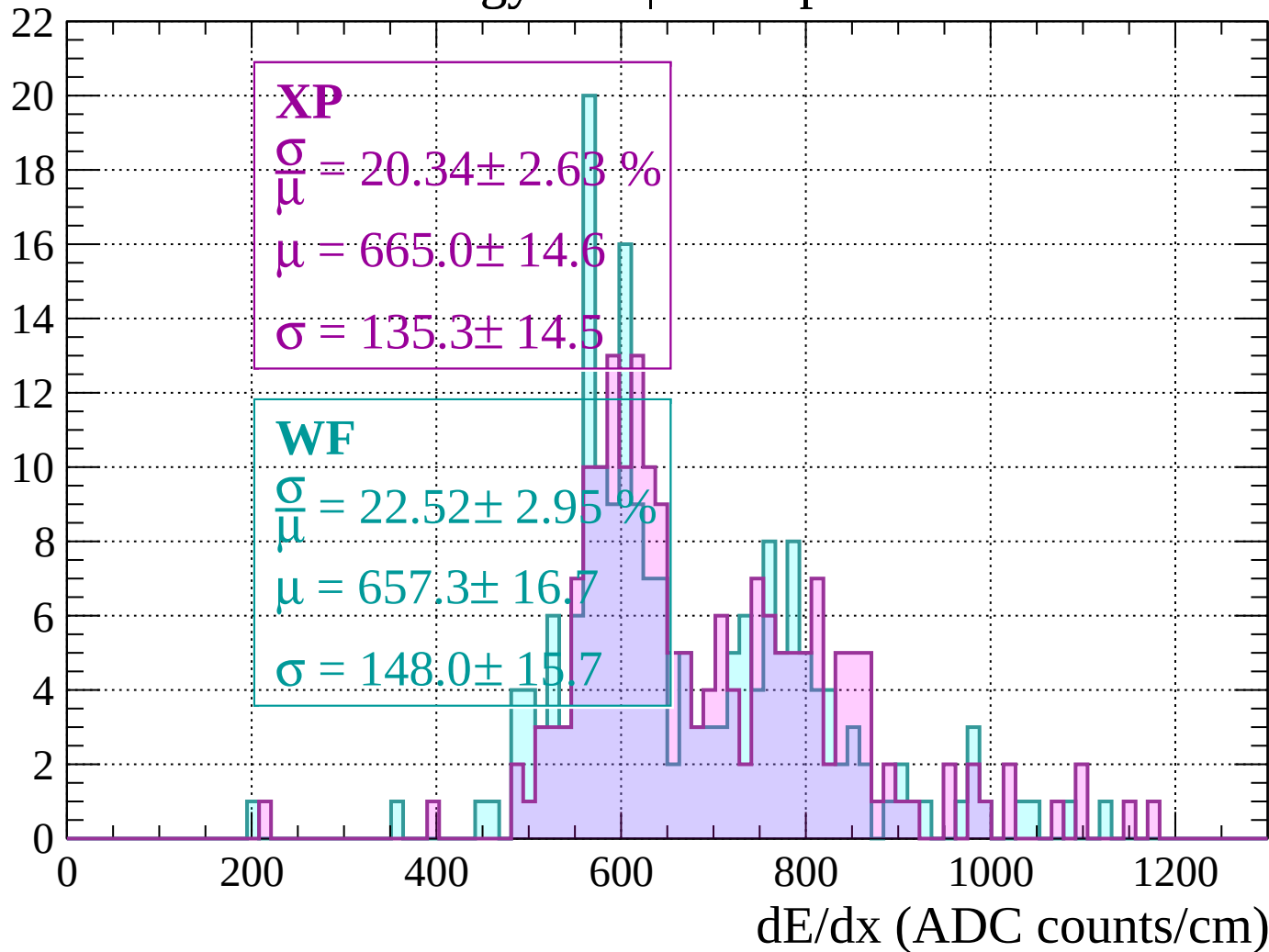
Energy loss | $-120 < p < -80$

Count



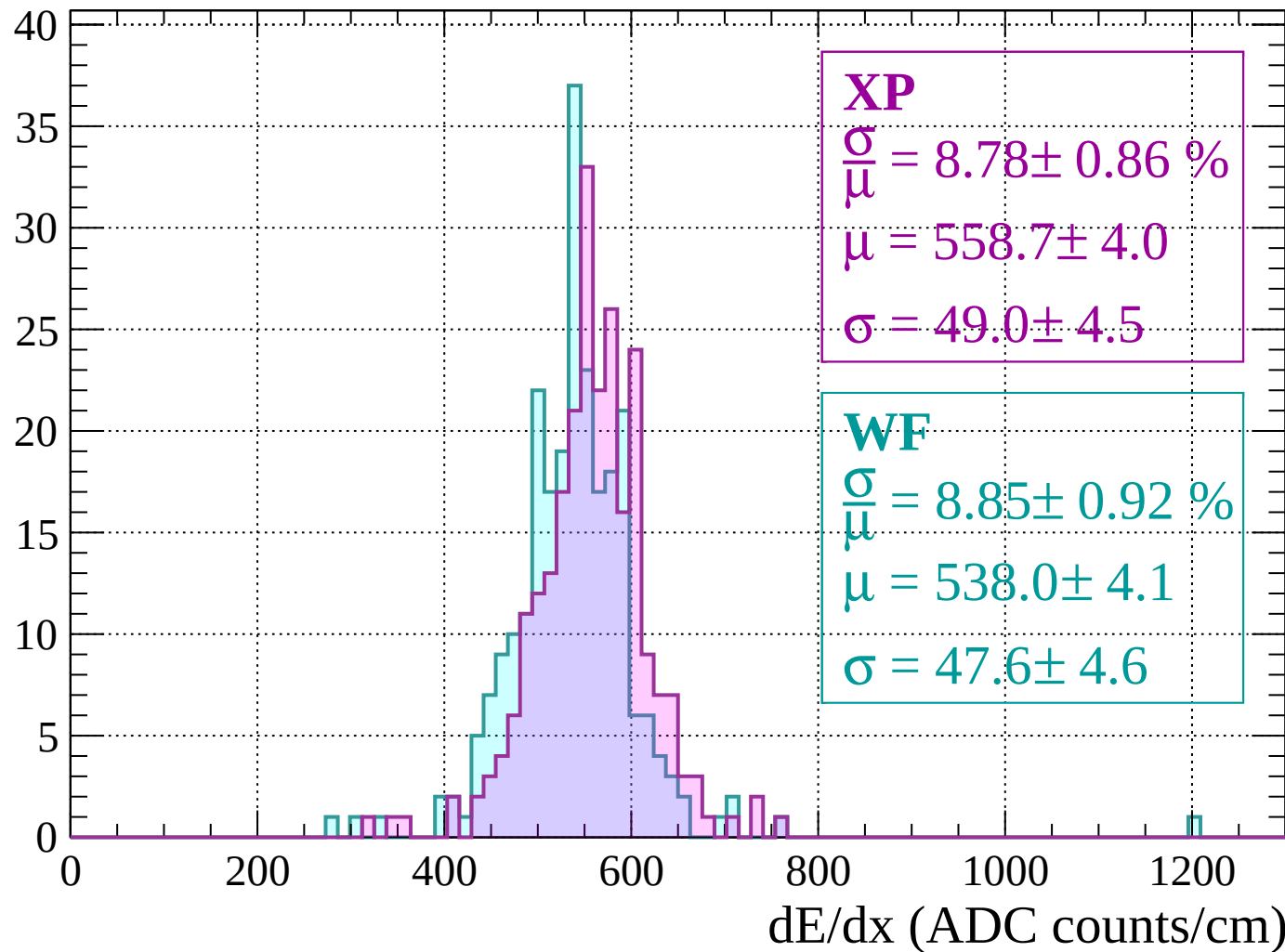
Energy loss | -80 < p < -40

Count



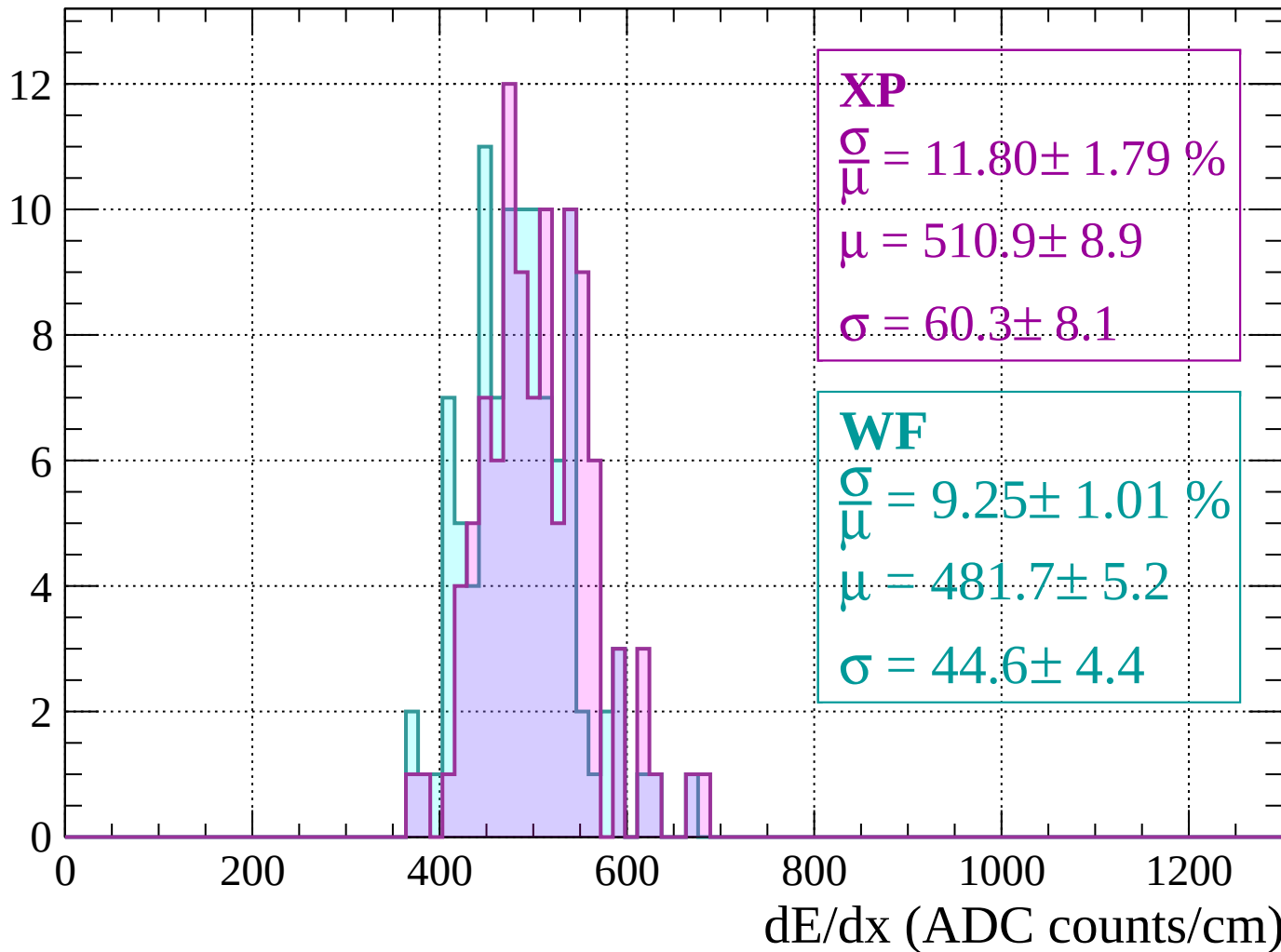
Energy loss | $-40 < p < 0$

Count



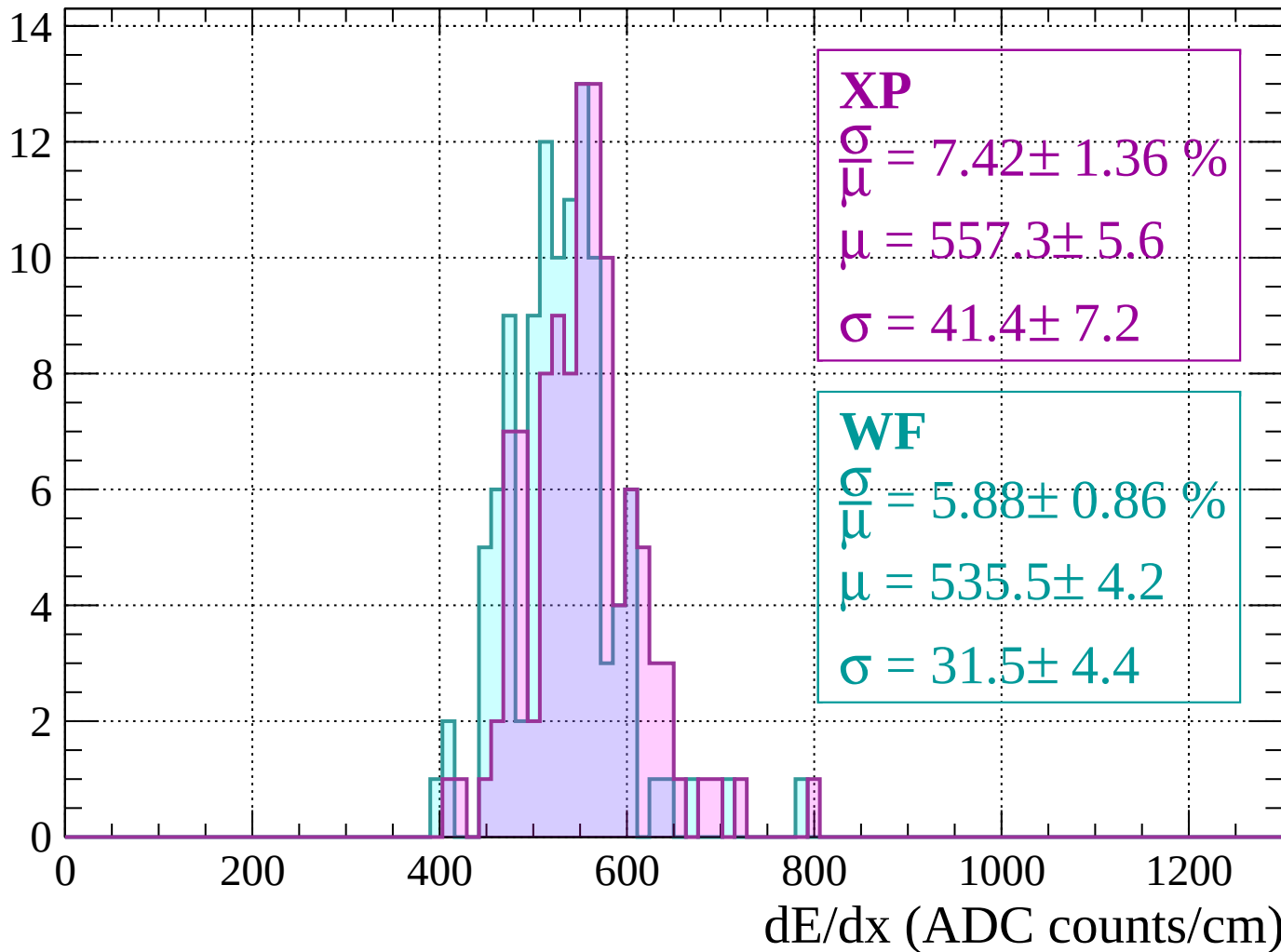
Energy loss | $0 < p < 40$

Count



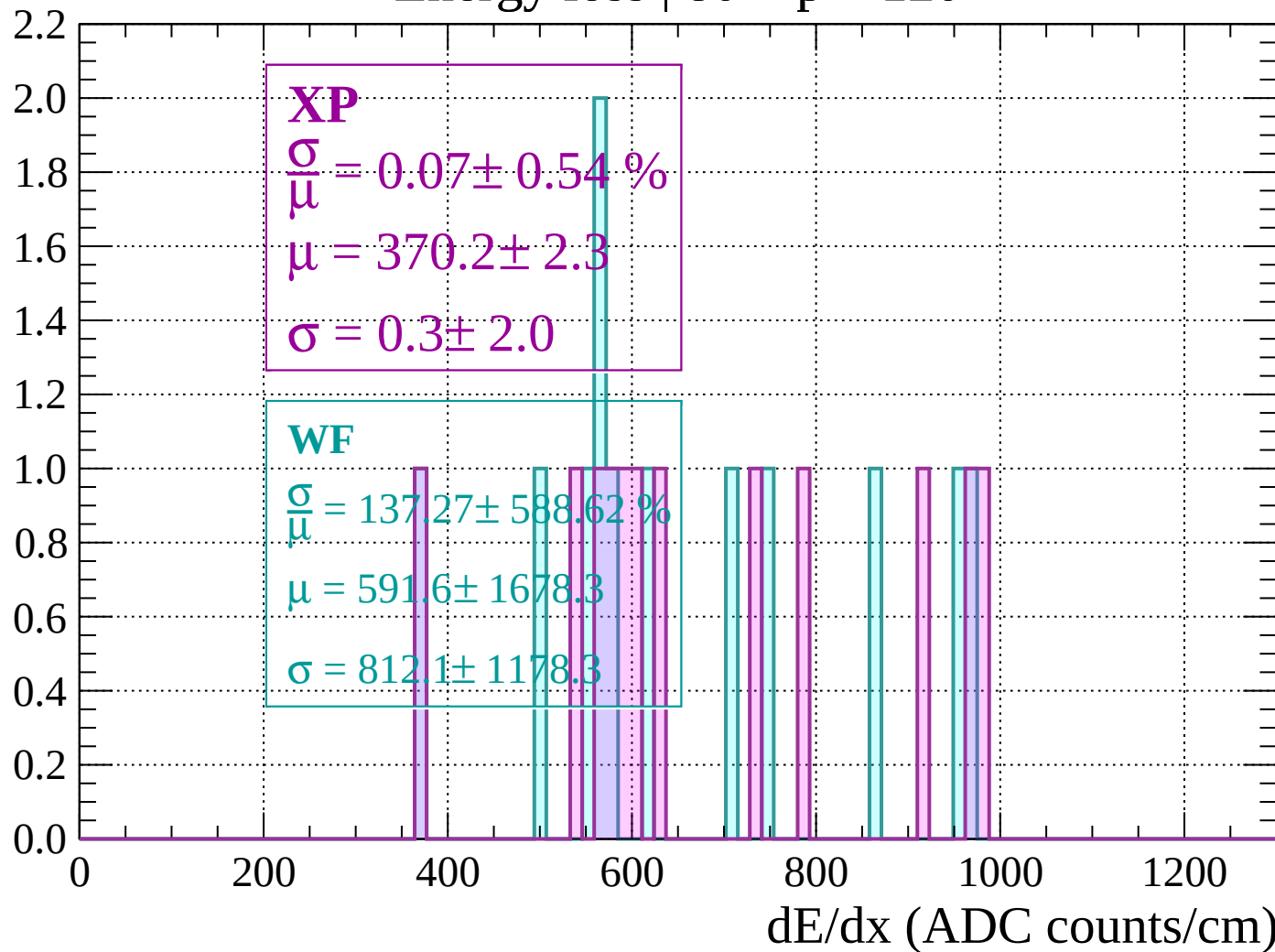
Energy loss | $40 < p < 80$

Count



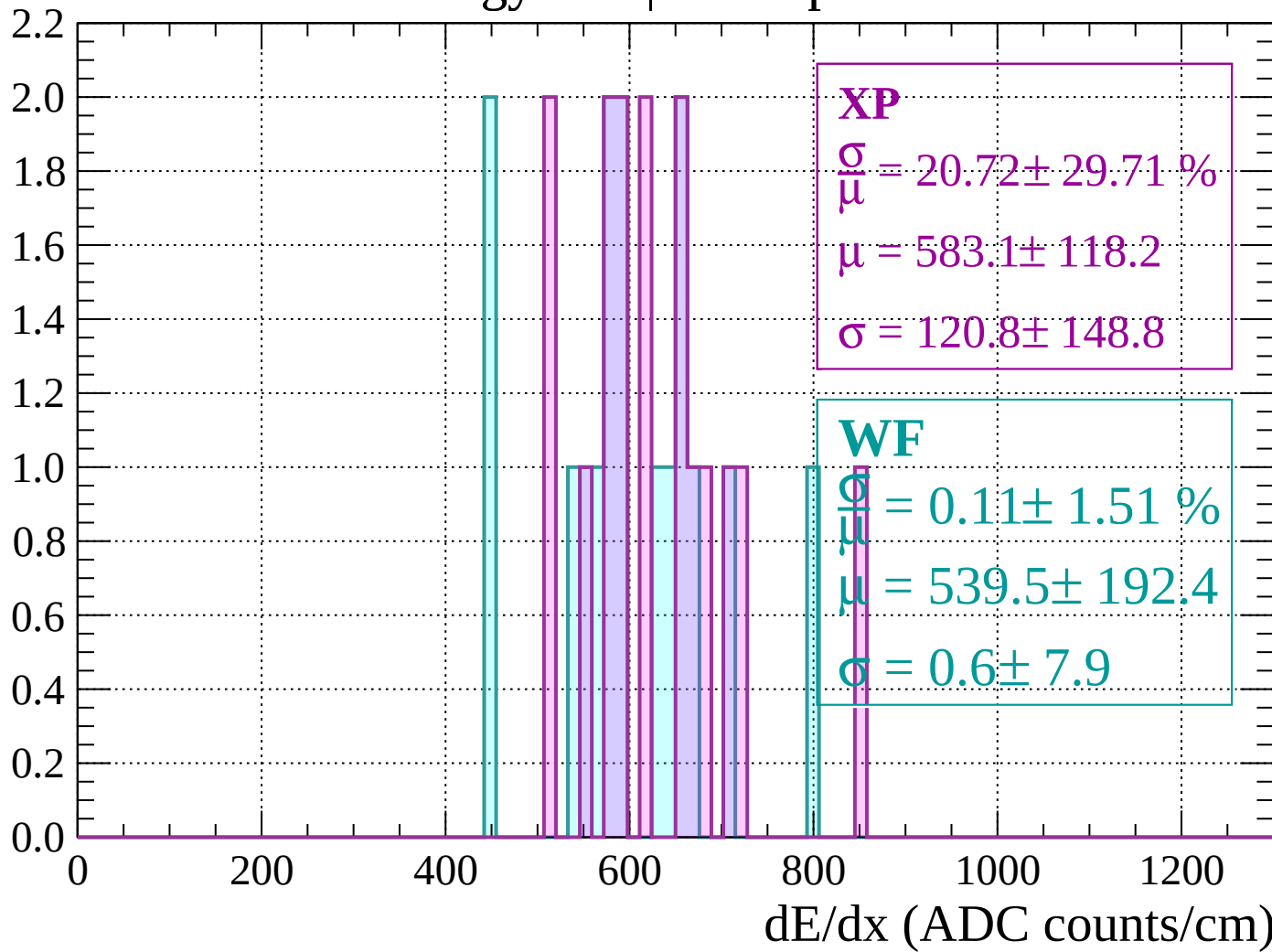
Energy loss | $80 < p < 120$

Count



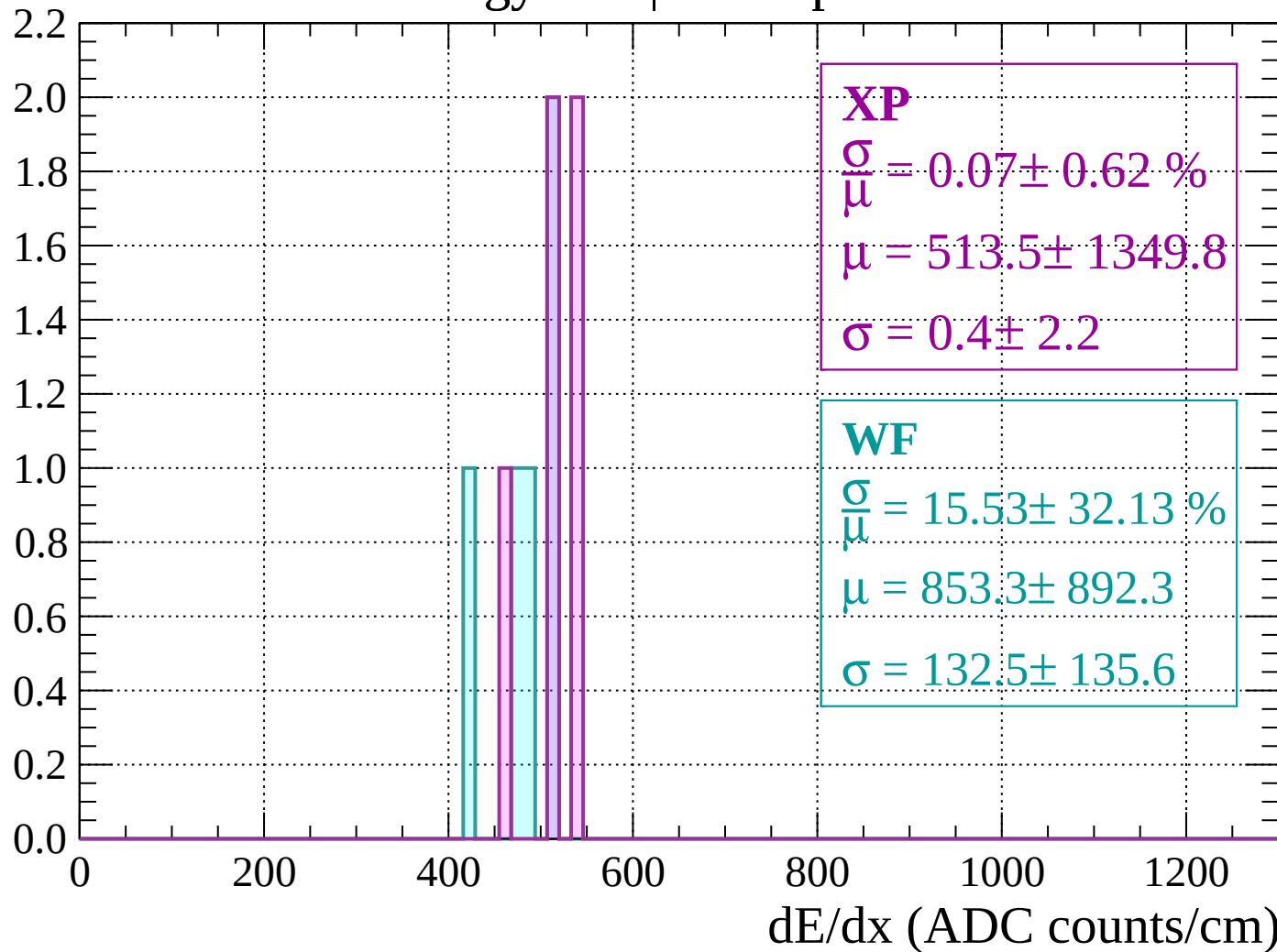
Energy loss | $120 < p < 160$

Count



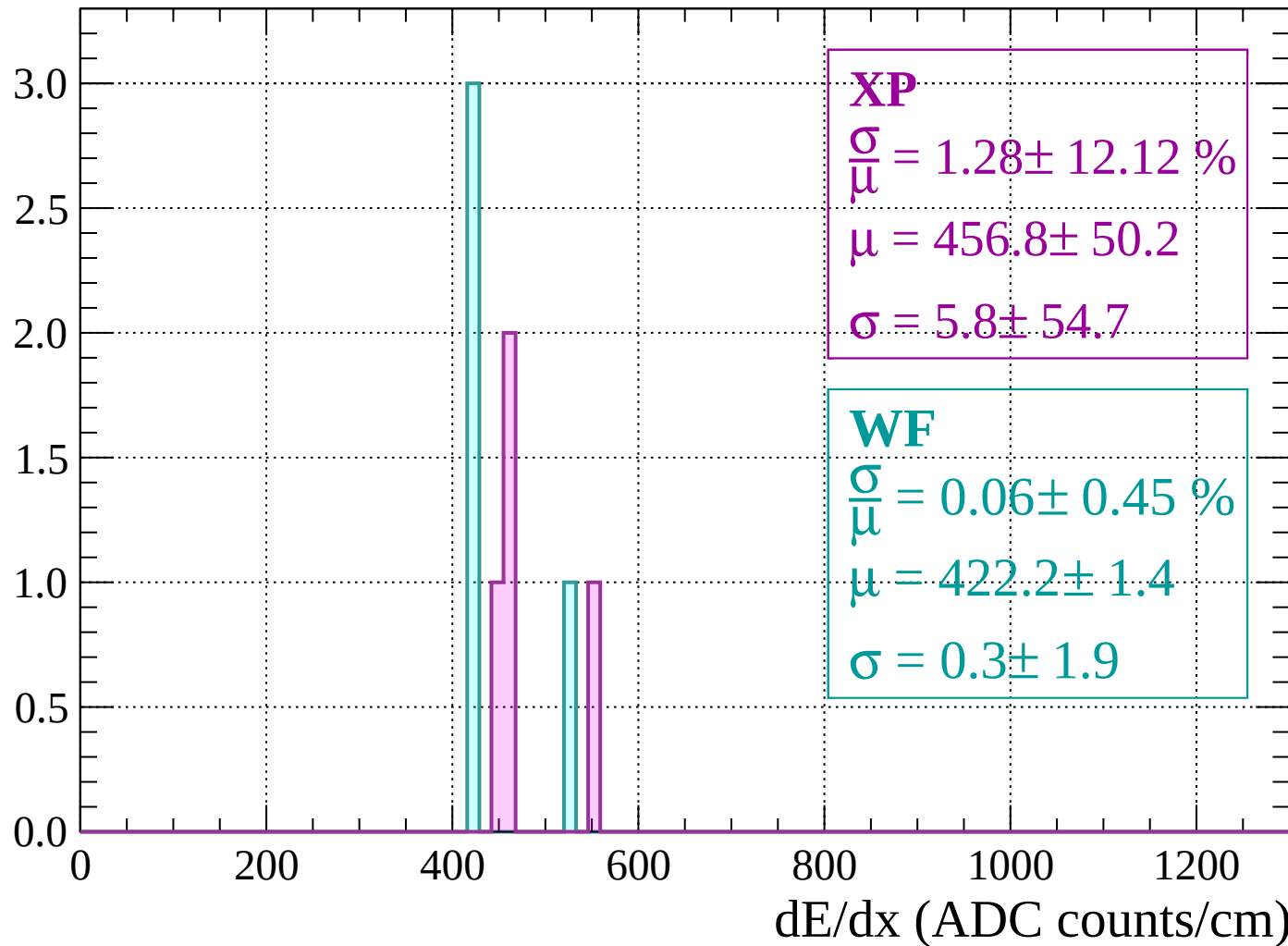
Energy loss | $160 < p < 200$

Count



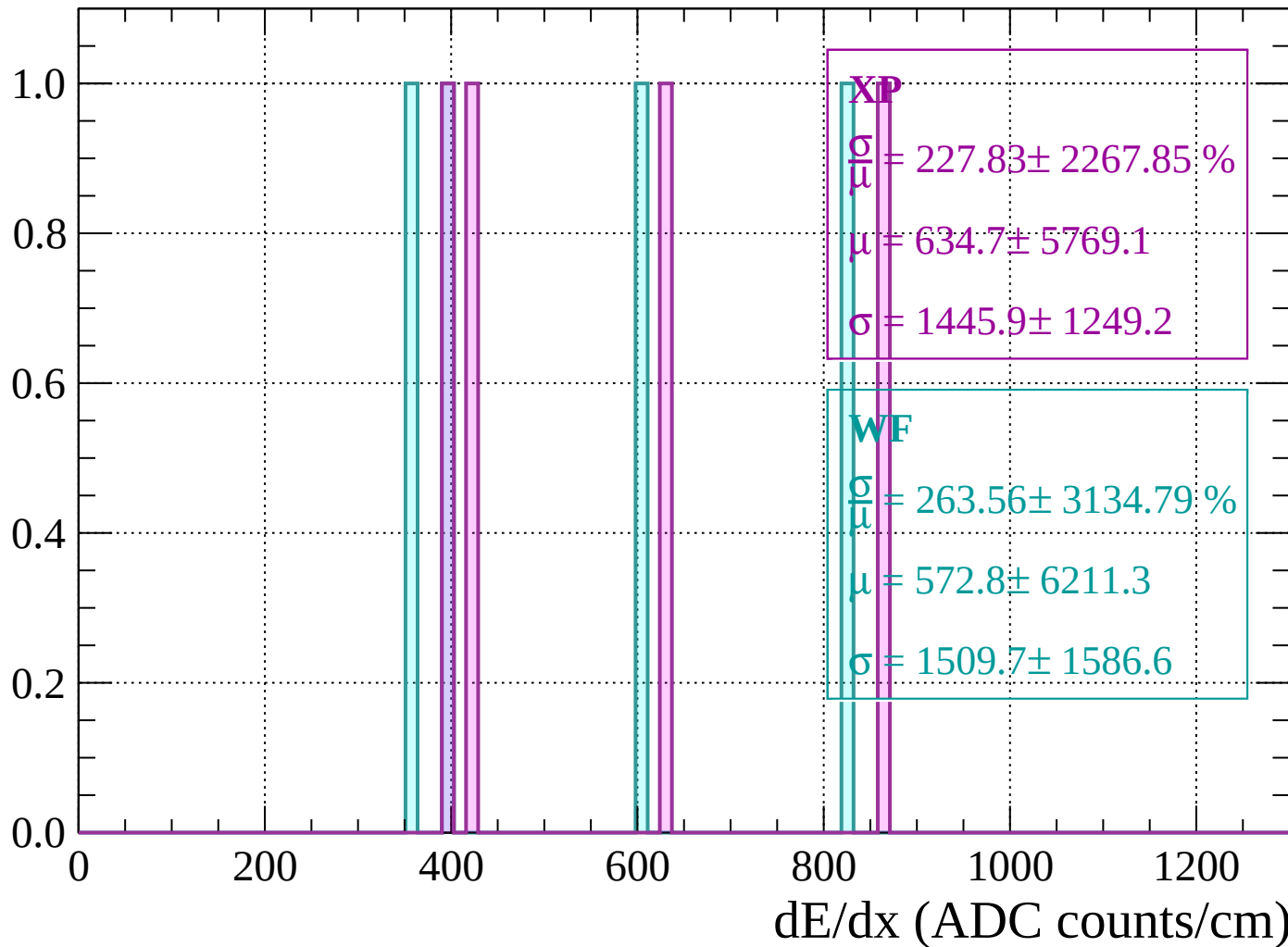
Energy loss | $200 < p < 240$

Count



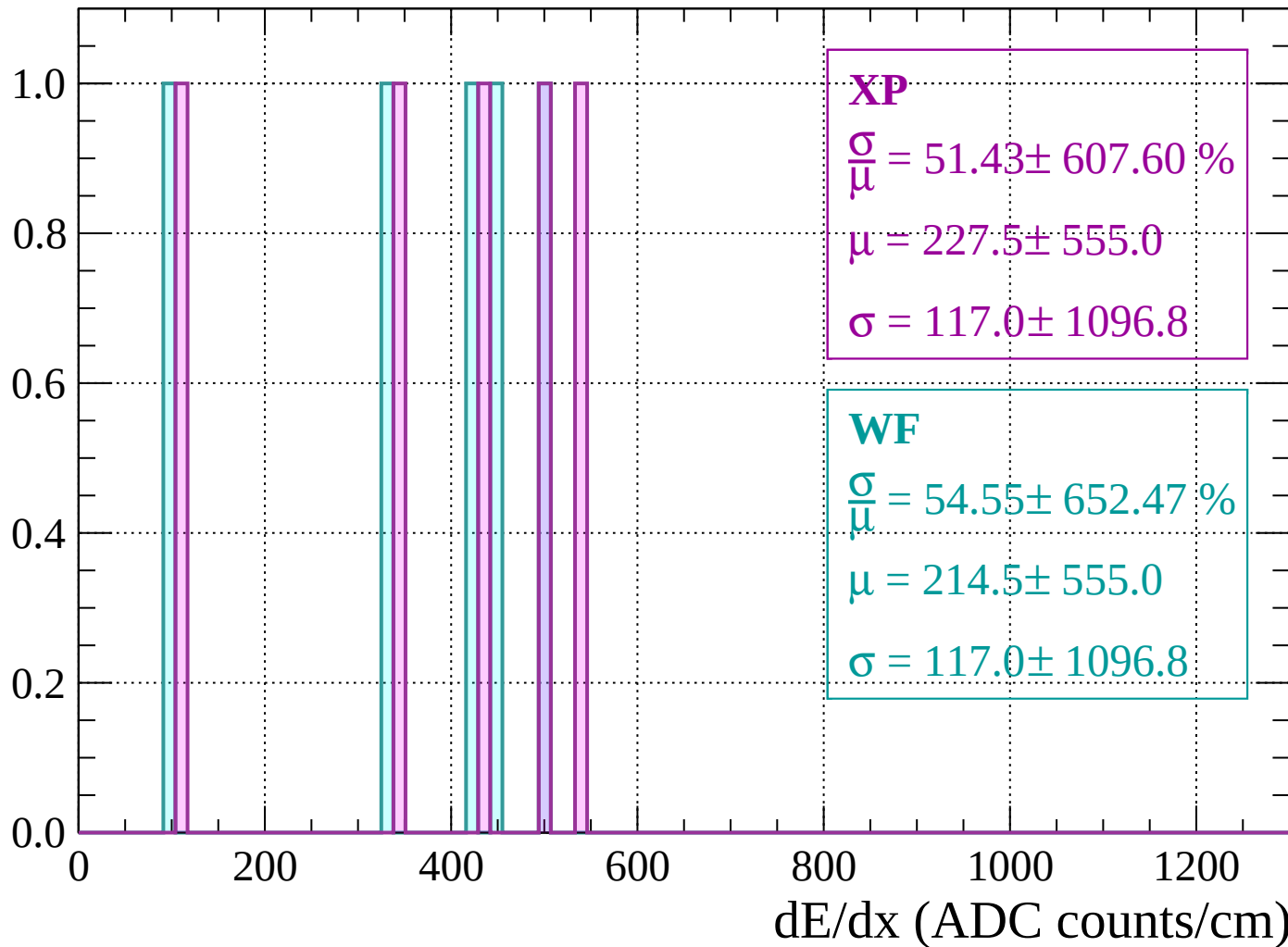
Energy loss | $240 < p < 280$

Count



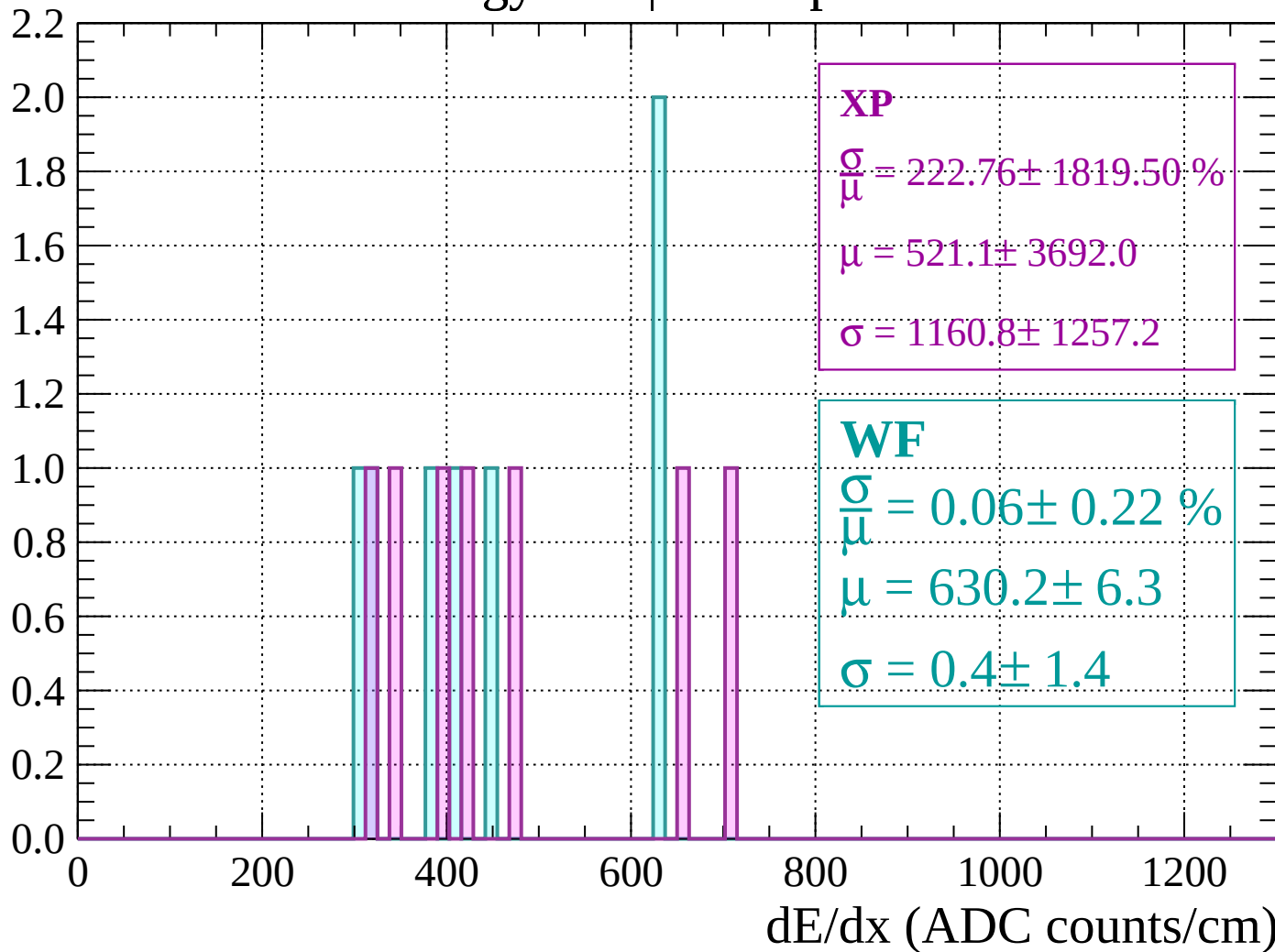
Energy loss | $280 < p < 320$

Count



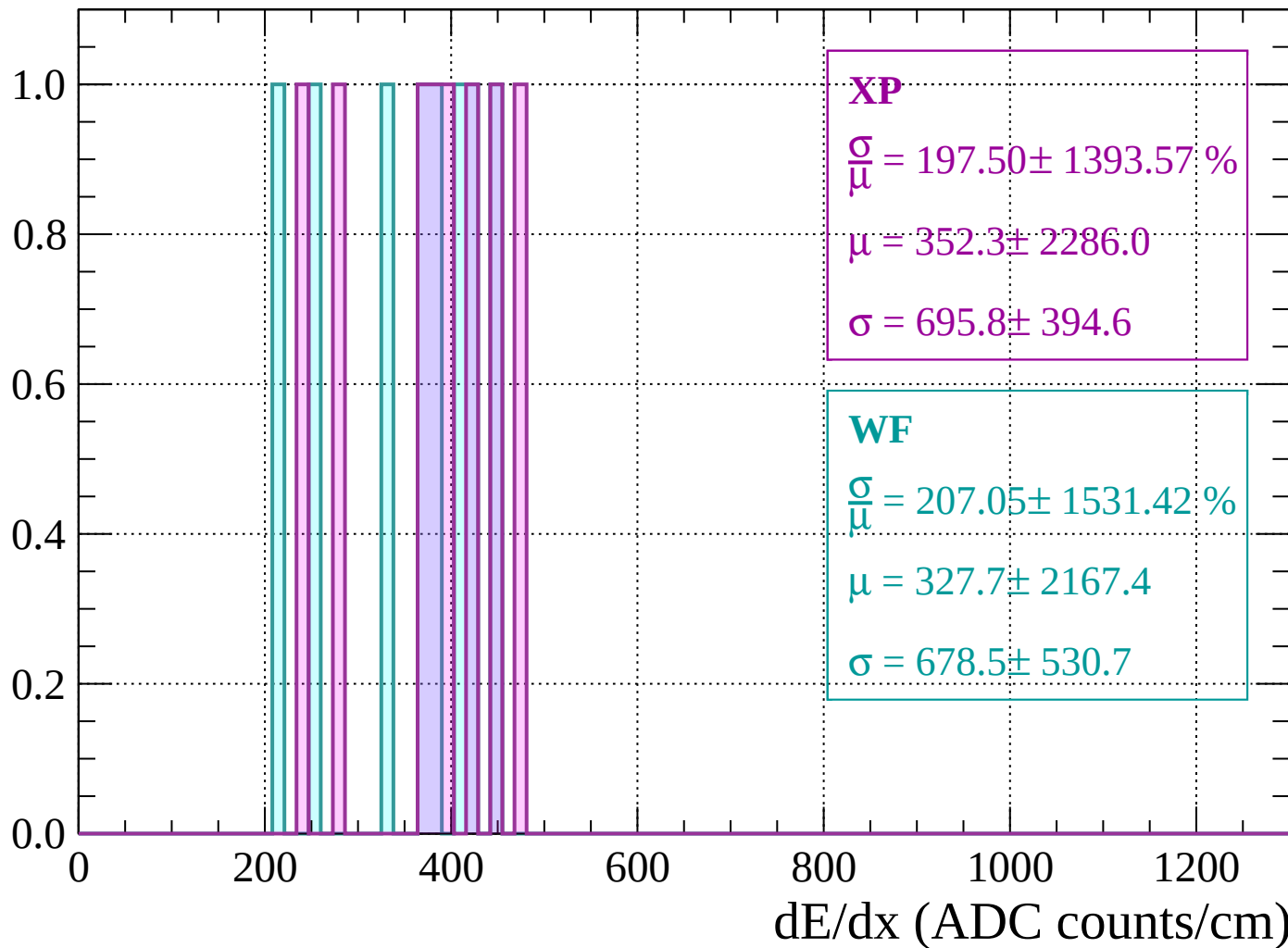
Energy loss | $320 < p < 360$

Count



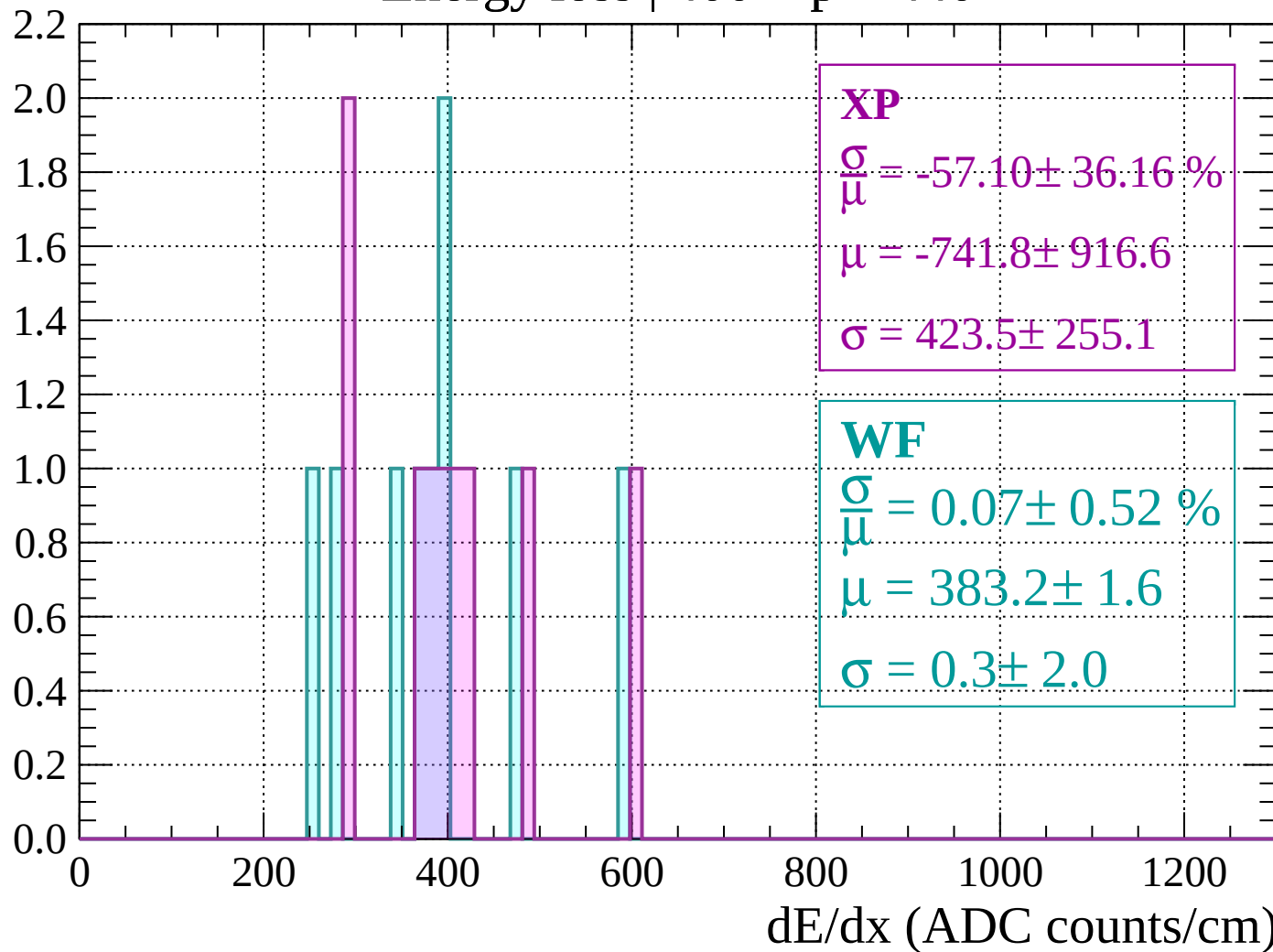
Energy loss | $360 < p < 400$

Count



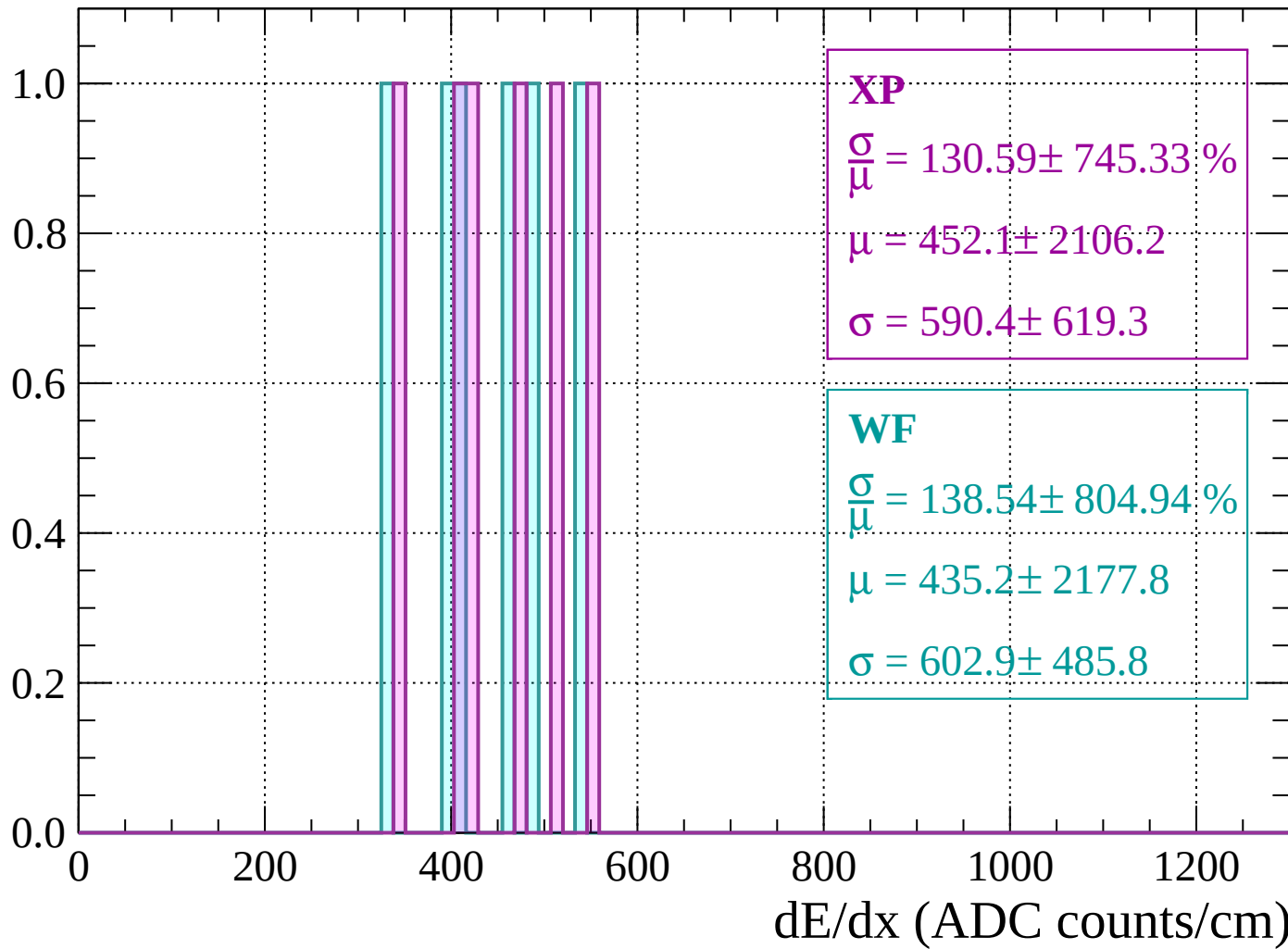
Energy loss | $400 < p < 440$

Count



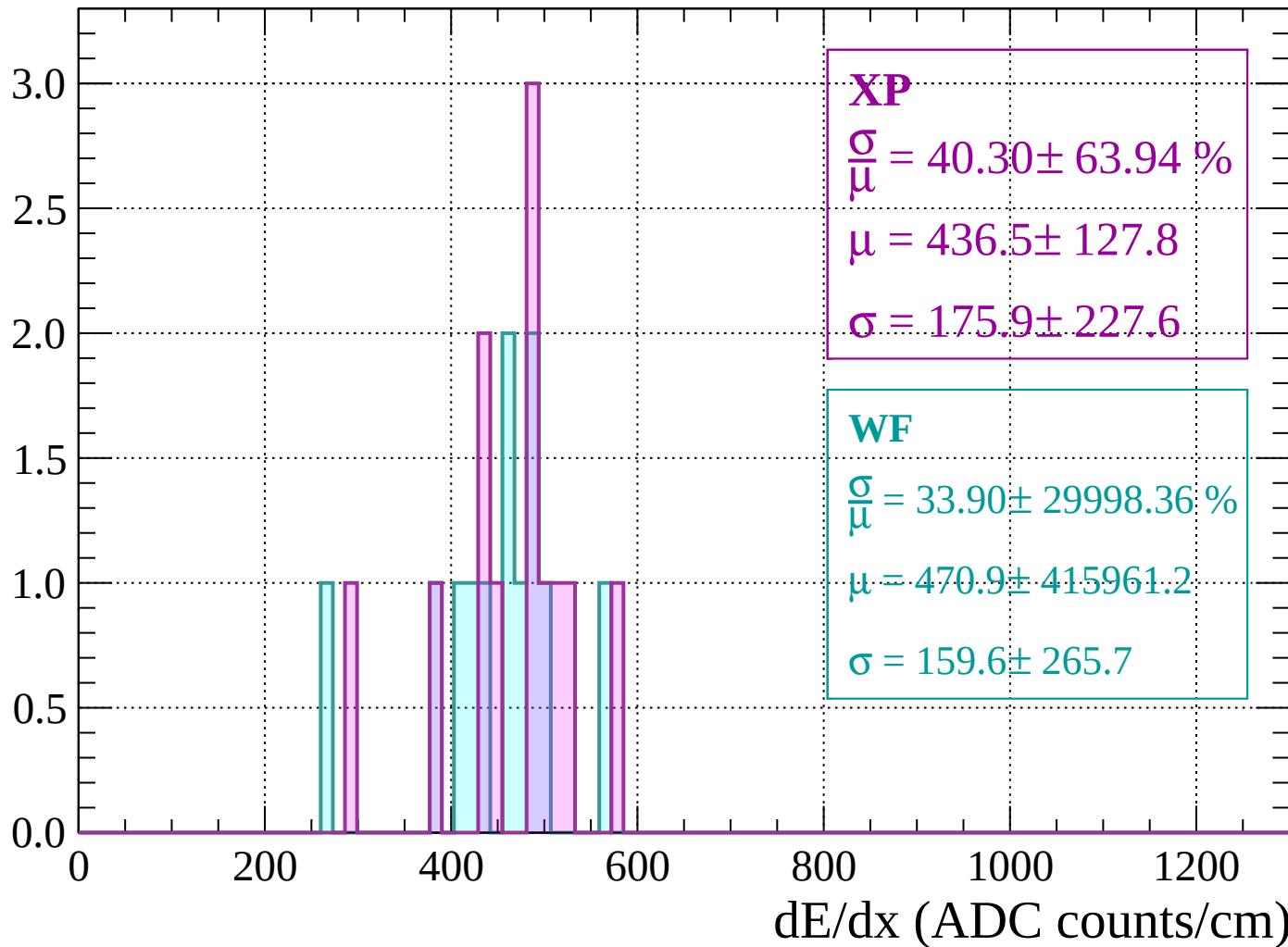
Energy loss | $440 < p < 480$

Count



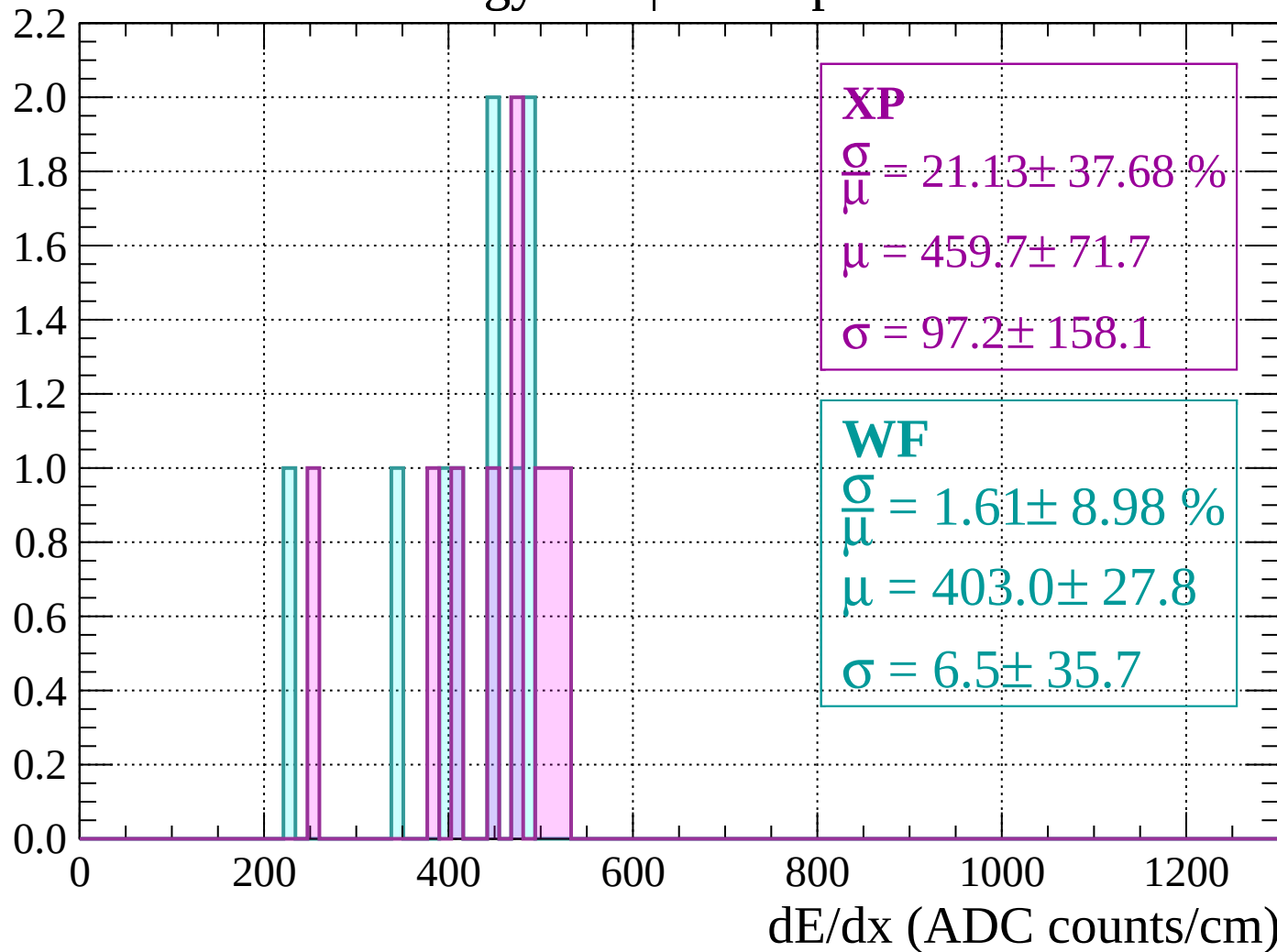
Energy loss | $480 < p < 520$

Count



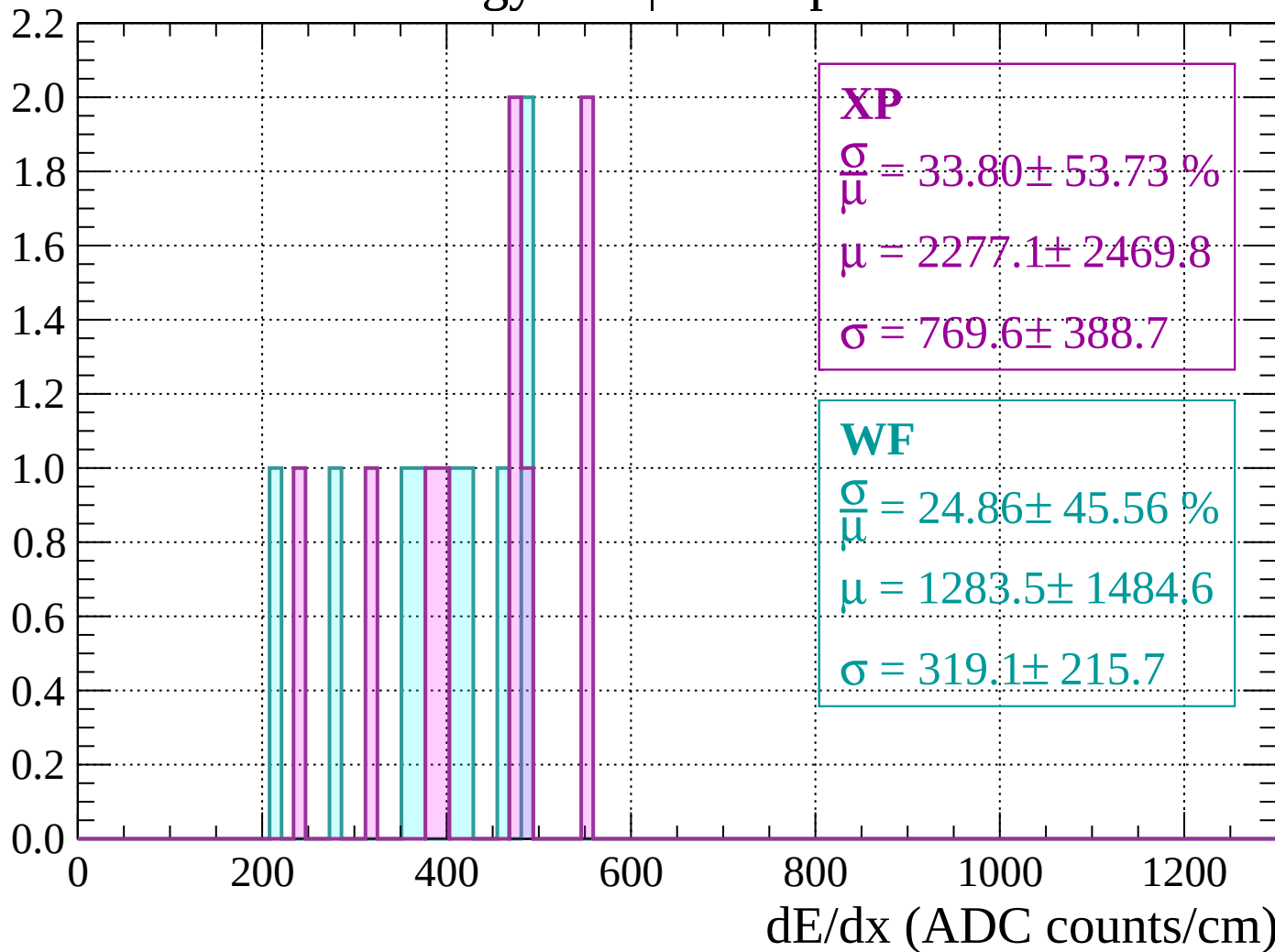
Energy loss | $520 < p < 560$

Count



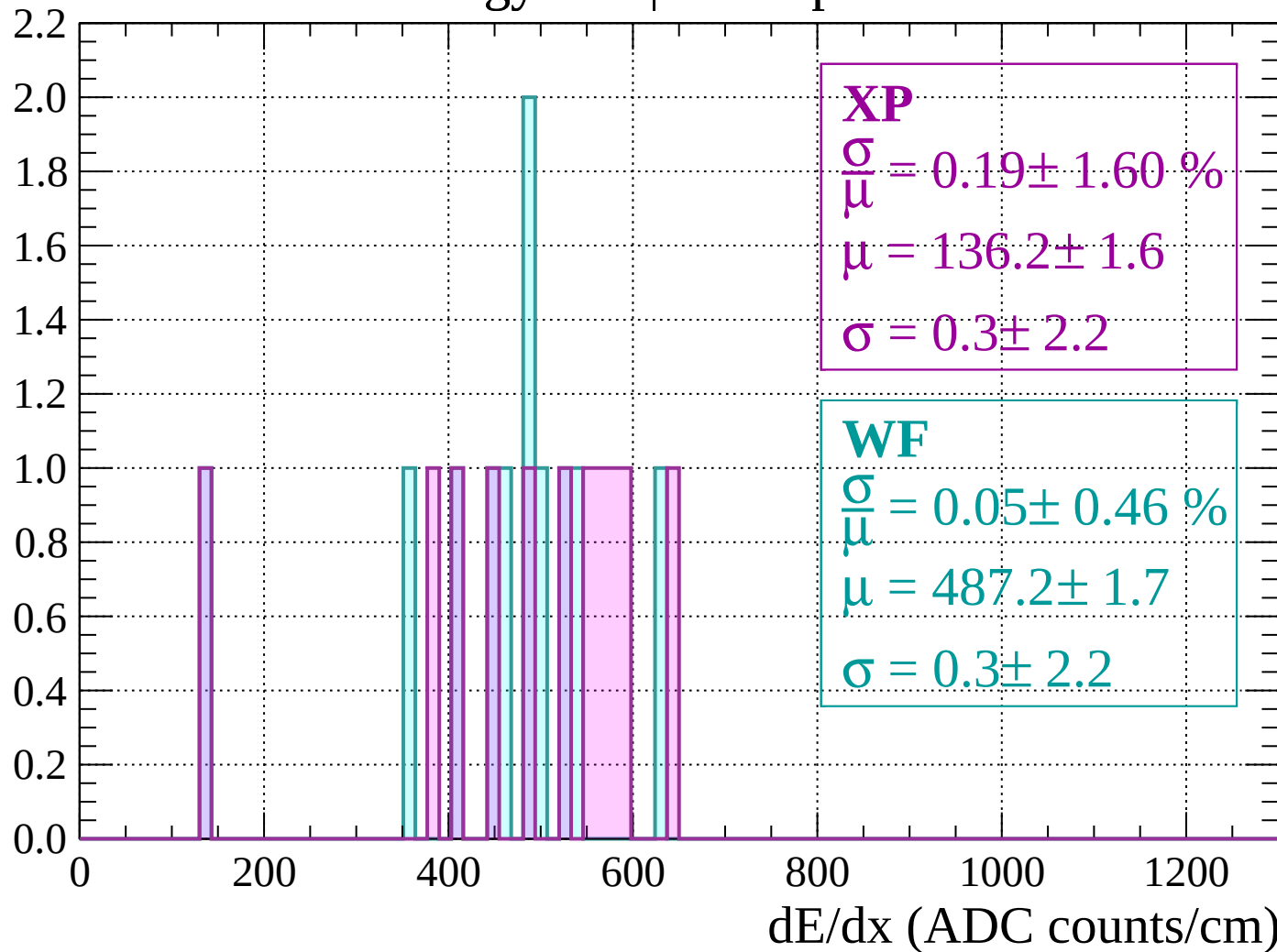
Energy loss | $560 < p < 600$

Count



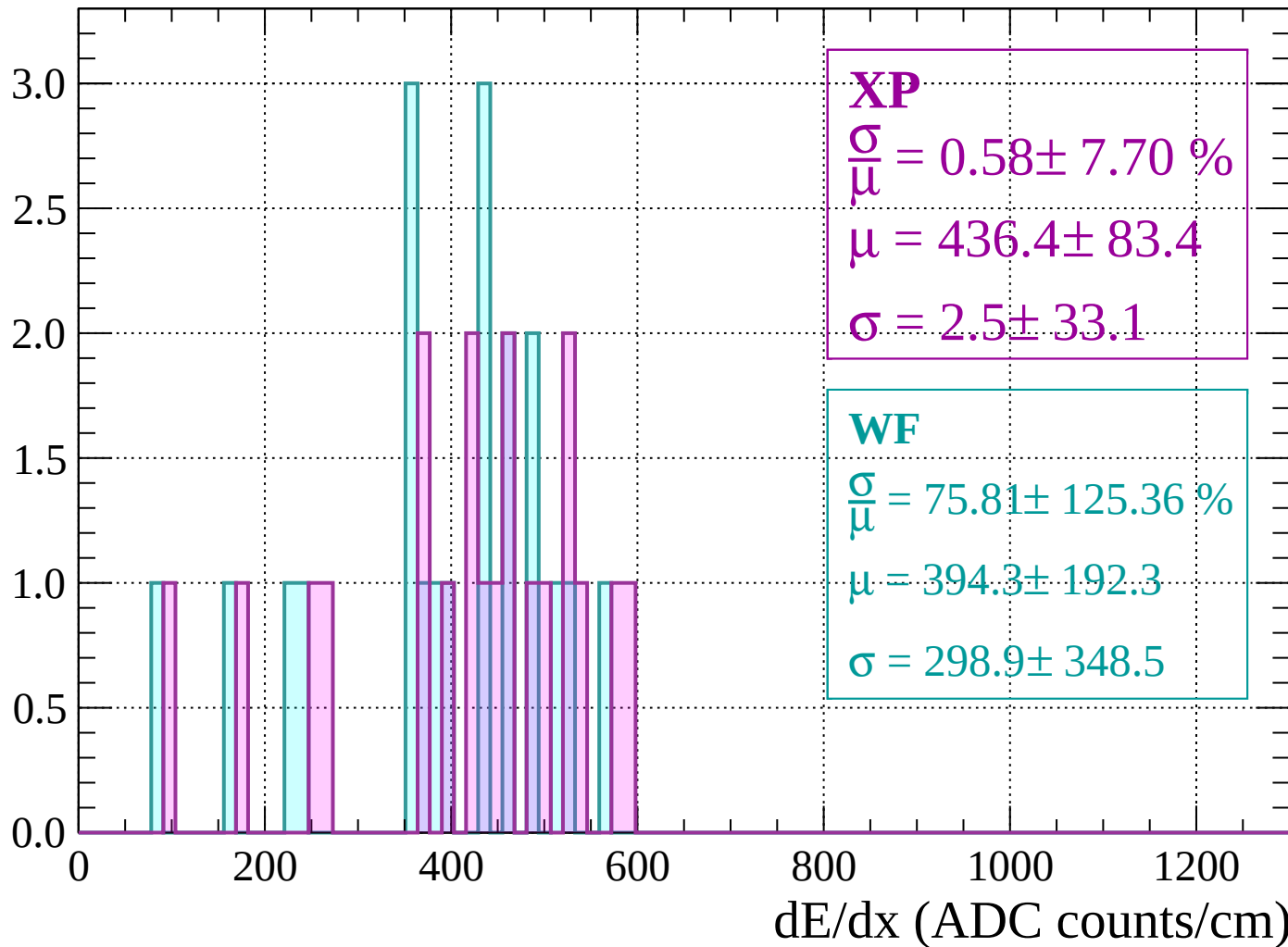
Energy loss | $600 < p < 640$

Count



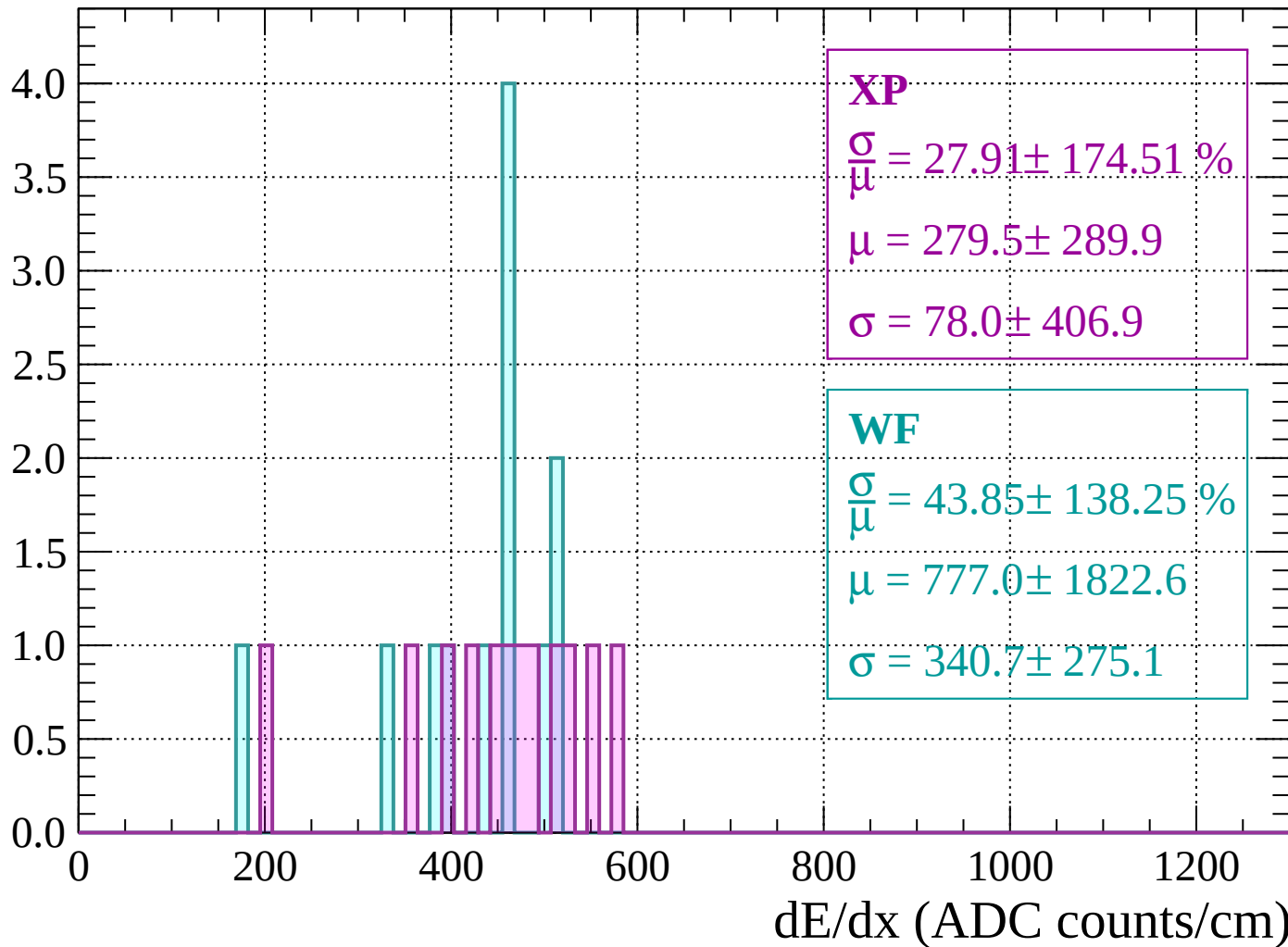
Energy loss | $640 < p < 680$

Count



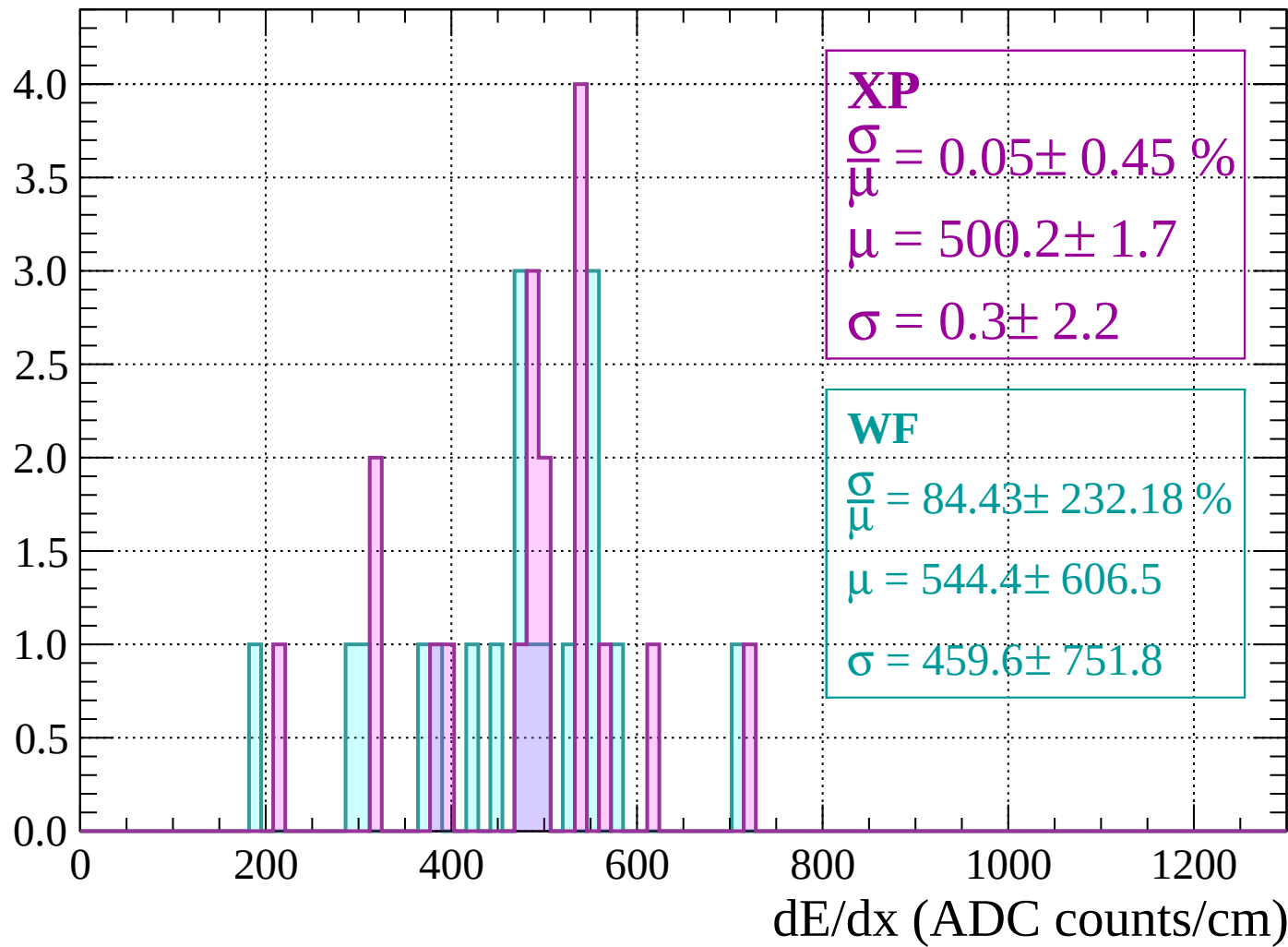
Energy loss | $680 < p < 720$

Count



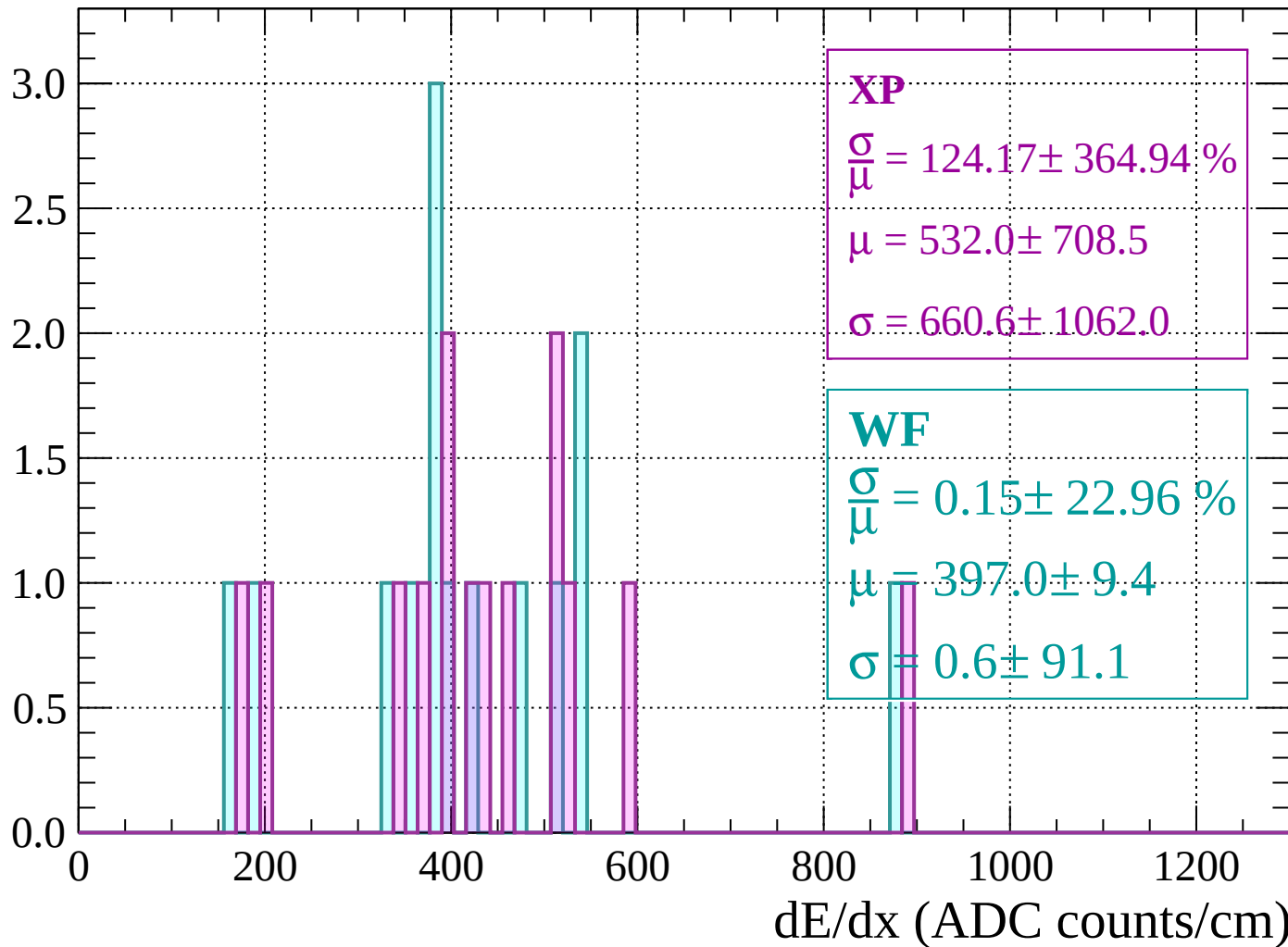
Energy loss | $720 < p < 760$

Count



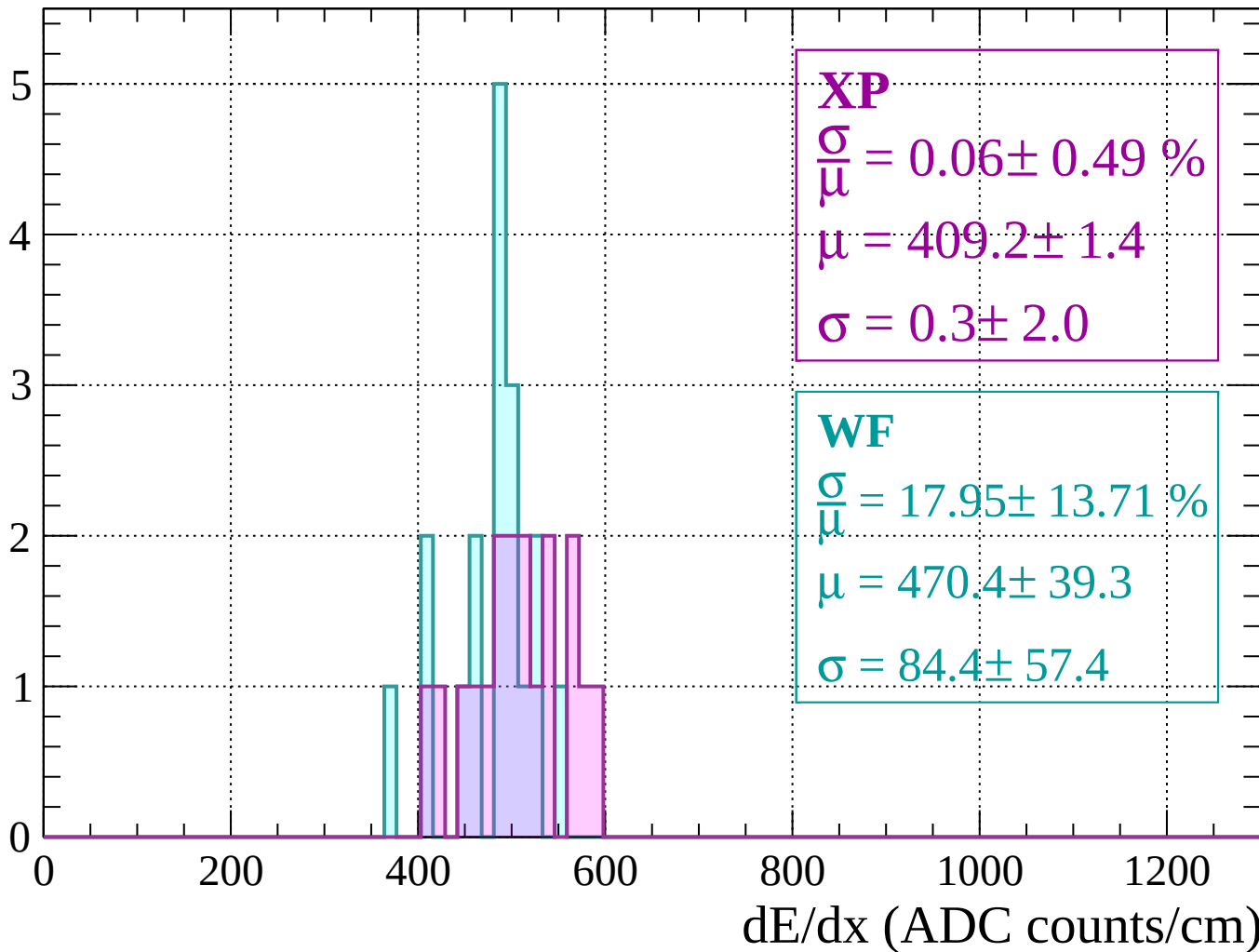
Energy loss | $760 < p < 800$

Count



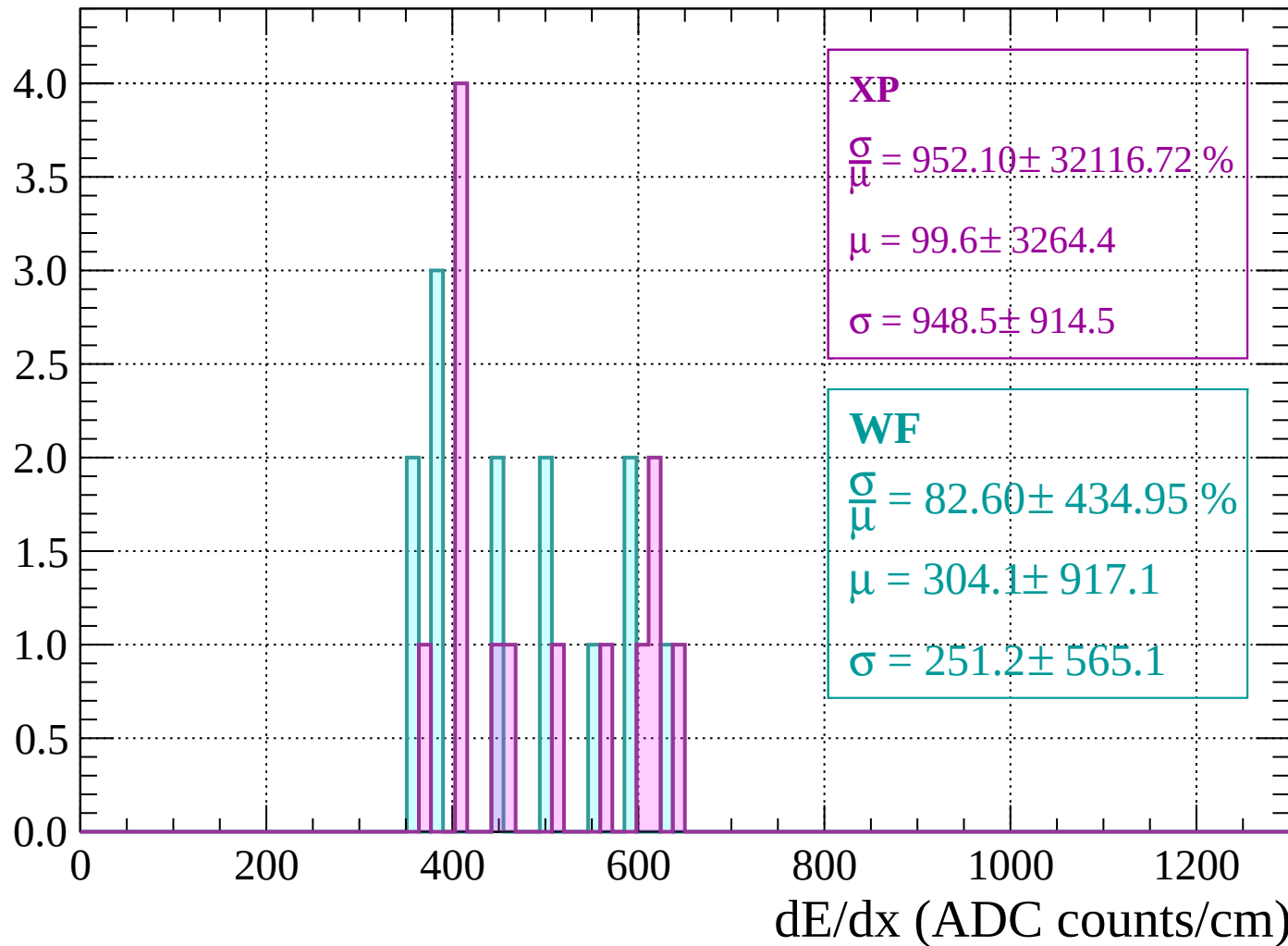
Energy loss | $800 < p < 840$

Count



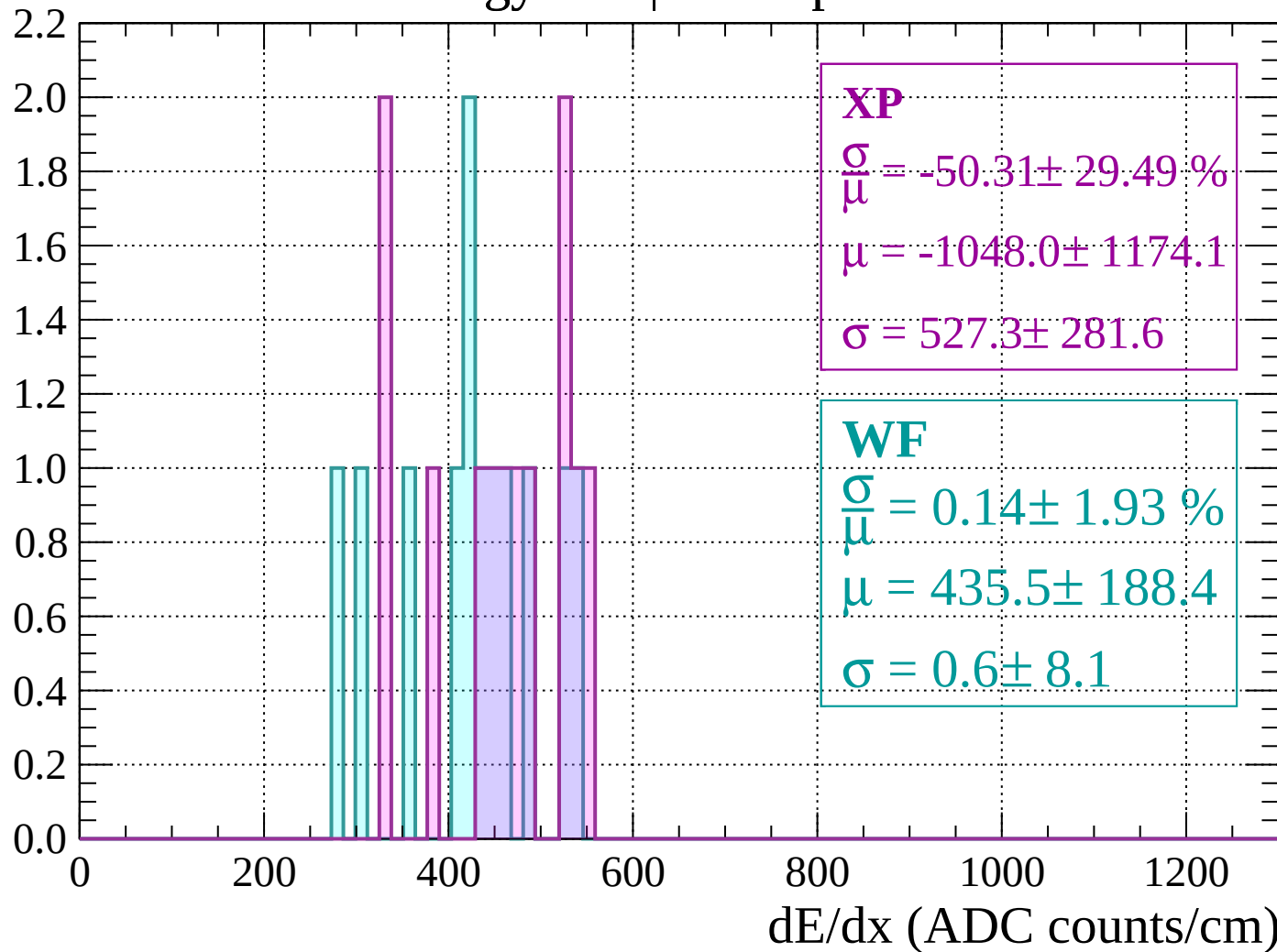
Energy loss | $840 < p < 880$

Count



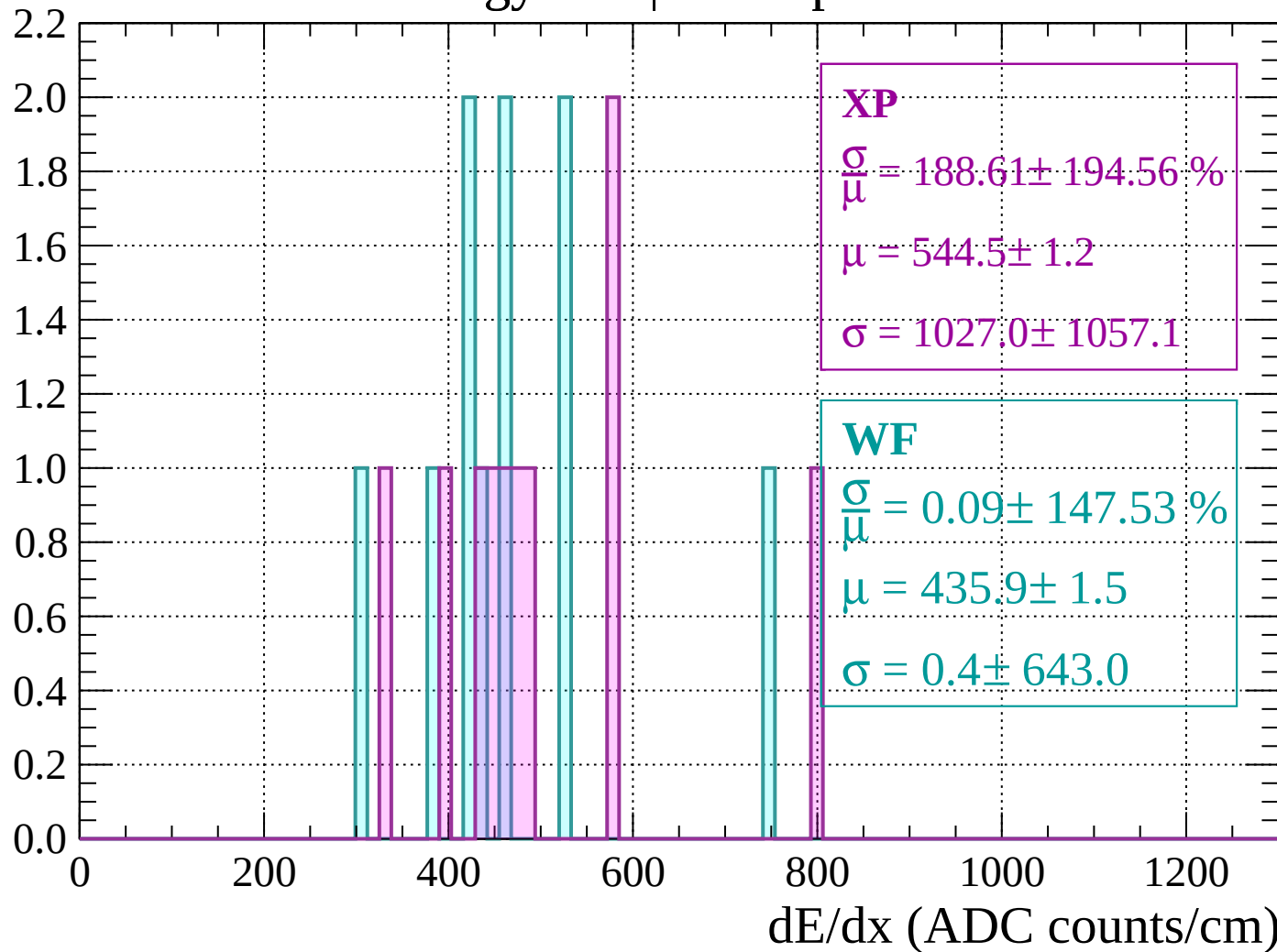
Energy loss | $880 < p < 920$

Count



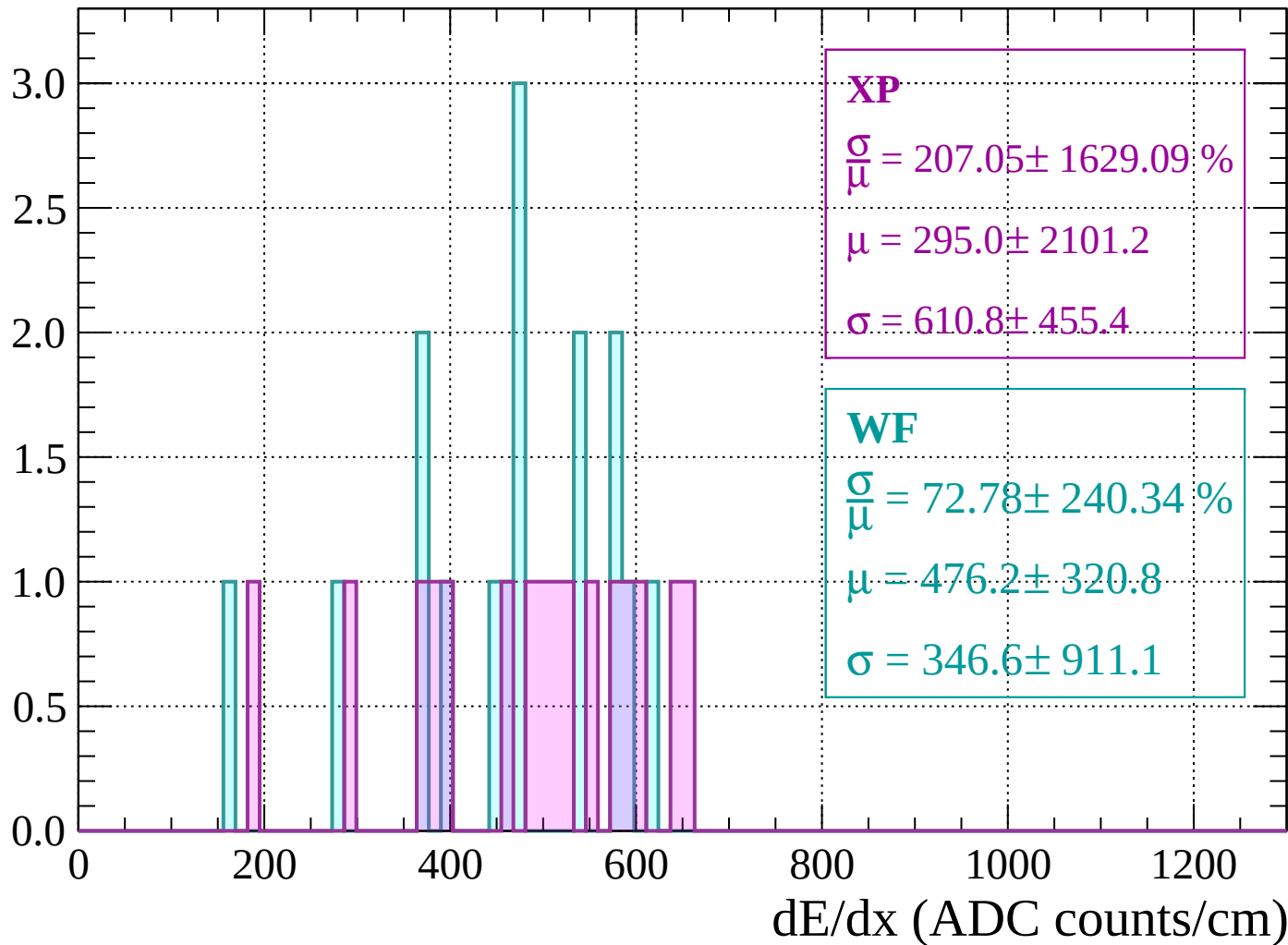
Energy loss | $920 < p < 960$

Count



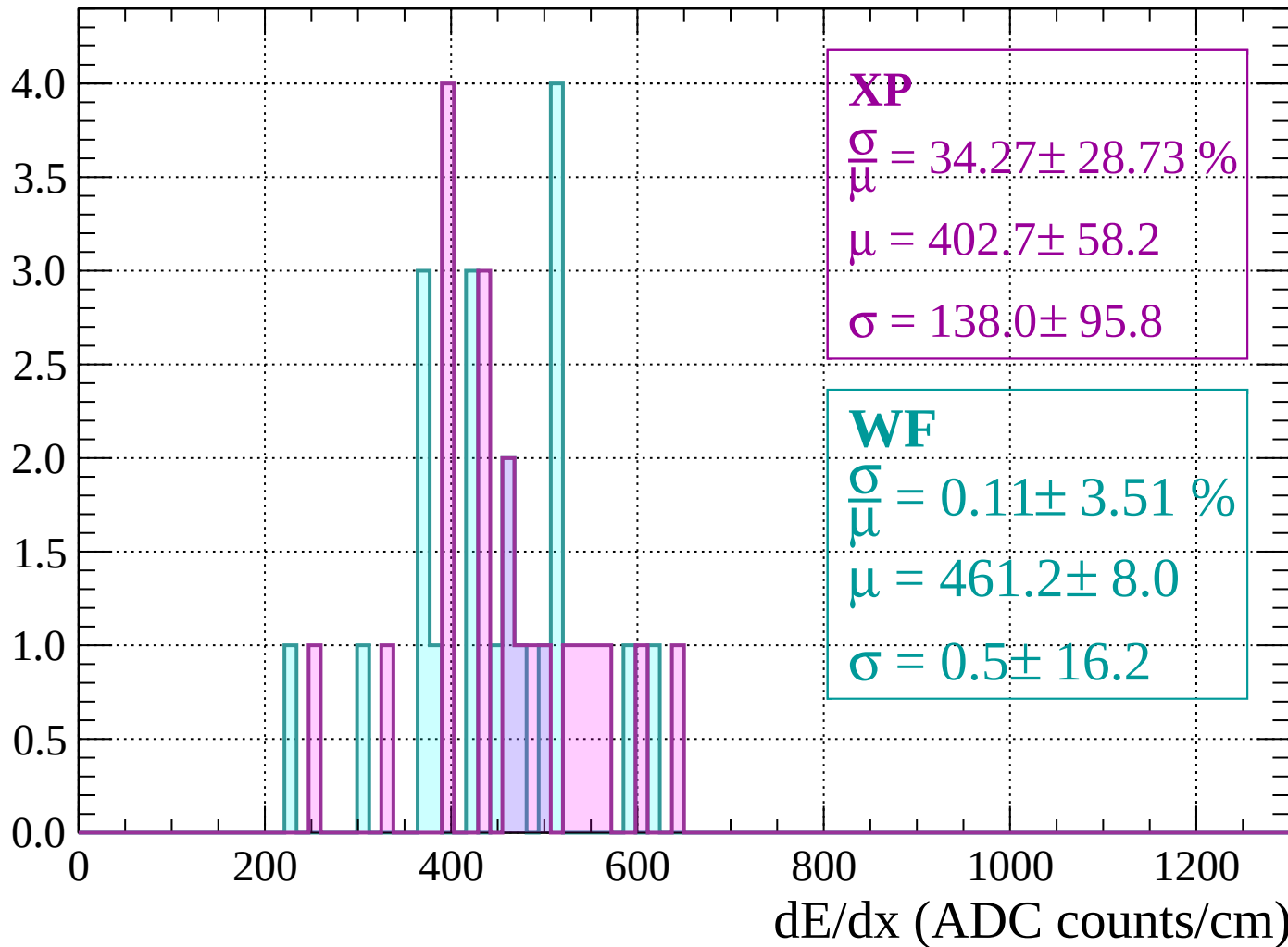
Energy loss | $960 < p < 1000$

Count



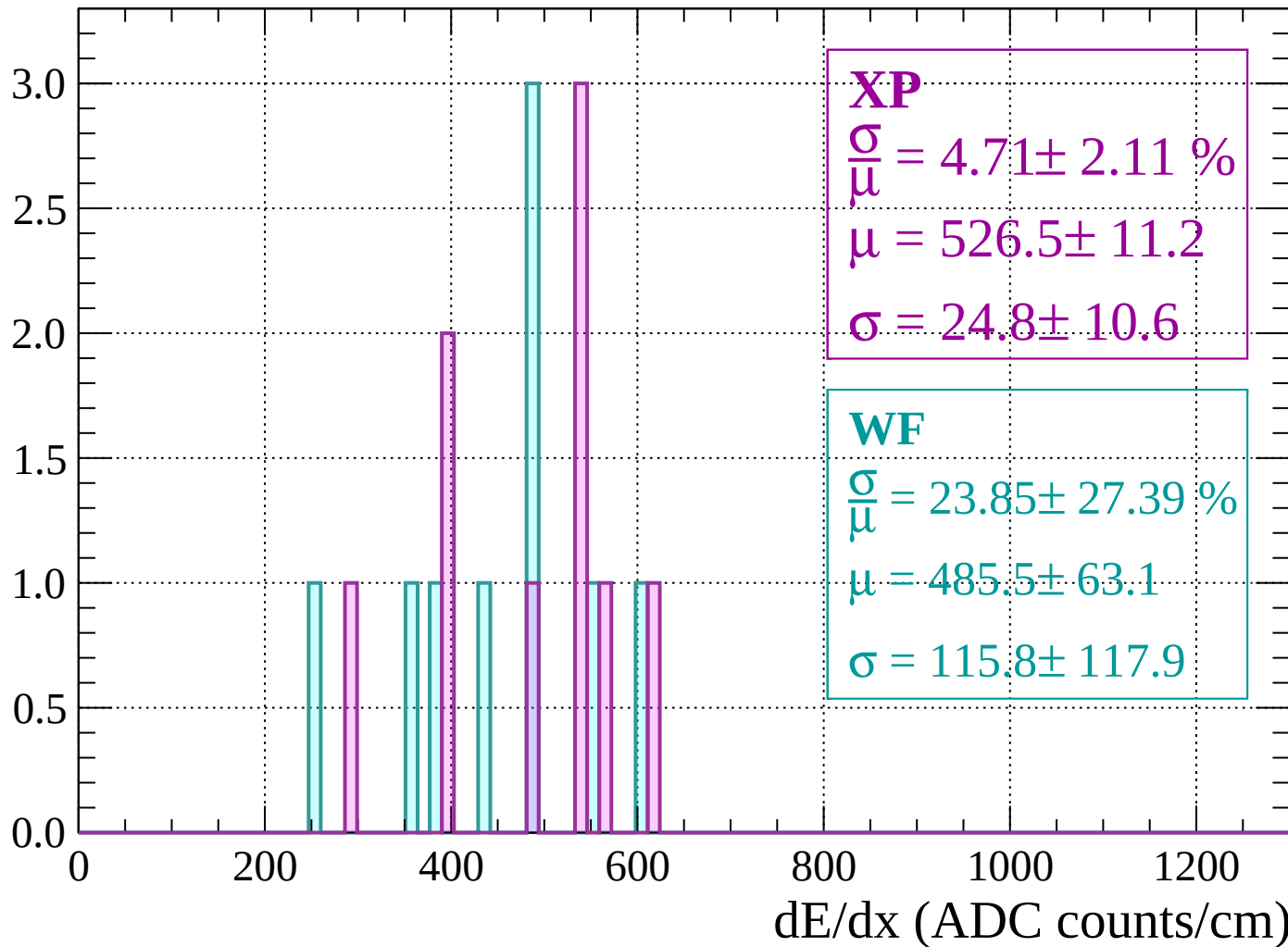
Energy loss | $1000 < p < 1040$

Count



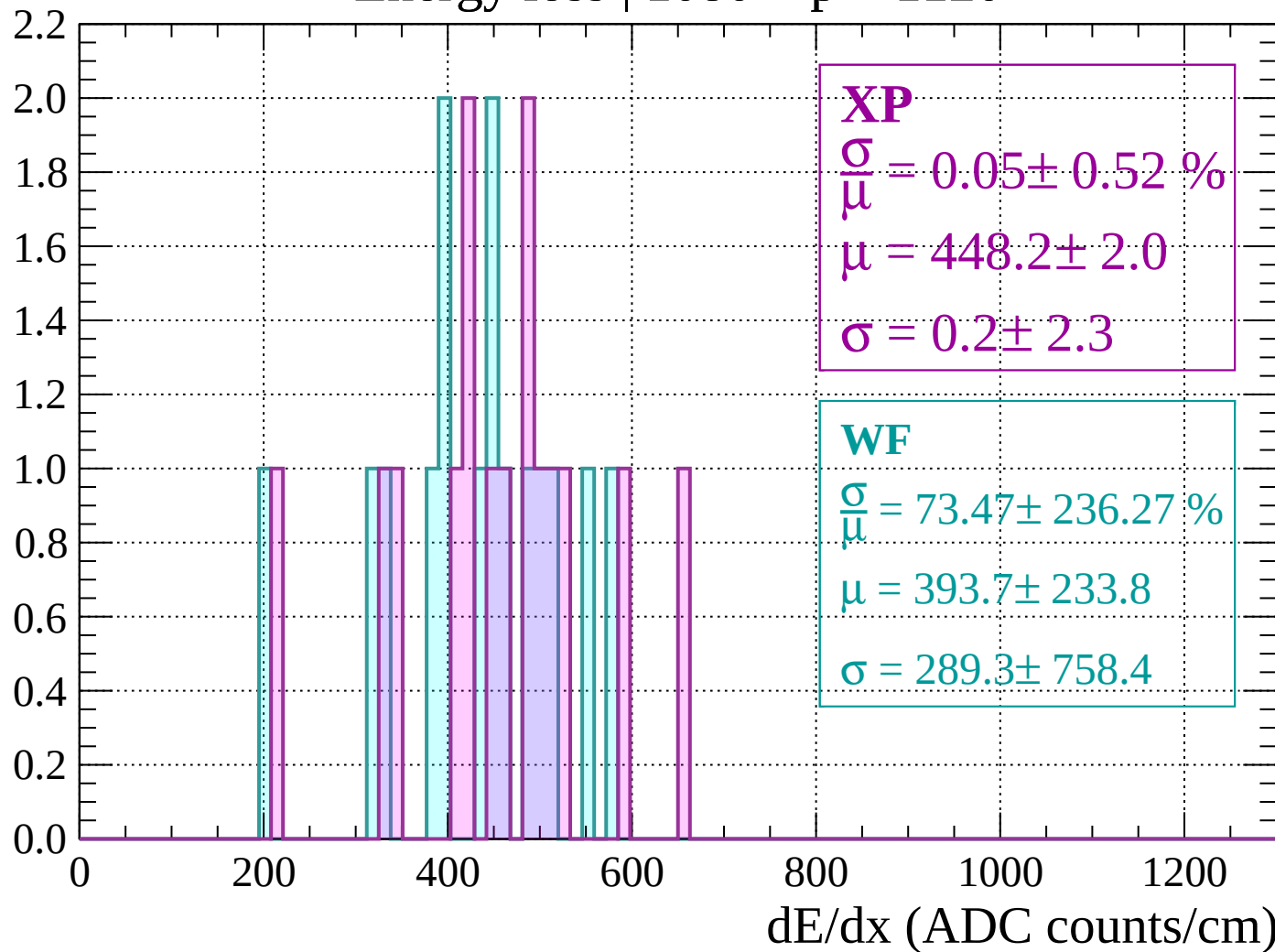
Energy loss | $1040 < p < 1080$

Count



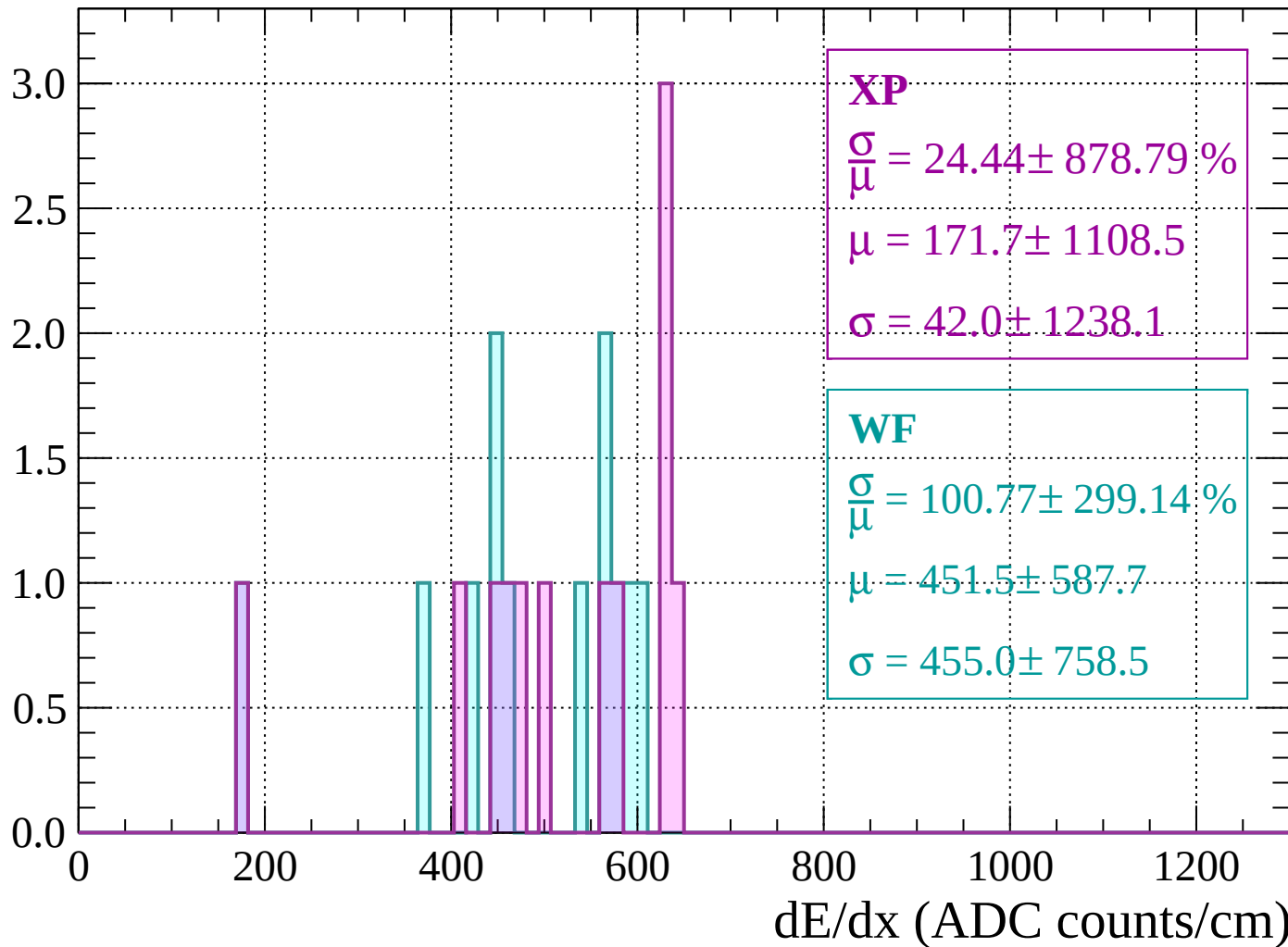
Energy loss | $1080 < p < 1120$

Count



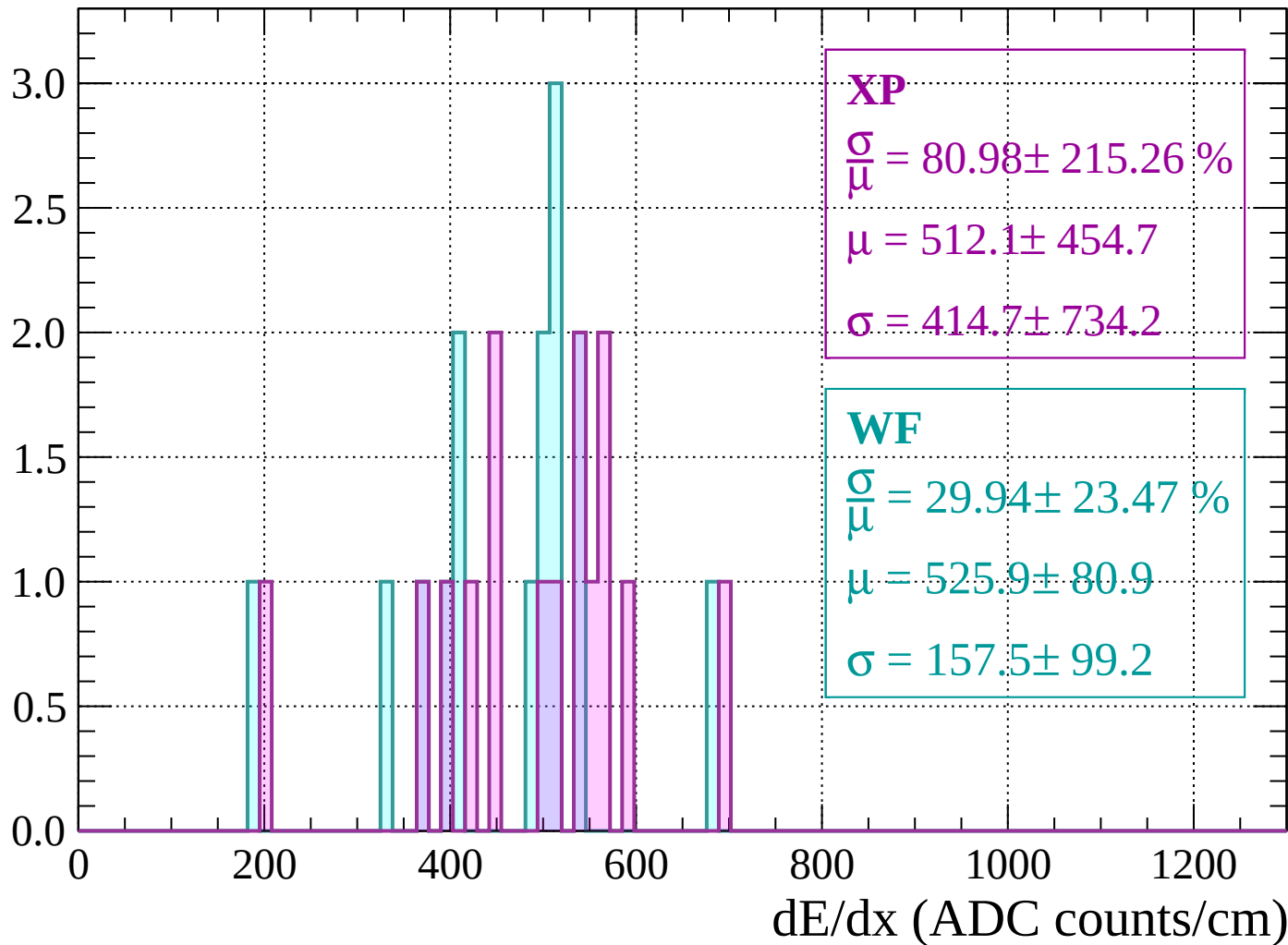
Energy loss | $1120 < p < 1160$

Count



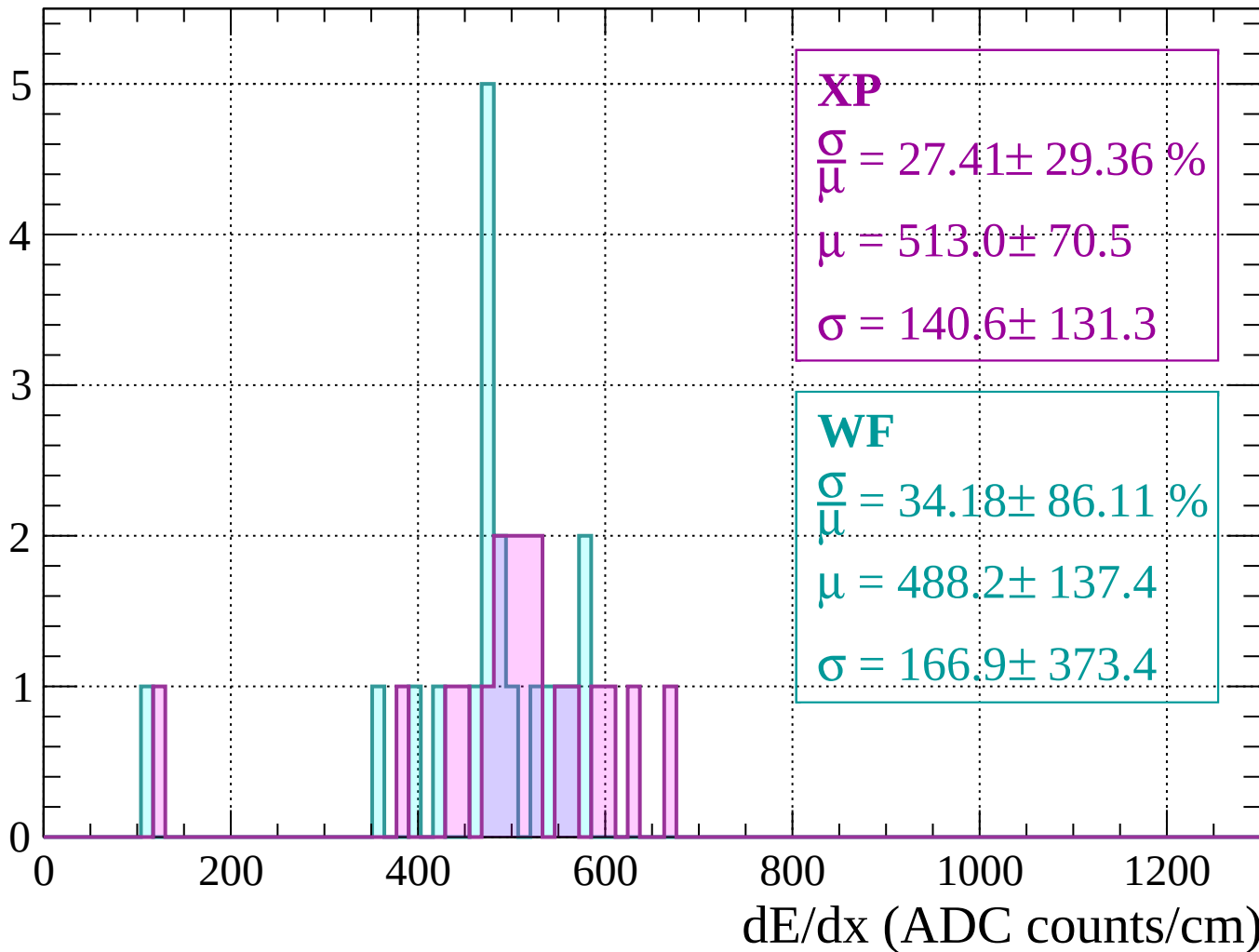
Energy loss | $1160 < p < 1200$

Count



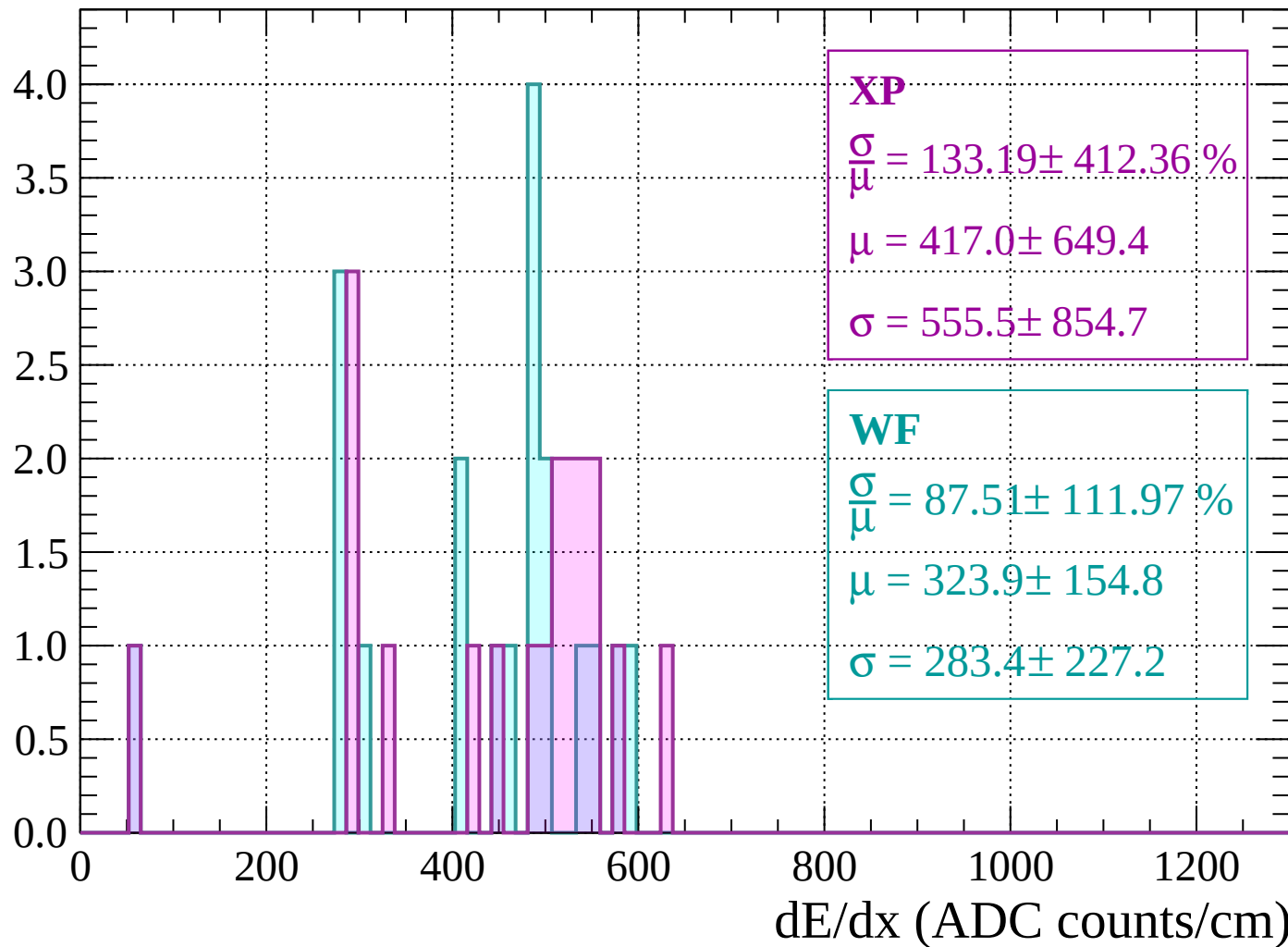
Energy loss | $1200 < p < 1240$

Count



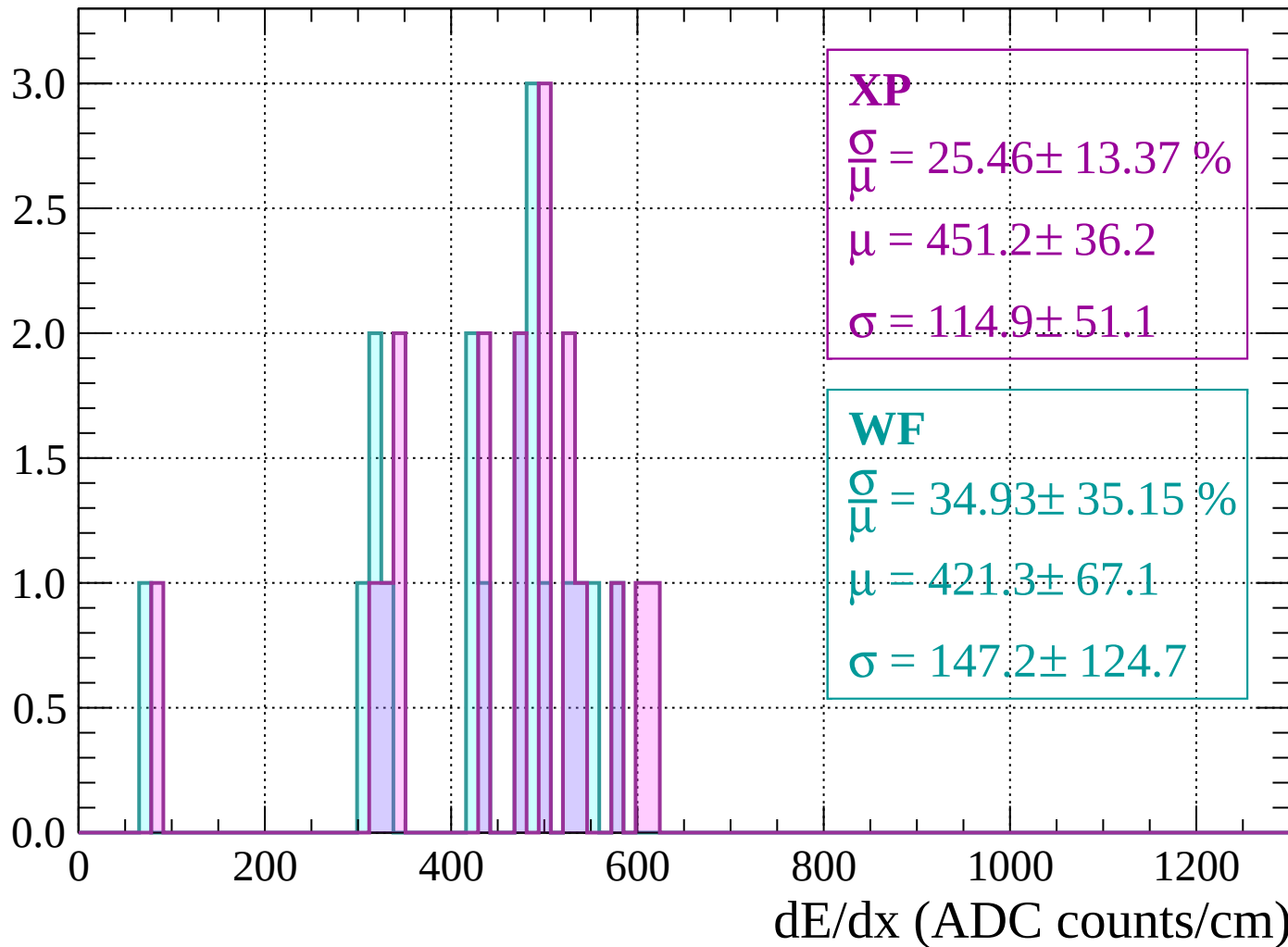
Energy loss | $1240 < p < 1280$

Count



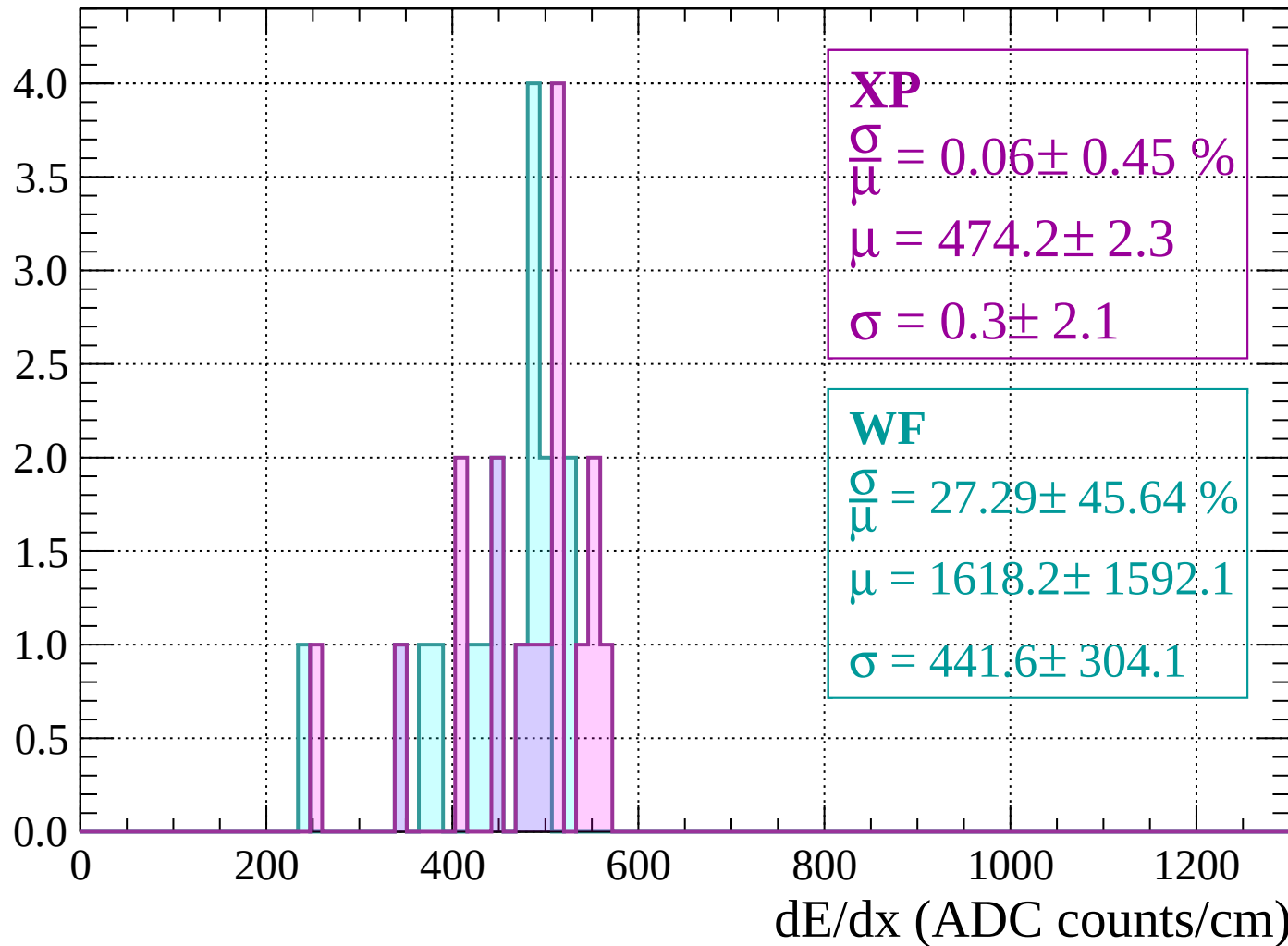
Energy loss | $1280 < p < 1320$

Count



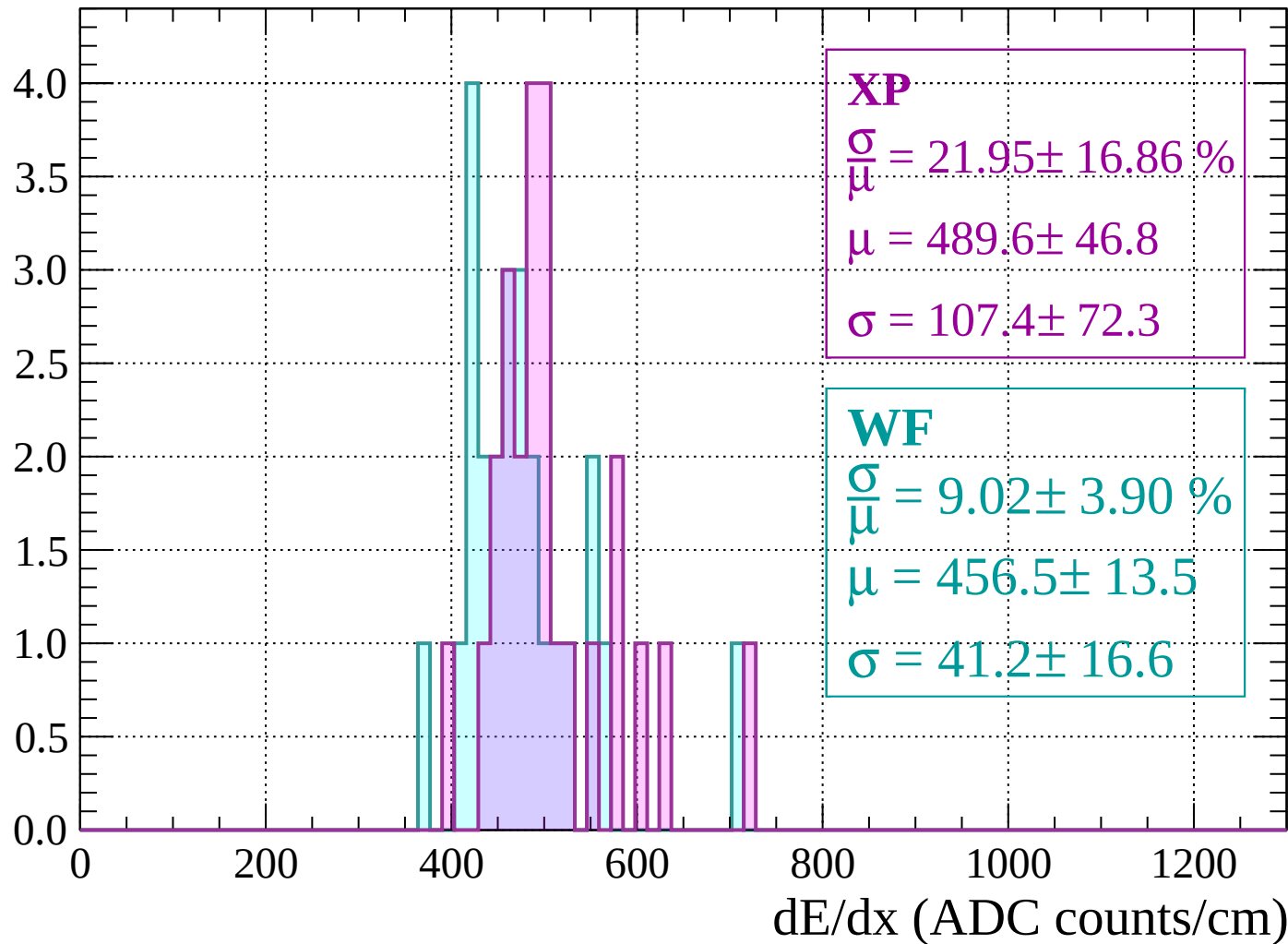
Energy loss | $1320 < p < 1360$

Count



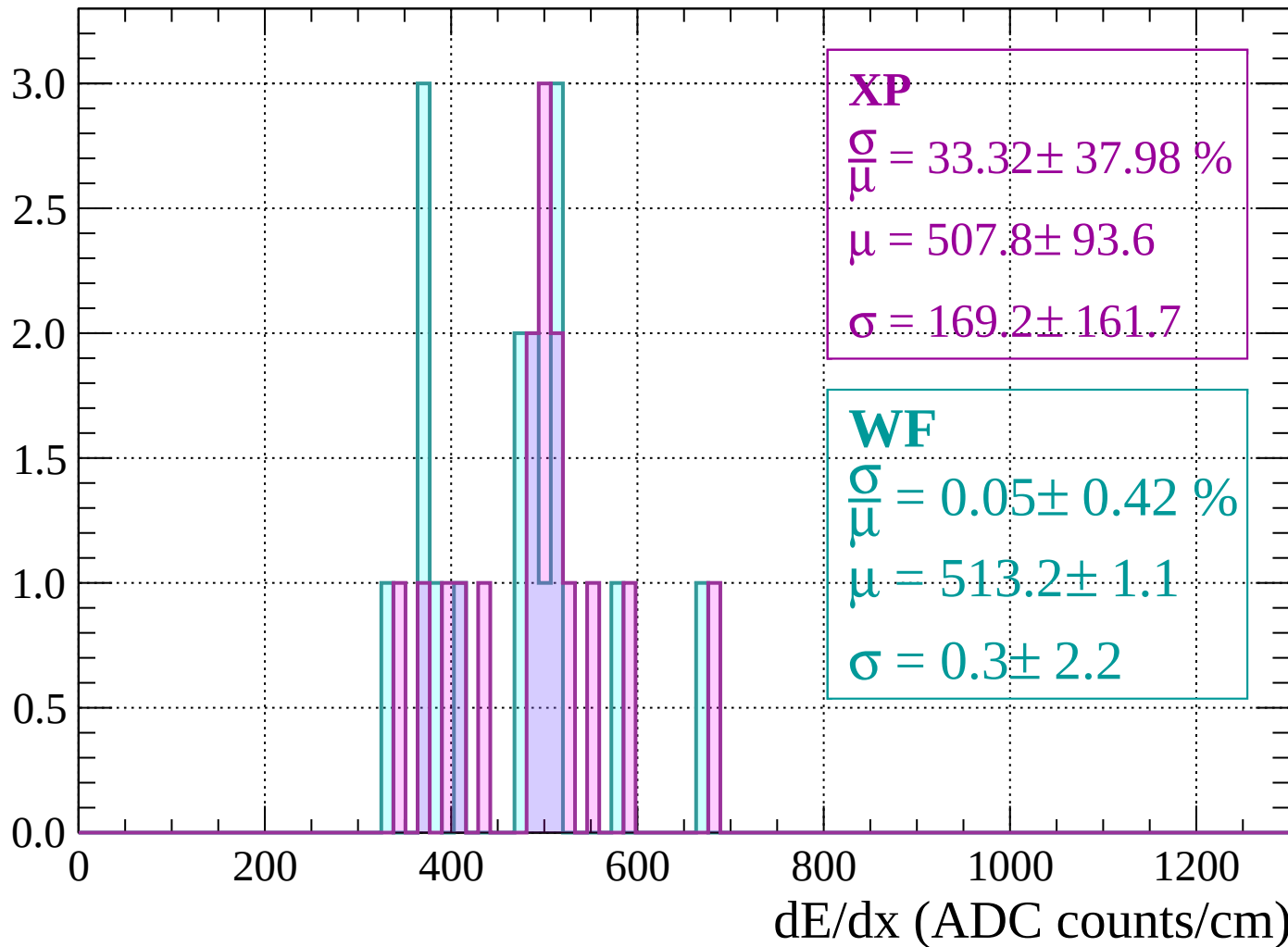
Energy loss | $1360 < p < 1400$

Count



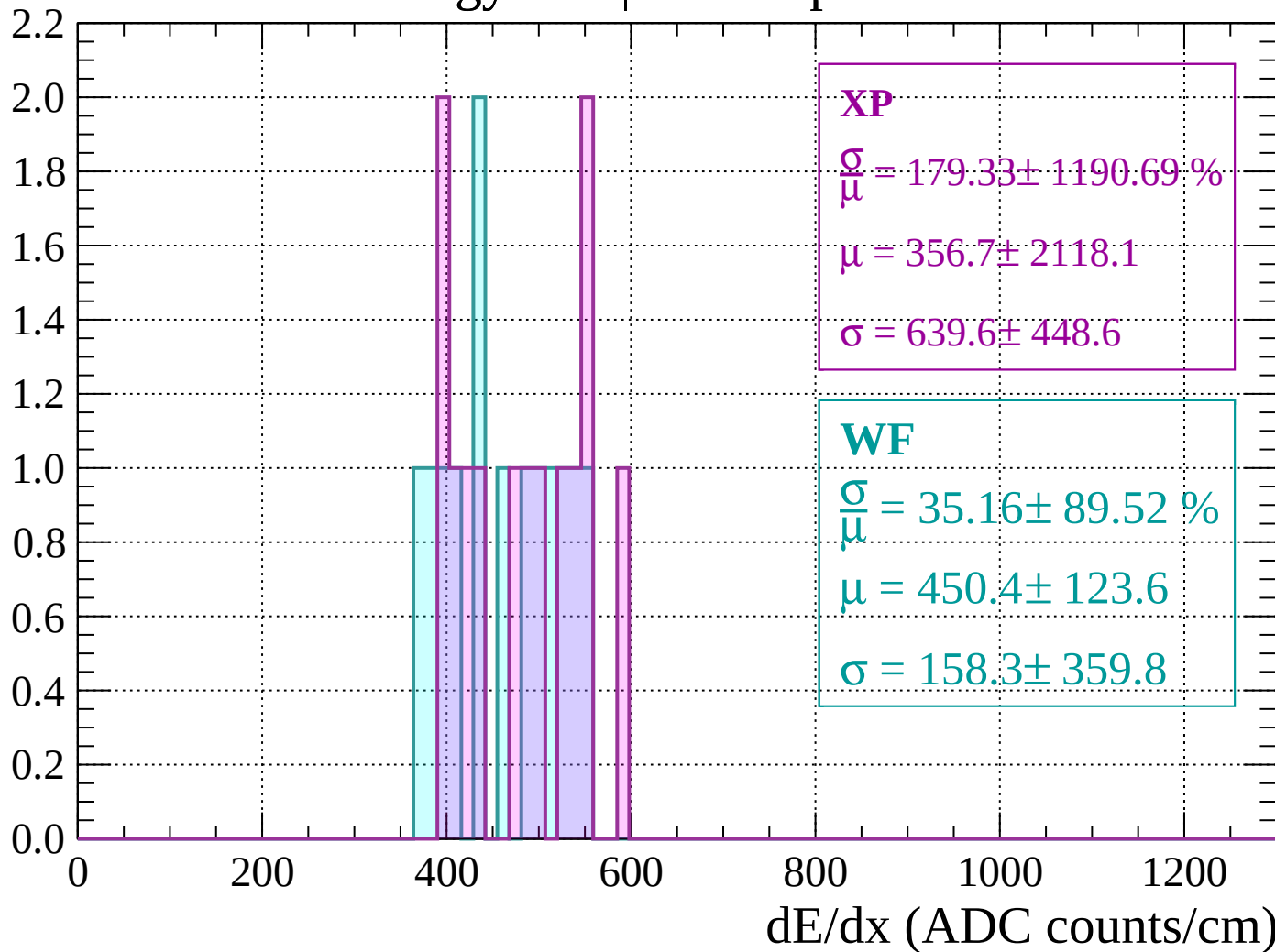
Energy loss | $1400 < p < 1440$

Count



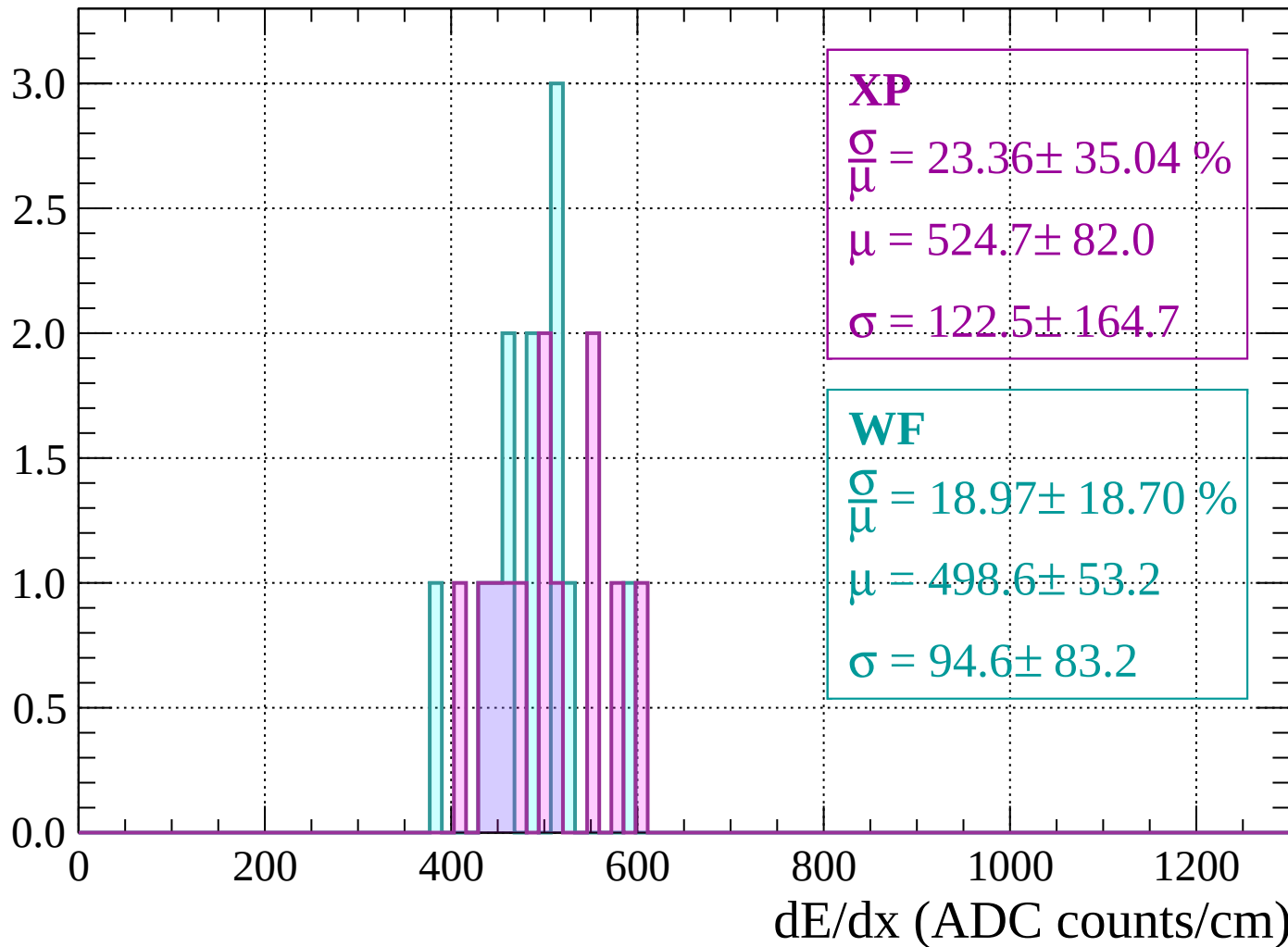
Energy loss | $1440 < p < 1480$

Count



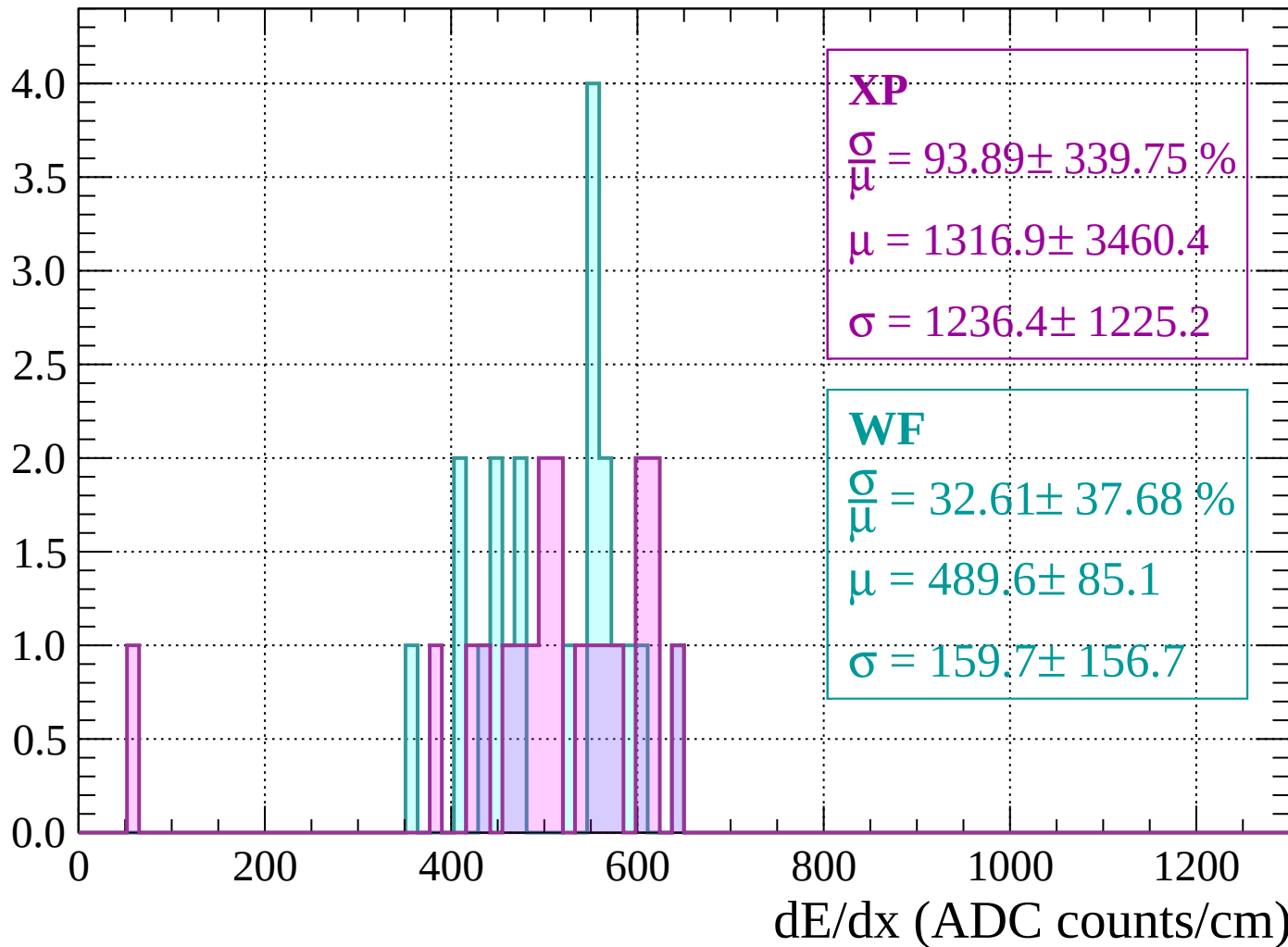
Energy loss | $1480 < p < 1520$

Count



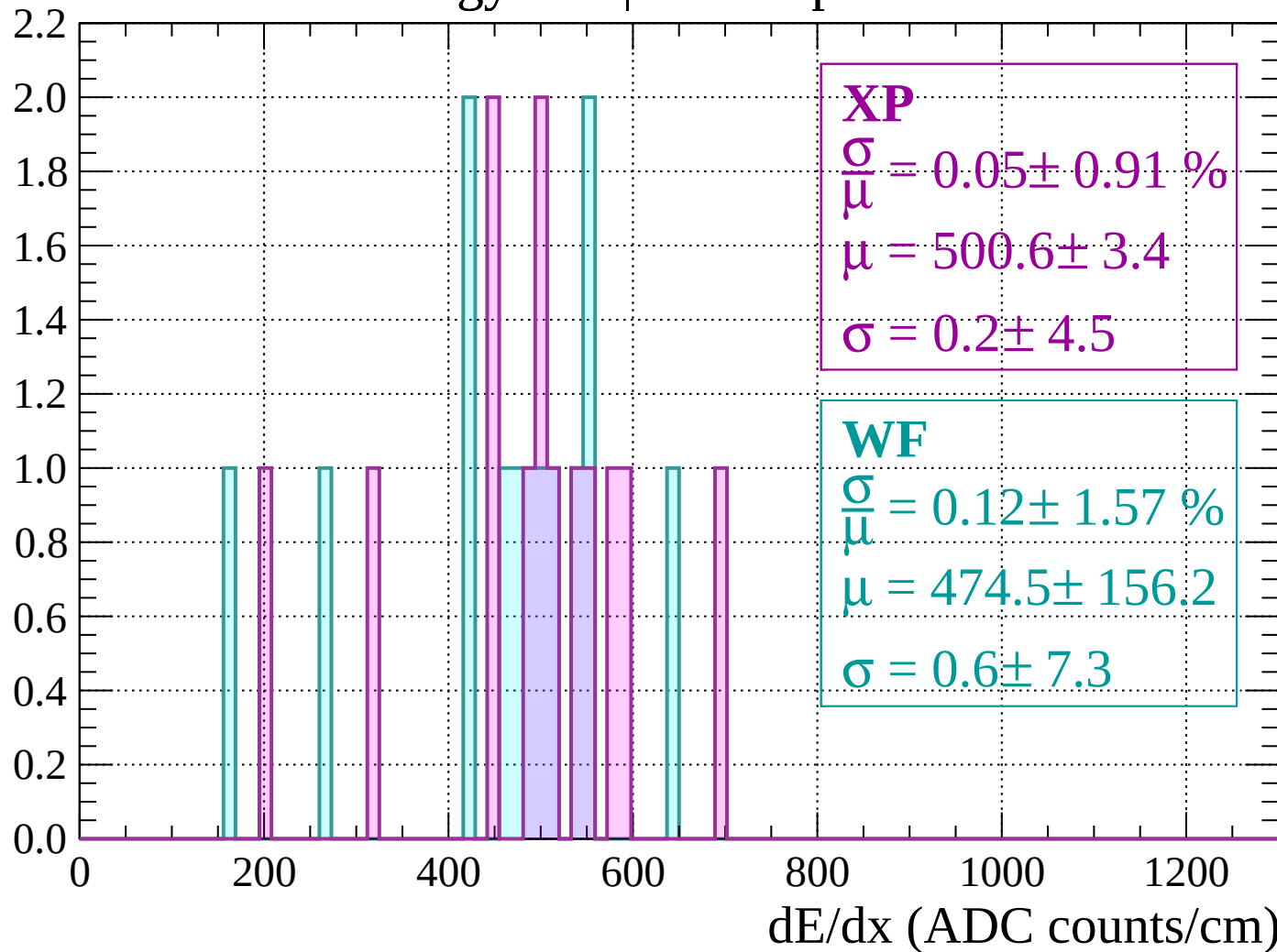
Energy loss | $1520 < p < 1560$

Count



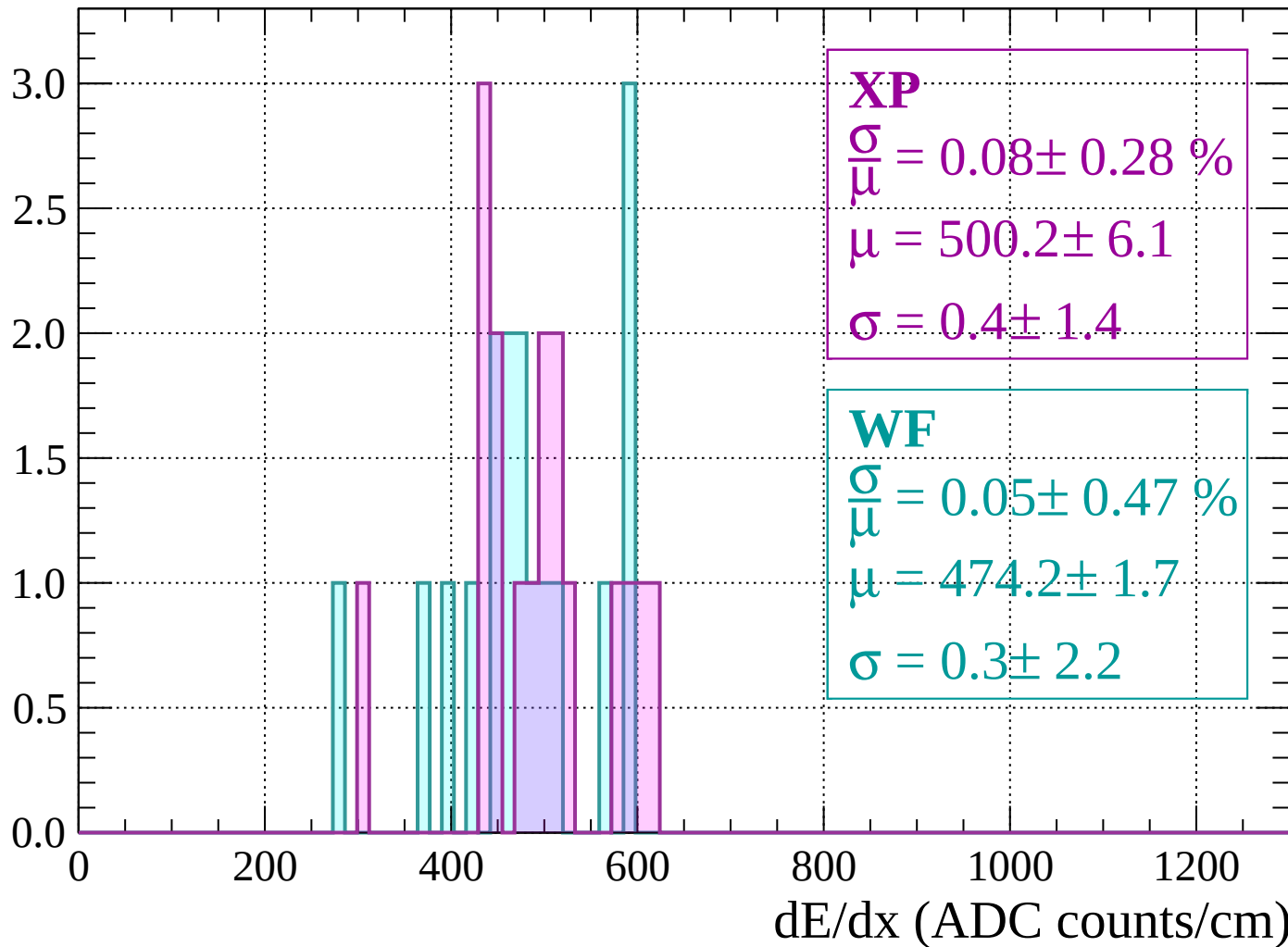
Energy loss | $1560 < p < 1600$

Count



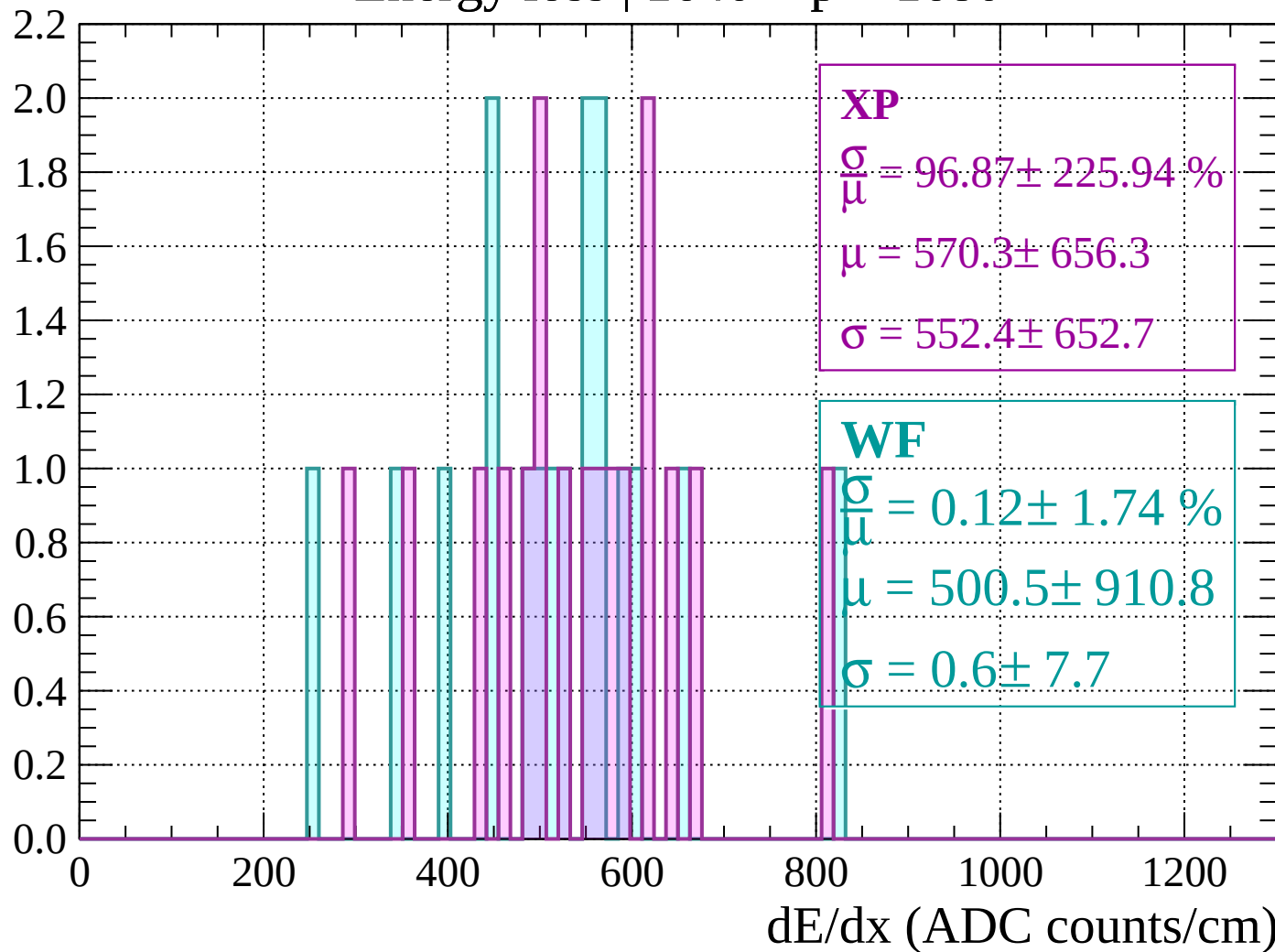
Energy loss | $1600 < p < 1640$

Count



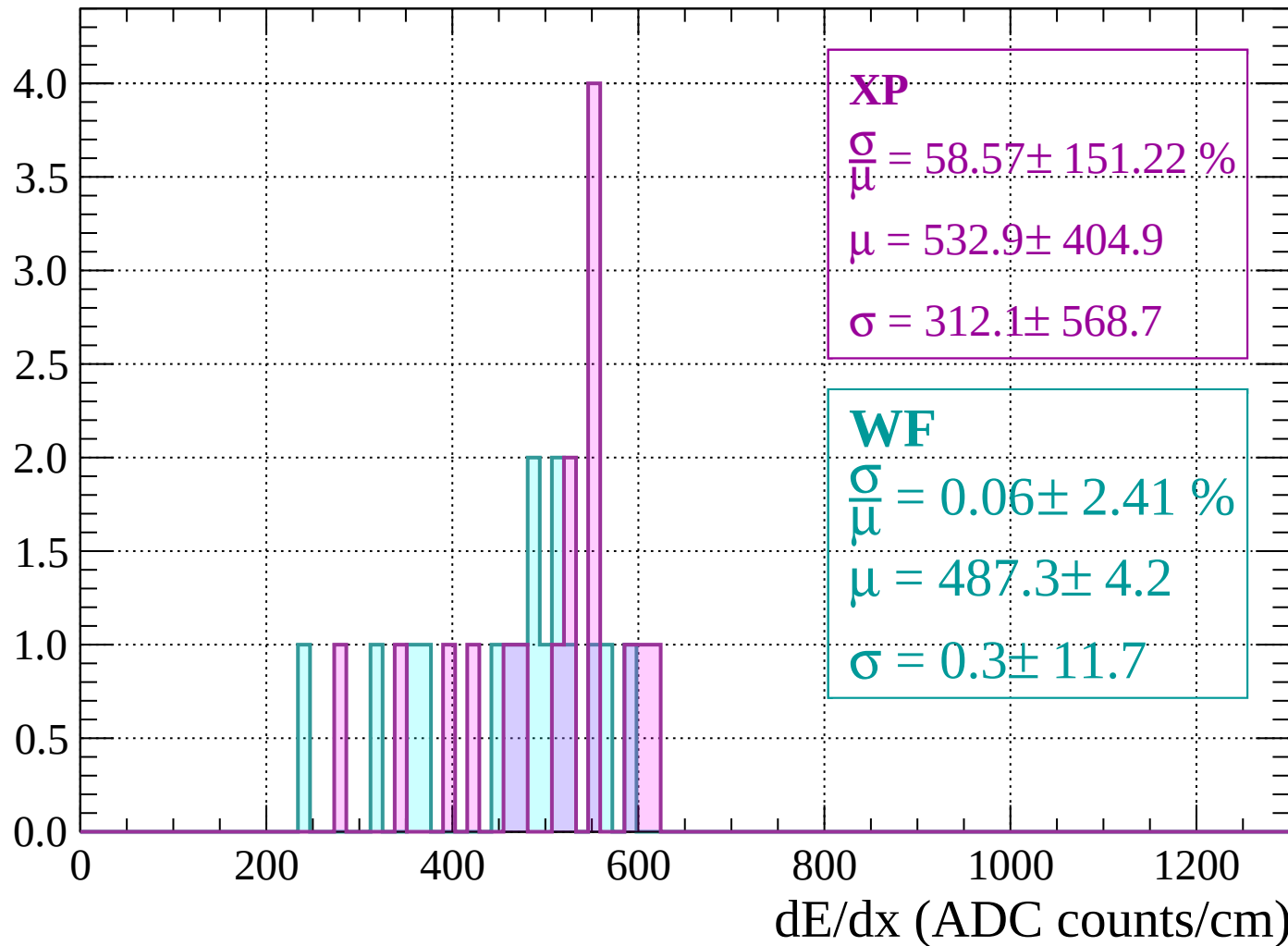
Energy loss | $1640 < p < 1680$

Count



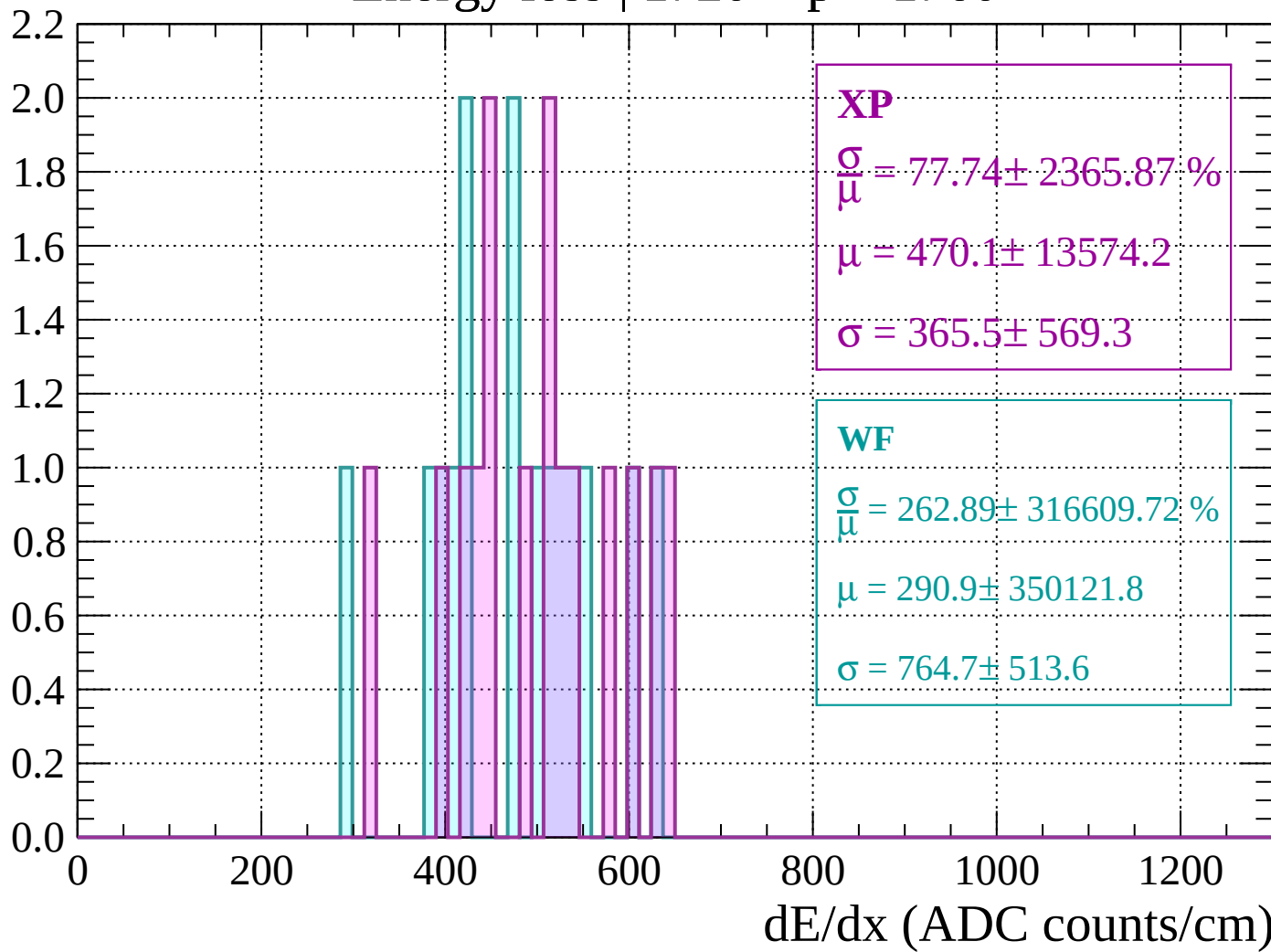
Energy loss | $1680 < p < 1720$

Count



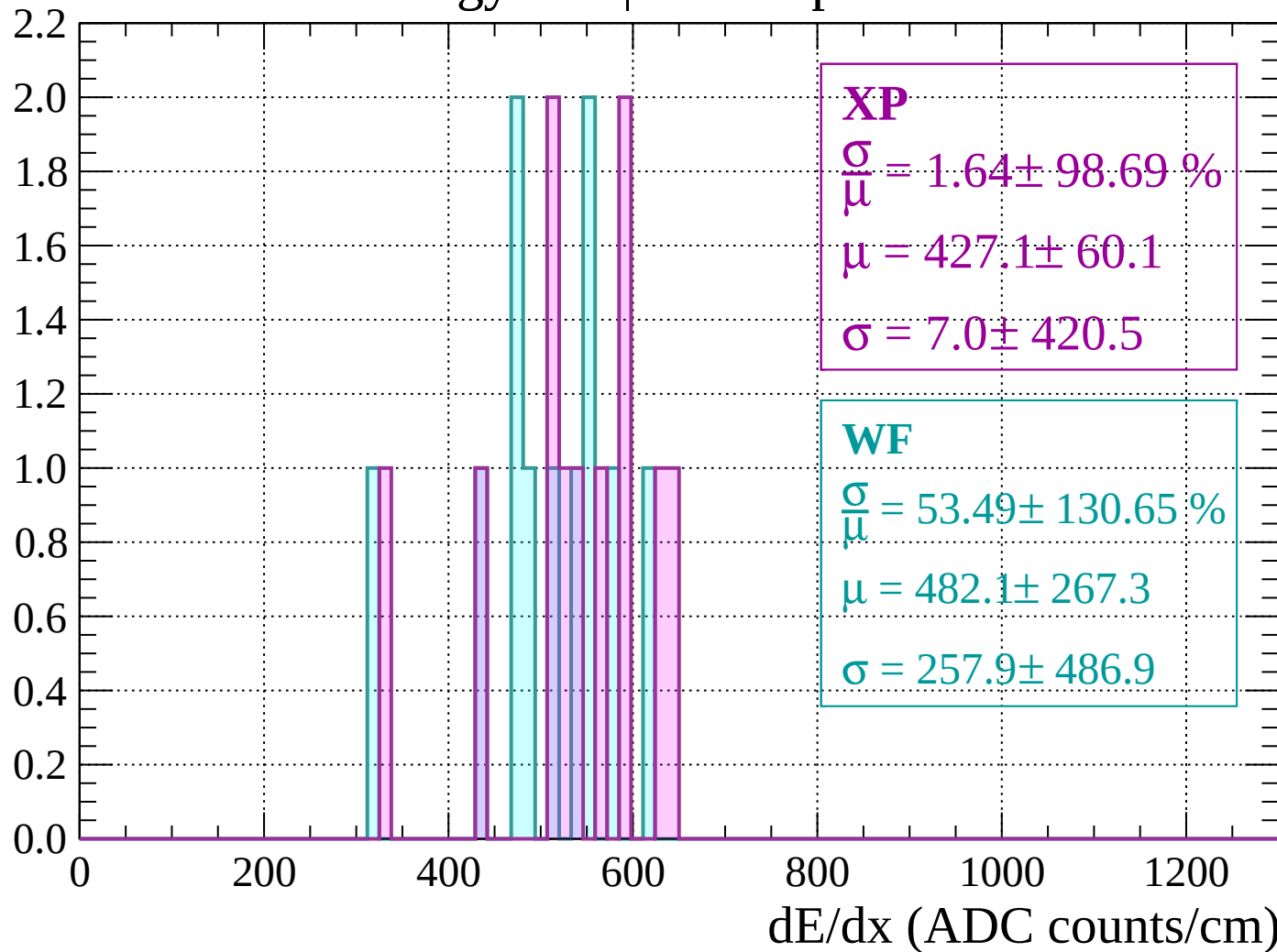
Energy loss | $1720 < p < 1760$

Count



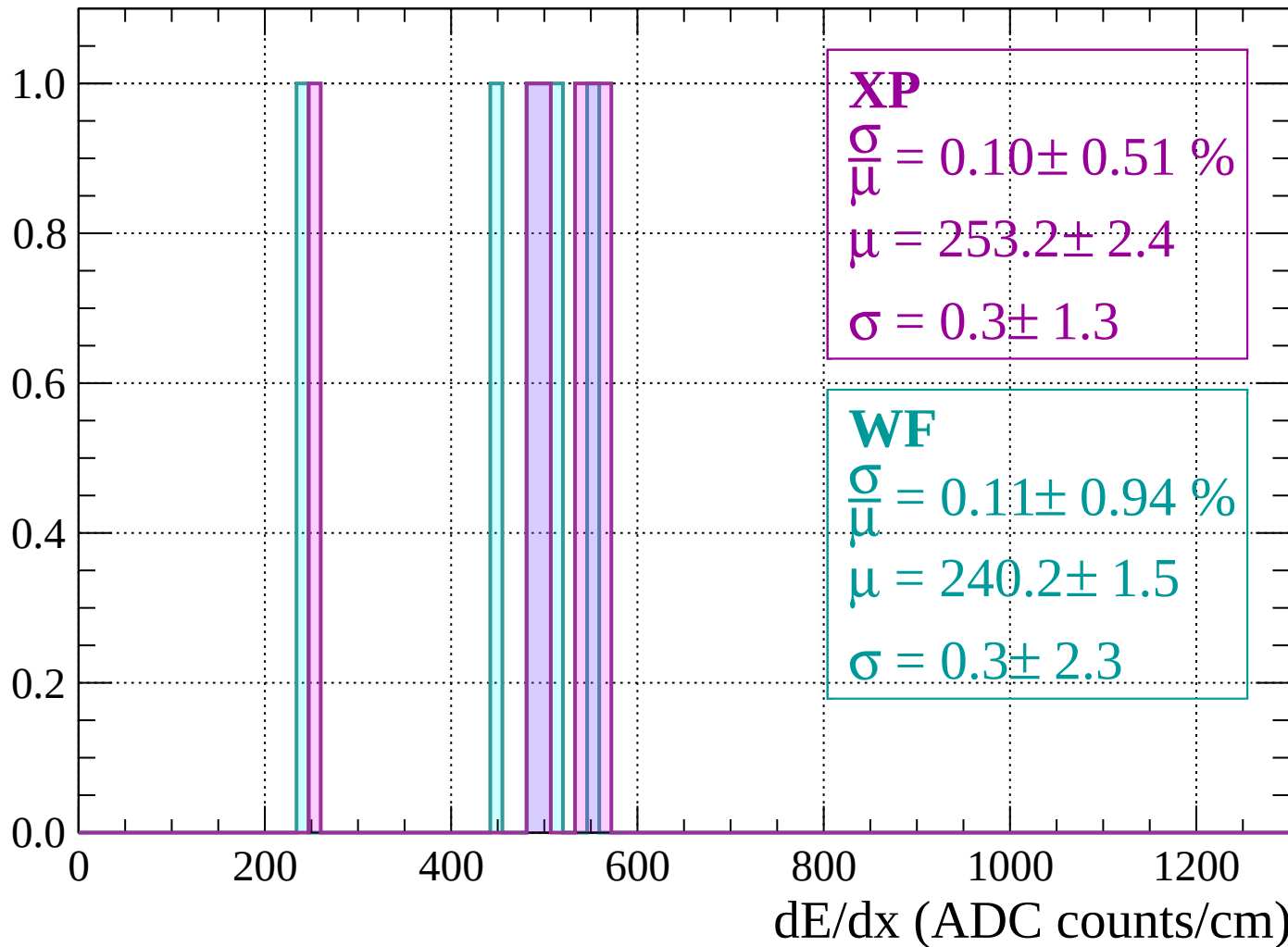
Energy loss | $1760 < p < 1800$

Count



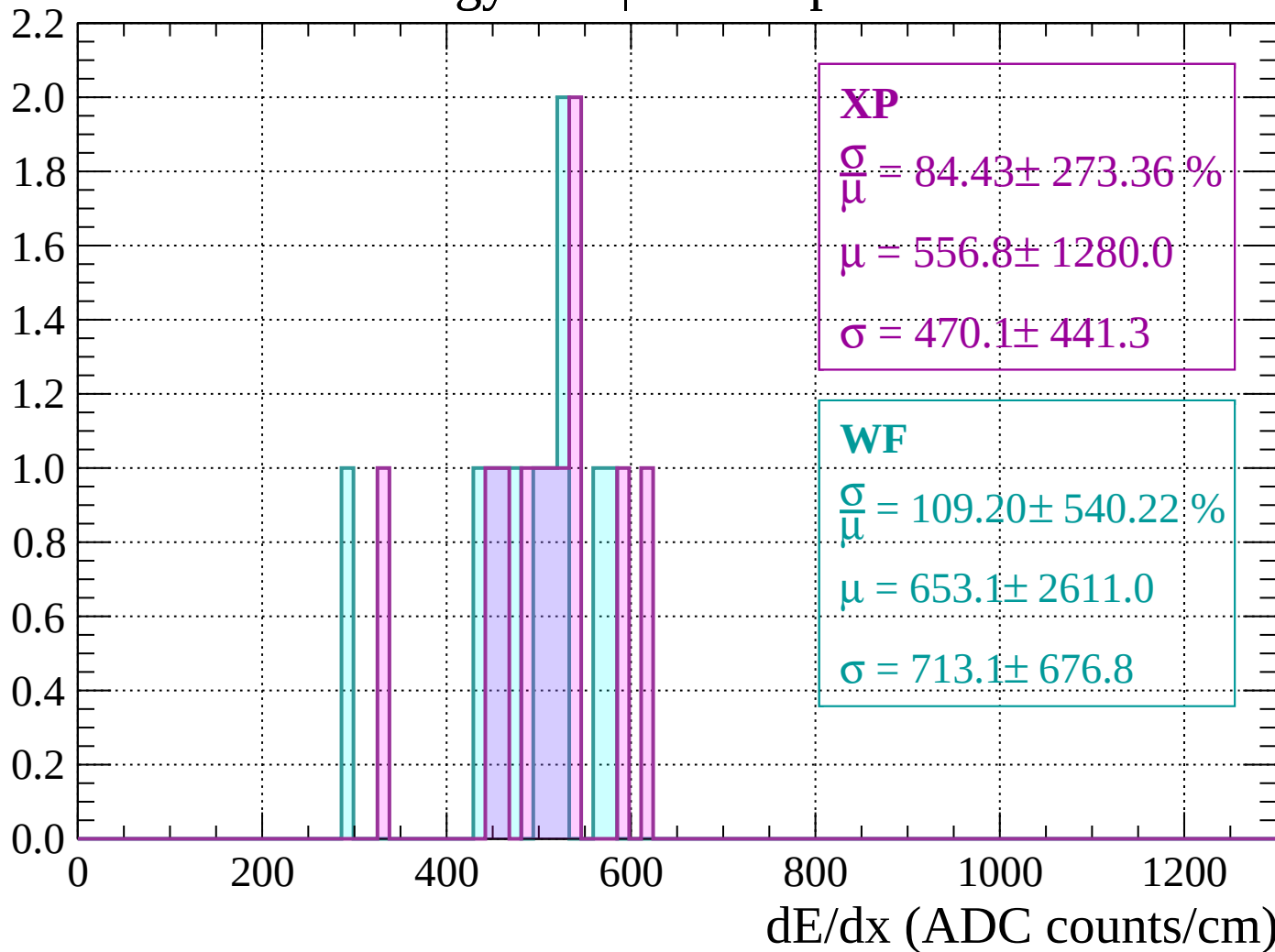
Energy loss | $1800 < p < 1840$

Count



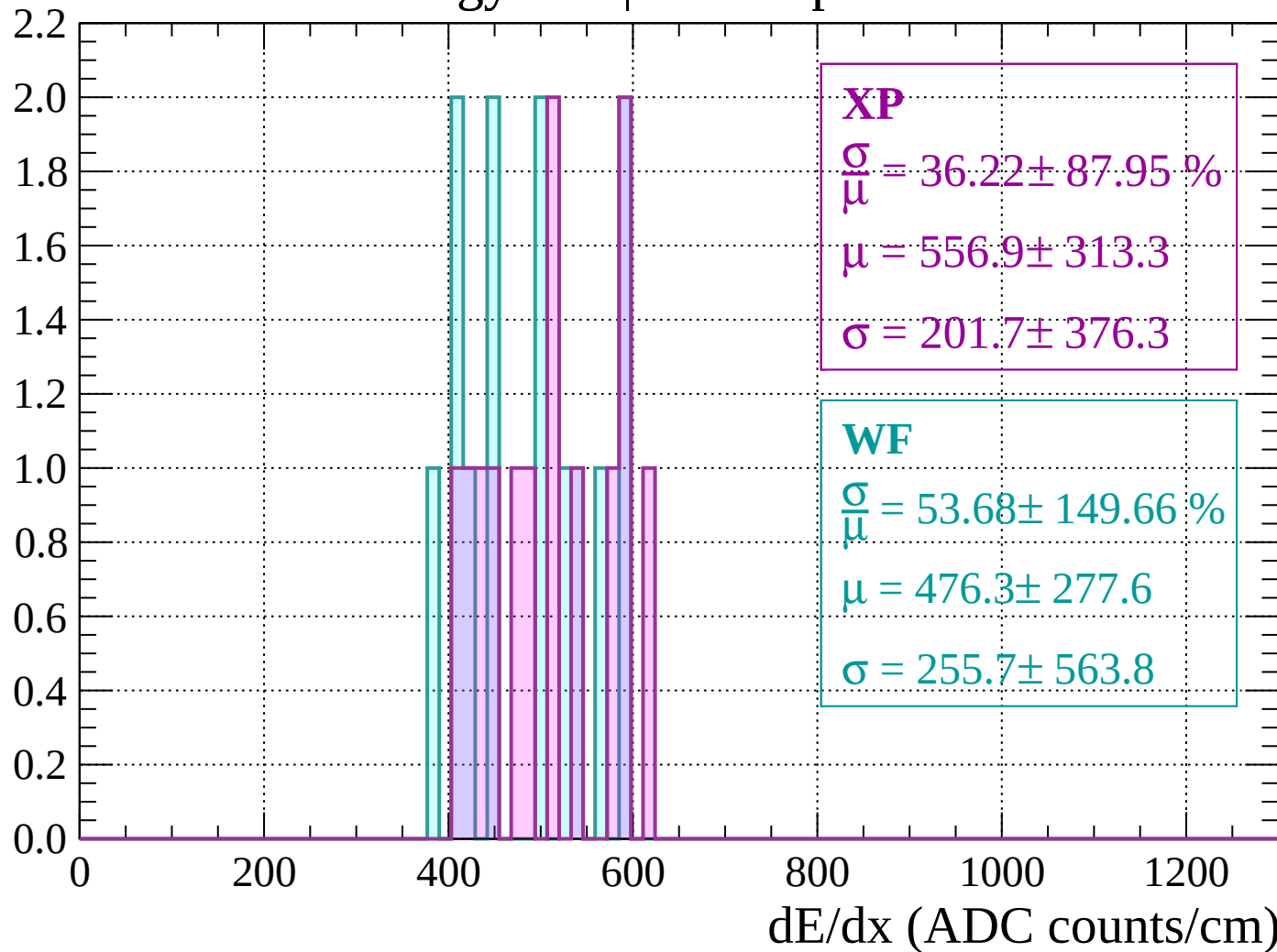
Energy loss | $1840 < p < 1880$

Count



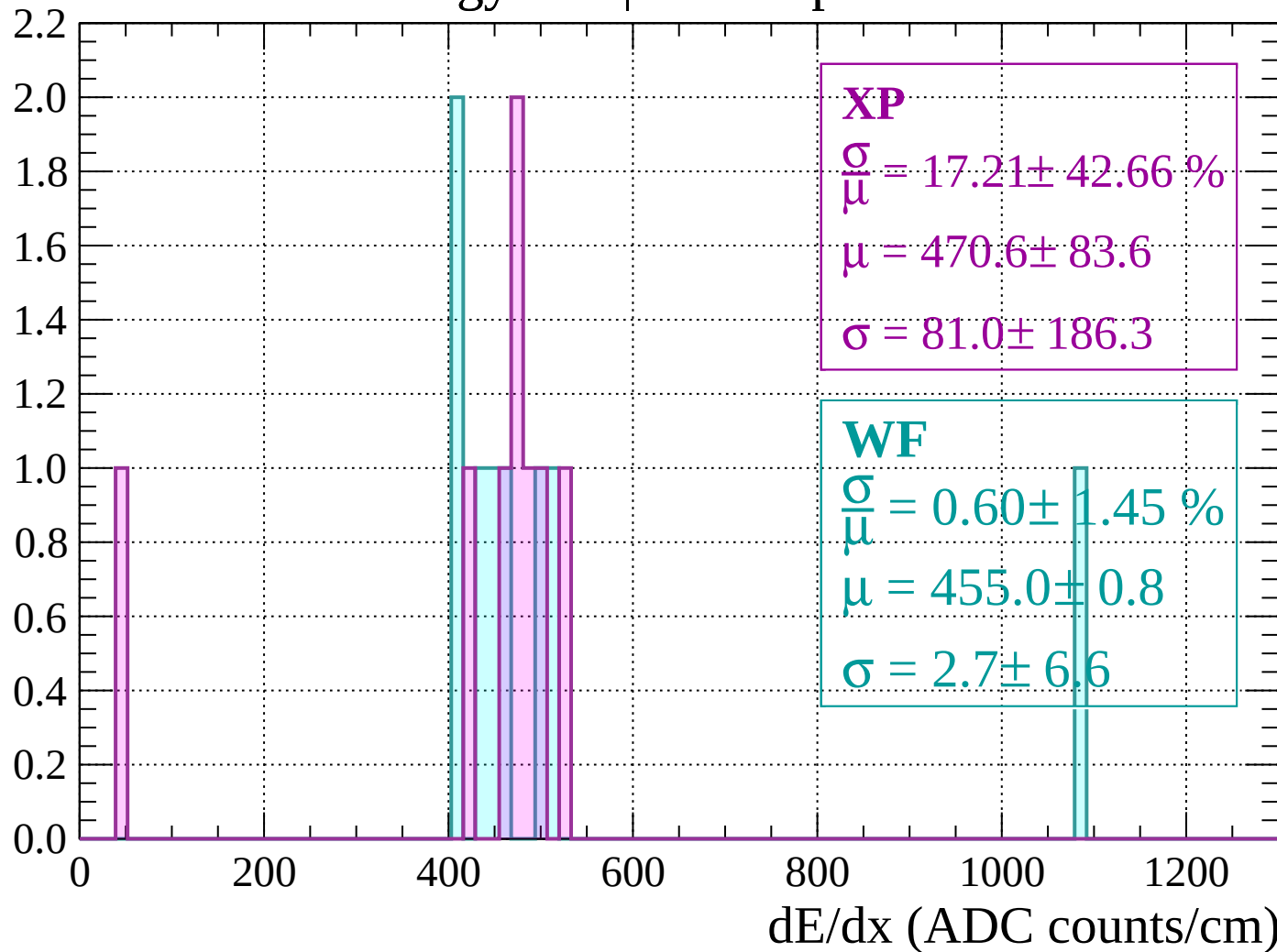
Energy loss | $1880 < p < 1920$

Count



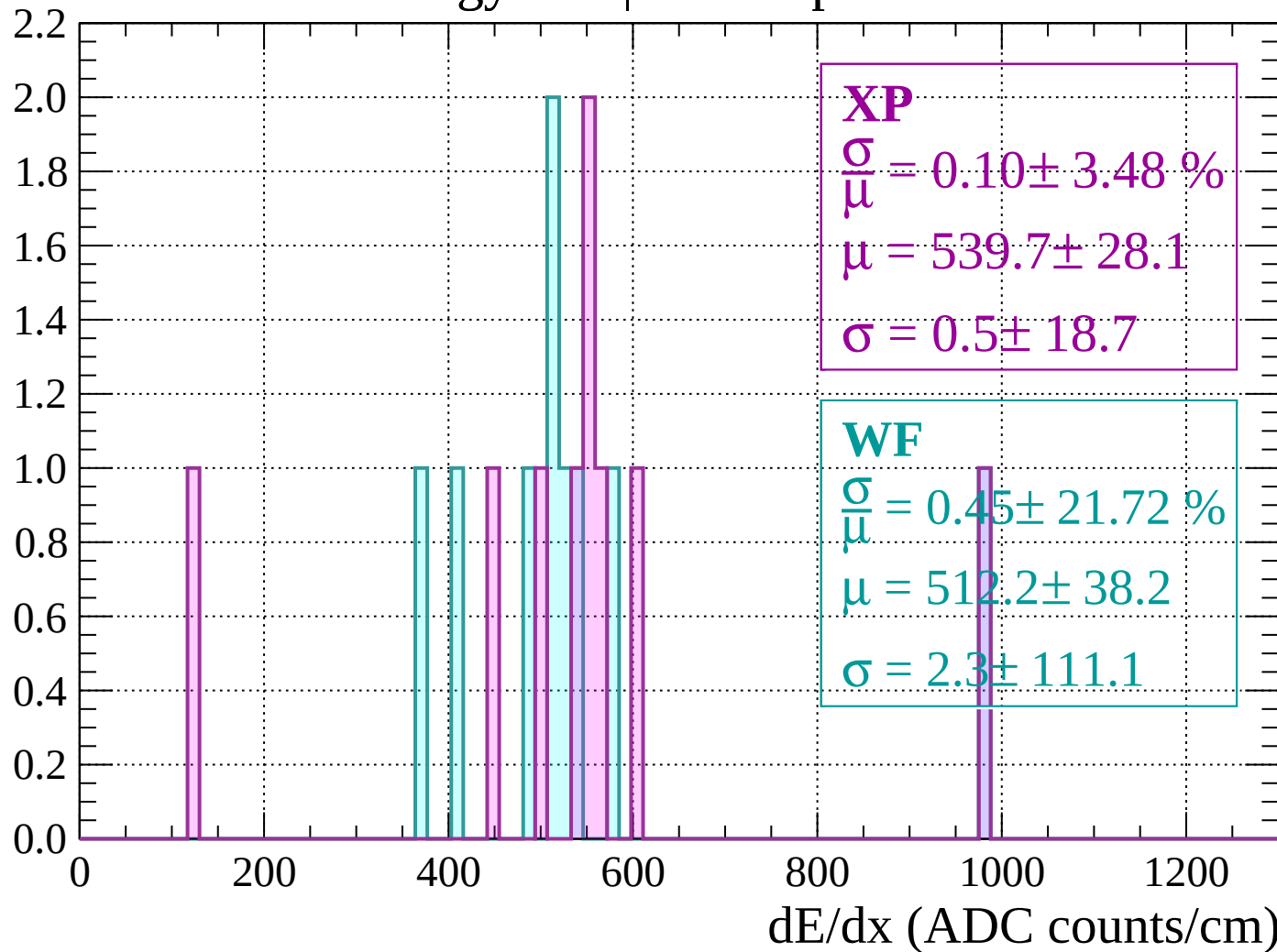
Energy loss | $1920 < p < 1960$

Count



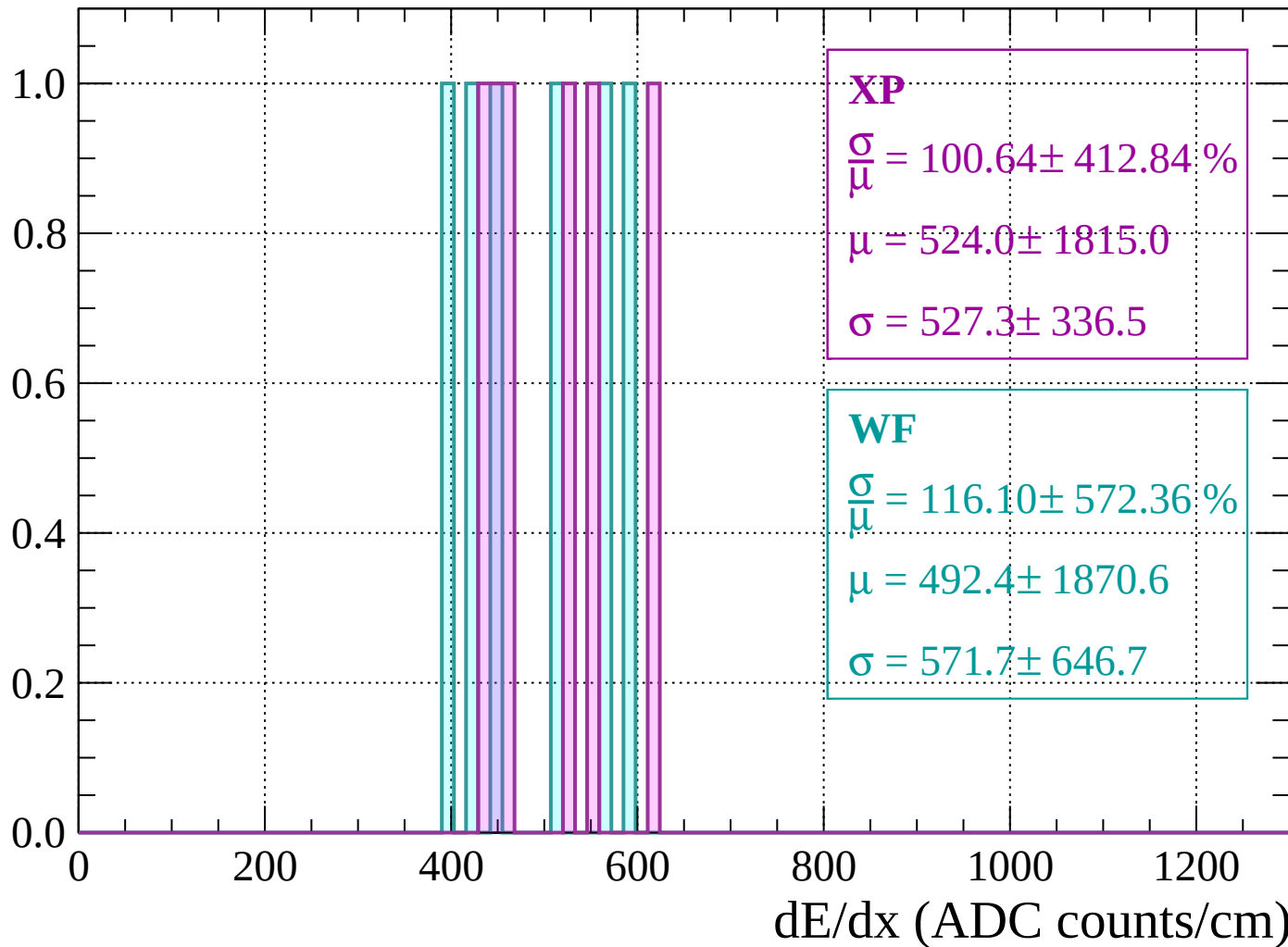
Energy loss | 1960 < p < 2000

Count



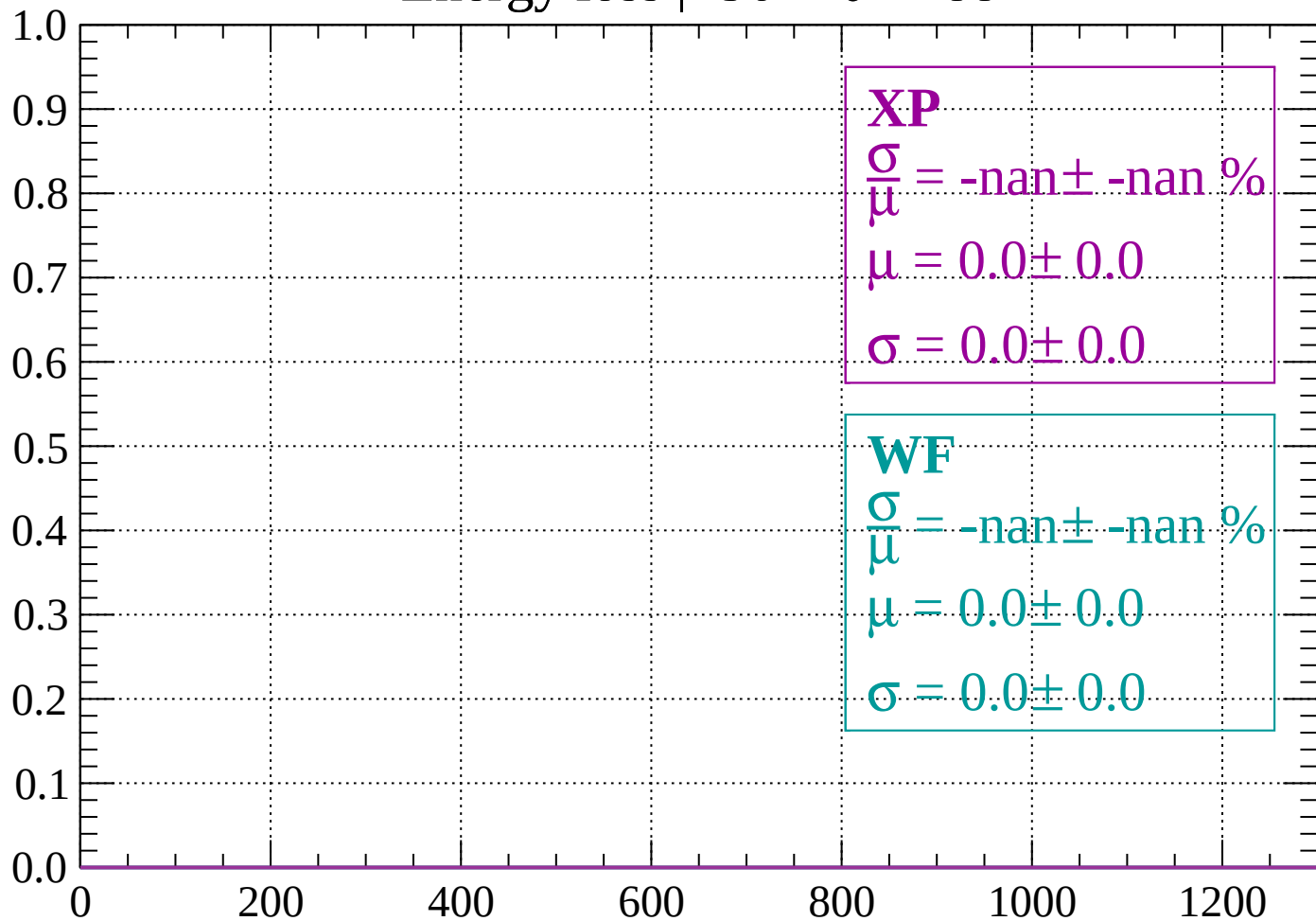
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

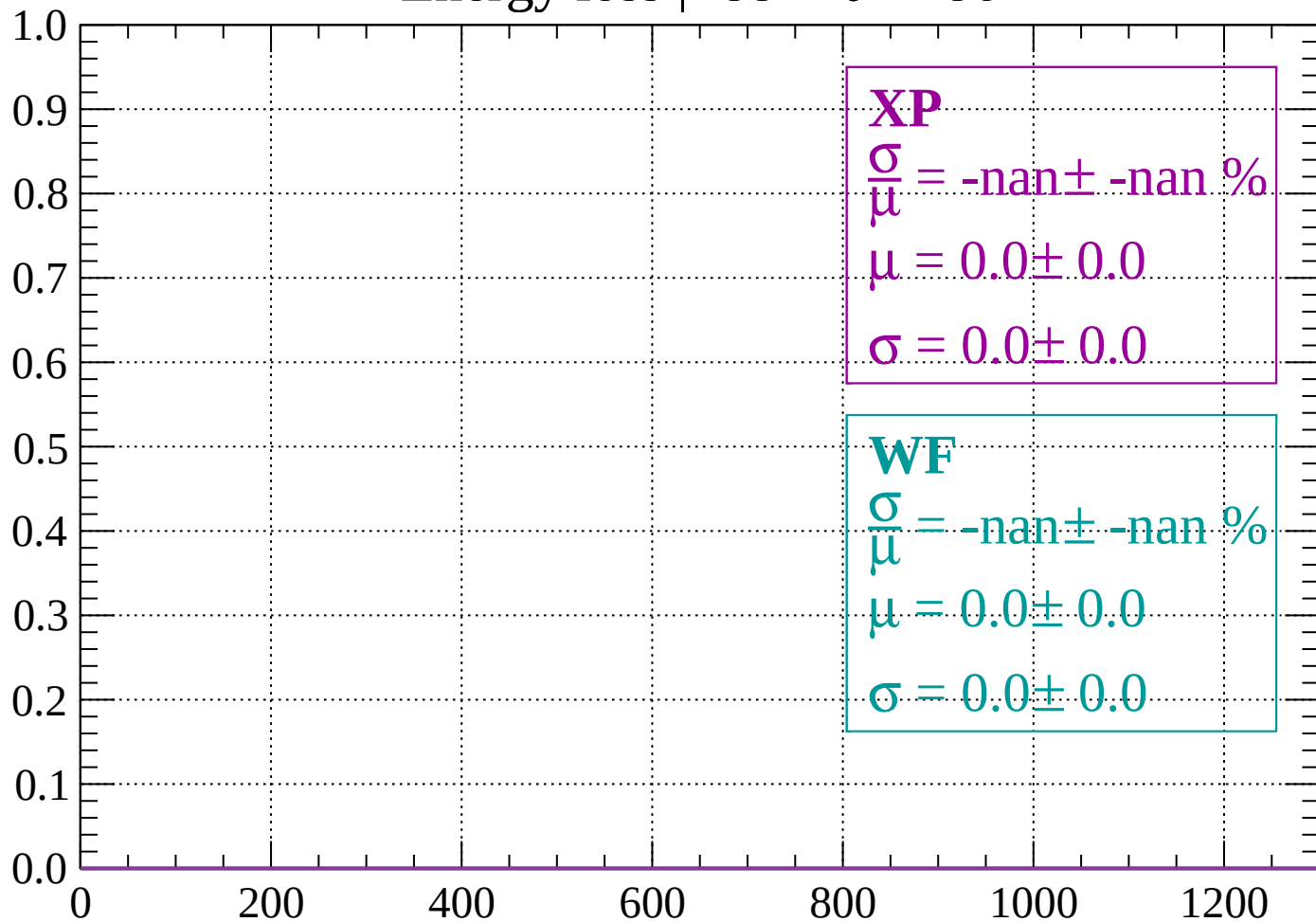
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

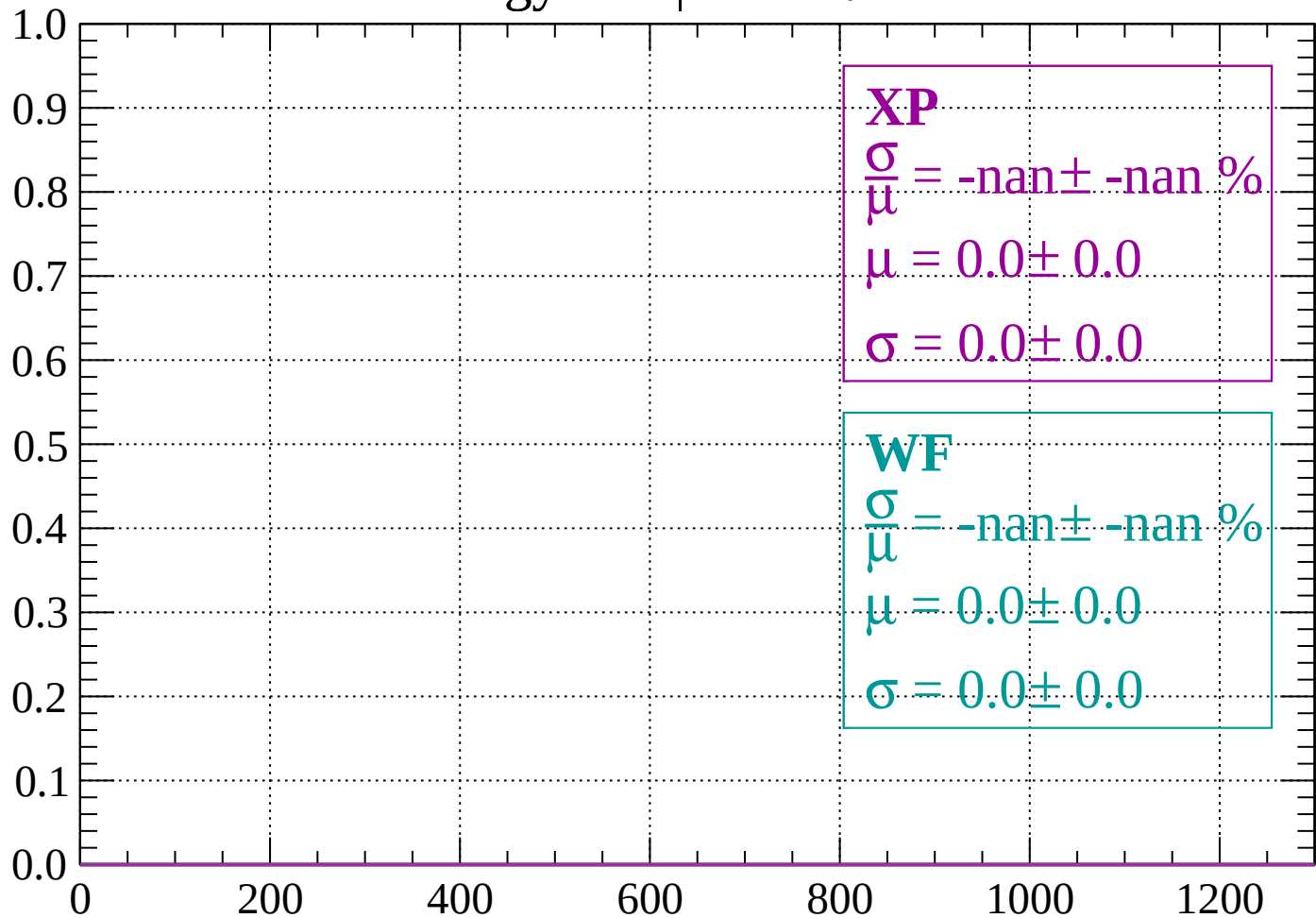
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

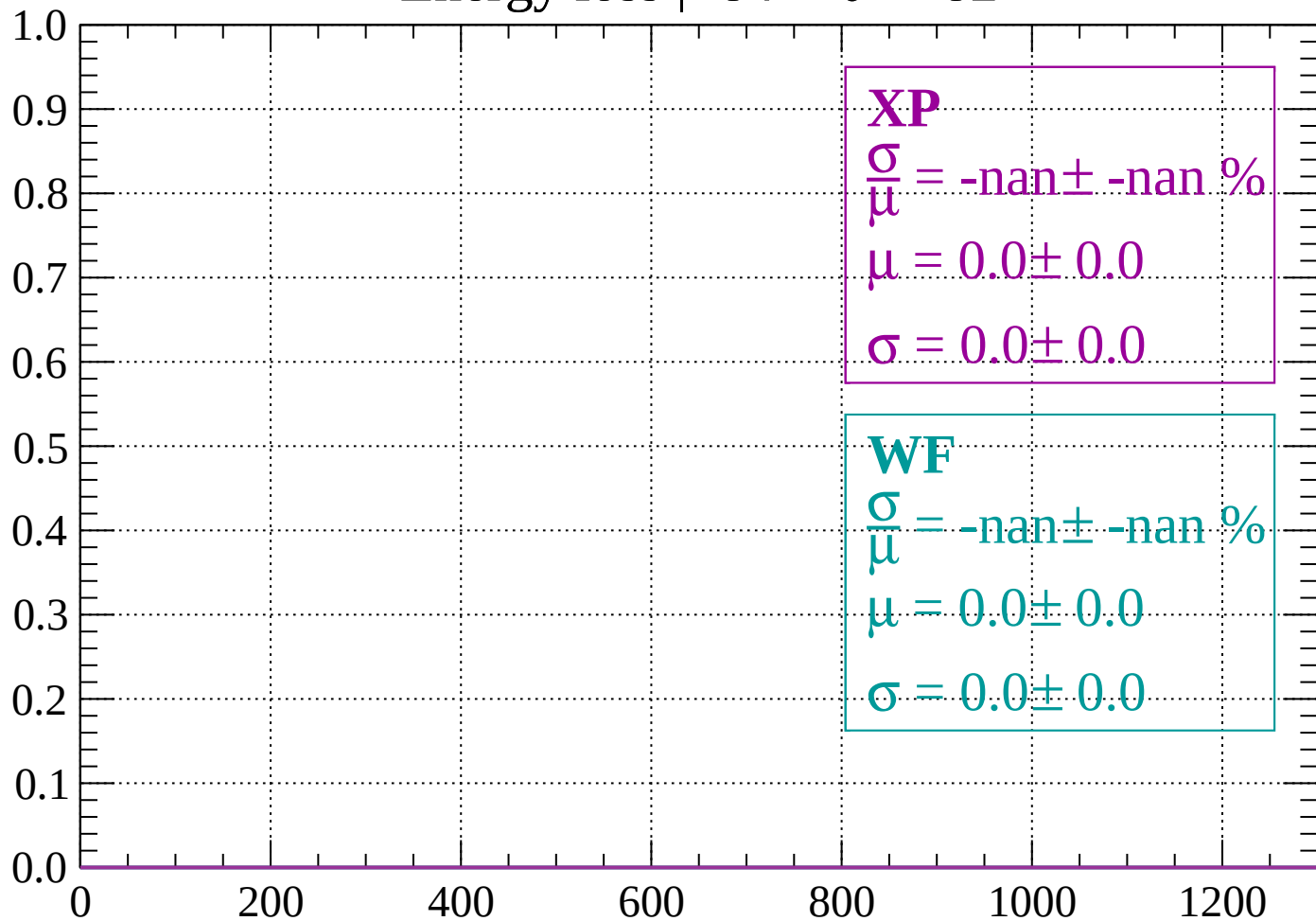
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

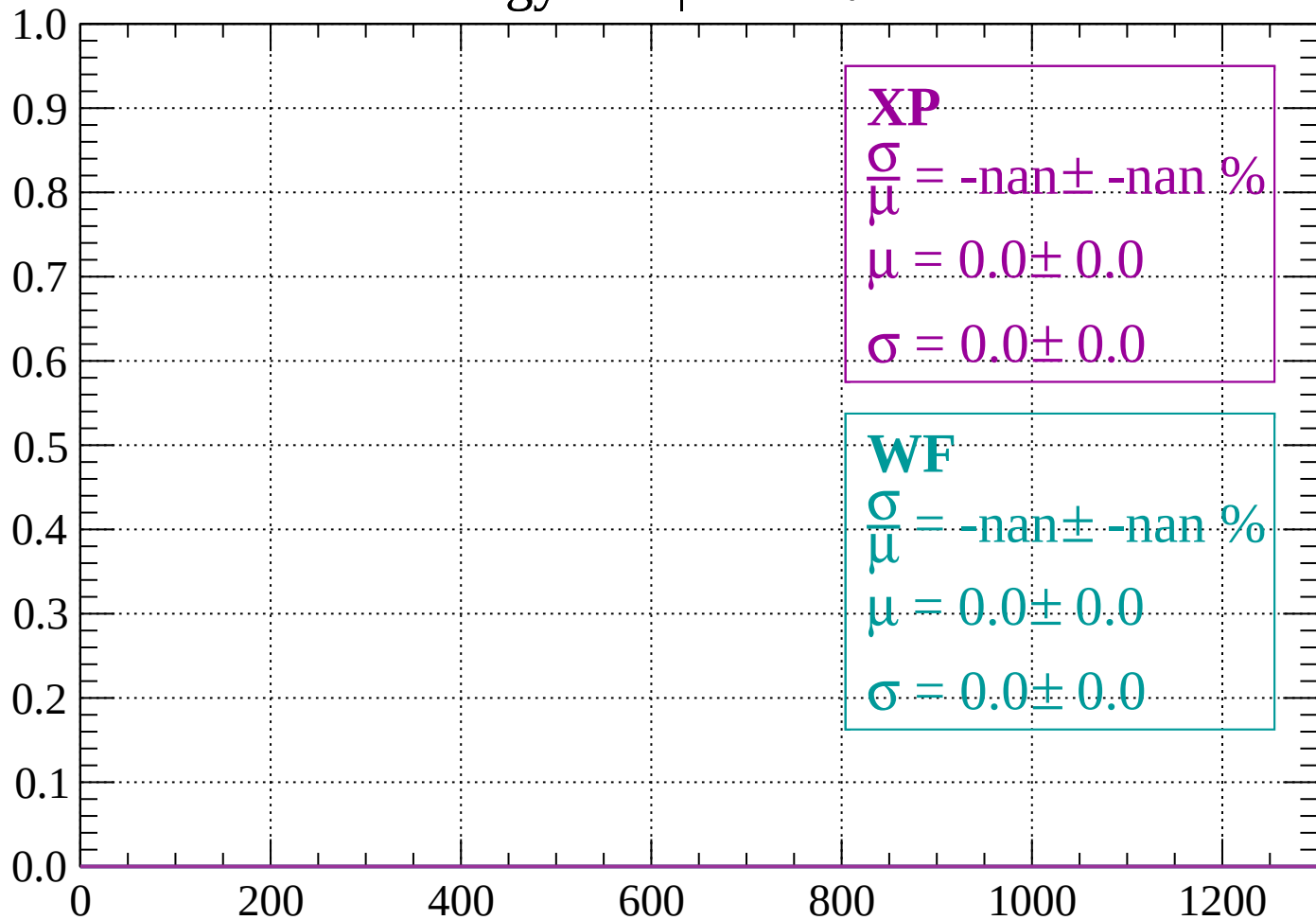
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

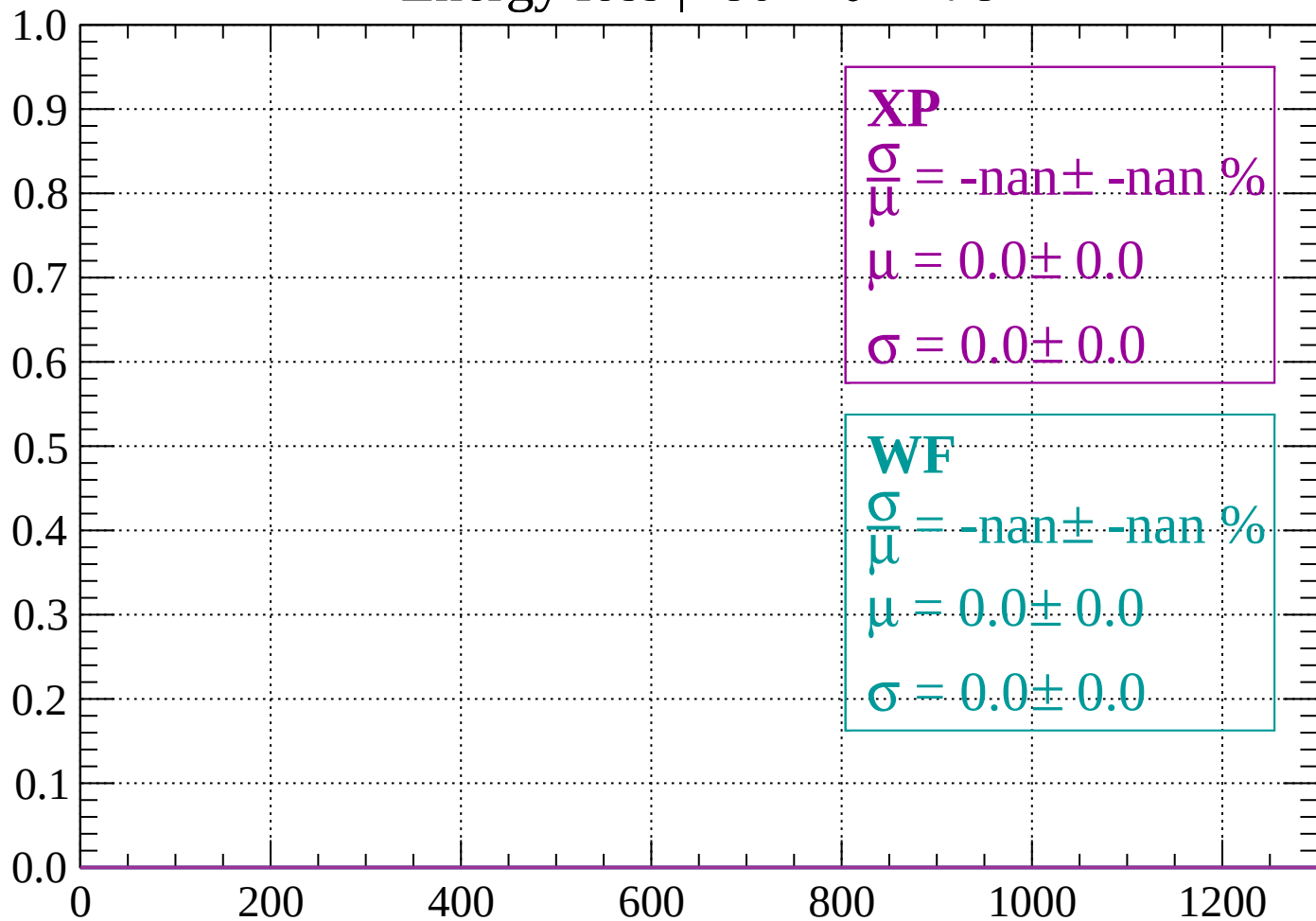
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

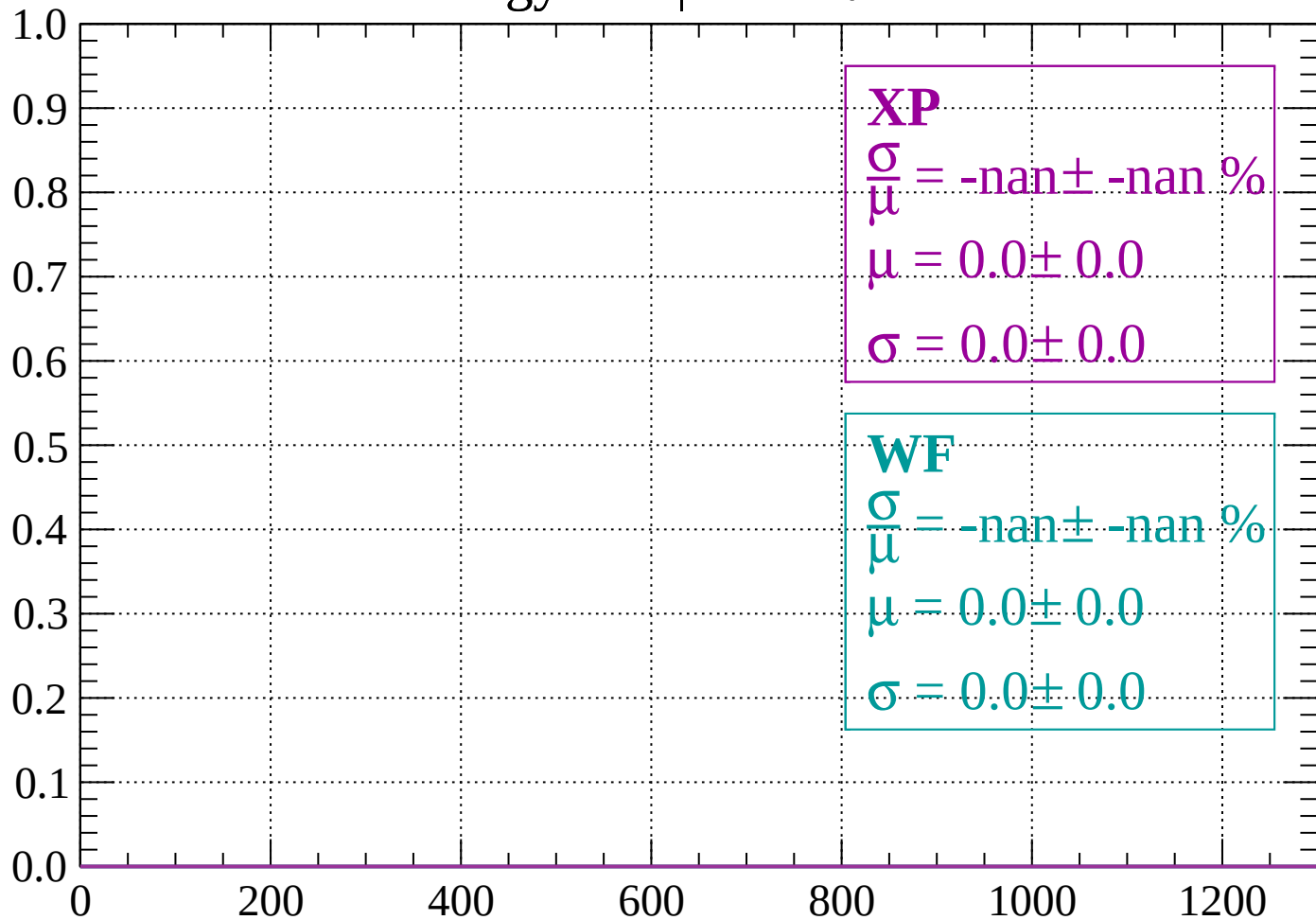
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

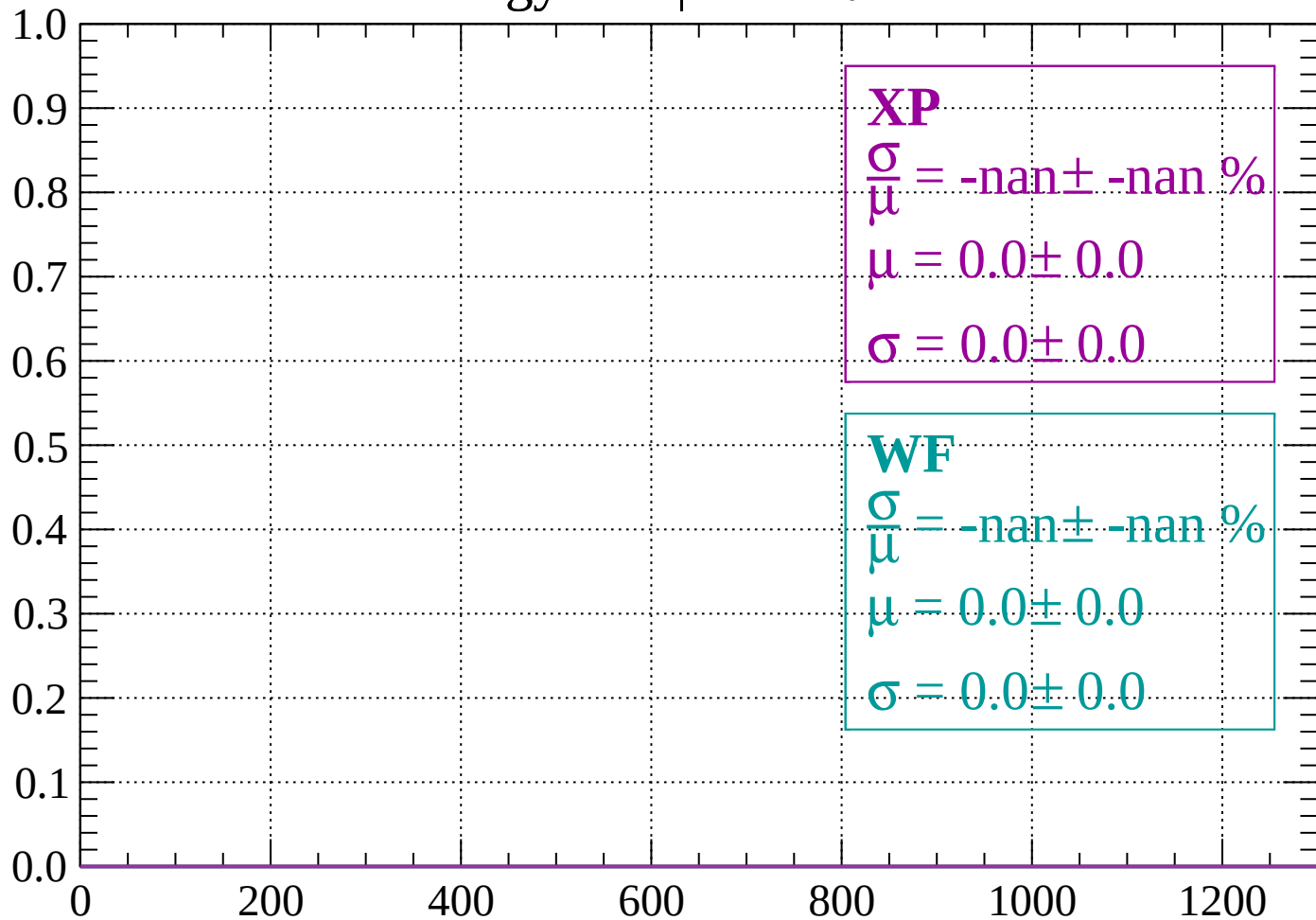
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

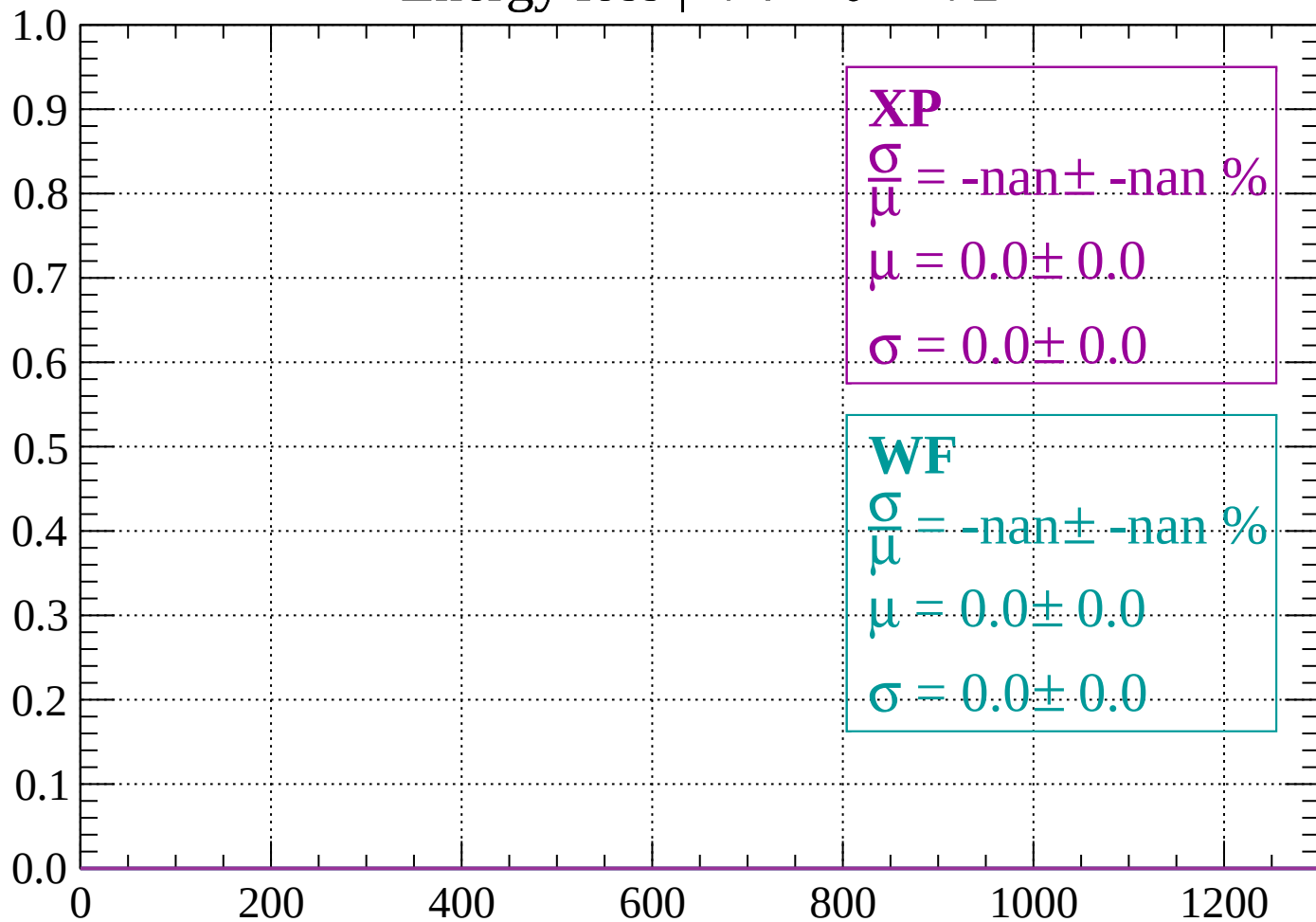
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

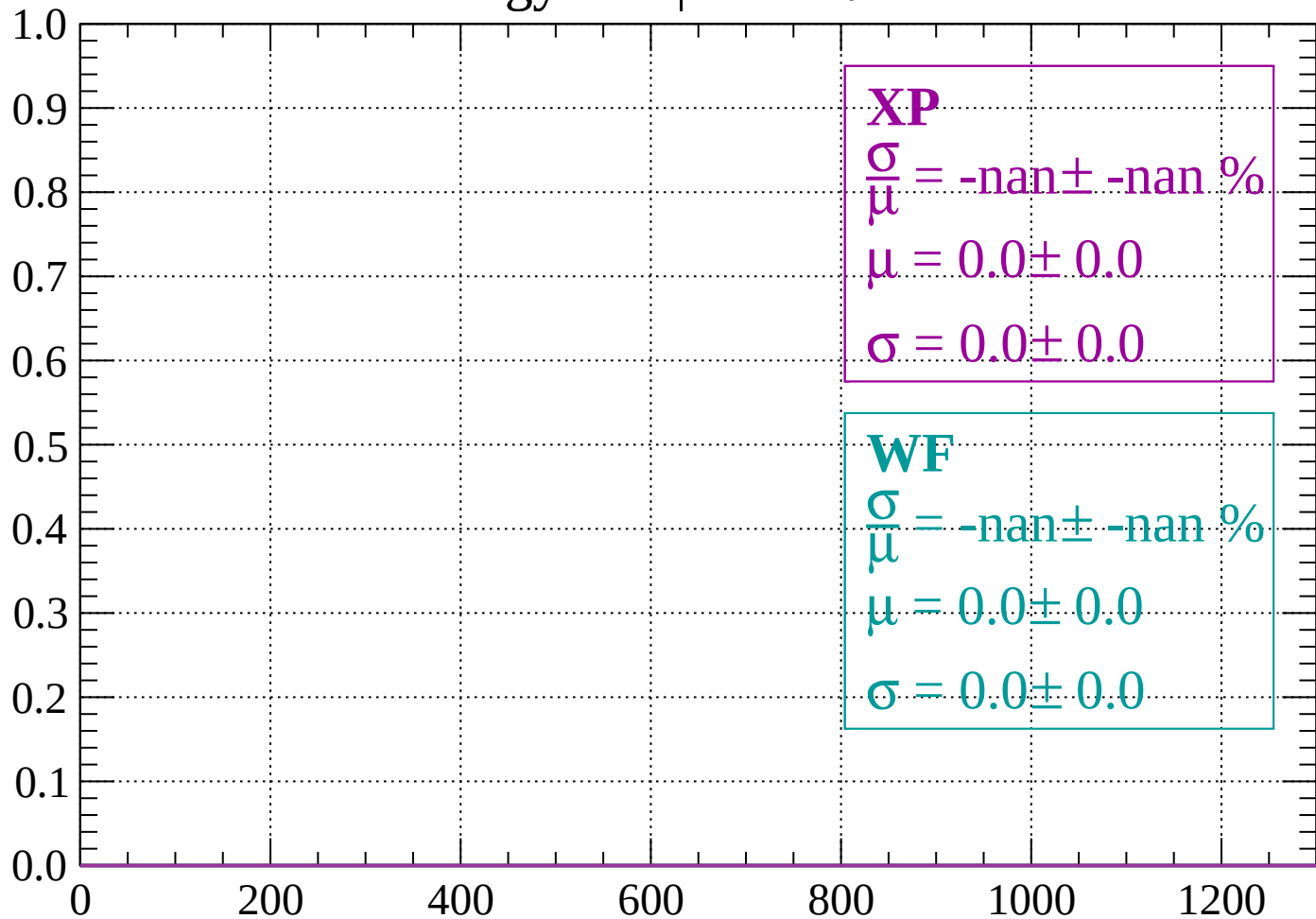
Count



dE/dx (ADC counts/cm)

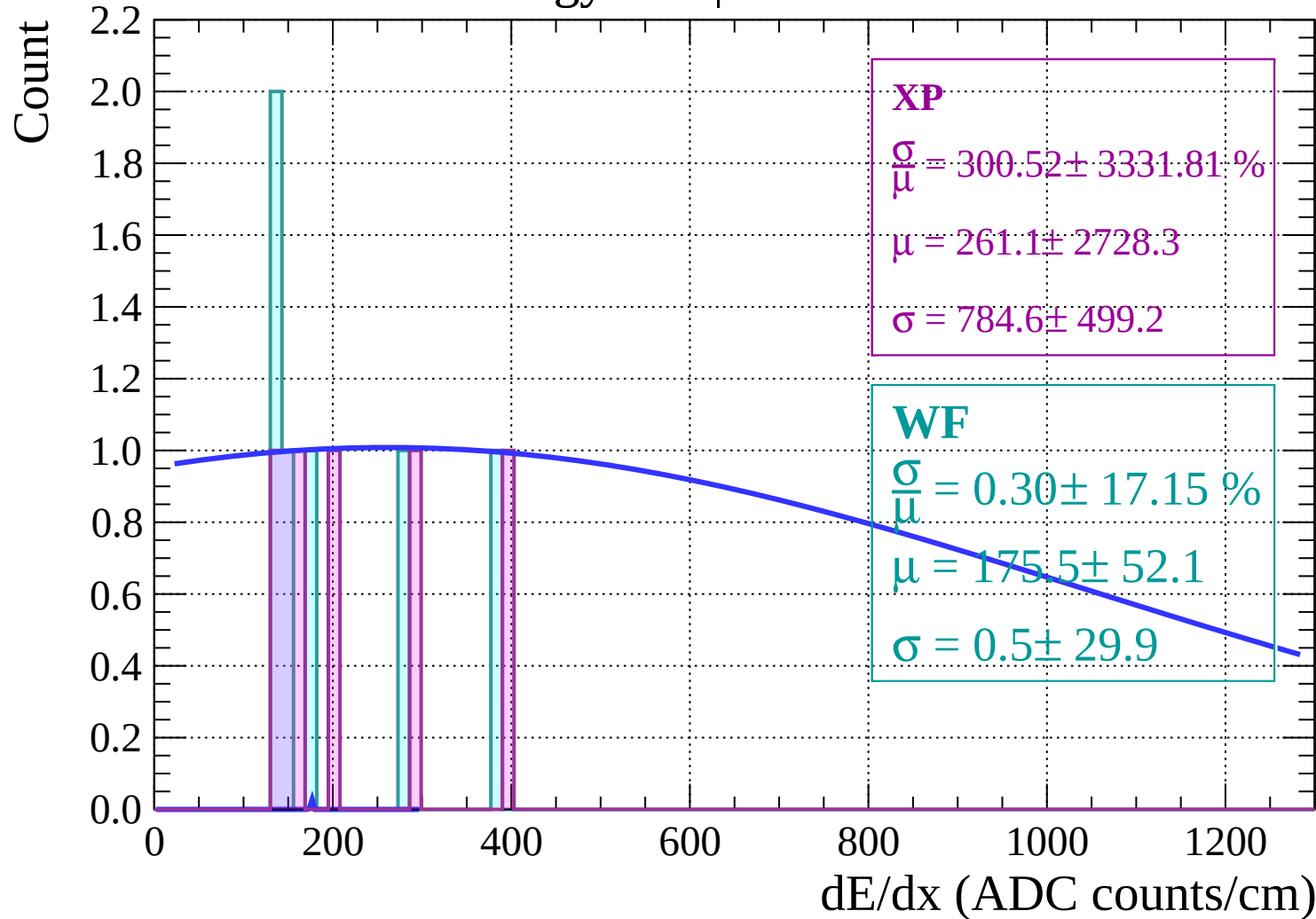
Energy loss | $-72 < \theta < -70$

Count



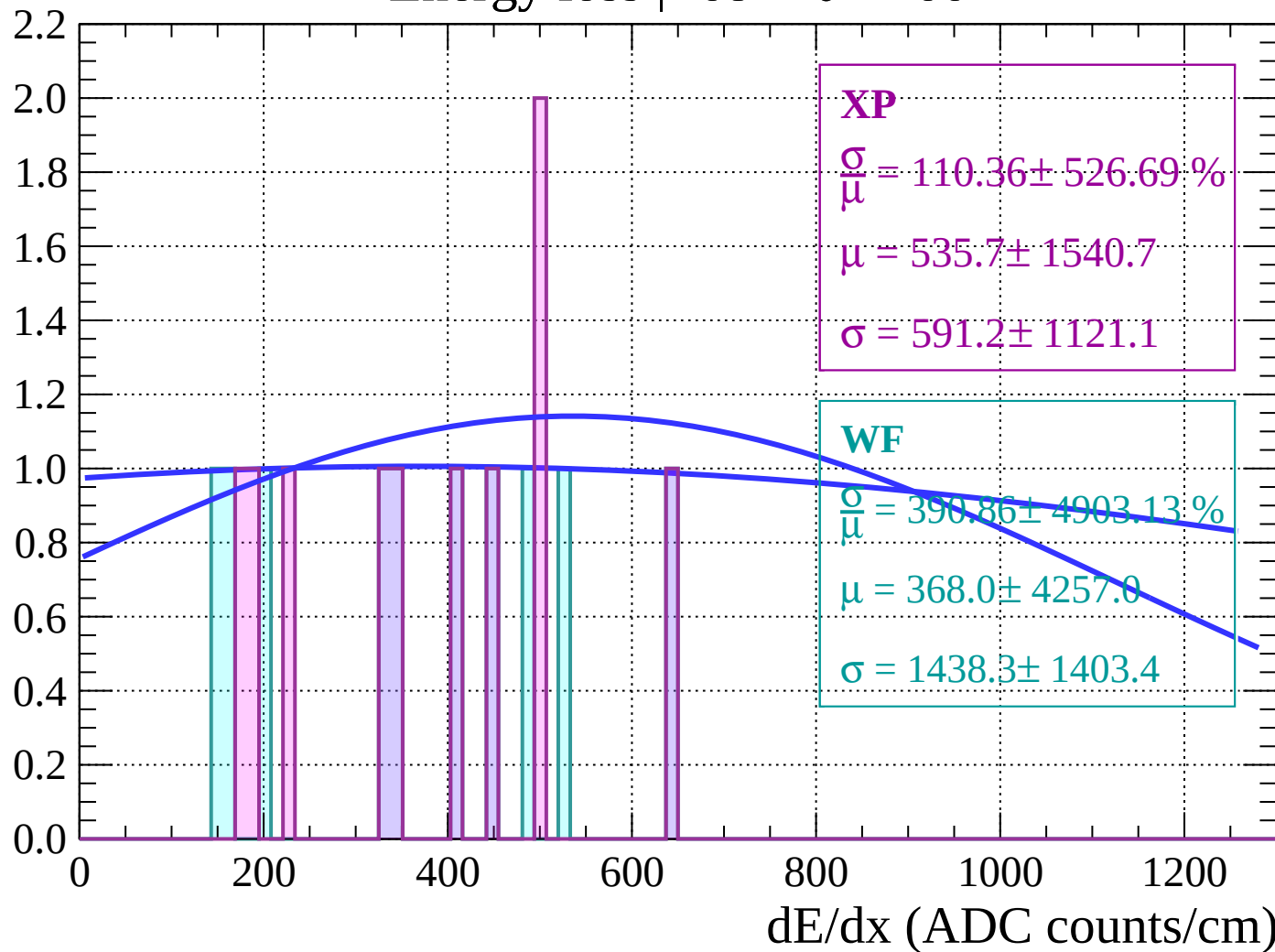
dE/dx (ADC counts/cm)

Energy loss | $-70 < \theta < -68$



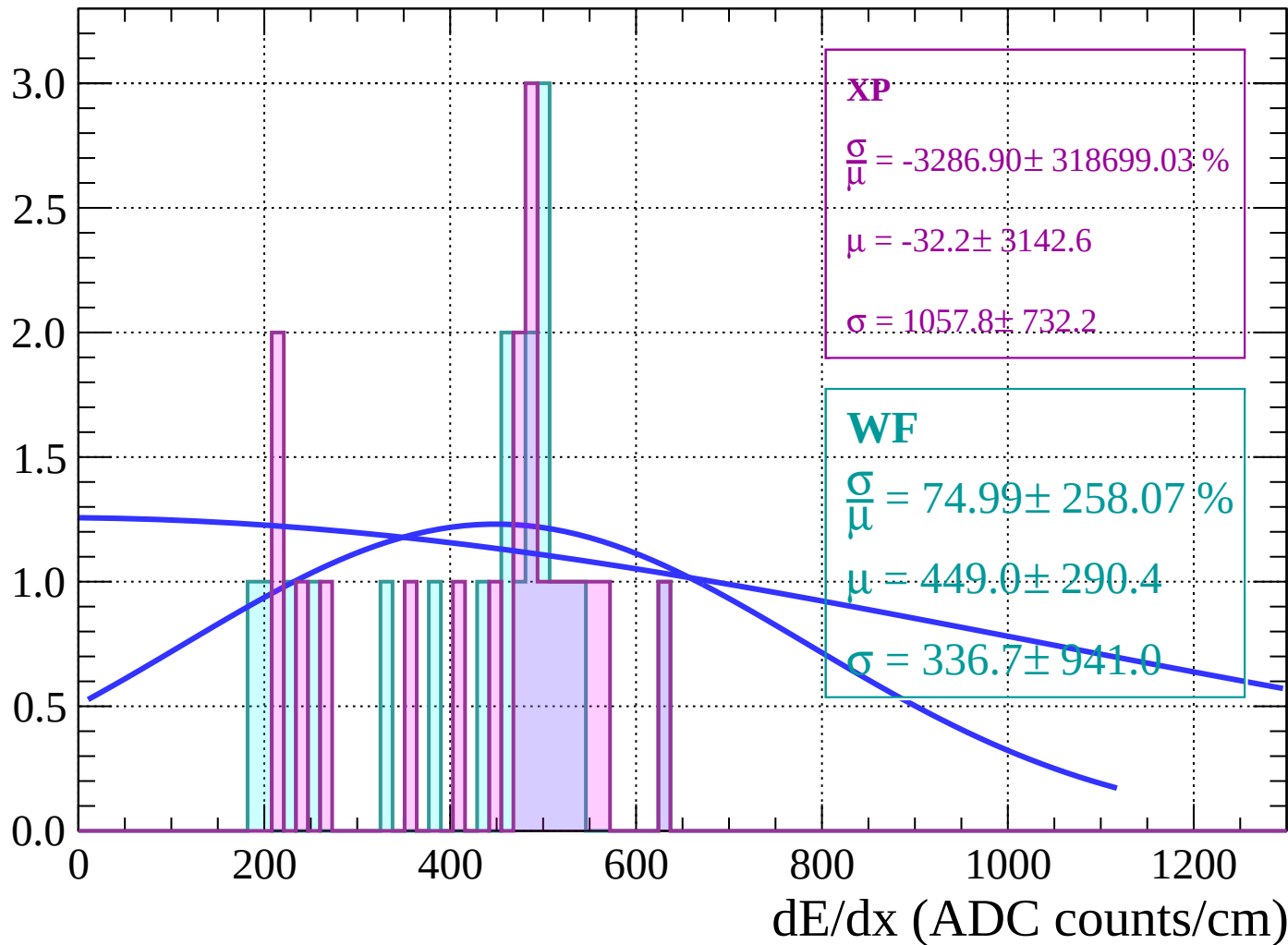
Energy loss | $-68 < \theta < -66$

Count



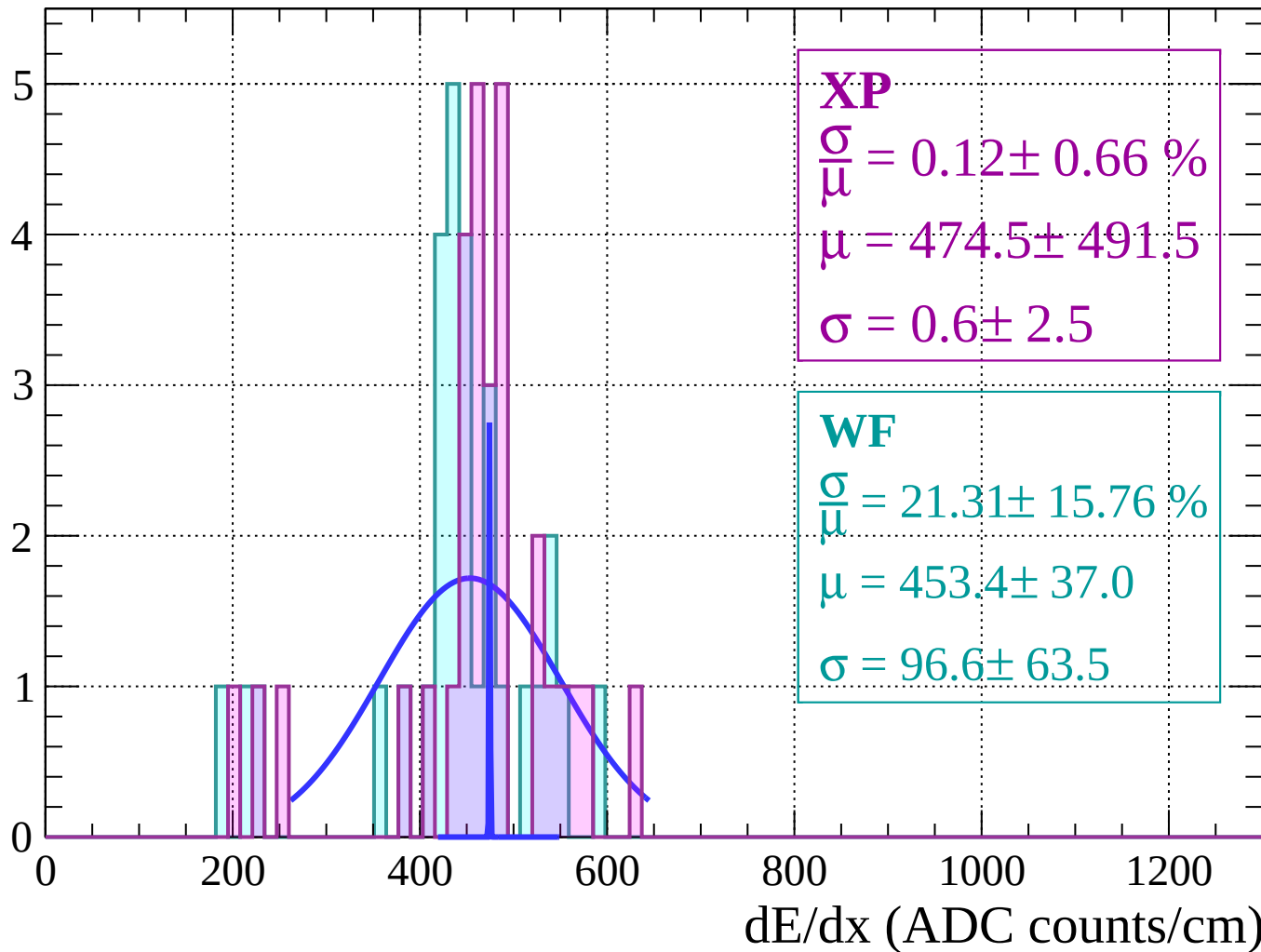
Energy loss | $-66 < \theta < -64$

Count



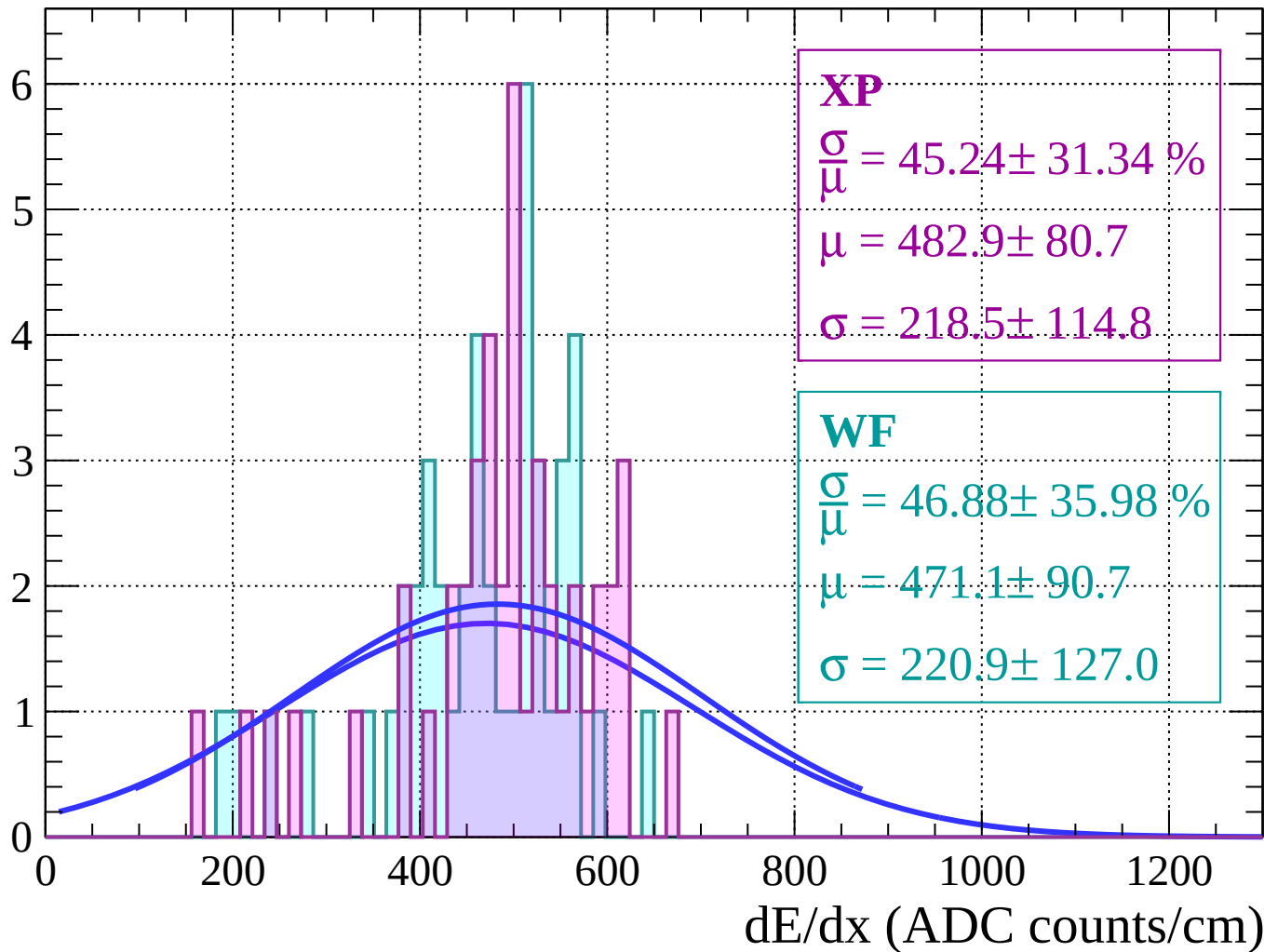
Energy loss $|-64 < \theta < -62$

Count



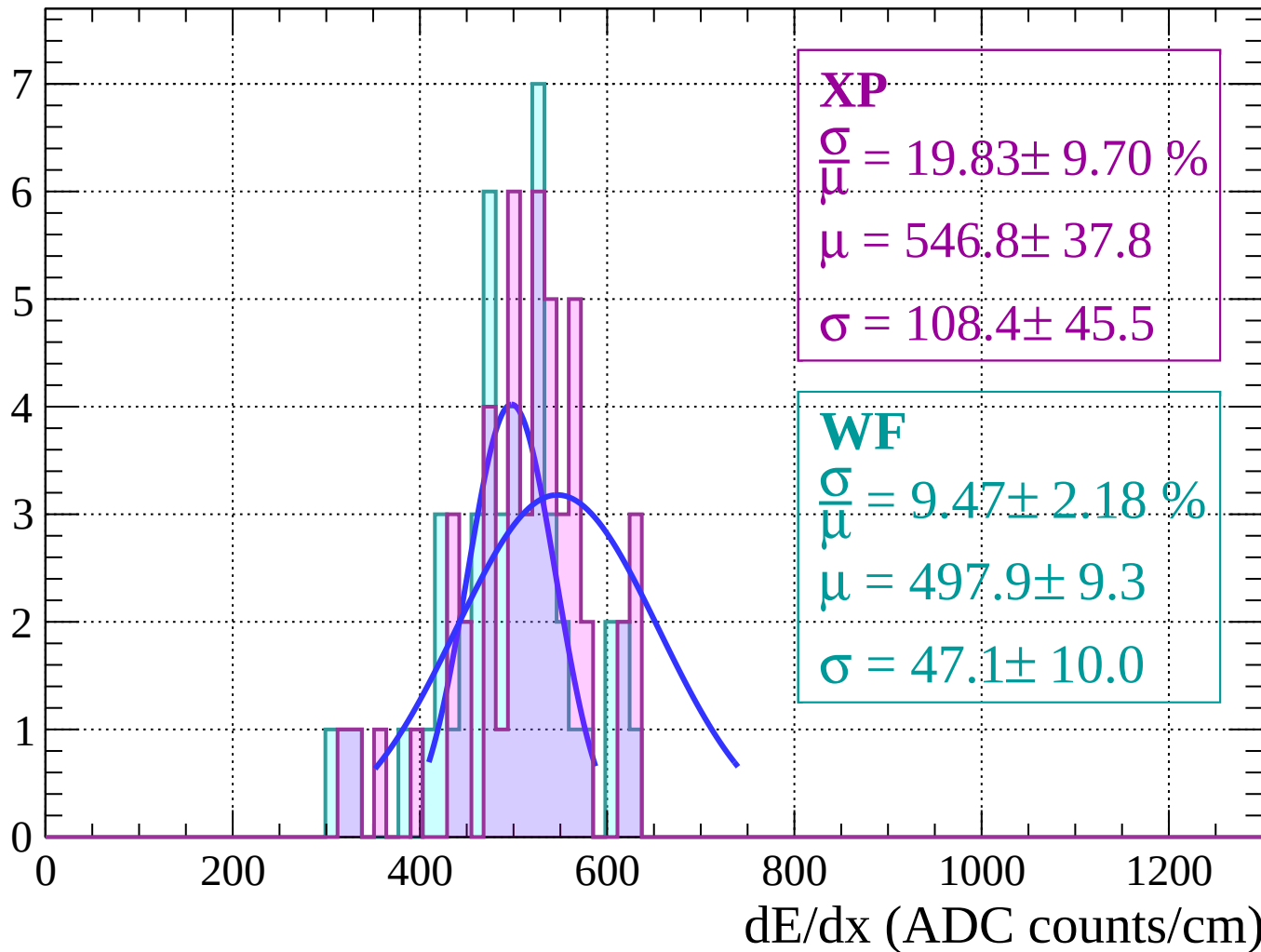
Energy loss | $-62 < \theta < -60$

Count



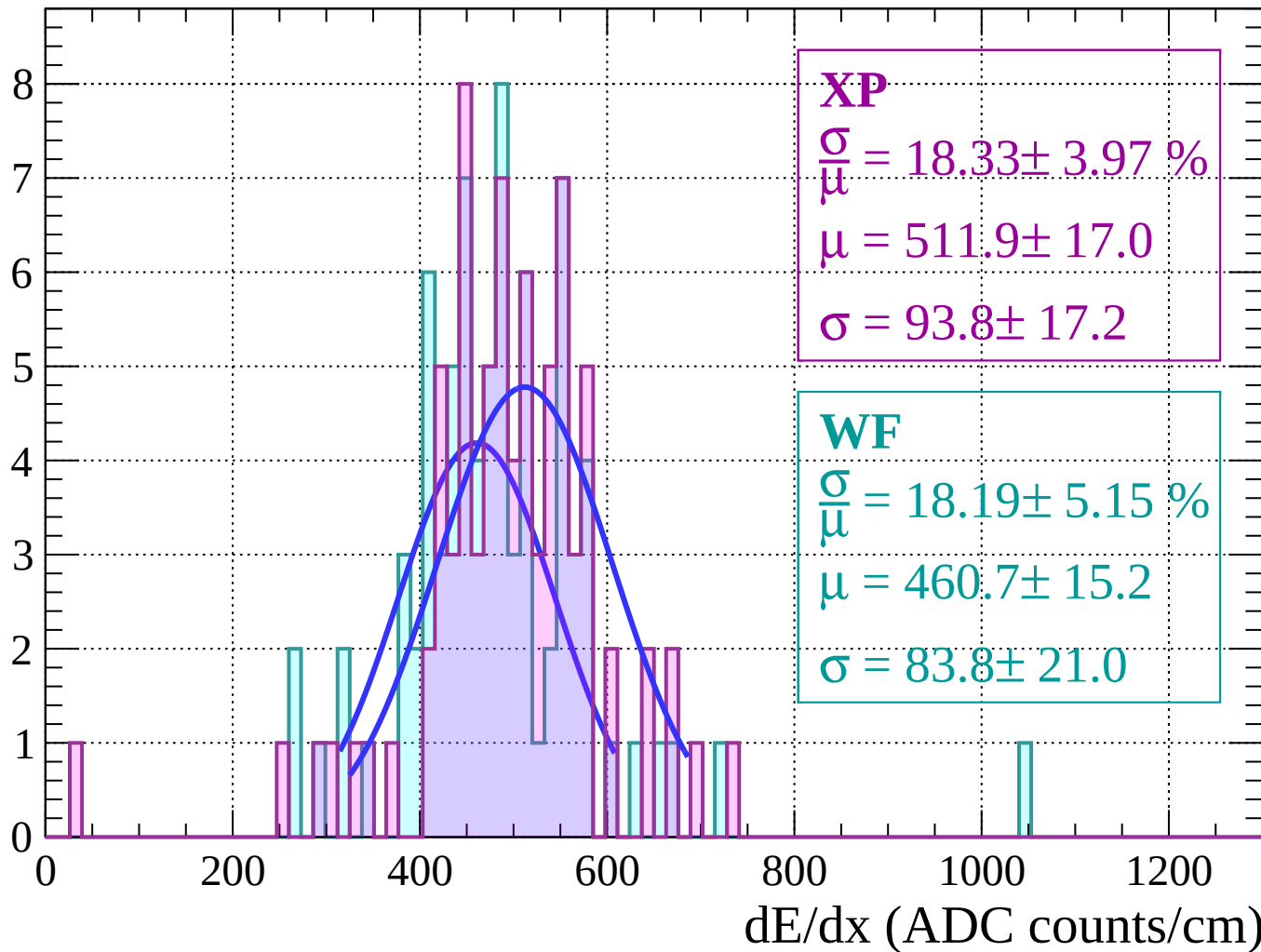
Energy loss | $-60 < \theta < -58$

Count



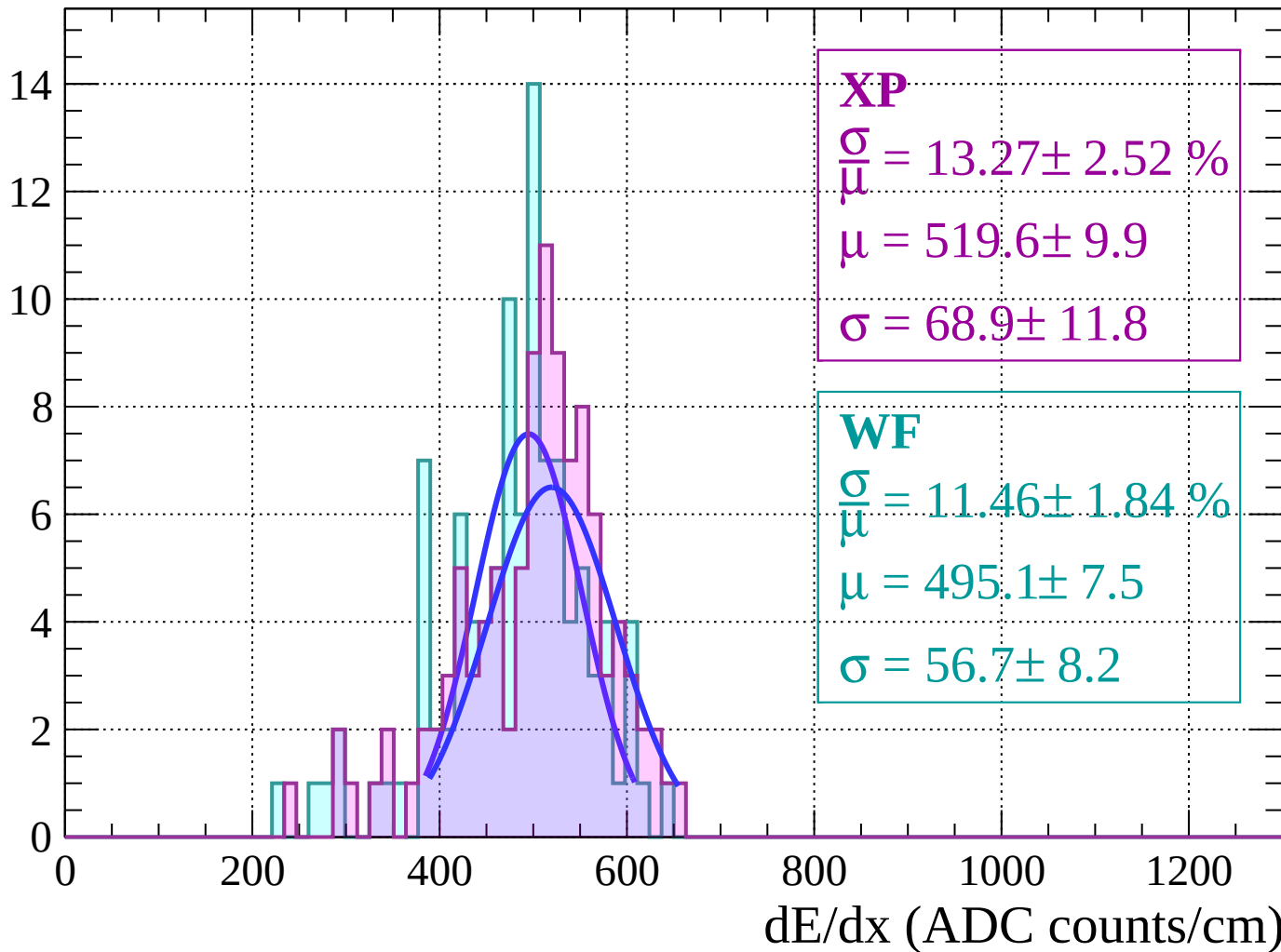
Energy loss | $-58 < \theta < -56$

Count



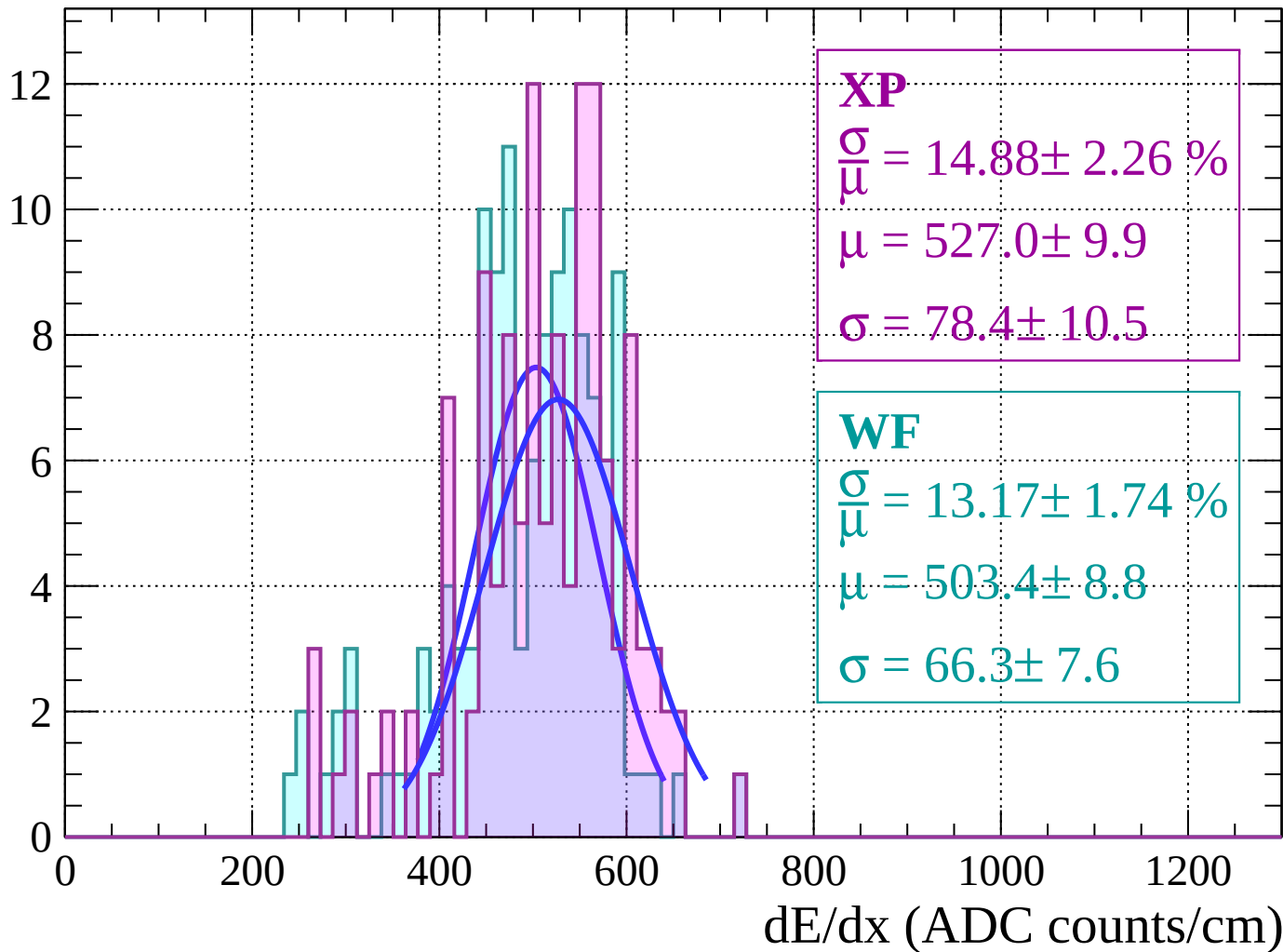
Energy loss $|-56 < \theta < -54$

Count



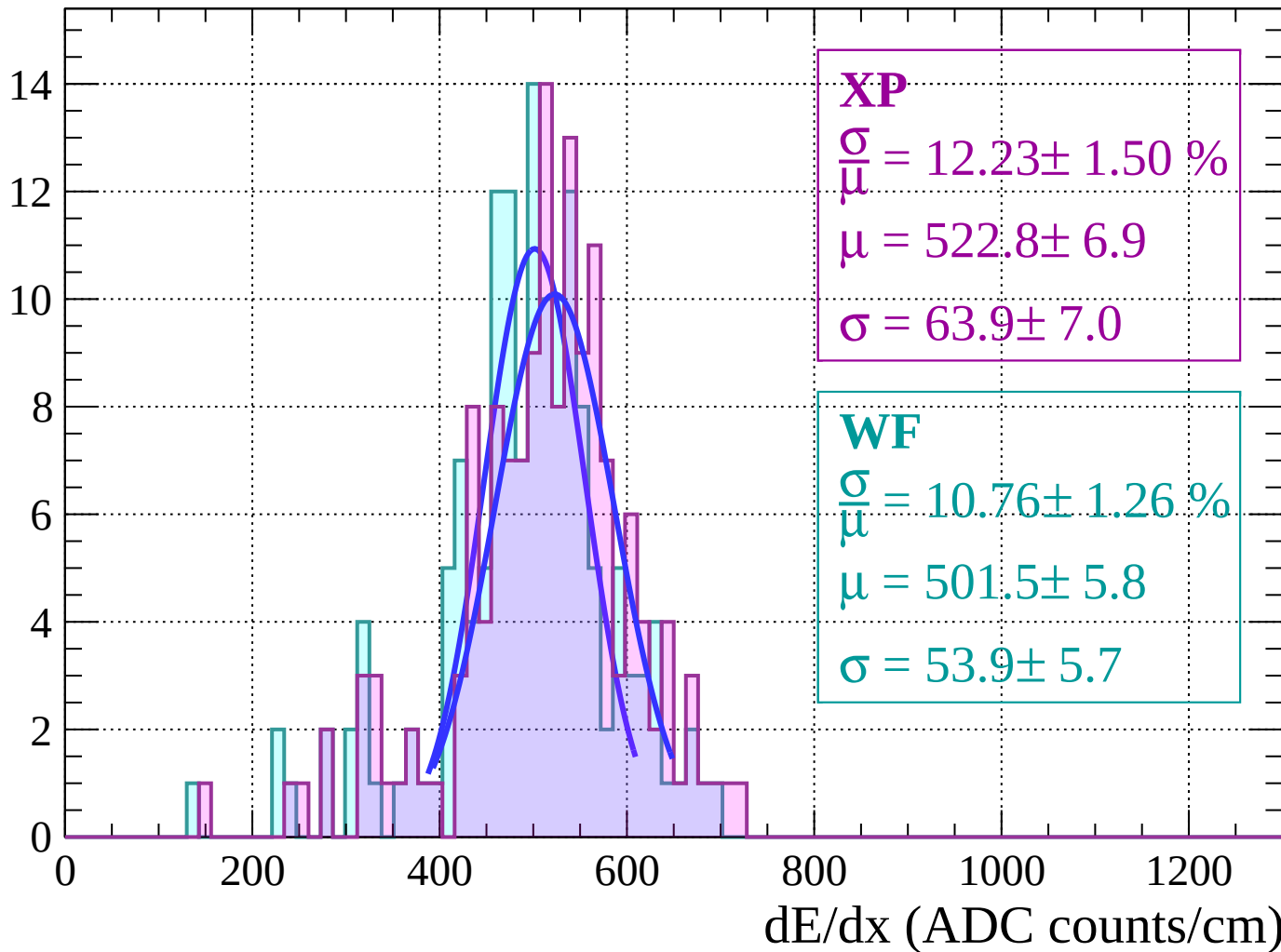
Energy loss | $-54 < \theta < -52$

Count



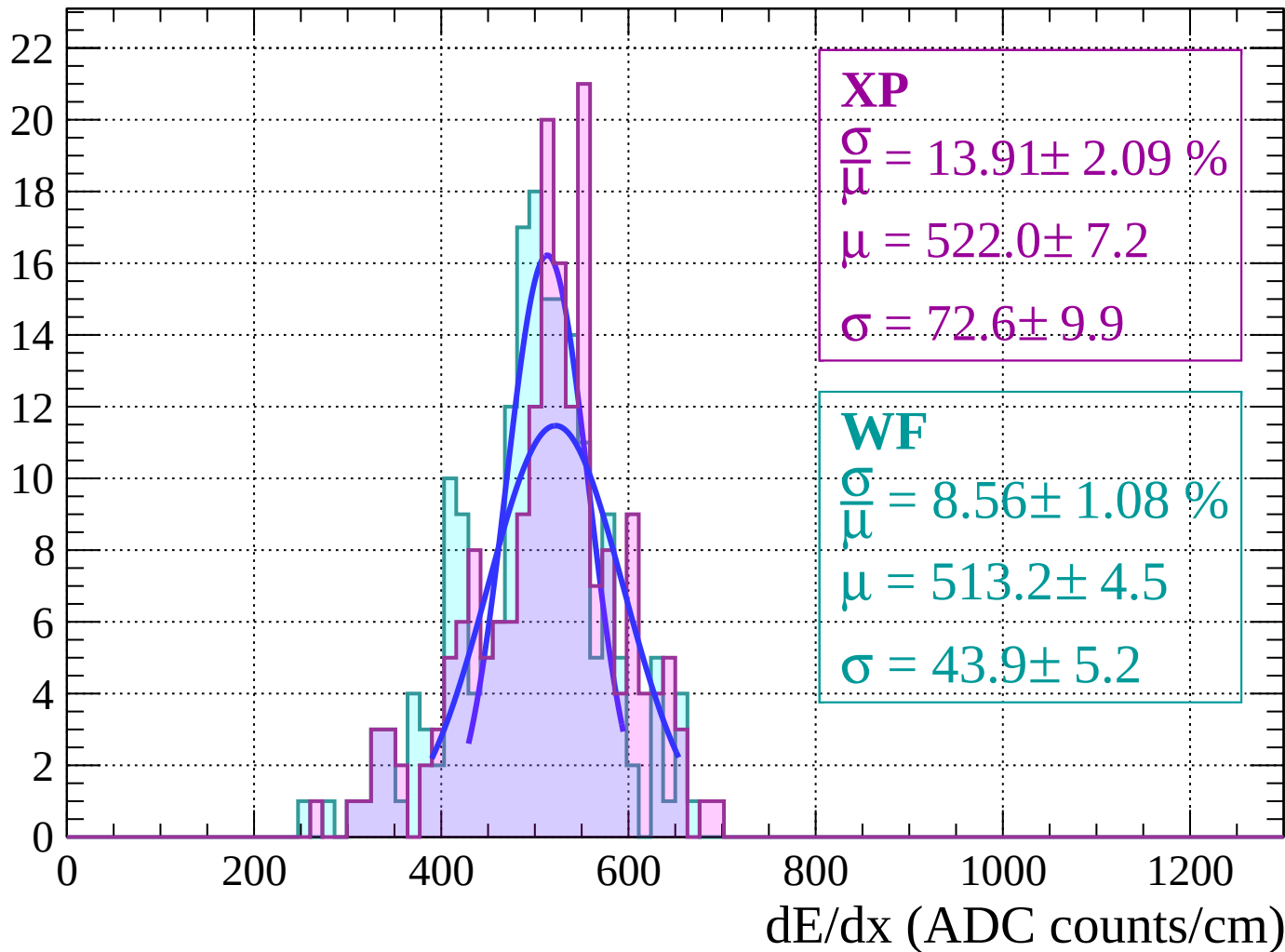
Energy loss | $-52 < \theta < -50$

Count



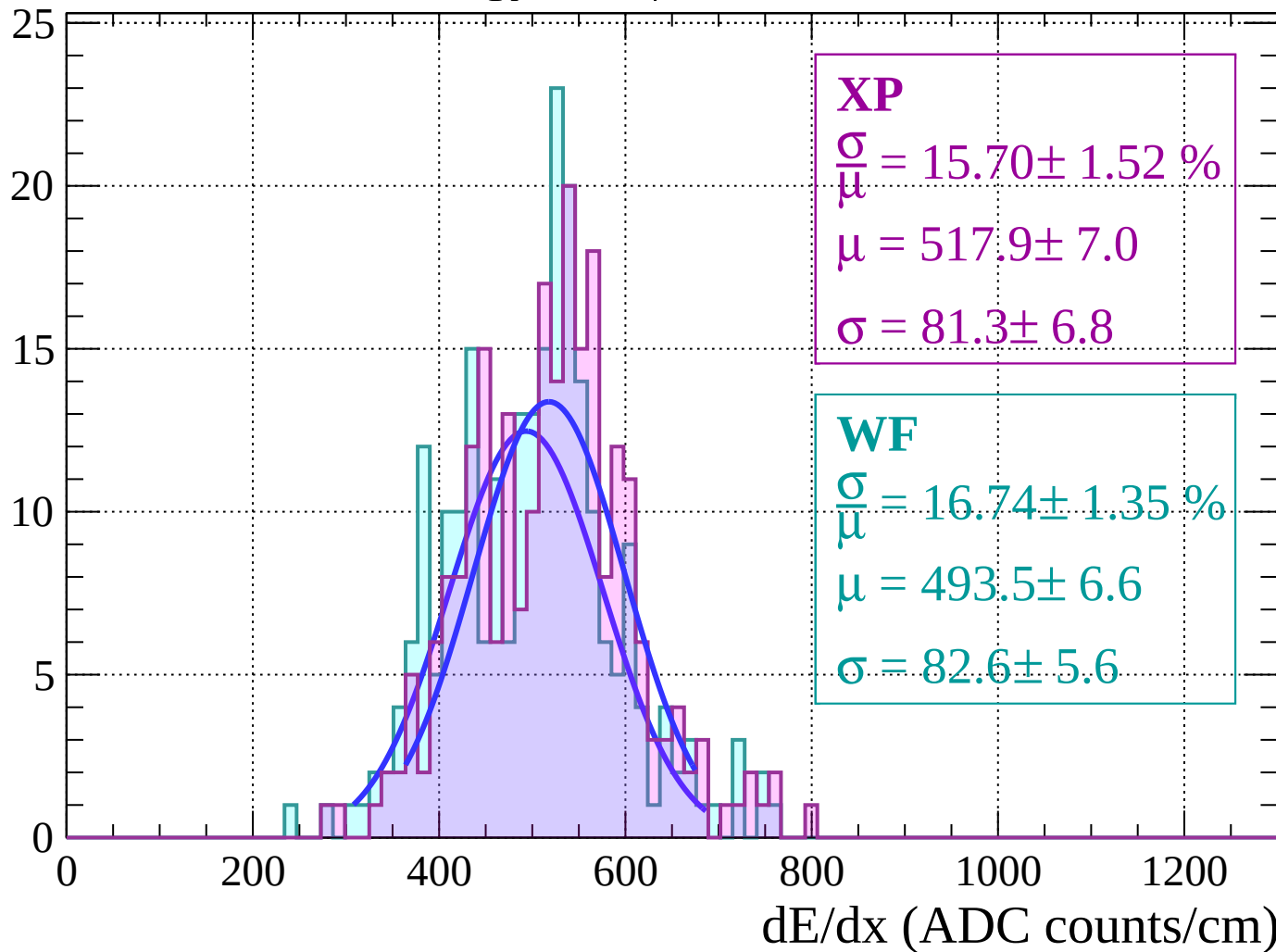
Energy loss $|-50 < \theta < -48$

Count



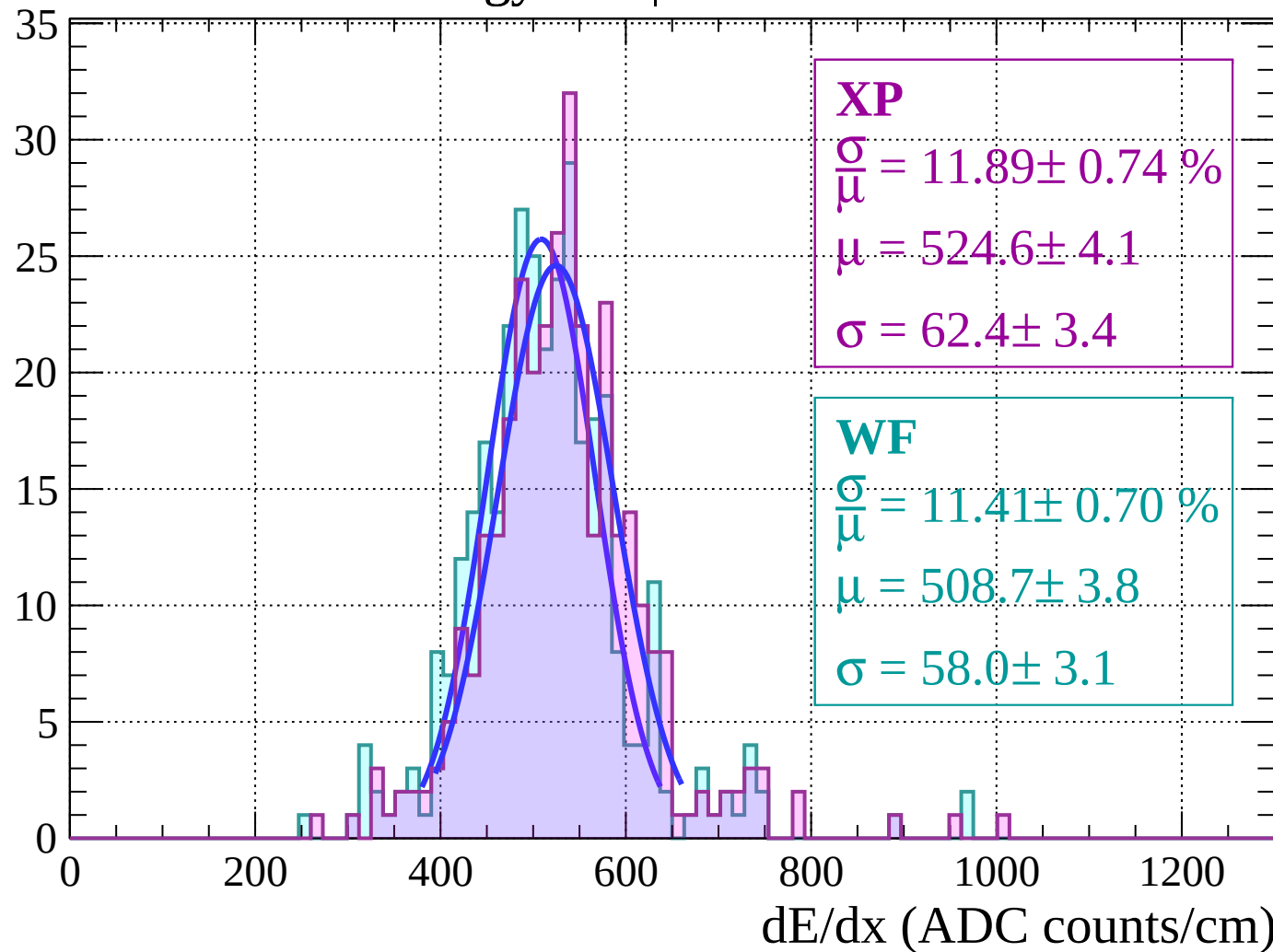
Energy loss | $-48 < \theta < -46$

Count



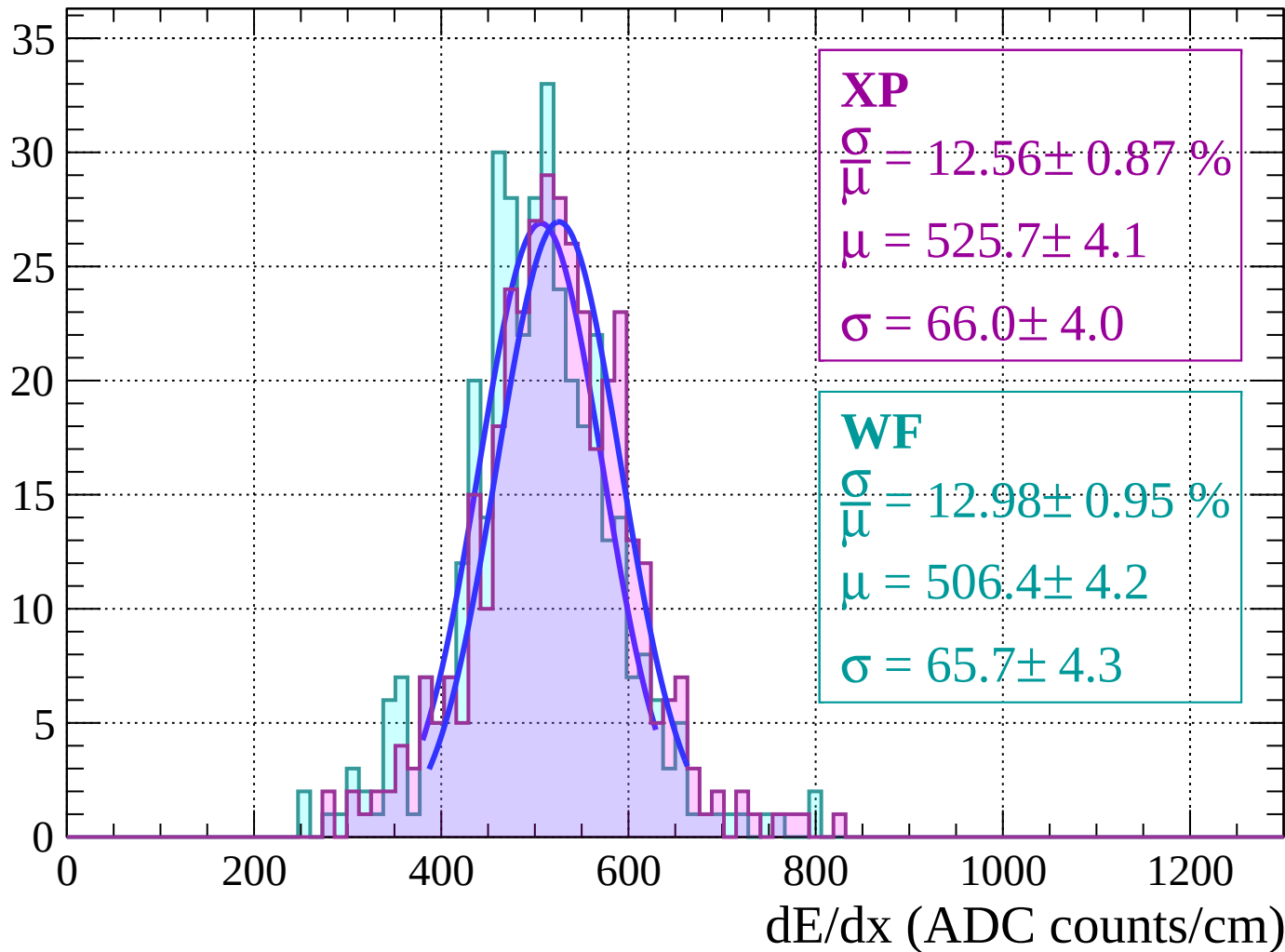
Energy loss $|-46 < \theta < -44$

Count



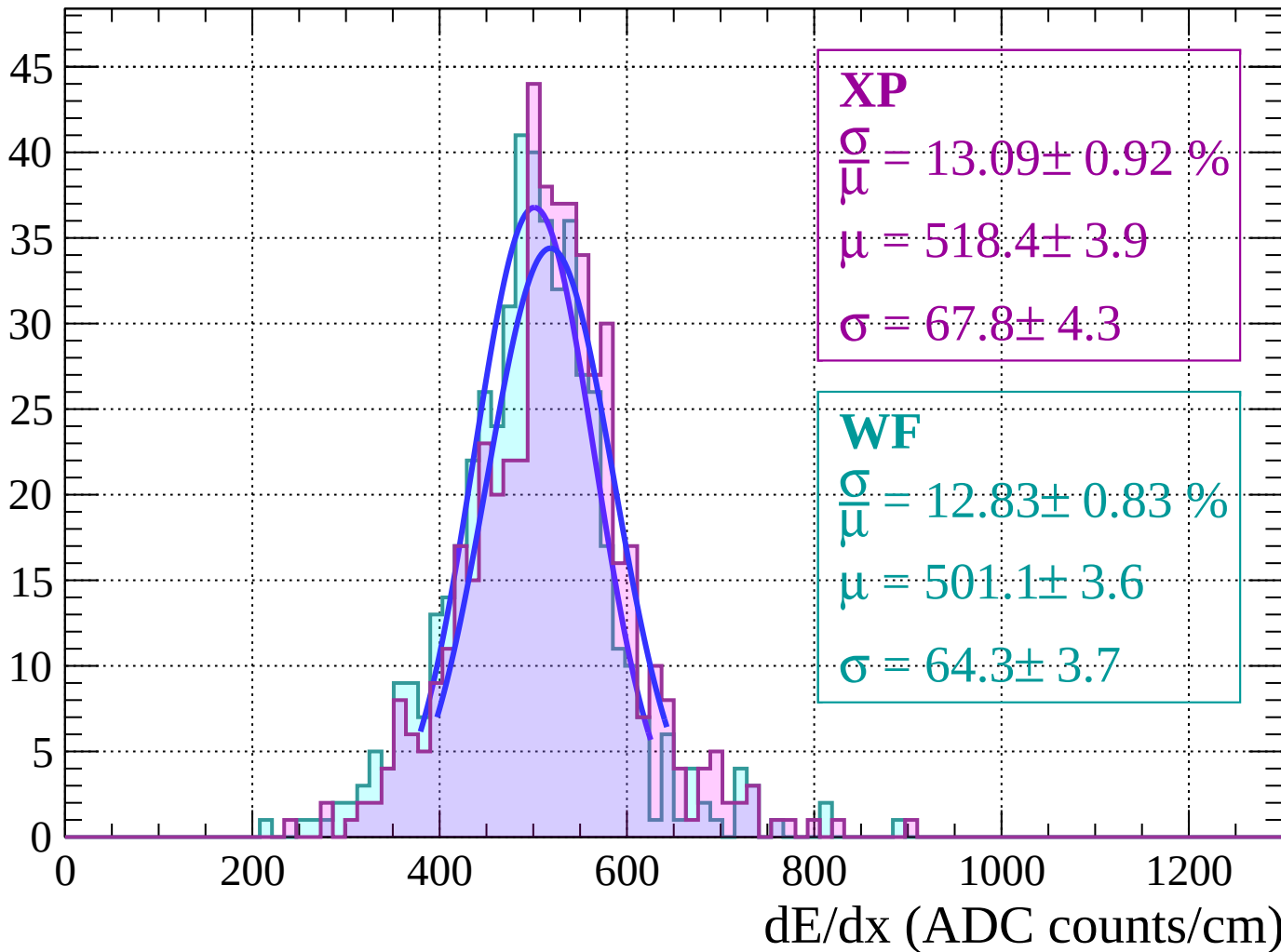
Energy loss $|-44 < \theta < -42$

Count



Energy loss | $-42 < \theta < -40$

Count



Energy loss | $-40 < \theta < -38$

Count

XP

$$\frac{\sigma}{\mu} = 12.33 \pm 0.66 \%$$

$$\mu = 515.4 \pm 3.2$$

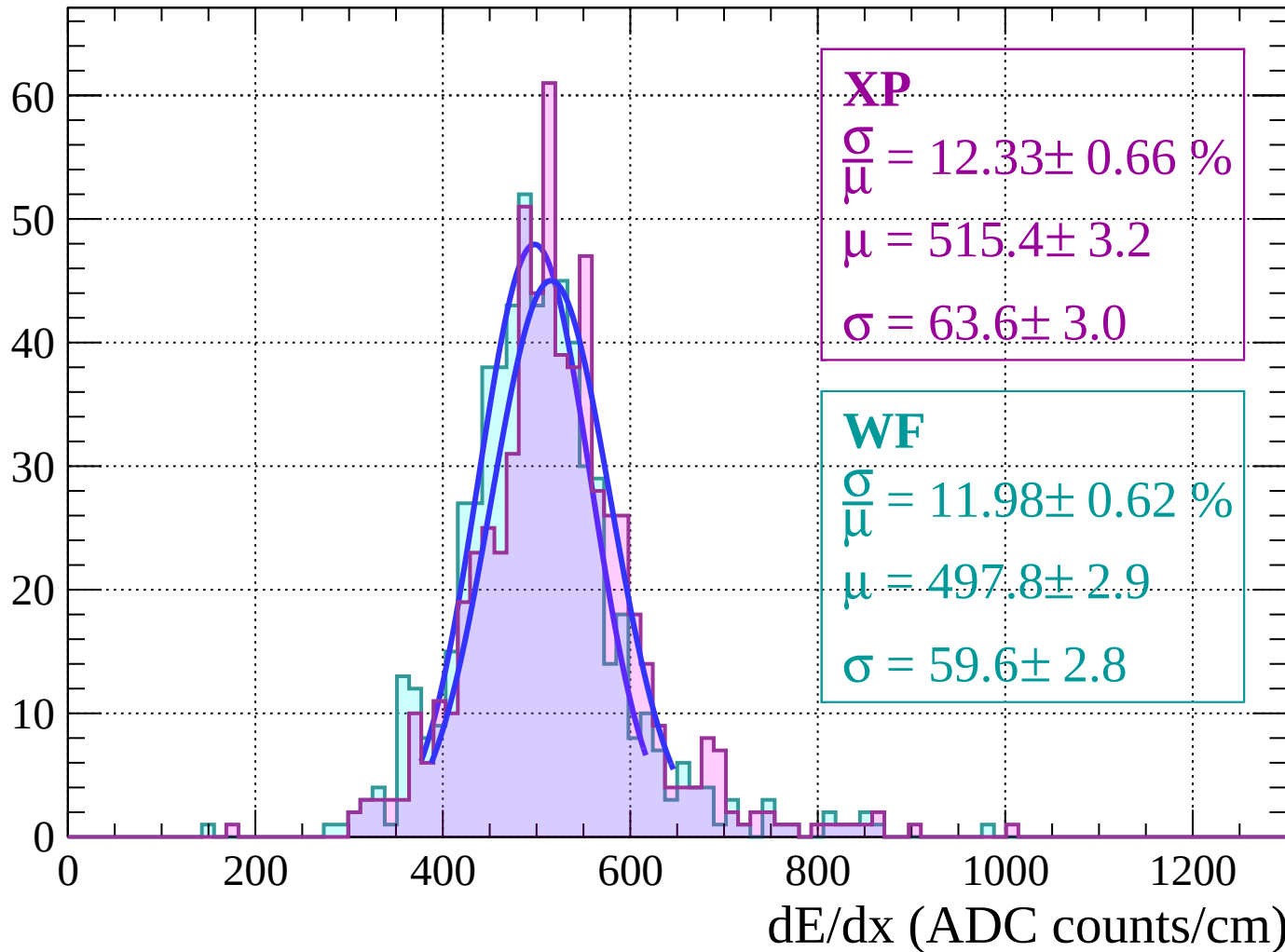
$$\sigma = 63.6 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 11.98 \pm 0.62 \%$$

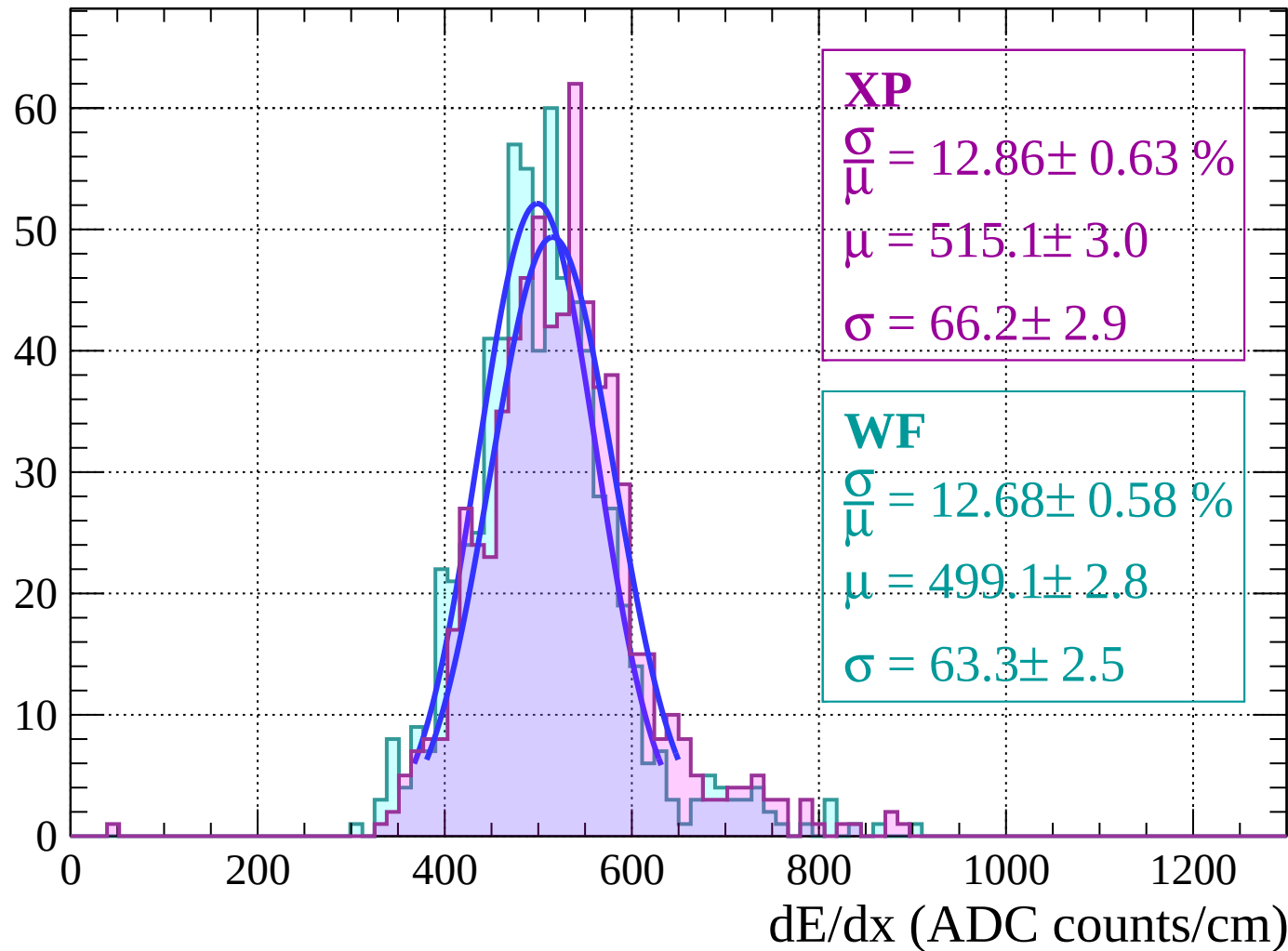
$$\mu = 497.8 \pm 2.9$$

$$\sigma = 59.6 \pm 2.8$$



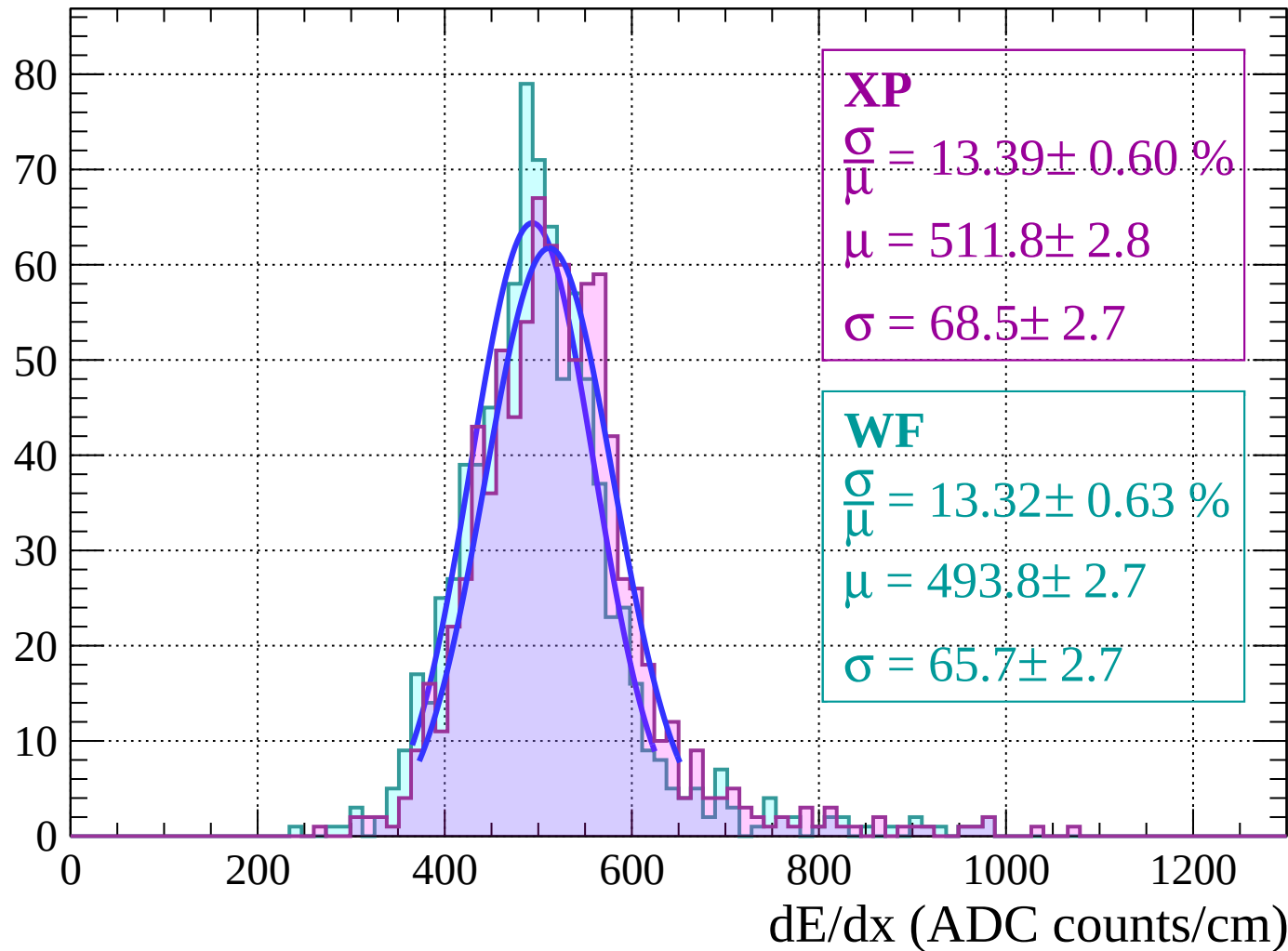
Energy loss $|-38 < \theta < -36$

Count



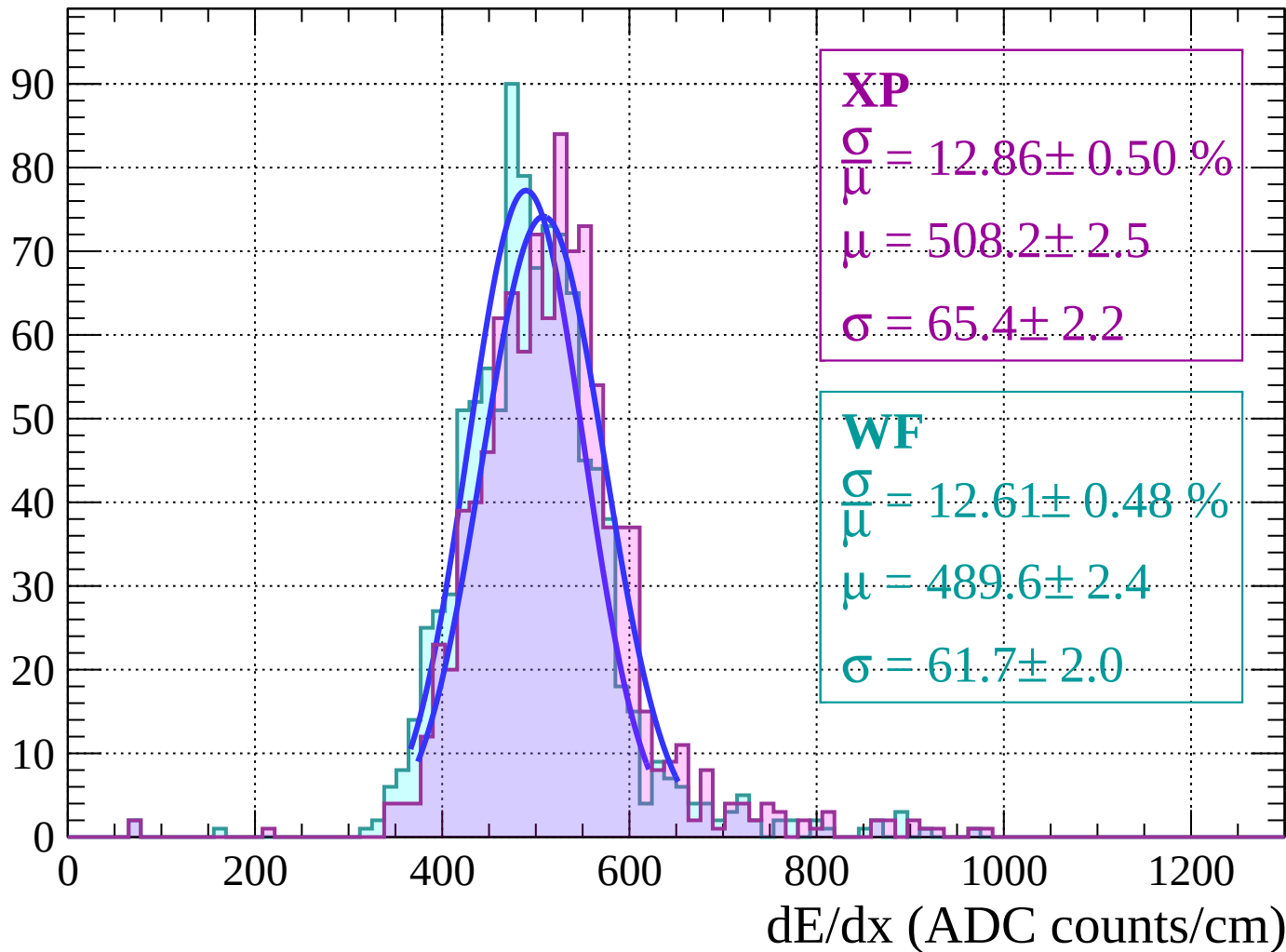
Energy loss $|-36 < \theta < -34$

Count



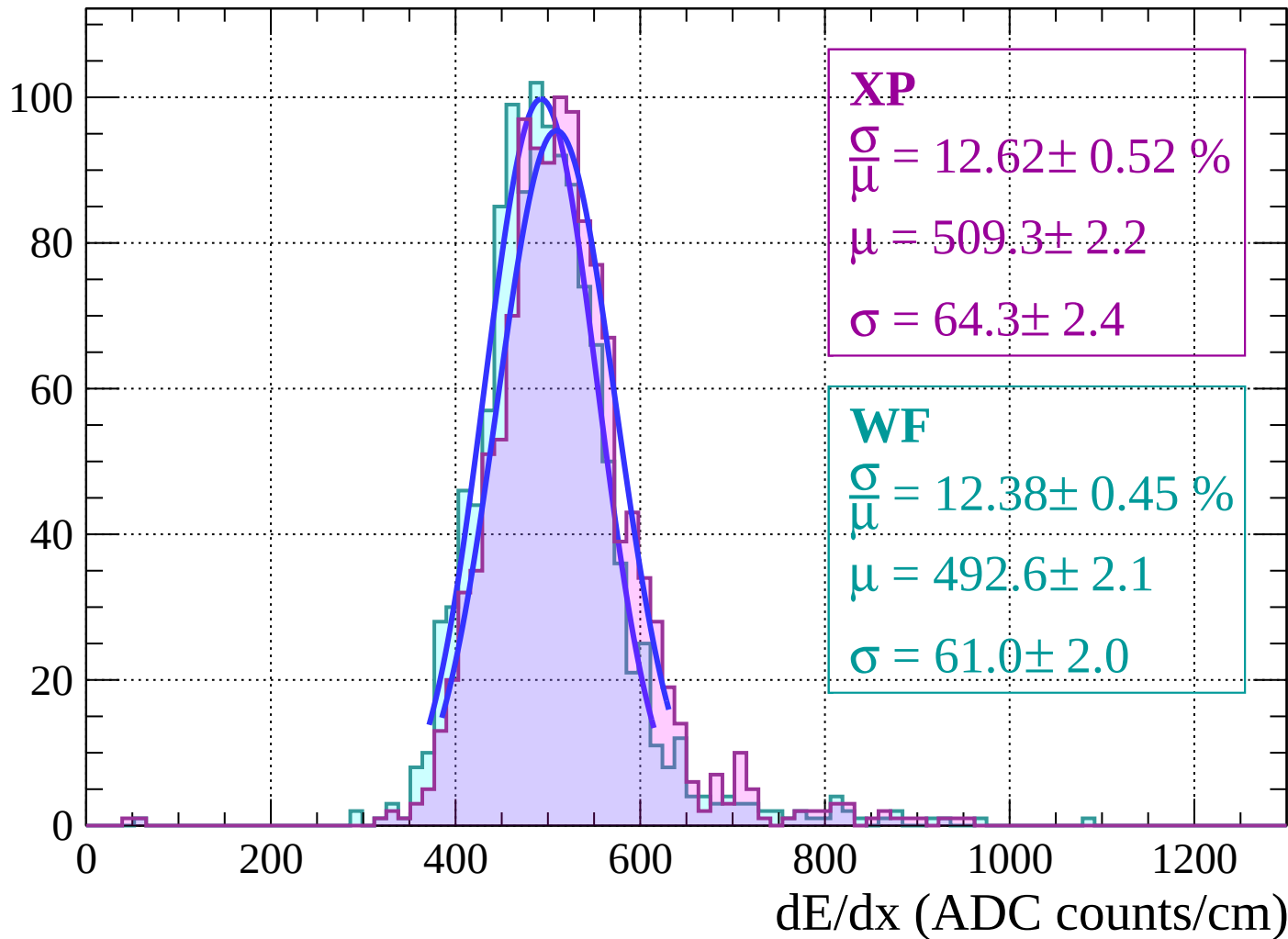
Energy loss $|-34 < \theta < -32$

Count

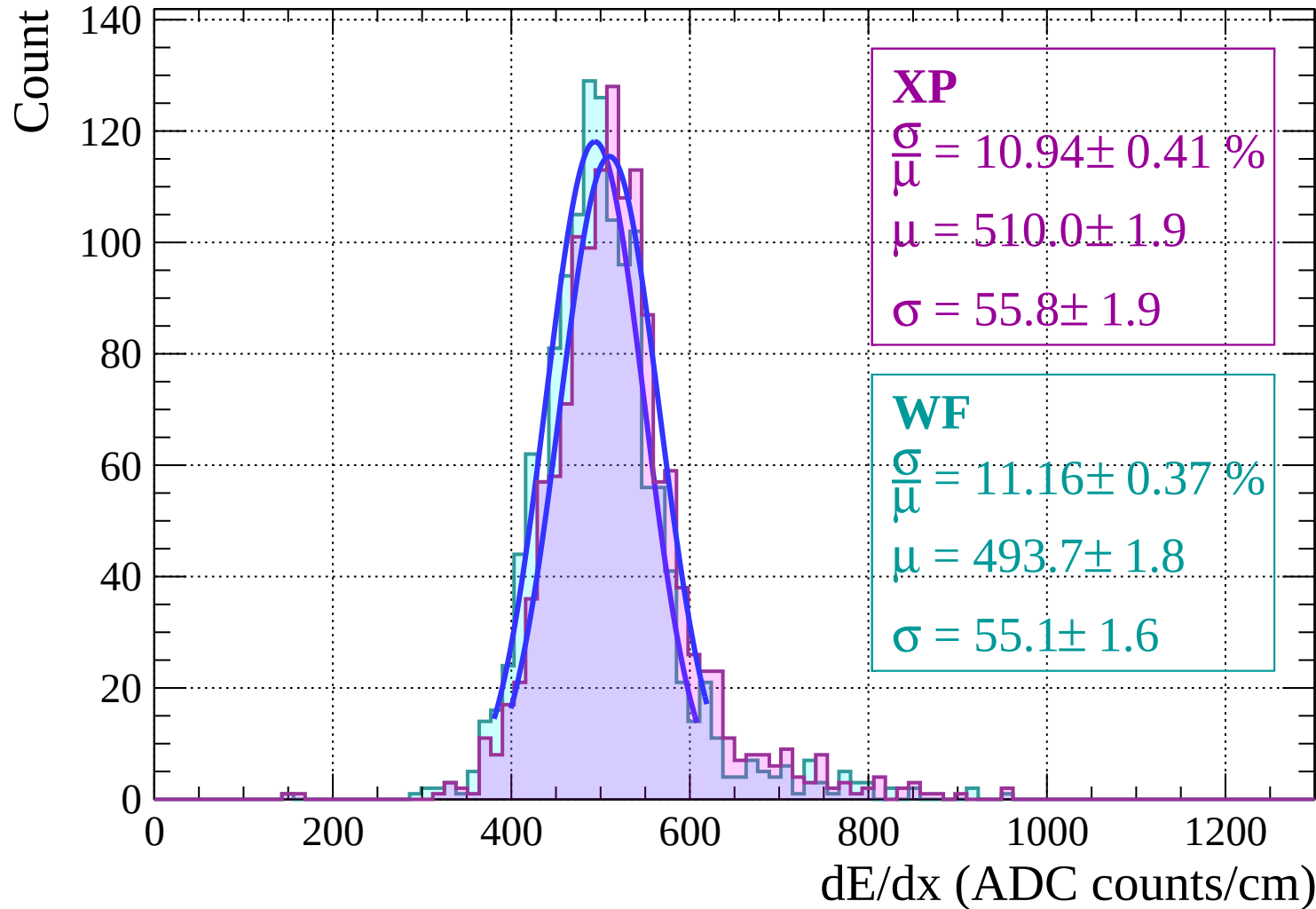


Energy loss | $-32 < \theta < -30$

Count

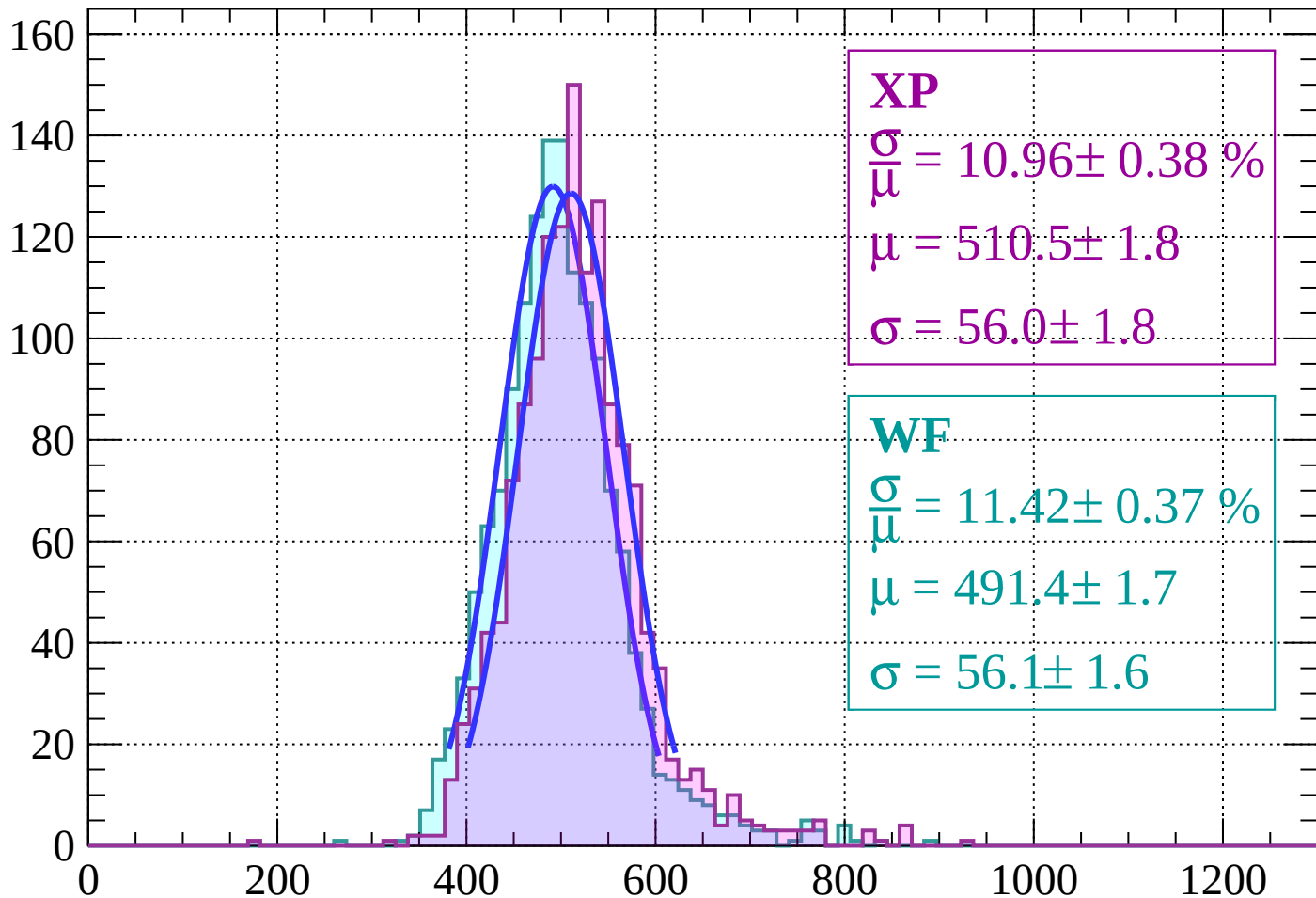


Energy loss | $-30 < \theta < -28$



Energy loss $|-28 < \theta < -26$

Count



XP

$$\frac{\sigma}{\mu} = 10.96 \pm 0.38 \%$$

$$\mu = 510.5 \pm 1.8$$

$$\sigma = 56.0 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 11.42 \pm 0.37 \%$$

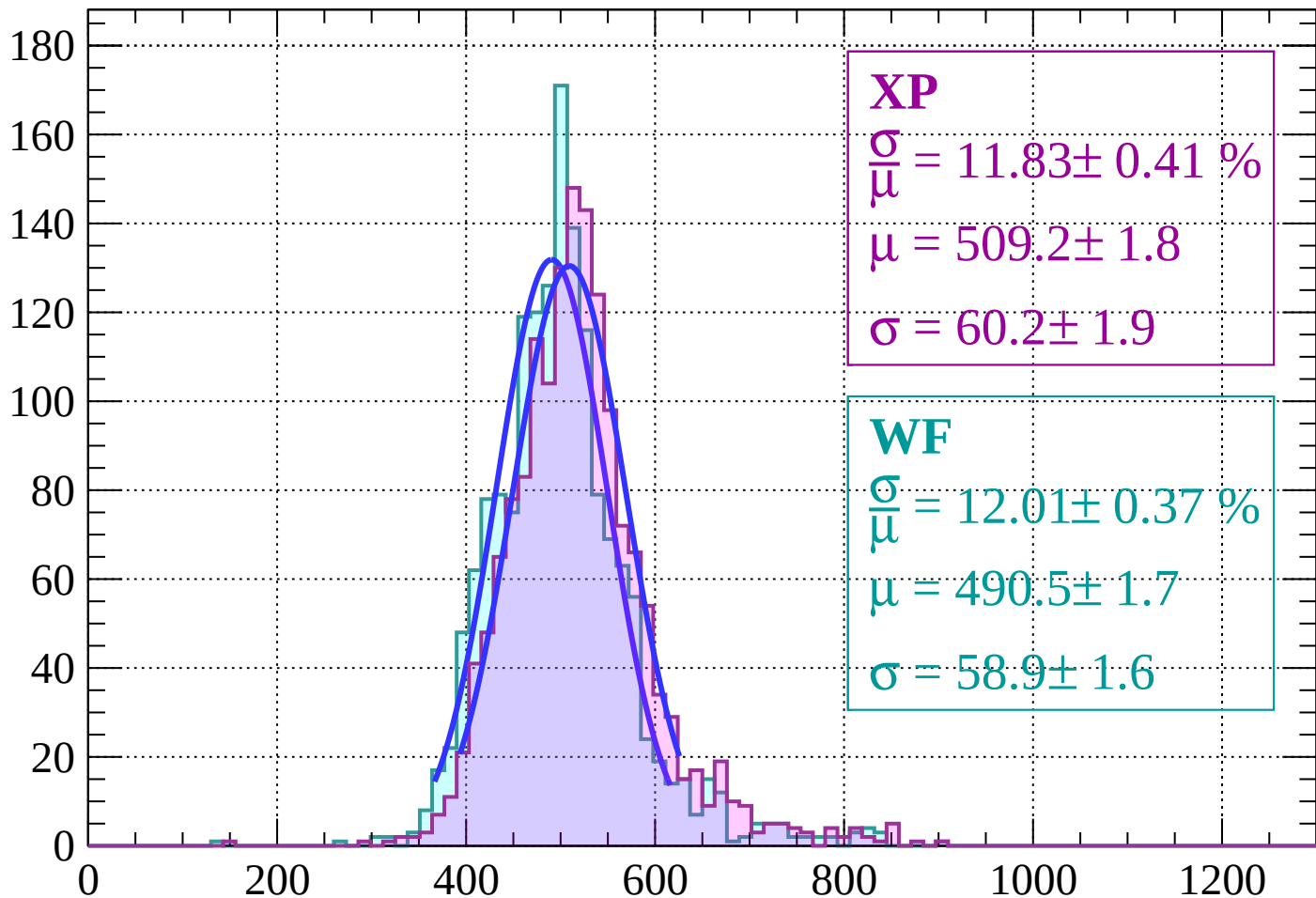
$$\mu = 491.4 \pm 1.7$$

$$\sigma = 56.1 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss $|-26 < \theta < -24$

Count



XP

$$\frac{\sigma}{\mu} = 11.83 \pm 0.41 \%$$

$$\mu = 509.2 \pm 1.8$$

$$\sigma = 60.2 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 12.01 \pm 0.37 \%$$

$$\mu = 490.5 \pm 1.7$$

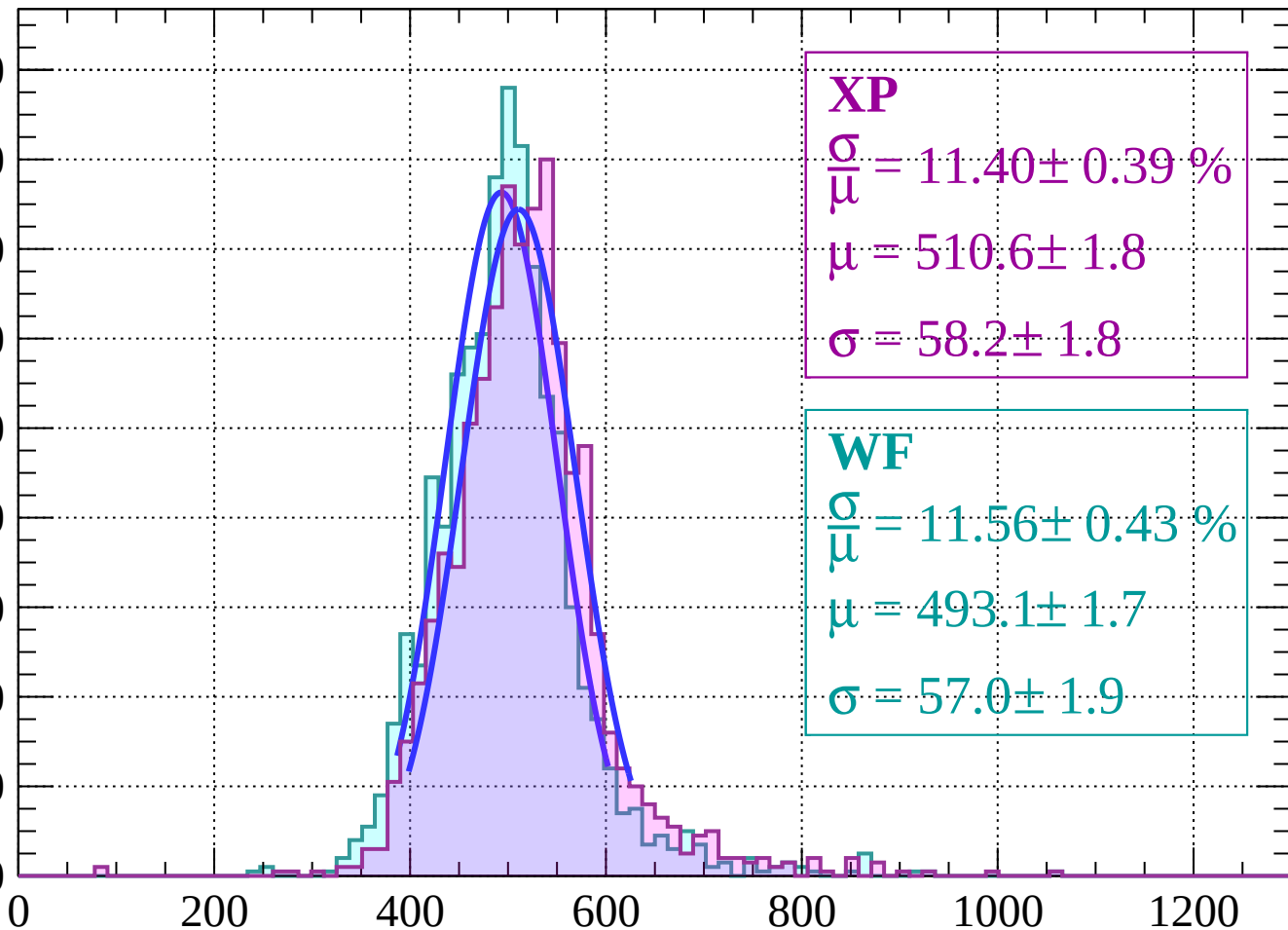
$$\sigma = 58.9 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-24 < \theta < -22$

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\bar{\mu}} = 11.40 \pm 0.39 \%$$

$$\mu = 510.6 \pm 1.8$$

$$\sigma = 58.2 \pm 1.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 11.56 \pm 0.43 \%$$

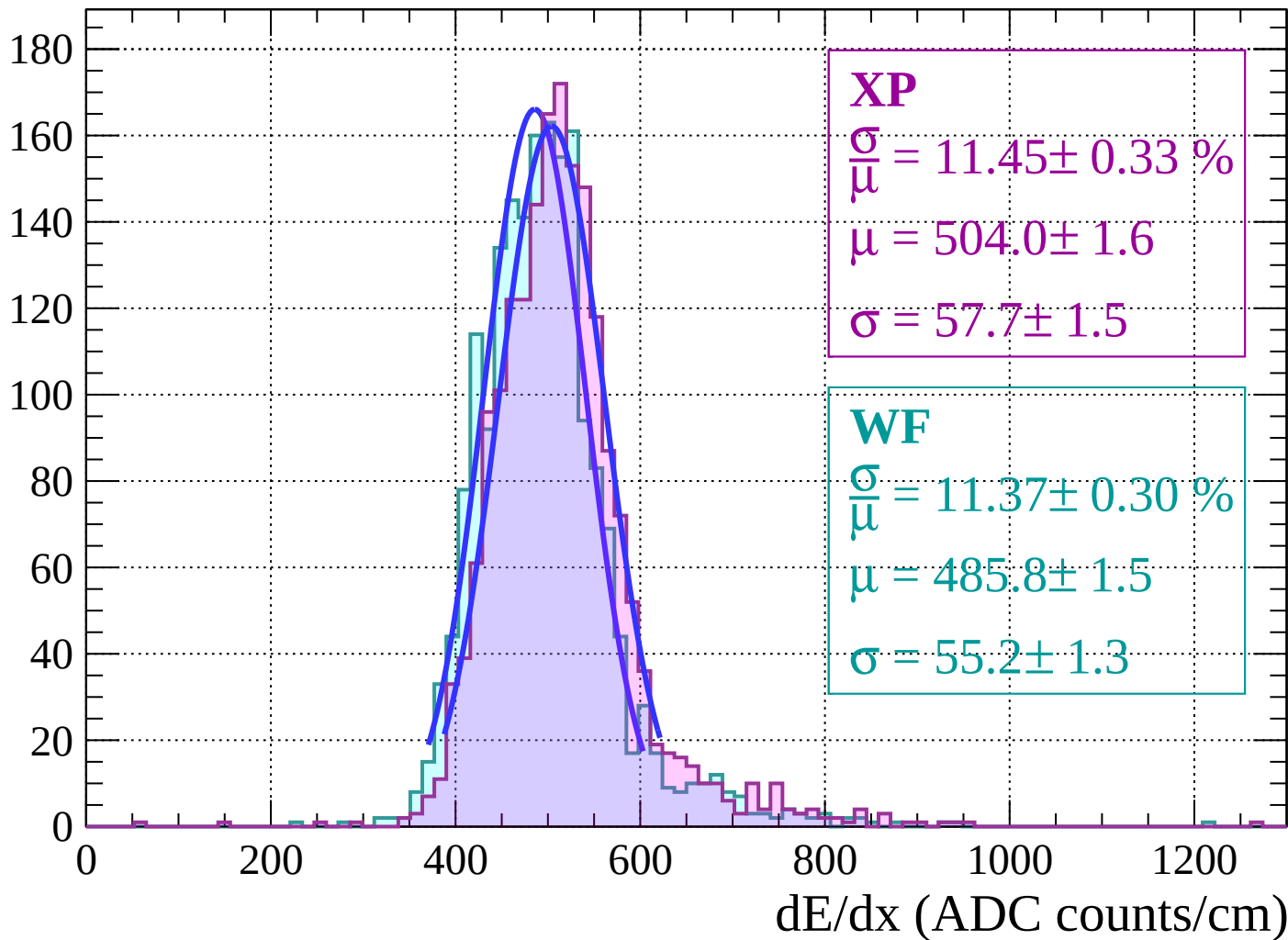
$$\mu = 493.1 \pm 1.7$$

$$\sigma = 57.0 \pm 1.9$$

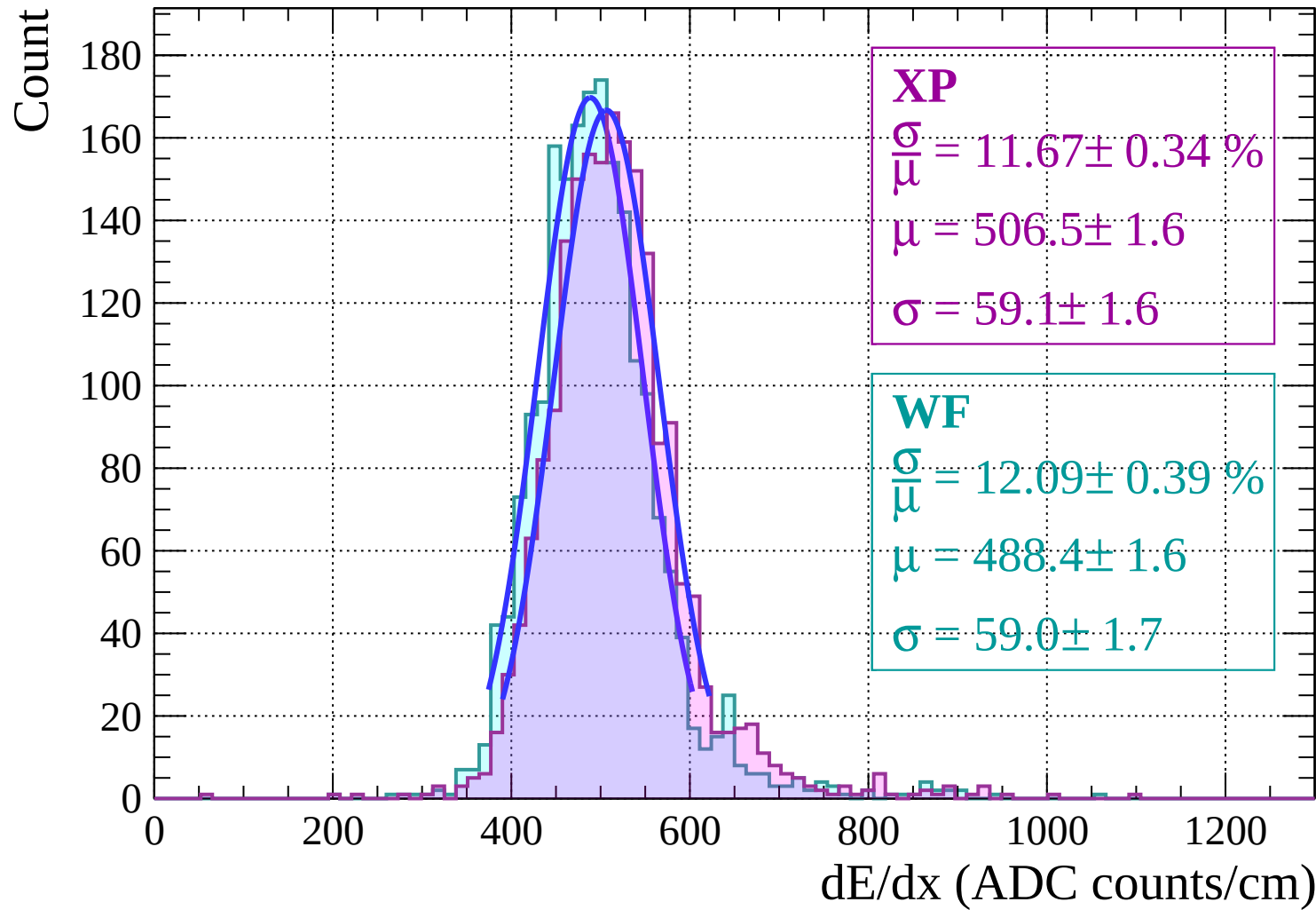
dE/dx (ADC counts/cm)

Energy loss | $-22 < \theta < -20$

Count

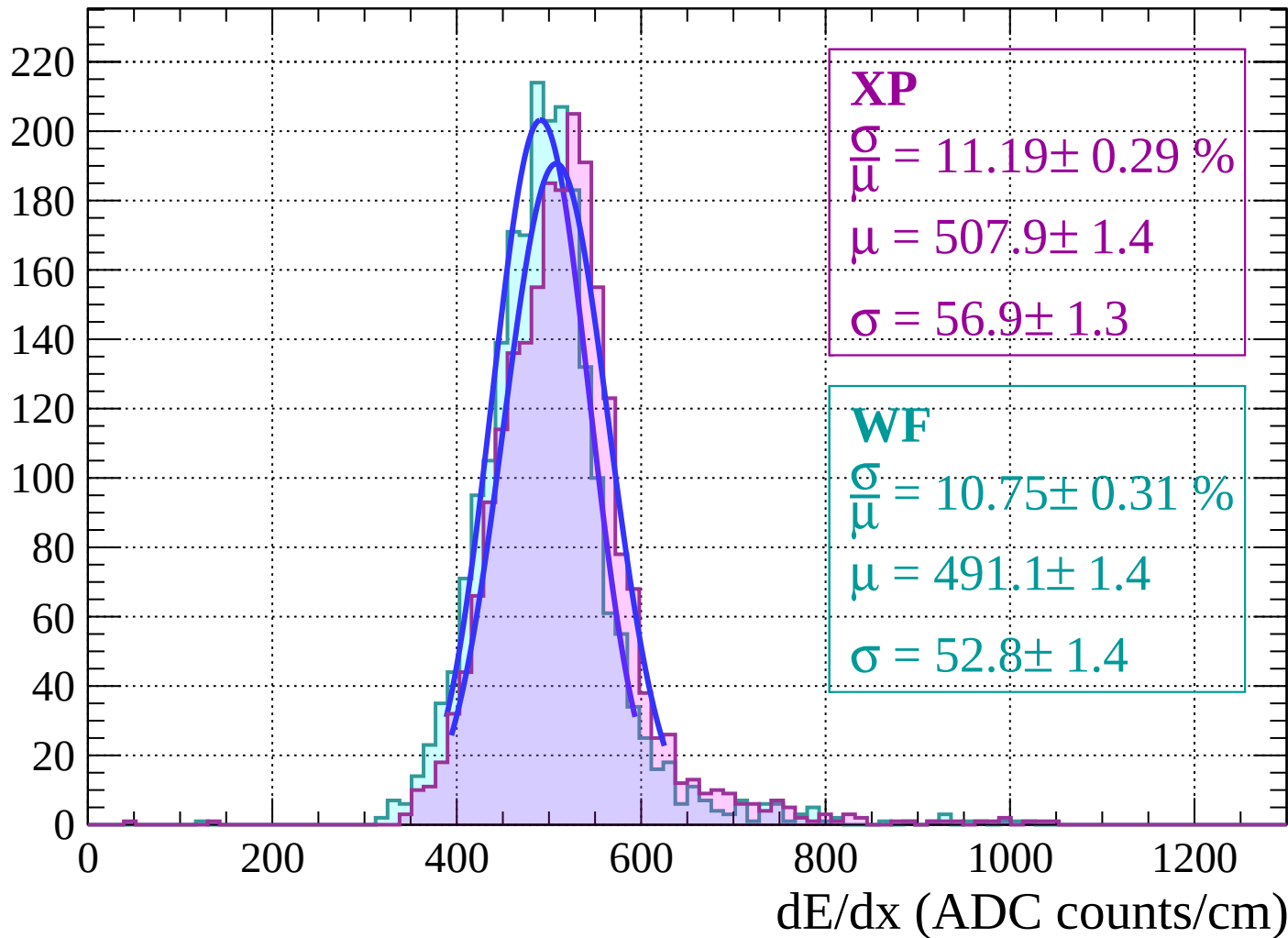


Energy loss | $-20 < \theta < -18$



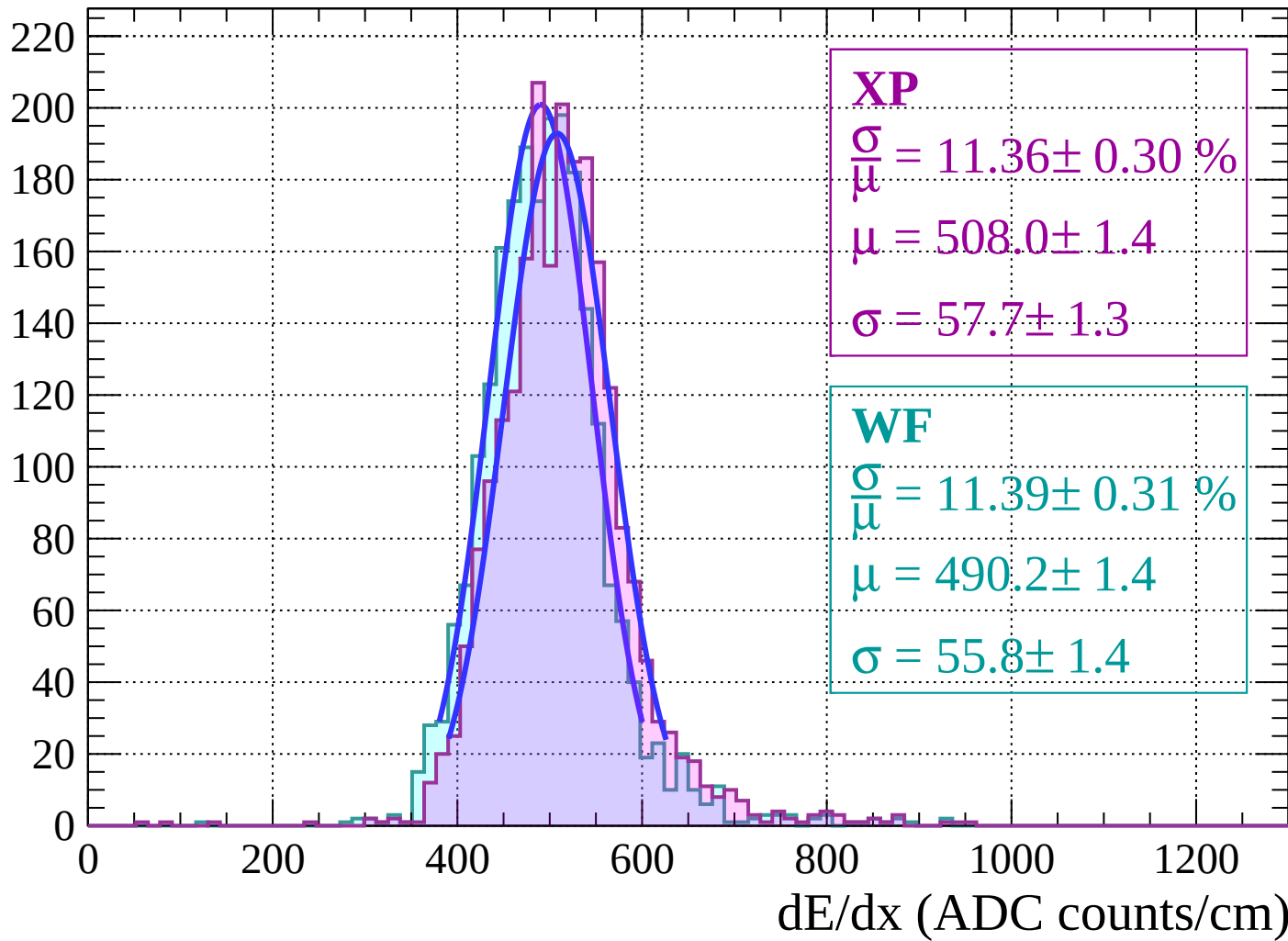
Energy loss | $-18 < \theta < -16$

Count



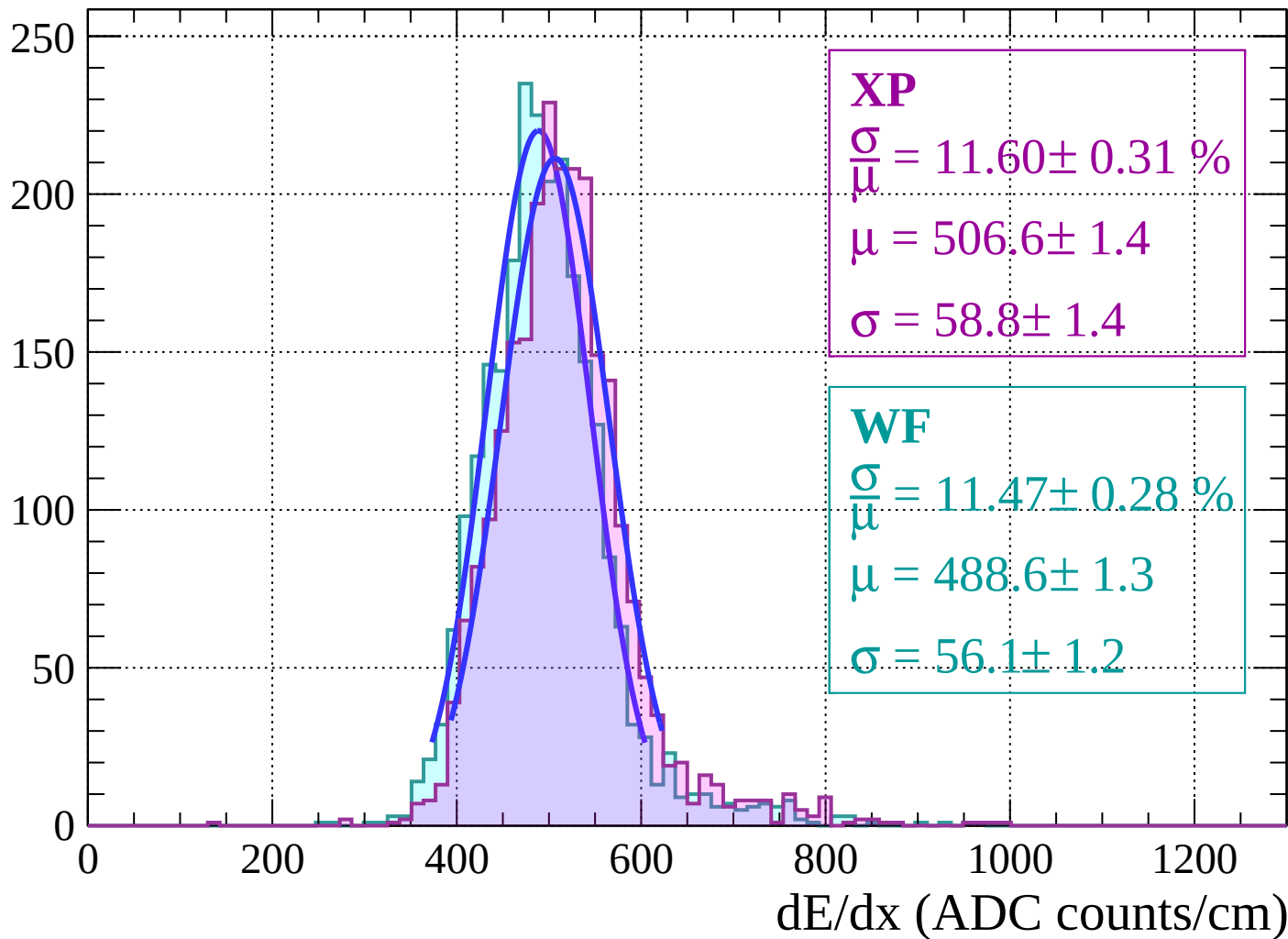
Energy loss | $-16 < \theta < -14$

Count



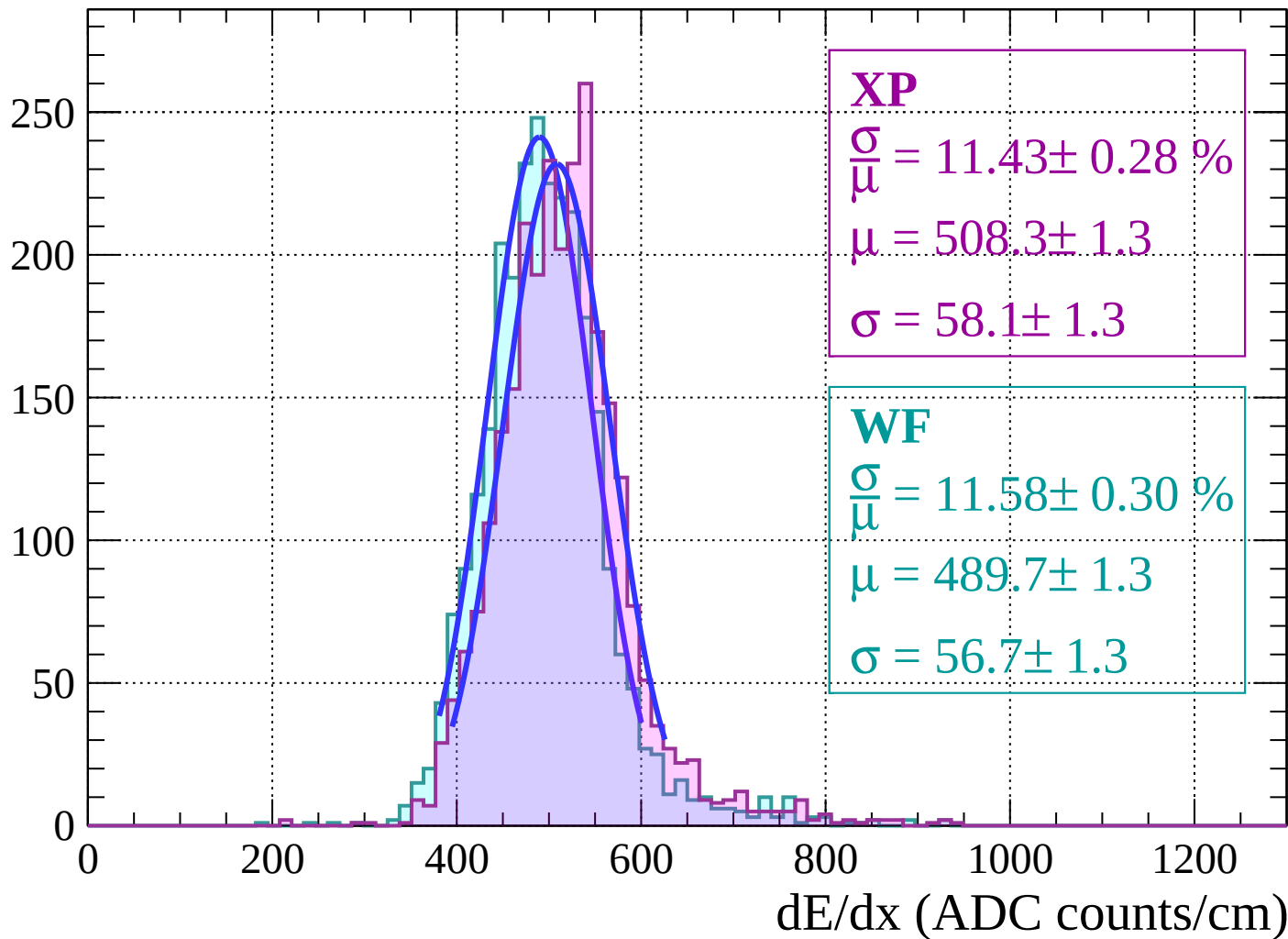
Energy loss $|-14 < \theta < -12$

Count



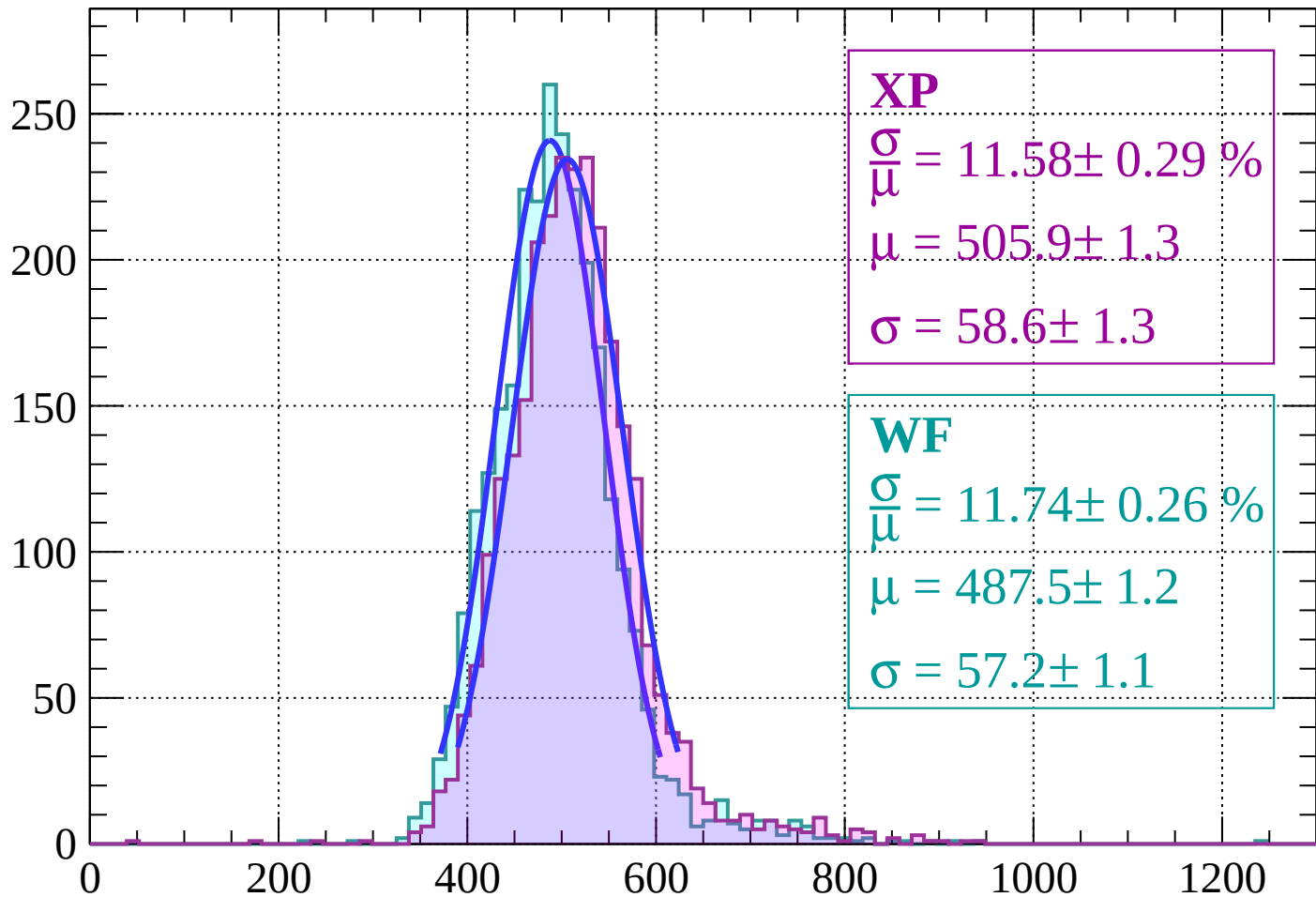
Energy loss | $-12 < \theta < -10$

Count



Energy loss | $-10 < \theta < -8$

Count



XP

$$\frac{\sigma}{\mu} = 11.58 \pm 0.29 \%$$

$$\mu = 505.9 \pm 1.3$$

$$\sigma = 58.6 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.74 \pm 0.26 \%$$

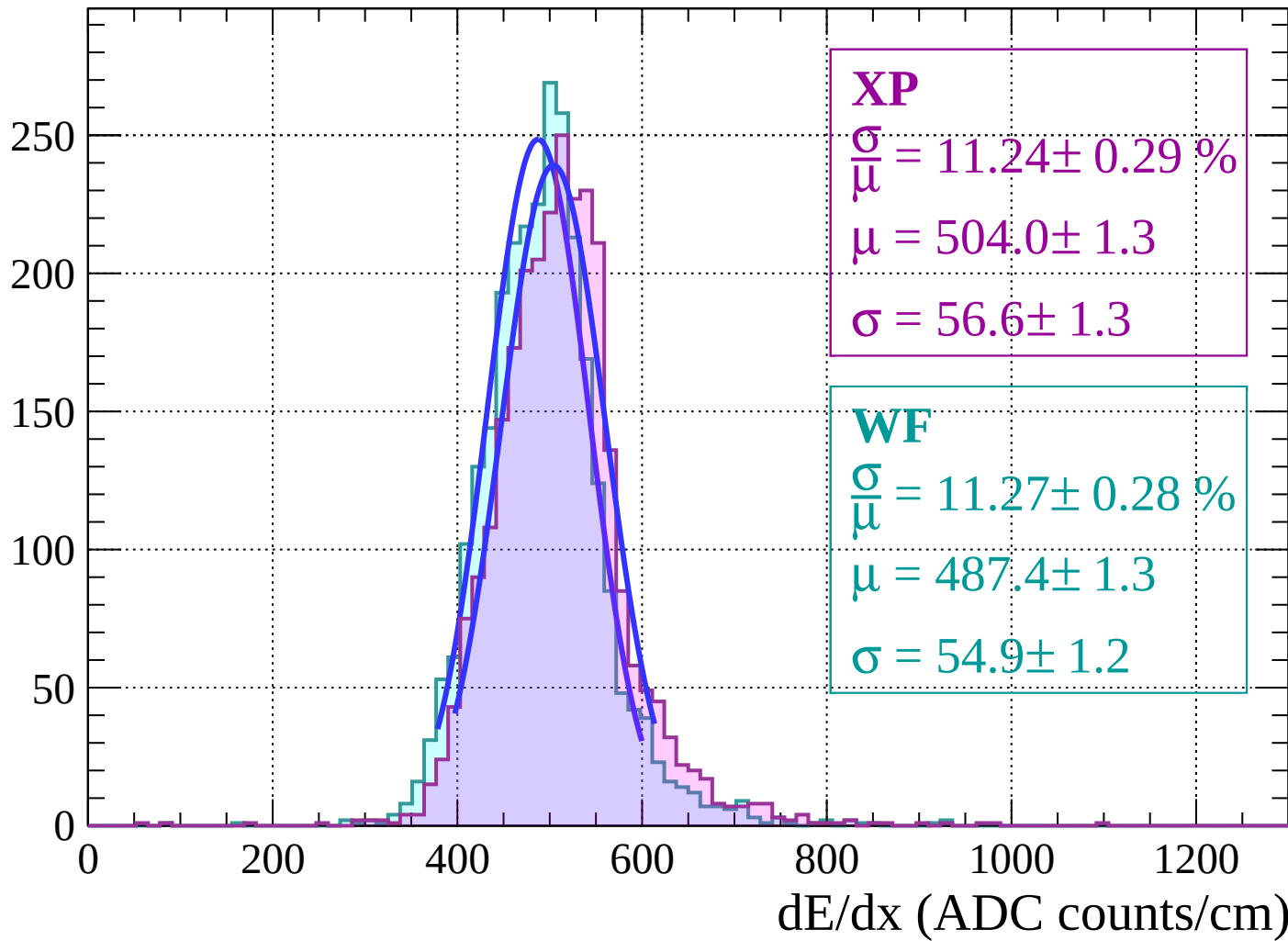
$$\mu = 487.5 \pm 1.2$$

$$\sigma = 57.2 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-8 < \theta < -6$

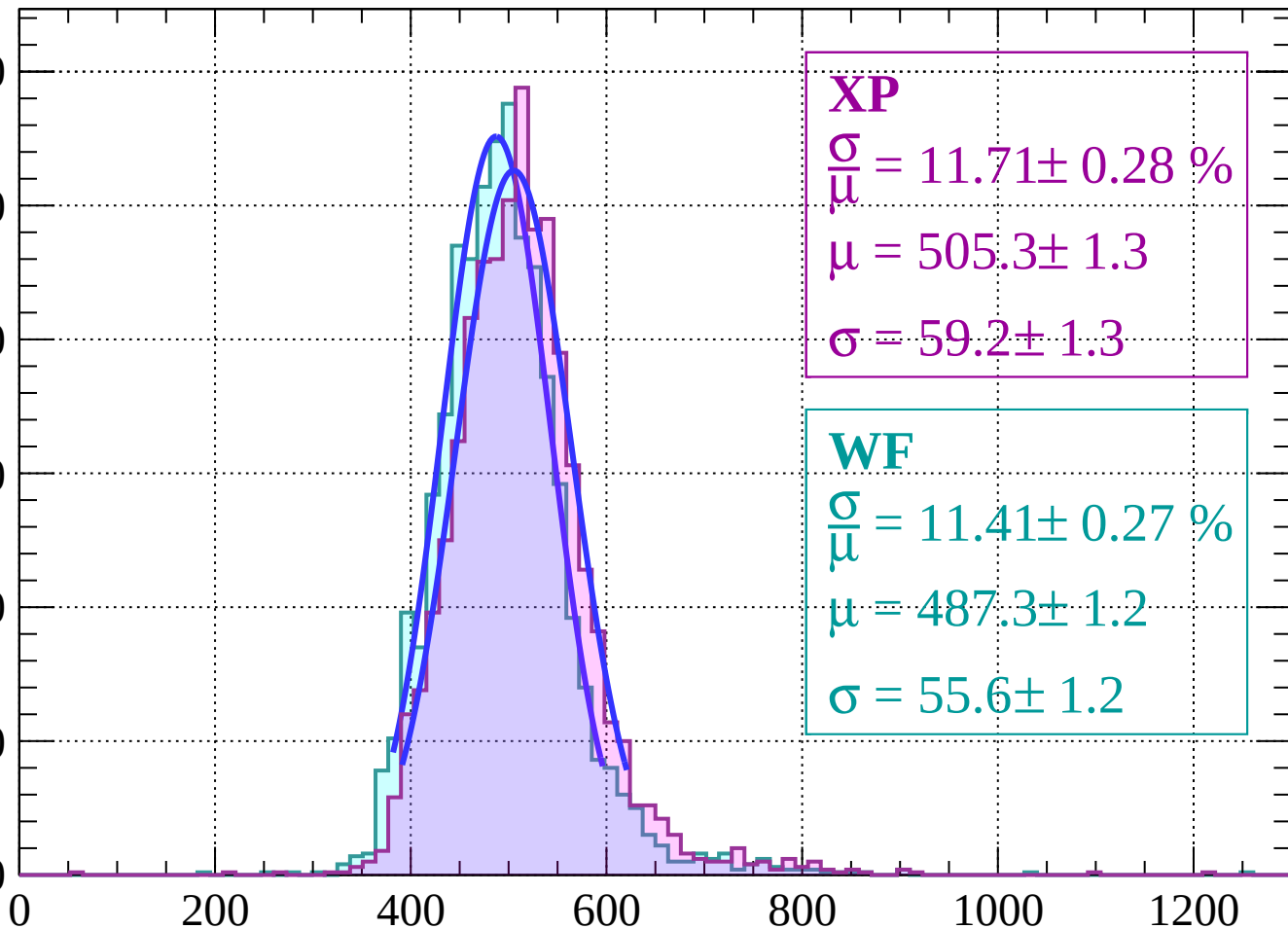
Count



Energy loss $-6 < \theta < -4$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.71 \pm 0.28 \%$$

$$\mu = 505.3 \pm 1.3$$

$$\sigma = 59.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.41 \pm 0.27 \%$$

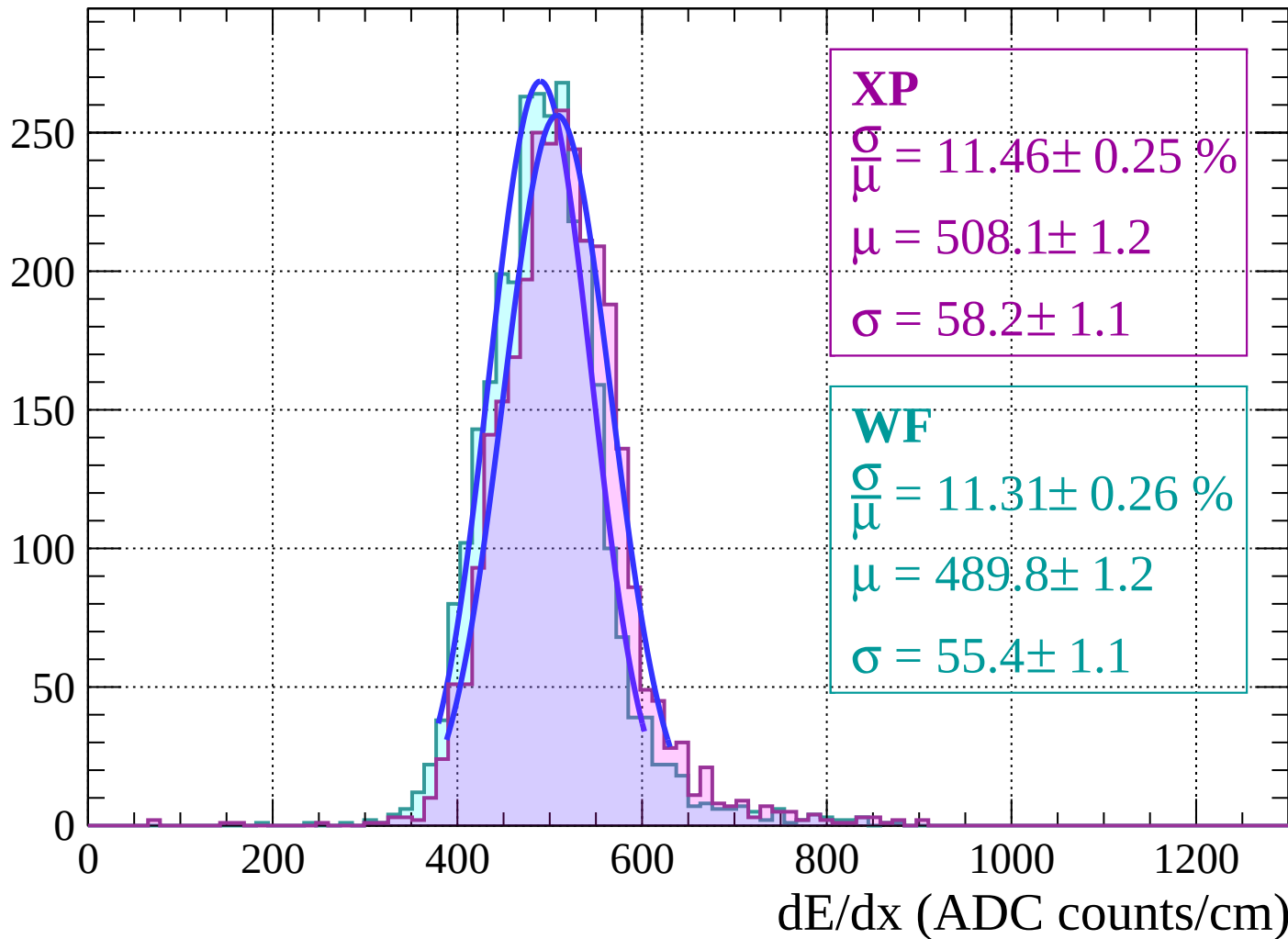
$$\mu = 487.3 \pm 1.2$$

$$\sigma = 55.6 \pm 1.2$$

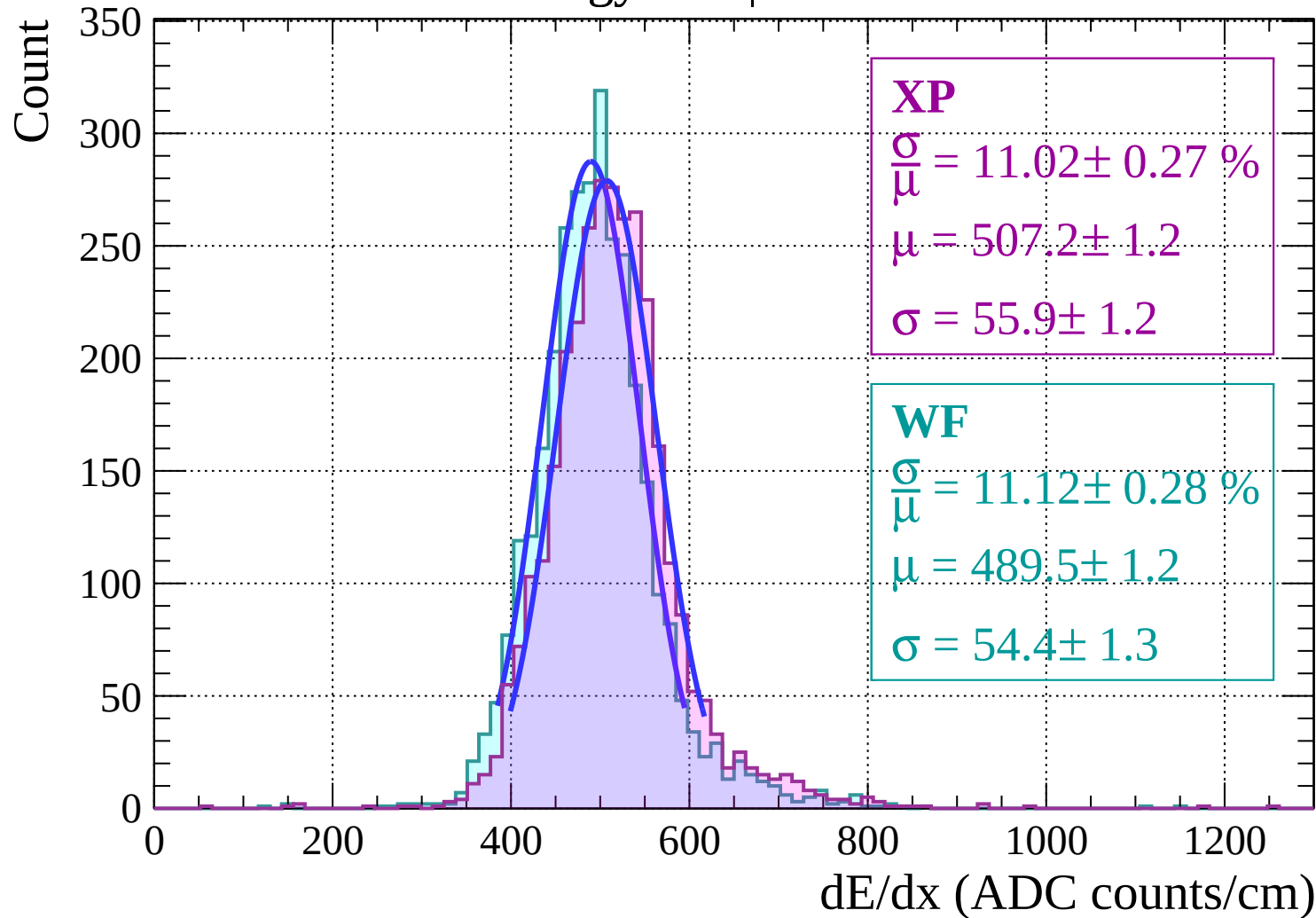
dE/dx (ADC counts/cm)

Energy loss | $-4 < \theta < -2$

Count

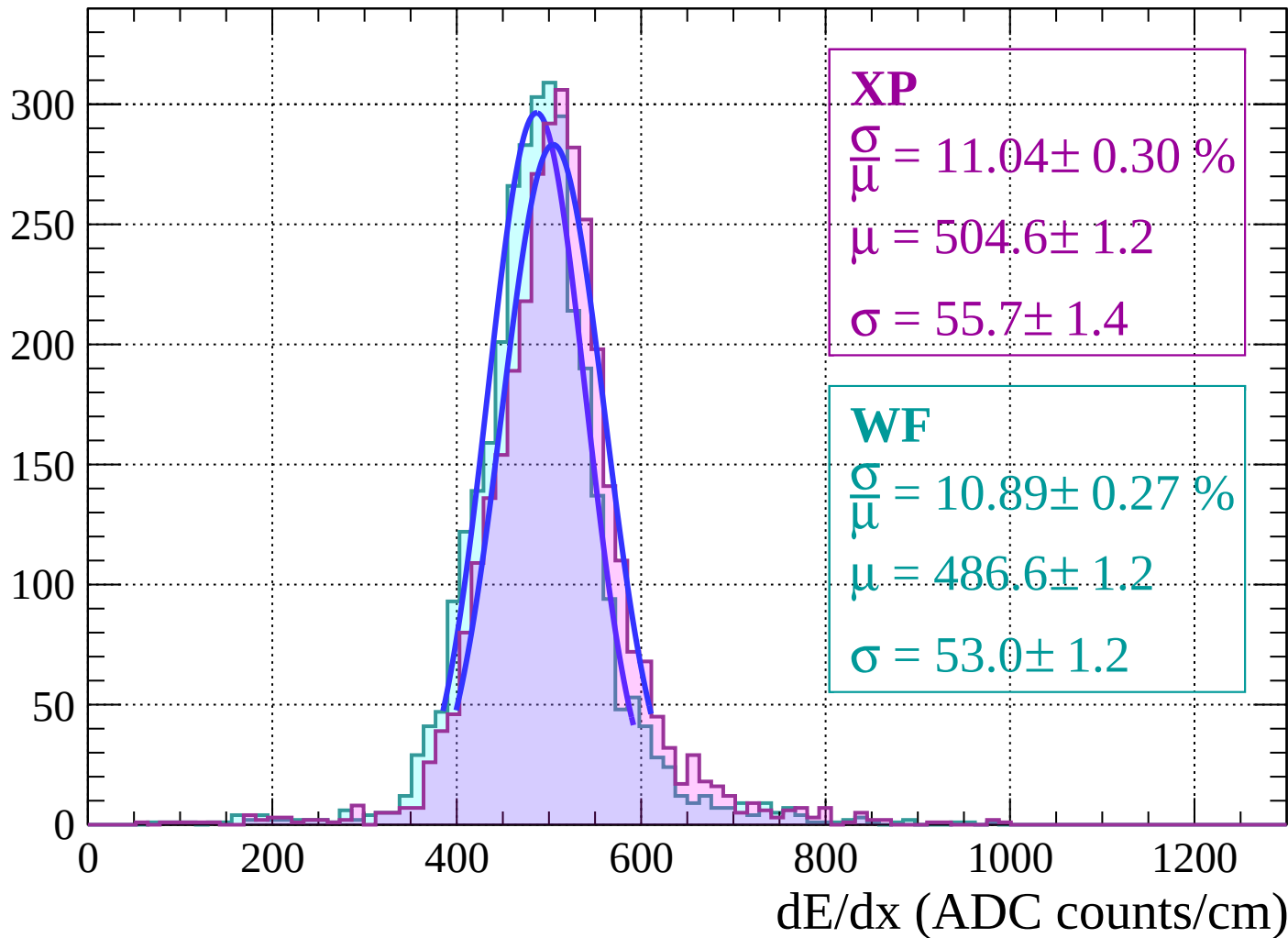


Energy loss | $-2 < \theta < 0$

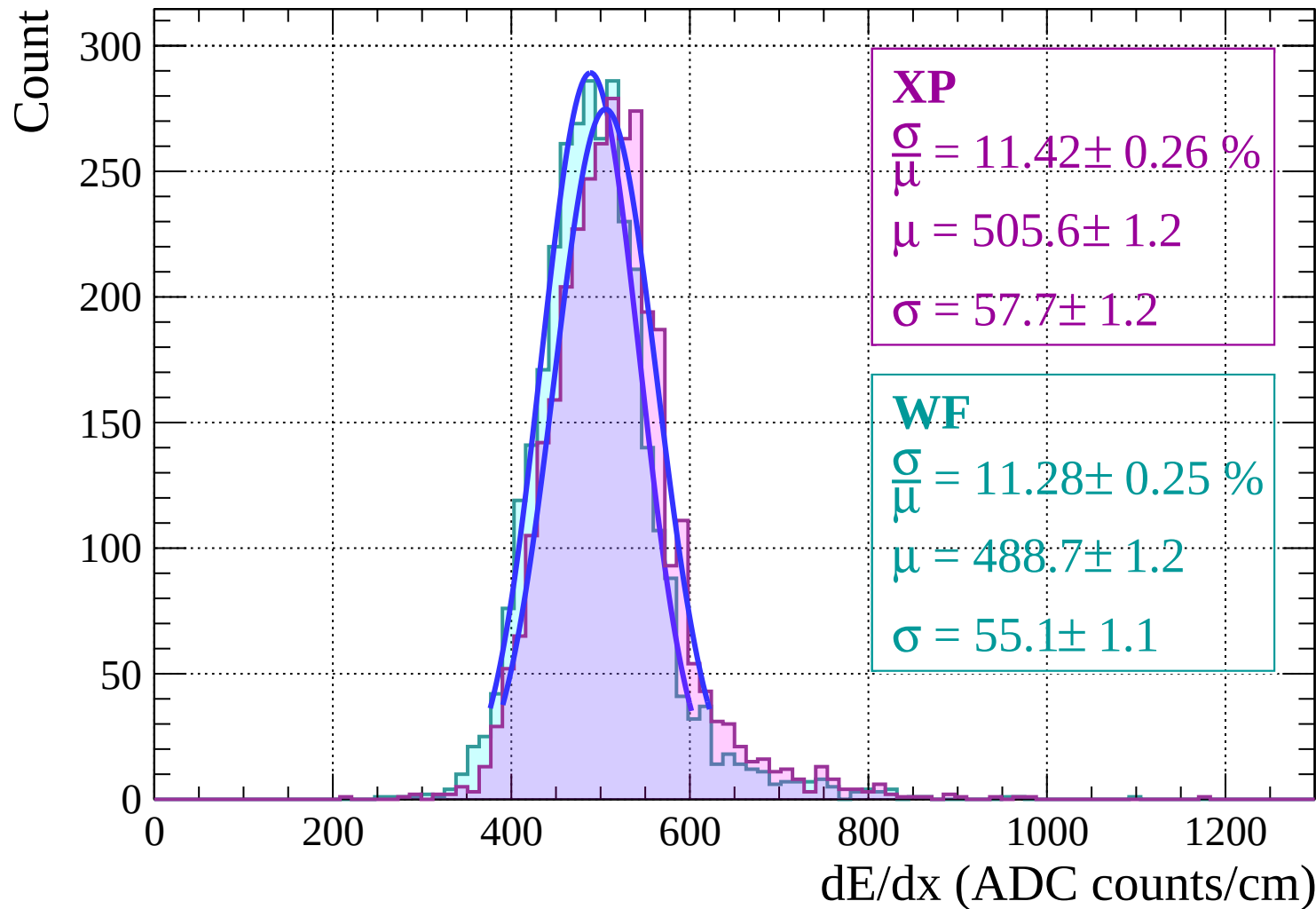


Energy loss | $0 < \theta < 2$

Count



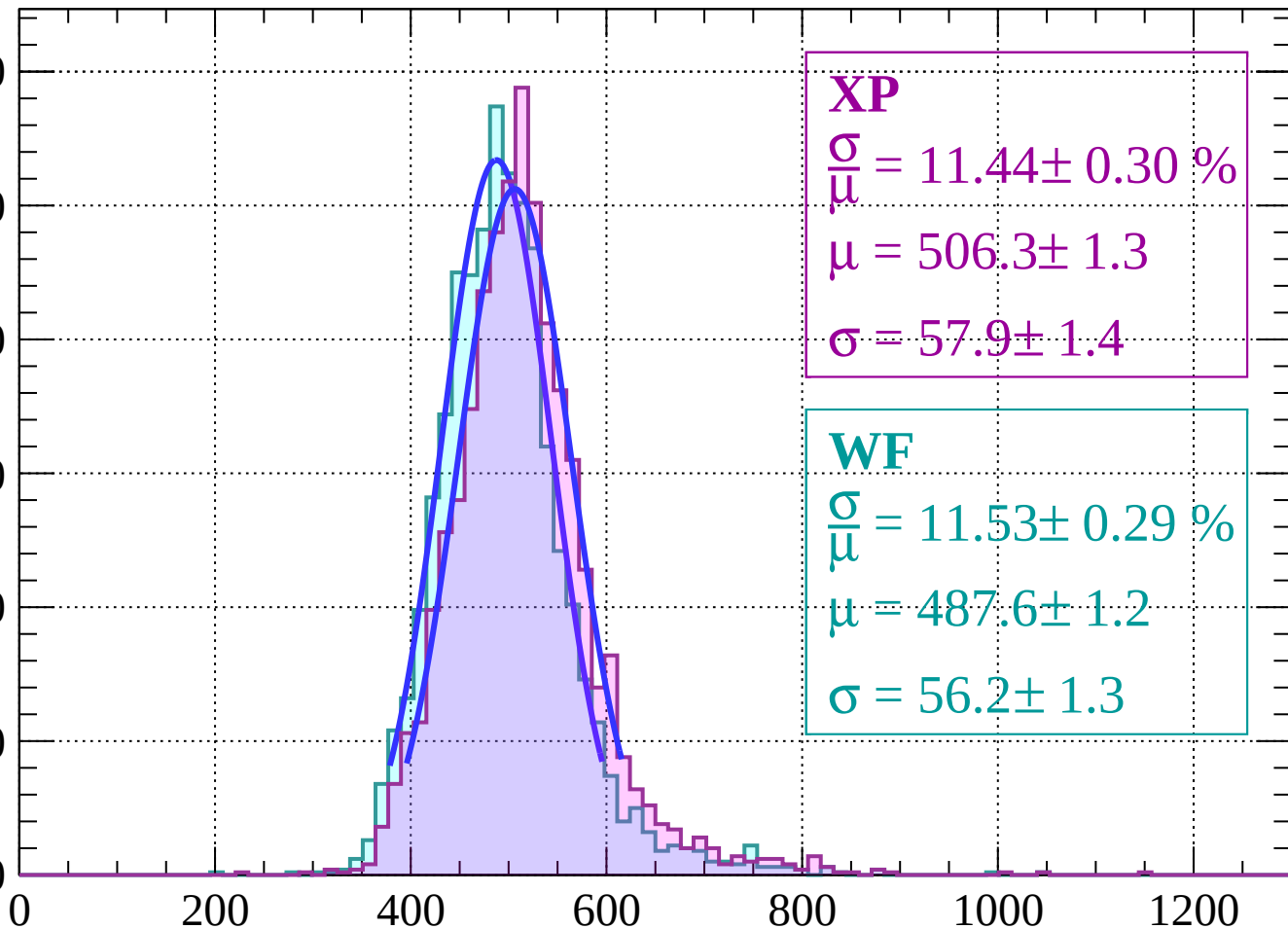
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.44 \pm 0.30 \%$$

$$\mu = 506.3 \pm 1.3$$

$$\sigma = 57.9 \pm 1.4$$

WF

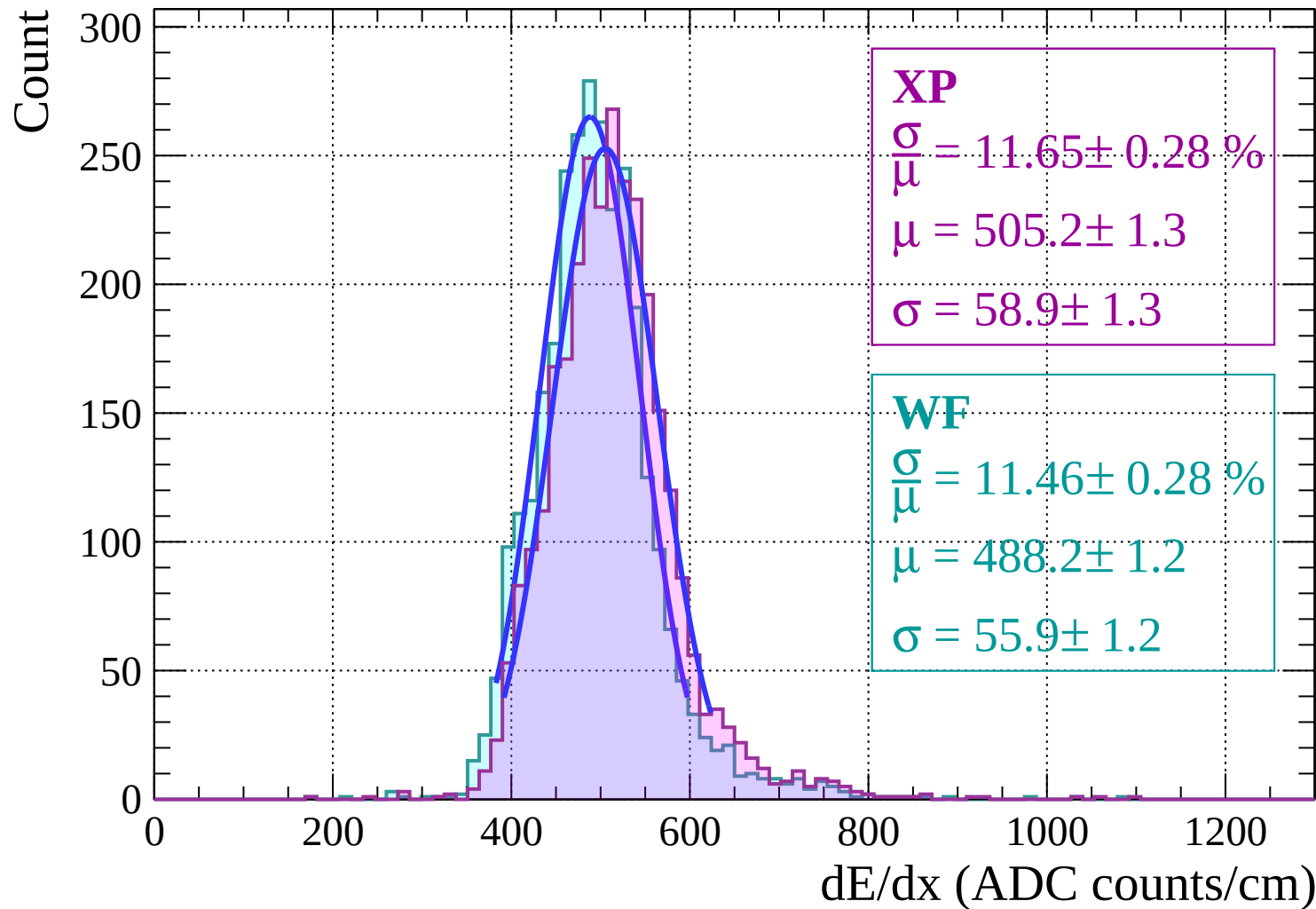
$$\frac{\sigma}{\mu} = 11.53 \pm 0.29 \%$$

$$\mu = 487.6 \pm 1.2$$

$$\sigma = 56.2 \pm 1.3$$

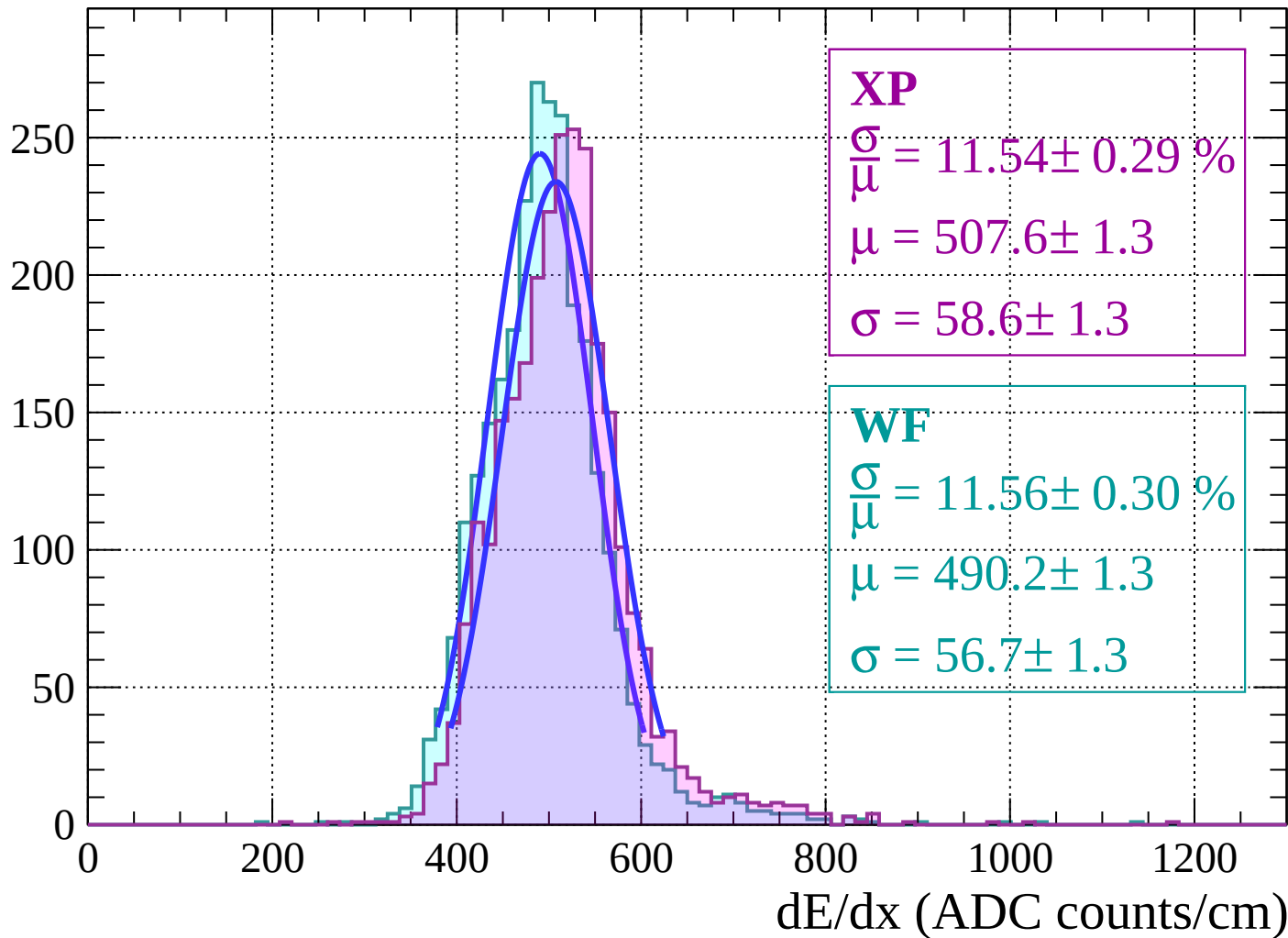
dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 8$



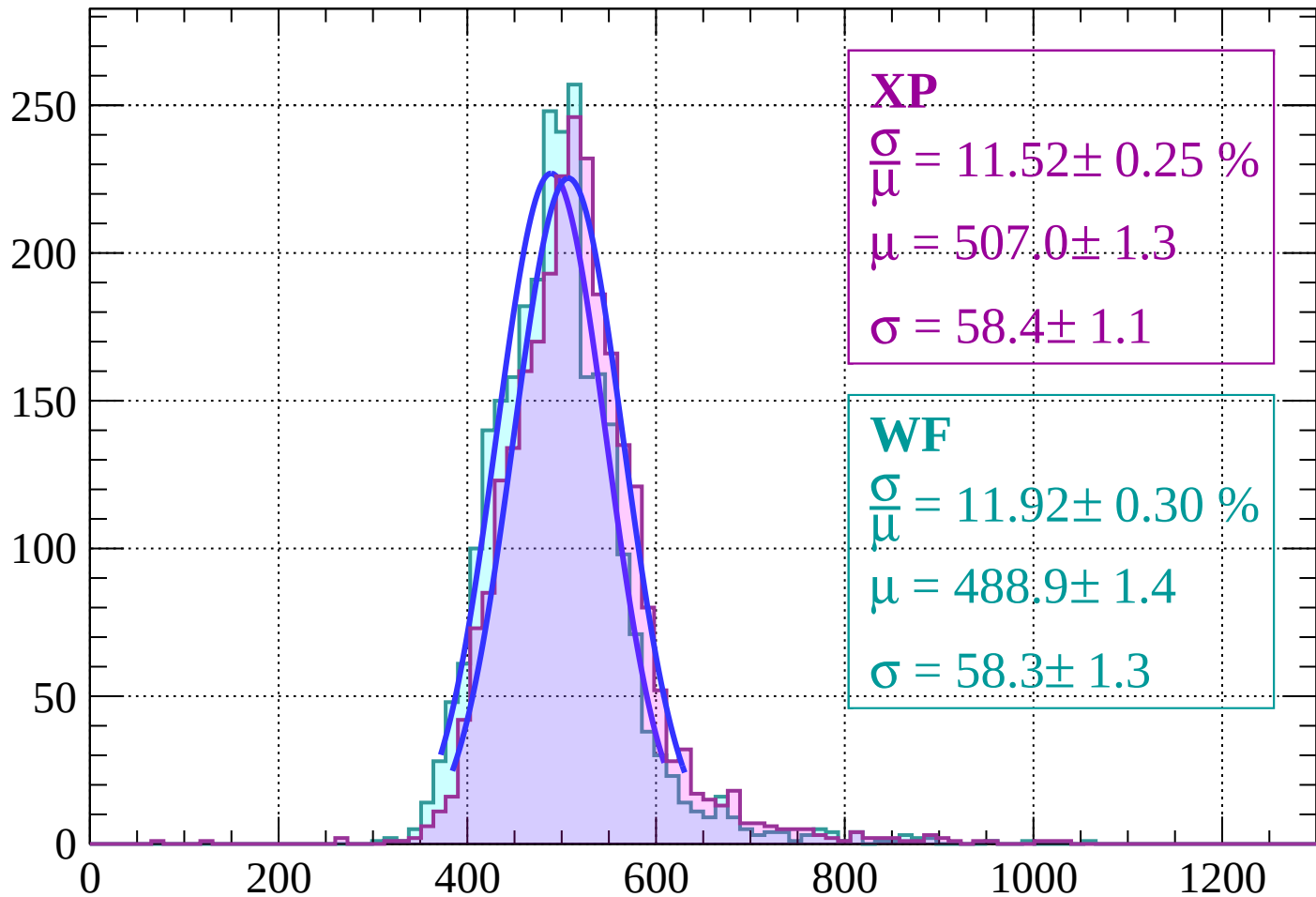
Energy loss | $8 < \theta < 10$

Count



Energy loss | $10 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 11.52 \pm 0.25 \%$$

$$\mu = 507.0 \pm 1.3$$

$$\sigma = 58.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.92 \pm 0.30 \%$$

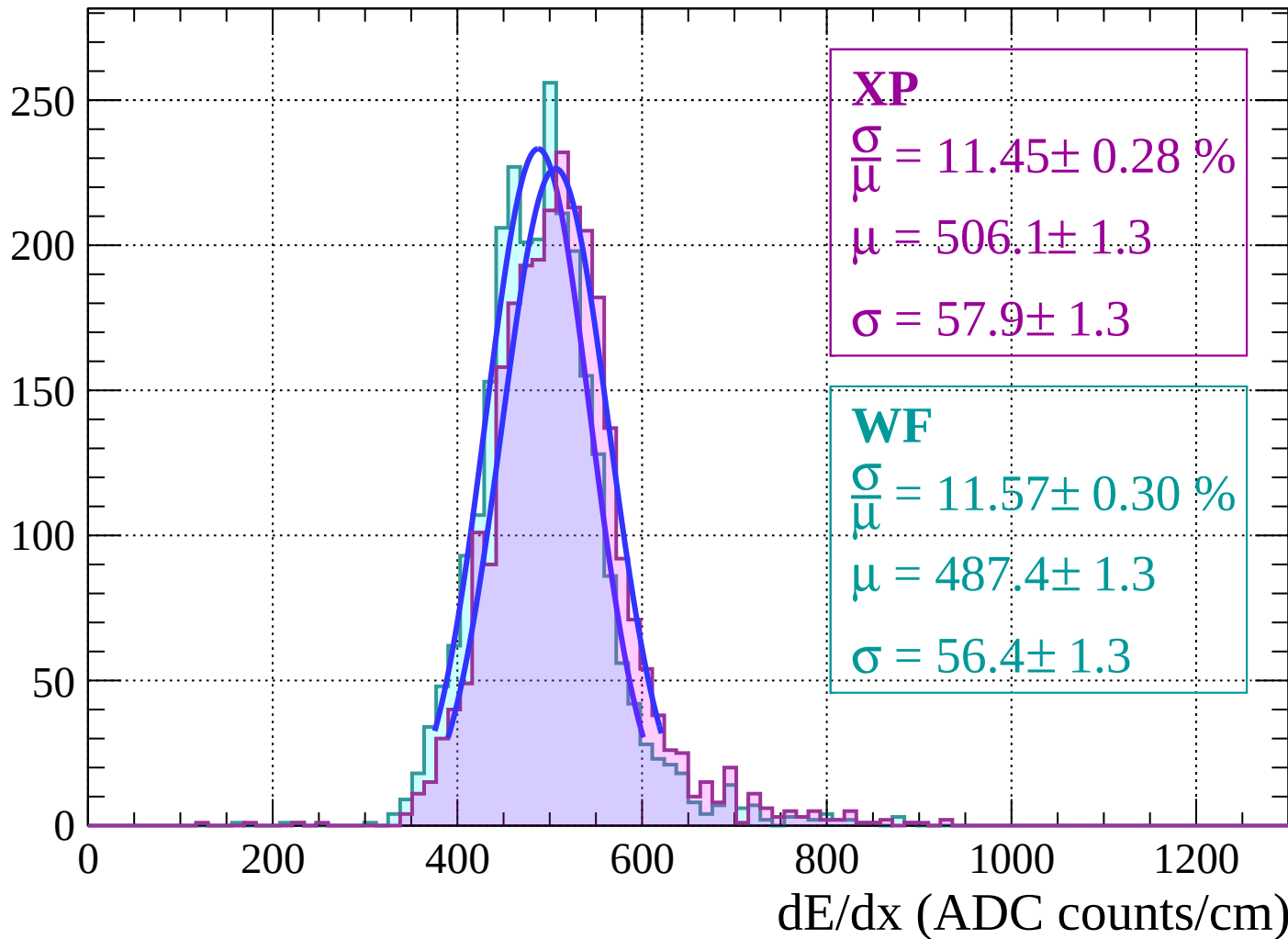
$$\mu = 488.9 \pm 1.4$$

$$\sigma = 58.3 \pm 1.3$$

dE/dx (ADC counts/cm)

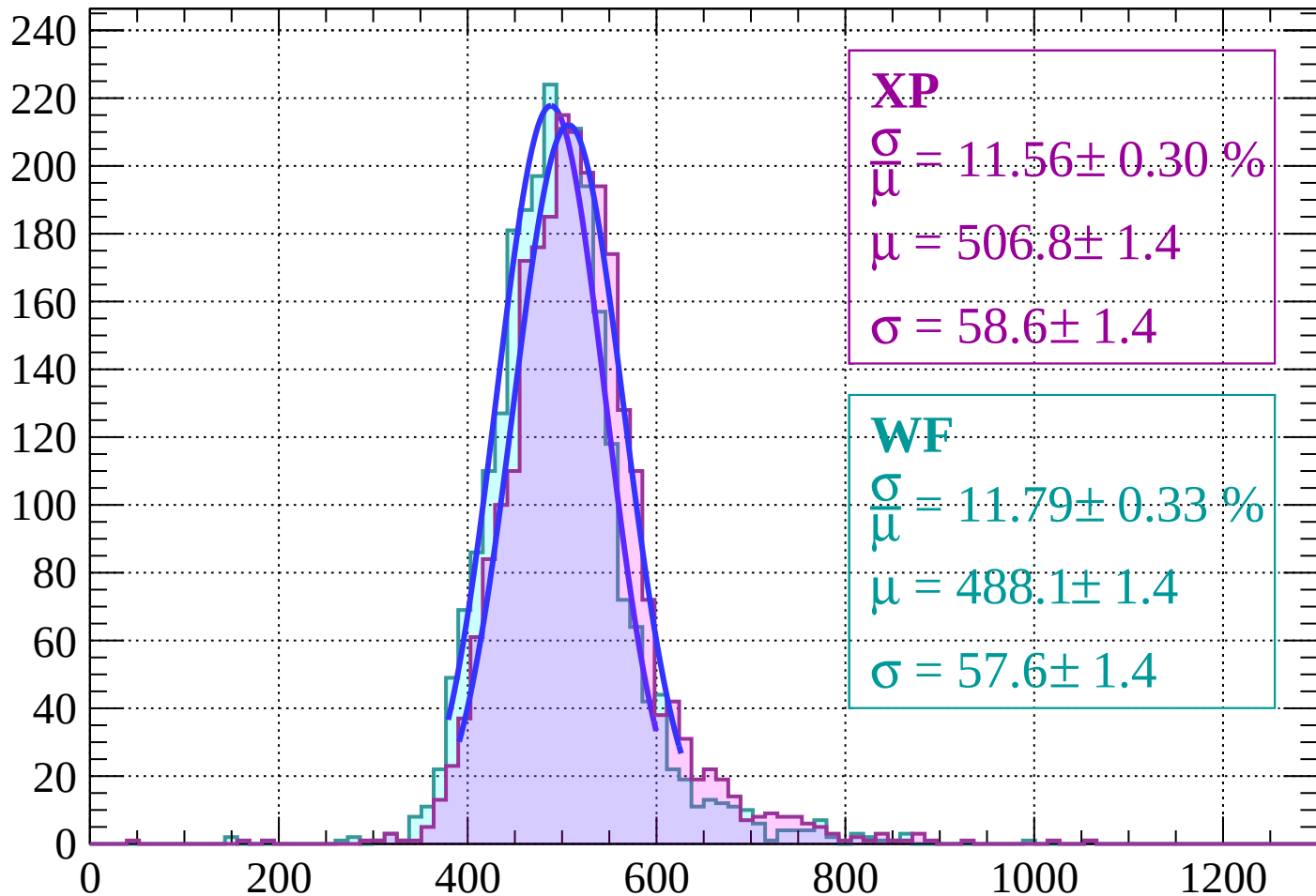
Energy loss | $12 < \theta < 14$

Count



Energy loss | $14 < \theta < 16$

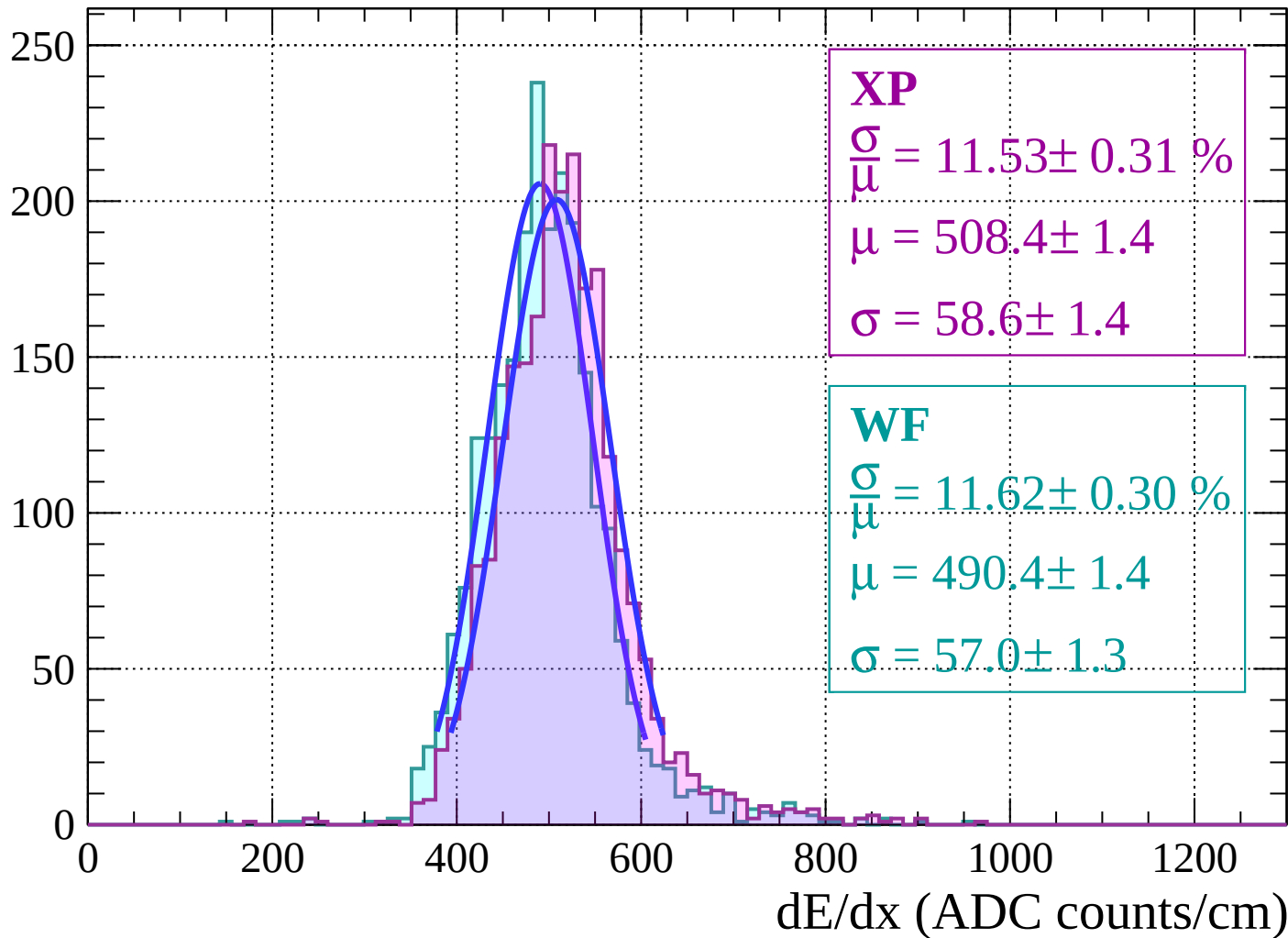
Count



dE/dx (ADC counts/cm)

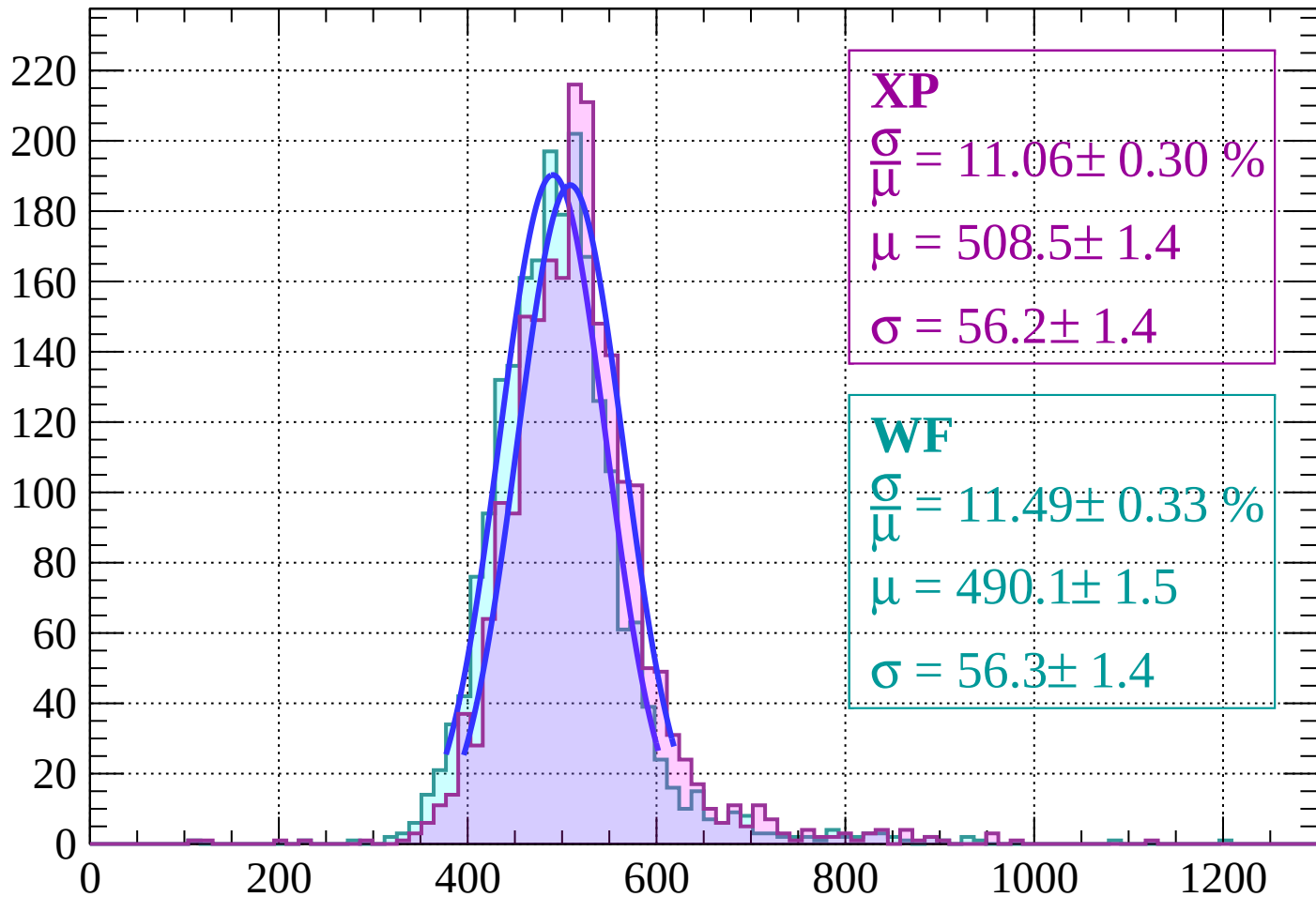
Energy loss | $16 < \theta < 18$

Count



Energy loss | $18 < \theta < 20$

Count



XP

$$\frac{\sigma}{\mu} = 11.06 \pm 0.30 \%$$

$$\mu = 508.5 \pm 1.4$$

$$\sigma = 56.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.49 \pm 0.33 \%$$

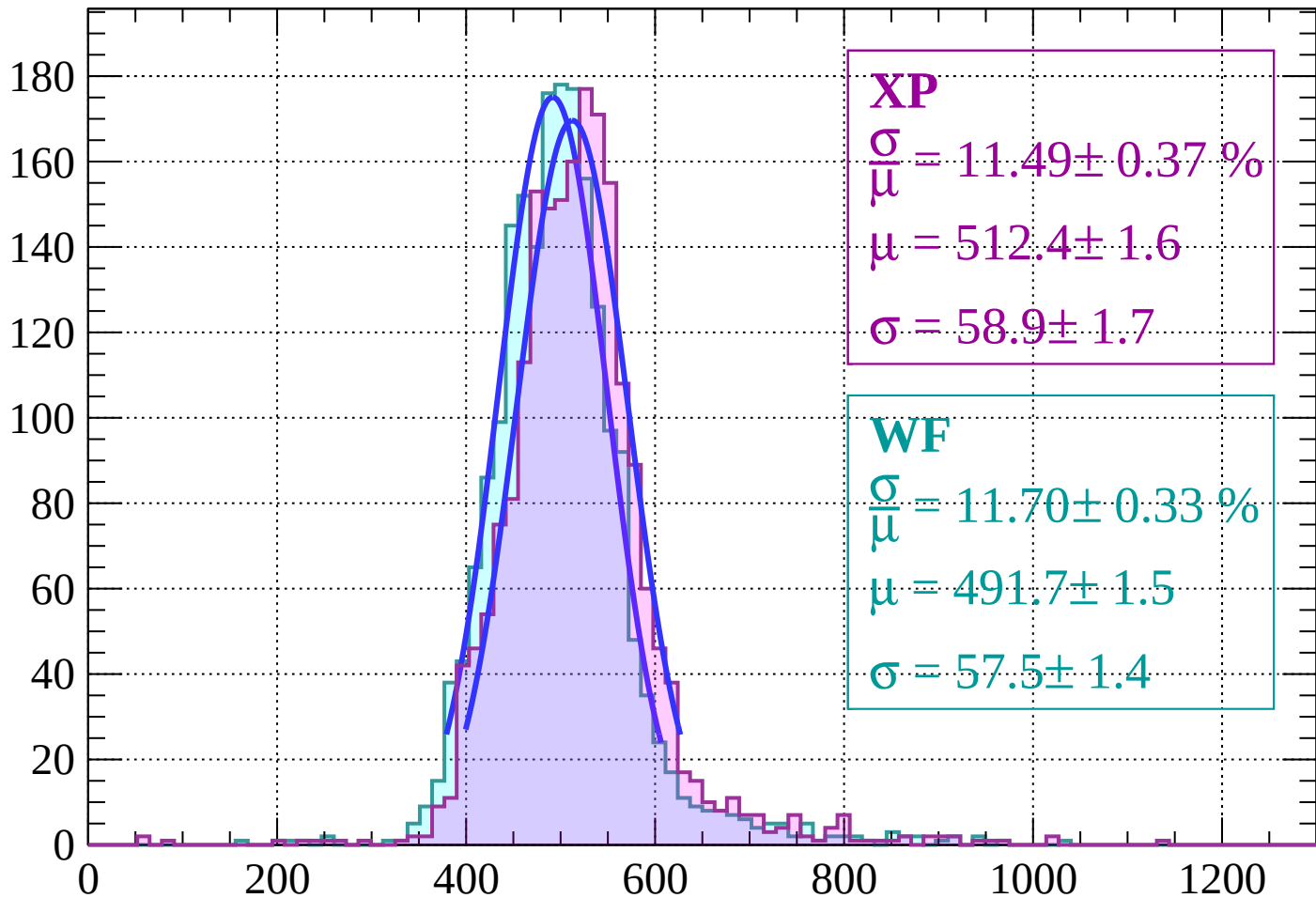
$$\mu = 490.1 \pm 1.5$$

$$\sigma = 56.3 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $20 < \theta < 22$

Count



XP

$$\frac{\sigma}{\mu} = 11.49 \pm 0.37 \%$$

$$\mu = 512.4 \pm 1.6$$

$$\sigma = 58.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 11.70 \pm 0.33 \%$$

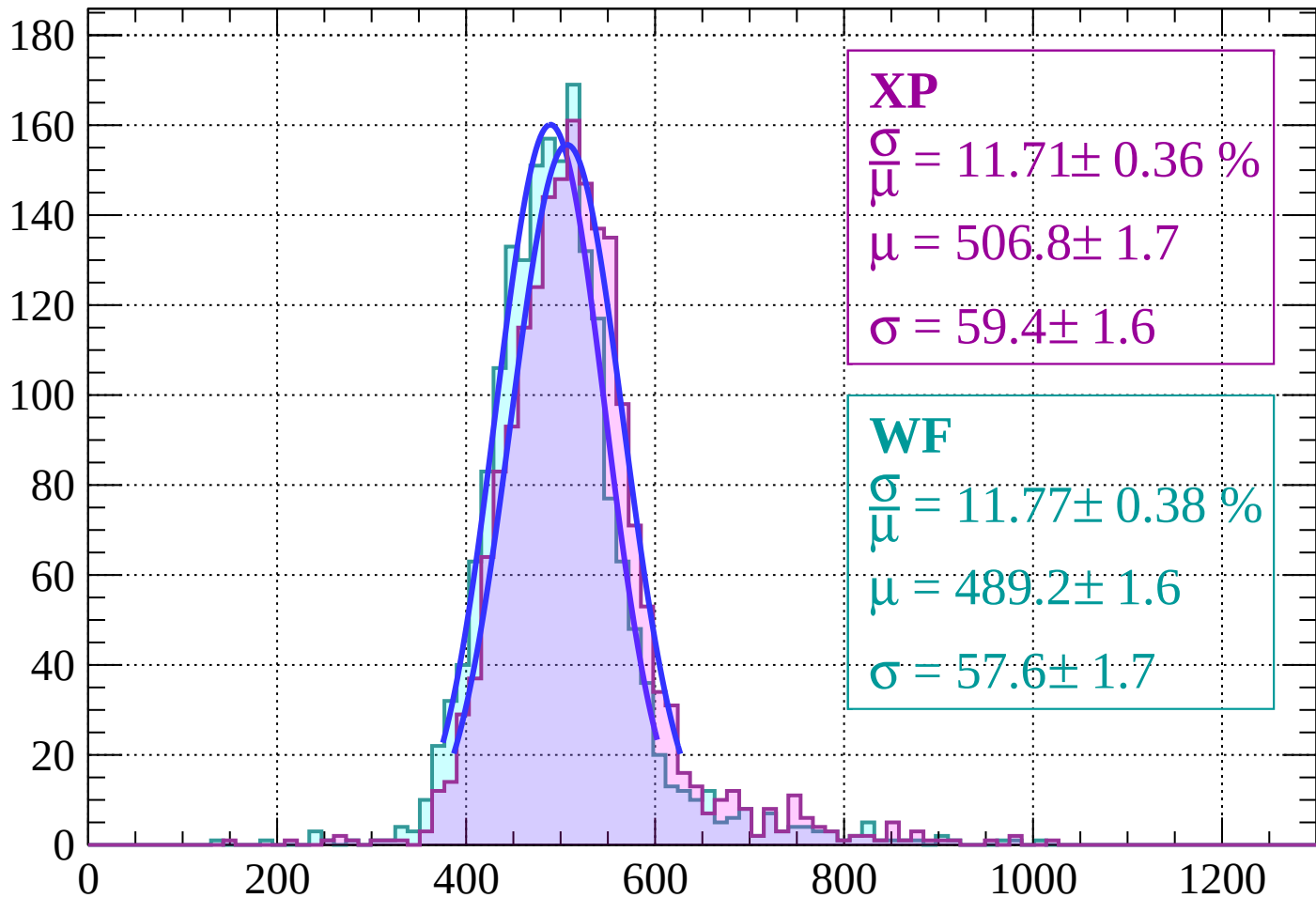
$$\mu = 491.7 \pm 1.5$$

$$\sigma = 57.5 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $22 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 11.71 \pm 0.36 \%$$

$$\mu = 506.8 \pm 1.7$$

$$\sigma = 59.4 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 11.77 \pm 0.38 \%$$

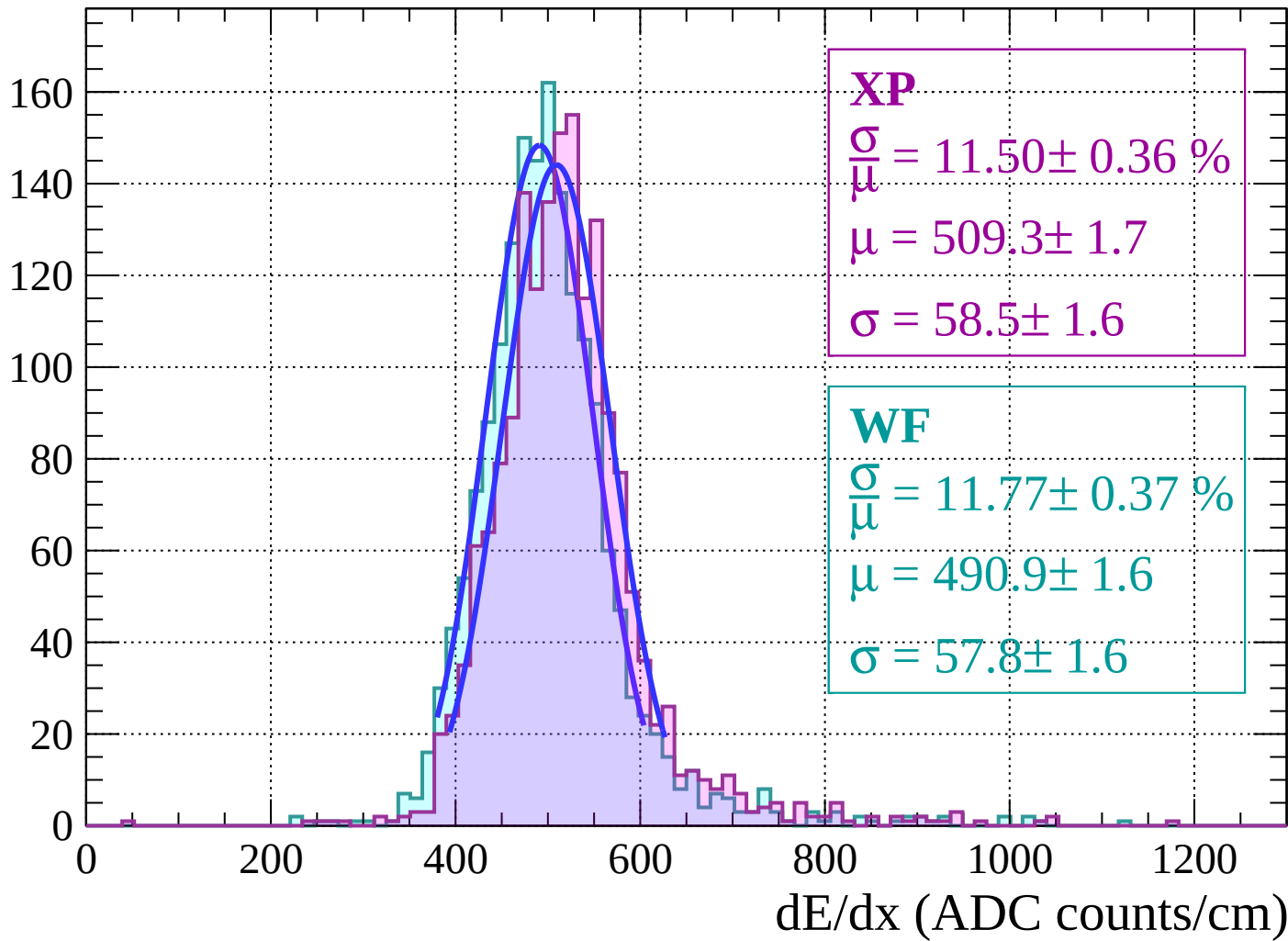
$$\mu = 489.2 \pm 1.6$$

$$\sigma = 57.6 \pm 1.7$$

dE/dx (ADC counts/cm)

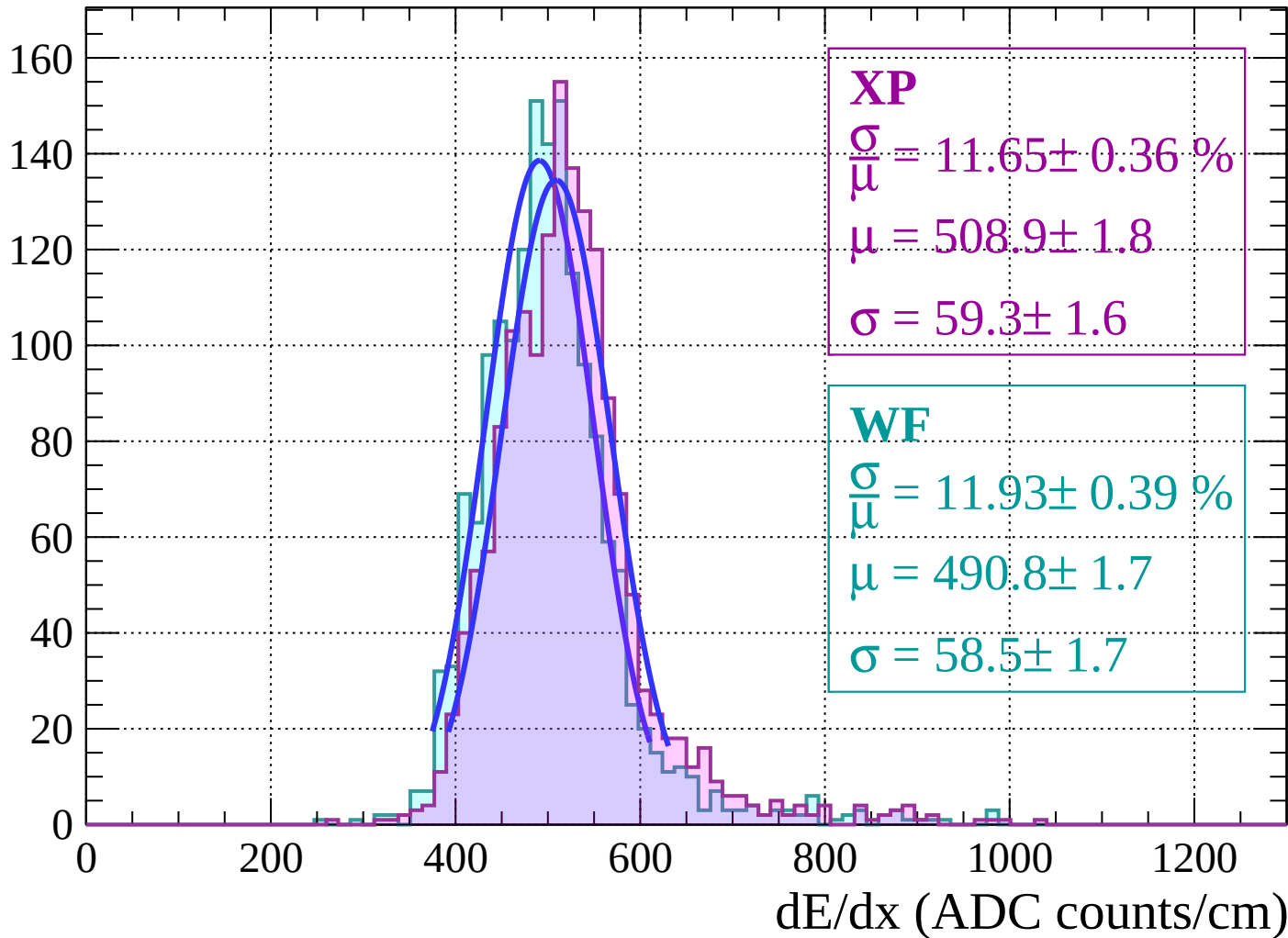
Energy loss | $24 < \theta < 26$

Count



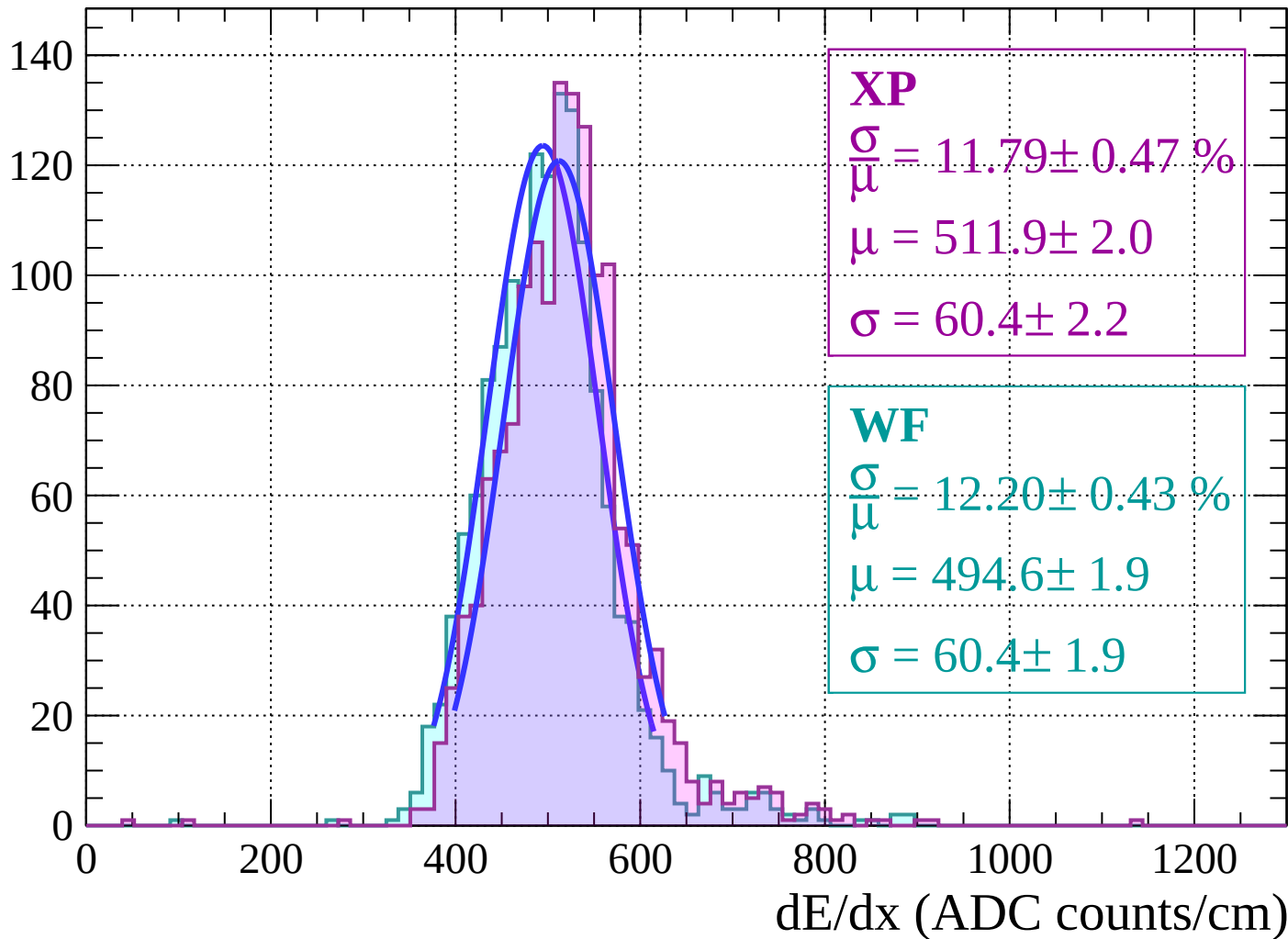
Energy loss | $26 < \theta < 28$

Count



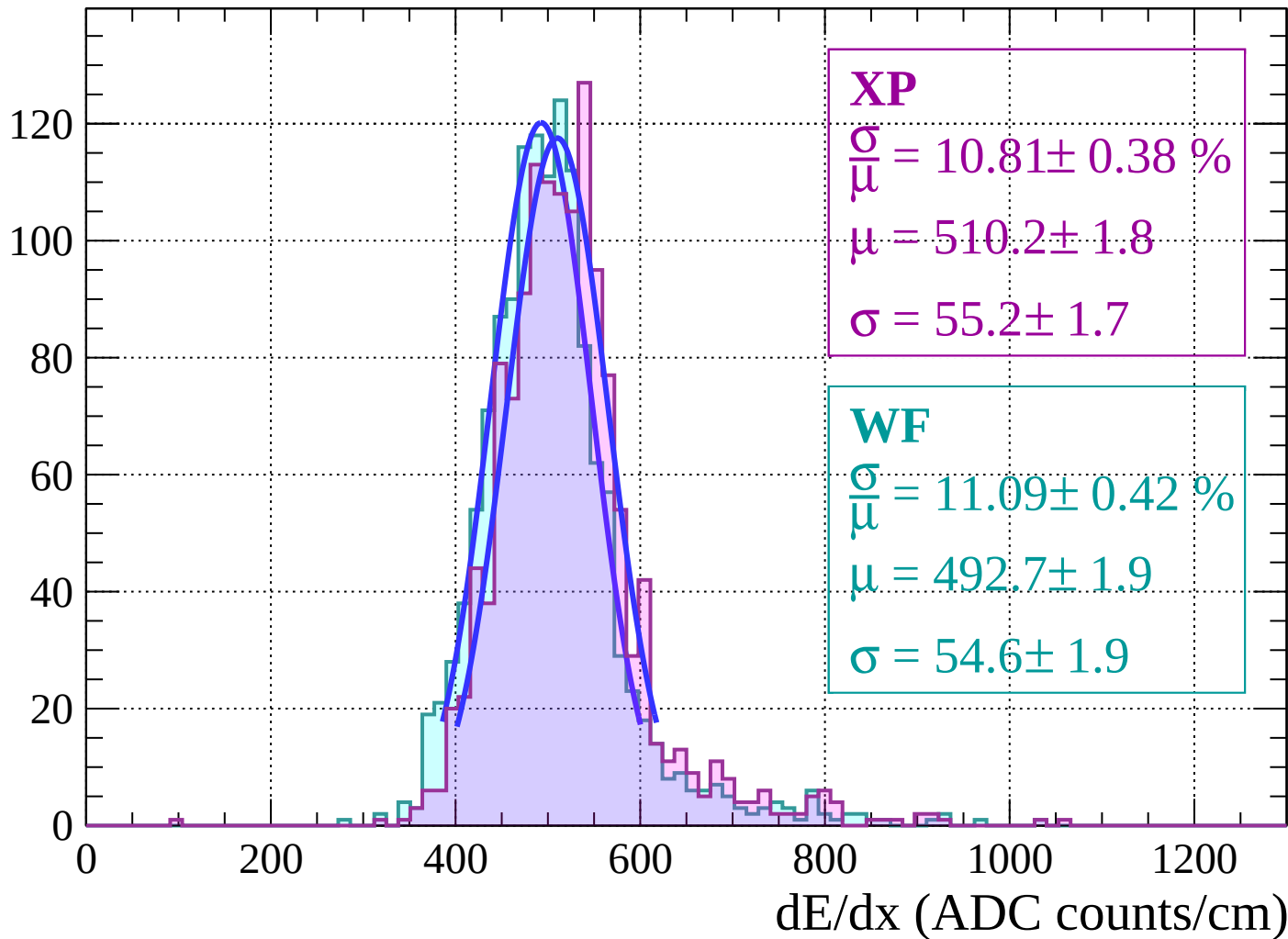
Energy loss | $28 < \theta < 30$

Count



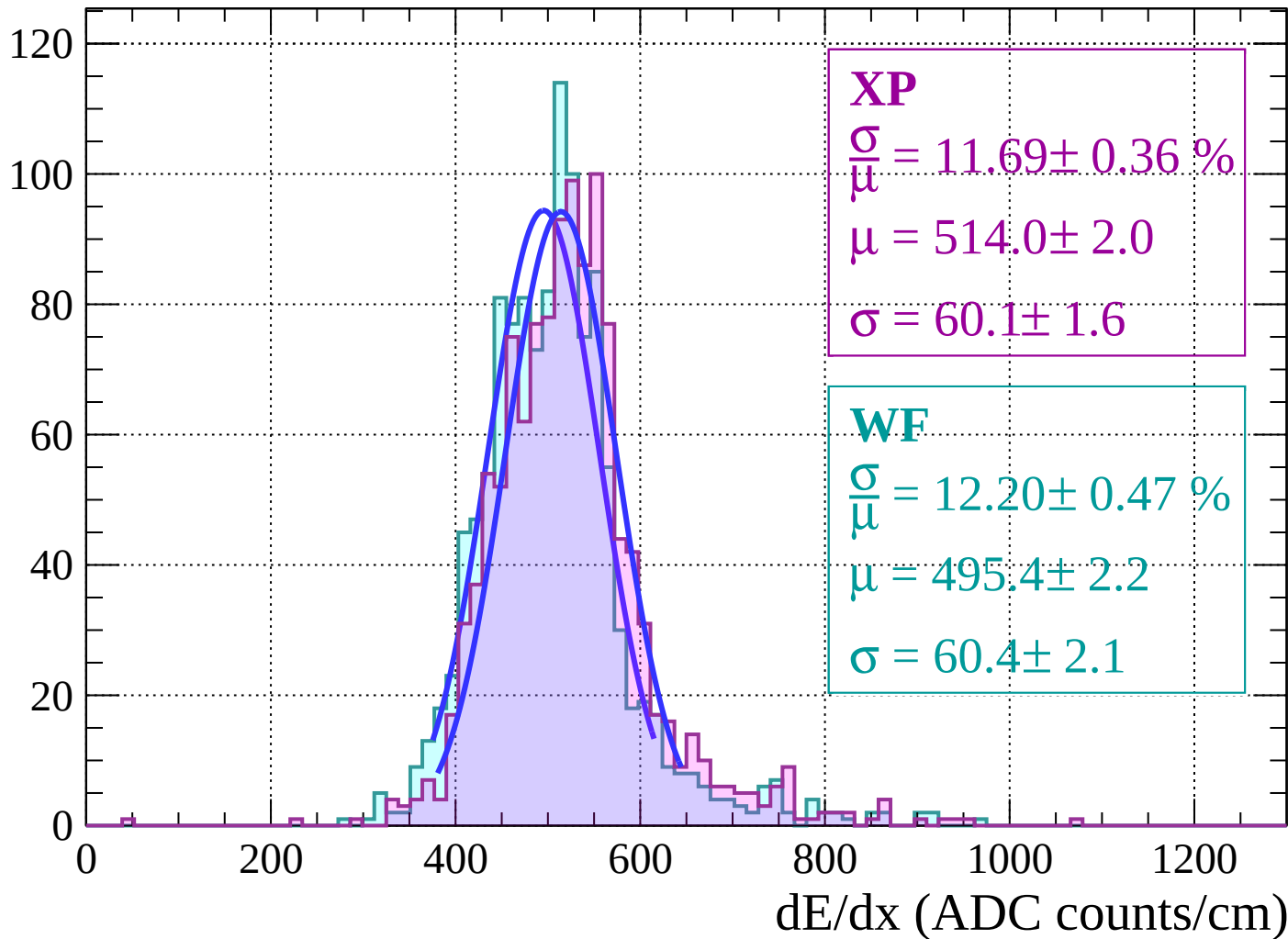
Energy loss | $30 < \theta < 32$

Count



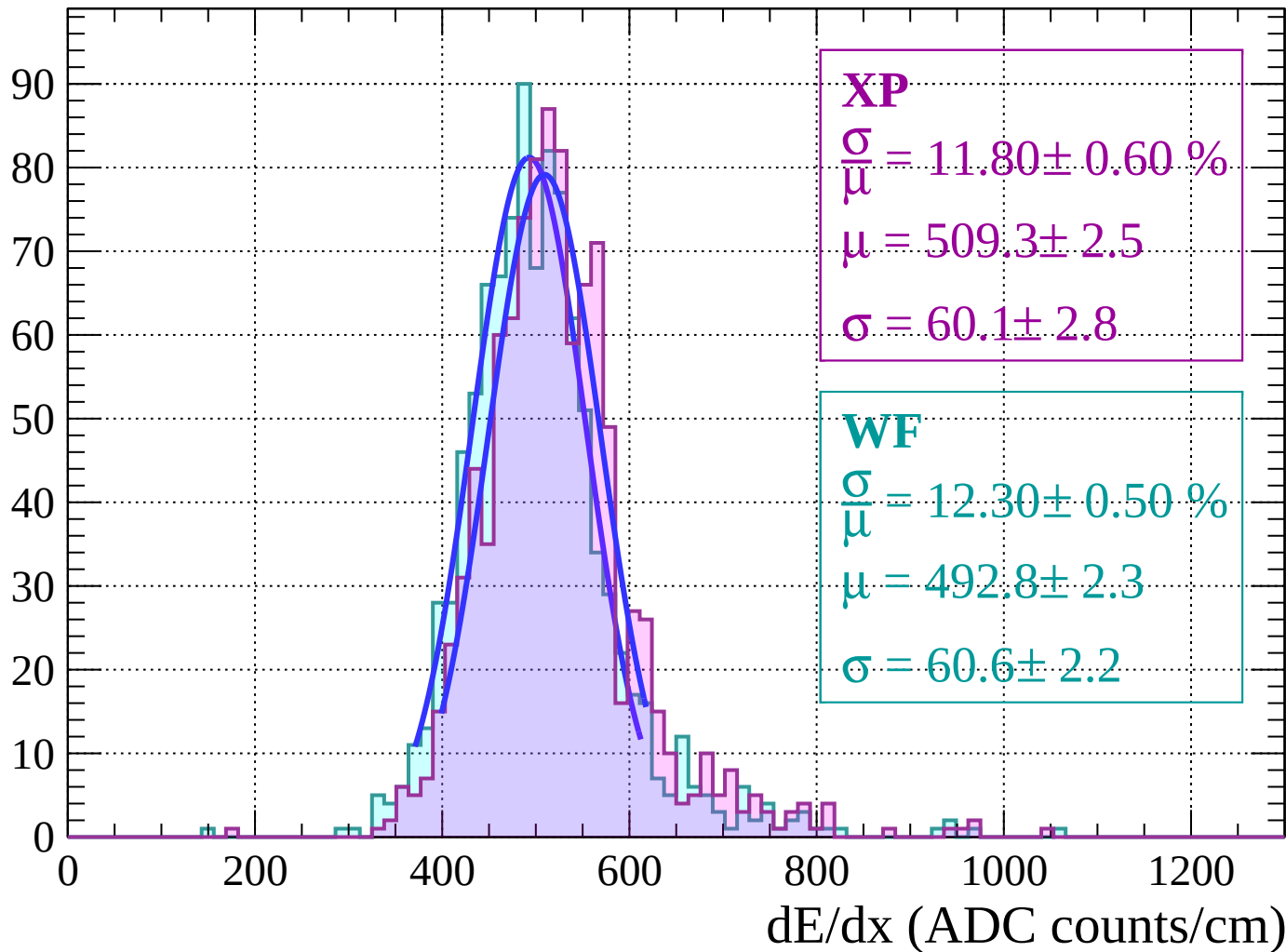
Energy loss | $32 < \theta < 34$

Count



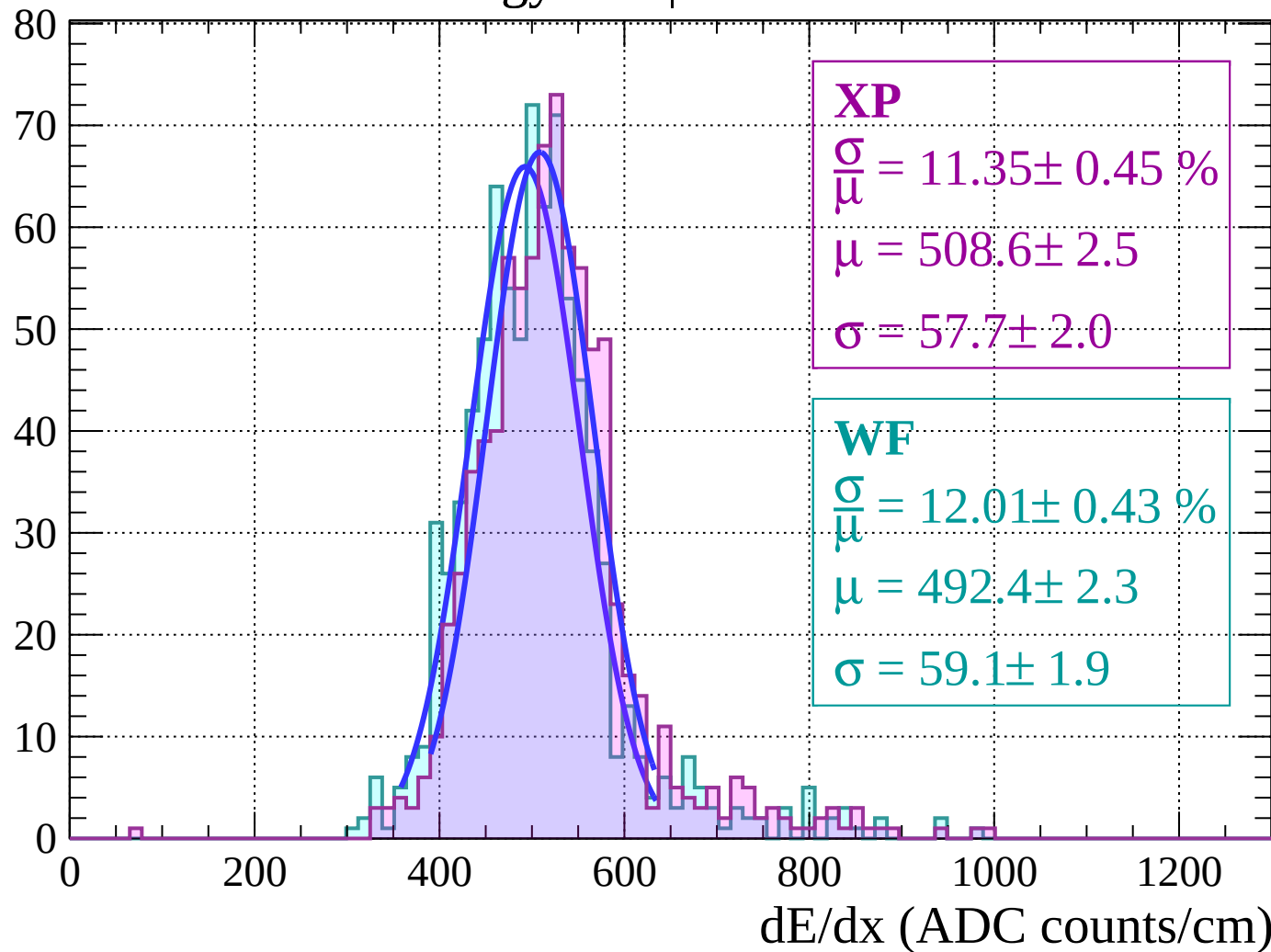
Energy loss | $34 < \theta < 36$

Count



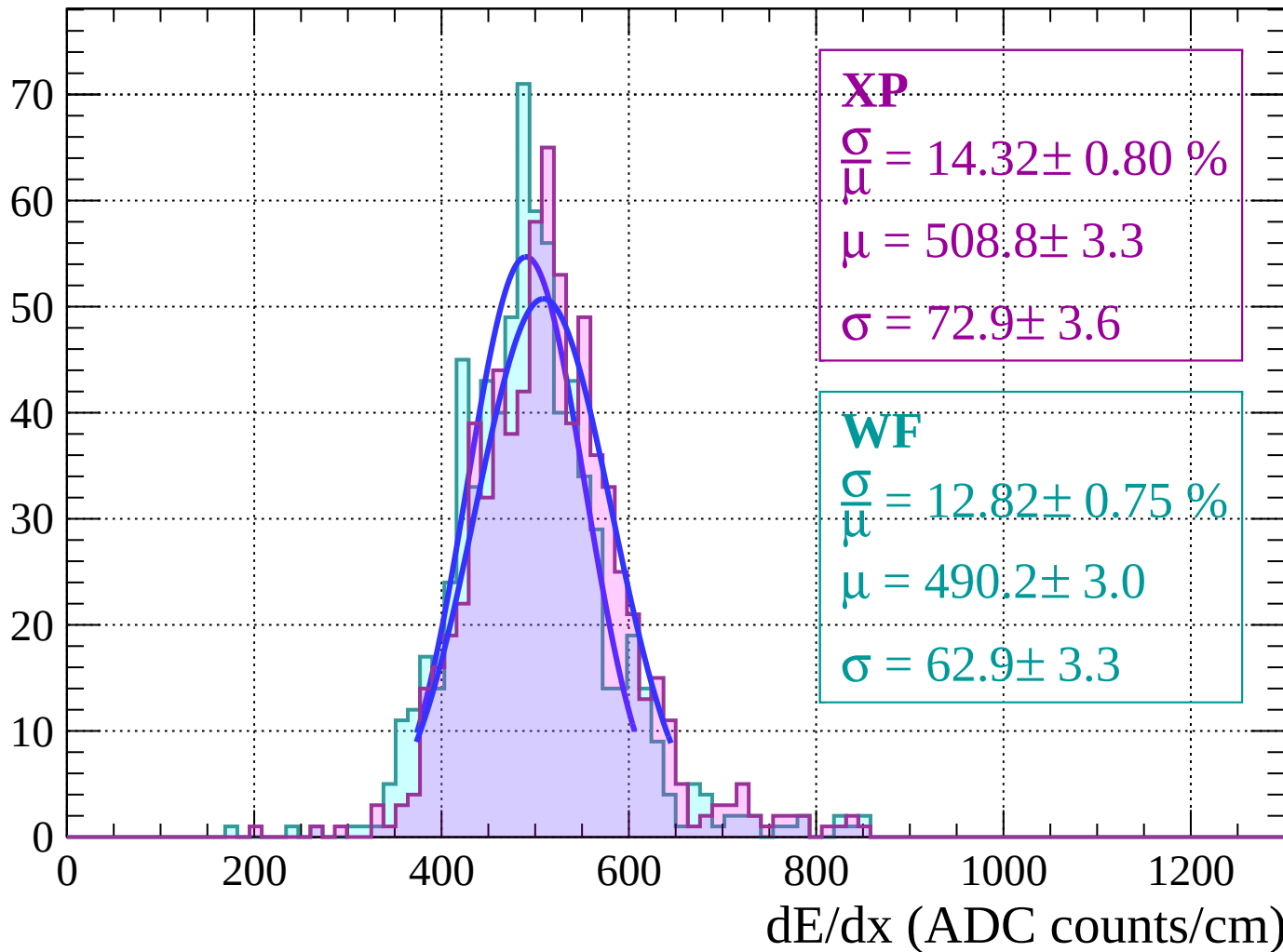
Energy loss | $36 < \theta < 38$

Count



Energy loss | $38 < \theta < 40$

Count



Energy loss | $40 < \theta < 42$

Count

XP

$$\frac{\sigma}{\mu} = 11.73 \pm 0.75 \%$$

$$\mu = 519.0 \pm 3.3$$

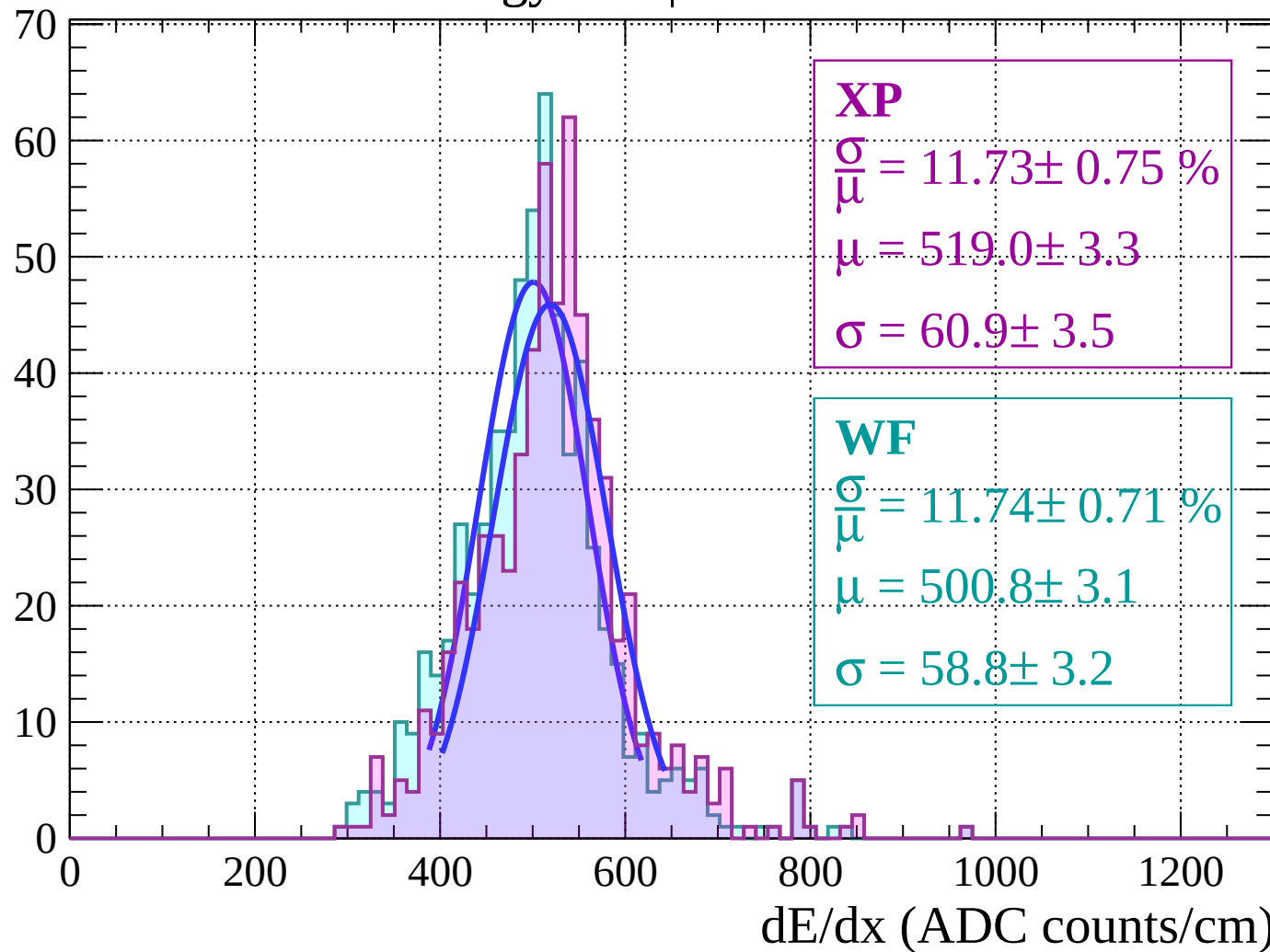
$$\sigma = 60.9 \pm 3.5$$

WF

$$\frac{\sigma}{\mu} = 11.74 \pm 0.71 \%$$

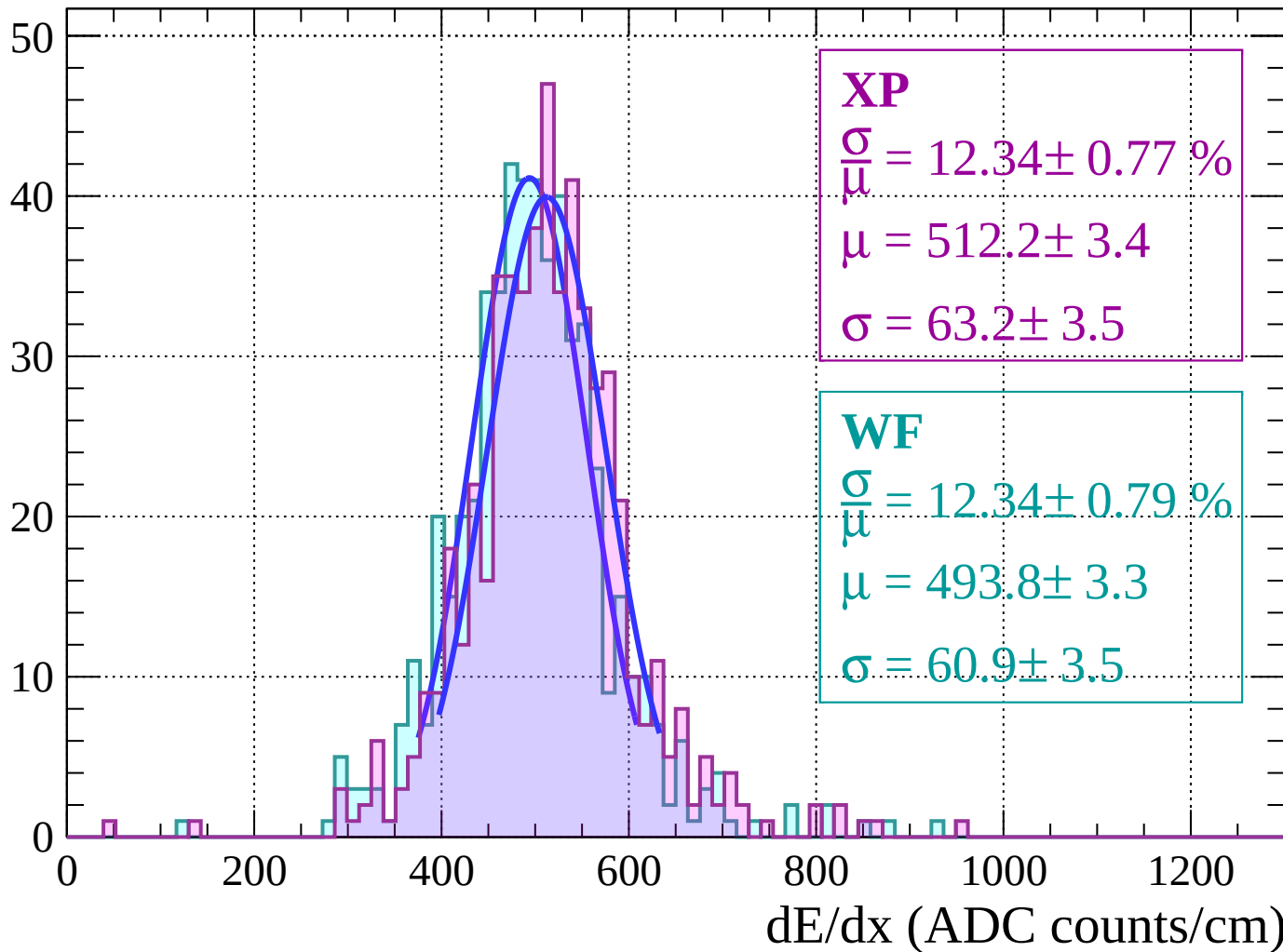
$$\mu = 500.8 \pm 3.1$$

$$\sigma = 58.8 \pm 3.2$$



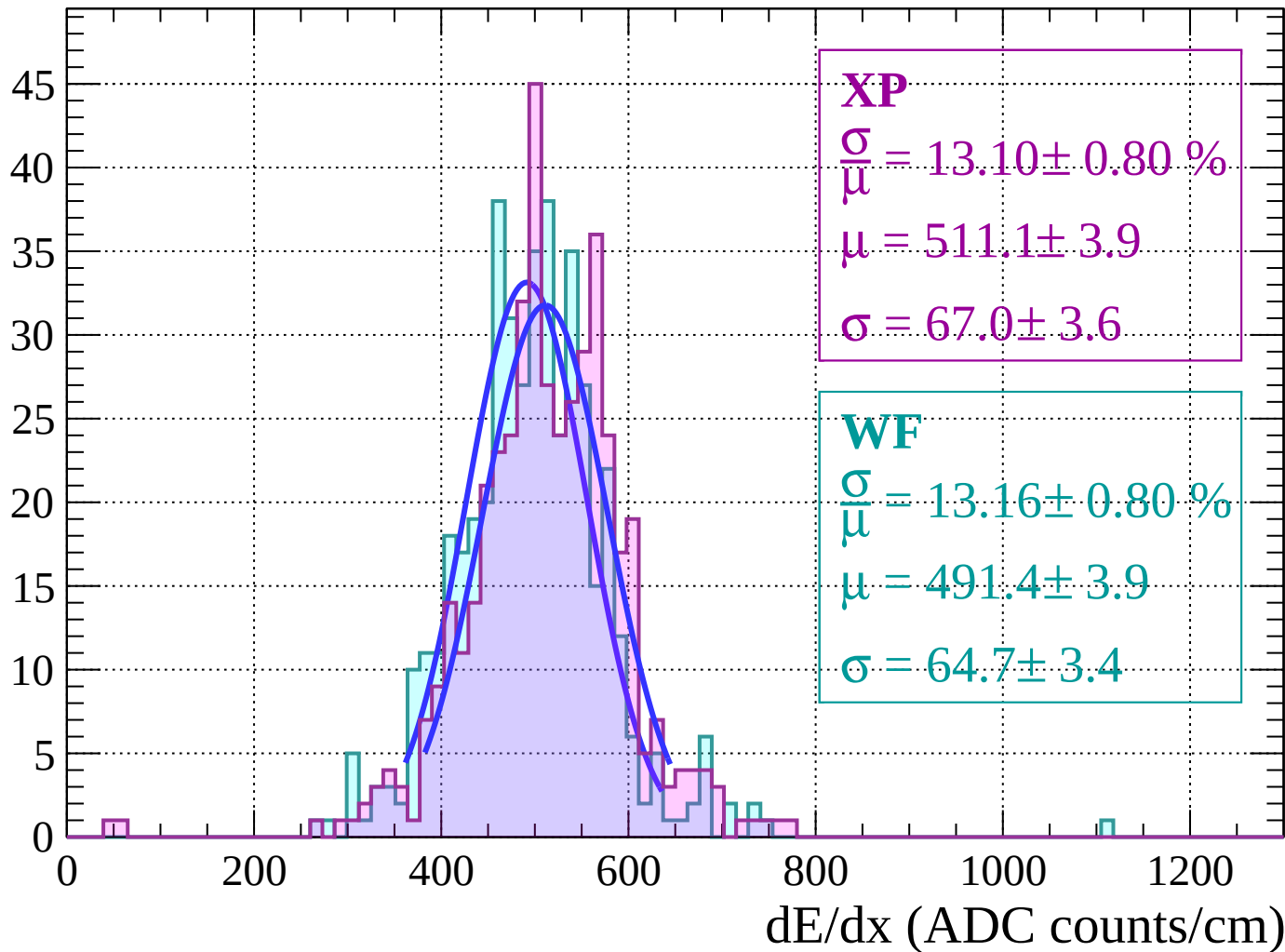
Energy loss | $42 < \theta < 44$

Count



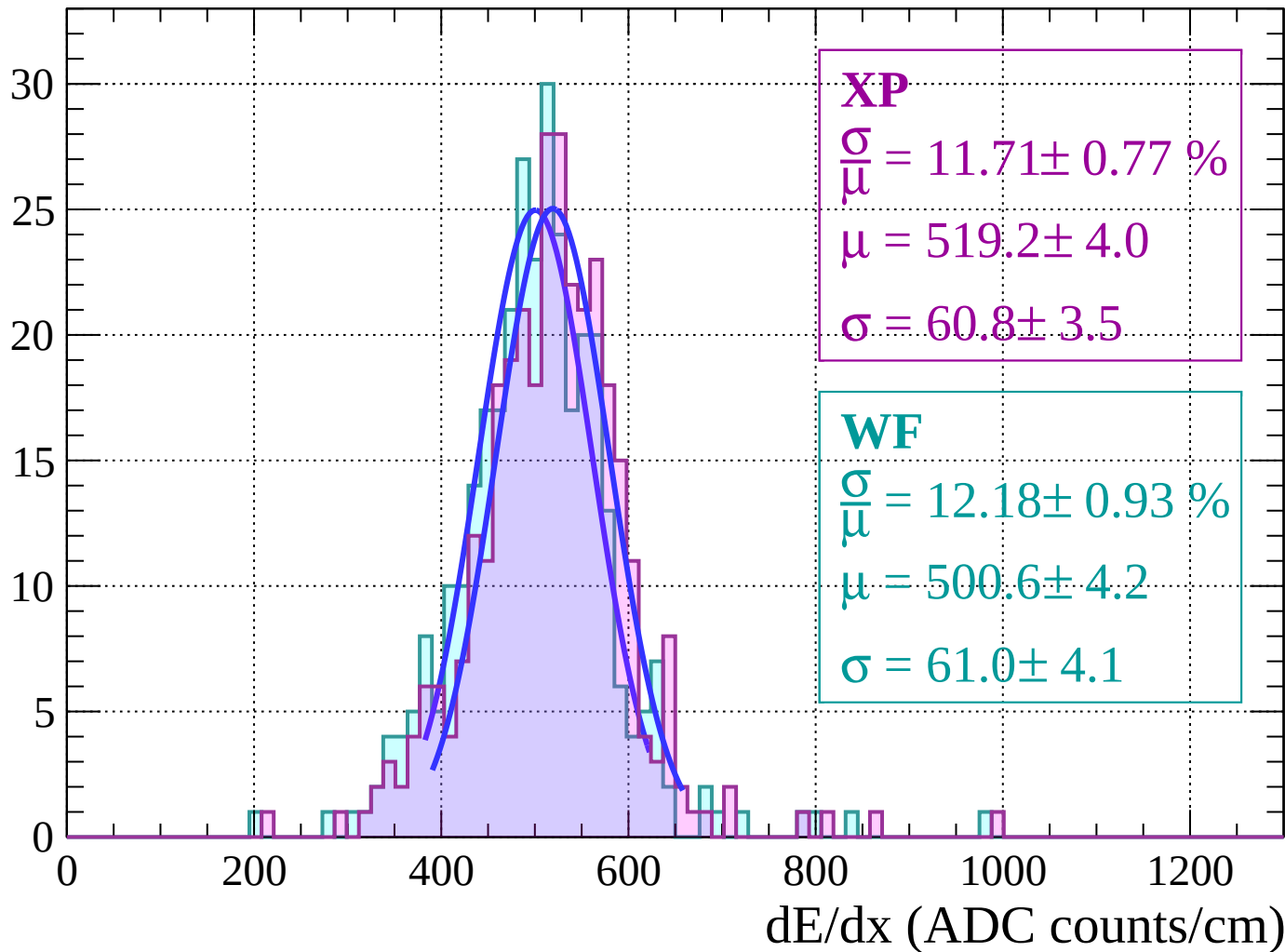
Energy loss | $44 < \theta < 46$

Count



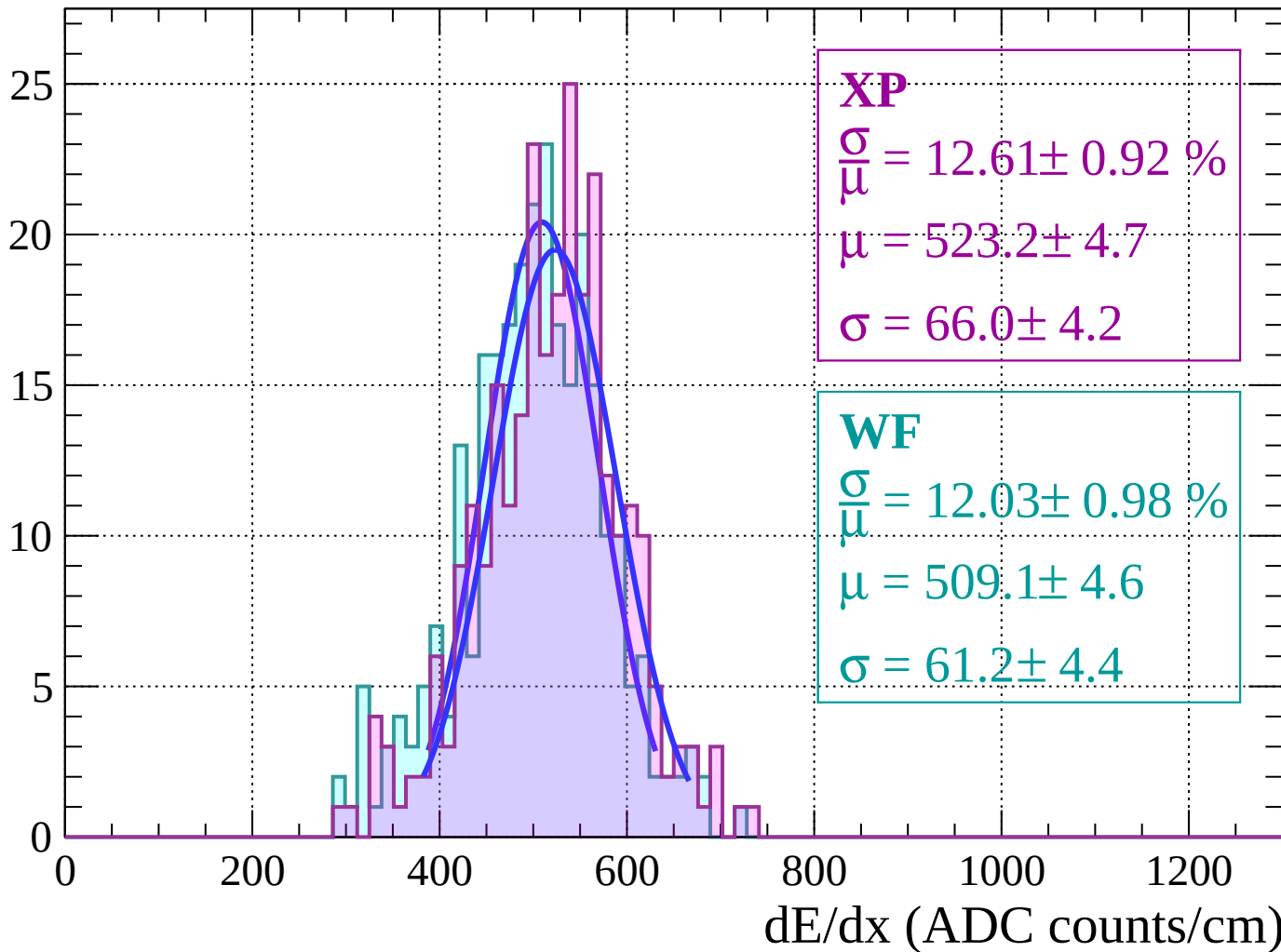
Energy loss | $46 < \theta < 48$

Count



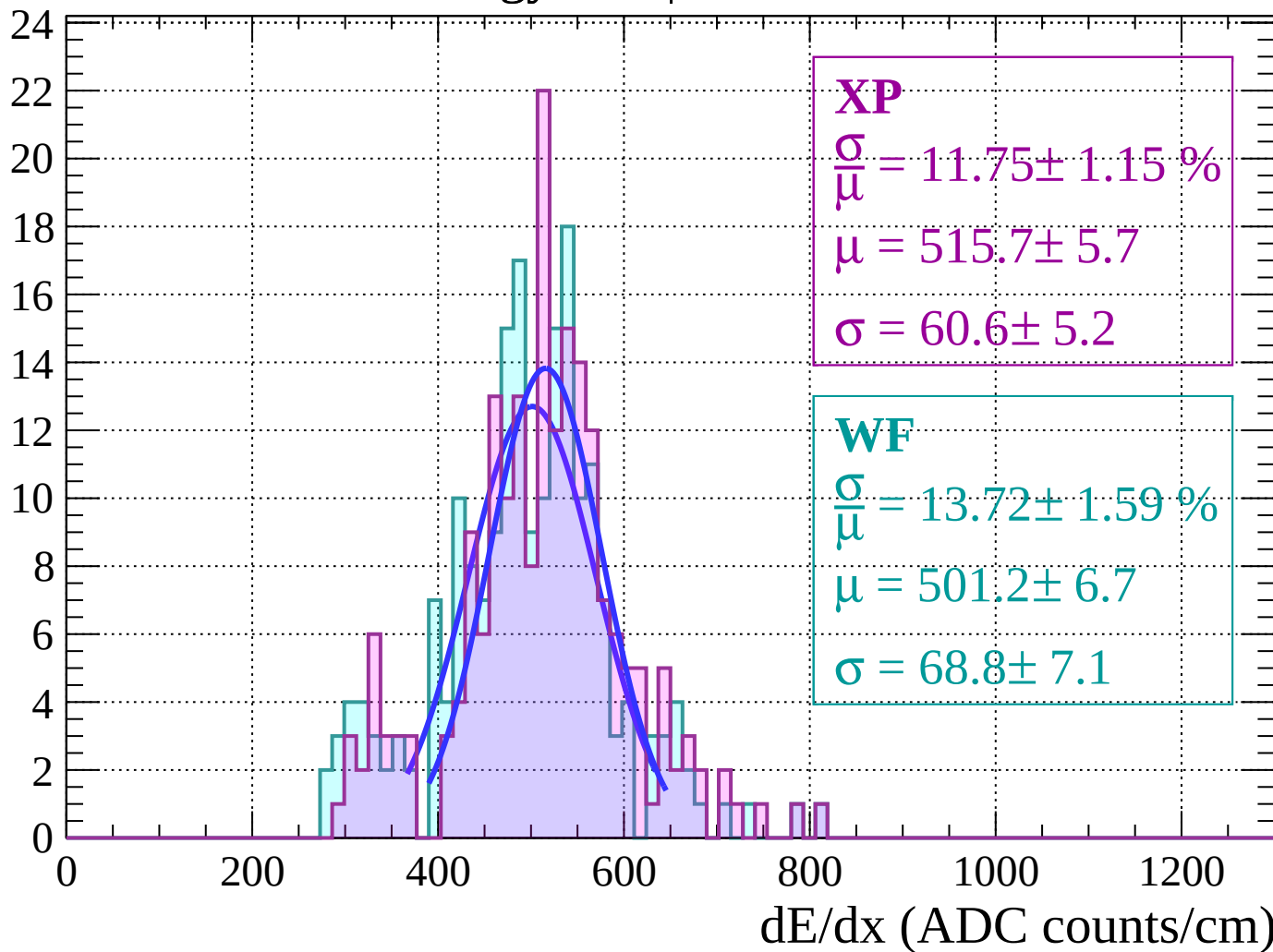
Energy loss | $48 < \theta < 50$

Count



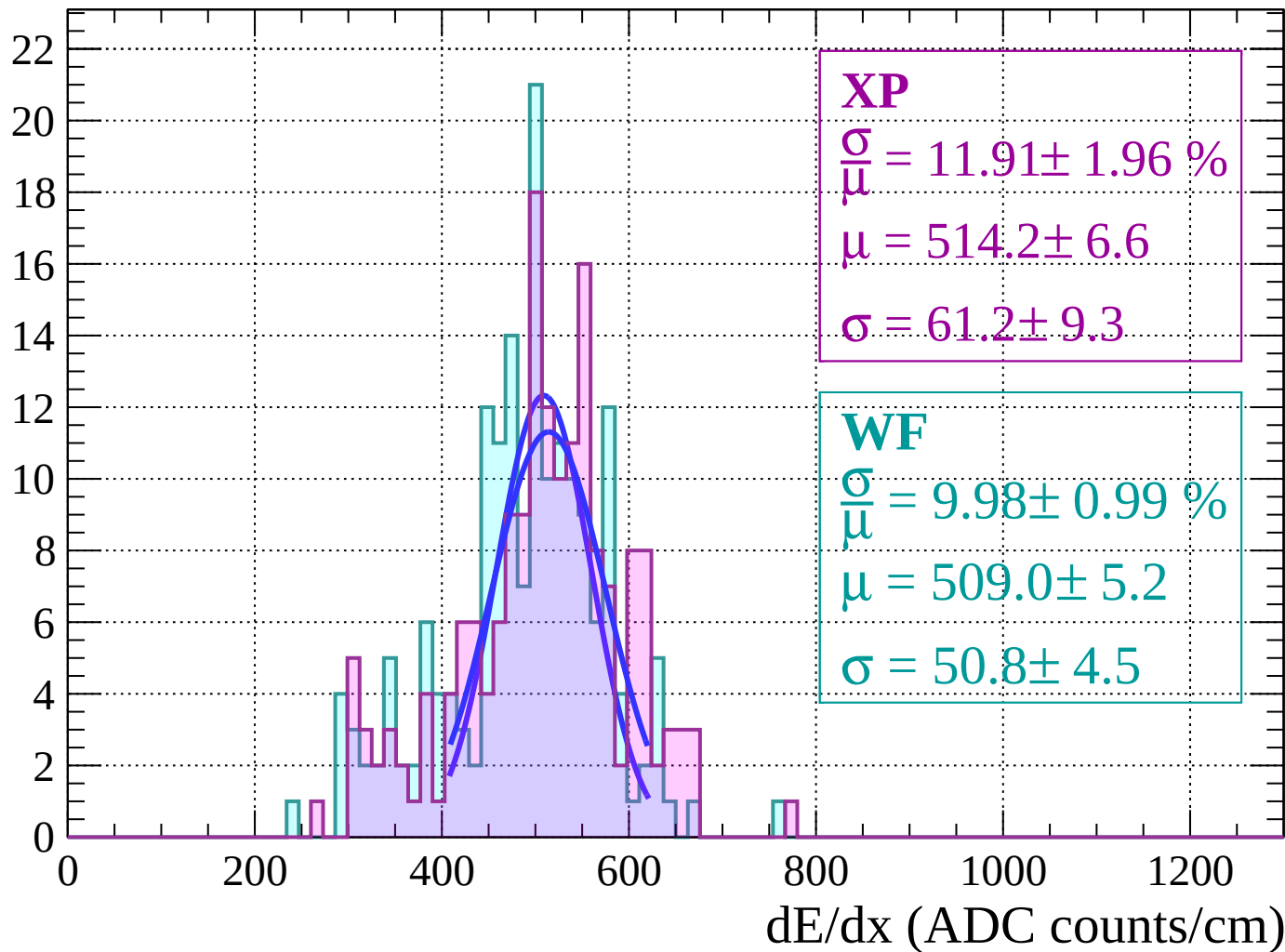
Energy loss | $50 < \theta < 52$

Count



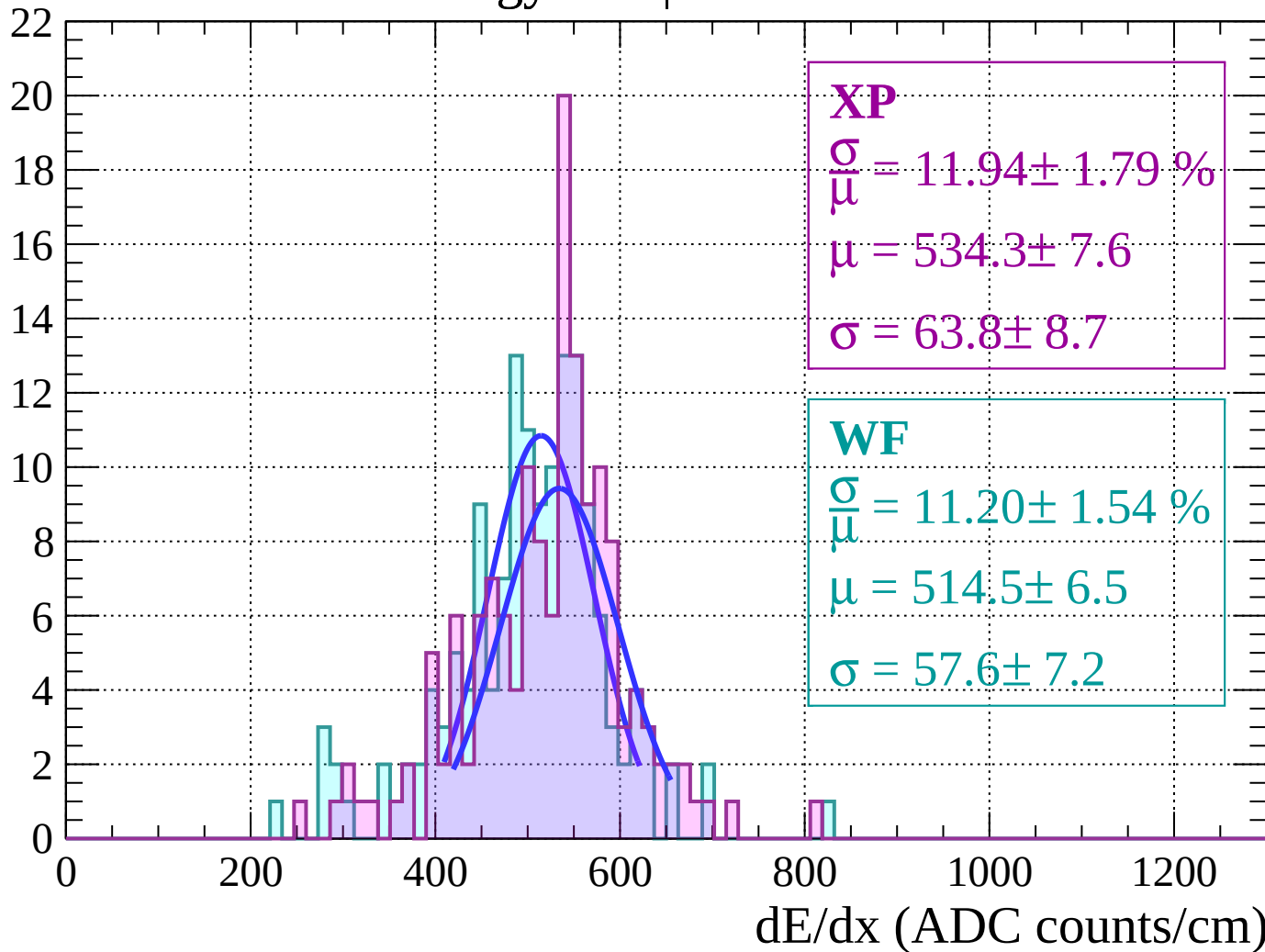
Energy loss | $52 < \theta < 54$

Count



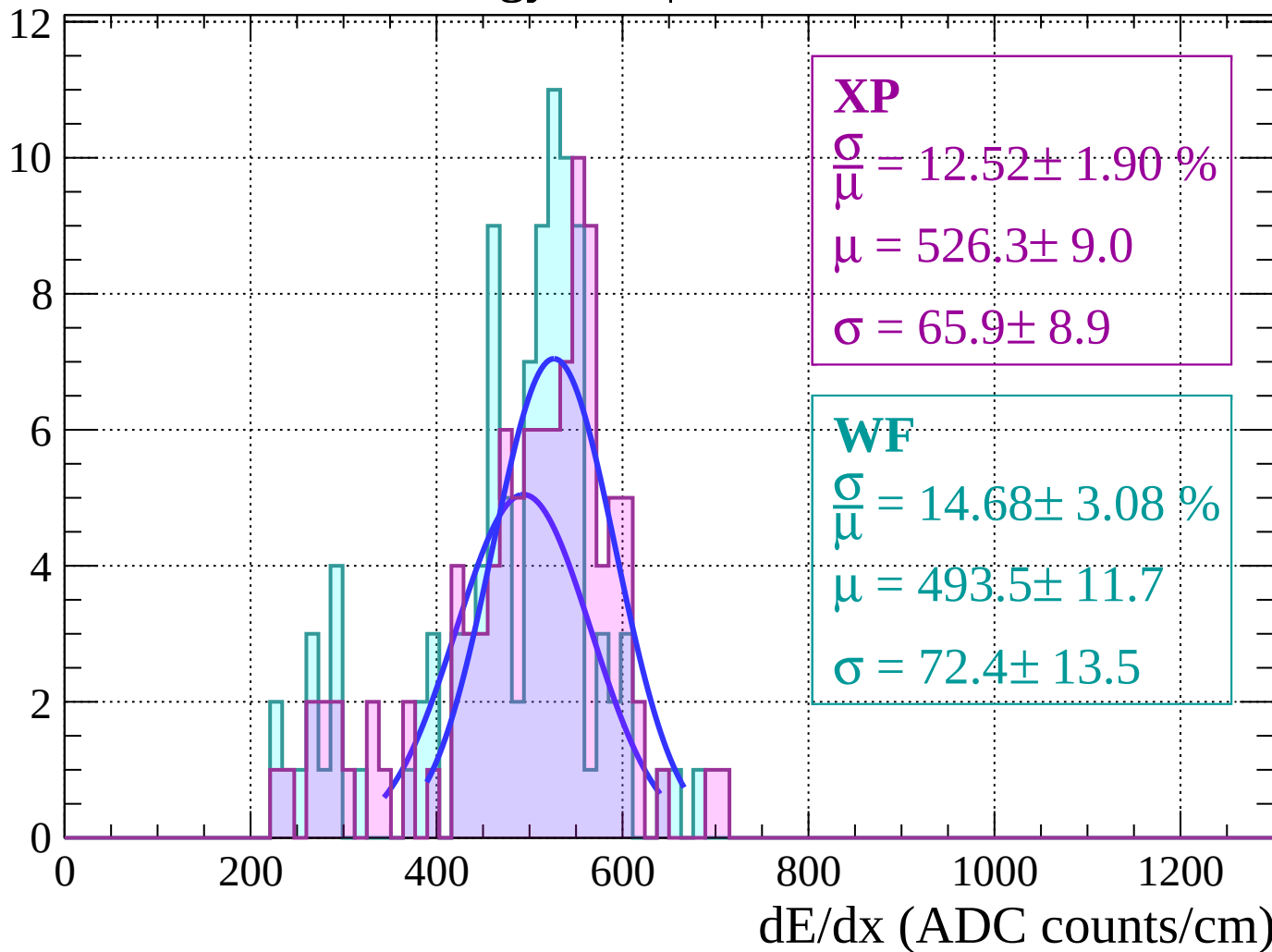
Energy loss | $54 < \theta < 56$

Count



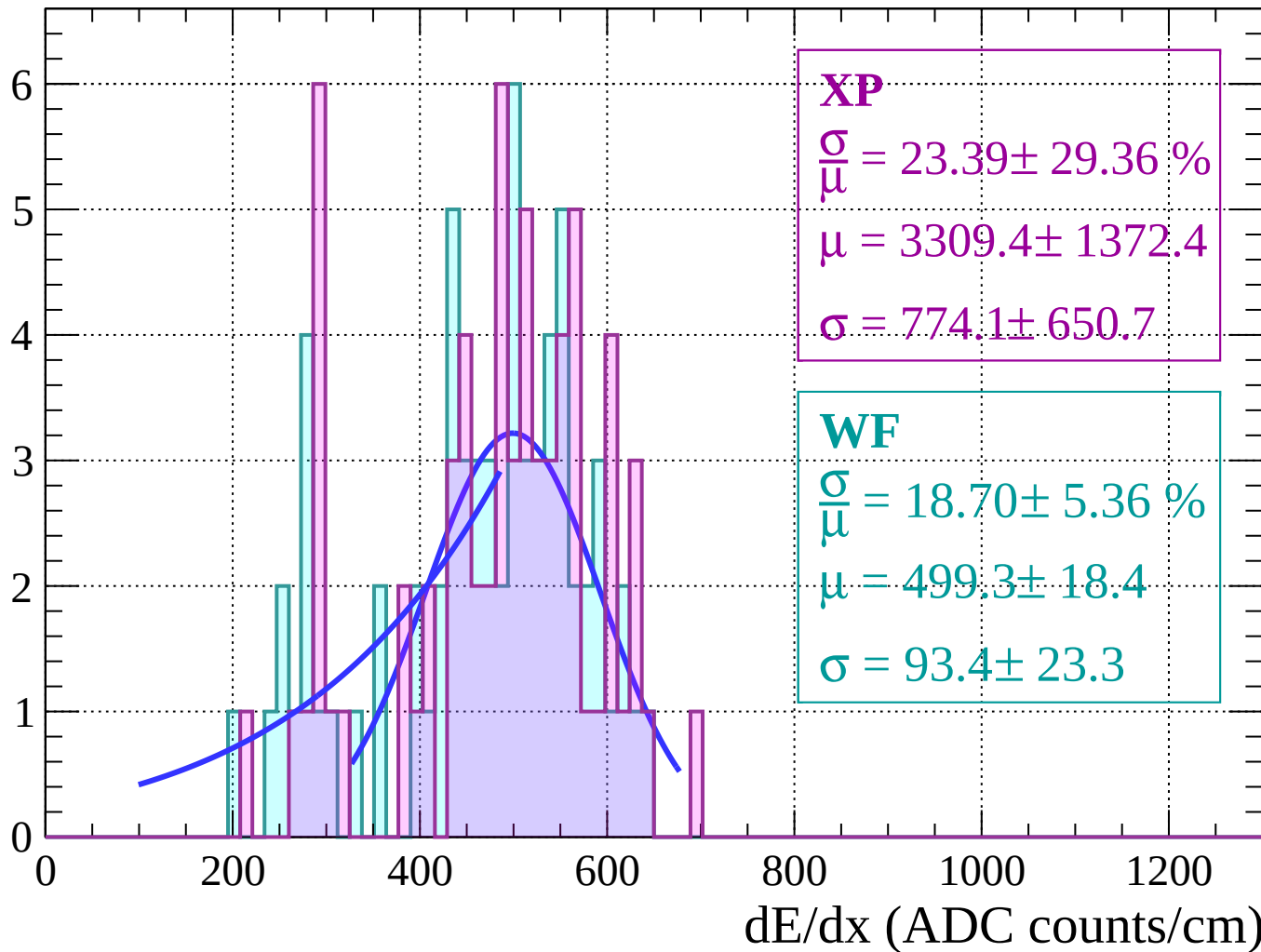
Energy loss | $56 < \theta < 58$

Count



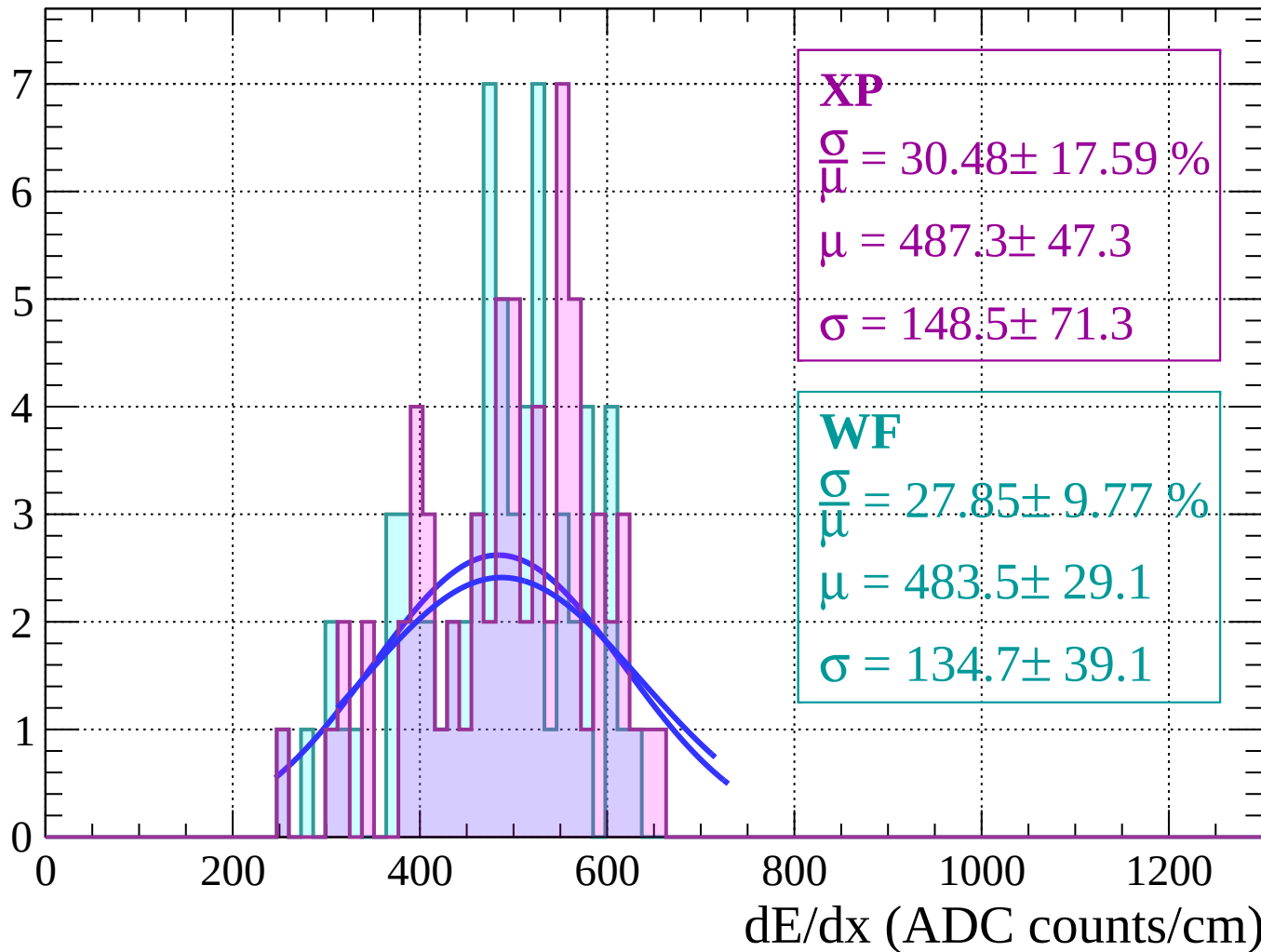
Energy loss | $58 < \theta < 60$

Count



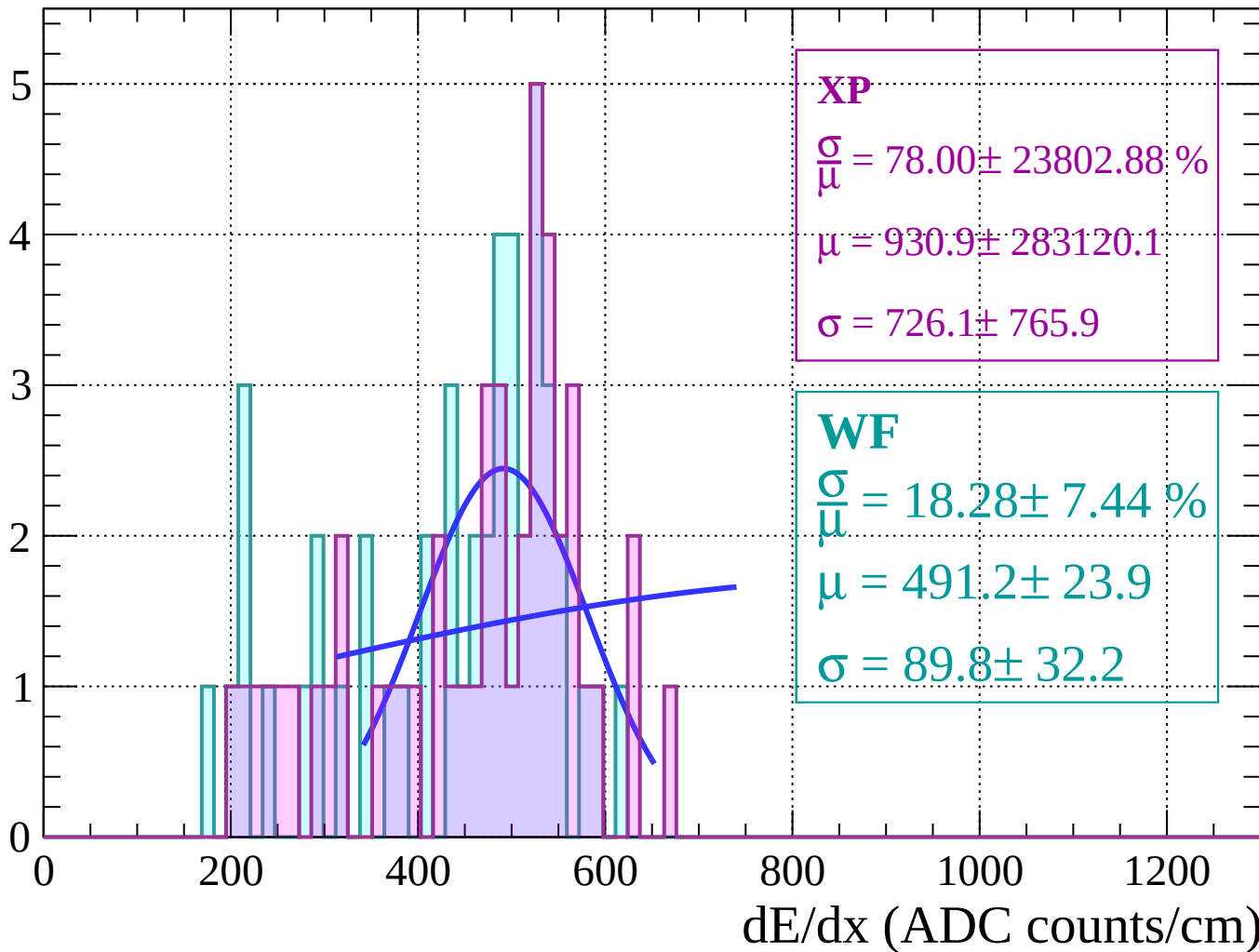
Energy loss | $60 < \theta < 62$

Count



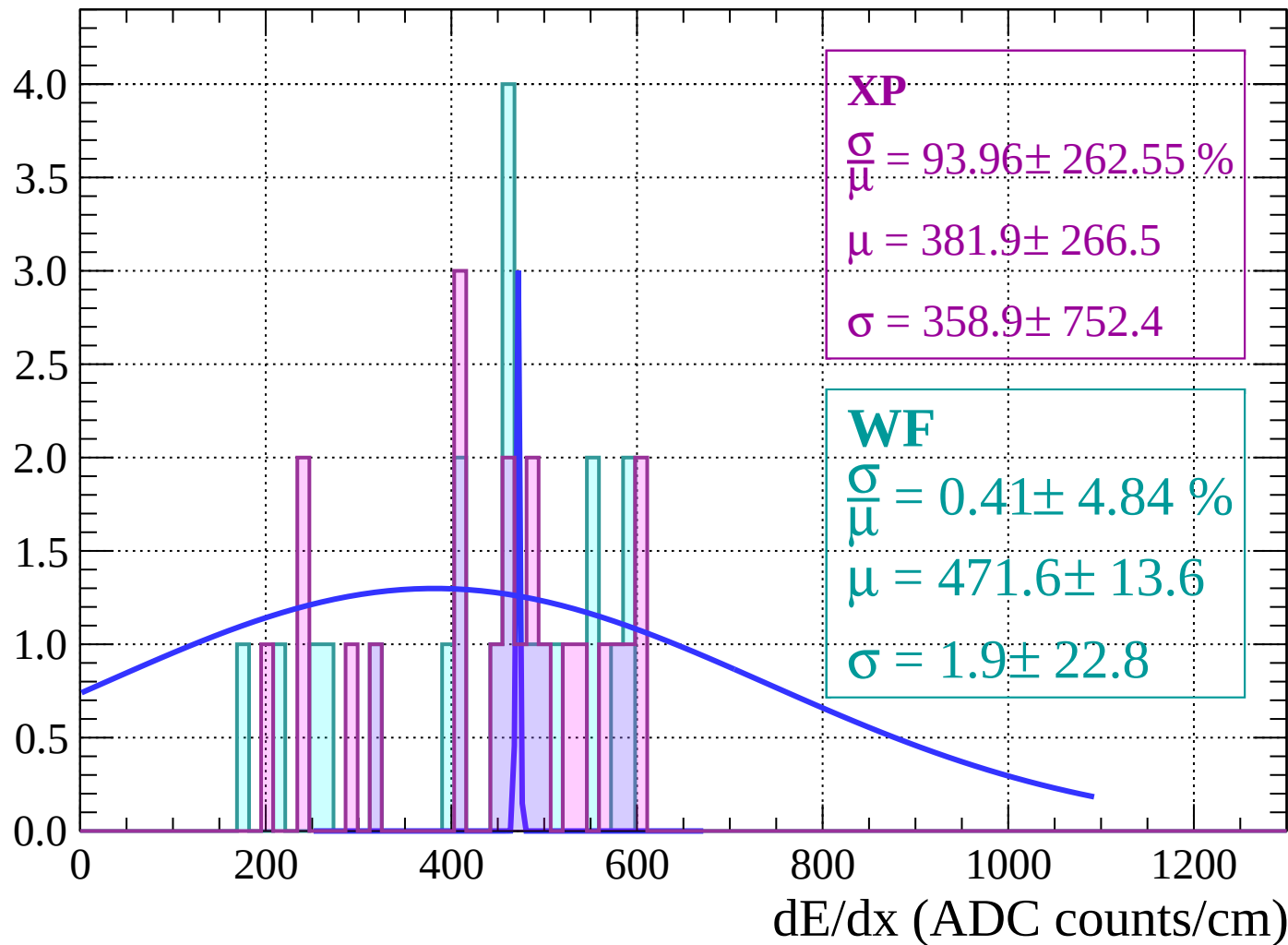
Energy loss | $62 < \theta < 64$

Count



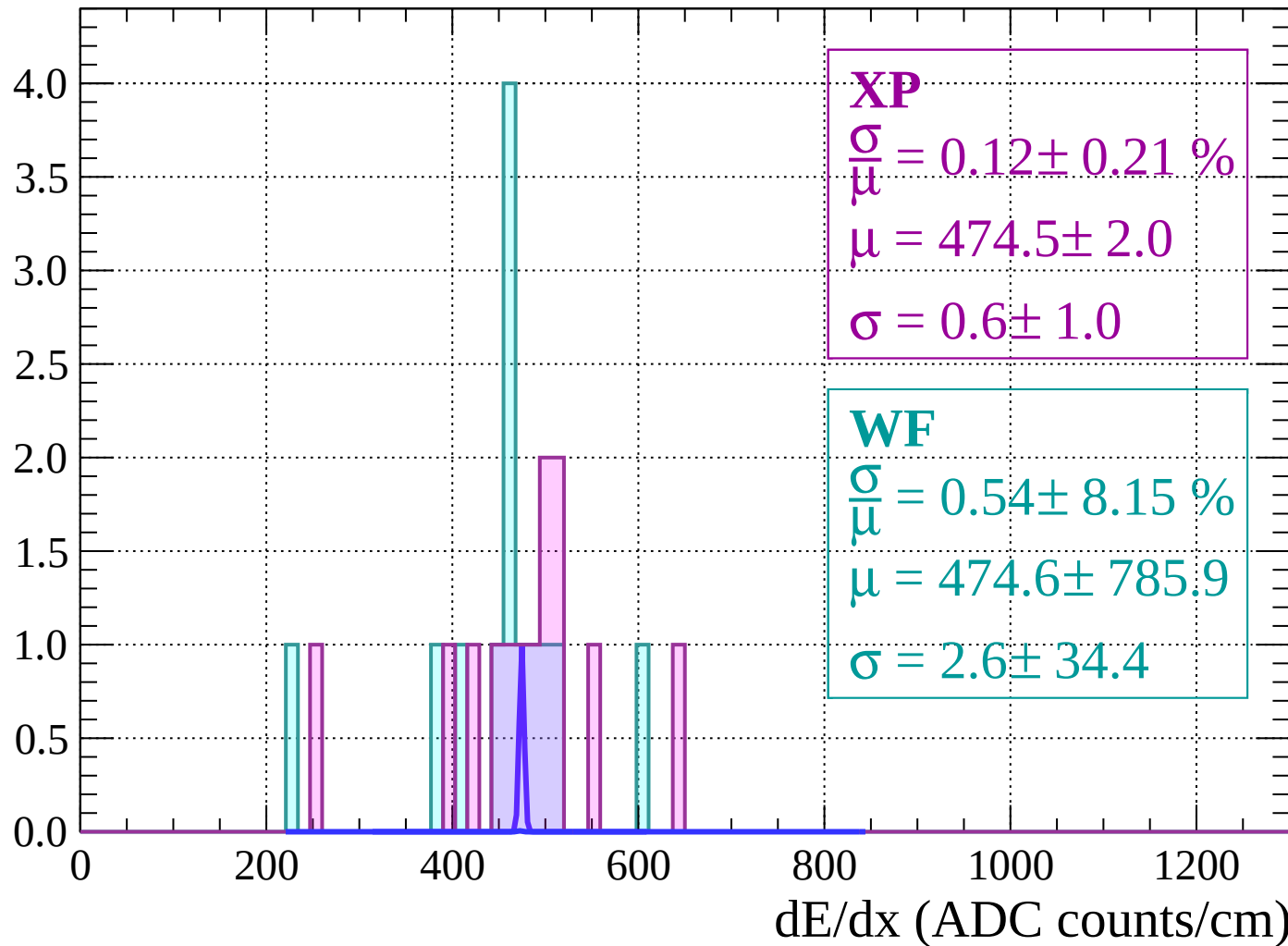
Energy loss | $64 < \theta < 66$

Count



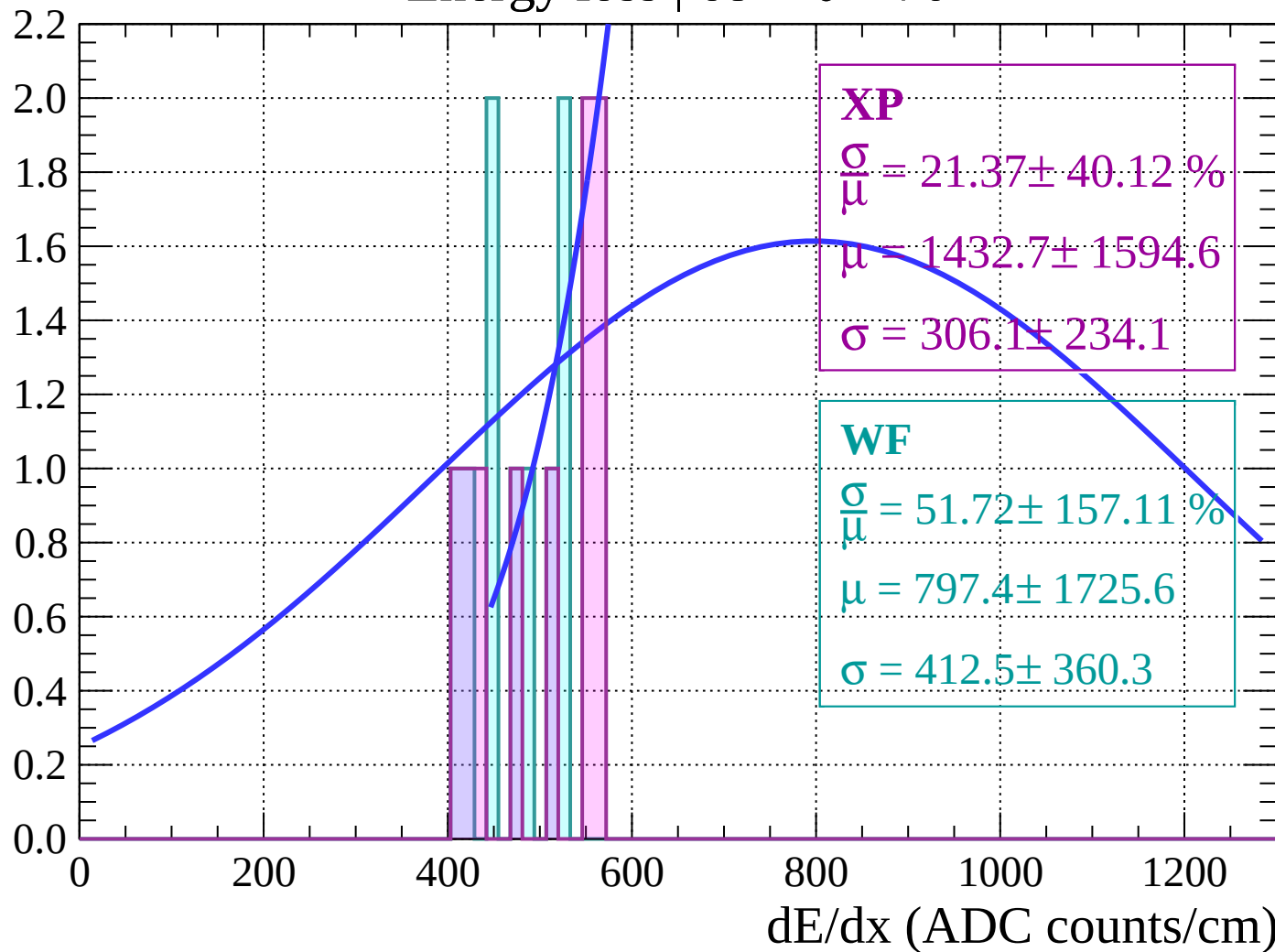
Energy loss | $66 < \theta < 68$

Count



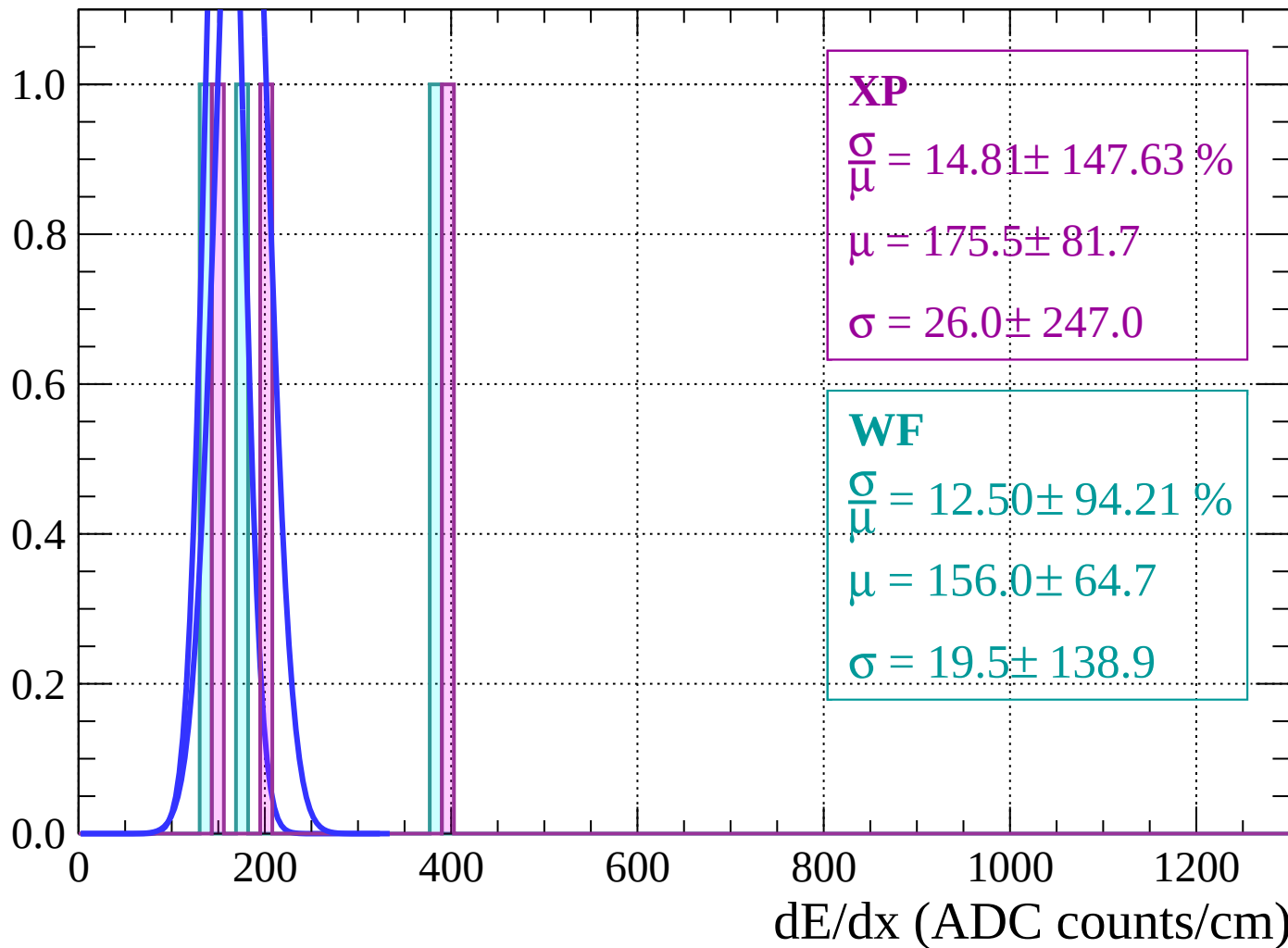
Energy loss | $68 < \theta < 70$

Count



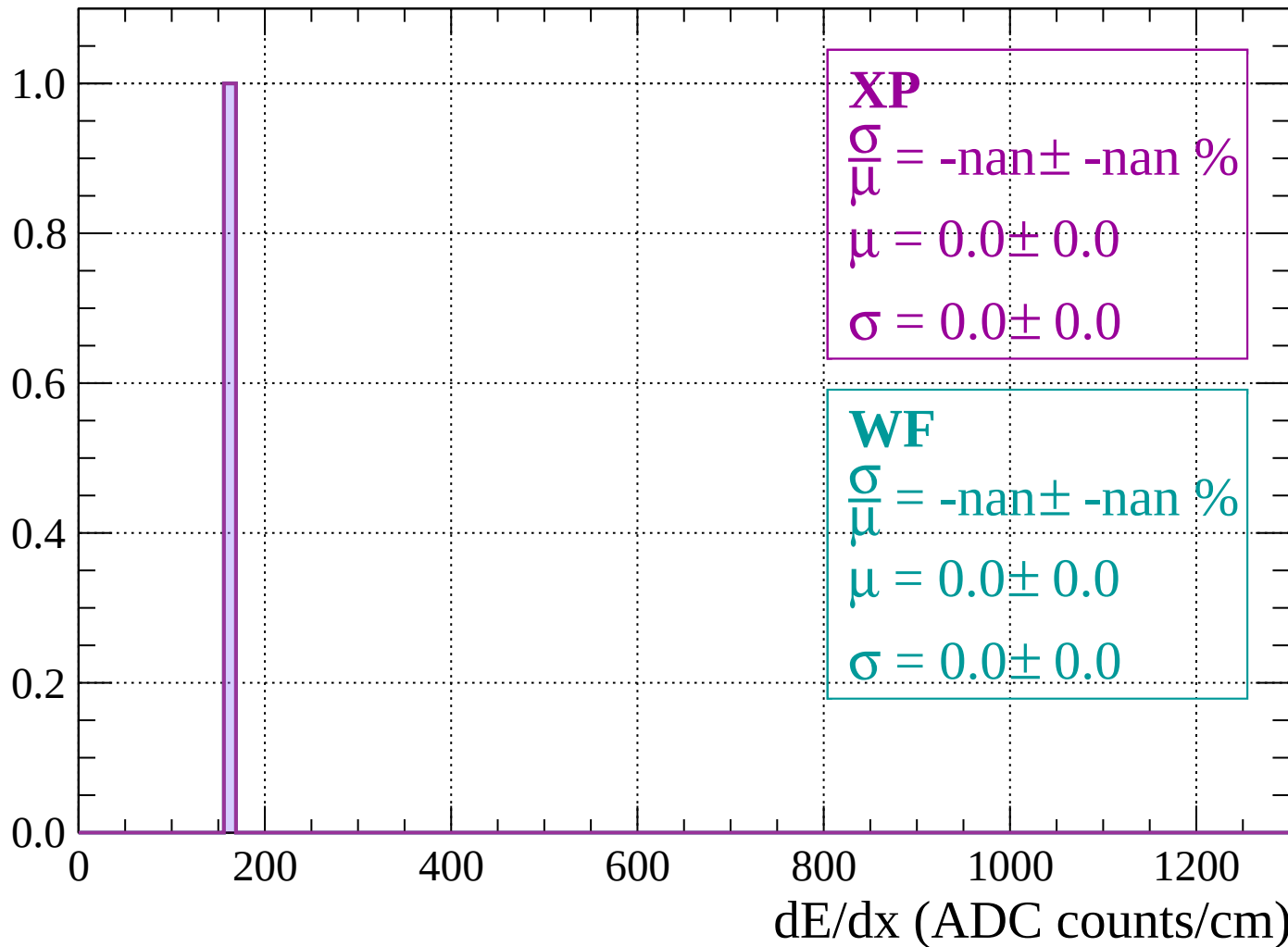
Energy loss | $70 < \theta < 72$

Count



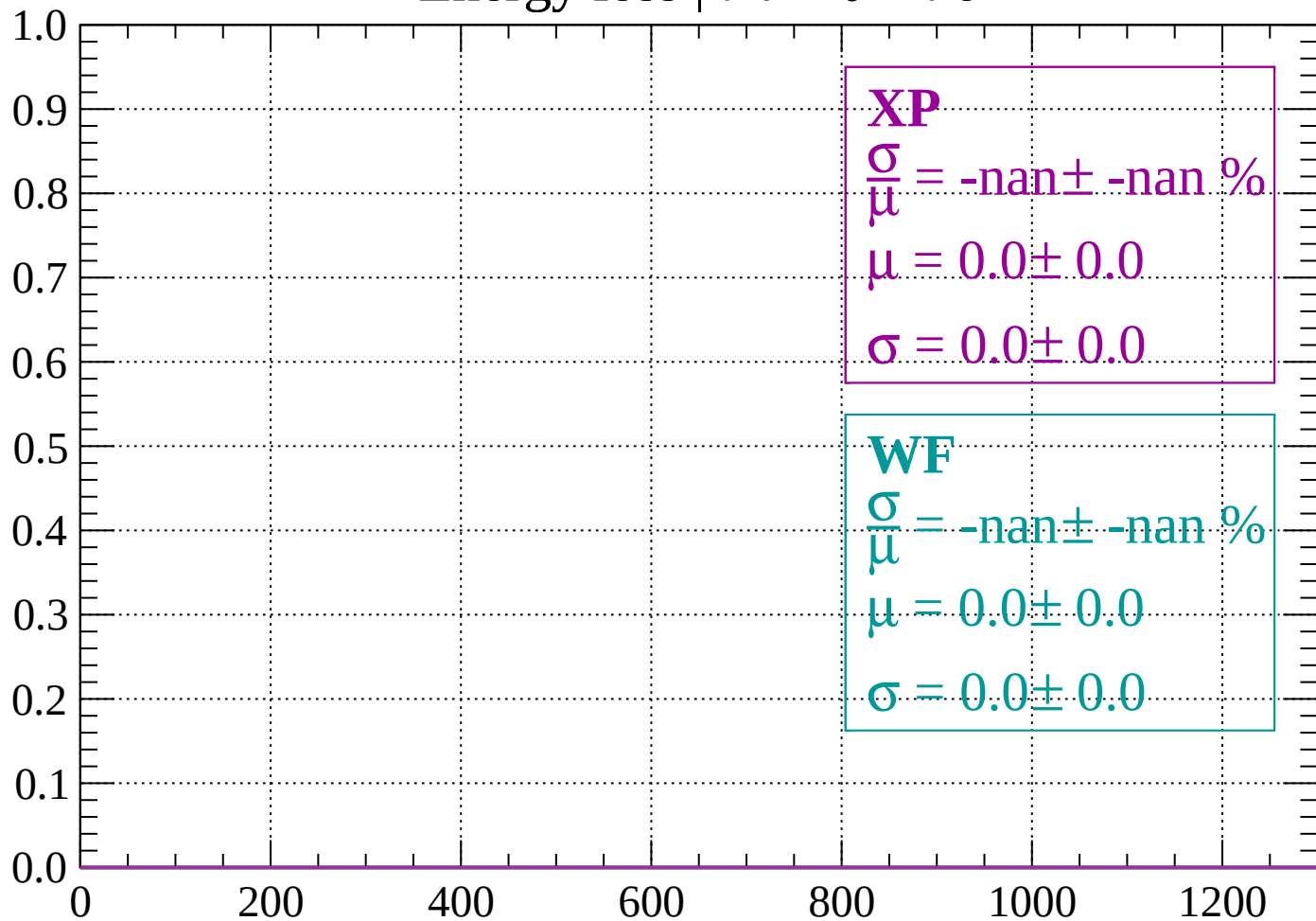
Energy loss | $72 < \theta < 74$

Count



Energy loss | $74 < \theta < 76$

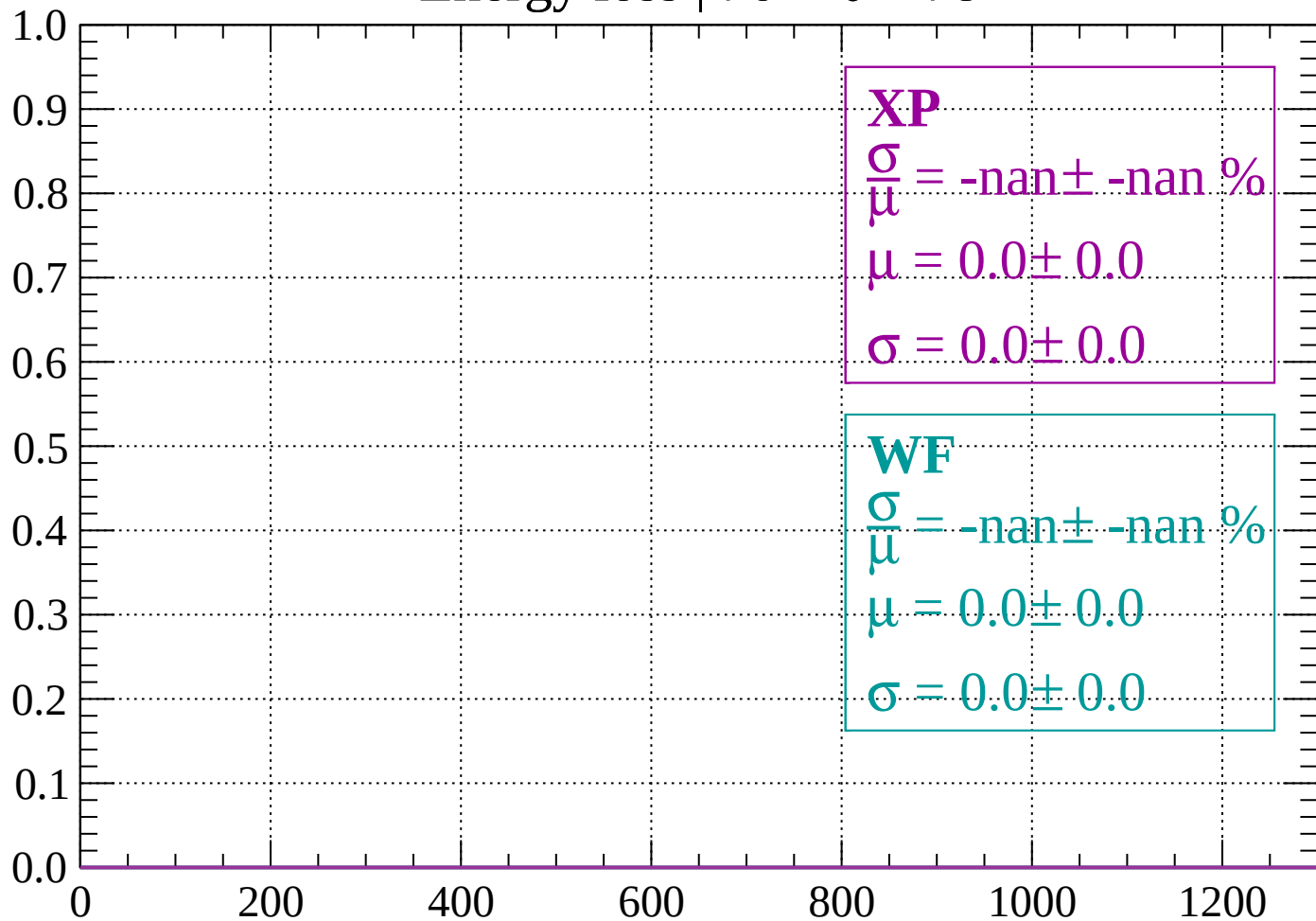
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

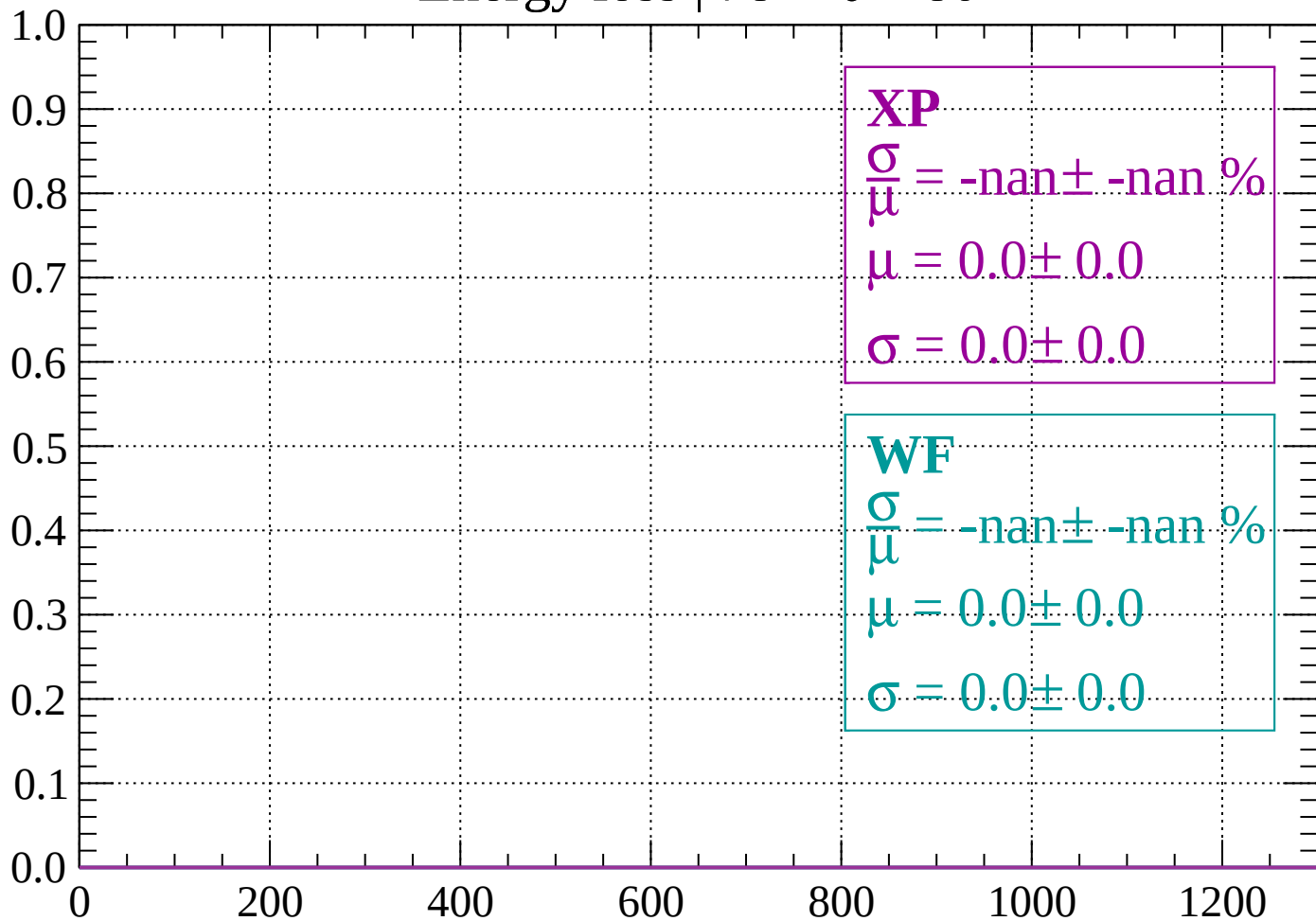
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

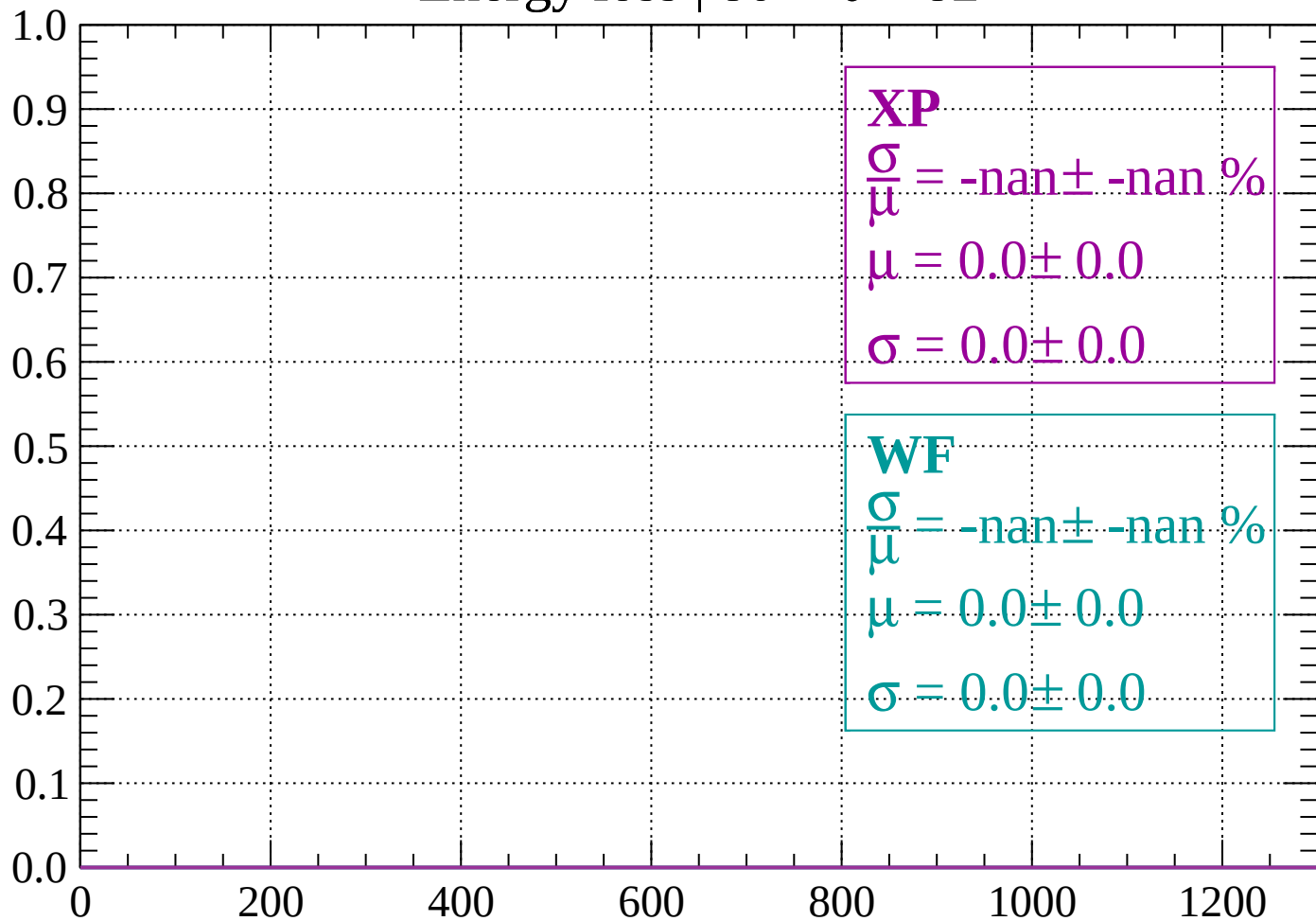
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

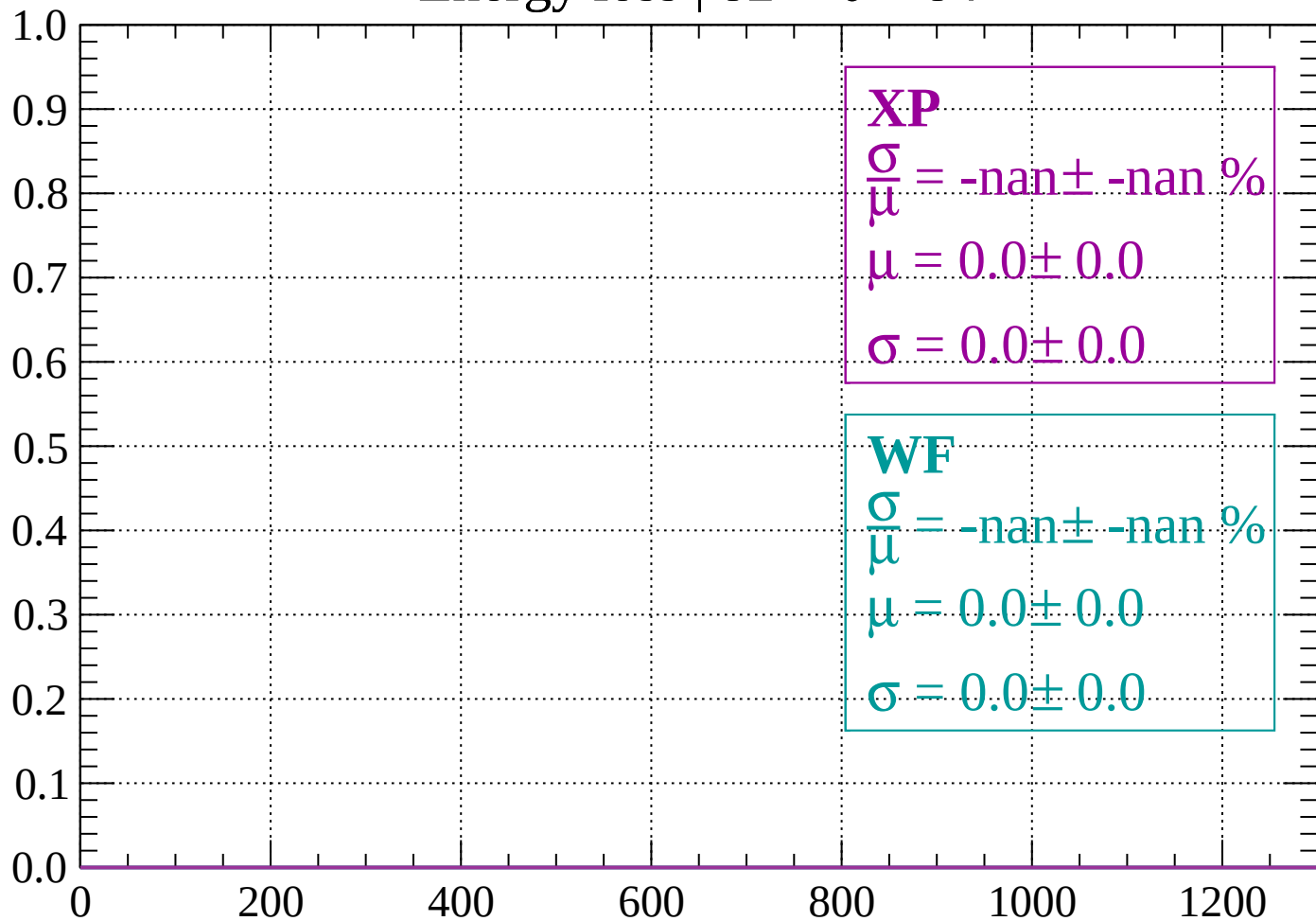
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

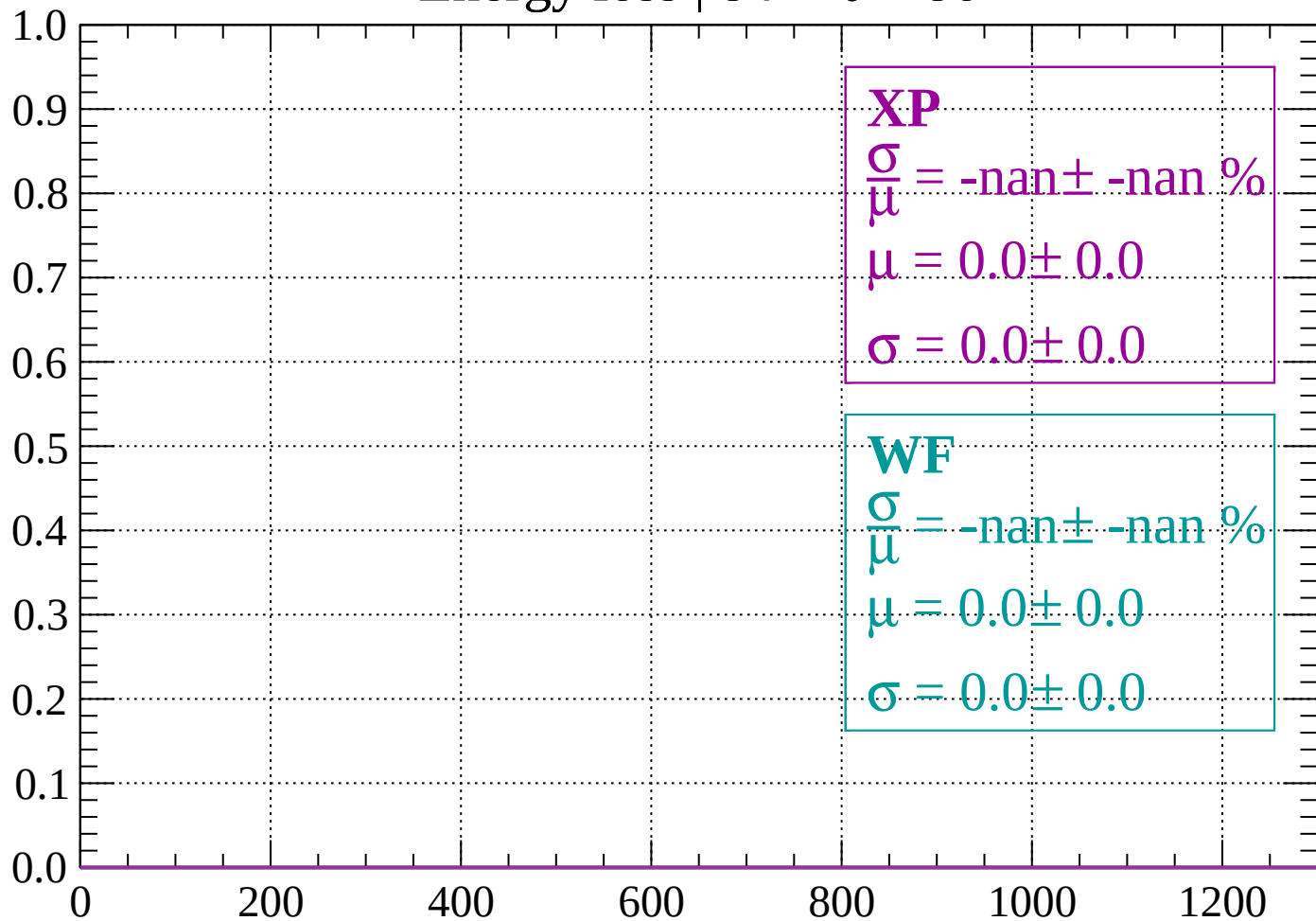
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

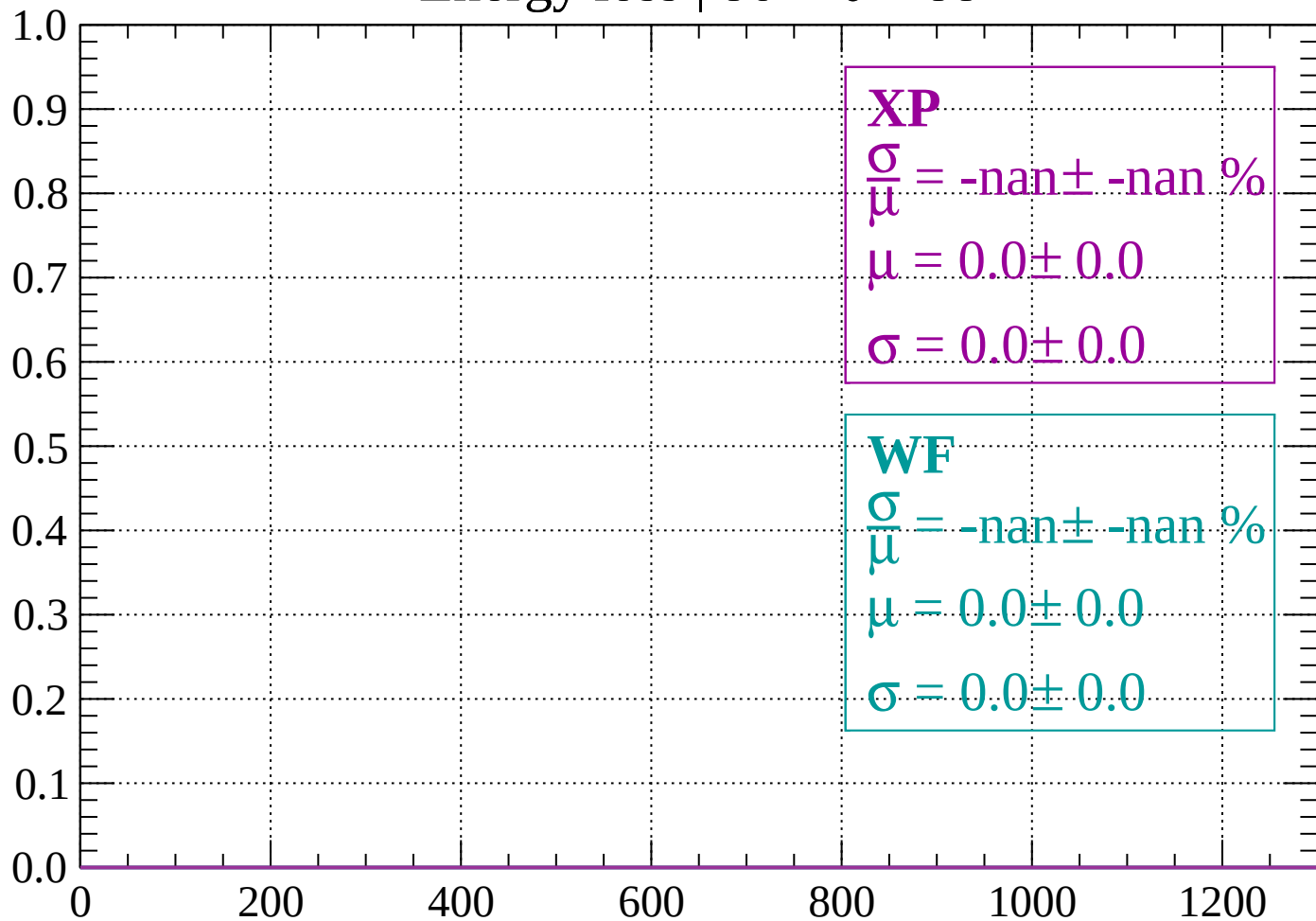
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

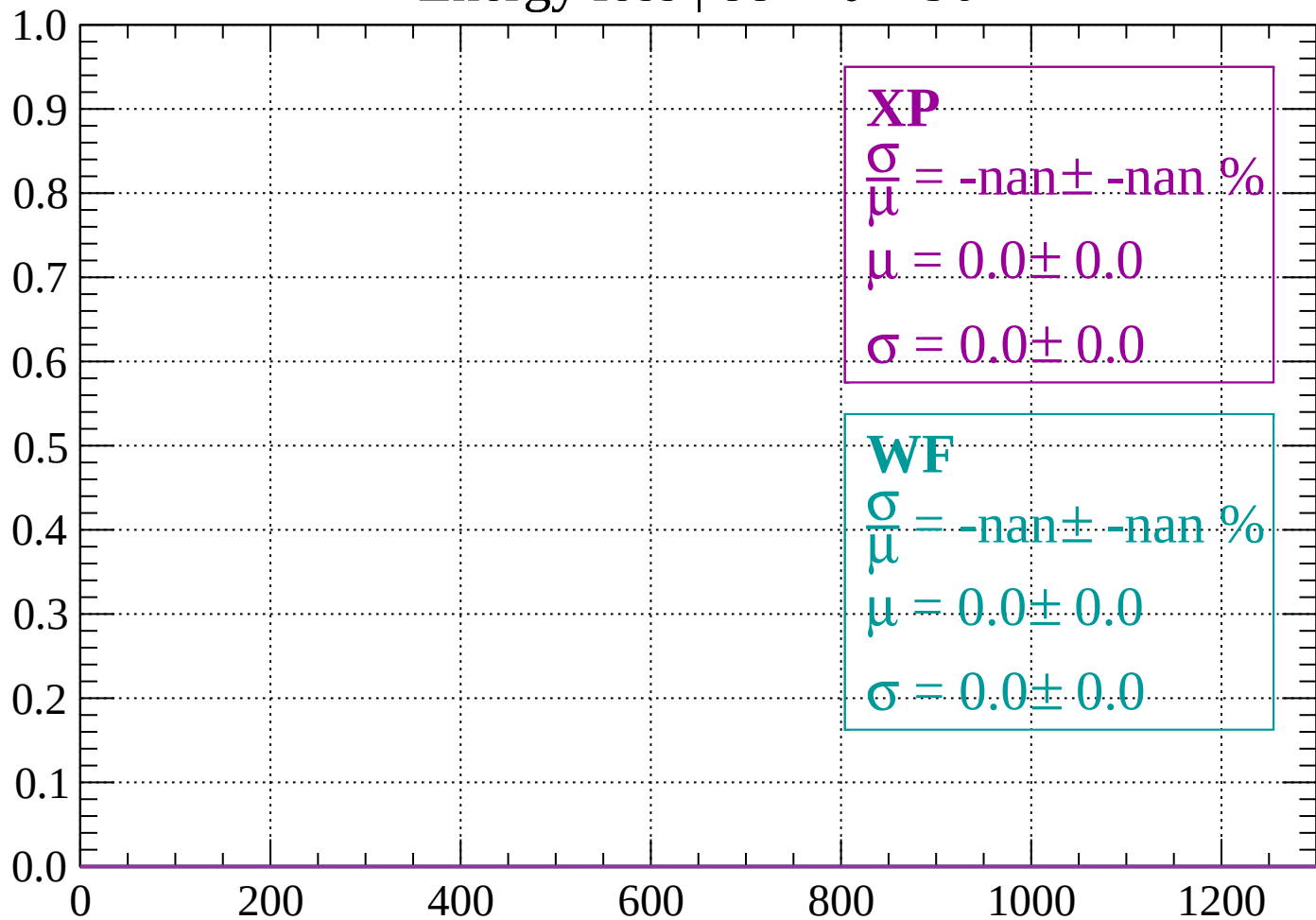
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

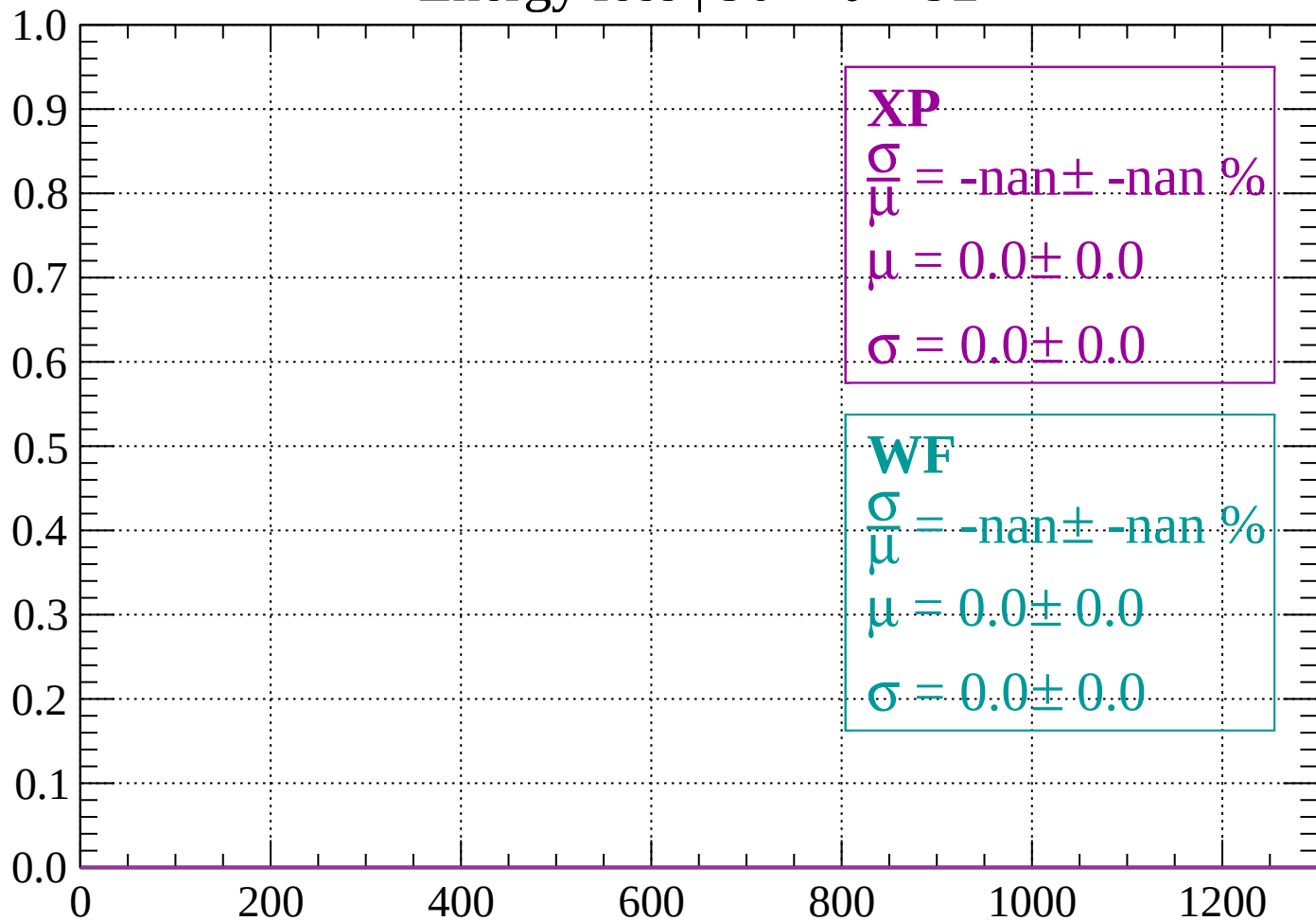
Count



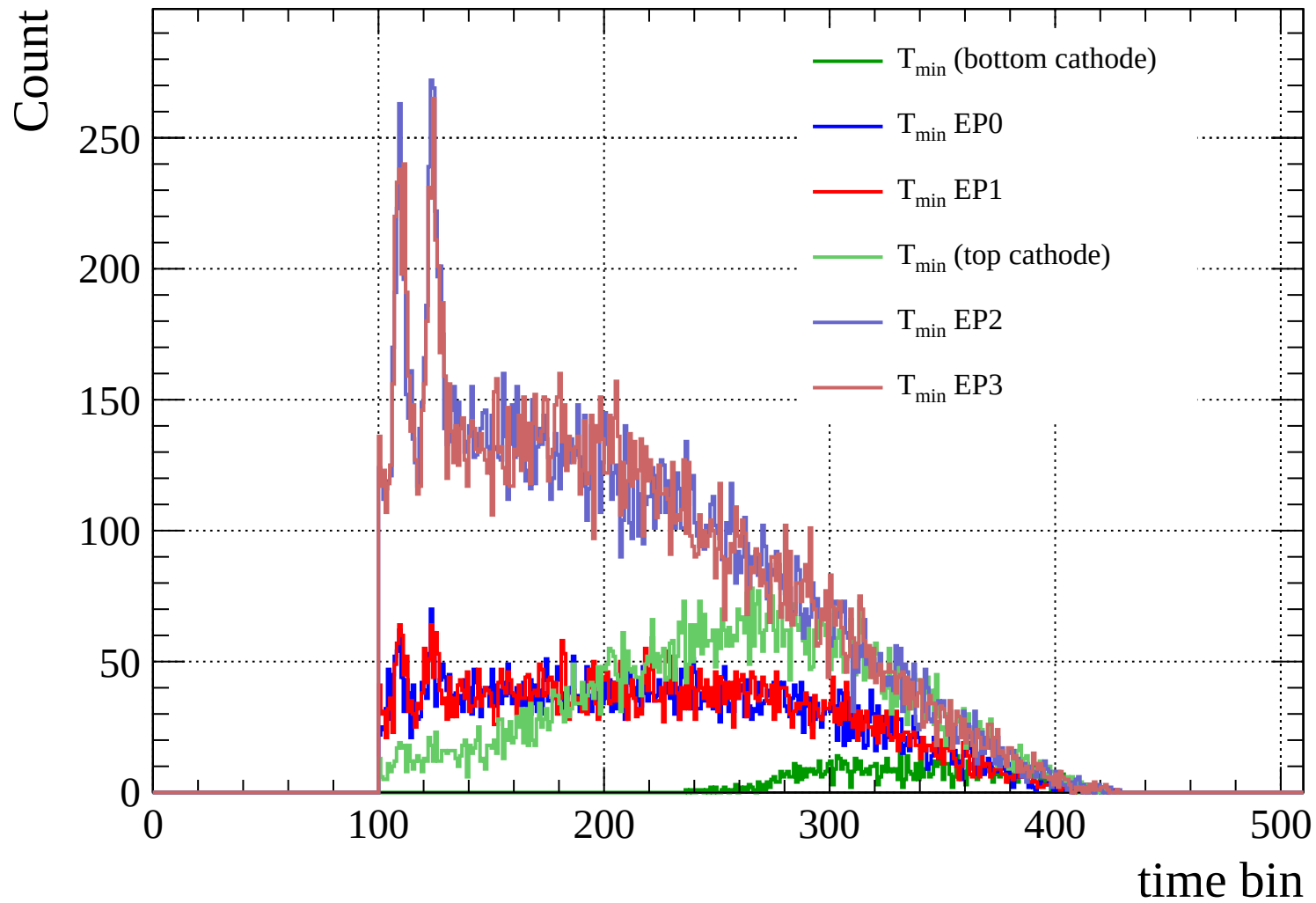
dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)



Count

